

Algebraic Curves — Units 1–30

Thirty lectures and worksheets, independent English edition

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This is an independent English edition of the complete classical 30-unit sequence of Holger Brenner’s *Algebraische Kurven*. Its scope is thirty lectures and thirty worksheets, together with every public solution available at the frozen source boundary. The source material comes from two distinct course boundaries. Units 1–23 follow the frozen revisions of *Algebraische Kurven* (Osnabrück 2025–2026). At the frozen boundary, that course has no Unit 24 or later units. Units 24–30 therefore follow the official lectures and worksheets of the earlier complete course, *Algebraische Kurven* (Osnabrück 2012). They are not an official continuation of the 2025–2026 edition, and the two courses are not treated as one source edition.

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canonical binary endpoints, so no local bytes or SHA-256 are claimed. The controlling semantic pages for Unit 30 were updated in 2026; the edition records that difference and does not claim that the two surfaces have identical revisions.

How to use this edition

Read each lecture, then work through the worksheet with the same number. Units 1–30 contain 693 exercises. Stars follow the source and do not by themselves mean that all starred exercises have the same solution coverage. The translation scope includes all 122 public solutions available at the frozen source boundary. Unit 30 has public solutions only for Exercises 30.3 and 30.4; missing source solutions are not invented. Formulae, numbering, point values, hints, stars and unit order are preserved so that the edition can be checked against the respective source revisions.

Edition notes openly identify corrections to source errors or omissions that would otherwise mislead the reader. Necessary additions bridging a proof step or prerequisite are labelled `Editorial bridge`; these additions are not part of the source text and are not attributed to the source author.

Lecture 1: Plane Algebraic Curves

Plane algebraic curves

What is an algebraic curve? For example, the objects shown in the following beautiful pictures.



Figure 1: Graph of a linear function

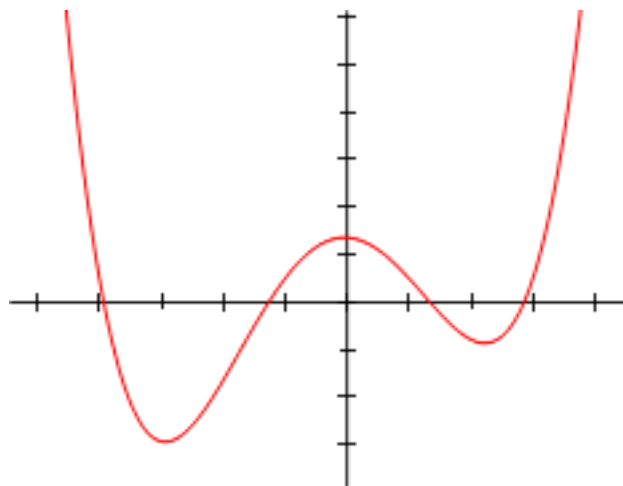


Figure 2: Graph of a polynomial of degree four

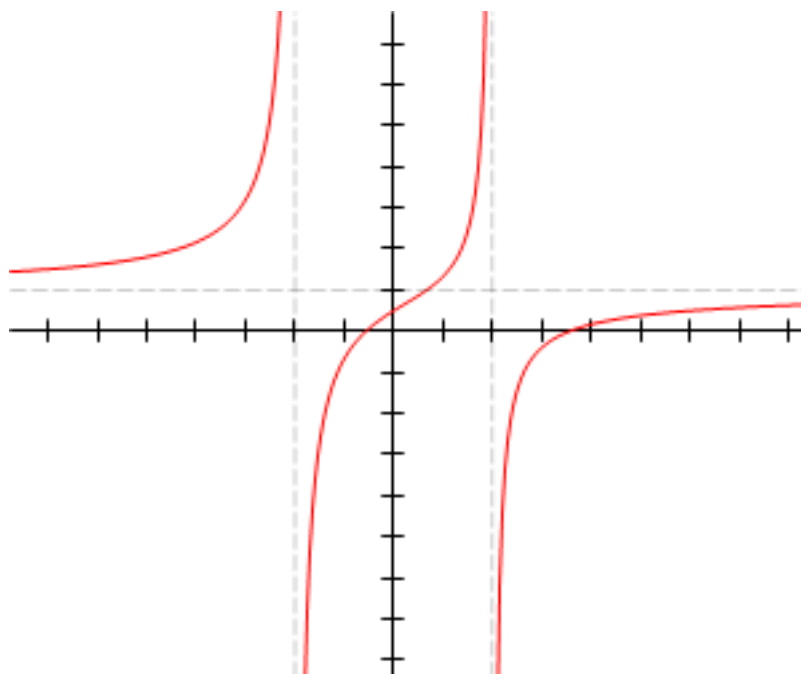


Figure 3: Graph of a rational function

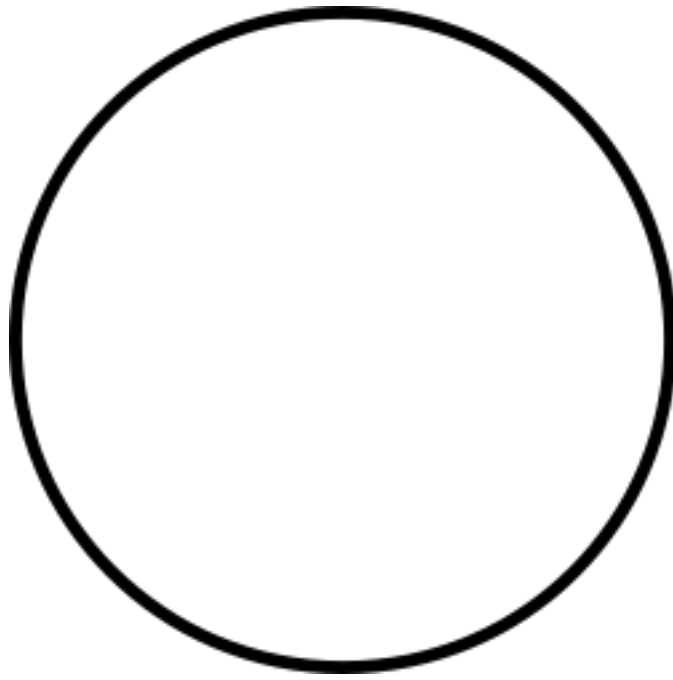


Figure 4: Unit circle



Figure 5: Ellipse

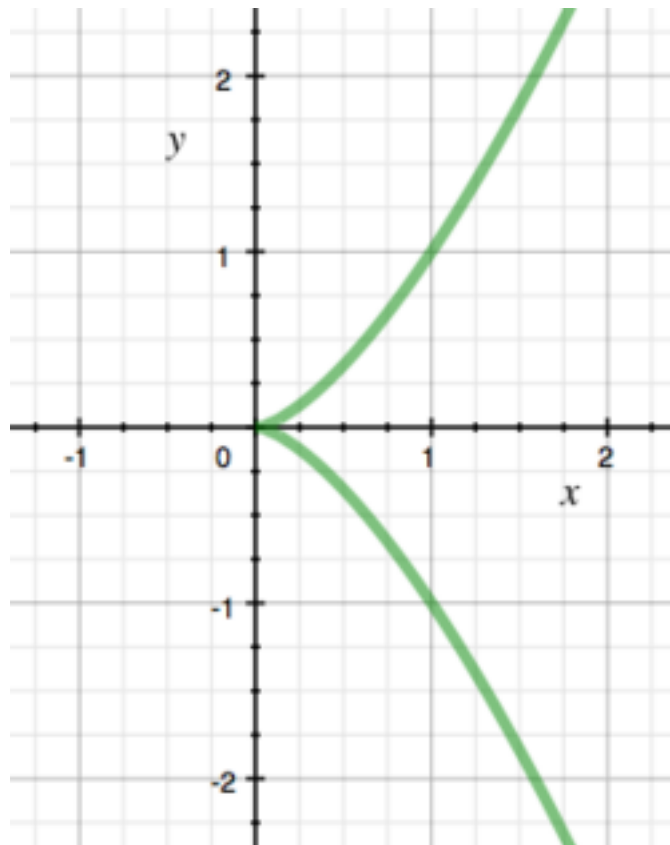


Figure 6: Curve with a cusp

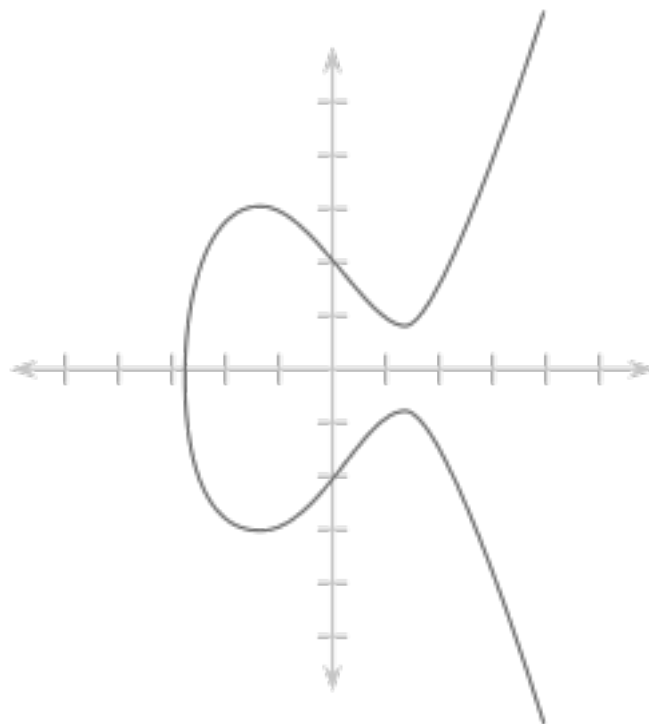


Figure 7: Example of an elliptic curve

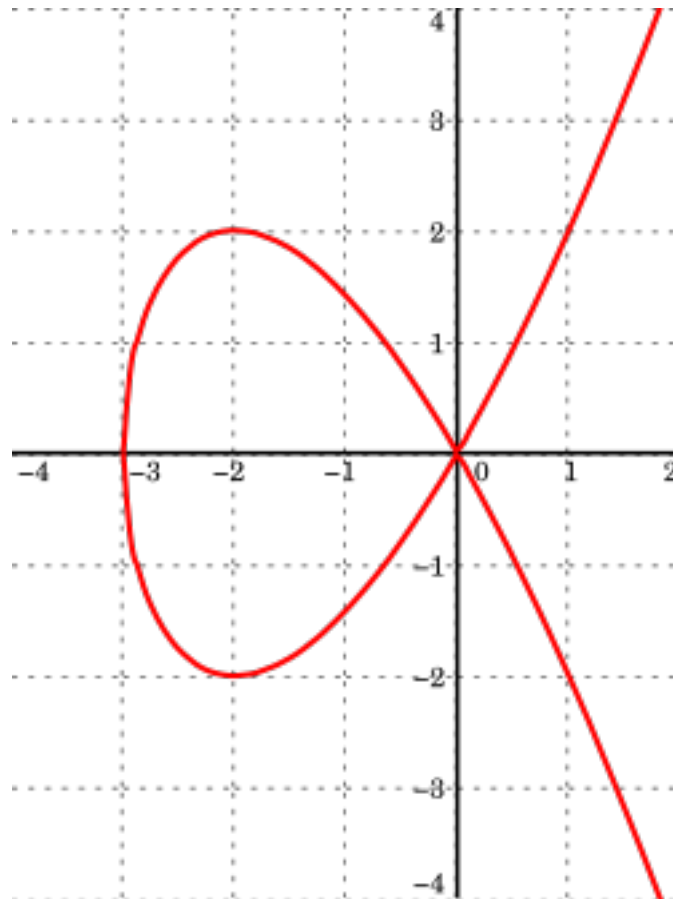


Figure 8: Tschirnhausen cubic

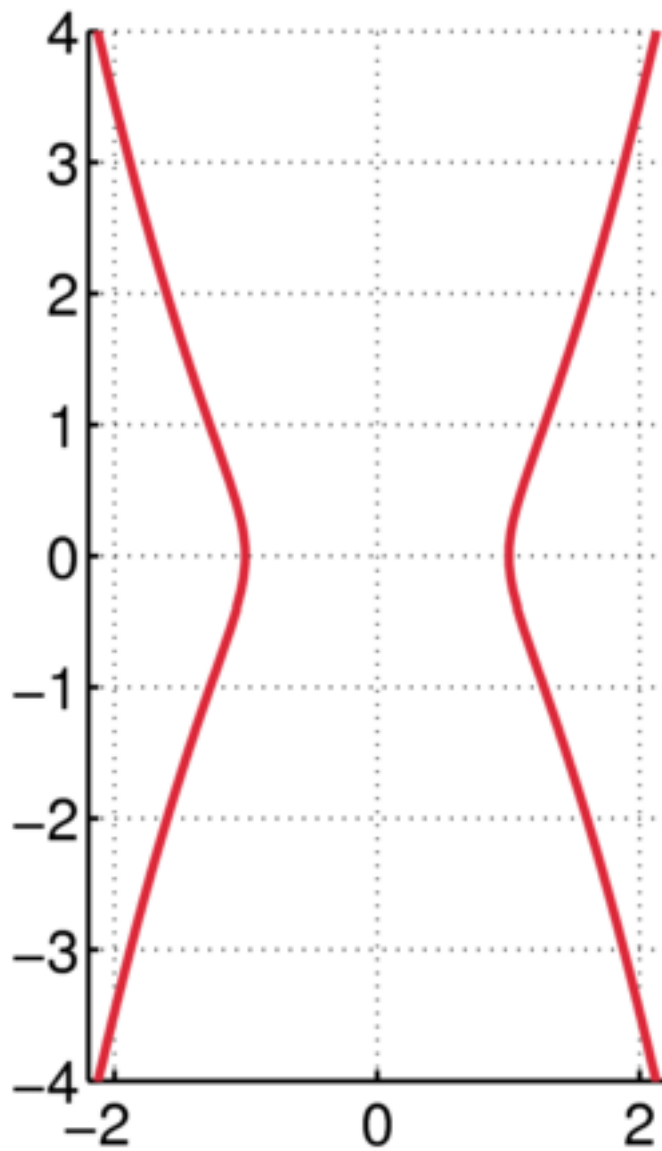


Figure 9: Kampyle of Eudoxus

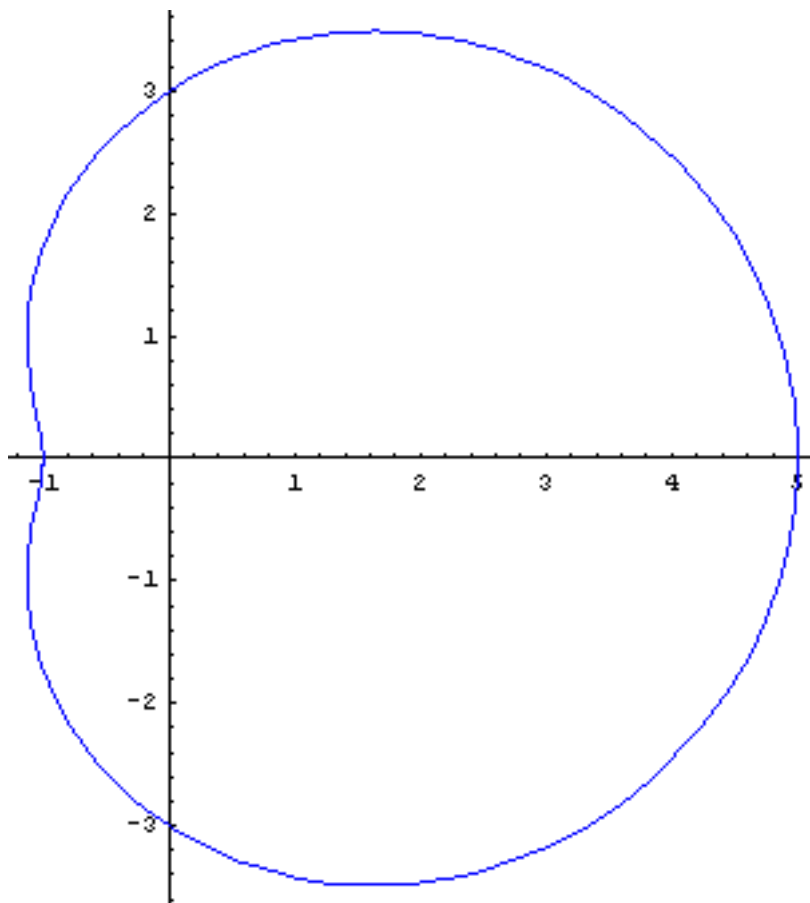


Figure 10: Conchoid of Pascal

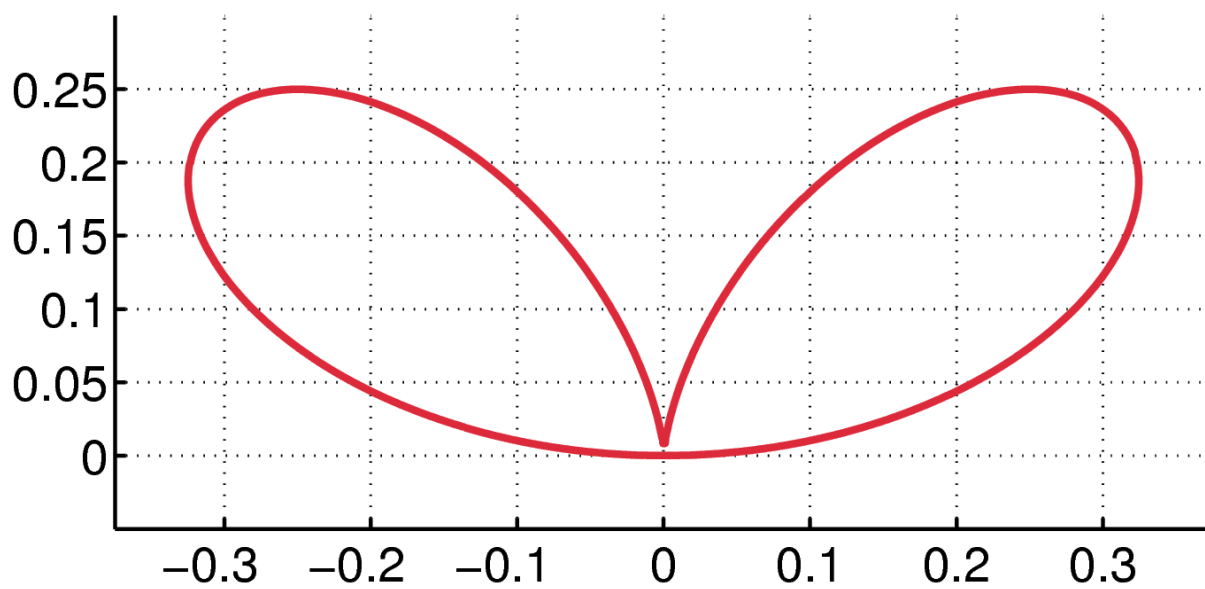


Figure 11: Bifolium

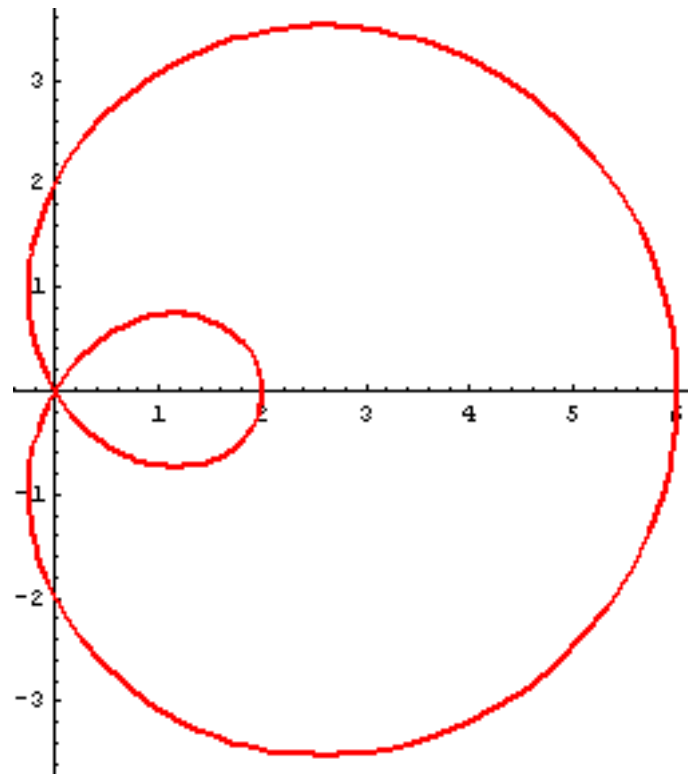


Figure 12: Limaçon

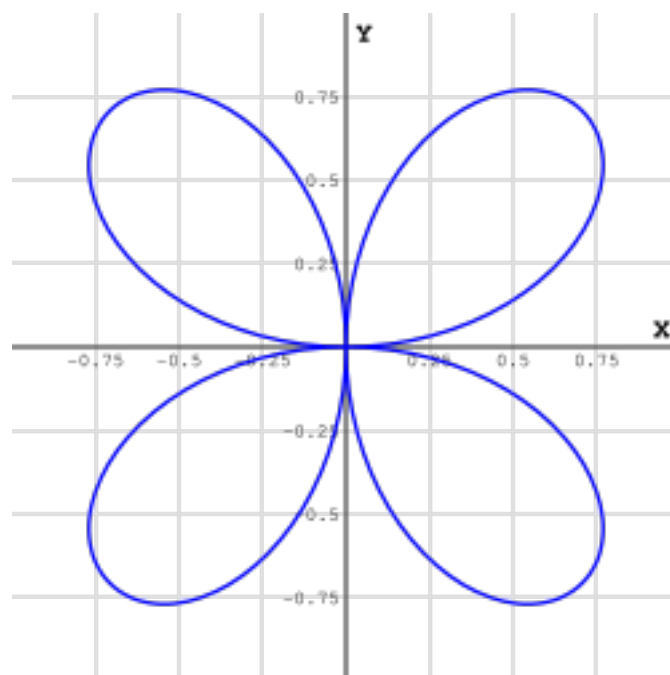


Figure 13: Quadrifolium

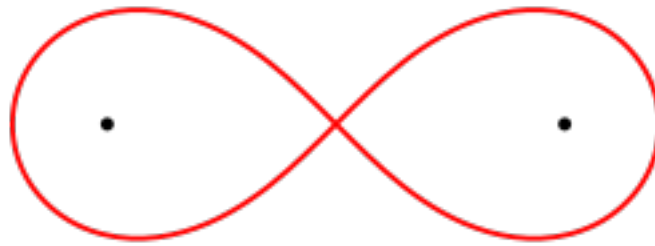


Figure 14: Lemniscate of Bernoulli

Of course, we can draw all sorts of things. The following curves are beautiful too, but they are not algebraic curves.



Figure 15: Cycloid

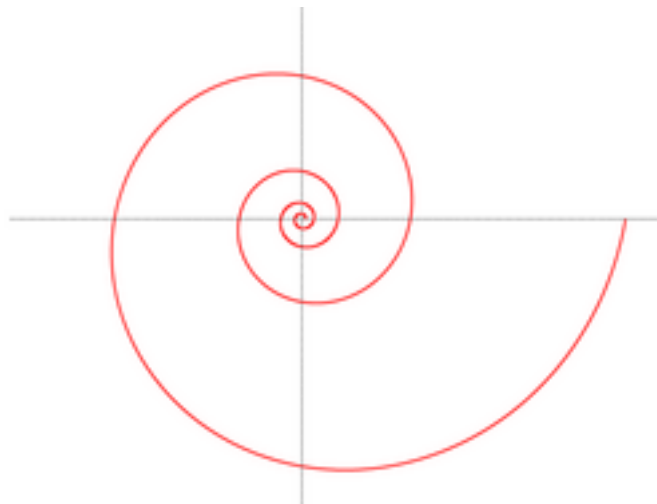


Figure 16: Logarithmic spiral

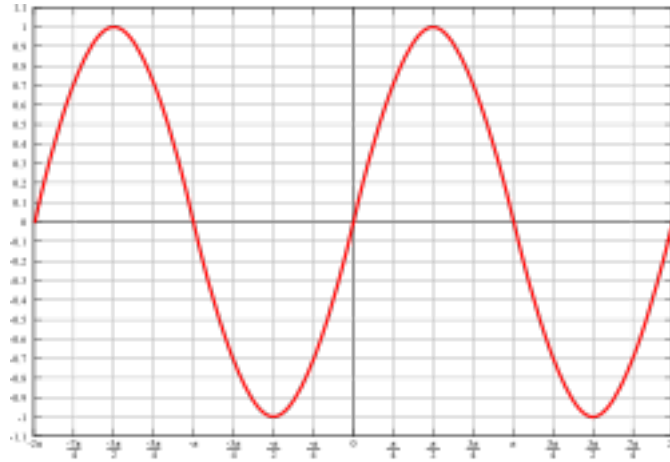


Figure 17: Sine graph

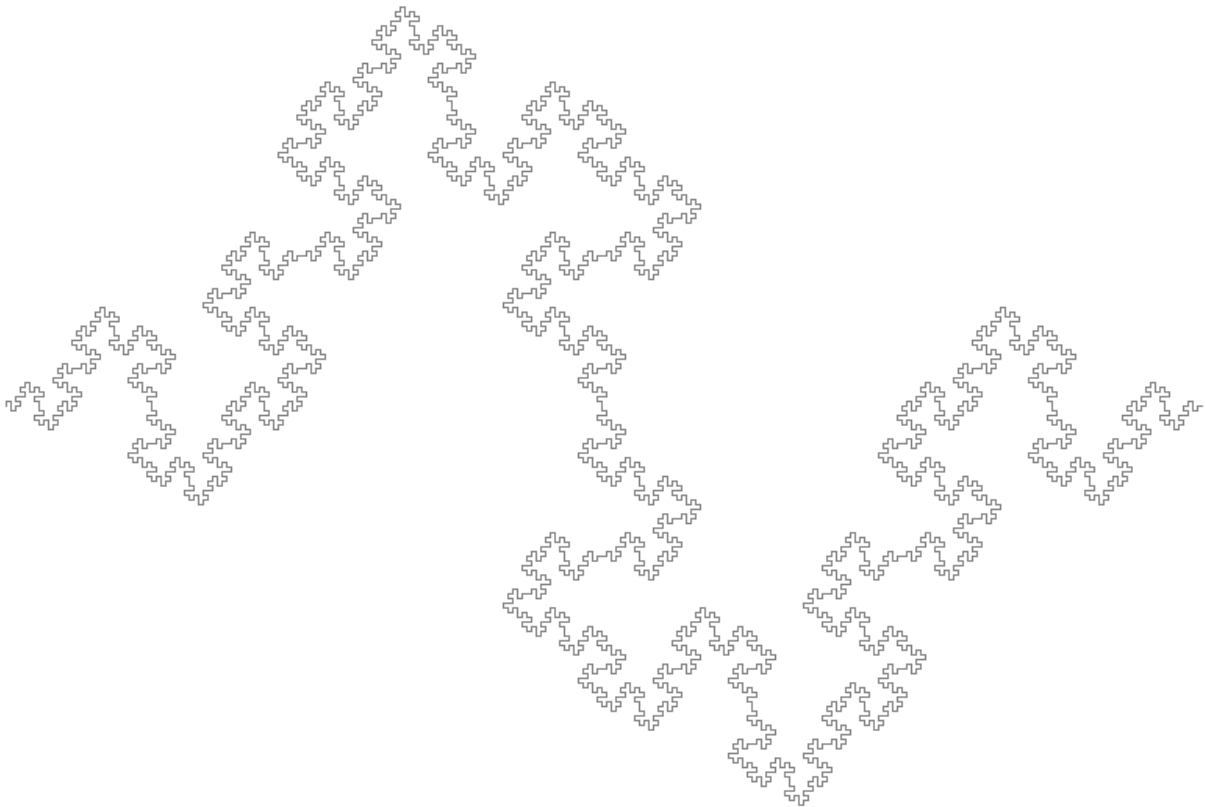


Figure 18: Quadratic Koch curve

The word “algebraic” in *algebraic curve* comes from the requirement that only algebraic operations may be used in its definition: addition and multiplication, but not analytic processes such as taking limits, infinite sums, approximation, differentiation or integration. The maps allowed in our context are given by polynomials in several variables. The pictures above show plane algebraic curves defined by a polynomial in two variables. The first two pictures are *graphs* of a polynomial function in one variable; they are described by

$$Y = P(X).$$

In the first picture, $P(X) = X$ (so the polynomial is linear), whereas in the second the polynomial has a form such as

$$P(X) = a_4X^4 + a_3X^3 + a_2X^2 + a_1X + a_0,$$

with coefficients a_i in a field K . In algebraic geometry we fix a *base field* K . Important fields for us are the real numbers (the pictures are primarily to be understood in this sense) and the complex numbers $\mathbb{C}$. Such a graph is a simple object in that each value of X has exactly one corresponding value of Y , namely the function value, which is also easy to calculate if we can calculate in the given field. In a certain sense, the graph is a “curved” copy of the base line, the X -axis.

Now consider the third picture. It is the graph of a *rational function*: we take two polynomials P, Q in the variable X and consider their quotient $P(X)/Q(X)$. This expression makes sense only where the denominator is nonzero. The rational function is undefined at the zeros of the denominator polynomial. If numerator and denominator vanish at the same point, cancellation sometimes gives the quotient a meaning there as well. If the denominator vanishes but the numerator does not, the undefined point is a *pole*: the real graph tends to $+\infty$ or $-\infty$. It is tempting to say that the rational function takes the value “infinity” at these points; in projective geometry this idea really does make sense, as we shall see later.

Because of these undefined points, however, the “graph equation” $Y = P(X)/Q(X)$ is not an ideal description of the curve. Multiplying by the denominator instead gives the condition, or *equation*,

$$YQ(X) = P(X), \quad \text{or, more precisely,} \quad \{(x, y) \in K^2 \mid yQ(x) = P(x)\},$$

whose two sides are well-defined polynomials. The *set satisfying the equation* (or *solution set*) is uniquely defined. For an x with $Q(x) = 0$, the left-hand side is zero. If $P(x) \neq 0$, there is no solution at this x , as in the picture; if $P(x) = 0$, every value of Y is allowed. In the latter case, the object therefore contains the line through $(x, 0)$ perpendicular to the X -axis.

Example: the hyperbola

A typical and important example of a rational function is $Y = 1/X$. Its graph is called a *hyperbola* H . Written without a denominator, the equation becomes

$$XY = 1, \quad \text{or} \quad H = \{(x, y) \mid xy = 1\}.$$

On $K^\times = K \setminus \{0\}$ this rational function is an ordinary function, with graph H , and it gives a “natural” bijection

$$K^\times \longrightarrow H, \quad x \longmapsto \left(x, \frac{1}{x}\right).$$

Thus $K^\times$ and H are “equivalent” or “isomorphic” in a sense that will be made precise later.

Both descriptions have advantages. The description as $K^\times \subset K$ takes place on a line (if we think of $K = \mathbb{R}$), but the point 0, which is a *limit point* of $K^\times$, does not belong to $K^\times$. In other words, $K^\times$ is not *closed*. The hyperbola, by contrast, is closed in $\mathbb{R}^2$; realising the object as a closed set therefore requires moving to a higher dimension. The question of what constitutes a good description of an algebraic-geometric object will recur throughout the course.

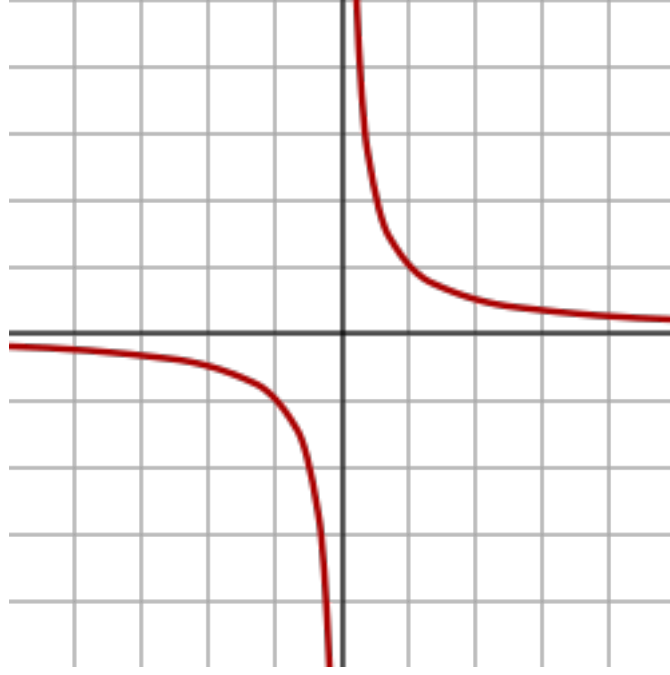


Figure 19: Rectangular hyperbola

In the real case, $K = \mathbb{R}$, the set $\mathbb{R}^\times$ (and likewise $H_{\mathbb{R}}$) consists of two disjoint “branches”, so it is not *connected*. In the complex case, $K = \mathbb{C}$, the set $\mathbb{C}^\times$ (and likewise $H_{\mathbb{C}}$) is a punctured real plane and is therefore connected. This is a typical phenomenon in algebraic geometry: important properties may depend on the base field. Nevertheless, properties that depend only on the defining equations and hold for their solution sets over every field have particular significance.

The fourth picture shows a *circle*, with equation

$$K = \{(x, y) \mid x^2 + y^2 = r^2\},$$

where r denotes its radius. The picture already shows that this object cannot be the graph of a function, since on a graph each x -value is paired with exactly one y -value. There is no function $y = \varphi(x)$ satisfying

$$K = \{(x, \varphi(x)) \mid x \in \mathbb{R}\}.$$

The question of whether an algebraic solution set can be realised as a graph is equivalent to asking whether its defining equation can be “solved” for y . In this example we can write

$$y^2 = r^2 - x^2, \quad y = \sqrt{r^2 - x^2} = \sqrt{(r - x)(r + x)}.$$

Is the circle a graph after all? There are two interpretations.

1. If we restrict ourselves to real numbers and positive square roots, the last step is not an equivalent transformation: we have “added” information that was not in the original equation. Taking the positive square root means restricting to the upper semicircle. Adding information or conditions makes the solution set smaller.
2. If instead $\sqrt{}$ is understood to include all solutions—in the real case, both the positive and the negative square root, often written $\pm\sqrt{}$ —we have added no information, but we have not solved for a function either; we have only obtained what is sometimes called a “multivalued function”.

Both viewpoints are useful. The attempt to describe part of a geometric object, such as the upper arc, simply as a graph reappears in the implicit function theorem, power-series methods, parametrisations and local theory.

Equations of the form $Y^2 = G(X)$



Figure 20: Cubic curves studied by Newton

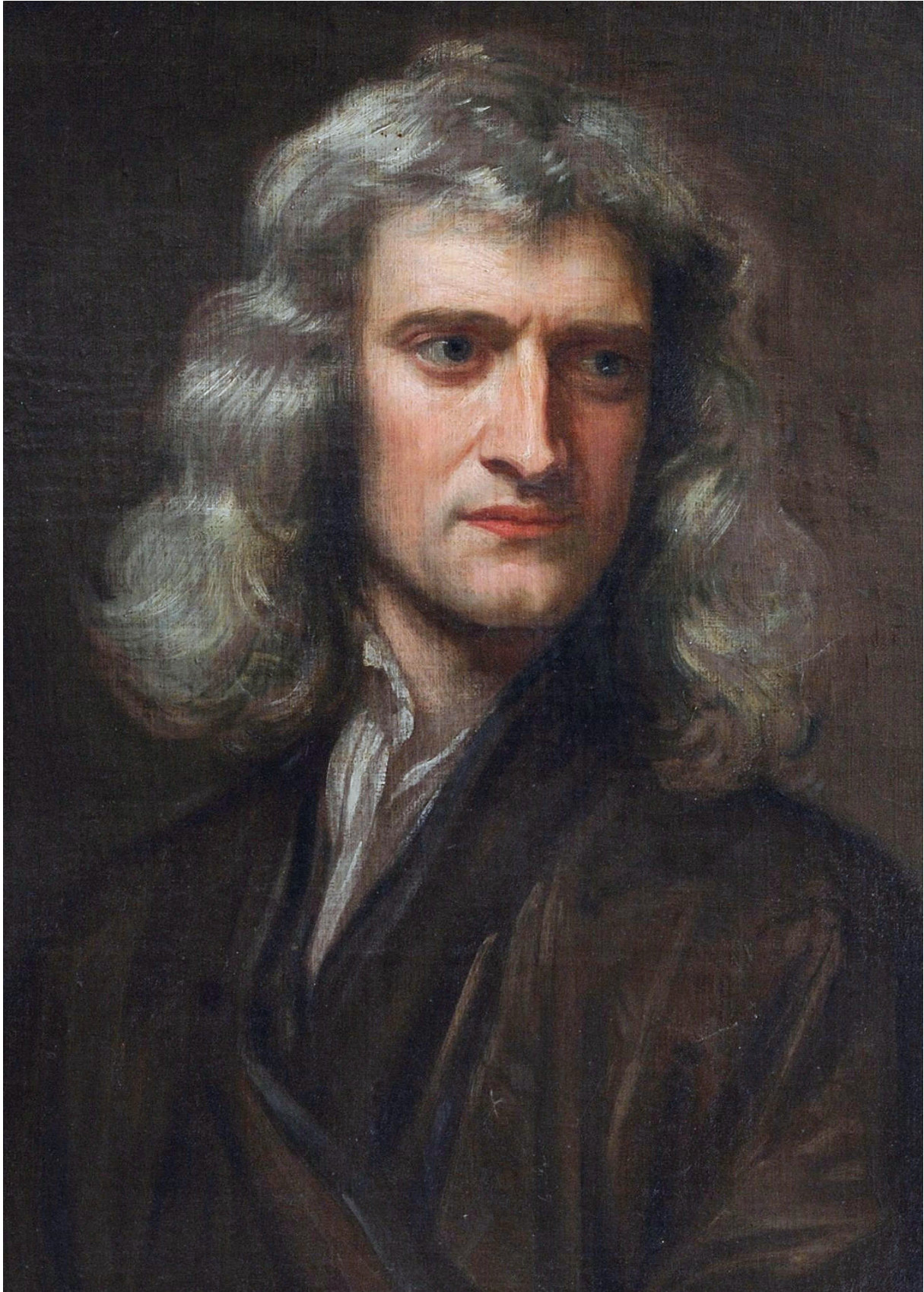


Figure 21: Isaac Newton (1643–1727)

A circle equation can be viewed as an equation of the form

$$Y^2 = G(X),$$

where G is a polynomial in the single variable X ; for the circle, $G = -X^2 + 1$. This is not a graph, but the “square root” of a graph. More generally, allow $G(X)$ to be more complicated. The zero set (or *zero locus*) represents the square root $\sqrt{G(X)}$. For any chosen value x of X , there are three possibilities for the corresponding real solutions y .

1. If $G(x) < 0$, there is no solution.
2. If $G(x) = 0$, there is exactly one solution, $y = 0$.
3. If $G(x) > 0$, there are two solutions, $y = \pm\sqrt{G(x)}$.

This also suggests how to visualise the real picture: for each x , calculate $G(x)$ and, if the radicand is nonnegative, mark the points $(x, \pm\sqrt{G(x)})$.

Over the complex numbers, we need distinguish only $G(x) = 0$ from $G(x) \neq 0$. If G has degree two, the resulting curve is a *conic section*, a subject studied since antiquity (see Lecture 7).

Isaac Newton studied intensively the case in which $G(X)$ is a real cubic polynomial, that is, a polynomial of degree three. Even this collection of examples is already very rich.

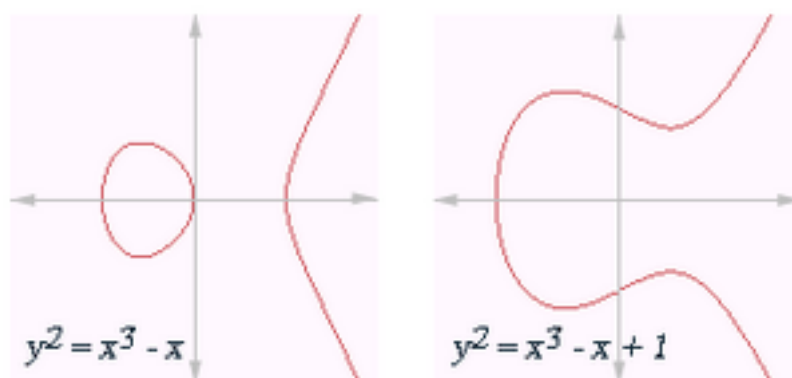


Figure 22: Examples of real elliptic curves

Consider the case $G(X) = X^3$, the object described by

$$\{(x, y) \mid y^2 = x^3\}.$$

This object is called *Neil’s parabola*. A new phenomenon appears here: the origin is different from all the other points. We call it a *singularity*; the other points, by contrast, are called *smooth* or *nonsingular*. Giving a precise definition is part of this course. As a first, imprecise formulation, a curve near a smooth point looks, in suitable coordinates, like the possibly rotated graph of a differentiable function. The singularity of Neil’s parabola is called a *cusp*—the source uses the German words *Spitze* and *Kuspe*, both meaning a pointed tip. By contrast, the singularity in the eighth picture is a *crossing point* or *double point*.

In the seventh picture at the beginning, and in the picture above, we also see zero loci of the form $Y^2 = G(X)$ with $G(X)$ of degree three. What must $G(X)$ look like to produce such a curve? The last examples also show that the presence of a singularity depends on the precise form of $G(X)$.

Let us stay with Neil’s parabola C . If t is any real or complex number, the point with coordinates

$$(x, y) = (t^2, t^3)$$

always lies on Neil’s parabola, since $(t^2)^3 = t^6 = (t^3)^2$. Conversely, one can show (see [Exercise 1.6](#)) that every point of Neil’s parabola has this form: for each (x, y) satisfying $y^2 = x^3$, there is exactly one t with $(x, y) = (t^2, t^3)$. The map

$$\mathbb{R} \longrightarrow C, \quad t \longmapsto (t^2, t^3),$$

is called a (bijective polynomial) *parametrisation* of Neil's parabola. Determining which algebraic curves admit a polynomial parametrisation is a nontrivial question. A smooth curve of the form $Y^2 = G(X)$ with $\deg G = 3$ has no such parametrisation. In elementary number theory, we learn that all *Pythagorean triples* can be written in a simple uniform form; see [Theorem 10.6 \(Number Theory, Osnabrück 2025\)](#). An equivalent statement is the existence of a rational parametrisation of the rational unit circle; see [Theorem 10.4](#). We shall discuss this in greater generality in [Theorem 7.6](#).

We now come to the first general definition.

Definition: affine plane algebraic curve

Let K be a field. An *affine plane algebraic curve* over K is the zero locus $V(F) \subseteq K^2$ of a nonconstant polynomial F in two variables, that is,

$$F = \sum_{0 \leq i, j \leq m} a_{ij} X^i Y^j \quad (a_{ij} \in K).$$

In other words,

$$V(F) = \left\{ (x, y) \in K^2 \mid F(x, y) = \sum_{0 \leq i, j \leq m} a_{ij} x^i y^j = 0 \right\}.$$

The following favourite polynomials in the variables X and Y were suggested in class:

1. $X^2 - Y^2$;
2. $2X + 4Y + 3$;
3. $X^2 + Y^2 - 3$;
4. $X^2 + Y^2$;
5. $5X^2 + 12Y^2 - 26$;
6. $3X^2 - 15Y^2 - 3$;
7. $X^3 + Y^3 + XY$;
8. $X^3 - 4Y^2 - XY$;
9. X^4 ;
10. $X^2Y^2 - X^2$.

The corresponding zero loci $V(F)$ vary in how difficult they are to understand. By the difference-of-squares identity,

$$X^2 - Y^2 = (X + Y)(X - Y),$$

and a product of two field elements is zero exactly when one factor is zero. This zero locus is therefore simply the union of the two diagonals: a union of two affine lines. The zero locus of $2X + 4Y + 3$ is the solution set of this linear equation, an affine line. The real zero locus of $X^2 + Y^2 - 3$ is a circle centred at the origin with radius $\sqrt{3}$. By contrast, the real zero locus of $X^2 + Y^2$ consists only of the origin $(0, 0)$; over $\mathbb{C}$ the situation is different. The zero locus of $5X^2 + 12Y^2 - 26$ is an ellipse with axes parallel to the coordinate axes, whereas the zero locus of $3X^2 - 15Y^2 - 3$ is a compressed hyperbola.

We shall discuss and classify the zero loci of quadratic polynomials in detail in Lecture 7. The two polynomials $X^3 + Y^3 + XY$ and $X^3 - 4Y^2 - XY$ have degree 3 and are much harder to understand. The first question is whether their curves are smooth or have singularities. The zero locus $V(X^4)$ equals $V(X)$ and is therefore the y -axis. The last polynomial factors as

$$X^2Y^2 - X^2 = X^2(Y^2 - 1) = X^2(Y - 1)(Y + 1),$$

so it is easy to understand. Its zero locus is the union of three lines: the y -axis and two lines parallel to the x -axis.

We shall prove a lemma that immediately shows why the four nonalgebraic curves pictured above are not algebraic.

Lemma: intersection with a line

Let C be an affine plane algebraic curve and L a line in K^2 .

Then $C \cap L$ is either the whole line L or a finite set of points.

Proof By definition, a plane algebraic curve $C = V(F)$ is always the zero locus of a polynomial F in two variables. Suppose the line L is given by

$$aX + bY + c = 0.$$

Without loss of generality, assume $a \neq 0$. Solving for X gives $X = \alpha Y + \beta$. An intersection point $P \in C \cap L$ must satisfy both $F(P) = 0$ and the line equation. Using the line equation, replace X in F by $\alpha Y + \beta$. This turns F into a polynomial in the single variable Y , which we call $\tilde{F}$.

Now $P \in C \cap L$ is equivalent to $P \in L$ and $\tilde{F}(P) = 0$. Thus the intersection is described by $\tilde{F}$. If $\tilde{F} = 0$, the whole line is the intersection. If $\tilde{F} \neq 0$, [Corollary 19.9 \(Linear Algebra, Osnabrück 2024–2025\)](#) says that it has only finitely many zeros. $\square$

For the four nonalgebraic examples above, there are lines meeting the curves in infinitely many points. The curves are therefore not algebraic.

Polynomial rings

After these introductory examples, we fix some terminology that is probably already familiar.

Definition: polynomial ring in one variable

The *polynomial ring* over a commutative ring R consists of all *polynomials*

$$P = a_0 + a_1X + a_2X^2 + \cdots + a_nX^n,$$

with $a_i \in R$ for $i = 0, \dots, n$ and $n \in \mathbb{N}$, equipped with componentwise addition and multiplication defined by extending the following rule distributively:

$$X^n \cdot X^m := X^{n+m}.$$

From this definition we can also define polynomial rings in several variables. Set

$$K[X, Y] := (K[X])[Y], \quad K[X, Y, Z] := (K[X, Y])[Z],$$

and so on. A polynomial in n variables has the form

$$F = \sum_{(\nu_1, \dots, \nu_n)} a_{(\nu_1, \dots, \nu_n)} X_1^{\nu_1} \cdots X_n^{\nu_n}.$$

The sum runs over a finite family of *exponent tuples* $(\nu_1, \dots, \nu_n)$. Expressions of the form $X_1^{\nu_1} \cdots X_n^{\nu_n}$ are also called *monomials*. A polynomial is usually abbreviated as $F = \sum_{\nu} a_{\nu} X^{\nu}$. Multiplying two monomials means adding their exponent tuples:

$$(X_1^{\nu_1} \cdots X_n^{\nu_n}) (X_1^{\mu_1} \cdots X_n^{\mu_n}) := X_1^{\nu_1 + \mu_1} \cdots X_n^{\nu_n + \mu_n}.$$

In algebraic geometry, the case of greatest interest to us is when the base ring R is a field. Algebraic geometry studies the shape of zero loci of polynomials in several variables. We shall see later that the relationship between algebraic and geometric properties is particularly strong when the base field is algebraically closed.

Definition: algebraically closed field

A field K is called *algebraically closed* if every nonconstant polynomial $F \in K[X]$ has a zero in K .



Figure 23: Carl Friedrich Gauss (1777–1855)

The *fundamental theorem of algebra* was first proved by Gauss.

Theorem: fundamental theorem of algebra

The field of complex numbers $\mathbb{C}$ is **algebraically closed**.

Proof We shall not prove this theorem here. Its proofs use topological or analytic methods. $\square$

Source navigation: [course](#) · [Lecture 2](#) · [worksheet for this lecture](#)

Worksheet 1

Practice exercises

Exercise 1.1

Linear algebra often deals with systems of linear equations. What nonlinear equations or systems of equations arise in linear algebra?

Exercise 1.2

Sketch the solution sets in $\mathbb{R}^2$ of the following equations.

1. $x^2 - y^2 - 1 = 0$;
2. $x^2 + xy + y^2 = 0$;
3. $x^2 + y^2 + 1 = 0$;
4. $x^2 + y^2 = 0$;
5. $x^2 + y^3 = 0$;
6. $x^3 - y^5 = 0$;
7. $x^2 - x^3 = 0$;
8. $x^3 + y^3 = 1$;
9. $x^4 + y^4 = 1$;
10. $-5 + 3x + 4x^2 + x^3 - y^2 = 1$.

Exercise 1.3

Calculate the intersection of each curve in Exercise 1.2 with each of the following lines.

1. $x = 0$;
2. $y = 0$;
3. $x = 1$;
4. $y = -2$;
5. $x = y$;
6. $x = -y$;
7. $2x - 3y + 4 = 0$.

Exercise 1.4 ★

1. Find an integer solution $(x, y) \in \mathbb{Z} \times \mathbb{Z}$ of

$$x^2 - y^3 + 2 = 0.$$

2. Show that

$$\left(\frac{383}{1000}, \frac{129}{100} \right)$$

is a solution of

$$x^2 - y^3 + 2 = 0.$$

Exercise 1.5 ★

Find a point on the plane algebraic curve

$$V(X^3 - Y^3 + 4X^2 - 2XY + Y + 3) \subset \mathbb{C}^2.$$

Exercise 1.6

Let K be a field. The image of the curve defined by

$$K \longrightarrow K^2, \quad t \longmapsto (t^2, t^3),$$

is called *Neil's parabola*. Show that a point $(x, y) \in K^2$ belongs to this image if and only if it satisfies $x^3 = y^2$.

Exercise 1.7

Let $C \subseteq \mathbb{R}^2$ be the image of the polynomial map

$$\mathbb{R} \longrightarrow \mathbb{R}^2, \quad t \longmapsto (t^2 - t + 2, t^2 - 3).$$

Find a nonzero polynomial F in two variables such that C lies in the zero locus of F .

Exercise 1.8

Let $C \subseteq \mathbb{R}^2$ be the image of the polynomial map

$$\mathbb{R} \longrightarrow \mathbb{R}^2, \quad t \longmapsto (t^3 - 1, t^2 - 1).$$

Find a nonzero polynomial F in two variables such that C lies in the zero locus of F .

Exercise 1.9

Consider the curve

$$\mathbb{R} \longrightarrow \mathbb{R}^2, \quad t \longmapsto (t^2 - 1, t^3 - t).$$

- a. Show that the image points (x, y) of this curve satisfy

$$y^2 = x^2 + x^3.$$

- b. Show that every point $(x, y) \in \mathbb{R}^2$ satisfying $y^2 = x^2 + x^3$ belongs to the image of the curve.
 c. Show that there are exactly two parameter values t_1 and t_2 with the same image point, and that the map is otherwise injective.

Exercise 1.10

Discuss the relationship between plane algebraic curves and the implicit function theorem.

Exercise 1.11

Let $K = \mathbb{Z}/(7)$. Find all points in $K^2 = K \times K$ lying on the curve defined by

$$X^2Y + 2Y^3 + 3Y^2 = 0.$$

How many solutions are there?

Exercise 1.12 ★

Find a line $G \subseteq \mathbb{A}_{\mathbb{C}}^2$ meeting the curve

$$C = V(X^3 + Y^3 + 1) \subseteq \mathbb{A}_{\mathbb{C}}^2$$

in exactly one point.

Exercise 1.13 ★

Show that Neil's parabola

$$C = V(Y^2 - X^3) \subseteq \mathbb{A}_{\mathbb{C}}^2$$

meets every line through $P = (1, 1) \in C$ in at least one further point.

Exercise 1.14 ★

Give an analytic argument showing that the unit circle $V(x^2 + y^2 - 1)$ and Neil's parabola $V(y^2 - x^3)$ have a real intersection point. Determine numerically the real x -coordinate of such a point with error ≤ 0.1 .

Exercise 1.15

Consider equations of the form

$$y^2 = G(x), \quad G(x) = x^3 + ax^2 + bx + c,$$

over $\mathbb{R}$. Sketch the different solution sets for coefficients $a, b, c \in \{1, -1, 0\}$.

The statement in the next exercise also follows from Theorem 6.1 using Lemma 1.3.

Exercise 1.16

Let K be a field and let

$$K \longrightarrow K^2, \quad t \longmapsto (P(t), Q(t))$$

be a map given by two polynomials $P(t), Q(t) \in K[t]$. Let B be its image and $G \subseteq K^2$ a line. Show that either $B \subseteq G$ or the intersection $B \cap G$ is finite.

Exercise 1.17

Multiply the following two polynomials in $\mathbb{Z}[x, y, z]$:

$$x^5 + 3x^2y^2 - xyz^3 \quad \text{and} \quad 2x^3yz + z^2 + 5xy^2z - x^2y.$$

Exercise 1.18

Multiply the following two polynomials in $(\mathbb{Z}/(5))[x, y]$:

$$x^4 + 2x^2y^2 - xy^3 + 2y^3 \quad \text{and} \quad x^4y + 4x^2y + 3xy^2 - x^2y^2 + 2y^2.$$

Exercise 1.19

Let R be an integral domain. Show that the polynomial ring $R[X]$ is also an integral domain.

Exercise 1.20 ★

Let R be an integral domain and $R[X]$ the polynomial ring over R . Show that the units of $R[X]$ are precisely the units of R .

Exercise 1.21 ★

Let K be a field. Show that the following two properties are equivalent.

1. K is algebraically closed.
2. Every nonconstant polynomial $F \in K[X]$ factors into linear factors.

Exercise 1.22

Let K be an algebraically closed field. Determine all irreducible polynomials in $K[X]$.

Exercise 1.23

Let K be an algebraically closed field. Show that K cannot be finite.

Exercises to hand in**Exercise 1.24 — 2 points**

Carry out the following polynomial division in $(\mathbb{Z}/(7))[X]$:

$$X^4 + 5X^2 + 3 \quad \text{divided by} \quad 2X^2 + X + 6.$$

Exercise 1.25 — 5 points

Find all monic irreducible polynomials of degree 4 in the polynomial ring $\mathbb{F}_3[X]$.

Exercise 1.26 — 3 points

Find all solutions of the circle equation

$$x^2 + y^2 = 1$$

for the fields $K = \mathbb{Z}/(2)$, $\mathbb{Z}/(5)$ and $\mathbb{Z}/(11)$.

Exercise 1.27 — 5 points

Let $C \subseteq \mathbb{C}^2$ be the image of the polynomial map

$$\mathbb{C} \longrightarrow \mathbb{C}^2, \quad t \longmapsto (t^3 - t^2 + 4t + 3, -t^2 + 5t - 1).$$

Find a nonzero polynomial F in two variables such that C lies in the zero locus of F .

Exercise 1.28 — 4 points

Consider the map

$$f : \mathbb{R} \longrightarrow S^1 \subseteq \mathbb{R}^2$$

assigning to each $t \in \mathbb{R}$ the unique intersection point other than $(0, -1)$ of the line G_t through $(t, 1)$ and $(0, -1)$ with the unit circle

$$S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}.$$

Show that this map is well defined and find the formulas describing it. Show that f is differentiable. Is f injective? Is f surjective?

Solutions to Worksheet 1

This section translates the seven public solutions linked from Worksheet 1. Their numbering and order follow the source exercises. No additional solutions are supplied for exercises without a public solution at the frozen source revision.

Solution to Exercise 1.4

1. $(5, 3)$ is an integer solution.
2. We have

$$\begin{aligned}\left(\frac{383}{1000}\right)^2 - \left(\frac{129}{100}\right)^3 + 2 &= \frac{146\,689}{1\,000\,000} - \frac{2\,146\,689}{1\,000\,000} + 2 \\ &= \frac{146\,689 - 2\,146\,689}{1\,000\,000} + 2 \\ &= \frac{-2\,000\,000}{1\,000\,000} + 2 \\ &= -2 + 2 \\ &= 0.\end{aligned}$$

[Back to Exercise 1.4.](#)

Solution to Exercise 1.5

Consider the intersection of the curve with the line $V(X - Y)$, imposing the additional condition $X = Y$. Substituting $Y = X$ into the curve equation gives

$$X^3 - X^3 + 4X^2 - 2X^2 + X + 3 = 2X^2 + X + 3 = 0.$$

The quadratic formula gives

$$X = -\frac{1}{4} \pm \sqrt{-\frac{23}{16}} = -\frac{1}{4} \pm \frac{\sqrt{23}}{4}i.$$

Therefore

$$\left(\frac{-1 + \sqrt{23}i}{4}, \frac{-1 + \sqrt{23}i}{4}\right)$$

is a point on the curve.

[Back to Exercise 1.5.](#)

Solution to Exercise 1.12

Choose the line

$$G = V(Y + 1).$$

To calculate $C \cap G$, substitute the equation $Y = -1$, which holds on G , into the curve equation. This gives

$$0 = X^3 + (-1)^3 + 1 = X^3.$$

The only solution is $X = 0$, so $(0, -1)$ is the unique intersection point of G and C .

[Back to Exercise 1.12.](#)

Solution to Exercise 1.13

Every line in the plane has an equation

$$ax + by = c,$$

where a and b are not both zero. If the line passes through $(1, 1)$, then $a + b = c$.

If $b = 0$, the line is $x = 1$ and has the further intersection point $(1, -1)$ with the curve. Hence assume $b \neq 0$. Solving the line equation for y gives

$$y = rx + s, \quad s = 1 - r.$$

On this line, the curve equation becomes

$$\begin{aligned} 0 &= y^2 - x^3 \\ &= (rx + (1 - r))^2 - x^3 \\ &= -x^3 + r^2x^2 + 2r(1 - r)x + (1 - r)^2. \end{aligned}$$

Since $x = 1$ is a zero, we can factor out $x - 1$:

$$-x^3 + r^2x^2 + 2r(1 - r)x + (1 - r)^2 = (x - 1)(-x^2 - (1 - r^2)x - (1 - r)^2).$$

After multiplication by -1 , the quadratic factor on the right is monic of degree 2, so it has roots in $\mathbb{C}$. We must show that at least one additional root differs from 1. Evaluating the displayed quadratic factor at $x = 1$ gives

$$-1 - (1 - r^2) - (1 - r)^2 = -3 + r^2 + 2r - r^2 = 2r - 3.$$

If $r \neq \frac{3}{2}$, then 1 is not a root of that quadratic factor. It remains to consider $r = \frac{3}{2}$. In this case,

$$x^2 + (1 - r^2)x + (1 - r)^2 = x^2 - \frac{5}{4}x + \frac{1}{4} = (x - 1)\left(x - \frac{1}{4}\right),$$

so there is another root, $x = \frac{1}{4}$. Thus every line through $(1, 1)$ meets Neil's parabola in at least one further point.

Edition note: The source calls the displayed quadratic factor monic, although its leading coefficient is -1 . The clarification above makes the multiplication by -1 explicit; its roots are unchanged.

[Back to Exercise 1.13.](#)

Solution to Exercise 1.14

Substitute $y^2 = x^3$ into the circle equation to obtain

$$x^3 + x^2 - 1 = 0.$$

At $x = 1$ the polynomial has value 1, whereas at $x = 0.5$ it has a negative value. By the intermediate value theorem it has a root x_0 in $[0.5, 1]$. Since x_0^3 is positive, its real square root

$$y_0 = \sqrt{x_0^3}$$

exists, and (x_0, y_0) is a real intersection point.

To approximate x_0 numerically, calculate

$$(0.7)^3 + (0.7)^2 - 1 < 0.49 + 0.49 - 1 < 0$$

and

$$\begin{aligned} (0.8)^3 + (0.8)^2 - 1 &= 0.64 \cdot 0.8 + 0.64 - 1 \\ &> 0.48 + 0.64 - 1 > 0. \end{aligned}$$

Thus an intersection point exists whose x -coordinate lies in $[0.7, 0.8]$.

Edition note: The source ends the second estimate with $0.48 + 0.64 - 1 = 0$. This edition uses the correct inequality $0.48 + 0.64 - 1 > 0$.

[Back to Exercise 1.14.](#)

Solution to Exercise 1.20

Let $a \in R$ be a unit. There is a $b \in R$ with $ab = 1$, and the same identity holds in the polynomial ring. Hence

$$R^\times \subseteq R[X]^\times.$$

Conversely, let

$$P = \sum_{i=0}^d a_i X^i, \quad a_d \neq 0,$$

be a unit in $R[X]$. Then there is a polynomial

$$Q = \sum_{j=0}^e b_j X^j, \quad b_e \neq 0,$$

with $PQ = 1$. Since R is an integral domain, $a_d b_e \neq 0$, and the product has the form

$$a_d b_e X^{d+e} + \text{terms of lower degree}.$$

Because $PQ = 1$, we must have $d + e = 0$ and $a_d b_e = 1$. Thus P is a constant unit. Consequently the units of $R[X]$ are precisely the units of R .

[Back to Exercise 1.20.](#)

Solution to Exercise 1.21

Suppose K is algebraically closed and $F \in K[X]$ is nonconstant. We prove by induction on $n = \deg F$ that F factors into linear factors.

For $n = 1$, we have $F = a_1 X + a_0$, so F is already a single linear factor. Assume every polynomial $G \in K[X]$ of degree $n - 1$ factors into linear factors. Since K is algebraically closed, F has a root x_0 . We can therefore write

$$F = G \cdot (X - x_0)$$

for a polynomial $G \in K[X]$ of degree $n - 1$. This degree assertion follows directly from the fact that a field is also an integral domain and a slight adaptation of the proof in Exercise 8. By the induction hypothesis, G factors into linear factors, so $G \cdot (X - x_0)$ does too. Hence every nonconstant polynomial $F \in K[X]$ factors into linear factors.

Conversely, if every nonconstant polynomial factors into linear factors, each has a root represented by one of its linear factors. Thus K is algebraically closed.

Edition note: “Exercise 8” is an unlinked reference in the frozen source and does not identify Exercise 1.8 of this worksheet. The degree assertion used here is $\deg(G(X)(X - x_0)) = \deg G + 1$ over a field; no renumbered source reference is inferred.

[Back to Exercise 1.21.](#)

Lecture 2: Affine Algebraic Sets

Affine algebraic sets

Definition: affine space

Let K be a field. The space

$$\mathbb{A}_K^n = K^n$$

is called *affine space* of dimension n over K .

Thus, to begin with, affine space is simply a set of points. A point in affine space is an n -tuple $(a_1, \dots, a_n)$ with coordinates in K . Why, then, introduce a new term? The term “affine space” indicates that we wish to regard K^n as an object of algebraic geometry. In other words, we regard n -dimensional affine space as the natural geometric object on which polynomials in n variables act as functions. We shall gradually equip affine space with further structures—the Zariski topology and the structure sheaf—which make it clear that it is “more” than “just” K^n . For $n = 1$ we speak of the *affine line*, and for $n = 2$ of the *affine plane*.

A polynomial $F \in K[X_1, \dots, X_n]$ can naturally be regarded as a function on affine space. To a point

$$P = (a_1, \dots, a_n) \in \mathbb{A}_K^n,$$

we assign the value

$$F(P) = F(a_1, \dots, a_n)$$

by replacing each variable X_i by a_i and carrying out all the operations in K . Given a polynomial $F \in K[X_1, \dots, X_n]$, we can ask whether $F(P) = 0$. One object of particular interest associated with F is therefore the zero locus it defines,

$$V(F) = \{P \in \mathbb{A}_K^n \mid F(P) = 0\}.$$

We encountered several examples in the first lecture. It is also useful, however, to study the common, or simultaneous, zero locus of several polynomials. This is the intersection of their individual zero loci—for example, in the case of conic sections, where a cone in three-dimensional space is intersected with various planes.

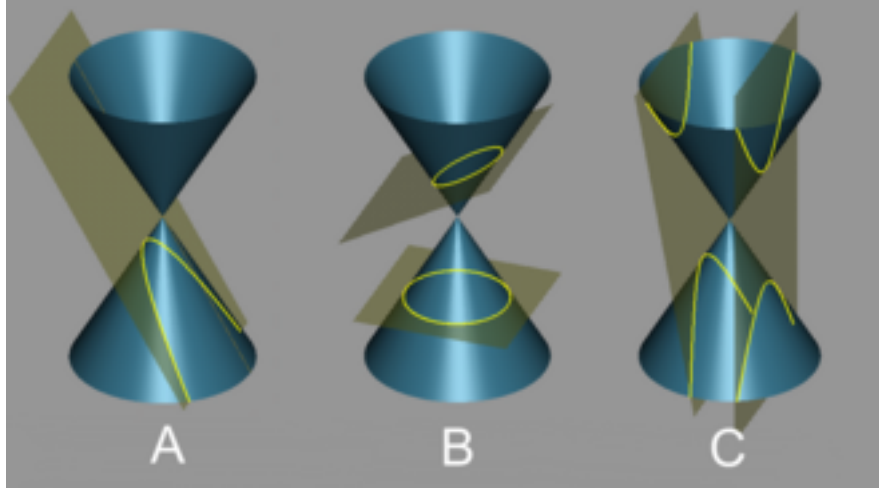


Figure 24: Conic sections

We therefore make the following general definition.

Definition: the zero locus of a family of polynomials

Let K be a field, and let

$$F_j \in K[X_1, \dots, X_n], \quad j \in J,$$

be a family of polynomials in n variables. The set

$$\{P \in \mathbb{A}_K^n \mid F_j(P) = 0 \text{ for every } j \in J\}$$

is called the *zero locus* (or *zero set*) defined by the family. It is denoted by $V(F_j, j \in J)$.

Those subsets of affine space that arise as zero sets deserve a name of their own.

Definition: affine algebraic set

Let K be a field and $K[X_1, \dots, X_n]$ the polynomial ring in n variables. A subset $V \subseteq \mathbb{A}_K^n$ is called an *affine algebraic set* if it is the zero set of a family of polynomials $F_j, j \in J$, with $F_j \in K[X_1, \dots, X_n]$; that is, if

$$V = V(F_j, j \in J).$$

The simplest examples are finite sets of points on the affine line $\mathbb{A}_K^1$, each given by a single polynomial, and affine linear subspaces of $\mathbb{A}_K^n$, which are the solution sets of inhomogeneous systems of linear equations over K .

Example: the axes and the origin

Consider the affine plane $\mathbb{A}_K^2$ and some affine algebraic subsets defined by the variables X and Y .

- The zero locus $V(X, Y)$ consists only of the *origin* $(0, 0)$, since both variables must be zero.
- The set $V(X)$ is the *Y-axis*: all points of the form $(0, y)$.
- The set $V(Y)$ is the *X-axis*.
- The set $V(X + Y)$ consists of all points (x, y) with $y = -x$: it is the *antidiagonal*.
- The set $V(XY)$ consists of the points (x, y) with $xy = 0$. Since K is a field, a product can be zero only if one of its factors is zero (Lemma 3.10 of Linear Algebra, Osnabrück 2024–2025). Thus

$$V(XY) = V(X) \cup V(Y),$$

the union of the two coordinate axes.

Points in affine space or on an affine algebraic set are often interpreted as representing more complicated mathematical objects. Properties of those objects are then reflected in whether their representing points satisfy certain algebraic equations or, equivalently, lie on certain affine algebraic sets. The following example illustrates this idea.

Example: matrices as points of affine space

A 2×2 matrix

$$\begin{pmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{pmatrix}$$

is uniquely determined by the four numbers $a_{11}, a_{21}, a_{12}, a_{22} \in K$. It can therefore be identified with a point of $\mathbb{A}_K^4$. In this interpretation, it is natural to denote the variables by $X_{11}, X_{21}, X_{12}, X_{22}$. We can now ask which properties of matrices can be described by algebraic equations. We discuss several typical properties.

A matrix is upper triangular precisely when $a_{12} = 0$. Thus the set of upper triangular matrices is the zero locus of X_{12} .

A matrix is invertible when

$$a_{11}a_{22} - a_{12}a_{21} \neq 0.$$

Consequently, the set of non-invertible matrices is described by the algebraic determinant condition

$$X_{11}X_{22} - X_{12}X_{21} = 0.$$

A matrix describes multiplication by a scalar if it is diagonal with equal diagonal entries. This set is described by the three equations

$$X_{12} = 0, \quad X_{21} = 0, \quad X_{11} - X_{22} = 0.$$

An element $\lambda \in K$ is an eigenvalue of a matrix precisely when it is a root of its characteristic polynomial (Theorem 23.2 of Linear Algebra, Osnabrück 2024–2025), that is, when

$$\det \begin{pmatrix} \lambda - a_{11} & -a_{21} \\ -a_{12} & \lambda - a_{22} \end{pmatrix} = \lambda^2 - \lambda(a_{11} + a_{22}) + a_{11}a_{22} - a_{12}a_{21} = 0.$$

In linear algebra, the matrix is usually given and we seek roots λ of this polynomial in one variable. We can also reverse the viewpoint: fix λ and study the zero locus

$$\lambda^2 - \lambda(X_{11} + X_{22}) + X_{11}X_{22} - X_{12}X_{21} = 0$$

in four variables. This equation describes all matrices having λ as an eigenvalue.

Similarly, a matrix has the two distinct eigenvalues $\lambda \neq \delta$ precisely when

$$\lambda^2 - \lambda(X_{11} + X_{22}) + X_{11}X_{22} - X_{12}X_{21} = 0$$

and

$$\delta^2 - \delta(X_{11} + X_{22}) + X_{11}X_{22} - X_{12}X_{21} = 0.$$

Subtracting the two equations gives

$$\lambda^2 - \delta^2 - (\lambda - \delta)(X_{11} + X_{22}) = 0,$$

which such a matrix must also satisfy. Since $\lambda \neq \delta$, we can write this as

$$X_{11} + X_{22} = \lambda + \delta.$$

The sum of a matrix's diagonal entries is called its *trace*. The last equation therefore says that the trace of a matrix with eigenvalues $\lambda \neq \delta$ must equal their sum.

The characteristic polynomial of a matrix can also be written as

$$\lambda^2 - \lambda \operatorname{Trace}(M) + \det(M),$$

where

$$\operatorname{Trace}(M) = X_{11} + X_{22}, \quad \det(M) = X_{11}X_{22} - X_{12}X_{21}.$$

Thus two matrices have the same characteristic polynomial precisely when they have the same trace and determinant. The set of matrices with a prescribed characteristic polynomial can therefore be regarded as a fibre of the map

$$\mathbb{A}_K^4 \longrightarrow \mathbb{A}_K^2, \quad M \longmapsto (\operatorname{Trace}(M), \det(M)).$$

This map is given by simple polynomial expressions. Is it surjective? Do its fibres always look alike—that is, does the set of matrices with a prescribed trace and determinant always have the same structure—or are there differences?

Fix s and d . We must study the solution set of the system

$$X_{11} + X_{22} = s, \quad X_{11}X_{22} - X_{12}X_{21} = d.$$

The variable X_{11} is uniquely determined by X_{22} , and conversely. We can therefore *eliminate* one variable by putting $X_{22} = s - X_{11}$. This gives an “equivalent” system in the three variables X_{11}, X_{12}, X_{21} with the single equation

$$X_{11}(s - X_{11}) - X_{12}X_{21} = d,$$

or

$$X_{11}^2 - sX_{11} + X_{12}X_{21} + d = 0.$$

Here “equivalent” means that the two solution sets are in bijection through maps given by polynomials. The last form shows that a solution always exists: we may choose any value of X_{11} and obtain an equation of the form $X_{12}X_{21} = a$, which has solutions.

A linear change of variables simplifies the equation further. Suppose that 2 is invertible in K , so the characteristic of K is not 2. With

$$X = X_{11} - \frac{s}{2}, \quad Y = X_{12}, \quad Z = X_{21},$$

we obtain

$$X^2 + YZ + c = 0, \quad c = -\frac{s^2}{4} + d.$$

Thus the shape of the set of matrices with a prescribed trace and determinant depends only on $-s^2/4 + d$. Indeed, the zero locus differs according to whether this expression is zero or nonzero. In the first case it has a singularity; in the second it does not, as we shall see later.

Ideals and zero loci

Since for now we allow arbitrary families of polynomials to define zero loci and hence affine algebraic sets, these objects initially seem difficult to get a handle on. Three important statements nevertheless hold, which we shall prove in stages.

1. The zero locus of a family of polynomials equals the zero locus of the ideal generated by that family.
2. Every ideal has a finite set of generators. Thus every zero locus can be described by finitely many polynomials (Hilbert's basis theorem).
3. Over an algebraically closed field, zero loci correspond bijectively to radical ideals, a special class of ideals (Hilbert's Nullstellensatz).

We can prove the first statement immediately. The other two require some algebraic preparation, which we shall develop in the following lectures.

Lemma: a family of polynomials and the ideal it generates

Let K be a field and $F_j \in K[X_1, \dots, X_n]$, $j \in J$, a family of polynomials in n variables. Let $\mathfrak{a}$ be the ideal of $K[X_1, \dots, X_n]$ generated by all the F_j . Then

$$V(F_j, j \in J) = V(\mathfrak{a}).$$

Proof The ideal $\mathfrak{a}$ consists of all finite linear combinations of the polynomials F_j and in particular contains every F_j . The inclusion

$$V(F_j, j \in J) \supseteq V(\mathfrak{a})$$

is therefore clear. For the reverse inclusion, take $P \in V(F_j, j \in J)$ and $H \in \mathfrak{a}$. There are polynomials $A_i \in K[X_1, \dots, X_n]$ and indices $j_1, \dots, j_k$ such that

$$H = \sum_{i=1}^k A_i F_{j_i}.$$

Then

$$H(P) = \sum_{i=1}^k A_i(P) F_{j_i}(P) = 0.$$

Thus every element of the ideal vanishes at P , so $P \in V(\mathfrak{a})$. $\square$

Henceforth, then, we may assume that every zero set is given by an ideal.

Lemma: inclusion of ideals reverses inclusion of zero loci

For ideals $\mathfrak{a} \subseteq \mathfrak{b}$ in $K[X_1, \dots, X_n]$, the corresponding zero loci satisfy

$$V(\mathfrak{a}) \supseteq V(\mathfrak{b}).$$

Proof Take $P \in V(\mathfrak{b})$. This means that $F(P) = 0$ for every $F \in \mathfrak{b}$. Since $\mathfrak{a} \subseteq \mathfrak{b}$, it follows in particular that $F(P) = 0$ for every $F \in \mathfrak{a}$. Thus $P \in V(\mathfrak{a})$. $\square$

Affine algebraic subsets of affine space have the following important structural properties.

Proposition: unions and intersections of affine algebraic sets

Let K be a field, $K[X_1, \dots, X_n]$ the polynomial ring in n variables, and $\mathbb{A}_K^n$ the corresponding affine space. The following properties hold.

1. $V(0) = \mathbb{A}_K^n$: the whole affine space is an affine algebraic set.
2. $V(1) = \emptyset$: the empty set is an affine algebraic set.
3. If $V_1, \dots, V_k$ are affine algebraic sets with $V_i = V(\mathfrak{a}_i)$, then

$$V_1 \cup V_2 \cup \dots \cup V_k = V(\mathfrak{a}_1 \mathfrak{a}_2 \dots \mathfrak{a}_k).$$

In particular, a finite union of affine algebraic sets is again an affine algebraic set.

4. If $V_i, i \in I$, are affine algebraic sets with $V_i = V(\mathfrak{a}_i)$, then

$$\bigcap_{i \in I} V_i = V\left(\sum_{i \in I} \mathfrak{a}_i\right).$$

In particular, an arbitrary intersection of affine algebraic sets is again an affine algebraic set.

Proof Statements (1) and (2) are clear: the constant polynomial 0 vanishes everywhere, whereas the constant polynomial 1 vanishes nowhere.

For (3), take a point in the union, say $P \in V(\mathfrak{a}_1)$. Then $f(P) = 0$ for every $f \in \mathfrak{a}_1$. Every element of the product ideal $\mathfrak{a}_1 \dots \mathfrak{a}_k$ has the form

$$h = \sum_{j=1}^m r_j f_{1j} f_{2j} \dots f_{kj},$$

with $f_{ij} \in \mathfrak{a}_i$. Since $f_{1j}(P) = 0$ in every term, we get $h(P) = 0$. Thus P belongs to the zero locus on the right.

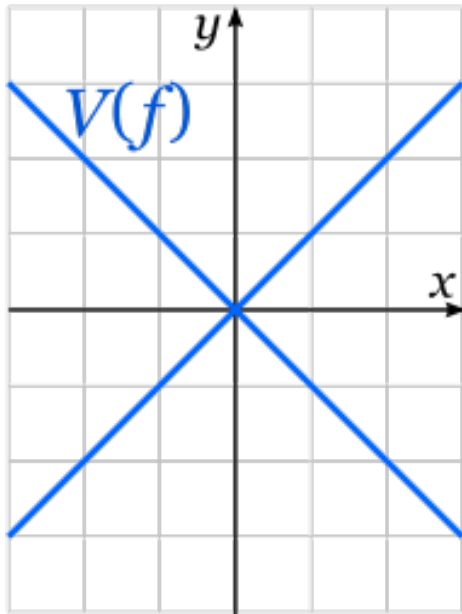
Conversely, suppose that P does not belong to the union on the left. Then $P \notin V(\mathfrak{a}_i)$ for every $i = 1, \dots, k$. For each i there is an $f_i \in \mathfrak{a}_i$ with $f_i(P) \neq 0$. Since K is a field,

$$(f_1 f_2 \dots f_k)(P) \neq 0,$$

while $f_1 f_2 \dots f_k \in \mathfrak{a}_1 \dots \mathfrak{a}_k$. Therefore P cannot belong to the zero locus on the right.

For (4), take $P \in \mathbb{A}_K^n$. The point P belongs to $V(\mathfrak{a}_i)$ for every $i \in I$ precisely when $f(P) = 0$ for every $f \in \mathfrak{a}_i$ and every $i \in I$. This holds precisely when $f(P) = 0$ for every f in the sum of these ideals. $\square$

$$f(x, y) = x^2 - y^2$$



$$g(x, y) = x^2 + y^2 - 4$$

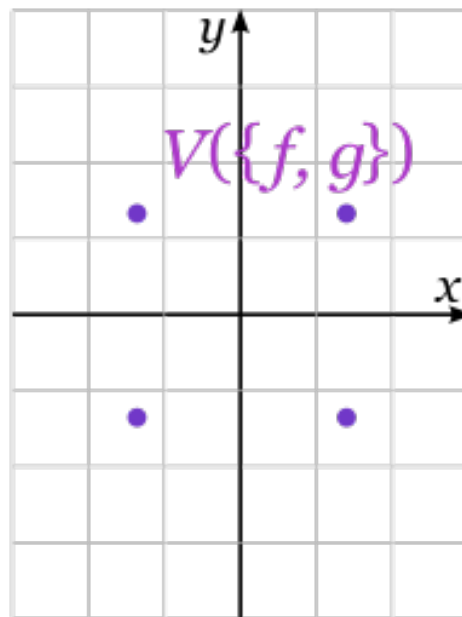
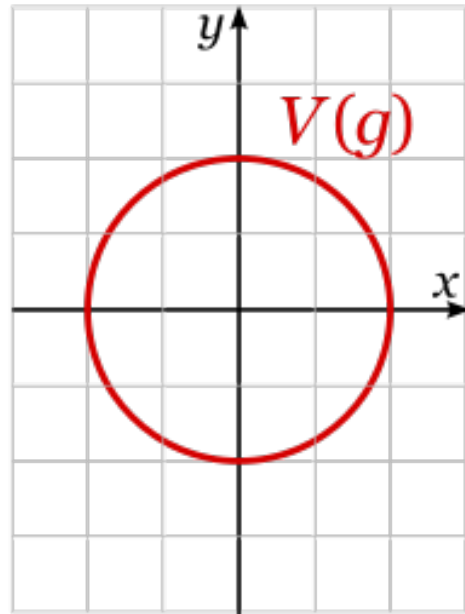


Figure 25: Examples of algebraic sets

Worksheet 2

Practice exercises

Exercise 2.1

Consider the two planes in $\mathbb{A}_K^3$,

$$E_1 = V(3x + 4y + 5z) \quad \text{and} \quad E_2 = V(2x - y + 3z).$$

Find a parametrisation of the intersection $E_1 \cap E_2$.

Exercise 2.2 ★

Find the intersection points of the unit circle and the line through the points $(-1, 1)$ and $(4, -2)$.

Exercise 2.3

Find the coordinates of the two intersection points of the line G and the circle K , where G is given by the equation

$$3y - 4x + 2 = 0,$$

and K has centre $(2, 5)$ and radius 7.

Exercise 2.4

Calculate the intersection points of the two curves

$$C = V(x^2 + 2y^2 + 3xy + x - 2) \quad \text{and} \quad L = V(4x + 3y - 5).$$

Exercise 2.5

Prove that the intersection of two distinct circles in the affine plane is the intersection of one of the circles with a line.

Exercise 2.6 ★

Find the intersection points of the unit circle E and the circle K with centre $(1, 0)$ and radius 2.

Exercise 2.7 ★

Find the intersection points of the two ellipses

$$\{(x, y) \in \mathbb{R}^2 \mid x^2 + xy + 3y^2 = 3\}$$

and

$$\{(x, y) \in \mathbb{R}^2 \mid 2x^2 - xy + y^2 = 4\}.$$

Exercise 2.8 ★

Find the intersection points of the unit circle and the standard parabola.

Exercise 2.9 ★

Let

$$P = \{(x, y) \in \mathbb{R}^2 \mid y = x^2\}$$

be the standard parabola, and let K be the circle with centre $(0, 1)$ and radius 1.

1. Sketch P and K .
2. Write down an equation for K .
3. Determine the intersection $P \cap K$.
4. Describe the lower semicircle as the graph of a function from $[-1, 1]$ to $\mathbb{R}$.
5. Describe the position of the parabola relative to the lower semicircle.

Exercise 2.10

Find all simultaneous solutions of the two equations

$$x^3 + y^2 = 2 \quad \text{and} \quad 2xy = 3$$

over the fields $K = \mathbb{Z}/(3)$, $\mathbb{Z}/(5)$, and $\mathbb{Z}/(7)$.

Exercise 2.11 ★

Consider the variety of commuting 2×2 matrices, that is, the set of pairs of matrices

$$V = \{(A, B) \mid A, B \in \text{Mat}_2(K), AB = BA\} \subseteq \text{Mat}_2(K) \times \text{Mat}_2(K) \cong \mathbb{A}_K^8.$$

1. Prove that V is an affine variety, and find equations describing it that are as simple as possible.
2. Prove that the map

$$\pi : V \longrightarrow \text{Mat}_2(K), \quad (A, B) \longmapsto A,$$

is surjective.

3. Determine the inverse image of

$$\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$$

under π .

Exercise 2.12

Prove that, for a point

$$P = (a_1, \dots, a_n) \in \mathbb{A}_K^n,$$

the corresponding ideal

$$(X_1 - a_1, X_2 - a_2, \dots, X_n - a_n)$$

is maximal.

Exercise 2.13

Let K be an algebraically closed field and $n \geq 2$. Prove that a point $P \in \mathbb{A}_K^n$ cannot be the zero set of a single polynomial.

Exercise 2.14

Let K be a finite field, and let

$$M = \{P_1, P_2, \dots, P_m\} \subseteq \mathbb{A}_K^n$$

be a finite set of points. Prove that M is the zero set of a single polynomial.

Exercise 2.15 ★

Let K be a field. Prove that the following three ideals in $K[X, Y, Z, W]$ are equal:

$$\mathfrak{a} = (Z^2 + W^2 - 1, X - 2Z^2 + 1, Y - 2ZW),$$

$$\mathfrak{b} = (Z^2 + W^2 - 1, X^2 + Y^2 - 1, (1 - X)Z - YW, YZ - (1 + X)W),$$

and

$$\mathfrak{c} = (Z^2 + W^2 - 1, (1 - X)Z - YW, YZ - (1 + X)W).$$

Exercise 2.16 ★

Consider the polynomial

$$F = X^4Y^2 + X^2Y^4 - 3X^2Y^2 + 1.$$

1. Find a real zero of F .
2. Verify the identity

$$\begin{aligned} F \cdot (X^2 + Y^2 + 1) &= (X^2Y - Y)^2 + (XY^2 - X)^2 + (X^2Y^2 - 1)^2 \\ &\quad + \frac{1}{4}(XY^3 - X^3Y)^2 + \frac{3}{4}(XY^3 + X^3Y - 2XY)^2. \end{aligned}$$

3. Let $K = \mathbb{R}$. Deduce that

$$F(x, y) \geq 0$$

for every point $(x, y) \in \mathbb{A}_{\mathbb{R}}^2$.

Source note. By a theorem of Artin—the solution to Hilbert’s seventeenth problem—every real polynomial that is nowhere negative can be written as a sum of squares of rational functions. The Motzkin polynomial F above gives a concrete example showing that such a polynomial cannot in general be written as a sum of squares of polynomials. Here, however, we prove only its non-negativity.

Exercise 2.17

Find generators for an ideal $\mathfrak{a} \subseteq \mathbb{R}[X, Y, Z]$ whose zero set consists of exactly the four points

$$(2, 3, 4), \quad (1, 1/5, 0), \quad (0, 0, 1), \quad (-1, -2, \sqrt{3}) \in \mathbb{A}_{\mathbb{R}}^3.$$

Exercise 2.18

Let $\mathfrak{a}, \mathfrak{b} \subseteq R$ be ideals in a commutative ring R . Prove the inclusion

$$\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{a} \cap \mathfrak{b}.$$

Exercise 2.19

Prove that a product of principal ideals is again a principal ideal.

Exercise 2.20 ★

Let I and J be ideals in a commutative ring R , and let $n \in \mathbb{N}$. Prove the equality

$$(I + J)^n = I^n + I^{n-1}J + I^{n-2}J^2 + \dots + I^2J^{n-2} + IJ^{n-1} + J^n.$$

Exercise 2.21

Let R be a commutative ring with ideals $\mathfrak{a}, \mathfrak{b} \subseteq R$. Let

$$S = R/\mathfrak{b}$$

and let $\tilde{\mathfrak{a}} = \mathfrak{a}S$ be the image ideal. Prove that

$$\mathfrak{a}^n S = \tilde{\mathfrak{a}}^n.$$

Exercise 2.22

Let K be a field. In $K[X, Y]$, consider the two prime ideals

$$\mathfrak{p} = (X) \subset (X, Y) = \mathfrak{m}.$$

Prove that there is no ideal $\mathfrak{a}$ with

$$\mathfrak{p} = \mathfrak{a}\mathfrak{m}.$$

Exercises to hand in**Exercise 2.23 — 3 points**

Find all solutions of the equation

$$x^2 + y^2 + xy = 1$$

over the fields $K = \mathbb{F}_2, \mathbb{F}_4$, and $\mathbb{F}_8$. You may use these [representations of the fields](#).

Exercise 2.24 — 3 points

Let

$$M = \{P_1, P_2, \dots, P_m\} \subseteq \mathbb{A}_{\mathbb{R}}^n$$

be a finite set of points. Prove that M is the zero set of a single polynomial.

Exercise 2.25 — 3 points

Prove that the set of real triangularisable 2×2 matrices, regarded as a subset of $\mathbb{A}_{\mathbb{R}}^4$, is not an affine algebraic set.

Exercise 2.26 – 4 points

Find the intersection points of the two ellipses

$$\{(x, y) \in \mathbb{R}^2 \mid 4x^2 - 3xy + 2y^2 = 7\}$$

and

$$\{(x, y) \in \mathbb{R}^2 \mid 3x^2 + 4xy + 5y^2 = 8\}.$$

Exercise 2.27 – 4 points

Let S be the zero locus in $\mathbb{A}_K^3$ given by the equation

$$x^2 + y^2 = z^2.$$

The intersection of S with a plane E is a curve, described in E by an equation in two suitable variables. Find such an equation for each of the planes

$$E_1 = V(x), \quad E_2 = V(z - 1), \quad E_3 = V(x + 2y + 3z), \quad E_4 = V(3x - 2z).$$

Source navigation: [Lecture 2](#) · [Worksheet 3 \(source\)](#) · [Lecture 1](#)

Solutions to Worksheet 2

This section translates all nine public solutions linked from Worksheet 2 at the source freeze. Their numbering and order follow the source exercises. No additional solutions are supplied for exercises without a public solution at that boundary.

Solution to Exercise 2.2

A direction vector of the line is

$$\begin{pmatrix} 5 \\ -3 \end{pmatrix}.$$

Its equation therefore has the form

$$3x + 5y = c.$$

Substituting either point gives $c = 2$. Thus

$$y = \frac{2 - 3x}{5}.$$

Substituting this into the circle equation

$$x^2 + y^2 = 1$$

gives

$$x^2 + \left(\frac{2 - 3x}{5}\right)^2 = 1,$$

or

$$x^2 + \frac{4 - 12x + 9x^2}{25} - 1 = \frac{34}{25}x^2 - \frac{12}{25}x - \frac{21}{25} = 0.$$

Normalising the equation gives

$$x^2 - \frac{6}{17}x - \frac{21}{34} = 0.$$

Consequently,

$$\begin{aligned} x_{1,2} &= \frac{\frac{6}{17} \pm \sqrt{\left(\frac{6}{17}\right)^2 + 4 \cdot \frac{21}{34}}}{2} \\ &= \frac{\frac{6}{17} \pm \sqrt{\left(\frac{6}{17}\right)^2 + \frac{42}{17}}}{2} \\ &= \frac{6 \pm \sqrt{6^2 + 714}}{34} \\ &= \frac{6 \pm \sqrt{750}}{34} \\ &= \frac{6 \pm 5\sqrt{30}}{34}, \end{aligned}$$

and

$$\begin{aligned} y_{1,2} &= \frac{2 - 3x_{1,2}}{5} \\ &= \frac{2 - 3\left(\frac{6 \pm 5\sqrt{30}}{34}\right)}{5} \\ &= \frac{68 - 3(6 \pm 5\sqrt{30})}{170} \\ &= \frac{50 \mp 15\sqrt{30}}{170} \\ &= \frac{10 \mp 3\sqrt{30}}{34}. \end{aligned}$$

The intersection points are therefore

$$\left(\frac{6 + 5\sqrt{30}}{34}, \frac{10 - 3\sqrt{30}}{34}\right) \quad \text{and} \quad \left(\frac{6 - 5\sqrt{30}}{34}, \frac{10 + 3\sqrt{30}}{34}\right).$$

[Back to Exercise 2.2.](#)

Solution to Exercise 2.6

The unit circle is the solution set of

$$x^2 + y^2 = 1,$$

while K is the solution set of

$$(x - 1)^2 + y^2 = x^2 - 2x + 1 + y^2 = 4.$$

Subtracting the first equation from the second gives

$$-2x + 1 = 3,$$

so $x = -1$. The unit circle equation then gives $y = 0$. The only intersection point is therefore $(-1, 0)$, which indeed satisfies both equations.

[Back to Exercise 2.6.](#)

Solution to Exercise 2.7

We seek the solutions of the system

$$x^2 + xy + 3y^2 = 3$$

and

$$2x^2 - xy + y^2 = 4.$$

Adding the two equations gives

$$3x^2 + 4y^2 = 7,$$

while twice the first equation minus the second gives

$$3xy + 5y^2 = 2.$$

From the latter equation,

$$x = \frac{2 - 5y^2}{3y};$$

there is certainly no solution with $y = 0$. Substituting this expression for x into the preceding equation gives

$$3 \left(\frac{2 - 5y^2}{3y} \right)^2 + 4y^2 = 7.$$

Multiplication by $3y^2$ yields

$$\begin{aligned} 0 &= (2 - 5y^2)^2 + 12y^4 - 21y^2 \\ &= 4 - 20y^2 + 25y^4 + 12y^4 - 21y^2 \\ &= 37y^4 - 41y^2 + 4. \end{aligned}$$

This is a biquadratic equation.

Edition note: The frozen public solution stops here, without giving the intersection coordinates.

[Back to Exercise 2.7.](#)

Solution to Exercise 2.8

The standard parabola is given by

$$y = x^2,$$

and the unit circle by

$$x^2 + y^2 = 1.$$

The intersection points must satisfy both equations simultaneously. Using the first equation to replace x^2 in the second, we obtain

$$y^2 + y - 1 = 0.$$

Thus

$$y = \frac{-1 \pm \sqrt{1+4}}{2} = \frac{-1 \pm \sqrt{5}}{2}.$$

The negative sign gives no real value of x , so

$$y = \frac{-1 + \sqrt{5}}{2}, \quad x = \pm \sqrt{\frac{-1 + \sqrt{5}}{2}}.$$

The two intersection points are

$$\left(-\sqrt{\frac{-1 + \sqrt{5}}{2}}, \frac{-1 + \sqrt{5}}{2} \right)$$

and

$$\left(\sqrt{\frac{-1 + \sqrt{5}}{2}}, \frac{-1 + \sqrt{5}}{2} \right).$$

[Back to Exercise 2.8.](#)

Solution to Exercise 2.9

1. **Edition note:** The frozen source leaves the sketch item blank.
2. We have

$$\begin{aligned} K &= \{(x, y) \in \mathbb{R}^2 \mid (y-1)^2 + x^2 = 1\} \\ &= \{(x, y) \in \mathbb{R}^2 \mid y^2 - 2y + 1 + x^2 = 1\} \\ &= \{(x, y) \in \mathbb{R}^2 \mid y^2 - 2y + x^2 = 0\}. \end{aligned}$$

3. We seek the common solution set of the two equations

$$y = x^2$$

and

$$y^2 - 2y + x^2 = 0.$$

Replacing x^2 by y in the second equation gives

$$0 = y^2 - 2y + y = y^2 - y = y(y - 1).$$

Thus $y = 0$ or $y = 1$. This gives the three intersection points $(0, 0)$, $(1, 1)$, and $(-1, 1)$.

4. The circle equation

$$y^2 - 2y + x^2 = 0$$

is equivalent to

$$y^2 - 2y = -x^2,$$

and hence to

$$(y - 1)^2 = 1 - x^2.$$

Therefore

$$y = 1 \pm \sqrt{1 - x^2}.$$

The lower semicircle is the graph of the function

$$[-1, 1] \longrightarrow \mathbb{R}, \quad x \longmapsto 1 - \sqrt{1 - x^2}.$$

5. We claim that on $[-1, 1]$ the parabola lies above the lower semicircle. We must show that

$$x^2 \geq 1 - \sqrt{1 - x^2}.$$

This is equivalent to

$$\sqrt{1 - x^2} \geq 1 - x^2.$$

Since both sides are non-negative on this interval, this is equivalent to

$$1 - x^2 \geq (1 - x^2)^2 = 1 + x^4 - 2x^2.$$

The last inequality is equivalent to $x^4 - x^2 \leq 0$, and then to $x^2 - 1 \leq 0$, which holds because $x \in [-1, 1]$.

Edition note: The source calls both sides positive on $[-1, 1]$. They are non-negative and vanish at the endpoints; non-negativity is the condition needed for squaring.

[Back to Exercise 2.9.](#)

Solution to Exercise 2.11

1. Let

$$A = \begin{pmatrix} X_1 & X_2 \\ X_3 & X_4 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} Y_1 & Y_2 \\ Y_3 & Y_4 \end{pmatrix}.$$

Then

$$AB = \begin{pmatrix} X_1 & X_2 \\ X_3 & X_4 \end{pmatrix} \begin{pmatrix} Y_1 & Y_2 \\ Y_3 & Y_4 \end{pmatrix} = \begin{pmatrix} X_1Y_1 + X_2Y_3 & X_1Y_2 + X_2Y_4 \\ X_3Y_1 + X_4Y_3 & X_3Y_2 + X_4Y_4 \end{pmatrix}$$

and

$$BA = \begin{pmatrix} Y_1 & Y_2 \\ Y_3 & Y_4 \end{pmatrix} \begin{pmatrix} X_1 & X_2 \\ X_3 & X_4 \end{pmatrix} = \begin{pmatrix} X_1Y_1 + X_3Y_2 & X_2Y_1 + X_4Y_2 \\ X_1Y_3 + X_3Y_4 & X_2Y_3 + X_4Y_4 \end{pmatrix}.$$

These product matrices are equal precisely when all four corresponding entries agree, that is, when

$$X_1Y_1 + X_2Y_3 = X_1Y_1 + X_3Y_2,$$

$$X_1Y_2 + X_2Y_4 = X_2Y_1 + X_4Y_2,$$

$$X_3Y_1 + X_4Y_3 = X_1Y_3 + X_3Y_4,$$

and

$$X_3Y_2 + X_4Y_4 = X_2Y_3 + X_4Y_4.$$

Thus the set is an affine variety. The first and fourth equations are equivalent to each other and to

$$X_2Y_3 = X_3Y_2.$$

The commuting matrices are therefore described by the system

$$X_2Y_3 = X_3Y_2,$$

$$X_1Y_2 + X_2Y_4 = X_2Y_1 + X_4Y_2,$$

$$X_3Y_1 + X_4Y_3 = X_1Y_3 + X_3Y_4.$$

2. The identity matrix E_2 commutes with every matrix. Hence (A, E_2) is a preimage of A .

3. We seek the matrices

$$B = \begin{pmatrix} Y_1 & Y_2 \\ Y_3 & Y_4 \end{pmatrix}$$

that satisfy the preceding system for

$$A = \begin{pmatrix} X_1 & X_2 \\ X_3 & X_4 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

The conditions become

$$Y_3 = 0, \quad Y_2 + Y_4 = Y_1, \quad 0 = Y_3,$$

and the third condition can be omitted. The inverse image of the given matrix is

$$\left\{ \left(\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} Y_2 + Y_4 & Y_2 \\ 0 & Y_4 \end{pmatrix} \right) \mid Y_2, Y_4 \in K \right\}.$$

[Back to Exercise 2.11.](#)

Solution to Exercise 2.15

First we prove $\mathfrak{b} \subseteq \mathfrak{a}$ by showing that every generator of $\mathfrak{b}$ is 0 in the quotient ring modulo $\mathfrak{a}$. In that ring,

$$\begin{aligned} X^2 + Y^2 &= (2Z^2 - 1)^2 + 4Z^2W^2 \\ &= (2Z^2 - 1)^2 + 4Z^2(1 - Z^2) \\ &= 4Z^4 - 4Z^2 + 1 + 4Z^2 - 4Z^4 \\ &= 1. \end{aligned}$$

Moreover,

$$\begin{aligned} YW &= 2ZW^2 \\ &= 2Z(1 - Z^2) \\ &= Z(2 - 2Z^2) \\ &= Z(1 - 2Z^2 + 1) \\ &= Z(1 - X), \end{aligned}$$

and

$$\begin{aligned} W(1 + X) &= W(2Z^2) \\ &= 2WZ^2 \\ &= ZY. \end{aligned}$$

The inclusion $\mathfrak{c} \subseteq \mathfrak{b}$ is clear, since one generator has been omitted.

Finally, we prove $\mathfrak{a} \subseteq \mathfrak{c}$ by showing that the generators of $\mathfrak{a}$ are 0 in the quotient ring modulo $\mathfrak{c}$. In this ring,

$$ZX = Z - YW \quad \text{and} \quad WX = YZ - W.$$

Consequently,

$$\begin{aligned} X &= X \cdot 1 \\ &= X(Z^2 + W^2) \\ &= Z^2 - ZYW + YZW - W^2 \\ &= Z^2 - W^2 \\ &= 2Z^2 - 1, \end{aligned}$$

and

$$\begin{aligned} Y &= Y \cdot 1 \\ &= Y(Z^2 + W^2) \\ &= WXZ + WZ + ZW - ZXW \\ &= 2ZW. \end{aligned}$$

Back to Exercise 2.15.

Solution to Exercise 2.16

1. Clearly $(1, 1)$ is a zero of F .
2. We have

$$\begin{aligned}
 F \cdot (X^2 + Y^2 + 1) &= (X^4Y^2 + X^2Y^4 - 3X^2Y^2 + 1)(X^2 + Y^2 + 1) \\
 &= X^6Y^2 + X^4Y^4 - 3X^4Y^2 + X^2 + X^4Y^4 + X^2Y^6 - 3X^2Y^4 + Y^2 \\
 &\quad + X^4Y^2 + X^2Y^4 - 3X^2Y^2 + 1 \\
 &= 2X^4Y^4 + X^6Y^2 + X^2Y^6 - 2X^4Y^2 - 2X^2Y^4 - 3X^2Y^2 + X^2 + Y^2 + 1.
 \end{aligned}$$

For the other side,

$$(X^2Y - Y)^2 + (XY^2 - X)^2 + (X^2Y^2 - 1)^2 + \frac{1}{4}(XY^3 - X^3Y)^2 + \frac{3}{4}(XY^3 + X^3Y - 2XY)^2,$$

we calculate the coefficient of each monomial. Only even degrees occur, and the highest degree is 8. Only the last three summands contribute at that degree, and the only monomials are X^6Y^2 , X^4Y^4 , and X^2Y^6 :

$$X^6Y^2 : \quad \frac{1}{4} + \frac{3}{4} = 1,$$

$$X^4Y^4 : \quad 1 + \frac{1}{4}(-2) + \frac{3}{4}(2) = 2,$$

$$X^2Y^6 : \quad \frac{1}{4} + \frac{3}{4} = 1.$$

In degree 6, only X^4Y^2 and X^2Y^4 occur:

$$X^4Y^2 : \quad 1 + \frac{3}{4}(-4) = -2,$$

$$X^2Y^4 : \quad 1 + \frac{3}{4}(-4) = -2.$$

In degree 4, only X^2Y^2 occurs:

$$X^2Y^2 : \quad -2 - 2 - 2 + \frac{3}{4}(4) = -3.$$

In degree 2, the coefficients of X^2 and Y^2 are both 1. In degree 0, the coefficient of 1 is also 1. Thus the two sides agree.

3. Dividing the identity in part (2) by $X^2 + Y^2 + 1$ in the field of fractions of $K[X, Y]$, we obtain

$$\begin{aligned}
 F &= \frac{1}{X^2 + Y^2 + 1} \left((X^2Y - Y)^2 + (XY^2 - X)^2 + (X^2Y^2 - 1)^2 \right. \\
 &\quad \left. + \frac{1}{4}(XY^3 - X^3Y)^2 + \frac{3}{4}(XY^3 + X^3Y - 2XY)^2 \right).
 \end{aligned}$$

Since $X^2 + Y^2 + 1$ has no real zero, this identity also holds as an identity of functions $\mathbb{R}^2 \rightarrow \mathbb{R}$. Squares are never negative, and all coefficients of the squares involved are positive. Hence the function is non-negative at every point.

Edition note: The source's introductory sentence for degree 2 mentions only X^2 , but its coefficient list includes both X^2 and Y^2 . The wording above follows that list and the displayed expansion.

[Back to Exercise 2.16.](#)

Solution to Exercise 2.20

To prove the inclusion $\subseteq$, take $f \in (I + J)^n$. Since a product of ideals consists of all sums of products, we can write

$$f = f_1 + f_2 + \cdots + f_k,$$

where

$$f_\ell = c_{\ell 1} c_{\ell 2} \cdots c_{\ell n},$$

with $c_{\ell r} \in I + J$. In turn,

$$c_{\ell r} = a_{\ell r} + b_{\ell r}$$

with $a_{\ell r} \in I$ and $b_{\ell r} \in J$. Thus

$$f_\ell = (a_{\ell 1} + b_{\ell 1})(a_{\ell 2} + b_{\ell 2}) \cdots (a_{\ell n} + b_{\ell n}).$$

Expanding this product by distributivity gives a sum of products with n factors each: s factors belong to I and $n - s$ to J . Each summand therefore belongs to the right-hand side, as do each f_ℓ and finally f .

To prove the inclusion $\supseteq$, it suffices to show

$$I^s J^{n-s} \subseteq (I + J)^n$$

for every s . Since $I, J \subseteq I + J$, we immediately have

$$I^s J^{n-s} \subseteq (I + J)^s (I + J)^{n-s} = (I + J)^n.$$

[Back to Exercise 2.20.](#)

Lecture 3: The Zariski Topology, Vanishing Ideals, and Radicals

The Zariski topology



Figure 26: Oscar Zariski (1899-1986)

In Proposition 2.8 we showed that the affine algebraic subsets of an affine space satisfy the axioms for the closed sets of a topology. This topology is called the *Zariski topology*.

Definition: the Zariski topology

On affine space $\mathbb{A}_K^n$, the *Zariski topology* is the topology in which the affine algebraic sets are declared to be closed. Thus the open sets of the Zariski topology are the complements of affine algebraic sets. For an ideal $\mathfrak{a}$, this complement is denoted by

$$D(\mathfrak{a}) = \mathbb{A}_K^n \setminus V(\mathfrak{a}).$$

The Zariski topology differs greatly from other topologies, especially those given by a metric. In particular, the Zariski topology is not Hausdorff. Generally speaking, nonempty open sets in the Zariski topology are very large (see Exercise 3.20), while the closed sets—the affine algebraic sets—are very thin, apart from the whole space itself.

Edition note: The source's non-Hausdorff assertion requires K to be infinite and $n \geq 1$. Over a finite field, affine space is a finite discrete space and is Hausdorff; $\mathbb{A}_K^0$ is also Hausdorff.

Example: the Zariski topology on the affine line

The Zariski topology on the affine line $\mathbb{A}_K^1$ over a field K is easy to describe. The whole affine line is a closed set, given by $V(0)$. All other closed subsets are given by $V(\mathfrak{a})$ with $\mathfrak{a} \neq 0$. Since $K[X]$ is a principal ideal domain, we can even write

$$\mathfrak{a} = (f), \quad f \neq 0.$$

The corresponding zero locus consists of only finitely many points. Conversely, each individual point P with coordinate a is the unique zero of the linear polynomial $X - a$, so

$$\{P\} = V(X - a)$$

is Zariski closed. A finite collection of points $P_1, \dots, P_k$ with coordinates $a_1, \dots, a_k$ is the zero locus of the polynomial

$$(X - a_1) \cdots (X - a_k).$$

The Zariski closed sets of the affine line are therefore all finite subsets, including the empty set, together with the whole affine line.



Figure 27: A line in the plane

Example: points are closed

Every point

$$P = (a_1, \dots, a_n) \in \mathbb{A}_K^n$$

is Zariski closed; more precisely,

$$P = V(X_1 - a_1, X_2 - a_2, \dots, X_n - a_n).$$

Apart from the empty set and the whole space, points are the simplest affine algebraic sets. The ideal

$$(X_1 - a_1, X_2 - a_2, \dots, X_n - a_n),$$

called the *point ideal*, is maximal; see Exercise 2.12.

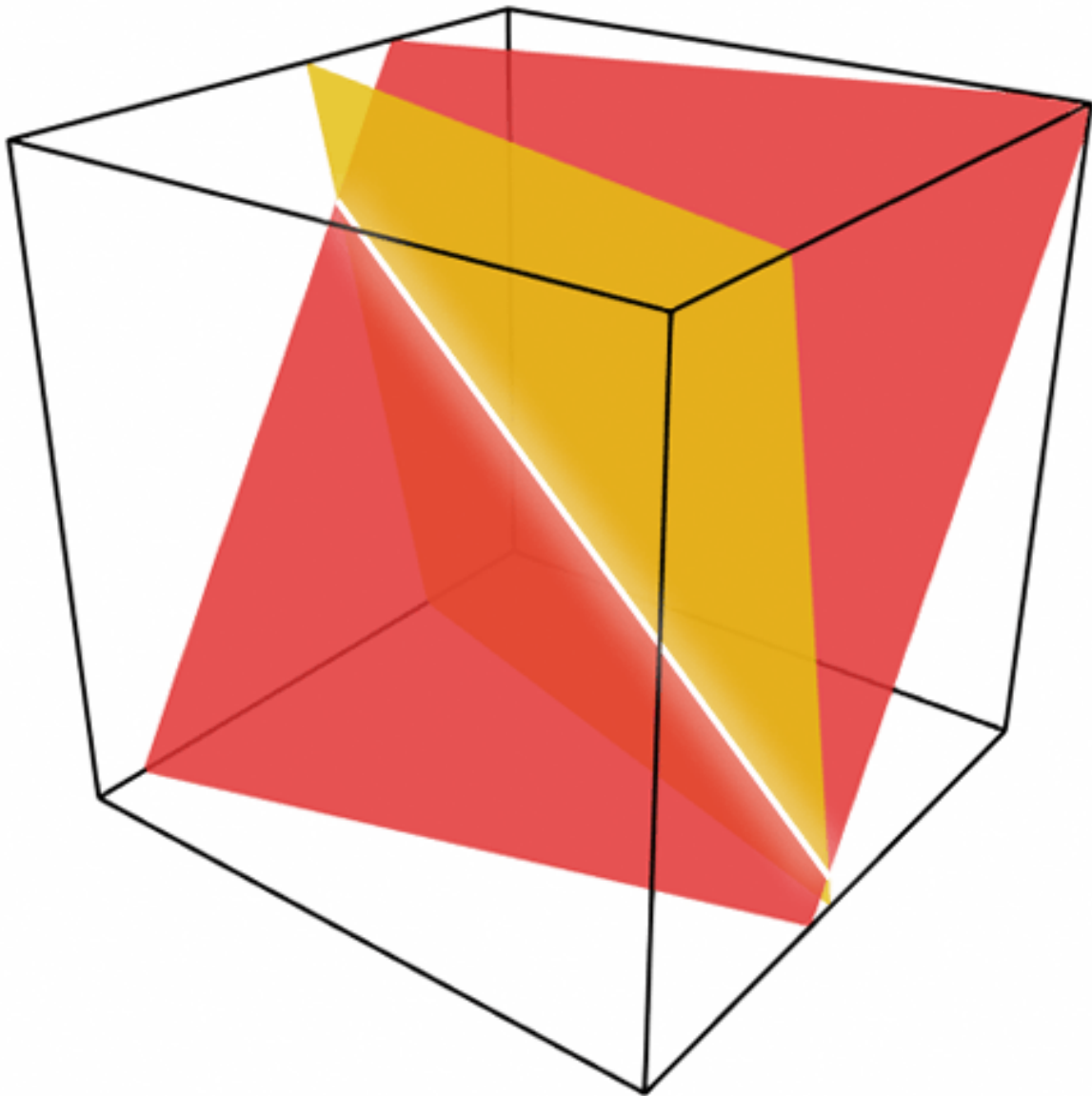


Figure 28: Intersection of two planes

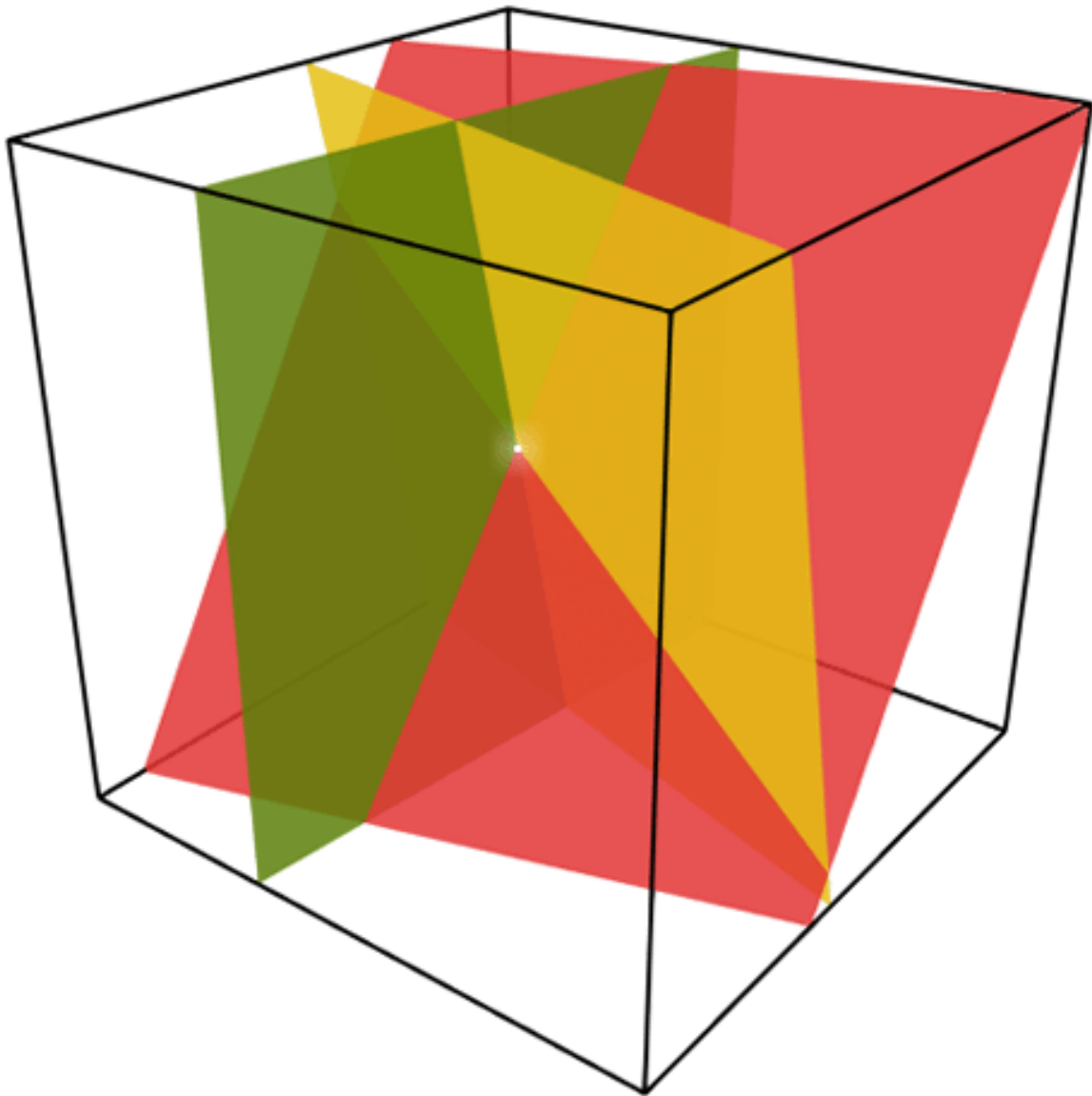


Figure 29: Intersection of three planes

By Proposition 2.8(3), every finite subset of affine space is Zariski closed. Thus, if E is a finite set of points, its complement

$$\mathbb{A}_K^n \setminus E$$

is Zariski open. Likewise, for a rational function P/Q with

$$P, Q \in K[X_1, \dots, X_n], \quad Q \neq 0,$$

its domain of definition, namely $D(Q)$, is open.

Vanishing ideals

Definition: vanishing ideal

Let $T \subseteq \mathbb{A}_K^n$ be a subset. The set

$$\text{Id}(T) = \{F \in K[X_1, \dots, X_n] \mid F(P) = 0 \text{ for every } P \in T\}$$

is called the *vanishing ideal* of T .

This set is indeed an ideal. If $F(P) = 0$ and $G(P) = 0$ for every $P \in T$, the same holds for the sum $F + G$ and every multiple HF .

We therefore have two assignments in opposite directions: a subset of affine space is assigned its vanishing ideal, while an ideal in the polynomial ring is assigned its zero locus. We wish to understand to what extent ideals and zero loci correspond to one another.

Example: the empty set and the whole space

The vanishing ideal of the empty set is the unit ideal, since there is no point at which the vanishing condition needs to be checked.

The vanishing ideal of the whole space $\mathbb{A}_K^n$ depends on the field. If K is infinite, only the zero polynomial vanishes everywhere, so the vanishing ideal is the zero ideal. This follows from Exercise 3.18.

If, on the other hand, K is a finite field with q elements, then

$$x^q - x = 0$$

for every $x \in K$. Thus the polynomial $X^q - X$ vanishes at every point of the affine line and belongs to its vanishing ideal. In higher dimensions,

$$\text{Id}(\mathbb{A}_K^n) = (X_1^q - X_1, X_2^q - X_2, \dots, X_n^q - X_n).$$

Example: the vanishing ideal of a point

Let

$$P = (a_1, \dots, a_n) \in \mathbb{A}_K^n.$$

Then

$$\text{Id}(P) = (X_1 - a_1, \dots, X_n - a_n).$$

First, the linear polynomials $X_i - a_i$ clearly vanish at P , since $(X_i - a_i)(P) = a_i - a_i = 0$. Hence the ideal they generate is contained in the vanishing ideal.

Conversely, let F be a polynomial with $F(P) = 0$. Express F in the “new variables”

$$\tilde{X}_1 = X_1 - a_1, \dots, \tilde{X}_n = X_n - a_n$$

by replacing X_i with $X_i - a_i + a_i$. In these new variables, write

$$F = \sum_{\nu} b_{\nu} \tilde{X}^{\nu}.$$

This polynomial has a constant term b_0 , while every other monomial contains at least one variable. Thus, for suitable polynomials F_i , we can write

$$F = F_1 \tilde{X}_1 + \dots + F_n \tilde{X}_n + c.$$

Since $F(P) = c = 0$, we obtain

$$F \in (\tilde{X}_1, \dots, \tilde{X}_n) = (X_1 - a_1, \dots, X_n - a_n).$$

Lemma: inclusion of subsets reverses inclusion of vanishing ideals

Let $V \subseteq W \subseteq \mathbb{A}_K^n$. Then

$$\text{Id}(W) \subseteq \text{Id}(V).$$

Proof Take $F \in \text{Id}(W)$. In other words, $F(P) = 0$ for every $P \in W$. Since $V \subseteq W$, in particular $F(P) = 0$ for every $P \in V$. Thus $F \in \text{Id}(V)$. $\square$

Lemma: relations between zero loci and vanishing ideals

Let $I \subseteq K[X_1, \dots, X_n]$ be an ideal and $T \subseteq \mathbb{A}_K^n$ a subset. The following statements hold.

1. $T \subseteq V(\text{Id}(T))$.
2. $I \subseteq \text{Id}(V(I))$.
3. $V(I) = V(\text{Id}(V(I)))$.
4. $\text{Id}(T) = \text{Id}(V(\text{Id}(T)))$.

Proof For (1), take $P \in T$. By definition, every polynomial $F \in \text{Id}(T)$ vanishes on T , so $P \in V(\text{Id}(T))$.

For (2), take $F \in I$. The polynomial F vanishes throughout $V(I)$, so $F \in \text{Id}(V(I))$.

For (3), apply (1) to $T = V(I)$ to obtain $V(I) \subseteq V(\text{Id}(V(I)))$. By (2), $I \subseteq \text{Id}(V(I))$; applying $V(-)$ and Lemma 2.7 gives the reverse inclusion.

Statement (4) is proved in the same way. $\square$

Example: strict inclusions

Both inclusions in Lemma 3.8(1) and (2) can be strict. For example, let $T \subsetneq \mathbb{A}_K^1$ be an infinite proper subset; this requires K to be infinite. Then

$$\text{Id}(T) = 0,$$

so $V(0) = \mathbb{A}_K^1$ is strictly larger than T .

For the inclusion in (2), take $R = K[X]$ and $I = (X^2)$. Then

$$V(I) = \{0\}, \quad \text{Id}(\{0\}) = (X),$$

but $X \notin (X^2)$. A more extreme example in $R = \mathbb{R}[X, Y]$ is $I = (X^2 + Y^2)$, with $V(I) = \{(0, 0)\}$. The vanishing ideal of that point is (X, Y) .

Lemma: Zariski closure

Let $T \subseteq \mathbb{A}_K^n$. The Zariski closure of T is

$$\overline{T} = V(\text{Id}(T)).$$

Proof The inclusion $T \subseteq V(\text{Id}(T))$ was proved in Lemma 3.8(1). Since $V(\text{Id}(T))$ is closed by definition, we obtain

$$\overline{T} \subseteq V(\text{Id}(T)).$$

Conversely, take $P \in V(\text{Id}(T))$ and suppose that $P \notin \overline{T}$. Then there is a Zariski open set U such that

$$P \in U, \quad U \cap T = \emptyset.$$

Write $U = D(\mathfrak{a})$. The condition $P \in U$ means that some $G \in \mathfrak{a}$ satisfies $G(P) \neq 0$. Then

$$P \in D(G) \subseteq U,$$

so $T \cap D(G) = \emptyset$. Hence $T \subseteq V(G)$ and $G \in \text{Id}(T)$. But $G(P) \neq 0$ contradicts $P \in V(\text{Id}(T))$. $\square$

Radicals

Definition: radical ideal

An ideal $\mathfrak{a}$ in a commutative ring R is called a *radical ideal* if the following holds: whenever $f^n \in \mathfrak{a}$ for some $n \in \mathbb{N}$, we already have $f \in \mathfrak{a}$.

Definition: the radical of an ideal

Let R be a commutative ring and $\mathfrak{a} \subseteq R$ an ideal. The set

$$\text{rad}(\mathfrak{a}) = \{f \in R \mid \text{there is an } r \text{ with } f^r \in \mathfrak{a}\}$$

is called the *radical* of $\mathfrak{a}$.

The radical of an ideal is itself a radical ideal.

Lemma: the radical of an ideal is a radical ideal

Let R be a commutative ring and $\mathfrak{a} \subseteq R$ an ideal. Then $\text{rad}(\mathfrak{a})$ is a radical ideal.

Proof First we show that the set is an ideal. Clearly 0 belongs to the radical. If $f \in \text{rad}(\mathfrak{a})$, say $f^r \in \mathfrak{a}$, then

$$(af)^r = a^r f^r \in \mathfrak{a},$$

so af belongs to the radical. For closure under addition, let $f, g \in \text{rad}(\mathfrak{a})$ with $f^r \in \mathfrak{a}$ and $g^s \in \mathfrak{a}$. Then

$$\begin{aligned} (f+g)^{r+s} &= \sum_{i+j=r+s} \binom{r+s}{i} f^i g^j \\ &= \sum_{\substack{i+j=r+s \\ i < r}} \binom{r+s}{i} f^i g^j + \sum_{\substack{i+j=r+s \\ i \geq r}} \binom{r+s}{i} f^i g^j \in \mathfrak{a}. \end{aligned}$$

Now suppose that $f^k \in \text{rad}(\mathfrak{a})$. For some r we have

$$(f^k)^r = f^{kr} \in \mathfrak{a},$$

so $f \in \text{rad}(\mathfrak{a})$. $\square$

Lemma: vanishing ideals are radical ideals

Let $T \subseteq \mathbb{A}_K^n$. Then the vanishing ideal $\text{Id}(T)$ is a radical ideal.

Proof Let $F \in K[X_1, \dots, X_n]$ and $F^s \in \text{Id}(T)$. Then

$$F^s(P) = 0$$

for every $P \in T$. Consequently $F(P) = 0$ for every $P \in T$, so $F \in \text{Id}(T)$. $\square$

Later we shall see that over an algebraically closed field, radical ideals and algebraic zero loci correspond to one another. This is the content of Hilbert's Nullstellensatz.

Source navigation: [course](#) · [Lecture 2](#) · [Lecture 4 \(source\)](#) · [Worksheet 3](#)

Worksheet 3

Practice exercises

Exercise 3.1

Prove that the nonempty Zariski open subsets of the affine line $\mathbb{A}_K^1$ are precisely the maximal domains of definition of rational functions.

Edition note: Assume K is infinite. The source omits this hypothesis: over a finite field with q elements, the rational expression $1/(X^q - X)$ has empty domain on K .

Exercise 3.2

Let K be an infinite field and $P \in K[X]$ a nonconstant polynomial. Prove that the function

$$P : K \longrightarrow K$$

defined by P takes infinitely many values.

Exercise 3.3

1. Sketch the real zero loci of $Y^n - X^n$.
2. Determine the vanishing ideals of the affine algebraic sets $V_n \subseteq \mathbb{A}_{\mathbb{R}}^2$ consisting of the union of all lines through the origin and a vertex of a regular n -gon with $(1, 0)$ as one vertex.

Exercise 3.4

Describe a map

$$\varphi : \mathbb{A}_K^1 \longrightarrow \mathbb{A}_K^1$$

that is continuous in the Zariski topology but is not given by a polynomial.

Edition note: The source omits the necessary assumption that K is infinite. Over a finite field, every map $K \rightarrow K$ is represented by a polynomial, so no such example exists.

Exercise 3.5

Let

$$V \subseteq \mathbb{A}_{\mathbb{C}}^n = \mathbb{C}^n$$

be an affine algebraic set. Prove that under the identification

$$\mathbb{C}^n = \mathbb{R}^{2n},$$

the subset V is also an affine algebraic set in $\mathbb{A}_{\mathbb{R}}^{2n}$. Prove that the converse does not hold.

Exercise 3.6

Let (M, d) be a metric space and $T \subseteq M$ a nonempty subset. Prove that

$$d_T(x) := \inf\{d(x, y) \mid y \in T\}$$

defines a well-defined continuous function $M \rightarrow \mathbb{R}$.

Exercise 3.7

Let (M, d) be a metric space and $T \subseteq M$ a subset. Prove that T is closed if and only if there is a continuous function

$$f : M \longrightarrow \mathbb{R}$$

with

$$f^{-1}(0) = T.$$

Exercise 3.8

Prove that, for $r > 0$, the open ball $U(P, r)$ in $\mathbb{R}^n$ is not Zariski open, and the closed ball $B(P, r)$ is not Zariski closed.

Exercise 3.9

Characterise the radical ideals in $\mathbb{Z}$ using prime factorisation.

Preliminary definition: nilpotent element

An element a of a commutative ring R is called *nilpotent* if

$$a^n = 0$$

for some natural number n .

Exercise 3.10

Let R be a commutative ring and $f, g \in R$ nilpotent elements. Prove that their sum $f + g$ is also nilpotent.

Exercise 3.11 ★

Let R be a commutative ring and $f \in R$ a nilpotent element. Prove that $1 + f$ is a unit.

Exercise 3.12

Let R be a commutative ring and $r \in R$ a nilpotent element. Construct a linear polynomial in $R[X]$ that is a unit, and give its inverse.

Edition note: If “linear” means degree exactly 1, assume $r \neq 0$. The source allows $r = 0$; in that case a polynomial of degree at most 1 is the appropriate formulation, and the construction may be constant.

Preliminary definition: reduced ring

A commutative ring R is called *reduced* if 0 is its only nilpotent element.

Exercise 3.13 ★

Prove that an ideal $\mathfrak{a}$ in a commutative ring R is radical if and only if the quotient ring $R/\mathfrak{a}$ is reduced.

Exercise 3.14

Let $\mathfrak{a} \subseteq R$ be an ideal in a commutative ring R . Prove that all the powers

$$\mathfrak{a}^n, \quad n \in \mathbb{N}_+,$$

have the same radical.

Exercise 3.15

Prove that every prime ideal is a radical ideal.

Exercise 3.16

Let R and S be commutative rings, $\varphi : R \rightarrow S$ a ring homomorphism, and $\mathfrak{a}$ a radical ideal in S . Prove that the inverse image $\varphi^{-1}(\mathfrak{a})$ is a radical ideal in R .

Exercise 3.17

Let $\mathfrak{a}$ and $\mathfrak{b}$ be ideals in $K[X_1, \dots, X_n]$ with the same radical. Prove that their zero loci are also equal. Give an example showing that the converse does not hold.

Exercise 3.18

Let K be an infinite field, $F \in K[X_1, \dots, X_n]$ a polynomial, and $U \subseteq \mathbb{A}_K^n$ a nonempty Zariski open subset. Suppose that $F|_U = 0$ as a function. Prove that F is the zero polynomial.

Exercises to hand in**Exercise 3.19 — 3 points**

Let

$$\varphi : \mathbb{A}_K^n \longrightarrow \mathbb{A}_K^m$$

be a map given by m polynomials in n variables. Prove that φ is continuous in the Zariski topology.

Exercise 3.20 — 4 points

Let K be an infinite field. Prove that every nonempty Zariski open subset

$$U \subseteq \mathbb{A}_K^n$$

is dense.

Source hint: Reduce to the case $n = 1$. Do not use Exercise 3.18.

Exercise 3.21 — 5 points

Determine the Zariski closure of each of the following subsets of the affine plane $\mathbb{A}_K^2$.

1. $\{(x, \sin x) \mid x \in \mathbb{R}\}$.
2. $\{(\cos x, \sin x) \mid x \in \mathbb{R}\}$.
3. $\{(x, x^3) \mid 0 \leq x \leq 1, x \in \mathbb{R}\}$.
4. $\{(x, x^3) \mid 0 \leq x \leq 1, x \in \mathbb{Q}\}$.
5. $\{(x, x^3) \mid x \in \mathbb{Z}/(5)\}$.

The next exercise uses some more advanced topological concepts.

Exercise 3.22 — 4 points

Let K be a field.

1. Prove that for both $K = \mathbb{R}$ and $K = \mathbb{C}$, the standard topology (the metric or Euclidean topology) is finer than the Zariski topology on $\mathbb{A}_K^1$.
2. Prove that the Zariski topology on $\mathbb{A}_K^1$ equals the cofinite topology. Does this also hold on $\mathbb{A}_K^n$ for $n \geq 2$?
3. When does the Zariski topology on $\mathbb{A}_K^n$ satisfy T_1 ? When is it Hausdorff?
4. What does the Zariski topology on $\mathbb{A}_K^n$ look like when K is a finite field?

Source navigation: [Lecture 3](#) · [public solutions for Unit 3](#) · [Worksheet 2](#) · [Worksheet 4 \(source\)](#)

Public Solutions to Worksheet 3

The source provides public solutions only to Exercises 3.11 and 3.13 at the frozen revision boundary. No additional solutions have been created for this edition.

Solution to Exercise 3.11

Suppose that $f^k = 0$. Then

$$\begin{aligned} & (1 + f)(1 - f + f^2 - f^3 + f^4 \pm \dots \pm f^{k-1}) \\ &= 1 - f + f^2 - f^3 + f^4 \pm \dots \pm f^{k-1} + f - f^2 + f^3 - f^4 \mp \dots \mp f^{k-1} \\ &= 1. \end{aligned}$$

Thus the element

$$1 - f + f^2 - f^3 + f^4 \pm \dots \pm f^{k-1}$$

is the inverse of $1 + f$, so $1 + f$ is a unit.

[Back to Exercise 3.11.](#)

Solution to Exercise 3.13

Let $\mathfrak{a}$ be a radical ideal and $f \in R/\mathfrak{a}$ nilpotent. Then

$$f^r = 0$$

in $R/\mathfrak{a}$ for some r . Interpreted back in R , this means $f^r \in \mathfrak{a}$, using the same letter for a representative. Since $\mathfrak{a}$ is radical, $f \in \mathfrak{a}$, so $f = 0$ in the quotient ring. Thus the quotient ring is reduced.

Conversely, suppose that an ideal

$$\mathfrak{a} \subseteq R$$

has reduced quotient ring $R/\mathfrak{a}$. Suppose that $f^r \in \mathfrak{a}$. Then the residue class of f^r is 0. Since the quotient ring is reduced, the residue class of f itself is already 0. This means that $f \in \mathfrak{a}$, so the ideal is radical.

[Back to Exercise 3.13.](#)

Lecture 4: Irreducibility, Components, and Intersections of Curves

Irreducible affine algebraic sets

Definition: irreducible set

An affine algebraic set

$$V \subseteq \mathbb{A}_K^n$$

is called *irreducible* if $V \neq \emptyset$ and there is no decomposition

$$V = Y \cup Z$$

with affine algebraic sets $Y, Z \subsetneq V$.

Thus a Zariski closed set V is irreducible precisely when $V \neq \emptyset$ and every decomposition $V = Y \cup Z$ forces $V = Y$ or $V = Z$. The same immediately follows for every finite expression as a union of closed sets.

Irreducibility is a purely topological property. In a general topological space, the preceding definition is formulated using closed sets in place of affine algebraic sets, which are the closed sets of the Zariski topology.

The following pictures show some irreducible and some reducible affine algebraic subsets. What are their irreducible components (see the definition below)?

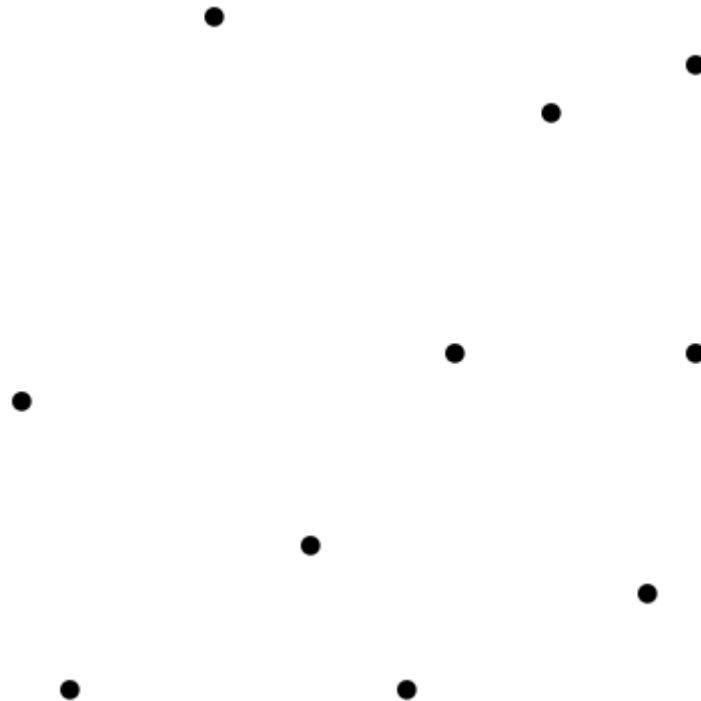


Figure 30: Points for a Delaunay triangulation

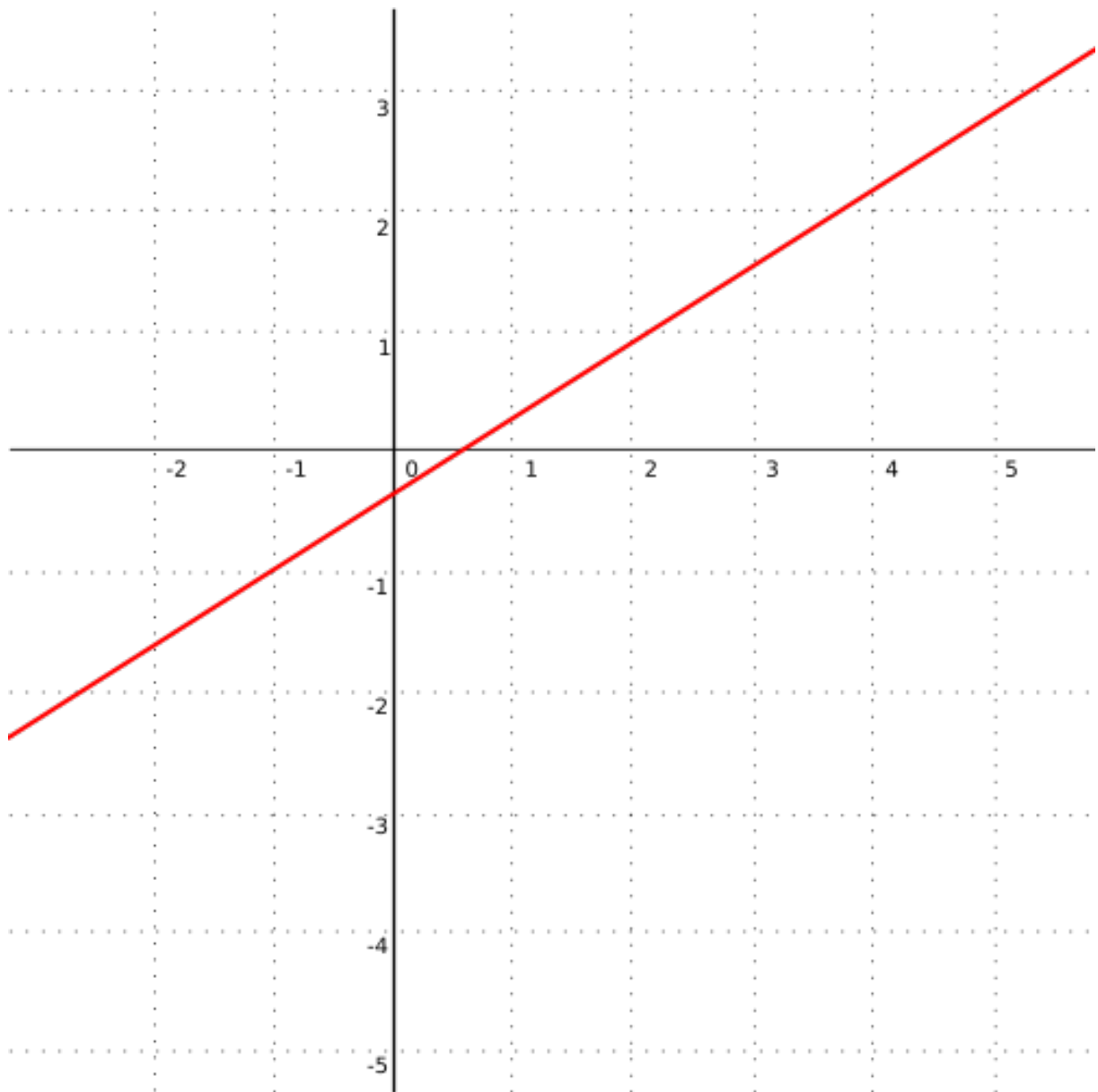


Figure 31: A line

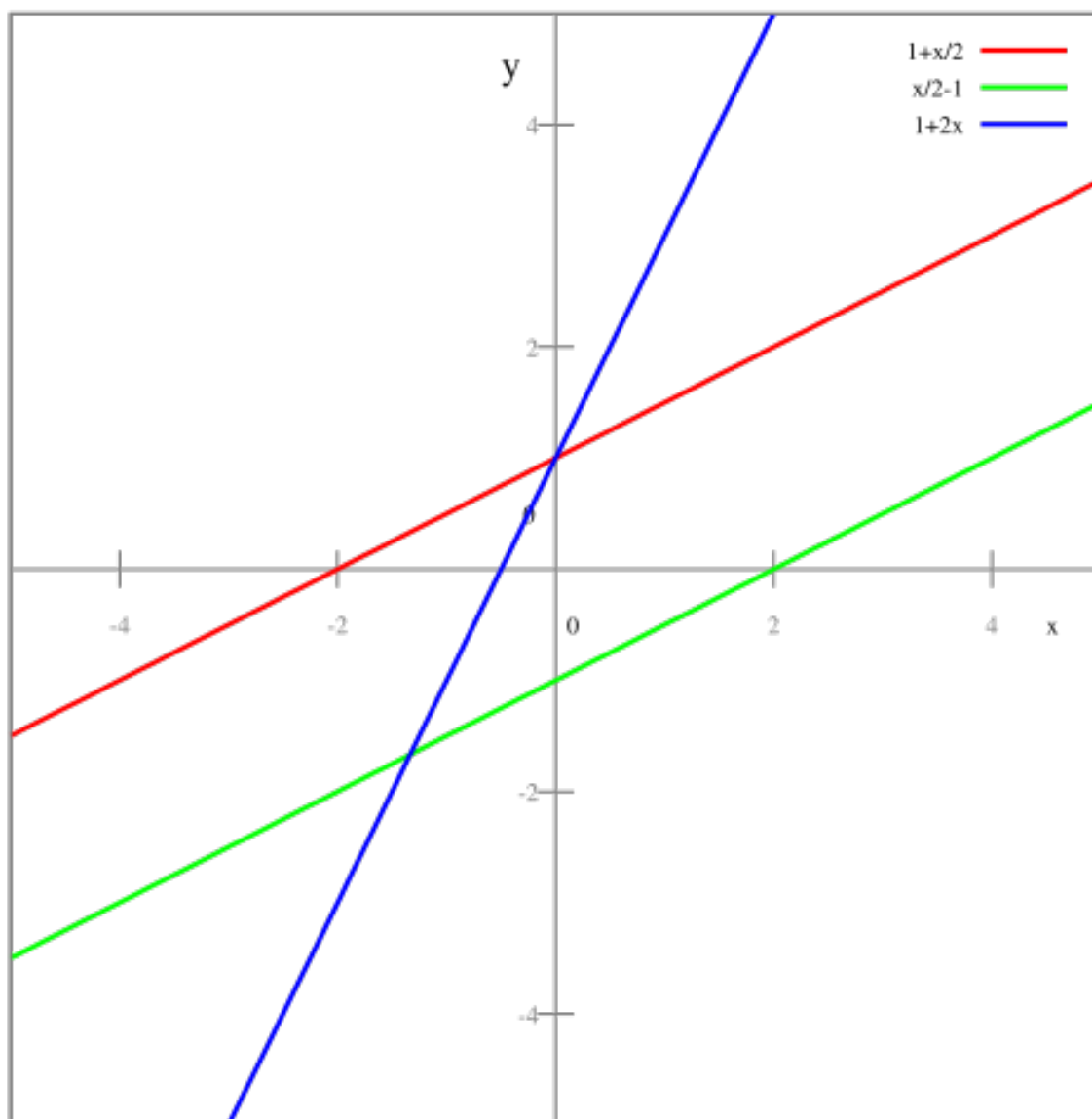


Figure 32: Several straight lines

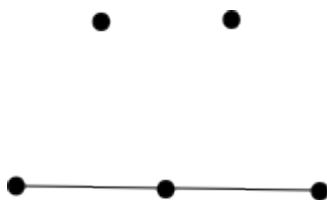


Figure 33: A linear space

Example: affine space

Consider affine space $\mathbb{A}_K^n$. If K is finite, this space consists of only finitely many points, and only its one-point subsets are irreducible. In particular, except when $n = 0$, affine space is not irreducible.

If K is infinite, on the other hand, affine space $\mathbb{A}_K^n$ is irreducible. Suppose that

$$\mathbb{A}_K^n = Y \cup Z$$

with Y and Z both proper affine algebraic subsets. Their open complements,

$$U = \mathbb{A}_K^n \setminus Y, \quad W = \mathbb{A}_K^n \setminus Z,$$

satisfy $U, W \neq \emptyset$, but $U \cap W = \emptyset$. This contradicts Exercise 3.20.

Lemma: irreducibility and prime ideals

Let $V \subseteq \mathbb{A}_K^n$ be an affine algebraic set with vanishing ideal $\text{Id}(V)$. Then V is irreducible if and only if $\text{Id}(V)$ is a prime ideal.

Proof First suppose that $\text{Id}(V)$ is not prime. If

$$\text{Id}(V) = K[X_1, \dots, X_n],$$

then $V = \emptyset$, so V is not irreducible by definition. Otherwise there are polynomials

$$F, G \in K[X_1, \dots, X_n]$$

with

$$FG \in \text{Id}(V), \quad F, G \notin \text{Id}(V).$$

Hence there are $P, Q \in V$ with $F(P) \neq 0$ and $G(Q) \neq 0$. Form the two ideals

$$\mathfrak{a}_1 = \text{Id}(V) + (F), \quad \mathfrak{a}_2 = \text{Id}(V) + (G).$$

By Lemma 3.8(3),

$$V(\mathfrak{a}_1), V(\mathfrak{a}_2) \subseteq V(\text{Id}(V)) = V.$$

Both inclusions are proper because $P \notin V(\mathfrak{a}_1)$ and $Q \notin V(\mathfrak{a}_2)$. On the other hand,

$$V(\mathfrak{a}_1) \cup V(\mathfrak{a}_2) = V(\mathfrak{a}_1 \mathfrak{a}_2) = V(\text{Id}(V)) = V.$$

Thus V has a nontrivial decomposition and is not irreducible.

Now suppose that V is not irreducible. If $V = \emptyset$, then $\text{Id}(V)$ is the whole ring and is not prime. Suppose, then, that $V \neq \emptyset$ and

$$V = Y \cup Z$$

is a nontrivial decomposition. Write

$$Y = V(\mathfrak{a}_1), \quad Z = V(\mathfrak{a}_2).$$

Since $Y \subsetneq V$, there is a point

$$P \in V = V(\text{Id}(V)), \quad P \notin V(\mathfrak{a}_1).$$

There is therefore an $F \in \mathfrak{a}_1$ with $F(P) \neq 0$, so $F \notin \text{Id}(V)$. Similarly, there is a $G \in \mathfrak{a}_2$ with $G \notin \text{Id}(V)$. For every $Q \in V = Y \cup Z$, we have $(FG)(Q) = 0$, since F vanishes on Y and G on Z . Thus

$$FG \in \text{Id}(V),$$

although neither factor belongs to the ideal. Consequently $\text{Id}(V)$ is not prime. $\square$

Definition: irreducible component

Let V be an affine algebraic set. An affine algebraic subset $W \subseteq V$ is called an *irreducible component* of V if W is irreducible and there is no irreducible subset W' with

$$W \subsetneq W' \subseteq V.$$

If V is irreducible, then V itself is its only irreducible component. In Theorem 9.11 we shall prove that every affine algebraic set can be written as a finite union of irreducible components.

Example: behaviour over the real and complex numbers

Consider the equation

$$F = Y^2 + X^2(X + 1)^2 = 0.$$

Over the real numbers, this equation has two solutions. Since a real square is never negative, F can be zero only when both summands are zero. Hence $Y = 0$ and either $X = 0$ or $X = -1$. In particular, the real solution set is neither connected nor irreducible; its vanishing ideal in the real setting is also very large.

Over the complex numbers, there is a factorisation

$$F = (Y + iX(X + 1))(Y - iX(X + 1))$$

into irreducible polynomials. This also shows that F , as a polynomial in $\mathbb{R}[X, Y]$, is irreducible, even though its real zero locus is not irreducible. Its complex zero locus consists of the two graphs

$$Y = \pm iX(X + 1),$$

which intersect at $(0, 0)$ and $(-1, 0)$.

For the equation

$$Y^2 + Z^2 + X^2(X + 1)^2 = 0$$

there are again only two real solution points, whereas the polynomial is irreducible over both the real and the complex numbers.



Figure 34: Hydrant on the island of Krk, Croatia

Example: the intersection of two congruent cylinders

In affine space $\mathbb{A}_K^3$ with $K = \mathbb{R}$, consider the two cylinders

$$S_1 = \{(x, y, z) \mid x^2 + y^2 = 1\}, \quad S_2 = \{(x, y, z) \mid y^2 + z^2 = 1\}.$$

Both are irreducible sets, as we shall see later for infinite K . What does their intersection look like? It is described by the ideal $\mathfrak{a}$ generated by $X^2 + Y^2 - 1$ and $Y^2 + Z^2 - 1$. Subtracting one equation from the other gives

$$X^2 - Z^2 = (X - Z)(X + Z) \in \mathfrak{a}.$$

Neither factor itself belongs to $\mathfrak{a}$. For example, $(1, 0, -1)$ is a point of the intersection at which $X - Z$ does not vanish (in characteristic $\neq 2$), while $(1, 0, 1)$ is a point at which $X + Z$ does not vanish. The components of the intersection are instead described by

$$\mathfrak{b}_1 = \mathfrak{a} + (X - Z), \quad \mathfrak{b}_2 = \mathfrak{a} + (X + Z).$$

Both are prime ideals, and the first quotient ring is

$$\begin{aligned} K[X, Y, Z]/\mathfrak{b}_1 &= K[X, Y, Z]/(\mathfrak{a} + (X - Z)) \\ &\cong K[X, Y]/(X^2 + Y^2 - 1). \end{aligned}$$

To see the last isomorphism, eliminate Z using $X - Z = 0$; the two cylinder equations then become identical. The argument for the other ideal is the same. Geometrically, every point of $S_1 \cap S_2$ lies in the plane

$$E_1 = V(Z - X) \quad \text{or} \quad E_2 = V(Z + X).$$

Moreover,

$$E_1 \cap S_1 = E_1 \cap S_1 \cap S_2 = E_1 \cap S_2,$$

and likewise for E_2 , since on each of these planes the two cylinder equations become identical.

Edition note: At this point the source repeats E_1 ; the second plane is $E_2 = V(Z + X)$, as used here.

What do these intersections look like within their planes? On E_1 , use the coordinates Y and $U = Z + X$. Since

$$X = \frac{1}{2}((Z + X) - (Z - X)),$$

the first cylinder equation can be written as

$$\left(\frac{1}{2}((Z + X) - (Z - X))\right)^2 + Y^2 = 1.$$

On the plane E_1 , where $Z = X$, this becomes

$$\left(\frac{1}{2}U\right)^2 + Y^2 = 1,$$

or

$$\frac{1}{4}U^2 + Y^2 = 1.$$

This is the equation of an ellipse, as is also geometrically apparent. The earlier calculation of $K[X, Y, Z]/\mathfrak{b}_1$, however, gave a circle equation. There is no contradiction: a circle and an ellipse can be transformed into each

other by a linear change of variables, so their quotient rings are isomorphic. As metric objects they are different, and the intersection of these two cylinders consists of two ellipses. An orthonormal change of variables preserves the metric structure, but the variables Y , $X + Z$, and $X - Z$ do not define an orthonormal transformation.

Thus

$$S_1 \cap S_2 = V(\mathfrak{b}_1) \cup V(\mathfrak{b}_2),$$

where

$$\mathfrak{b}_1 = (X^2 + Y^2 - 1, X - Z), \quad \mathfrak{b}_2 = (X^2 + Y^2 - 1, X + Z),$$

describe two ellipses. To determine how the ellipses intersect, calculate the sum of their ideals:

$$\begin{aligned} \mathfrak{b}_1 + \mathfrak{b}_2 &= (X^2 + Y^2 - 1, X - Z, X + Z) \\ &= (Y^2 - 1, X, Z). \end{aligned}$$

Its zero locus consists of the two points $(0, 1, 0)$ and $(0, -1, 0)$.

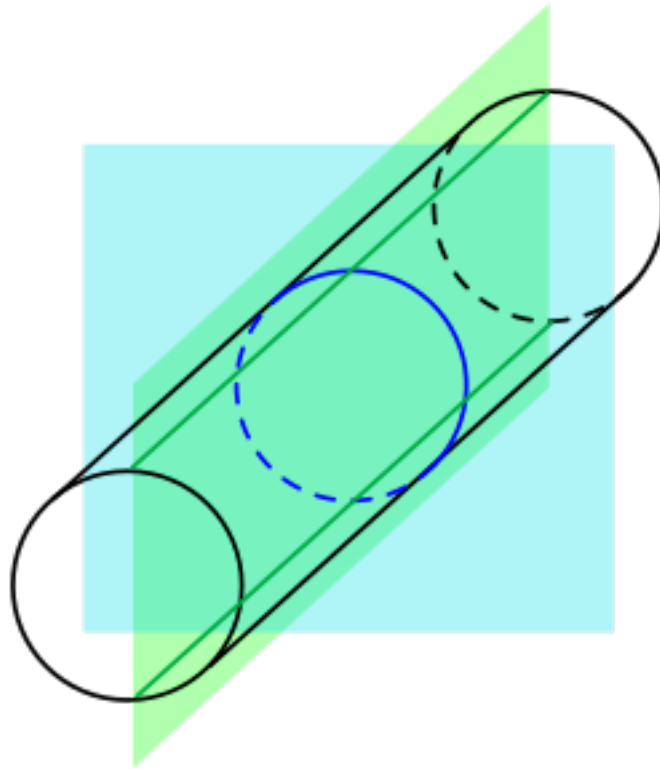


Figure 35: Principal directions on a cylinder

The number of points on curves

We have already seen that the intersection of a curve and a line consists of only finitely many points, unless the line itself is a component of the curve; see Lemma 1.3. We shall now generalise this to the intersection of two arbitrary plane curves. We need the following definition.

Definition: rational function field

Let K be a field and $K[X]$ the polynomial ring in one variable over K . The field of fractions $Q(K[X])$ is called the *rational function field* (or *field of rational functions*) over K and is denoted by

$$K(X).$$

Theorem: intersection of curves without a common component

Let K be a field, and let

$$F, G \in K[X, Y]$$

be two polynomials without a common nonconstant factor. Then $V(F, G)$ contains only finitely many points $P_1, \dots, P_n$.

Proof Regard $F, G \in K[X, Y]$ as elements of $K(X)[Y]$, where $K(X)$ is the field of rational functions in X . By Exercise 4.27, F and G also have no common nonconstant factor in $K(X)[Y]$. Since this ring is a principal ideal domain, they generate the unit ideal. Thus there are

$$A, B \in K(X)[Y]$$

with

$$AF + BG = 1.$$

Multiplying by a common denominator of A and B gives, in $K[X, Y]$,

$$\tilde{A}F + \tilde{B}G = H, \quad 0 \neq H \in K[X].$$

Every common zero of F and G in $\mathbb{A}_K^2$ must be a zero of H . Thus only finitely many X -values can occur at common zeros. Interchanging X and Y shows that only finitely many Y -values can occur as well. Consequently, there are only finitely many common zeros altogether. $\square$

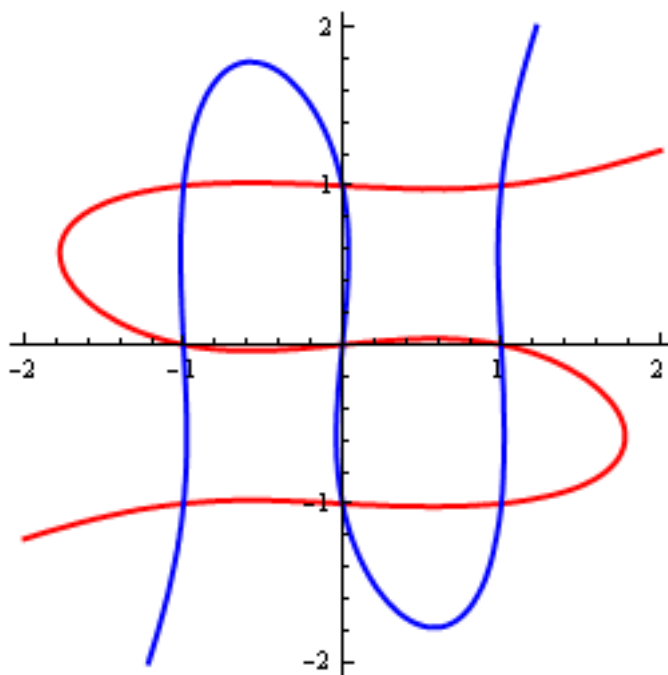


Figure 36: Two cubic curves

Corollary: a prime curve with infinitely many points

Let K be a field and $F \in K[X, Y]$ a prime polynomial. Suppose that the curve $V(F)$ has infinitely many points. Then its vanishing ideal is the principal ideal (F) , and $V(F)$ is irreducible.

Proof Clearly

$$(F) \subseteq \text{Id}(V(F)).$$

Take $G \in \text{Id}(V(F))$. By Lemma 3.8(3),

$$V(F) = V(\text{Id}(V(F))) \subseteq V(F, G).$$

If G were not a multiple of F , Theorem 4.8 would immediately contradict the assumption that $V(F)$ has infinitely many points. Hence

$$\text{Id}(V(F)) = (F).$$

This ideal is prime, and by Lemma 4.3, $V(F)$ is irreducible. $\square$

Source navigation: [course](#) - [Lecture 3](#) - [Lecture 5 \(source\)](#) - [Worksheet 4](#)

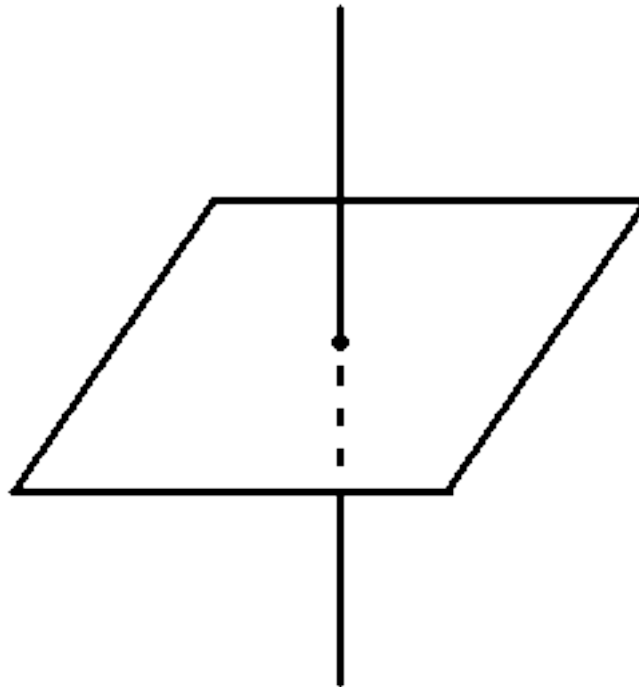
Worksheet 4**Practice exercises****Exercise 4.1**

Figure 37: A surface and a vertical line

Find an ideal whose zero locus is the object shown above.

Exercise 4.2

Let $V \subseteq \mathbb{A}_K^n$ be a subset consisting of finitely many points. Prove that V is irreducible if and only if it consists of a single point.

Exercise 4.3

Sketch an example of an affine algebraic subset that is connected but not irreducible.

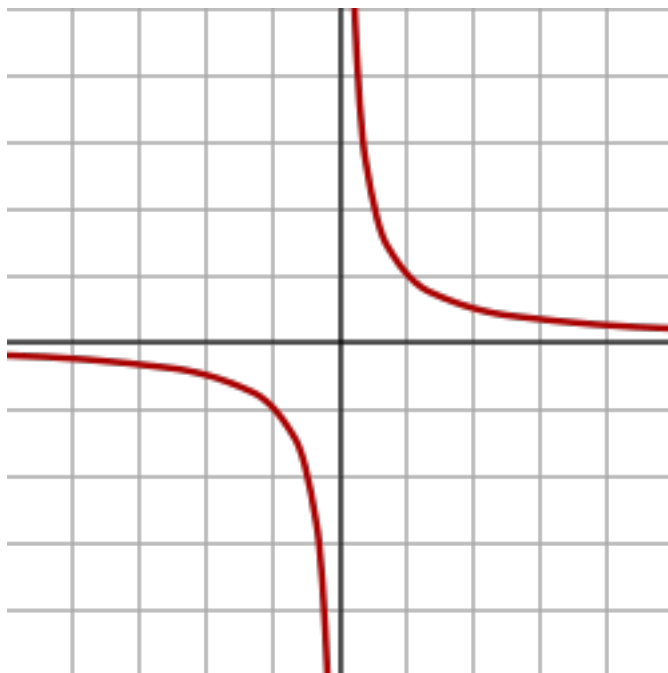
Exercise 4.4

Figure 38: Rectangular hyperbola

Determine the irreducible components of the real hyperbola.

Exercise 4.5

Let K be a field of characteristic $\neq 2$ and $a \in K$ nonzero. Prove that the polynomial

$$X^2 + Y^2 + a \in K[X, Y]$$

is irreducible.

Exercise 4.6

Let K be a field and

$$\mathfrak{p} = (p) \subseteq K[X]$$

a prime ideal. Prove that the zero locus $V(\mathfrak{p}) \subseteq \mathbb{A}_K^1$ is either the whole of $\mathbb{A}_K^1$ (whose irreducibility depends on the field) or a single point (and hence irreducible).

Edition note: As written, the source statement omits the case $V(\mathfrak{p}) = \emptyset$, for example $K = \mathbb{R}$ and $\mathfrak{p} = (X^2 + 1)$. This note does not alter the source exercise.

Exercise 4.7

Let K be a finite field and $\mathfrak{p} \subset K[X_1, \dots, X_n]$ a prime ideal. Prove that its zero locus

$$V(\mathfrak{p}) \subseteq \mathbb{A}_K^n$$

can be irreducible only if it consists of a single point.

Exercise 4.8

Prove that the real polynomial

$$P = X^2(X-1)^2 + Y^2(X^2 + (X-1)^2) \in \mathbb{R}[X, Y]$$

is a prime polynomial, whereas its zero locus

$$V(P) \subseteq \mathbb{R}^2$$

is nonempty but reducible.

Exercise 4.9

In $\mathbb{A}_{\mathbb{R}}^3$, calculate the intersection of the cylinder

$$V(x^2 + y^2 - 1)$$

with the sphere of centre $P = (0, 0, 0)$ and radius r , as a function of r . When is the intersection empty, and when is it irreducible?

You may use the fact that the real circle is irreducible.

The next exercise says that the intersection of the graphs of two one-variable polynomials agrees, not only pointwise but also algebraically, with the intersection of the graph of their difference and the x -axis. See also Exercise 26.3.

Exercise 4.10 ★

Let $F, G \in K[X]$ be polynomials in one variable over a field K . Prove that there is a K -algebra isomorphism

$$K[X, Y]/(Y - F, Y - G) \cong K[X]/(F - G).$$

Exercise 4.11 ★

Let two distinct circles in the plane be given by circle equations F and G .

1. Prove that the quotient ring $K[X, Y]/(F, G)$ is isomorphic to $K[X, Y]/(F, H)$, where H has degree at most 1.
2. Prove that $K[X, Y]/(F, G)$ is isomorphic to a ring of the form $K[U]/(Q)$, where $Q \in K[U]$ has degree at most 2.

Exercise 4.12 ★

Let R be a commutative ring and $R[X]$ the polynomial ring over R . Let $\mathfrak{a} \subseteq R[X]$ be an ideal with generators

$$\mathfrak{a} = (F_0, F_1, \dots, F_n),$$

where $F_0 = X - r$ for some $r \in R$. For $i \geq 1$, let $G_i \in R$ be obtained from F_i by replacing X with r . Prove the isomorphism of quotient rings

$$R[X]/\mathfrak{a} \cong R/(G_1, \dots, G_n).$$

Exercise 4.13

Consider the real numbers $\mathbb{R}$ with the metric topology. Is $\mathbb{R}$ irreducible?

Exercise 4.14 ★

Let p be a prime number and $\mathbb{Z}/(p)$ the corresponding residue field. Prove that every quadratic equation of the form

$$F = aX^2 + bY^2 + c = 0, \quad a, b \neq 0,$$

has at least one solution in $\mathbb{Z}/(p)$.

Exercise 4.15 ★

Let $\mathfrak{a}$ be an ideal in a commutative ring R . Prove that $\mathfrak{a}$ is prime if and only if it is the kernel of a ring homomorphism

$$\varphi : R \longrightarrow K$$

to a field K .

Exercise 4.16

Prove that every maximal ideal $\mathfrak{m}$ in a commutative ring R is prime.

Exercise 4.17 ★

Let R be a commutative ring and $\mathfrak{p}$ an ideal. Prove that $\mathfrak{p}$ is prime if and only if the quotient ring $R/\mathfrak{p}$ is an integral domain.

Exercise 4.18

Let $\mathfrak{p} \subseteq R$ be a prime ideal in a commutative ring R . Prove that

$$\mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{p}$$

implies $\mathfrak{a} \subseteq \mathfrak{p}$ or $\mathfrak{b} \subseteq \mathfrak{p}$.

Exercise 4.19

Let R and S be commutative rings, $\varphi : R \rightarrow S$ a ring homomorphism, and $\mathfrak{p}$ a prime ideal in S . Prove that the inverse image $\varphi^{-1}(\mathfrak{p})$ is a prime ideal in R .

Give an example showing that the inverse image of a maximal ideal need not be maximal.

Exercise 4.20

Let K be a field and $L = K(X)$ the field of fractions of the polynomial ring $K[X]$. Prove that there are infinitely many intermediate fields between K and L .

Exercise 4.21

Explain where the proof of Theorem 4.8 breaks down if one tries to extend it to more than two variables.

Exercise 4.22

Let $P(X)$ and $Q(Y)$ be nonconstant polynomials in the indicated variables. Give a bound (under what condition?) for the number of intersection points of the two curves

$$V(Y - P(X)) \quad \text{and} \quad V(X - Q(Y)).$$

The following exercises use the terms *closed map* and *open map*. A continuous map $f : X \rightarrow Y$ between topological spaces is called *closed* if the image of every closed set is closed. It is called *open* if the image of every open set is open.

Exercise 4.23

Prove that the projection

$$\mathbb{A}_K^2 \longrightarrow \mathbb{A}_K^1, \quad (x, y) \longmapsto x,$$

is not a closed map in the Zariski topology.

Edition note: Assume K is infinite. The source omits this hypothesis; over a finite field, both spaces are discrete and every map between them is closed.

Exercise 4.24

Prove that a plane algebraic curve over the complex numbers $\mathbb{C}$ is not compact in the metric topology.

Exercises to hand in**Exercise 4.25 - 6 points**

Let $p \geq 3$ be a prime number and $K = \mathbb{Z}/(p)$ the corresponding residue field. Consider the polynomial

$$F = \alpha X^2 + \beta XY + \gamma Y^2 + \delta X + \epsilon Y + \eta \in \mathbb{Z}/(p)[X, Y].$$

If α, β, γ are not all zero, this is a quadratic polynomial. Prove that exactly one of the following three alternatives holds for its zero locus $V(F) \subseteq \mathbb{A}_K^2$.

1. $V(F)$ has at least one point.
2. $F = c$ for a constant $c \neq 0$.
3. There is a change of variables such that, in the new coordinates, the polynomial has the form $Z^2 - u$ with $u \in \mathbb{Z}/(p)$ a nonsquare.

Edition note: In alternative (3), one must also allow multiplication of the polynomial by a nonzero scalar, which does not change its zero locus. The source omits this normalisation: for example, $2X^2 + 2$ over $\mathbb{F}_3$ has no zero, but an invertible linear or affine change of variables alone cannot turn its nonsquare leading coefficient into 1.

Source hint: Exercise 9.20 in *Number Theory (Osnabrück 2025)* is useful for one important case.

Exercise 4.26 - 3 points

Let V be an irreducible affine algebraic set with at least two points, and let $P_1, \dots, P_m \in V$ be finitely many points. Prove that

$$V \setminus \{P_1, \dots, P_m\}$$

is also irreducible in the induced topology.

Exercise 4.27 - 4 points

Let R be a unique factorisation domain with field of fractions $Q(R)$. Prove that if $F, G \in R[X]$ have no common nonconstant factor, then they also have no common nonconstant factor when regarded as elements of $Q(R)[X]$.

Source note: You may restrict attention to the case where R is a principal ideal domain.

Exercise 4.28 - 3 points

Let $K = \mathbb{Q}$ be the field of rational numbers. Determine, with justification, whether

$$V(X^2 + Y^2 - 1) \subseteq \mathbb{A}_{\mathbb{Q}}^2$$

is irreducible.

Source hint: Use Exercise 1.28 on the rational parametrisation of the unit circle and Corollary 4.9.

Exercise 4.29 - 4 points

Prove that the projection

$$\mathbb{A}_K^2 \longrightarrow \mathbb{A}_K^1, \quad (x, y) \longmapsto x,$$

is an open map in the Zariski topology.

Exercise 4.30 - 4 points

Prove that the affine plane $\mathbb{A}_K^2$ with the Zariski topology is compact.

Terminology note: Here “compact” means that every open cover has a finite subcover; no Hausdorff condition is included (this is also called *quasi-compact*).

Source navigation: [Lecture 4 - public solutions for Unit 4](#) - [Worksheet 3](#) - [Worksheet 5 \(source\)](#)

Public Solutions to Worksheet 4

The source provides public solutions only to Exercises 4.10, 4.11, 4.12, 4.14, 4.15, and 4.17 at the frozen revision boundary. No additional solutions have been created for this edition.

Solution to Exercise 4.10

We prove the isomorphism

$$K[X, Y]/(Y - F, Y - G) \cong K[X]/(F - G).$$

Consider the K -algebra homomorphism

$$\varphi : K[X, Y] \longrightarrow K[X]/(F - G)$$

that sends $X \mapsto X$ and $Y \mapsto F$. We have

$$\varphi(Y - F) = F - F = 0$$

and

$$\varphi(Y - G) = F - G = 0.$$

The homomorphism theorem for rings gives an induced K -algebra homomorphism

$$\overline{\varphi} : K[X, Y]/(Y - F, Y - G) \longrightarrow K[X]/(F - G).$$

Now consider the K -algebra homomorphism

$$\psi : K[X] \longrightarrow K[X, Y]/(Y - F, Y - G)$$

with $X \mapsto X$. In the target ring,

$$\psi(F - G) = F - G = (Y - G) - (Y - F) = 0.$$

This therefore induces a homomorphism

$$\overline{\psi} : K[X]/(F - G) \longrightarrow K[X, Y]/(Y - F, Y - G).$$

Both compositions $\overline{\psi} \circ \overline{\varphi}$ and $\overline{\varphi} \circ \overline{\psi}$ are the respective identity maps.

Back to Exercise 4.10.

Solution to Exercise 4.11

1. A circle equation has the form

$$(X - a)^2 + (Y - b)^2 - c = 0.$$

Expanding, write

$$F = X^2 + Y^2 + rX + sY + t$$

and similarly

$$G = X^2 + Y^2 + \tilde{r}X + \tilde{s}Y + \tilde{t}.$$

Then

$$H = F - G = (r - \tilde{r})X + (s - \tilde{s})Y + (t - \tilde{t}).$$

Since the circles are distinct, H has degree 1 or 0. Moreover,

$$(F, G) = (F, H),$$

so their quotient rings are isomorphic.

2. Use the description from the first part:

$$R = K[X, Y]/(F, H).$$

If H is a nonzero constant, R is the zero ring. Otherwise H is linear, so one variable can be expressed in terms of the other; say

$$Y = \alpha X + \beta.$$

Hence

$$\begin{aligned}
K[X, Y]/(F, H) &\cong K[X, Y]/(X^2 + Y^2 + rX + sY + t, Y - \alpha X - \beta) \\
&\cong K[X]/(X^2 + (\alpha X + \beta)^2 + rX + s(\alpha X + \beta) + t) \\
&\cong K[X]/(uX^2 + vX + w).
\end{aligned}$$

Back to Exercise 4.11.

Solution to Exercise 4.12

Consider the surjective evaluation homomorphism

$$R[X] \longrightarrow R/(G_1, \dots, G_n), \quad X \mapsto [r].$$

The generator $F_0 = X - r$ maps to 0, and for $i \geq 1$, the generator F_i maps to $G_i = 0$. The homomorphism theorem gives a surjective ring homomorphism

$$\varphi : R[X]/\mathfrak{a} \longrightarrow R/(G_1, \dots, G_n).$$

It remains to prove injectivity. Suppose that

$$P = a_0 + a_1X + \dots + a_mX^m \in R[X]$$

maps to 0 under φ . This means that

$$P(r) = a_0 + a_1r + \dots + a_mr^m \in (G_1, \dots, G_n)$$

in R . Furthermore,

$$\begin{aligned}
P - P(r) &= \sum_{i=0}^m a_i X^i - \sum_{i=0}^m a_i r^i \\
&= \sum_{i=0}^m a_i (X^i - r^i) \\
&= \sum_{i=1}^m a_i (X^i - r^i) \\
&= \sum_{i=1}^m (X - r) H_i,
\end{aligned}$$

since $X^i - r^i$ is always divisible by $X - r$. Thus

$$P - P(r) \in (X - r),$$

and altogether

$$P \in (X - r, G_1, \dots, G_n).$$

For suitable elements B_i , we also have

$$F_i - G_i = F_i - F_i(r) = (X - r)B_i.$$

Consequently,

$$\begin{aligned}
P &\in (X - r, G_1, \dots, G_n) \\
&= (X - r, F_1, \dots, F_n) \\
&= \mathfrak{a}.
\end{aligned}$$

Thus φ is injective and is the required isomorphism.

[Back to Exercise 4.12.](#)

Solution to Exercise 4.14

For $p = 2$, the statement can be checked directly. Suppose that $p \geq 3$. Write the equation as

$$aX^2 = -bY^2 - c.$$

Since a and b are units, the theorem on the number of quadratic residues shows that the sets of values on the left and on the right each contain $(p+1)/2$ elements. The field $\mathbb{Z}/(p)$ has only p elements, so the two sets cannot be disjoint. Thus there is a $d \in \mathbb{Z}/(p)$ that can be written as

$$d = aX^2 = -bY^2 - c$$

for suitable $X, Y \in \mathbb{Z}/(p)$. This pair solves the original equation.

[Back to Exercise 4.14.](#)

Solution to Exercise 4.15

First let $\mathfrak{a}$ be a prime ideal. By the [characterisation of prime ideals by quotient rings](#), $R/\mathfrak{a}$ is an integral domain and therefore has a field of fractions $Q(R/\mathfrak{a})$. The composition of the canonical projection with the inclusion into this field,

$$\varphi : R \longrightarrow Q(R/\mathfrak{a}), \quad x \longmapsto [x],$$

is a ring homomorphism to a field with

$$\ker \varphi = \mathfrak{a}.$$

Conversely, the kernel of a ring homomorphism

$$\varphi : R \longrightarrow K$$

is always an ideal, by the [kernel-ideal theorem](#). If $ab \in \ker \varphi$, then

$$0 = \varphi(ab) = \varphi(a)\varphi(b).$$

Since [a field is an integral domain](#), K has no zero divisors, so either $\varphi(a) = 0$ or $\varphi(b) = 0$. Equivalently, $a \in \ker \varphi$ or $b \in \ker \varphi$. Thus $\ker \varphi$ is a prime ideal.

Edition note: The source calls the map to $Q(R/\mathfrak{a})$ the canonical projection. More precisely, it is the quotient projection followed by the inclusion into the field of fractions, as stated above.

[Back to Exercise 4.15.](#)

Solution to Exercise 4.17

First let $\mathfrak{p}$ be a prime ideal. In particular, $\mathfrak{p} \subsetneq R$, so $R/\mathfrak{p}$ is not the zero ring. Suppose that $fg = 0$ in $R/\mathfrak{p}$, with f and g represented by elements of R . Then $fg \in \mathfrak{p}$, so $f \in \mathfrak{p}$ or $g \in \mathfrak{p}$. In $R/\mathfrak{p}$, this means exactly that $f = 0$ or $g = 0$. Thus $R/\mathfrak{p}$ is an integral domain.

Conversely, suppose that $R/\mathfrak{p}$ is an integral domain. This quotient is not the zero ring, so $\mathfrak{p} \neq R$. If $f, g \notin \mathfrak{p}$, their classes are both nonzero in $R/\mathfrak{p}$. Since the ring is an integral domain, their product is nonzero. Hence

$$fg \notin \mathfrak{p}.$$

Taking the contrapositive, $fg \in \mathfrak{p}$ forces $f \in \mathfrak{p}$ or $g \in \mathfrak{p}$. Thus $\mathfrak{p}$ is prime.

[Back to Exercise 4.17.](#)

Lecture 5: Homogeneous Components, Noether Normalisation, and Polynomial Maps

Homogeneous components

We discuss the degree of a polynomial in several variables and its decomposition into homogeneous components.

Definition: degree

Let S be a commutative ring and

$$R = S[X_1, \dots, X_n]$$

the polynomial ring in n variables over S . For a monomial

$$G = X^\nu = X_1^{\nu_1} \cdots X_n^{\nu_n},$$

the number

$$|\nu| = \sum_{j=1}^n \nu_j$$

is called the *degree* of G . For a nonzero polynomial

$$F = \sum_{\nu} a_{\nu} X^{\nu},$$

the number

$$\max\{|\nu| : a_{\nu} \neq 0\}$$

is called the *degree* of F .

Definition: homogeneous decomposition

Let S and $R = S[X_1, \dots, X_n]$ be as above. For a polynomial

$$F = \sum_{\nu} a_{\nu} X^{\nu} \in R,$$

the decomposition

$$F = \sum_{i=0}^d F_i,$$

where

$$F_i = \sum_{|\nu|=i} a_\nu X^\nu,$$

is called the *homogeneous decomposition* of F . The polynomial F_i is called the *homogeneous component* of F of degree i . The polynomial F itself is called *homogeneous* if its homogeneous decomposition has only one nonzero component.

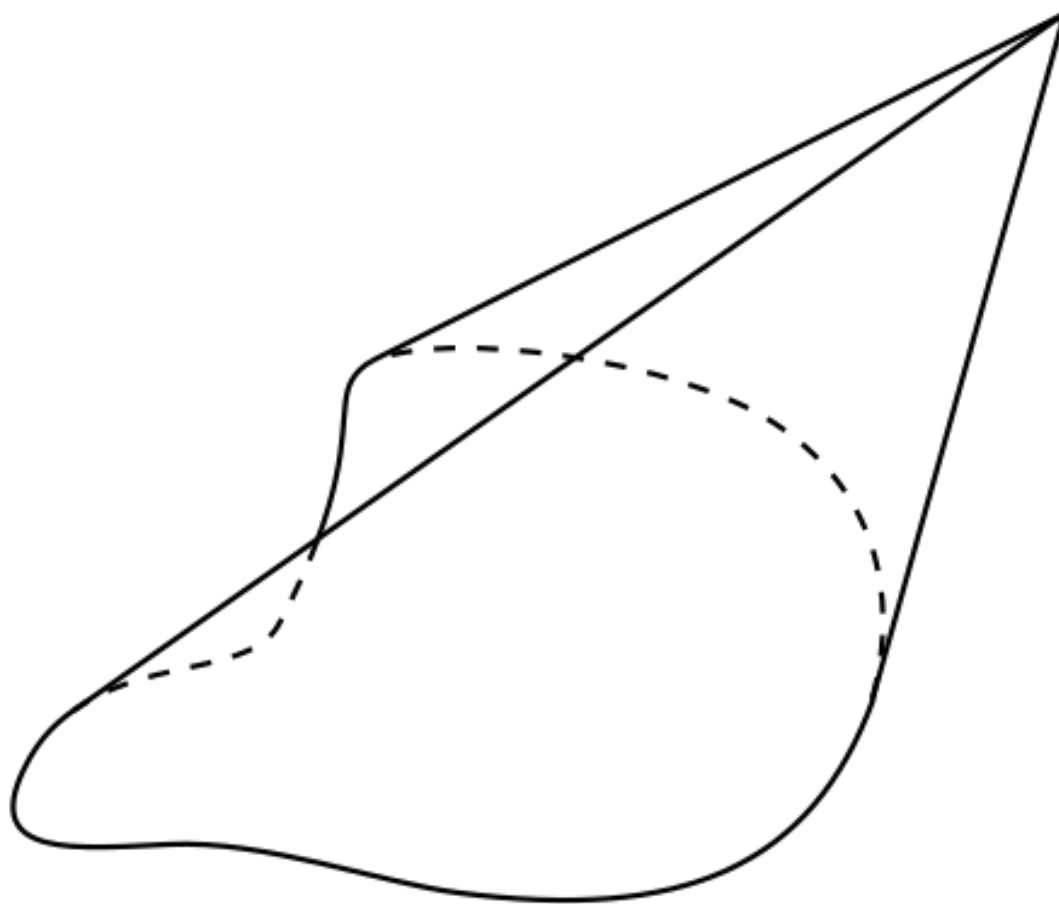


Figure 39: A cone: the zero set of a homogeneous polynomial

The zero set of a homogeneous polynomial F is a cone of lines through the origin: if a point P belongs to $V(F)$, the entire line through P and 0 also belongs to $V(F)$.

Example: total degree and degree in one variable

The polynomial

$$F = 4X^3YZ^2 + 2X^2Y^5 + 5XYZ^7 - 3X^4YZ^4 + X^8 - Y^7 + 2Y^6Z^3 + X + 5$$

has degree 9, with homogeneous components

$$F_9 = 5XYZ^7 - 3X^4YZ^4 + 2Y^6Z^3,$$

$$F_8 = X^8,$$

$$F_7 = 2X^2Y^5 - Y^7,$$

$$F_6 = 4X^3YZ^2,$$

$$F_5 = F_4 = F_3 = F_2 = 0,$$

and

$$F_1 = X, \quad F_0 = 5.$$

If we regard F as a polynomial in $(K[Y, Z])[X]$ and consider only the powers of X , we speak of its X -degree. The X -degree of F is 8. There is also a homogeneous decomposition with respect to the X -grading: the component of X -degree zero is

$$-Y^7 + 2Y^6Z^3 + 5,$$

while the component of X -degree one is

$$5XYZ^7 + X.$$

The number of points on curves II



Figure 40: Emmy Noether (1882-1935)

The following theorem is called *Noether normalisation* in the case of plane curves.

Theorem: Noether normalisation for plane curves

Let K be an algebraically closed field and $F \in K[X, Y]$ a nonconstant polynomial of degree d defining the algebraic curve

$$C = V(F).$$

There is a linear change of coordinates such that, in the new coordinates $\tilde{X}, \tilde{Y}$, the transformed polynomial has the form

$$\tilde{F} = \tilde{X}^d + \text{terms of lower degree in } \tilde{X}.$$

Proof Write the homogeneous decomposition

$$F = F_d + F_{d-1} + \cdots + F_1 + F_0,$$

with

$$F_i = \sum_{a+b=i} c_{a,b} X^a Y^b.$$

A homogeneous polynomial in two variables has the same factorisation properties as a polynomial in one variable. Since K is algebraically closed, there is a factorisation

$$F_d = c(Y - e_1 X) \cdots (Y - e_k X) X^{d-k}.$$

Since c has a d th root, scaling the variables allows us to assume $c = 1$. In particular, K is infinite, so we can choose $e \in K$ distinct from all the e_j . Use the new coordinates

$$\tilde{Y} = Y - eX, \quad \tilde{X} = X.$$

In these coordinates, each linear factor becomes

$$\begin{aligned} Y - e_j X &= Y - eX + eX - e_j X \\ &= \tilde{Y} - (e_j - e)X \\ &= \tilde{Y} - (e_j - e)\tilde{X}, \end{aligned}$$

with $e_j - e \neq 0$, while the factor X becomes $\tilde{X}$. On expanding, $\tilde{X}^d$ occurs with a nonzero coefficient in K , which can again be made 1 by scaling. The top homogeneous component consequently has the form $\tilde{X}^d$ plus terms of $\tilde{X}$ -degree at most $d - 1$. Since the lower-degree homogeneous components retain their degrees, all other monomials also have $\tilde{X}$ -degree at most $d - 1$. $\square$

Corollary: plane curves have infinitely many points

Let K be an algebraically closed field and $F \in K[X, Y]$ a nonconstant polynomial defining the algebraic curve $C = V(F)$. Then C has infinitely many elements.

Proof By Noether normalisation, we may assume that

$$F = X^d + P_{d-1}(Y)X^{d-1} + \cdots + P_1(Y)X + P_0(Y),$$

with $P_i(Y) \in K[Y]$. For every prescribed value $a \in K$ of Y , substituting $Y = a$ gives a monic polynomial of degree d in X . Since K is algebraically closed, this polynomial has at least one root $b \in K$. Thus the point with coordinates $X = b$ and $Y = a$ lies on C . Since K is infinite, the curve has infinitely many points. $\square$

Edition note: After setting $Y = a$ and choosing a root $X = b$, the source writes the point as (a, b) . This edition specifies both coordinates explicitly to retain the order $X = b, Y = a$.

Polynomial maps between affine spaces

Consider the map

$$\begin{aligned} \varphi : \mathbb{A}_K^r &\longrightarrow \mathbb{A}_K^n, \\ (t_1, \dots, t_r) &\longmapsto (\varphi_1(t_1, \dots, t_r), \dots, \varphi_n(t_1, \dots, t_r)) = (x_1, \dots, x_n), \end{aligned}$$

where each component function $\varphi_i \in K[T_1, \dots, T_r]$ is a polynomial. Thus each component of the map is given by a polynomial in r variables. The case $n = 1$ is a polynomial in r variables; the case $r = 1$ and $n = 2$ is a

parametrisation of an algebraic curve. Later we shall define morphisms between affine algebraic sets in greater generality.

Edition note: The source prints the ring of component functions as $K[X_1, \dots, X_r]$, although the map's parameters and the target ring of its substitution homomorphism use $T_1, \dots, T_r$. This edition consistently uses $K[T_1, \dots, T_r]$.

An important accompanying feature of a polynomial map $\varphi : \mathbb{A}_K^r \rightarrow \mathbb{A}_K^n$ is that it induces a K -algebra homomorphism between the polynomial rings in the opposite direction. This substitution homomorphism is determined by $X_i \mapsto \varphi_i$ and is denoted by

$$\begin{aligned}\tilde{\varphi} : K[X_1, \dots, X_n] &\longrightarrow K[T_1, \dots, T_r], \\ F &\longmapsto F \circ \varphi = F(\varphi_i/X_i).\end{aligned}$$

The notation φ_i/X_i means replacing the variable X_i with φ_i . As a function, $F \circ \varphi$ is the composite map

$$\mathbb{A}_K^r \xrightarrow{\varphi} \mathbb{A}_K^n \xrightarrow{F} \mathbb{A}_K^1.$$

For the zero locus $V(F) \subseteq \mathbb{A}_K^n$, we have

$$\varphi^{-1}(V(F)) = V(\tilde{\varphi}(F)).$$

Apart from constant maps, the simplest polynomial maps are affine-linear maps, whose component functions are affine-linear polynomials:

$$\varphi_i = a_{i1}T_1 + \dots + a_{ir}T_r + c_i.$$

These maps need not be linear, since the origin need not map to the origin: translations are allowed. An affine-linear map is the composition of a linear map and a translation. For $r = n$, a bijective affine-linear map is regarded as a coordinate transformation, or change of variables.

Definition: affine-linear change of variables

Let K be a field. A map $\varphi : \mathbb{A}_K^n \rightarrow \mathbb{A}_K^n$ of the form

$$\varphi(x_1, \dots, x_n) = M \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + (v_1, \dots, v_n),$$

where M is an invertible matrix, is called an *affine-linear change of variables*.

One can debate whether a linear change of variables actually moves anything in space or merely changes the coordinates. In either case, such transformations are important tools for putting a polynomial, a system of algebraic equations, or an affine algebraic set into a simpler form. Under a change of variables, the set

$$V = V(F_1, \dots, F_m)$$

becomes

$$\tilde{V} = V(\tilde{F}_1, \dots, \tilde{F}_m), \quad \tilde{F}_i = \tilde{\varphi}(F_i),$$

and $\tilde{V}$ is the inverse image of V under φ .

Definition: affine-linear equivalence

Two affine algebraic sets

$$V, \tilde{V} \subseteq \mathbb{A}_K^n$$

are called *affine-linearly equivalent* if there is an affine-linear change of variables $\varphi : \mathbb{A}_K^n \rightarrow \mathbb{A}_K^n$ such that

$$\varphi^{-1}(V) = \tilde{V}.$$

This notion depends on how the objects are embedded. Later we shall see that a parabola and a line in the plane are isomorphic, since both are isomorphic to the affine line, but they are not affine-linearly equivalent.

The essential algebraic and topological properties of an affine algebraic set are preserved under an affine-linear change of variables: irreducibility, singularities, intersections, connectedness, and compactness. By contrast, properties typical of real metric geometry may change: angles, lengths and ratios of lengths, volumes, and shapes. These latter notions are not relevant to algebraic geometry. Henceforth we shall transform a situation into a desired form without special emphasis whenever such a transformation is available.

Theorem: quotient rings under affine-linear equivalence

Let K be a field and $V, \tilde{V} \subseteq \mathbb{A}_K^n$ two affine-linearly equivalent affine algebraic sets. Let $\text{Id}(V)$ and $\text{Id}(\tilde{V})$ be their vanishing ideals. Then there is a K -algebra isomorphism

$$K[X_1, \dots, X_n]/\text{Id}(V) \cong K[X_1, \dots, X_n]/\text{Id}(\tilde{V}).$$

Proof By definition, there is an affine-linear change of variables

$$\mathbb{A}_K^n \longrightarrow \mathbb{A}_K^n, \quad P \longmapsto \varphi(P),$$

with $\varphi^{-1}(V) = \tilde{V}$. Let $\tilde{\varphi}$ be the corresponding automorphism of $K[X_1, \dots, X_n]$. Then

$$\tilde{\varphi}^{-1}(\text{Id}(\tilde{V})) = \text{Id}(V).$$

The isomorphism theorem gives the isomorphism of the two quotient rings. $\square$

Remark: the coordinate ring as an intrinsic invariant

The preceding theorem expresses an important principle of algebraic geometry: the algebraic object attached to a zero locus is the quotient of the polynomial ring by its vanishing ideal. This is an *intrinsic invariant* of the zero locus, independent of its embedding.

From this perspective, Noether normalisation for plane curves takes on new meaning. We may assume that the curve equation has the form

$$F = X^d + P_{d-1}(Y)X^{d-1} + \dots + P_1(Y)X + P_0(Y).$$

The equation $F = 0$ is an equation of integral dependence for the residue class of X . More precisely, the residue class of X in $K[X, Y]/(F)$ is integral over $K[Y]$. These notions may be familiar from elementary number theory and will again play an important role here. Since X generates the ring as an algebra over $K[Y]$, there is an integral, indeed finite, ring extension

$$K[Y] \longrightarrow K[X, Y]/(X^d + P_{d-1}(Y)X^{d-1} + \dots + P_1(Y)X + P_0(Y)).$$

Thus Noether normalisation also says that, for every algebraic curve over an algebraically closed field, its coordinate ring can be realised as a finite extension of the principal ideal domain $K[Y]$. This is a direct analogy with rings of integers in number theory, which are likewise finite extensions of the principal ideal domain $\mathbb{Z}$.

Under general polynomial maps between affine spaces, unlike affine-linear transformations, many algebraic properties may change: dimension may change, singularities may arise, and so on. Irreducibility, however, passes to the Zariski closure of the image.

Theorem: the closure of the image of a polynomial map is irreducible

Let K be an infinite field and

$$\varphi : \mathbb{A}_K^r \longrightarrow \mathbb{A}_K^n$$

a map given by n polynomials in r variables. Then the Zariski closure of the image of φ is irreducible.

Proof Let

$$B = \varphi(\mathbb{A}_K^r)$$

be the image of the map. By Lemma 3.10,

$$\overline{B} = V(\text{Id}(B)).$$

For $P = \varphi(Q)$ with $Q \in \mathbb{A}_K^r$ and $F \in K[X_1, \dots, X_n]$, we have

$$F(P) = F(\varphi(Q)) = (F \circ \varphi)(Q),$$

where $F \circ \varphi \in K[T_1, \dots, T_r]$ is obtained by replacing X_i with the i th component function $\varphi_i \in K[T_1, \dots, T_r]$. Consequently, F vanishes throughout B precisely when $F \circ \varphi$ vanishes throughout $\mathbb{A}_K^r$. Since K is infinite, the latter condition means that $F \circ \varphi$ is the zero polynomial.

Thus

$$F \in \text{Id}(B)$$

precisely when F maps to zero under the homomorphism

$$\widetilde{\varphi} : K[X_1, \dots, X_n] \longrightarrow K[T_1, \dots, T_r].$$

Hence $\text{Id}(B)$ is the inverse image of a prime ideal, namely the zero ideal in $K[T_1, \dots, T_r]$, and so is itself prime by Exercise 4.19. Lemma 4.3 then shows that $V(\text{Id}(B))$ is irreducible. $\square$

Worksheet 5

Practice exercises

Exercise 5.1

Let $F \in K[X_1, \dots, X_n]$ be a homogeneous polynomial with zero locus $V(F)$. Prove that for every point $P \in V(F)$ and every scalar $\lambda \in K$, we also have $\lambda P \in V(F)$.

Exercise 5.2

Determine the factorisation of the polynomial

$$X^n - Y^n \in \mathbb{C}[X, Y]$$

for $n \in \mathbb{N}_+$.

Exercise 5.3 ★

Let K be an algebraically closed field and $F \in K[X, Y]$ a homogeneous polynomial. Prove that F splits into linear factors.

Exercise 5.4

Let K be a field, $F \in K[X_1, \dots, X_n]$ a homogeneous polynomial, and $F = GH$ a factorisation. Prove that G and H are also homogeneous.

Exercise 5.5

Let R be a commutative ring and

$$P = R[X_1, \dots, X_m]$$

the polynomial ring in m variables over R . Let

$$\mathfrak{m} = (X_1, \dots, X_m)$$

be the ideal generated by the variables. Prove that

$$\mathfrak{m}^n = P_{\geq n},$$

where $P_{\geq n}$ denotes the ideal of P generated by all homogeneous polynomials of degree $\geq n$.

Exercise 5.6

Prove that a homogeneous polynomial remains homogeneous of the same degree under a linear change of variables, whereas this need not hold under an affine-linear change of variables.

Exercise 5.7

Prove that affine-linear equivalence is an equivalence relation on the affine algebraic sets $V, V' \subseteq \mathbb{A}_K^n$.

Exercise 5.8

Let $P = (a, b)$ be a point in the affine plane, and L, L' two distinct lines through P . Let $C = V(F)$, with $F \in K[X, Y]$, be a plane algebraic curve. Explicitly describe a change of variables (a change of coordinates) such that, in the new coordinates, P is the origin and the two lines are the coordinate axes. What is the curve equation in the new coordinates?

Exercise 5.9

Let C and D each be a plane affine algebraic curve consisting of the union of three distinct lines, with the three lines of each curve meeting at one point. Prove that an affine-linear change of coordinates takes C to D .

Exercise 5.10

Let C and D each be a plane affine algebraic curve consisting of the union of four distinct lines, with the four lines of each curve meeting at one point. Prove that in general no affine-linear change of coordinates takes C to D .

The next two exercises are intended to aid understanding of Theorem 5.4 and Corollary 5.5.

Exercise 5.11

Apply the proof of Theorem 5.4 to the polynomial Y .

Exercise 5.12

Apply the proof of Theorem 5.4 to the hyperbola $XY - 1$.

Exercise 5.13

Apply the proof of Theorem 5.4 to the polynomial

$$X^2Y^3 + 5X^3Y^2 - X^2Y^2 + 3Y + 7 \in \mathbb{C}[X, Y].$$

Exercise 5.14

Let $F \in \mathbb{C}[X, Y]$ be a nonconstant polynomial. Prove that the corresponding algebraic curve $C = V(F)$ has uncountably many elements.

Exercise 5.15 ★

Let

$$M = \{P_1, \dots, P_n\} \subseteq K^2$$

be a finite set of points in the plane over an infinite field K .

1. Prove that M can be obtained as the intersection of two algebraic curves.
2. Prove that M can be obtained as the intersection of two irreducible algebraic curves.

Exercise 5.16

Calculate the image $\tilde{F}$ of the polynomial

$$F = X^2Y + 3XY - Y^3$$

under the substitution homomorphism

$$K[X, Y] \longrightarrow K[S, T]$$

determined by

$$X \mapsto T^2 + S - 3, \quad Y \mapsto 3TS + S^2 - T.$$

Exercise 5.17

Let K be an infinite field and $F \in K[X_1, \dots, X_n]$ a polynomial with associated map

$$F : \mathbb{A}_K^n \longrightarrow \mathbb{A}_K^1.$$

Prove, both with and without Theorem 5.10, that the image of F consists of one point or infinitely many points.

Exercise 5.18

Let K be a finite field with q elements, and let

$$P_1, \dots, P_n \in \mathbb{A}_K^2$$

be n distinct points in the affine plane. Prove that there is a polynomial map

$$\varphi : \mathbb{A}_K^1 \longrightarrow \mathbb{A}_K^2$$

with

$$\text{bild } \varphi = \{P_1, \dots, P_n\}$$

if and only if $1 \leq n \leq q$.

Edition note. The source's count is understood to be a count of distinct points; otherwise repetitions would make the criterion in terms of n false. Its notation bild is retained here.

Exercise 5.19 ★

Give an example of a polynomial map

$$\mathbb{A}_K^2 \longrightarrow \mathbb{A}_K^1$$

such that the inverse image of one point is reducible, while the inverse image of every other point is irreducible.

Exercise 5.20 ★

Let K be a field. For each $n \in \mathbb{N}_+$, consider the map

$$\begin{aligned} \varphi : K^n &\longrightarrow K^n, \\ (\lambda_1, \lambda_2, \dots, \lambda_n) &\longmapsto (c_0, c_1, \dots, c_{n-1}), \end{aligned}$$

which sends a tuple of roots $(\lambda_1, \dots, \lambda_n)$ to the tuple of coefficients $(c_0, \dots, c_{n-1})$, omitting the coefficient 1, of the monic polynomial

$$(X - \lambda_1)(X - \lambda_2) \cdots (X - \lambda_n) = P = c_0 + c_1X + \cdots + c_{n-1}X^{n-1} + X^n.$$

1. Describe φ explicitly for $n = 2$.
2. Describe φ explicitly for $n = 3$.
3. Explain why these maps φ are polynomial.
4. Prove that the fibres of φ are finite.
5. When is the fibre over a tuple $(c_0, c_1, \dots, c_{n-1})$ empty?
6. What is the maximum number of elements in a fibre? Give examples showing that this maximum is attained for $K = \mathbb{R}$.
7. Now let K be algebraically closed. Prove that φ is surjective.

Exercise 5.21

Let

$$\varphi : \mathbb{A}_K^r \longrightarrow \mathbb{A}_K^n$$

be a polynomial map and $T \subseteq \mathbb{A}_K^r$ a subset. Prove that

$$\overline{\varphi(T)} = \overline{\varphi(\overline{T})}.$$

Exercise 5.22

Prove that the statement of Exercise 5.21 does not hold without the assumption that the map is polynomial.

Exercises to hand in

Exercise 5.23 - 3 points

How many monomials of degree d are there in the polynomial ring in one, two, and three variables?

Exercise 5.24 - 3 points

Apply the proof of Theorem 5.4 to the algebraic curve corresponding to the rational function

$$Y = \frac{X^2 - 2X}{X^2 - 1}.$$

Exercise 5.25 - 3 points

Consider the map

$$\mathbb{A}_K^2 \longrightarrow \mathbb{A}_K^2, \quad (x, y) \longmapsto (x, xy).$$

Determine the image and the fibres of this map.

Exercise 5.26 - 3 points

Consider the ellipsoid

$$E = V(2x^2 + 3y^2 + 4z^2 - 5) = \{(x, y, z) : 2x^2 + 3y^2 + 4z^2 = 5\}.$$

Find an affine-linear change of variables over $\mathbb{R}$ such that the image of E under the map is the standard unit sphere

$$V(x^2 + y^2 + z^2 - 1).$$

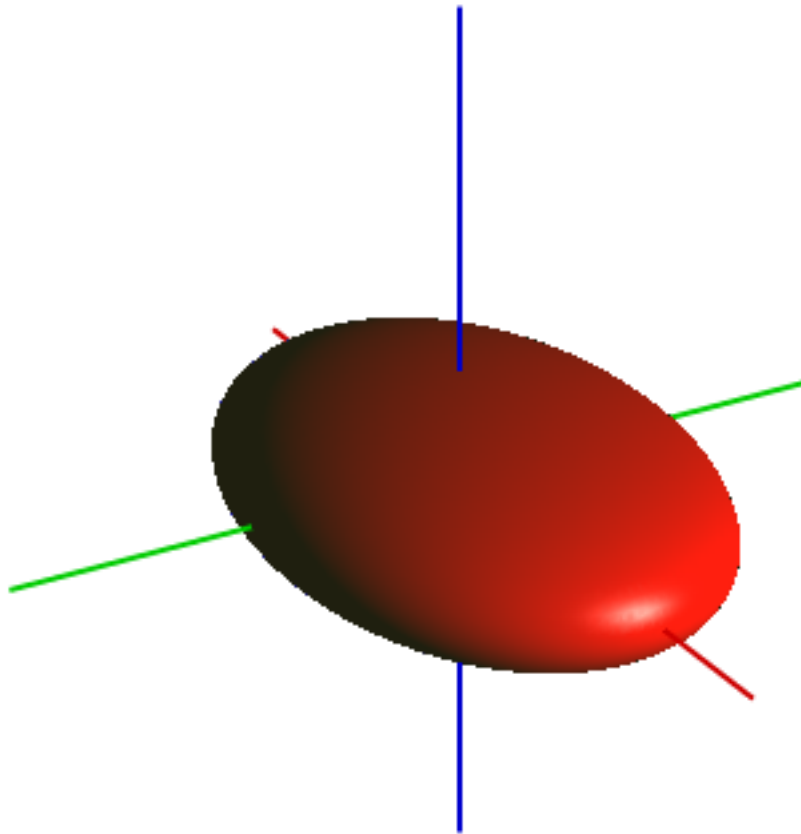


Figure 41: An ellipsoid; in algebraic geometry this means its surface

Exercise 5.27 - 4 points

Let V and $\tilde{V}$ be affine algebraic sets in $\mathbb{A}_K^2$ for $K = \mathbb{Z}/(2)$. Prove that they are affine-linearly equivalent if and only if they have the same cardinality.

Also prove that this statement fails for $K = \mathbb{Z}/(p)$ with $p \geq 3$, and in $\mathbb{A}_{\mathbb{Z}/(2)}^n$ for $n \geq 3$.

Public Solutions to Worksheet 5

The source provides public solutions only to Exercises 5.3, 5.15, 5.19, and 5.20 at the frozen source revisions. No additional solutions have been created for this edition.

Solution to Exercise 5.3

Let

$$F = \sum_{i=0}^n a_i X^i Y^{n-i}.$$

Its dehomogenisation is the one-variable polynomial

$$\tilde{F} = \sum_{i=0}^n a_i X^i.$$

Write $m = \deg \tilde{F}$ and suppose $\tilde{F} \neq 0$. Since the field is algebraically closed, this polynomial has a factorisation

$$\tilde{F} = a_m \prod_{i=1}^m (X - c_i).$$

Homogenising again gives the factorisation

$$F = a_m Y^{n-m} \prod_{i=1}^m (X - c_i Y).$$

If $\tilde{F} = 0$, then $F = 0$, so the statement is immediate.

Edition note: The source formula places a_n inside the product, which would produce a factor a_n^n , and also tacitly assumes that $\deg \tilde{F} = n$. The formula above places the leading coefficient a_m once outside the product and includes the factor Y^{n-m} required when $m < n$.

[Back to Exercise 5.3.](#)

Solution to Exercise 5.15

1. Choose a sufficiently general linear form taking distinct values at the given points. We may therefore assume that coordinates have been chosen so that, for

$$P_i = (a_i, b_i),$$

all the first coordinates a_i are distinct. Let

$$F = (X - a_1) \cdots (X - a_n).$$

By the interpolation theorem, choose a polynomial H in the one variable X with

$$H(a_i) = b_i$$

for $i = 1, \dots, n$. With $G = Y - H$, we obtain

$$M = V(F) \cap V(G),$$

expressing M as the intersection of two curves.

2. Replace F by

$$F' = Y - H + F.$$

Then

$$V(G) \cap V(F') = V(G) \cap V(F) = M.$$

Both curves are graphs and are therefore irreducible.

[Back to Exercise 5.15.](#)

Solution to Exercise 5.19

Let K be an algebraically closed field. Consider the map

$$\mathbb{A}_K^2 \longrightarrow \mathbb{A}_K^1, \quad (x, y) \longmapsto xy.$$

The fibre over zero is the union of the coordinate axes,

$$V(xy) = V(x) \cup V(y),$$

which is reducible. The fibre over a point $\lambda \in K$ with $\lambda \neq 0$ is $V(xy - \lambda)$. It suffices to show that $xy - \lambda$ is a prime polynomial. This follows from the isomorphism

$$K[x, y]/(xy - \lambda) \longrightarrow K[u]_u, \quad x \longmapsto u, \quad y \longmapsto \lambda u^{-1},$$

with inverse $u \mapsto x$. The universal properties of quotient rings and localisation ensure that these maps really are inverse to one another.

[Back to Exercise 5.19.](#)

Solution to Exercise 5.20

1. For $n = 2$, since

$$(X - \lambda_1)(X - \lambda_2) = X^2 - (\lambda_1 + \lambda_2)X + \lambda_1\lambda_2,$$

the map is

$$\begin{aligned} \varphi : K^2 &\longrightarrow K^2, \\ (\lambda_1, \lambda_2) &\longmapsto (\lambda_1\lambda_2, -(\lambda_1 + \lambda_2)). \end{aligned}$$

2. For $n = 3$, since

$$\begin{aligned} (X - \lambda_1)(X - \lambda_2)(X - \lambda_3) \\ = X^3 - (\lambda_1 + \lambda_2 + \lambda_3)X^2 + (\lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3)X - \lambda_1\lambda_2\lambda_3, \end{aligned}$$

the map is

$$\begin{aligned} \varphi : K^3 &\longrightarrow K^3, \\ (\lambda_1, \lambda_2, \lambda_3) &\longmapsto (-\lambda_1\lambda_2\lambda_3, \lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3, -(\lambda_1 + \lambda_2 + \lambda_3)). \end{aligned}$$

3. Fix $n \in \mathbb{N}_+$ and $0 \leq k \leq n-1$. By distributivity, the coefficient c_k of $\prod_{i=1}^n (X - \lambda_i)$ is

$$\pm \sum_{1 \leq i_1 < i_2 < \dots < i_{n-k} \leq n} \lambda_{i_1} \lambda_{i_2} \dots \lambda_{i_{n-k}},$$

with sign determined by the parity of $n - k$. Thus every component function is polynomial.

4. A tuple $(\lambda_1, \dots, \lambda_n)$ belongs to the fibre over the coefficient tuple $(c_0, \dots, c_{n-1})$ precisely when

$$\prod_{i=1}^n (X - \lambda_i) = \sum_{j=0}^{n-1} c_j X^j + X^n = P.$$

In particular, all the λ_i must be roots of P . Since a polynomial has only finitely many roots, there are only finitely many possible permutations.

5. The fibre over a tuple $(c_0, \dots, c_{n-1})$ is empty precisely when the polynomial

$$\sum_{j=0}^{n-1} c_j X^j + X^n$$

does not split completely into linear factors.

6. Every fibre has at most $n!$ elements. If the polynomial given by the coefficient tuple splits completely and its distinct roots have multiplicities $m_1, \dots, m_r$, the fibre consists of all orderings of those roots with those multiplicities, and hence has cardinality

$$\frac{n!}{m_1! \dots m_r!}.$$

If the polynomial does not split completely, the fibre is empty. Thus the bound $n!$ is attained when K contains n distinct elements and the polynomial has n distinct roots. In particular, for $K = \mathbb{R}$, the root tuple

$$(1, 2, 3, \dots, n)$$

maps to a coefficient tuple whose fibre consists of all permutations of that root tuple.

Edition note: The source calls $n!$ the maximum without restricting the field K . The multiplicity count above is an editorial clarification: in general, $n!$ is an upper bound, while equality requires at least n distinct elements in K . The example requested in the exercise is specifically for $K = \mathbb{R}$.

7. If K is algebraically closed, every monic polynomial splits into monic linear factors. By part 5, each corresponding fibre is nonempty. Thus φ is surjective.

[Back to Exercise 5.20.](#)

Lecture 6: Polynomial and Rational Parametrisations

Polynomial parametrisations in the plane

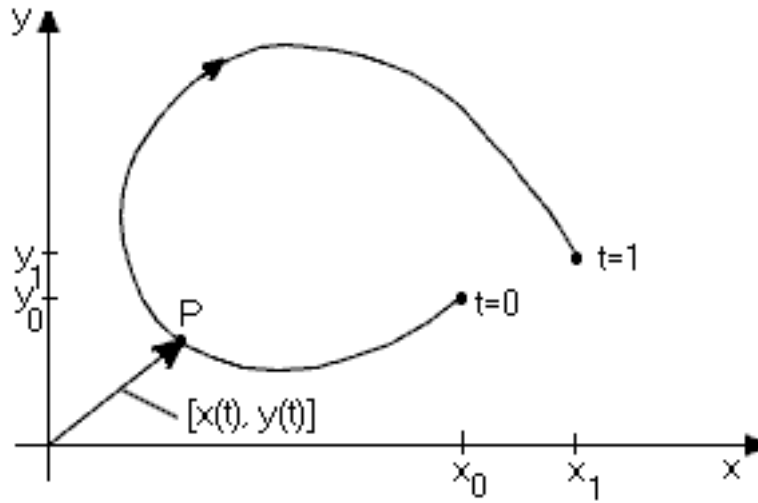


Figure 42: A parametrised curve can be thought of as a trajectory of motion

We now consider maps

$$\varphi : \mathbb{A}_K^1 \longrightarrow \mathbb{A}_K^2$$

given by two polynomials in one variable

$$P, Q \in K[T].$$

The image of such a map lies in an affine algebraic curve, as the following theorem shows. We also speak of *parametrised curves*, or more precisely *polynomially parametrised curves*.

Here two ways of describing an algebraic curve compete. The points of a curve given by a curve equation are specified only implicitly. For each point of the plane, it is easy to check whether it lies on the curve, but in general it is difficult to find or explicitly specify points on the curve. A parametrised curve, by contrast, is given explicitly: for each point of the affine line, its image point can easily be calculated, giving the points of the curve explicitly. However, not every algebraic curve can be parametrised by polynomials.

Theorem: an equation for a polynomially parametrised curve

Let K be a field and let $P, Q \in K[T]$ be polynomials. Then there is a polynomial

$$F \in K[X, Y], \quad F \neq 0,$$

such that

$$F(P, Q) = 0.$$

In other words, the image of a polynomially parametrised curve lies in a plane algebraic curve

$$C = V(F).$$

If K is infinite and P, Q are not both constant, the Zariski closure of the image is an irreducible curve C .

Proof *Edition note.* The source's degree argument below treats nonzero P, Q . If $P = 0$, take $F = X$; if $Q = 0$, take $F = Y$. Thus the zero-polynomial cases of the theorem are covered as well.

Let d and e be the degrees of P and Q , respectively. Consider the monomials

$$P^i Q^j.$$

These are polynomials in T of degree $di + ej$. For $i \leq n$ and $j \leq m$, there are $(n+1)(m+1)$ such monomials. They all lie in the $(dn + em + 1)$ -dimensional K -vector space spanned by

$$1 = T^0, T^1, T^2, \dots, T^{dn+em}.$$

If

$$(n+1)(m+1) > dn + em + 1,$$

there must be a nontrivial linear dependence among the $P^i Q^j$. This gives a polynomial $F(X, Y) \neq 0$ with $F(P, Q) = 0$. The numerical condition above can be met by choosing n, m sufficiently large.

From now on, let K be infinite. By Lemma 3.10, the Zariski closure of the image

$$B = \varphi(\mathbb{A}_K^1)$$

is $V(\text{Id}(B))$, and by Theorem 5.10 this set is irreducible. Since K is infinite and the map is nonconstant, irreducibility also forces $V(\text{Id}(B))$ to contain infinitely many points. By Lemma 4.3, $\text{Id}(B)$ is a prime ideal; by the first part it contains an element

$$F \in \text{Id}(B), \quad F \neq 0.$$

Since $K[X, Y]$ is a unique factorisation domain, a prime factor of F also belongs to this ideal. We may therefore assume that F is a prime polynomial. We have the inclusions

$$B \subseteq \overline{B} = V(\text{Id}(B)) \subseteq V(F).$$

For $H \in \text{Id}(B)$, the set

$$V(\text{Id}(B)) \subseteq V(H) \cap V(F)$$

is infinite. By Theorem 4.8, H and F must have a common nonconstant factor. Since F is prime, H must be a multiple of F . Thus

$$\text{Id}(B) = (F).$$

□

Example: eliminating the parameter

Consider the curve given by the parametrisation

$$x = t^2 + t + 1, \quad y = 2t^2 + 3t - 1.$$

We have

$$x - 1 = t^2 + t, \quad y + 1 = 2t^2 + 3t.$$

A simple subtraction gives

$$(y + 1) - 2(x - 1) = 3t - 2t = t.$$

Thus

$$x - 1 = t^2 + t = (y - 2x + 3)^2 + (y - 2x + 3).$$

Expanding gives the curve equation

$$y^2 + 4x^2 - 4xy - 15x + 7y + 13 = 0.$$

Example: a curve with one self-intersection

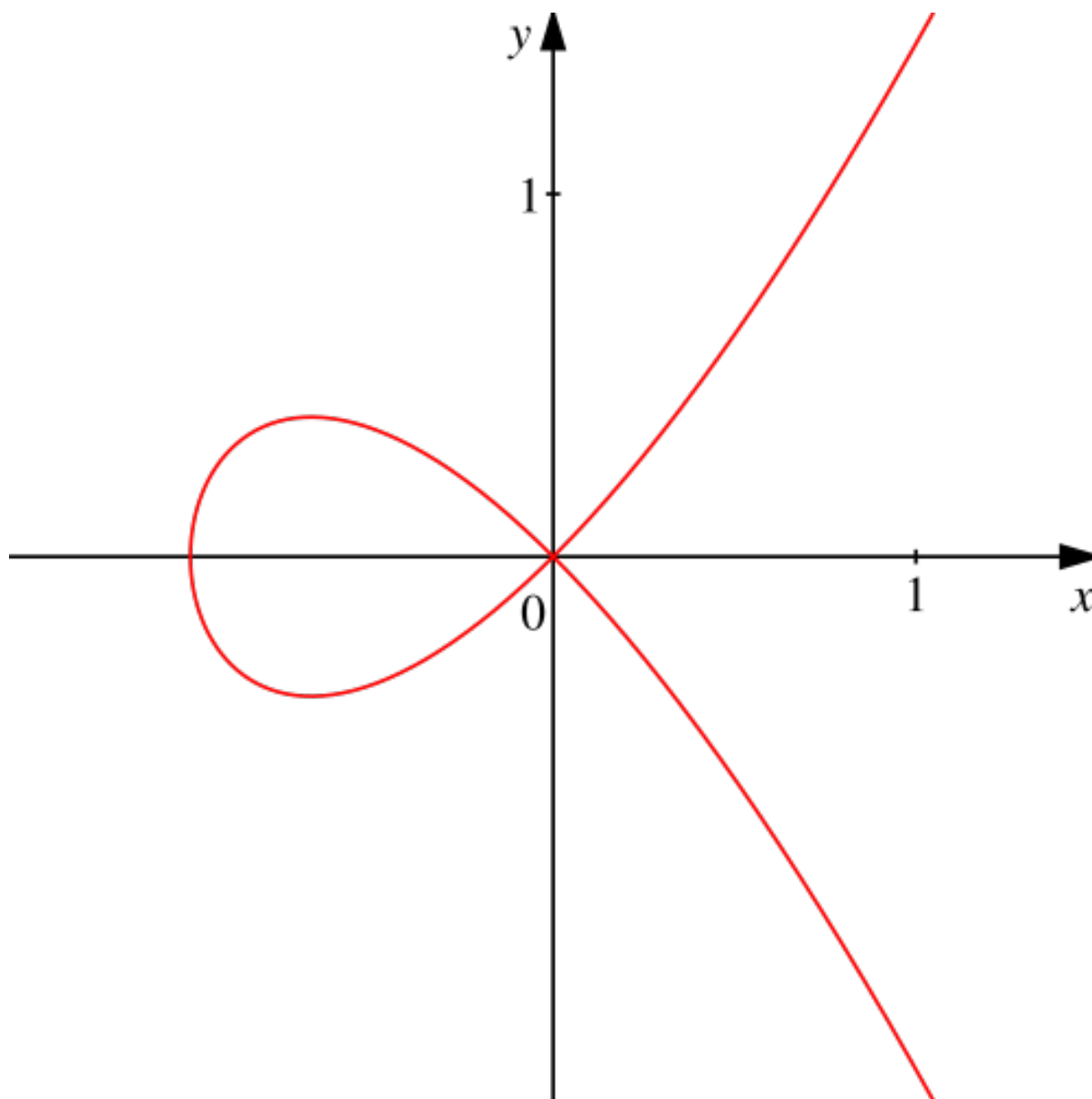


Figure 43: A cubic curve with a double point

Consider the map $\mathbb{A}_K^1 \rightarrow \mathbb{A}_K^2$ given by

$$P = t^2 - 1, \quad Q = t^3 - t = t(t^2 - 1).$$

Edition note. The source's two-branch description assumes $\text{char } K \neq 2$. In characteristic 2, $1 = -1$, so there is only one parameter value over $(0, 0)$ and the self-intersection conclusion below does not apply; the displayed algebraic identities remain valid.

Both parameter values $t = \pm 1$ give the point $(0, 0)$. For every other value $t \neq \pm 1$, we can write

$$t = \frac{t^3 - t}{t^2 - 1} = \frac{Q(t)}{P(t)}.$$

Thus the parameter t can be reconstructed from its image, which means that the map is injective away from those two values. The image curve therefore intersects itself at exactly one point.

To determine the curve equation, write $x = t^2 - 1$ and $y = t^3 - t$. Then

$$t^2 = x + 1$$

and

$$\begin{aligned} y^2 &= t^2(t^2 - 1)^2 \\ &= t^2 x^2 \\ &= (x + 1)x^2 \\ &= x^3 + x^2. \end{aligned}$$

The polynomial describing the curve is therefore

$$Y^2 - X^3 - X^2.$$

Rational parametrisations

Consider a rational function

$$Y = \frac{P}{Q}, \quad P, Q \in K[T].$$

This immediately gives a new form of parametrisation through the map

$$\begin{aligned} \mathbb{A}_K^1 &\supseteq D(Q) \longrightarrow \mathbb{A}_K^2, \\ t &\longmapsto \left(t, \frac{P(t)}{Q(t)} \right). \end{aligned}$$

Here $D(Q)$ is the domain of definition of the map, namely

$$D(Q) = \mathbb{A}_K^1 \setminus V(Q),$$

consisting of all points where the denominator polynomial Q is nonzero. This map clearly reaches every point of the graph of the rational function, so, like a polynomial parametrisation, it provides an explicit description of the curve. To describe curves, it is therefore natural to allow parametrisations whose component functions are rational as well.

Definition: rational parametrisation

Two rational functions

$$\varphi_1 = \frac{P_1}{Q_1}, \quad \varphi_2 = \frac{P_2}{Q_2},$$

with

$$P_1, P_2, Q_1, Q_2 \in K[T], \quad Q_1, Q_2 \neq 0,$$

are called a *rational parametrisation* of the algebraic curve

$$C = V(F), \quad F \in K[X, Y] \text{ nonconstant},$$

if

$$F(\varphi_1(T), \varphi_2(T)) = 0$$

and the pair (φ_1, φ_2) is nonconstant.

The equality in this definition is understood in the rational function field $K(T)$. If K is infinite, this is equivalent to the equality holding for every $t \in K$ where the denominators allow the functions to be defined.

Definition: rational curve

A plane algebraic curve

$$C = V(F)$$

is called *rational* if it is irreducible and has a rational parametrisation.

The following simple example shows that rational functions can parametrise more curves than polynomials can. However, we should already mention that this difference disappears again in the context of projective geometry.

Example: the hyperbola

Consider the hyperbola

$$H = V(XY - 1).$$

We claim that it has no polynomial parametrisation. For two polynomials $P(t)$ and $Q(t)$, the condition that their image always lie on H is

$$P(t)Q(t) = 1 \quad \text{for every } t \in \mathbb{A}_K^1,$$

or that $P(t)Q(t) = 1$ in the polynomial ring $K[t]$. These conditions are equivalent over an infinite field; over a finite field, the second identity is the appropriate condition. This identity means that P and Q are inverses of one another, so both are units. The only units in a polynomial ring are the nonzero constants. Thus both polynomials are constant, the map they define is constant, and there is no polynomial parametrisation.

By contrast,

$$\begin{aligned} \mathbb{A}_K^1 \setminus \{0\} &\longrightarrow \mathbb{A}_K^2, \\ t &\longmapsto \left(t, \frac{1}{t}\right) \end{aligned}$$

is a rational parametrisation of the hyperbola.

We want to show that the image of a nonconstant rational map always satisfies an algebraic equation, and thus always gives a rational parametrisation of an algebraic curve. In the polynomial case, an algebraic equation followed from a counting argument: the number of monomials in two variables grows faster with the degree than the number of monomials in one variable. We will use a similar argument together with an additional trick, *homogenisation*. This makes an inhomogeneous situation homogeneous by adding another variable (that is the price we have to pay). Here we use this process purely algebraically, but behind it lies the interplay between affine and projective geometry.

Definition: homogenisation

Let

$$F \in K[X_1, \dots, X_n], \quad F \neq 0,$$

be a polynomial with homogeneous decomposition

$$F = \sum_{i=0}^d F_i,$$

and let Z be an additional variable. The homogeneous polynomial of degree d

$$\widehat{F} = \sum_{i=0}^d F_i Z^{d-i} \in K[X_1, \dots, X_n, Z]$$

is called the *homogenisation* of F .

The original polynomial can be recovered from its homogenisation by setting the additional variable $Z = 1$. This process is called *dehomogenisation*.

Lemma: a homogeneous relation for three homogeneous polynomials

Let

$$P_1, P_2, P_3 \in K[S, T]$$

be three homogeneous polynomials of the same degree. Then there is a homogeneous polynomial

$$F \in K[X, Y, Z], \quad F \neq 0,$$

such that

$$F(P_1, P_2, P_3) = 0.$$

Proof This follows from a counting argument similar to the proof of Theorem 6.1; see Exercise 6.7. $\square$

Example: a monomial relation

Consider the map

$$(S, T) \mapsto (S^2, T^2, ST) = (X, Y, Z),$$

given by homogeneous polynomials, indeed by monomials. An algebraic relation for its image is easy to find:

$$Z^2 = (ST)^2 = S^2 T^2 = XY.$$

Thus the image lies in $V(Z^2 - XY)$. See also Exercise 6.29.

Theorem: the image of a rational map satisfies an algebraic equation

Suppose two rational functions

$$\varphi_1 = \frac{P_1}{Q_1}, \quad \varphi_2 = \frac{P_2}{Q_2},$$

with $P_1, P_2, Q_1, Q_2 \in K[T]$, $Q_1, Q_2 \neq 0$, are given and are not both constant. Then there is a nonconstant polynomial $F \in K[X, Y]$ such that

$$F(\varphi_1(T), \varphi_2(T)) = 0.$$

Thus φ_1 and φ_2 define a rational parametrisation.

Proof By passing to a common denominator, we may assume that the rational map is given by

$$\varphi_1 = \frac{P_1}{Q}, \quad \varphi_2 = \frac{P_2}{Q},$$

with $P_1, P_2, Q \in K[T]$ and $Q \neq 0$. Let

$$H'_1, H'_2, H'_3 \in K[T, S]$$

be the homogenisations of these three polynomials with the new variable S , and let e be their largest degree. *Edition note:* if a numerator is zero, set its H'_i and H_i to zero and take e over the nonzero polynomials; the degree formula below is applied only to nonzero H'_i . The zero polynomial is homogeneous of the required degree. Set

$$H_i = S^{e-\deg(H'_i)} H'_i.$$

The polynomials H_1, H_2, H_3 all have degree e , while their dehomogenisations at $S = 1$ remain P_1, P_2, Q . By Lemma 6.8, there is a homogeneous polynomial

$$F \in K[U, V, W], \quad F \neq 0,$$

of degree d in U, V, W , such that

$$F(H_1, H_2, H_3) = 0.$$

Now consider

$$\frac{1}{W^d} F(U, V, W) = F\left(\frac{U}{W}, \frac{V}{W}, \frac{W}{W}\right),$$

which is a polynomial in the two rational functions U/W and V/W . The homogeneity of F is crucial for this step. Substituting the three homogeneous polynomials gives

$$0 = F\left(\frac{H_1}{H_3}, \frac{H_2}{H_3}, 1\right).$$

This is an equality in the fraction field of $K[S, T]$. Setting $S = 1$, that is, dehomogenising, and writing

$$G(X, Y) = F(X, Y, 1),$$

we obtain a nonzero polynomial $G \in K[X, Y]$ such that

$$0 = G\left(\frac{P_1}{Q}, \frac{P_2}{Q}\right),$$

which is an equation for the two original rational functions. $\square$

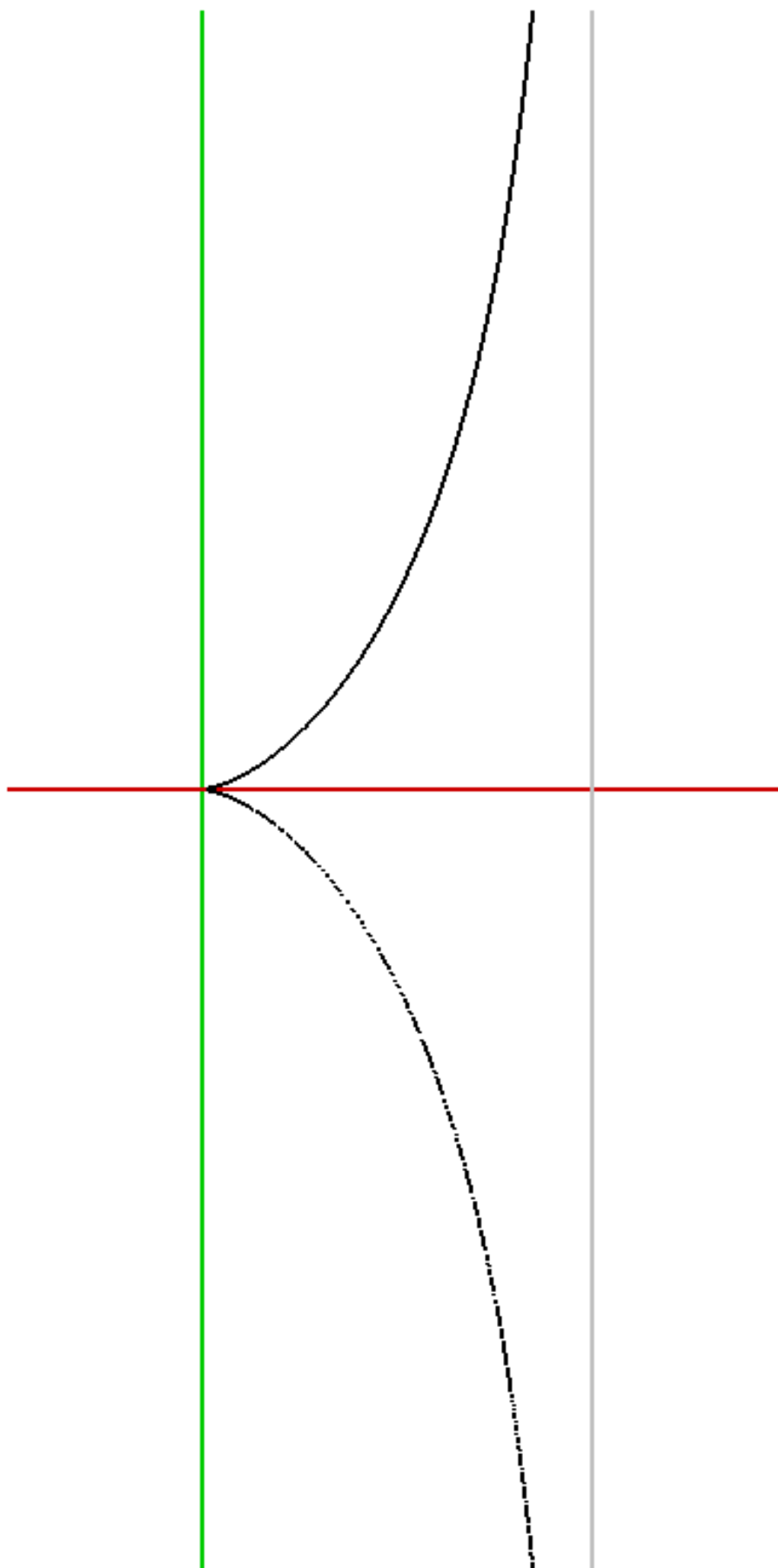


Figure 44: The cissoid of Diocles (black in the image) can be parametrised rationally

Remark: local differentiable parametrisations

We can go a step further and ask whether there are other ways to describe an algebraic curve

$$C = V(F)$$

by a map $\varphi : K \rightarrow K^2$, allowing φ to belong to a larger class of functions. An important result here is the implicit function theorem. For $K = \mathbb{R}$ or $K = \mathbb{C}$, it says that if the two partial derivatives of F do not both vanish at a point of the curve, then there is an infinitely differentiable, indeed analytic, map describing the curve in a small open neighbourhood of that point. An algebraic version of the implicit function theorem reappears in the power-series approach that we will discuss later.

Worksheet 6**Practice exercises****Exercise 6.1**

Let $F(x) \in K[x]$ be a polynomial in one variable over a field K . Parametrise the graph of F by polynomials.

Exercise 6.2

Determine a curve equation for the parametrised curve

$$x = -3t^2 + 4t - 2, \quad y = 2t^2 + 5t - 3.$$

Exercise 6.3 ★

Let K be a field. Consider the parametrisation

$$\begin{aligned} \mathbb{A}_K^1 &\longrightarrow \mathbb{A}_K^2, \\ t &\longmapsto (t + t^2, t^3) = (x, y). \end{aligned}$$

Determine a nontrivial algebraic equation satisfied by every image point of this map. Also give a point in the affine plane that does not lie on the image curve.

Exercise 6.4 ★

Let K be a field. Consider the polynomial map

$$\begin{aligned} \mathbb{A}_K^1 &\longrightarrow \mathbb{A}_K^2, \\ t &\longmapsto (t^2 + 1, t^3 - t). \end{aligned}$$

Determine a nontrivial algebraic equation satisfied by every image point of this map.

The phenomenon described in the following exercise cannot occur over an algebraically closed field.

Exercise 6.5

Give an example of a polynomial parametrisation

$$\varphi : \mathbb{A}_{\mathbb{R}}^1 \longrightarrow C \subset \mathbb{A}_{\mathbb{R}}^2,$$

where C is the Zariski closure of the image, such that infinitely many points of C are not in the image.

Exercise 6.6

Show that, in the situation of Example 6.3, the given parametrisation of the curve is surjective for $K = \mathbb{R}$.

Exercise 6.7

Let

$$P_1, P_2, P_3 \in K[S, T]$$

be three homogeneous polynomials of the same degree. Show that there is a homogeneous polynomial

$$F \in K[X, Y, Z], \quad F \neq 0,$$

such that

$$F(P_1, P_2, P_3) = 0.$$

Exercise 6.8 ★

Is there a homogeneous polynomial $F \in K[X, Y, Z]$, $F \neq 0$, such that

$$F(S, T, ST) = 0?$$

Exercise 6.9 ★

Determine an algebraic equation for the image curve of the map

$$\begin{aligned} \mathbb{A}_K^1 \setminus \{0\} &\longrightarrow \mathbb{A}_K^2, \\ t &\longmapsto \left(\frac{t^2 + 1}{t}, \frac{t + 1}{t} \right). \end{aligned}$$

Check the resulting equation directly.

Exercise 6.10

Determine a nontrivial algebraic equation for the image of the map

$$\begin{aligned} \mathbb{A}_K^1 \setminus \{-1, 0, 1\} &\longrightarrow \mathbb{A}_K^2, \\ t &\longmapsto \left(\frac{t}{t^2 - 1}, \frac{1}{t} \right). \end{aligned}$$

The following exercises use the notion of a differentiable curve as discussed in Analysis 2.

Exercise 6.11

Give an example of a non-algebraic (non-polynomial) differentiable curve in $\mathbb{R}^2$ whose image nevertheless coincides with an algebraic curve.

Exercise 6.12

Show that the trajectory of the Archimedean spiral

$$\begin{aligned} f : \mathbb{R}_{\geq 0} &\longrightarrow \mathbb{R}^2, \\ t &\longmapsto (t \cos t, t \sin t) \end{aligned}$$

is not algebraic.

Exercise 6.13

Let

$$C = V(F) \subseteq \mathbb{A}_{\mathbb{R}}^2$$

be a rationally parametrised curve. Show that the curve also has a non-algebraic differentiable parametrisation.

Exercise 6.14

Give a differentiable curve $f : \mathbb{R} \rightarrow \mathbb{R}^2$ whose image is exactly the union of the two coordinate axes.

Exercise 6.15

Let $\varphi : \mathbb{A}_K^2 \rightarrow \mathbb{A}_K^2$ be a polynomial map and let C be a plane rational curve. Suppose that φ does not map C to a single point. Show that

$$\overline{\varphi(C)}$$

is also a rational curve.

Exercise 6.16

Let K be an algebraically closed field of characteristic zero, and suppose a polynomial map

$$\begin{aligned} \varphi : \mathbb{A}_K^1 &\longrightarrow \mathbb{A}_K^2, \\ t &\longmapsto (t^2, \psi(t)), \end{aligned}$$

with $\psi(t) \in K[t]$ is given. Show that φ is injective if and only if ψ has the form

$$\psi(t) = at^n + \theta(t),$$

where $a \neq 0$, n is odd, and $\theta(t)$ is a polynomial containing only even powers.

Exercise 6.17 ★

Let $p \in [0, 1]$ be the probability that a certain event A occurs in an experiment, so that $1 - p$ is the probability that it does not occur ($\neg A$). The experiment is performed twice independently. The combined events

$$(A, A), (A, \neg A), (\neg A, A), (\neg A, \neg A)$$

have probabilities depending on p . Regard this dependence as the polynomial map

$$\begin{aligned} \varphi : \mathbb{A}_K^1 &\longrightarrow \mathbb{A}_K^4, \\ p &\longmapsto (p^2, p(1-p), (1-p)p, (1-p)^2) = (x, y, z, w). \end{aligned}$$

1. Show that this map is injective.
2. Describe its image completely by polynomial equations.

Exercise 6.18 ★

Let $p \in [0, 1]$ be the probability that event A occurs in a first experiment, and let $q \in [0, 1]$ be the probability that event B occurs in a second experiment. The two experiments are performed independently. The combined events

$$(A, B), (A, \neg B), (\neg A, B), (\neg A, \neg B)$$

have probabilities depending on p and q . Regard this dependence as the polynomial map

$$\begin{aligned}\varphi : \mathbb{A}_K^2 &\longrightarrow \mathbb{A}_K^4, \\ (p, q) &\longmapsto (pq, p(1-q), (1-p)q, (1-p)(1-q)) = (x, y, z, w).\end{aligned}$$

1. Show that this map is injective.
2. Describe its image completely by polynomial equations.

Exercise 6.19

Consider the map

$$\begin{aligned}\mathbb{A}_K^2 \supseteq D(s) &\longrightarrow \mathbb{A}_K^3, \\ (s, t) &\longmapsto \left(s, \frac{t^2}{s}, t\right) = (x, y, z).\end{aligned}$$

Determine an algebraic equation F for its image. Investigate the injectivity and surjectivity of this map as a map to $V(F)$. Compare it with the maps discussed in Exercise 6.29.

Exercise 6.20

Let K be a field, $K[X_1, \dots, X_n]$ the polynomial ring over K in n variables, and $K[X_1, \dots, X_n, Z]$ the polynomial ring in $n+1$ variables. For $F \in K[X_1, \dots, X_n]$, let $\widehat{F} \in K[X_1, \dots, X_n, Z]$ be its homogenisation with respect to Z . For $G \in K[X_1, \dots, X_n, Z]$, let $\widetilde{G}$ be its dehomogenisation given by $Z \mapsto 1$. Show that

$$\widetilde{\widehat{F}} = F,$$

but that it need not be the case that

$$\widehat{\widetilde{G}} = G.$$

Exercise 6.21 ★

Let K be a field and $K[X_1, \dots, X_n, Z]$ the polynomial ring over K in $n+1$ variables. Let

$$G, H \in K[X_1, \dots, X_n, Z]$$

be homogeneous polynomials of the same degree. If their dehomogenisations with respect to Z satisfy

$$\widetilde{G} = \widetilde{H},$$

show that $G = H$.

Exercise 6.22 ★

Let K be a field, $K[X_1, \dots, X_n]$ the polynomial ring over K in n variables, and $K[X_1, \dots, X_n, Z]$ the polynomial ring in $n+1$ variables. Show that homogenisation with respect to Z is compatible with multiplication.

Exercise 6.23

Let K be a field, $K[X_1, \dots, X_n]$ the polynomial ring over K in n variables, and $K[X_1, \dots, X_n, Z]$ the polynomial ring in $n+1$ variables. Describe dehomogenisation with respect to Z as a substitution homomorphism.

Exercise 6.24

Formulate and prove a division-with-remainder statement for homogeneous polynomials in two variables over a field.

Exercise 6.25 ★

Let

$$F = X^4 + 9X^3Y + 7X^2Y^2 + XY^3 + 8Y^4$$

and

$$G = X^3 + 5X^2Y.$$

Find homogeneous polynomials $Q, R \in \mathbb{Q}[X, Y]$, with $Q \neq 0$, such that

$$F = GQ + R.$$

Exercises for submission**Exercise 6.26 - 3 points**

Determine the area enclosed by the loop of the real curve

$$V(y^2 - x^3 - x^2) \subseteq \mathbb{A}_{\mathbb{R}}^2.$$

The following exercises may require the use of a computer.

Exercise 6.27 - 6 points

Determine an algebraic equation for the image curve of the map

$$\begin{aligned} \mathbb{A}_K^1 &\longrightarrow \mathbb{A}_K^2, \\ t &\longmapsto (t^2 + t^3, 2t^2 - t^4). \end{aligned}$$

Exercise 6.28 - 5 points

Determine an algebraic equation for the image curve of the map

$$\begin{aligned} \mathbb{A}_K^1 \setminus \{0\} &\longrightarrow \mathbb{A}_K^2, \\ t &\longmapsto \left(\frac{t^2 - 1}{t}, \frac{t + 3}{t} \right). \end{aligned}$$

Check the resulting equation directly.

Exercise 6.29 - 5 points

Consider the two maps

$$(s, t) \longmapsto (s^2, t^2, st) = (x, y, z)$$

and

$$(s, t) \longmapsto (s, st^2, st) = (x, y, z).$$

Show that the images of both maps satisfy the same algebraic equation F . Investigate the injectivity and surjectivity of the two maps as maps to $V(F)$. Which map gives a “better” description of $V(F)$?

Exercise 6.30 - 4 points

Let K be a field and $F \in K[X, Y]$ an irreducible polynomial. If the zero locus $V(F)$ is infinite, show that $V(F)$ is an irreducible affine algebraic set.

Also give an example showing that this statement is false in three variables.

Source navigation: [Lecture 6 - public solutions for Unit 6](#) - [Worksheet 5](#) - [Worksheet 7 \(source\)](#)

Public Solutions to Worksheet 6

At the frozen revision boundary, the source provides public solutions only for Exercises 6.3, 6.4, 6.8, 6.9, 6.17, 6.18, 6.21, 6.22, and 6.25. No additional solutions have been created for this edition.

Solution to Exercise 6.3

We calculate the first few monomials in X and Y :

$$X^0 Y^0 = 1,$$

$$X = t^2 + t,$$

$$Y = t^3,$$

$$XY = t^5 + t^4,$$

$$X^2 = t^4 + 2t^3 + t^2,$$

$$Y^2 = t^6,$$

and

$$X^3 = t^6 + 3t^5 + 3t^4 + t^3.$$

We seek a nontrivial relation among these polynomials in $K[t]$. Since

$$X^3 - Y^2 - Y = 3t^5 + 3t^4 = 3XY,$$

an algebraic relation for the image curve is

$$X^3 - Y^2 - Y - 3XY = 0.$$

For example, the point $(1, 0)$ in the affine plane does not lie on the image curve, since

$$1^3 - 0^2 - 0 - 3 \cdot 1 \cdot 0 = 1 \neq 0.$$

[Back to Exercise 6.3.](#)

Solution to Exercise 6.4

We calculate some monomials in X and Y :

$$X^0Y^0 = 1,$$

$$X = t^2 + 1,$$

$$X^2 = t^4 + 2t^2 + 1,$$

$$X^3 = t^6 + 3t^4 + 3t^2 + 1,$$

and

$$Y^2 = t^6 - 2t^4 + t^2.$$

These are five polynomials containing only the powers t^0, t^2, t^4, t^6 , so they must be linearly dependent. One linear relation is

$$-Y^2 + X^3 - 5X^2 + 8X - 4 = 0.$$

[Back to Exercise 6.4.](#)

Solution to Exercise 6.8

No such polynomial exists. Suppose

$$F = \sum_{\alpha+\beta+\gamma=d} a_{(\alpha,\beta,\gamma)} X^\alpha Y^\beta Z^\gamma$$

is a homogeneous polynomial of degree d . The required equation would be

$$F(S, T, ST) = \sum_{\alpha+\beta+\gamma=d} a_{(\alpha,\beta,\gamma)} S^{\alpha+\gamma} T^{\beta+\gamma} = 0.$$

Suppose two monomials in this sum are equal:

$$S^{\alpha_1+\gamma_1} T^{\beta_1+\gamma_1} = S^{\alpha_2+\gamma_2} T^{\beta_2+\gamma_2}.$$

Then

$$\alpha_1 + \gamma_1 = \alpha_2 + \gamma_2$$

and

$$\beta_1 + \gamma_1 = \beta_2 + \gamma_2.$$

Adding the equations gives

$$\alpha_1 + \beta_1 + 2\gamma_1 = \alpha_2 + \beta_2 + 2\gamma_2.$$

Since

$$\alpha_1 + \beta_1 + \gamma_1 = \alpha_2 + \beta_2 + \gamma_2 = d,$$

we obtain $\gamma_1 = \gamma_2$, and then $\alpha_1 = \alpha_2$ and $\beta_1 = \beta_2$. Thus all the monomials $S^{\alpha+\gamma}T^{\beta+\gamma}$ in the sum above are pairwise distinct. For $F(S, T, ST) = 0$, every coefficient $a_{(\alpha, \beta, \gamma)}$ must be zero. Hence $F = 0$, contrary to the requirement of the exercise.

[Back to Exercise 6.8.](#)

Solution to Exercise 6.9

We use the homogenisations of equal degree

$$H_1 = T^2 + S^2, \quad H_2 = S(T + S) = ST + S^2, \quad H_3 = ST.$$

These give six monomials of degree 4 in S, T . Since there are only five monomials of degree 4 in two variables, a linear dependence must exist. Explicitly,

$$F_1 = H_1^2 = T^4 + 2S^2T^2 + S^4,$$

$$F_2 = H_2^2 = S^2T^2 + 2S^3T + S^4,$$

$$F_3 = H_3^2 = S^2T^2,$$

$$F_4 = H_1H_2 = ST^3 + S^2T^2 + S^3T + S^4,$$

$$F_5 = H_1H_3 = ST^3 + S^3T,$$

and

$$F_6 = H_2H_3 = S^2T^2 + S^3T.$$

Since T^4 occurs only in F_1 , we can seek a linear relation among F_2, F_3, F_4, F_5, F_6 . Since F_3 is a monomial, we focus on the relevant monomials ST^3, S^3T, S^4 and on F_2, F_4, F_5, F_6 . We have

$$F_2 - F_4 + F_5 - 2F_6 = -2S^2T^2.$$

Thus

$$F_2 + 2F_3 - F_4 + F_5 - 2F_6 = 0.$$

Consequently,

$$F(U, V, W) = V^2 + 2W^2 - UV + UW - 2VW$$

is a homogeneous polynomial of degree 2 that vanishes when U, V, W are replaced by H_1, H_2, H_3 , respectively. The corresponding equation, obtained by dividing $F(U, V, W) = 0$ by W^2 , is

$$\left(\frac{V}{W}\right)^2 + 2 - \frac{U}{W} \frac{V}{W} + \frac{U}{W} - 2\frac{V}{W} = 0.$$

After substituting H_1, H_2, H_3 and then setting $S = 1$, the ratios U/W and V/W become the original rational functions. An annihilating polynomial is therefore

$$Y^2 - XY + X - 2Y + 2.$$

As a direct check,

$$\begin{aligned} & \left(\frac{t+1}{t} \right)^2 - \frac{t^2+1}{t} \frac{t+1}{t} + \frac{t^2+1}{t} - 2 \frac{t+1}{t} + 2 \\ &= \frac{t^2 + 2t + 1 - (t^3 + t^2 + t + 1) + t^3 + t - 2t^2 - 2t + 2t^2}{t^2} \\ &= 0. \end{aligned}$$

Back to Exercise 6.9.

Solution to Exercise 6.17

1. From

$$(x, y, z, w) = (p^2, p(1-p), (1-p)p, (1-p)^2) = (p^2, p-p^2, p-p^2, p^2-2p+1),$$

we immediately obtain

$$p = p^2 + (p - p^2) = x + y.$$

Thus the input variable can be reconstructed from the component polynomials, proving injectivity.

2. Clearly $y = z$, which gives a first equation and allows z to be eliminated. Using

$$u = p = x + y,$$

we obtain

$$y = u - u^2$$

and

$$w = u^2 - 2u + 1.$$

Thus y and w can also be eliminated, and the image is described completely by the three equations

$$y - z = 0,$$

$$y - (x + y) + (x + y)^2 = 0,$$

and

$$w - (x + y)^2 + 2(x + y) - 1 = 0.$$

Back to Exercise 6.17.

Solution to Exercise 6.18

1. From

$$(x, y, z, w) = (pq, p(1 - q), (1 - p)q, (1 - p)(1 - q)) = (pq, p - pq, q - pq, pq - p - q + 1),$$

we immediately obtain

$$p = pq + (p - pq) = x + y$$

and

$$q = pq + (q - pq) = x + z.$$

Thus both input variables can be reconstructed, proving injectivity.

2. Using

$$u = p = x + y, \quad v = q = x + z,$$

we obtain

$$x = uv$$

and

$$w = uv - u - v + 1.$$

Thus x and w can be eliminated, and the image is described completely by the two equations

$$x - (x + y)(x + z) = 0$$

and

$$w - (x + y)(x + z) + (x + y) + (x + z) - 1 = 0.$$

[Back to Exercise 6.18.](#)

Solution to Exercise 6.21

Let d be the common degree of G and H , and write

$$G = \sum_{\nu \in \mathbb{N}^n} a_\nu X^\nu Z^{d-|\nu|}$$

and

$$H = \sum_{\nu \in \mathbb{N}^n} b_\nu X^\nu Z^{d-|\nu|}.$$

Their dehomogenisations are

$$\sum_{\nu \in \mathbb{N}^n} a_\nu X^\nu \quad \text{and} \quad \sum_{\nu \in \mathbb{N}^n} b_\nu X^\nu,$$

which are equal by assumption. Hence $a_\nu = b_\nu$ for every ν , and therefore the original polynomials are also equal.

[Back to Exercise 6.21.](#)

Solution to Exercise 6.22

Let

$$F = F_d + F_{d-1} + \cdots + F_1 + F_0$$

and

$$G = G_e + G_{e-1} + \cdots + G_1 + G_0$$

be polynomials of degrees d and e , written in their homogeneous decompositions. Their homogenisations are

$$\widehat{F} = F_d + F_{d-1}Z + \cdots + F_1Z^{d-1} + F_0Z^d$$

and

$$\widehat{G} = G_e + G_{e-1}Z + \cdots + G_1Z^{e-1} + G_0Z^e.$$

Their product has the form

$$\widehat{F}\widehat{G} = \sum_{k=0}^{d+e} P_k Z^{d+e-k},$$

where

$$P_k = \sum_{i=0}^d F_i G_{k-i},$$

with components whose indices lie outside the range understood to be zero. On the other hand,

$$FG = \sum_{k=0}^{d+e} H_k$$

has homogeneous components

$$H_k = \sum_{i=0}^d F_i G_{k-i}.$$

Therefore,

$$\begin{aligned} \widehat{F}\widehat{G} &= \sum_{k=0}^{d+e} H_k Z^{d+e-k} \\ &= \sum_{k=0}^{d+e} \left(\sum_{i=0}^d F_i G_{k-i} \right) Z^{d+e-k} \\ &= \sum_{k=0}^{d+e} P_k Z^{d+e-k} \\ &= \widehat{F}\widehat{G}. \end{aligned}$$

[Back to Exercise 6.22.](#)

Solution to Exercise 6.25

Division with remainder in the homogeneous case gives

$$\begin{aligned} X^4 + 9X^3Y + 7X^2Y^2 + XY^3 + 8Y^4 \\ = (X^3 + 5X^2Y)(X + 4Y) - 13X^2Y^2 + XY^3 + 8Y^4. \end{aligned}$$

Thus

$$Q = X + 4Y$$

and

$$R = -13X^2Y^2 + XY^3 + 8Y^4.$$

[Back to Exercise 6.25.](#)

Lecture 7: Conic Sections and Quadrics

Conic sections and quadrics

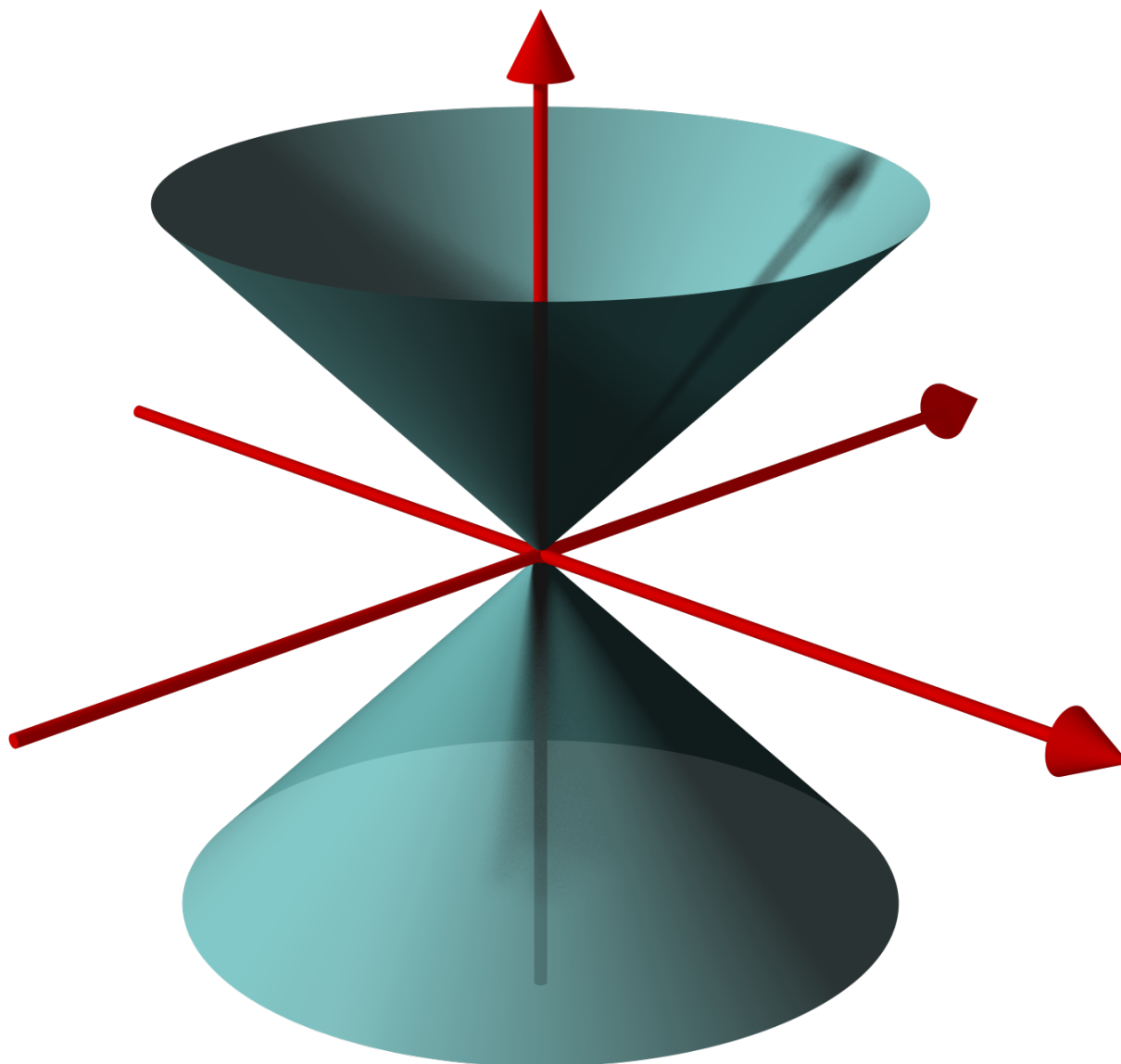


Figure 45: The standard cone

The *standard cone* in three-dimensional affine space is given by the homogeneous equation

$$Z^2 = X^2 + Y^2.$$

One can picture this by thinking of z as specifying the radius of a circle (*edition clarification*: over $\mathbb{R}$, the radius is $|z|$, not a negative z) in the plane parallel to the x - y plane through the point $(0, 0, z)$. Every intersection of this cone with an affine plane E is called a *conic section*.

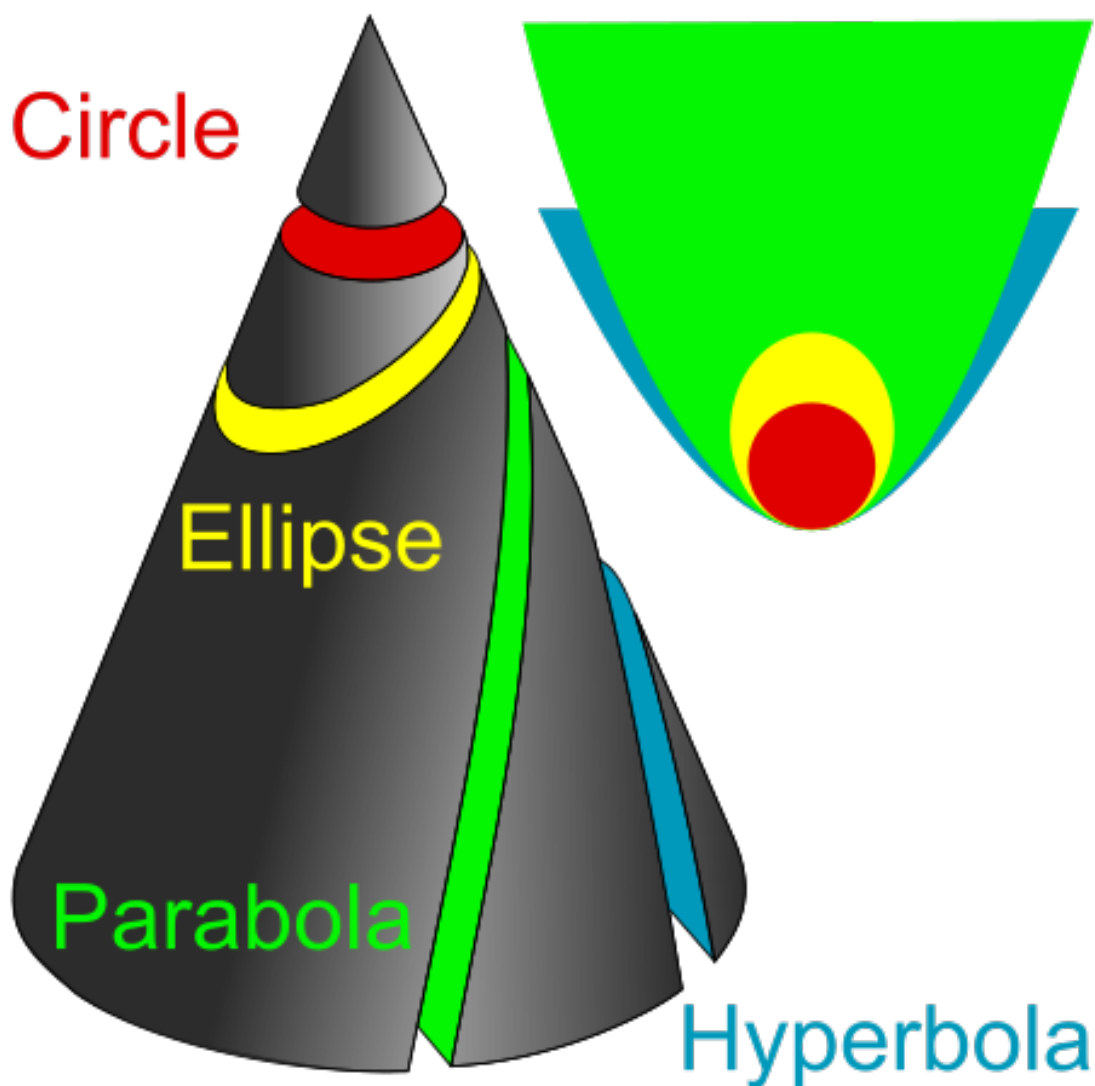


Figure 46: Sections of the standard cone by affine planes

Definition: conic section

A *conic section* C is the intersection of the standard cone $V(Z^2 - X^2 - Y^2)$ with an affine plane $V(aX + bY + cZ + d)$, where a, b, c are not all zero; thus

$$C = V(Z^2 - X^2 - Y^2) \cap V(aX + bY + cZ + d).$$

The theory of conic sections is a classical subject, on which [Apollonius of Perga](#) already wrote a treatise. Since the plane is given by an equation

$$aX + bY + cZ + d = 0,$$

we can solve linearly for one variable and obtain a new equation in two variables for the conic section. This is an affine-linear substitution of variables, so the new equation also has degree two.

We therefore consider *affine quadrics* in two variables in general.

Definition: a quadric in two variables

A polynomial of the form

$$F = \alpha X^2 + \beta XY + \gamma Y^2 + \delta X + \epsilon Y + \eta, \quad \alpha, \beta, \gamma, \delta, \epsilon, \eta \in K,$$

where at least one of the coefficients α, β, γ is nonzero, is called a *quadratic form in two variables* (over K), or a *quadric in two variables*. The corresponding zero locus

$$V(F) \subseteq \mathbb{A}_K^2$$

is also called a *quadric*.

Terminology note. Here the source uses “quadratic form” for a possibly inhomogeneous degree-two polynomial. In the usual homogeneous sense, the quadratic form is only $\alpha X^2 + \beta XY + \gamma Y^2$.

We want to know how many different types of quadrics there are. The answer depends on the ground field. We must also specify which notion of equivalence we wish to use. For two quadrics

$$F, G \in K[X, Y],$$

the following notions of equivalence are worth investigating.

1. F and G are *affinely equivalent* as polynomials: there is a (bijective) affine-linear change of variables

$$\begin{aligned} \varphi : K[X, Y] &\longrightarrow K[X, Y], \\ X &\longmapsto rX + sY + t, \\ Y &\longmapsto \tilde{r}X + \tilde{s}Y + \tilde{t}, \end{aligned}$$

such that $G = \varphi(F)$.

2. The principal ideals (F) and (G) are *affinely equivalent*: there is a (bijective) affine-linear change of variables φ such that

$$(G) = (\varphi(F)).$$

3. The quotient rings

$$K[X, Y]/(F) \quad \text{and} \quad K[X, Y]/(G)$$

are isomorphic as K -algebras.

4. The zero loci $V(F)$ and $V(G)$ are *affine-linearly equivalent*.

The first notion is stronger than the second, and the second is stronger than the last two. An essential difference between (1) and (2) is that in (2) we may always multiply by a unit (which does not change the zero locus either). Over a field that is not algebraically closed, equivalence in (4) can be very coarse, since all F with an empty zero locus are equivalent in the sense of (4).

For $K = \mathbb{R}$ and $K = \mathbb{C}$, we are also interested in whether the corresponding zero loci have the same topological properties. Here we will consider the different notions of equivalence for two quadrics F and G in parallel, but our main interest is in (2).

Lemma: first reduction of affine quadrics

Let K be a field of characteristic $\neq 2$, and let

$$F = \alpha X^2 + \beta XY + \gamma Y^2 + \delta X + \epsilon Y + \eta$$

be a quadric. Then there is a change of variables in the affine plane such that, in the new variables, the transformed polynomial has the form

$$G = \gamma Y^2 + H(X), \quad H(X) = aX^2 + bX + c,$$

with $\gamma \neq 0$. If $a \neq 0$, we can arrange that $b = 0$.

Over an algebraically closed field, we can arrange that $\gamma = 1$ by a change of variables.

If we are interested in the generated ideal or the zero locus, we can also arrange that $\gamma = 1$ by division.

Proof First we reduce to the case $\gamma \neq 0$. If $\gamma = 0$ and $\alpha \neq 0$, we may interchange X and Y . If $\alpha = \gamma = 0$, then $\beta \neq 0$. In this case, the change

$$X \mapsto X + Y, \quad Y \mapsto Y,$$

makes the coefficient of Y^2 nonzero. Henceforth we therefore assume that $\gamma \neq 0$.

We write the polynomial as

$$\gamma Y^2 + (\beta X + \epsilon)Y + \tilde{H}(X),$$

where $\tilde{H}$ is a polynomial in X of degree ≤ 2 . Completing the square gives

$$\gamma \left(Y + \frac{\beta X + \epsilon}{2\gamma} \right)^2 + \tilde{H}(X) - \frac{(\beta X + \epsilon)^2}{4\gamma}.$$

In the new variables

$$Y + \frac{\beta X + \epsilon}{2\gamma} \quad \text{and} \quad X,$$

the equation has the form

$$G = \gamma Y^2 + H(X), \quad H(X) = aX^2 + bX + c.$$

If K is algebraically closed, γ has a square root, so $Y \mapsto Y/\sqrt{\gamma}$ makes the coefficient equal to 1. The other additional assertion is clear. $\square$

Classification of real and complex quadrics

Example: classification of real quadrics

Let $K = \mathbb{R}$. We want to classify real quadrics, mainly with respect to affine-linear equivalence of their generated principal ideals. In other words, we may make affine changes of variables and divide by -1 . By Lemma 7.3, we may assume that the defining equation has the form

$$Y^2 = aX^2 + bX + c.$$

If $a = b = 0$, the change $Y \mapsto \sqrt{c}Y$ for $c > 0$, or $Y \mapsto \sqrt{-c}Y$ for $c < 0$, followed by division by $\pm c$, lets us make the right-hand side equal to 1, -1 , or 0.

If $a = 0$ and $b \neq 0$, we may take $bX + c$ as a new variable and obtain the equation

$$Y^2 = X.$$

Now let $a \neq 0$. The change $X \mapsto X/\sqrt{a}$ or $X \mapsto X/\sqrt{-a}$ lets us arrange that $a = \pm 1$. Completing the square makes $b = 0$. If $c = 0$, we can transform the equation into

$$Y^2 = \pm X^2.$$

So let $c \neq 0$. By the simultaneous change

$$X \mapsto uX, \quad Y \mapsto uY, \quad u = \sqrt{\pm c},$$

followed by division, we can arrange that $c = \pm 1$. The remaining possibilities to consider are therefore

$$Y^2 = \pm X^2 \pm 1,$$

where the two equations

$$Y^2 - X^2 = \pm 1$$

are equivalent to one another.

We now know that every real quadric can be brought into one of the following nine forms.

I. $Y^2 = 0$. This is a *double line*.

II. $Y^2 = 1$. This means $Y = \pm 1$, giving *two parallel lines*.

III. $Y^2 = -1$. This locus is *empty*.

IV. $Y^2 = X$. This is a *parabola*.

V. $Y^2 = X^2$. This means $(Y - X)(Y + X) = 0$, giving *two intersecting lines*.

VI. $Y^2 = -X^2$. The only solution is the *point* $(0, 0)$.

VII. $Y^2 = X^2 + 1$. This means $(Y - X)(Y + X) = 1$, giving a *hyperbola*.

VIII. $Y^2 = -X^2 + 1$. This is a *unit circle*.

IX. $Y^2 = -X^2 - 1$. This locus is again *empty*.

Are these nine types all different from one another? That depends on the notion of equivalence used. Types III and IX are both empty, and thus have identical zero loci. On the other hand, the corresponding quotient rings

$$\mathbb{R}[X, Y]/(Y^2 + 1) \quad \text{and} \quad \mathbb{R}[X, Y]/(X^2 + Y^2 + 1)$$

are not isomorphic, and over the complex numbers their zero loci are not the same. We therefore regard them as different here too. Apart from this exception, the zero loci are usually already different for topological reasons. For example, the unit circle is [compact](#), the hyperbola is noncompact with two connected components, and the parabola is noncompact with one connected component, and so on.

However, the double line and the parabola are the same in the real topology, as are the hyperbola and the two parallel lines. In each pair, the quotient rings differ; in the second pair, the complex versions also differ. For example, $K[X, Y]/(Y^2)$ is not reduced, whereas

$$K[X, Y]/(Y^2 - X) \cong K[Y]$$

is an integral domain. The complex hyperbola is connected because it is isomorphic to

$$\mathbb{C}^\times = \mathbb{C} \setminus \{0\},$$

that is, to the punctured complex line $\mathbb{A}_{\mathbb{C}}^1 \setminus \{0\}$.

The following images show the rotation and translation of a quadric.

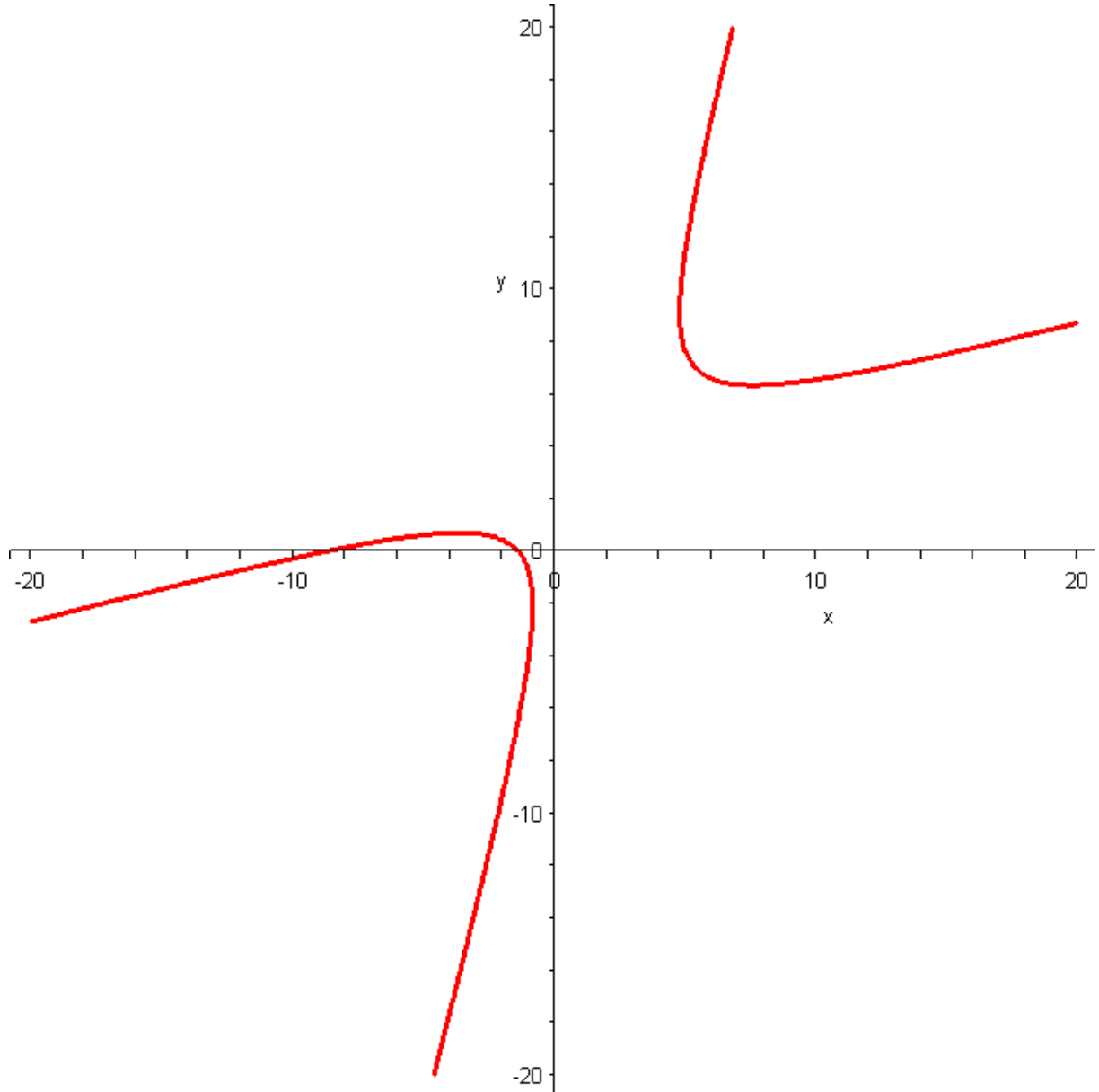


Figure 47: First stage of a principal-axis transformation of a quadric

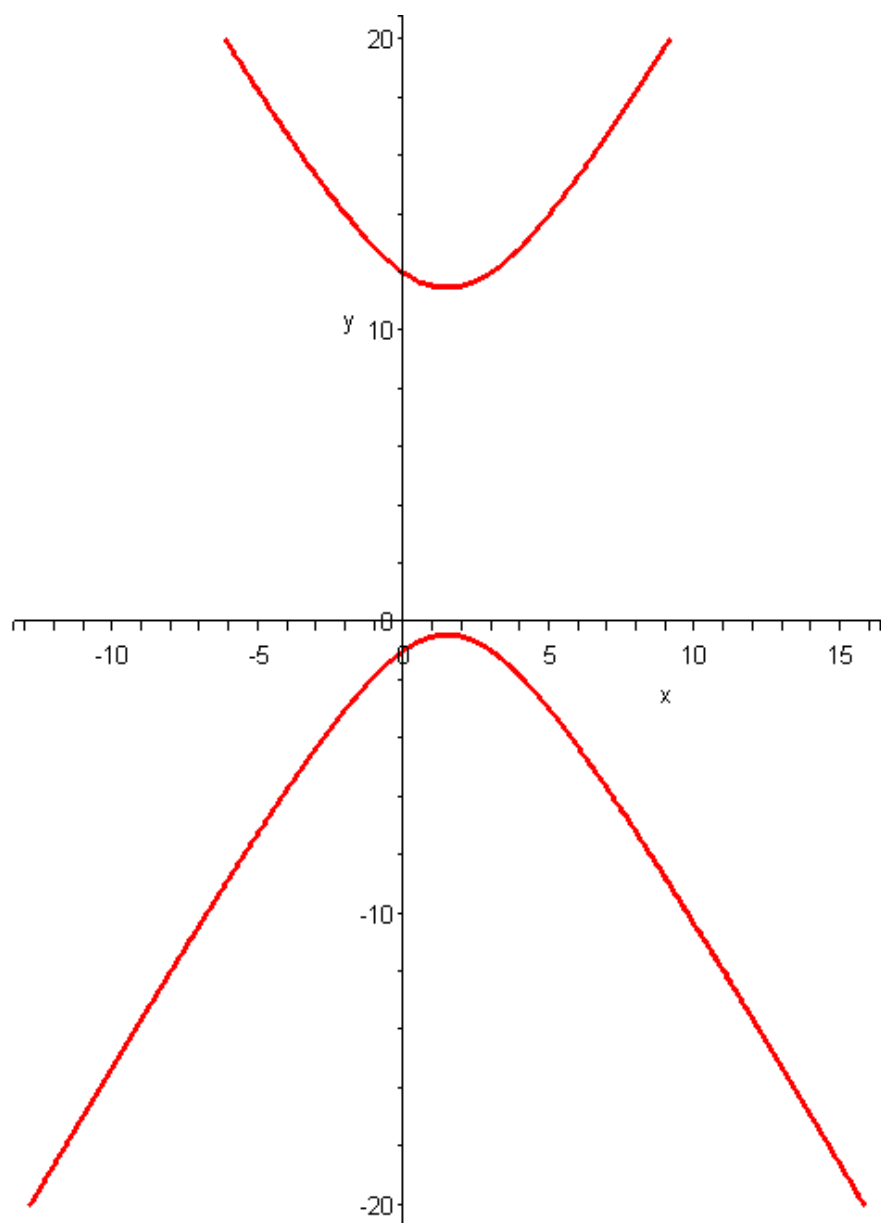


Figure 48: Second stage of a principal-axis transformation of a quadric

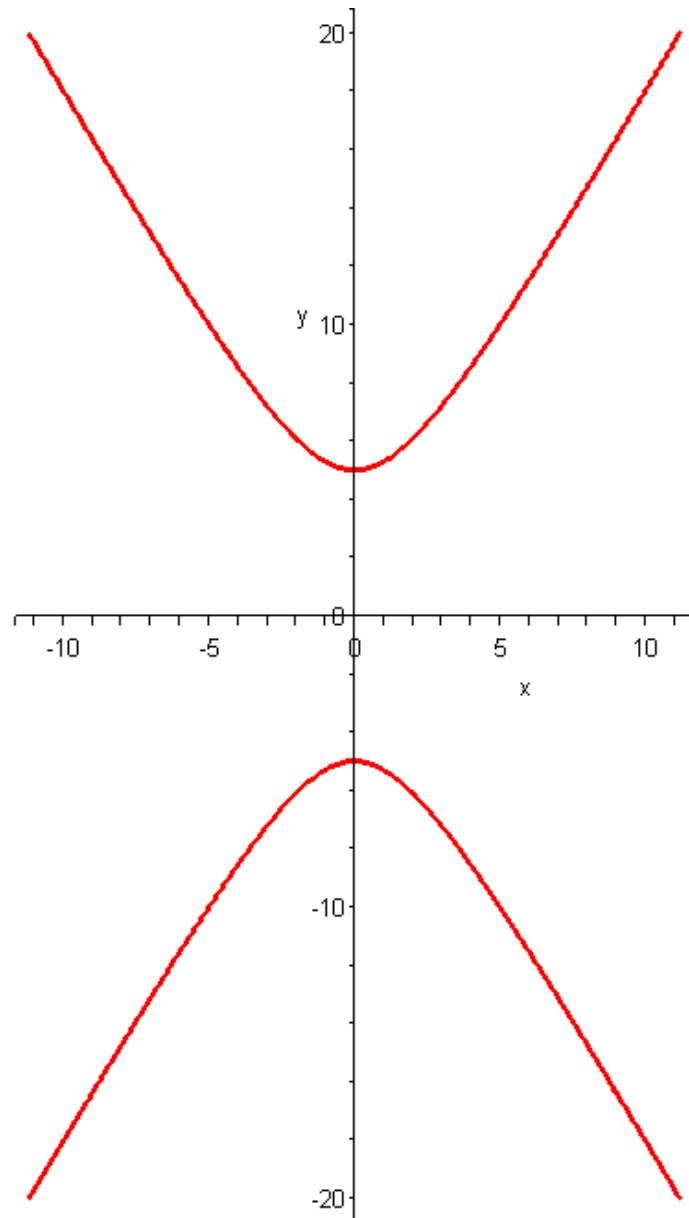


Figure 49: Third stage of a principal-axis transformation of a quadric

Example: classification of complex quadrics

Let $K = \mathbb{C}$. We want to classify complex quadrics. By [Lemma 7.3](#), we may assume that the defining equation has the form

$$Y^2 = aX^2 + bX + c.$$

If $a = b = 0$ and $c = 0$, we retain the equation $Y^2 = 0$. If $a = b = 0$ and $c \neq 0$, scaling the variable Y and dividing by a nonzero constant lets us make the equation $Y^2 = 1$.

If $a = 0$ and $b \neq 0$, we may take $bX + c$ as a new variable and obtain the equation

$$Y^2 = X.$$

Now let $a \neq 0$. The change $X \mapsto X/\sqrt{a}$ lets us arrange that $a = 1$. Completing the square makes $b = 0$. Finally, a simultaneous change $X \mapsto uX$, $Y \mapsto uY$, followed by division, lets us arrange that $c = 1$ if $c \neq 0$; if $c = 0$, the form $Y^2 = X^2$ is retained.

Edition note: In both normalisation steps above, the source uses scaling or division requiring $c \neq 0$ without separating the case $c = 0$. This edition makes the distinction explicit: $c = 0$ gives form I when $a = b = 0$, and form IV, $Y^2 = X^2$, when $a \neq 0$.

We now know that every complex quadric can be brought into one of the following five forms.

I. $Y^2 = 0$. This is a *double line*.

II. $Y^2 = 1$. This means $Y = \pm 1$, giving *two parallel complex lines*.

III. $Y^2 = X$. This is a *complex parabola*.

IV. $Y^2 = X^2$. This means $(Y - X)(Y + X) = 0$, giving *two complex lines* intersecting at one point.

V. $Y^2 = X^2 + 1$. This means $(Y - X)(Y + X) = 1$, giving a *complex hyperbola*.

In the complex topology, Types I and III are a complex affine line, hence a real plane, and therefore topologically the same. Speaking of a “complex plane” is dangerous in algebraic geometry, since it may mean $\mathbb{C}$ or $\mathbb{C}^2$. Their quotient rings differ, however, so they are listed as distinct types. Apart from that pair, all types differ in the complex topology. Besides the real plane, we have the punctured complex affine line (the hyperbola, topologically a punctured real plane), two disjoint lines, and two lines intersecting at a point.

The classification of complex quadrics in the last example holds over every algebraically closed field of characteristic $\neq 2$.

Parametrisation of quadrics

In elementary number theory, we learn how to obtain all Pythagorean triples systematically. The reason is that the unit circle has a parametrisation by rational functions. Generalising [Exercise 1.28](#), we now show that every irreducible quadric can be parametrised rationally.

Theorem: rational parametrisation of quadrics

Let

$$C = V(F)$$

be a quadric in two variables, that is,

$$F = \alpha X^2 + \beta XY + \gamma Y^2 + \delta X + \epsilon Y + \eta,$$

with α, β, γ not all zero. Suppose that there is at least one point on the quadric. Then there are polynomials

$$P_1, P_2, Q \in K[T], \quad Q \neq 0,$$

such that the image of the rational map

$$\begin{aligned} \mathbb{A}_K^1 \supseteq D(Q) &\longrightarrow \mathbb{A}_K^2, \\ t &\longmapsto \left(\frac{P_1(t)}{Q(t)}, \frac{P_2(t)}{Q(t)} \right) \end{aligned}$$

lies in C .

If C has at least two points, the map is nonconstant and injective apart from finitely many exceptions.

Edition note. Over a finite field, “nonconstant” here refers to the pair of rational functions in $K(T)$, not necessarily to the induced function on the finite set $D(Q)(K)$. As the source’s final remark explains, that set can even be empty.

If C is also [irreducible](#), the map is surjective apart from finitely many exceptions. In particular, an irreducible quadric with at least two points is a [rational curve](#).

Proof By a change of variables, we can arrange that $\alpha \neq 0$. We can then divide by α and assume that $\alpha = 1$. By translation, we may assume that the origin $0 = (0, 0)$ lies on the curve. Then $\eta = 0$. If the quadric consists of two intersecting lines, we can translate so that the origin is not their intersection point (but still lies on one of the lines).

The idea is, for a point

$$H = (t, 1),$$

to consider the line through 0 and H and its intersection with C . This intersection consists of at most two points (unless it is the whole line). Since 0 is one of those points, the other point that must exist is uniquely determined.

So let $H = (t, 1)$ be given. The line through H and 0 consists of all points

$$(at, a), \quad a \in K.$$

Its intersection points with C are obtained by substituting $(x, y) = (at, a)$ into F and solving for a . Substitution gives the condition

$$F(at, a) = (at)^2 + \beta(at a) + \gamma a^2 + \delta at + \epsilon a.$$

The solution $a = 0$ corresponds to the origin, which we already know. The second solution is

$$a_2 = \frac{-\delta t - \epsilon}{t^2 + \beta t + \gamma}.$$

This expression is defined if

$$Q(t) = t^2 + \beta t + \gamma \neq 0,$$

which excludes at most two values of t . The point on C corresponding to a_2 is

$$\begin{aligned} a_2(t, 1) &= (a_2 t, a_2) \\ &= \left(t \frac{-\delta t - \epsilon}{t^2 + \beta t + \gamma}, \frac{-\delta t - \epsilon}{t^2 + \beta t + \gamma} \right). \end{aligned}$$

We must therefore set

$$P_1 = -t(\delta t + \epsilon), \quad P_2 = -\delta t - \epsilon.$$

This map is well-defined on the Zariski-open set $D(Q)$ (and that set is nonempty as soon as the field has at least three elements).

From now on, suppose that C has at least two points. If $\delta = \epsilon = 0$, then F has the form

$$F = X^2 + \beta XY + \gamma Y^2.$$

Since we assumed that C has at least two points, F is a product of two homogeneous linear forms (monic in X). If F is the square of a linear form, geometrically we simply have a “double line”, which can be parametrised bijectively directly. Otherwise, F is the product of two distinct homogeneous linear forms, and both corresponding lines pass through the origin, which we have excluded. Thus in this case δ and ϵ cannot both be 0.

We therefore need only consider the situation in which $\delta t + \epsilon$ is not the zero polynomial. It follows that the map on its domain of definition is injective apart from finitely many exceptions, since if $\delta t + \epsilon \neq 0$, the preimage t can be reconstructed from the image using

$$t = \frac{P_1}{Q} \cdot \frac{Q}{P_2}.$$

To show that the map is surjective apart from finitely many exceptions, we need the assumption that C is irreducible. In particular, this means that C is not the union of two lines. Let $P \in C$ have nonzero y -coordinate (there are at most two points with zero y -coordinate). Then the line through P and 0 intersects the parametrising line $V(Y - 1)$ at a point

$$H = (t, 1).$$

Apart from finitely many values of t , the map is defined at this point H , and P is then its image point. By irreducibility, only finitely many points of C lie on the exceptional lines; thus almost all points are reached. $\square$



Figure 50: Portrait of an unidentified man, formerly misidentified as Johannes Kepler

Translator’s note: The frozen source displays the file Johannes Kepler 1610 . jpg with a caption identifying the sitter as Kepler. The available Commons file is now titled PortraitConfusedWithJohannesKepler1610.jpg and identifies the sitter as an unidentified man formerly misidentified as Kepler. The local asset name and caption have been adjusted transparently.

The nonsingular conic sections are also the trajectories of celestial bodies. The possible celestial trajectories were first described by [Johannes Kepler](#). The underlying law states that at each instant, acceleration is proportional to the gravitational force between the central point mass (the star, the Sun) and the moving point mass (the planet, the comet). The attractive force itself depends on the two masses and the square of their distance. There are “bound” orbits (ellipses) and “unbound” orbits (parabolas, hyperbolas).

A circle and an ellipse can be transformed into one another by a linear change of variables. Note that rational parametrisations are not “physical parametrisations”. The latter truly describe the motion: the parameter is time, and the derivative at a given time is the instantaneous velocity. Rational parametrisations “only” describe the trajectory. As is well known, the circle is traversed uniformly (at constant speed) by

$$(x, y) = (\cos t, \sin t).$$

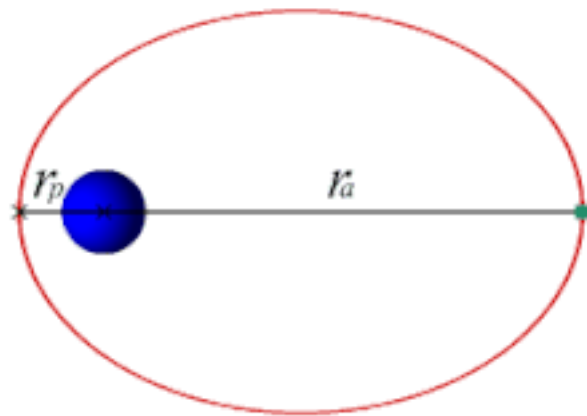


Figure 51: Elliptic orbit

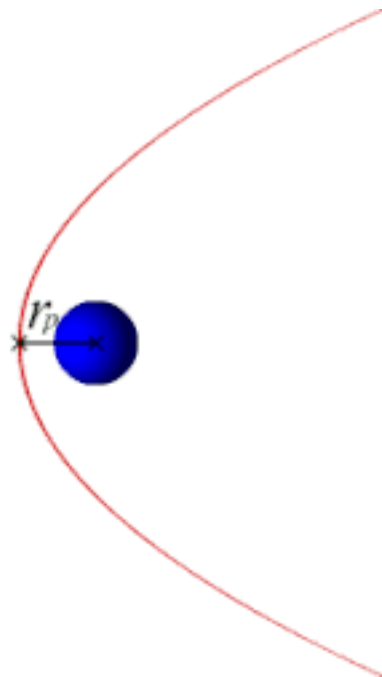


Figure 52: Parabolic orbit

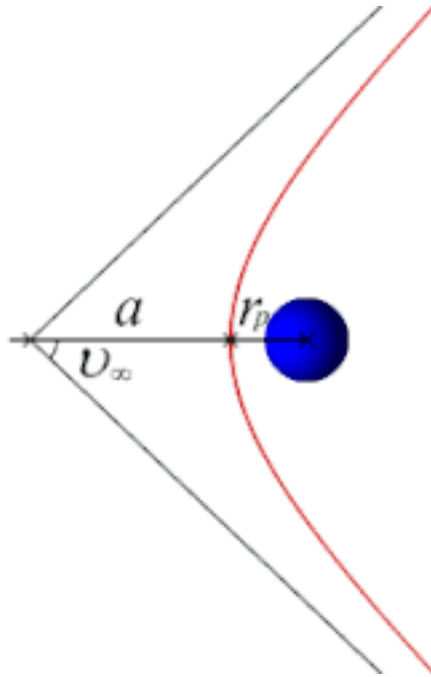


Figure 53: Hyperbolic orbit

Remark: the domain of a quadric parametrisation

The parametrisation of a quadric does not depend on the ground field, since the expressions defining the map are always the same. Over a finite field, however, the domain of definition of a rational map can be empty. Passing to a larger finite field $\mathbb{F}_q$ always gives the map a nonempty domain of definition.

Geometrically, the gaps in the domain of the parametrisation arise because the connecting lines constructed in the proof of [Theorem 7.6](#) have no other intersection with the quadric besides the origin; or, conversely, the entire line lies on the quadric (which can happen only in the reducible case or for a double line). The exceptional points of the quadric that do not lie in the image are the points on the x -axis (in particular the origin) and, in the reducible case, the points on the line lying entirely on the quadric and passing through the origin.

Worksheet 7

Practice exercises

Exercise 7.1

Determine the plane sections of a sphere. Which of them are conic sections?

Exercise 7.2

Determine the plane sections of a cylinder. Which of them are conic sections?

Exercise 7.3

Determine the conic sections obtained by intersecting with a plane through the origin.

Exercise 7.4

Show that parallel planes, neither of which passes through the origin, define the same type of conic section.

Exercise 7.5

Consider the standard cone

$$V(Z^2 - X^2 - Y^2) \subset \mathbb{A}_{\mathbb{R}}^3$$

and the planes through the axis of rotation $V(X, Z)$. Determine, as a function of the rotation angle α (measured anticlockwise in the x - z plane), the type of conic section given by the plane E_α .

Exercise 7.6

Consider the standard cone

$$V(Z^2 - X^2 - Y^2) \subset \mathbb{A}_{\mathbb{R}}^3$$

and the planes through the axis of rotation $V(X - 1, Z)$. Determine, as a function of the rotation angle α (measured anticlockwise in the x - z plane), the type of conic section given by the plane E_α .

Exercise 7.7

Consider the standard cone

$$V(Z^2 - X^2 - Y^2) \subset \mathbb{A}_{\mathbb{R}}^3$$

and the planes through the axis of rotation $V(X - 1, Z - 1)$. Determine, as a function of the rotation angle α (measured anticlockwise in the x - z plane), the type of conic section given by the plane E_α .

Exercise 7.8

Determine the quadrics associated with homogeneous quadratic polynomials in two variables.

Exercise 7.9

Transform the quadric

$$2x^2 + xy - 3y^2 + 5x - y + 3$$

into a real standard form.

Exercise 7.10 ★

Consider the two quadrics

$$X^2 + Y^2 = 1 \quad \text{and} \quad 4X^2 + 3Y^2 = 9.$$

Show that they are affine-linearly equivalent over $\mathbb{R}$, but not over $\mathbb{Q}$.

Hint. One possible argument follows from [Theorem 7.2 \(Measure and Integration Theory, Osnabrück 2022-2023\)](#).

Edition note: The source calls both loci “circles”, but $4X^2 + 3Y^2 = 9$ is an ellipse, not a circle. The requested equivalence statement is unchanged; the wording of the question has been corrected to “quadrics”.

Exercise 7.11 ★

Let

$$F = (0, 0)$$

be the origin in the real plane, and let

$$G = V(X - 1).$$

Let $e > 0$ be a real number. Determine an algebraic equation for the set of points $P = (x, y)$ such that the distance $d(P, F)$ is proportional, with proportionality factor $\sqrt{e}$, to the perpendicular distance $d(P, G)$.

By transforming the equation appropriately, show that it gives an ellipse for $e < 1$, a parabola for $e = 1$, and a hyperbola for $e > 1$.

Exercise 7.12

Parametrise the quadric defined by

$$F = 2x^2 - xy + 3y^2 + x - 5y$$

using the origin and the line $V(y - 1)$.

Exercise 7.13

Consider the algebraic curve

$$C = V(X^{d+1} - Y^d), \quad d \geq 1.$$

Show that the origin and the line $V(X - 1)$ give a parametrisation of C by the method described in the proof of [Theorem 7.6](#).

In the following exercises, we discuss a notion of automorphism that goes beyond affine-linear changes of coordinates.

Let K be a field. A polynomial map

$$\varphi : \mathbb{A}_K^n \longrightarrow \mathbb{A}_K^n$$

is called a **(polynomial) automorphism of affine space** if it has a polynomial inverse.

An automorphism of affine space is the same as a K -algebra automorphism of the polynomial ring $K[X_1, \dots, X_n]$ to itself. It is given by n polynomials in n variables.

Edition note. For the stated equivalence with polynomial-ring automorphisms, the polynomial inverse must satisfy the composition identities as polynomial identities. Over a finite field, pointwise identities on K^n alone do not suffice.

Exercise 7.14

Show that a K -algebra automorphism

$$\varphi : K[X] \longrightarrow K[X]$$

is given by

$$X \mapsto aX + b, \quad a \neq 0,$$

that is, by an affine-linear change of variables.

Exercise 7.15

Give an example of a bijective polynomial map

$$\mathbb{A}_K^1 \longrightarrow \mathbb{A}_K^1$$

whose inverse is not polynomial.

Edition note: The existence of such an example depends on K . For example, for $K = \mathbb{R}$, the map $x \mapsto x^3$ is bijective, while its inverse $x \mapsto \sqrt[3]{x}$ is not polynomial. Over $\mathbb{C}$, every bijective polynomial map of the affine line has degree one.

Exercise 7.16

Let $F \in K[X]$ be a polynomial. Show that the map

$$\begin{aligned} \mathbb{A}_K^2 &\longrightarrow \mathbb{A}_K^2, \\ (x, y) &\longmapsto (x, y + F(x)) \end{aligned}$$

is an automorphism of affine space. Determine an inverse explicitly.

Exercise 7.17

Determine the inverse of the map

$$\begin{aligned} \mathbb{R}^2 &\longrightarrow \mathbb{R}^2, \\ (x, y) &\longmapsto (x + y^2, -y^4 - 2xy^2 - x^2 + y^2 + x + y). \end{aligned}$$

Exercise 7.18

Let

$$\varphi : \mathbb{A}_{\mathbb{C}}^n \longrightarrow \mathbb{A}_{\mathbb{C}}^n$$

be an automorphism of affine space. Show that the Jacobian determinant of φ is constant and equal to some $c \neq 0$.

The **Jacobian problem** asks whether a polynomial map

$$\varphi : \mathbb{A}_{\mathbb{C}}^n \longrightarrow \mathbb{A}_{\mathbb{C}}^n,$$

whose Jacobian determinant is constantly equal to 1, has a polynomial inverse, that is, is an automorphism of affine space. This problem is open even for $n = 2$. By the inverse function theorem, under the given assumption there is a local differentiable inverse at every point.

Two affine algebraic sets

$$V, \widetilde{V} \subseteq \mathbb{A}_K^n$$

are called **affine-algebraically equivalent** if there is an automorphism of affine space

$$\varphi : \mathbb{A}_K^n \longrightarrow \mathbb{A}_K^n$$

such that

$$\varphi^{-1}(V) = \widetilde{V}.$$

Exercise 7.19

Let $F \in K[X]$ be a polynomial and let

$$C = V(Y - F(X)) \subset \mathbb{A}_K^2$$

be its graph. Show that C is affine-algebraically equivalent to the x -axis

$$V(Y) \subset \mathbb{A}_K^2,$$

but in general is not affine-linearly equivalent to it.

Exercise 7.20

Let

$$V, \widetilde{V} \subset \mathbb{A}_K^n$$

be two affine-algebraically equivalent affine algebraic sets, with vanishing ideals $\text{Id}(V)$ and $\text{Id}(\widetilde{V})$. Show that the quotient rings

$$K[X_1, \dots, X_n]/\text{Id}(V)$$

and

$$K[X_1, \dots, X_n]/\text{Id}(\widetilde{V})$$

are isomorphic.

Exercise 7.21

Which quadrics in $\mathbb{R}^2$ (in $\mathbb{C}^2$) are affine-algebraically equivalent to one another?

Exercise 7.22 ★

For the rational quadric

$$C = V(X^2 + Y^2 - 5) \subset \mathbb{A}_{\mathbb{Q}}^2,$$

carry out a rational parametrisation in the sense of [Theorem 7.6](#), using the auxiliary point $(1, 2)$ and a suitable line.

Exercise 7.23

Let p be a prime number with

$$p \bmod 4 = 3.$$

Referring to [Theorem 9.10 \(Number Theory, Osnabrück 2025\)](#), explain why the rational quadric

$$C = V(X^2 + Y^2 - p) \subset \mathbb{A}_{\mathbb{Q}}^2$$

is empty.

Exercise 7.24

Let R be an integral domain and $r \in R$ an element with no square root in R . Show that the polynomial

$$X^2 - r \in R[X]$$

is irreducible.

Exercise 7.25

Let R be an integral domain with $2 \neq 0$, and let $r \in R$ have no square root in R . Consider the quadratic ring extension

$$R \subset R[X]/(X^2 - r) =: S.$$

Show that the elements $f \in R$ having a square root in S have the form

$$f = y^2$$

with $y \in R$, or the form

$$f = rz^2$$

with $z \in R$.

Exercise 7.26

Let p and q be distinct prime numbers. Show that the quadrics defined over $\mathbb{Q}$,

$$C = V(X^2 + Y^2 - p), \quad D = V(X^2 + Y^2 - q) \subset \mathbb{A}_{\mathbb{Q}}^2,$$

are not affine-linearly equivalent.

Edition note: The universal claim in the source is false as written. For $p = 2$ and $q = 5$, the invertible rational transformation

$$(x, y) \mapsto \left(\frac{3x - y}{2}, \frac{x + 3y}{2} \right)$$

multiplies $x^2 + y^2$ by $5/2$ and therefore maps C to D . Thus this proof exercise requires additional hypotheses.

Source hint, with corrected ring notation. Apply [Exercise 7.25](#) to

$$R = \mathbb{Q}[Y]$$

and

$$S = (\mathbb{Q}[Y])[X]/(X^2 + Y^2 - p).$$

Edition note: The frozen hint replaces $\mathbb{Q}$ by K without explanation and writes $X^2 - Y^2 + p$, which is not the coordinate equation of C . The notation corrected above identifies S with the coordinate ring of C ; it does not remove the counterexample to the universal claim.

Exercises for submission**Exercise 7.27 - 6 points**

For the [various real quadrics](#), find a realisation as a conic section, that is, as the intersection of a plane with the cone

$$V(x^2 + y^2 - z^2),$$

or prove that no such realisation exists.

Exercise 7.28 - 9 points

Consider the map

$$\begin{aligned}\mathbb{A}_K^2 &\longrightarrow \mathbb{A}_K^2, \\ (u, v) &\longmapsto (u^2 + uv, v - u^2) = (x, y).\end{aligned}$$

For each of the following three families of parallel lines, determine the image curve of every line under this map; give both a parametrisation and a curve equation.

1. Lines parallel to the u -axis.
2. Lines parallel to the v -axis.
3. Lines parallel to the antidiagonal.

Also sketch typical image curves for $K = \mathbb{R}$. For each family, determine whether its image curves intersect one another.

Exercise 7.29 - 4 points

Let

$$F, G \in K[X_1, \dots, X_n]$$

be polynomials and $K \subseteq L$ a field extension. Discuss how the different notions of equivalence from Lecture 7 for F and G (and for $V(F)$ and $V(G)$) behave under this field extension.

Exercise 7.30 - 6 points

Consider the two quotient rings

$$R = \mathbb{R}[X, Y]/(X^2 + Y^2 - 1) \quad \text{and} \quad S = \mathbb{R}[X, Y]/(XY - 1).$$

Show that S is a principal ideal domain, whereas R is not.

(These are the rings associated with the real circle and the real hyperbola.)

Hint. For R , consider the ideal $(X - 1, Y)$.

Exercise 7.31 - 4 points

Parametrise the quadric

$$C = V(F), \quad F = x^2 + 2xy - y^2 + x - 3y + 4,$$

using the point $(1, 2) \in C$ and the y -axis. Do not make a change of variables.

Exercise 7.32 - 6 points

Consider the two zero loci in $\mathbb{A}_{\mathbb{R}}^2$

$$K = V(X^2 + Y^2 - 1) \quad \text{and} \quad C = V(X^4 + Y^4 - 1).$$

Show that there is a polynomial map in two variables mapping one of these zero loci surjectively onto the other. Show that this map is already defined over $\mathbb{Q}$, but is not surjective there. Show further that over $\mathbb{Q}$ there is no surjective polynomial map at all from C to K , and that the only polynomial maps from K to C are constant.

Exercise 7.33 - 6 points

Let K be an algebraically closed field and $F \in K[X, Y]$ an irreducible polynomial. Show that the curve $V(F)$ is rational if and only if there is an injective K -algebra homomorphism

$$Q(K[X, Y]/(F)) \longrightarrow K(T).$$

Here the left-hand side is the fraction field and the right-hand side is the rational function field.

Source navigation: [Lecture 7 - public solutions for Unit 7](#) - [Worksheet 6](#) - [Worksheet 8 \(source\)](#)

Public Solutions to Worksheet 7

At the frozen revision boundary, the source provides public solutions only for Exercises 7.10, 7.11, and 7.22. No additional solutions have been created for this edition.

Solution to Exercise 7.10

The equation

$$4X^2 + 3Y^2 = 9$$

is equivalent to

$$\left(\frac{2}{3}X\right)^2 + \left(\frac{1}{\sqrt{3}}Y\right)^2 = 1 = \widetilde{X}^2 + \widetilde{Y}^2.$$

Thus, over $\mathbb{R}$,

$$\widetilde{X} = \frac{2}{3}X, \quad \widetilde{Y} = \frac{1}{\sqrt{3}}Y$$

is an affine-linear transformation.

For the case $\mathbb{Q}$, we set

$$\widetilde{X} = aX + bY + c$$

and

$$\widetilde{Y} = dX + eY + f$$

with coefficients $a, b, c, d, e, f \in \mathbb{Q}$. We obtain

$$\begin{aligned} \widetilde{X}^2 + \widetilde{Y}^2 &= (aX + bY + c)^2 + (dX + eY + f)^2 \\ &= (a^2 + d^2)X^2 + (b^2 + e^2)Y^2 + 2(ab + de)XY + H(X, Y), \end{aligned}$$

where $H \in \mathbb{Q}[X, Y]$ and $\deg H \leq 1$.

Edition note: The source immediately compares coefficients at this step. The justification is that an affine equivalence maps the unique centre of either quadric to the unique centre of the other, so $c = f = 0$. After normalising the pullback equation to have constant term -1 , the two quadratic polynomials defining the same locus coincide; comparing the coefficients of Y^2 then gives the following equation.

With this justification, we must have

$$b^2 + e^2 = \frac{1}{3} \iff 3(b^2 + e^2) = 1.$$

Clearing denominators puts the equation in the form

$$3(r^2 + s^2) = t^2,$$

with $r, s, t \in \mathbb{Z}$. We will show that this equation has no nontrivial integer solution. Since the left-hand side is a multiple of 3, we obtain $3 \mid t$, so $9 \mid t^2$. Consequently, $3 \mid (r^2 + s^2)$. In $\mathbb{Z}/(3)$, the equation $r^2 + s^2 = 0$ holds exactly when $r = 0$ and $s = 0$. Hence $9 \mid (r^2 + s^2)$, and both sides of the equation can be divided by 9. Now set

$$r' = \frac{r}{3}, \quad s' = \frac{s}{3}, \quad t' = \frac{t}{3}.$$

An infinite descent completes the proof.

[Back to Exercise 7.10.](#)

Solution to Exercise 7.11

The distance from $P = (x, y)$ to the origin is $\sqrt{x^2 + y^2}$, while the perpendicular distance to the line $x = 1$ is $|x - 1|$. The proportionality is expressed by

$$\frac{d(P, F)}{d(P, G)} = \sqrt{e}.$$

Thus

$$\sqrt{x^2 + y^2} = \sqrt{e} |x - 1| \quad \text{or, equivalently,} \quad x^2 + y^2 = e(x - 1)^2.$$

Consequently,

$$(1 - e)x^2 + y^2 + 2ex - e = 0$$

is an algebraic equation for a curve containing every point satisfying the condition. If $e = 1$, the equation becomes

$$y^2 + 2ex - e = 0 \quad \text{or} \quad x = -\frac{1}{2e}y^2 + \frac{1}{2},$$

so the curve is a parabola in this case. Henceforth let $e \neq 1$. The general equation can be rewritten as

$$x^2 + \frac{1}{1-e}y^2 + \frac{2e}{1-e}x - \frac{e}{1-e} = 0$$

and, by completing the square, brought into the form

$$\left(x + \frac{e}{1-e}\right)^2 + \frac{1}{1-e}y^2 - \frac{e^2}{(1-e)^2} - \frac{e}{1-e} = 0.$$

We write this as

$$\begin{aligned} \left(x + \frac{e}{1-e}\right)^2 + \frac{1}{1-e}y^2 &= \frac{e^2}{(1-e)^2} + \frac{e}{1-e} \\ &= \frac{e^2 - e^2 + e}{(1-e)^2} \\ &=: c > 0. \end{aligned}$$

The factor $\frac{1}{1-e}$ is positive for $e < 1$ and negative for $e > 1$. In the first case, a change of coordinates gives an equation of the form

$$\tilde{x}^2 + \tilde{y}^2 = c,$$

that is, an ellipse. In the second case, we obtain

$$\tilde{x}^2 - \tilde{y}^2 = c,$$

that is, a hyperbola.

[Back to Exercise 7.11.](#)

Solution to Exercise 7.22

We translate the point $(1, 2)$ to the origin by introducing new variables

$$U = X - 1$$

and

$$V = Y - 2.$$

The equation then becomes

$$\begin{aligned} X^2 + Y^2 - 5 &= (U + 1)^2 + (V + 2)^2 - 5 \\ &= U^2 + V^2 + 2U + 4V. \end{aligned}$$

Write the translated curve as

$$\tilde{C} = V(U^2 + V^2 + 2U + 4V).$$

The parametrisation formulas using the line $V = 1$ give

$$P_1 = -t(2t + 4),$$

$$P_2 = -(2t + 4),$$

and

$$Q = t^2 + 1.$$

The parametrisation is therefore given by

$$\mathbb{Q} \longrightarrow \tilde{C} \subset \mathbb{A}_{\mathbb{Q}}^2, \quad t \longmapsto \left(\frac{-t(2t + 4)}{t^2 + 1}, \frac{-(2t + 4)}{t^2 + 1} \right).$$

Edition note: The source denotes the intermediate curve in coordinates (U, V) by the same symbol C as the original curve. The symbol $\tilde{C}$ is used here to distinguish the translation step from the formulas after translating back.

This gives a parametrisation for the original equation:

$$X = -t \frac{2t+4}{t^2+1} + 1 = \frac{-t^2 - 4t + 1}{t^2 + 1}$$

and

$$Y = \frac{-(2t+4)}{t^2+1} + 2 = \frac{2t^2 - 2t - 2}{t^2 + 1}.$$

Back to Exercise 7.22.

Lecture 8: Mechanically Defined Algebraic Curves

Mechanically defined algebraic curves

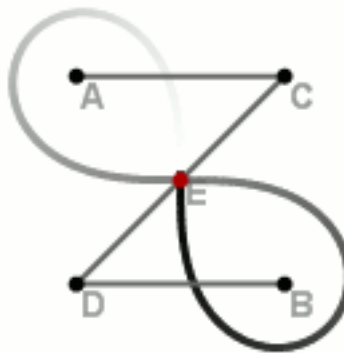


Figure 54: Construction of a lemniscate

Let S be a rigid rod (think of a mechanical machine component) with two fixed points

$$P_1, P_2 \in S$$

(think of joints). This rod can move in the plane (that is, $\mathbb{R}^2$), subject to the condition that the two points remain on prescribed paths B_1 and B_2 , respectively (think of rails). These paths can be quite simple, for example lines or circles. In a steam engine, a rotating wheel and a straight rail are coupled by a rod. How do we describe the resulting motion? What are the *allowed configurations of the system*? Since a configuration is determined by the positions of the two points, each specified by two plane coordinates, this is altogether a four-dimensional situation.

If we fix a point P on the rod (for example, by marking it in colour), what does the *path of motion* (or *trajectory*) of that point in the plane look like?

In the extreme cases

$$P = P_1 \quad \text{and} \quad P = P_2,$$

the trajectories are (usually proper) subsets of B_1 and B_2 . For points between them, we expect a *continuous deformation* of one path into the other.

Situation: a mechanical rod linkage

Let B_1 and B_2 be two plane algebraic curves described by the equations $F_1 = 0$ and $F_2 = 0$, with

$$F_1, F_2 \in K[X, Y].$$

Let S be a “moving line” (a rod) with two points

$$P_1, P_2 \in S, \quad P_1 \neq P_2,$$

at distance d from one another. The *mechanical system* given by all positions of S in the plane satisfying both

$$P_1 \in B_1 \quad \text{and} \quad P_2 \in B_2$$

is described as follows.

A position of the rod in the plane is uniquely determined once the positions of its two points are specified (this does not yet account for the distance condition), hence by four variables

$$(P_1, P_2) = (x_1, y_1, x_2, y_2).$$

An *allowed configuration* must satisfy the following three algebraic conditions.

1. $F_1(x_1, y_1) = 0$.
2. $F_2(x_2, y_2) = 0$.
3. $(x_2 - x_1)^2 + (y_2 - y_1)^2 = d^2$ (the distance condition).

Thus we have three algebraic equations in four variables, so we expect the solution set to be a curve in $\mathbb{A}_K^4$. A point

$$P \in S$$

is described by its distance from P_1 or P_2 . Since these points move in the mechanical system, we specify the *co-moving point* P by

$$P = P_1 + u(P_2 - P_1)$$

(so the distance of P from P_1 is $\|u(P_2 - P_1)\| = |ud|$), and write its coordinates as

$$\begin{aligned} (x, y) &= (x_1, y_1) + u(x_2 - x_1, y_2 - y_1) \\ &= (ux_2 + (1 - u)x_1, uy_2 + (1 - u)y_1). \end{aligned}$$

The entire mechanical system can then be expressed (by a linear transformation) in the four variables x_1, y_1, x, y . For $u \neq 0$, substitute

$$x_2 = \frac{x - (1 - u)x_1}{u} \quad \text{and} \quad y_2 = \frac{y - (1 - u)y_1}{u}$$

into the equations. In the new variables, we obtain the three equations

$$\begin{aligned}
F_1(x_1, y_1) &= 0, \\
F_2\left(\frac{x - (1-u)x_1}{u}, \frac{y - (1-u)y_1}{u}\right) &= 0, \\
(x - x_1)^2 + (y - y_1)^2 &= u^2 d^2.
\end{aligned}$$

In principle, the trajectory corresponding to P can be obtained by “eliminating” the variables x_1 and y_1 from this system, yielding an algebraic equation for x and y . This is easier said than done, however; it is often more useful to simplify the system by skilful manipulation.

Edition clarification. Elimination gives polynomial equations satisfied by the trajectory, not necessarily an exact description of its real points. Real projections can require inequalities; the line-segment trajectories in the two-line example below illustrate this distinction.

Remark: a co-moving plane

Sometimes we are also interested in the situation where an entire plane moves with the rod, and in the trajectories of points in that plane. This happens, for example, when further machine components are mounted on the rod. In that case, any point of the plane can be expressed relative to P_1 and P_2 as

$$(x, y) = (x_1, y_1) + u(x_2 - x_1, y_2 - y_1) + v(y_2 - y_1, -x_2 + x_1).$$

Thus P_1 is taken as the origin of the moving plane, the line connecting it to P_2 as the first coordinate axis, and the perpendicular axis as the second coordinate axis.

The whole mechanical rod system is therefore described by four variables with three equations. Its visible operation, however—the motion of a fixed point P on S —gives a trajectory in the affine plane.

We consider some examples.

Two lines as paths

Example: two lines as paths

Let L_1 and L_2 be two lines in the real plane $\mathbb{R}^2$, and let S be a moving line (a rod) with two points P_1, P_2 at distance d from one another. The allowed configurations of the system are the positions of S satisfying both

$$P_1 \in L_1 \quad \text{and} \quad P_2 \in L_2.$$

Let the lines be specified by

$$L_1 = \{(x, y) \mid a_1 x + b_1 y = c_1\}$$

and

$$L_2 = \{(x, y) \mid a_2 x + b_2 y = c_2\}.$$

By **Situation 8.1**, the allowed configurations are specified by three conditions

$$\begin{aligned}
a_1 x_1 + b_1 y_1 &= c_1, \\
a_2 x_2 + b_2 y_2 &= c_2, \\
(x_2 - x_1)^2 + (y_2 - y_1)^2 &= d^2.
\end{aligned}$$

The solution set of each linear equation is a three-dimensional affine subspace. The solution set of the third equation can be viewed as the product of a circle (in the variables $x_2 - x_1$ and $y_2 - y_1$) with an affine plane. This is a kind of cylinder, although its fibres are two-dimensional. How can we describe their common zero locus, and what trajectory does the mechanical system produce for a point

$$P \in S?$$

By a change of variables, we may assume that the first line is the x -axis, defined by

$$y = 0,$$

while the other line is defined by

$$ax + by = c.$$

For the system, this gives the condition $y_1 = 0$, so y_1 can be eliminated. We then obtain a system with three variables x_1, x_2, y_2 and two conditions

$$\begin{aligned}(x_2 - x_1)^2 + y_2^2 &= d^2, \\ ax_2 + by_2 &= c.\end{aligned}$$

Parallel lines.



Figure 55: Parallel lines

If the second line is parallel to the first, then $a = 0$, and the second equation can be solved for y_2 , giving

$$y_2 = \frac{c}{b} = e$$

(with $b \neq 0$, since otherwise the equation does not define a line). The number e is the distance between the parallel lines.

Edition note. This source wording and the comparisons below assume $e \geq 0$, which can be arranged by reflecting the y -axis coordinate if necessary. Without that convention, the distance is $|e|$ and the three cases compare $|e|$ with d . We can now eliminate y_2 as well, leaving the single equation

$$(x_2 - x_1)^2 + e^2 = d^2,$$

or

$$(x_2 - x_1)^2 = d^2 - e^2 = (d - e)(d + e).$$

If $e > d$, this has no solution (the constant distance between the parallel lines exceeds the coupling distance on the rod).

If $e = d$, we obtain the condition

$$x_1 = x_2.$$

This corresponds to the situation in which the distance between the parallel lines equals the coupling distance. The only allowed configurations are those in which the rod is perpendicular to both lines. The solution set is therefore a line. For each point on the rod, the trajectory is simply another parallel line.

Now let $e < d$. Then

$$x_2 - x_1 = \pm \sqrt{(d - e)(d + e)}.$$

The solution set consists of two disjoint lines. These correspond to the two different ways of attaching the rod, which cannot be transformed into one another. The mechanical system therefore has two connected components. For a point on the rod, however, both attachments give the same trajectory: a parallel line that is, in a sense, traversed twice. Thus the solution set of the complete mechanical system consists of two parallel affine lines in four-dimensional affine space, whereas their trajectories for a fixed point form just one line.

Nonparallel lines.

Now consider the case where the two lines are not parallel. They then intersect, and the solution set cannot be empty. By a further translation (called a linear transformation in the source), we may assume that their intersection is the origin $(0, 0)$. The second equation is then described by

$$x_2 = ey_2.$$

We can therefore eliminate x_2 , obtaining in the two variables x_1, y_2 the single equation

$$(ey_2 - x_1)^2 + y_2^2 = d^2.$$

Thus the configuration space of the mechanical system lies in a plane (defined by $y_1 = 0$ and $x_2 = ey_2$) and is described by a quadric. Taking $ey_2 - x_1$ as a new variable shows that this is an ellipse (in coordinates x_1, y_2 ; in coordinates $ey_2 - x_1, y_2$, it is a circle).

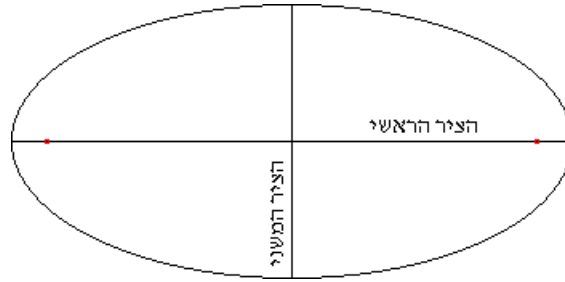


Figure 56: An ellipse

What do the trajectories look like? Let P be the point on the rod given by

$$P_1 + t(P_2 - P_1).$$

By the [description in Situation 8.1](#), P has coordinates

$$((1 - t)x_1 + tey_2, ty_2),$$

subject to

$$(ey_2 - x_1)^2 + y_2^2 = d^2.$$

In the extreme cases $t = 0$ and $t = 1$, the resulting solution sets are respectively $(x_1, 0)$ (with arbitrary x_1) and (ey_2, y_2) (with arbitrary y_2). The condition

$$(ey_2 - x_1)^2 + y_2^2 = d^2$$

must still be satisfied: for a given x_1 (or y_2), the equation must have a solution in the other variable. Such a solution exists when x_1 (or y_2) is sufficiently small. Altogether, we obtain certain segments on the original lines. The points P_1 and P_2 must stay on their paths and cannot move arbitrarily far from the other line.

Edition note: “Sufficiently small” in the source means small in absolute value. For $t = 0$, minimising the left-hand side over y_2 gives the exact condition $|x_1| \leq d\sqrt{1+e^2}$; for $t = 1$, minimising over x_1 gives the exact condition $|y_2| \leq d$.

So let

$$t \neq 0, 1.$$

The ansatz

$$(x, y) = ((1-t)x_1 + tey_2, ty_2)$$

gives

$$y_2 = \frac{y}{t}$$

and

$$x_1 = \frac{x - tey_2}{1-t} = \frac{x - ey}{1-t}$$

(so the preimage is uniquely determined). The equation then becomes

$$\left(\frac{ey}{t} - \frac{x - ey}{1-t}\right)^2 + \frac{y^2}{t^2} = d^2,$$

which is again the equation of an ellipse.

A line and a circle as paths

We now consider a mechanical curve for which one path is a line and the second is a circle. This is the situation in a steam engine (in particular when the line passes through the centre of the circle).

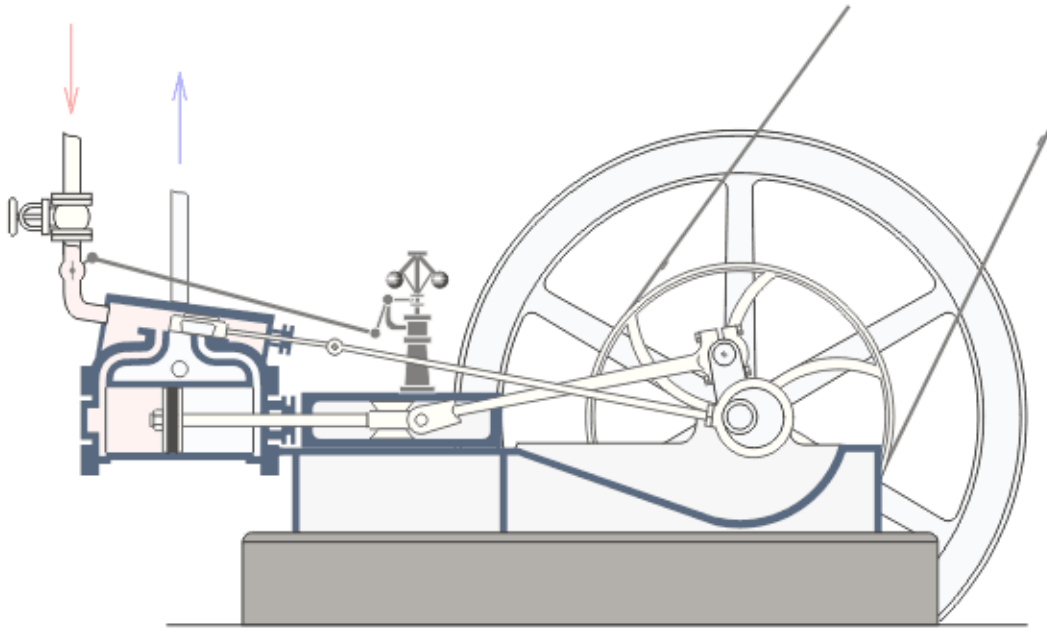


Figure 57: A steam engine in action

Without loss of generality, we may assume that the line is given by

$$y = 0.$$

The coordinates of the point on the line are then

$$P_1 = (x_1, y_1) = (x_1, 0).$$

We may assume that the circle has centre $(0, b)$ and radius r . The point on the circular path

$$P_2 = (x_2, y_2)$$

satisfies

$$x_2^2 + (y_2 - b)^2 = r^2.$$

Thus the entire mechanical system is described by the two conditions

$$\begin{aligned} x_2^2 + (y_2 - b)^2 &= r^2, \\ (x_2 - x_1)^2 + y_2^2 &= d^2, \end{aligned}$$

where d again denotes the coupling distance. Looking at these equations in the coordinates x_2, y_2 and $x_2 - x_1$ shows that they describe the intersection of two cylinders, as in [Example 4.6](#). Thus, in three suitable coordinates, the allowed rod configurations can be interpreted as the intersection of two cylinders. However, their radii need not agree, nor need their central axes intersect. Such an intersection and its associated trajectories can be quite complicated.

For the following examples, we need a lemma describing a simple *elimination situation*.

Lemma: elimination of two quadratic equations

Let R be an [integral domain](#), and let

$$F_1 = a_1X^2 + b_1X + c_1 \quad \text{and} \quad F_2 = a_2X^2 + b_2X + c_2$$

be two quadratic polynomials in one variable over R , with

$$a_1 \neq 0,$$

and with (a_1, b_1) and (a_2, b_2) linearly independent. Then the ideal

$$(F_1, F_2) \cap R$$

contains the element

$$\begin{aligned} & (a_2c_1 - a_1c_2)^2 \\ & - b_1(-a_2c_1b_2 - c_2a_2b_1 + a_1b_2c_2) \\ & + c_1(a_1b_2^2 - 2a_2b_1b_2). \end{aligned}$$

Proof First we have

$$a_2F_1 - a_1F_2 = (a_2b_1 - a_1b_2)X + a_2c_1 - a_1c_2.$$

Edition supplement. The source's substitution argument below does not by itself justify division inside an ideal. A direct identity supplies the justification. Put $A = a_2b_1 - a_1b_2$, $B = a_2c_1 - a_1c_2$, and let $E \in R$ denote the element displayed in the lemma. Expanding gives

$$E = (a_2B - b_2A - a_2AX)F_1 + (a_1AX + b_1A - a_1B)F_2.$$

Thus $E \in (F_1, F_2) \cap R$ without any division. The following is the source's original calculation.

This gives the expression (this argument is not entirely correct, but can also be carried out more rigorously)

$$X = -\frac{a_2c_1 - a_1c_2}{a_2b_1 - a_1b_2}.$$

Substituting into F_1 and multiplying by the square of the denominator gives

$$\begin{aligned} & a_1(a_2c_1 - a_1c_2)^2 \\ & - b_1(a_2c_1 - a_1c_2)(a_2b_1 - a_1b_2) \\ & + c_1(a_2b_1 - a_1b_2)^2. \end{aligned}$$

The second summand contains $-b_1a_2c_1a_2b_1$, and the third contains $c_1a_2^2b_1^2$; these two terms cancel. Every remaining monomial contains a_1 . We can therefore cancel a_1 , leaving

$$\begin{aligned} & (a_2c_1 - a_1c_2)^2 \\ & - b_1(-a_2c_1b_2 - c_2a_2b_1 + a_1b_2c_2) \\ & + c_1(a_1b_2^2 - 2a_2b_1b_2). \end{aligned}$$

□

Example: the unit circle and a tangent line

Consider the mechanical linkage defined by the unit circle and its tangent line at $(0, 1)$, with coupling distance

$$d = 2.$$

Thus the point on the straight path and the point on the circular path are

$$P_1 = (x_1, 1) \quad \text{and} \quad P_2 = (x_2, y_2),$$

with the two conditions

$$\begin{aligned} x_2^2 + y_2^2 &= 1, \\ (x_2 - x_1)^2 + (y_2 - 1)^2 &= 4. \end{aligned}$$

This is therefore the intersection of two cylinders, but with different radii and nonintersecting central axes. The difference of the equations is

$$x_1^2 - 2x_1x_2 - 2y_2 - 2 = 0,$$

which can replace one of them. This also shows that we can eliminate y_2 , obtaining a system with one equation in two variables; see [Exercise 8.9](#). The system is irreducible; see [Exercise 8.10](#).

The following two lines are of interest:

$$G_1 = V(x_2, y_2 + 1)$$

and

$$G_2 = V(x_2 - x_1, y_2 + 1).$$

They intersect at the point

$$P = (0, 0, -1).$$

The line G_1 lies on one cylinder and is tangent to the other, and conversely. Geometrically, the smaller cylinder punches a “bent figure eight” out of the larger cylinder, with P as the crossing point of the figure eight.

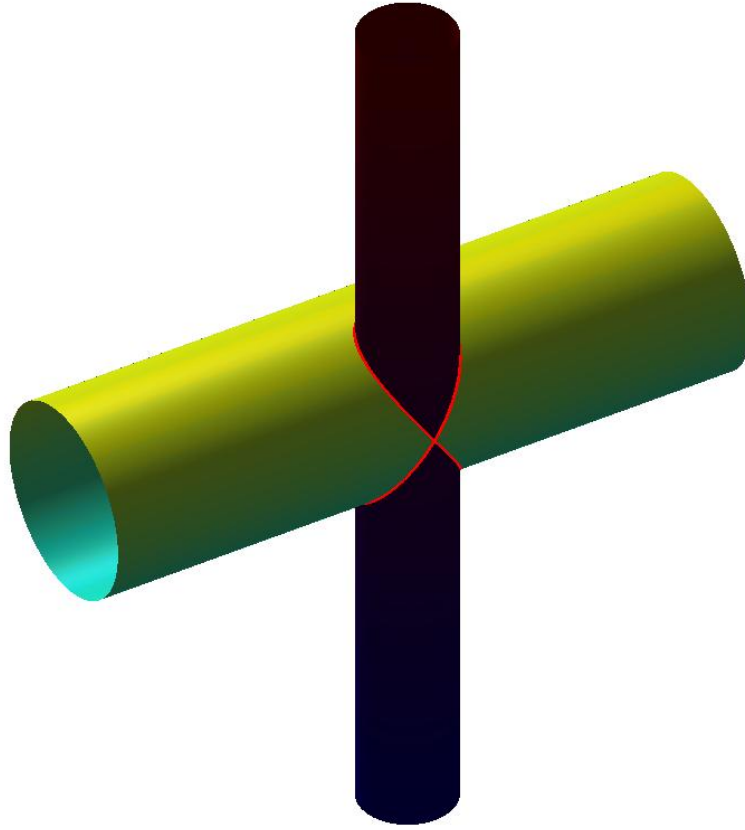


Figure 58: Intersection of two cylinders

The allowed rod configurations can be obtained as follows. For each point of the circle, the rod has two possible positions, except at the circular-path point $(0, -1)$, where the straight-path point must be $(0, 1)$.

Start with $(0, 1)$ as the circular-path point and $(-2, 1)$ as the straight-path point (so the rod lies to the left along the line), and let the circular-path point move clockwise around the circle. It pulls the straight-path point behind it until it reaches the bottom at $(0, -1)$. The rod is then the vertical diameter of the circle (the straight-path point is at $(0, 1)$ and the circular-path point is at the bottom). The circular-path point then moves upwards along the left-hand arc, pushing the rod further to the right until the straight-path point reaches $(2, 1)$.

The other possibility with $(0, 1)$ as the circular-path point has the rod lying to the right along the line (with $(2, 1)$ as the straight-path point). The circular-path point again moves clockwise. At first it pushes the straight-path point to the right until an extreme position is reached, where the rod is perpendicular to the circle at the circular-path point. It then pulls the straight-path point back to the left as the rod rises, until the rod occupies the vertical diameter of the circle. The circular-path point next moves upwards along the left-hand arc again, pushing the straight-path point leftwards to an extreme position, and finally pulling it back to $(-2, 1)$.

In particular, the rod occupies the vertical diameter twice; this rod configuration therefore corresponds to the crossing point of the figure eight.

We now want to calculate the trajectory of the midpoint of the rod, namely

$$\begin{aligned}
 P &= P_1 + \frac{1}{2}(P_2 - P_1) \\
 &= (x_1, 1) + \frac{1}{2}(x_2 - x_1, y_2 - 1) \\
 &= \left(\frac{1}{2}x_1 + \frac{1}{2}x_2, \frac{1}{2}y_2 + \frac{1}{2}\right) \\
 &:= (x, y).
 \end{aligned}$$

We seek an equation for x and y , and introduce the variable

$$z = \frac{1}{2}x_1 - \frac{1}{2}x_2.$$

Then

$$x_1 = x + z, \quad x_2 = x - z, \quad y_2 = 2y - 1,$$

and the system in the new variables becomes

$$(x - z)^2 + (2y - 1)^2 = 1 \quad \text{and} \quad (-2z)^2 + (2y - 2)^2 = 4.$$

The second equation can be written as

$$z^2 + (y - 1)^2 = 1,$$

or as

$$z^2 + y^2 - 2y = 0.$$

Expanding the first equation gives

$$z^2 - 2zx + x^2 + 4y^2 - 4y = 0.$$

By [Lemma 8.4](#), with

$$R = \mathbb{R}[x, y]$$

and the additional variable z (so $a_1 = a_2 = 1$ and $b_1 = 0$), we obtain the equation

$$\begin{aligned} (c_1 - c_2)^2 + c_1 b_2^2 &= (y^2 - 2y - x^2 - 4y^2 + 4y)^2 + (y^2 - 2y)(2x)^2 \\ &= (-3y^2 + 2y - x^2)^2 + (y^2 - 2y)(2x)^2 \\ &= 9y^4 + x^4 + 4y^2 - 12y^3 + 6x^2y^2 - 4x^2y \\ &\quad + 4x^2y^2 - 8x^2y \\ &= 9y^4 + 10x^2y^2 + x^4 - 12y^3 - 12x^2y + 4y^2. \end{aligned}$$

This is a quartic (a curve of degree four) with two singularities.

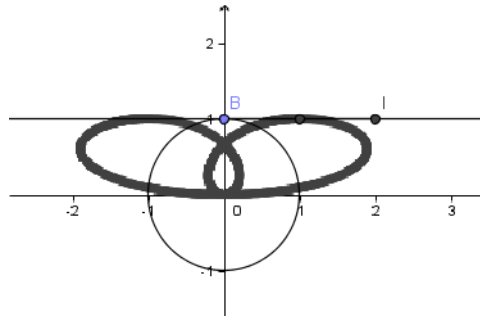


Figure 59: Figure for the exercise

Example: radius equal to the coupling distance

Consider the mechanical system consisting of the unit circle and the x -axis, with coupling distance

$$d = 1.$$

The mechanical system is described by the two equations

$$\begin{aligned}x_2^2 + y_2^2 &= 1, \\(x_2 - x_1)^2 + y_2^2 &= 1.\end{aligned}$$

This is the intersection of two cylinders with equal radii and intersecting central axes, so we may use the results of [Example 4.6](#). There we showed that the intersection consists of two ellipses meeting at two points. This description must also reappear in the context of the mechanical system. Which rod configurations correspond to the first ellipse, which to the second, and which lie on both?

Let us survey the allowed configurations. If the line point (the point on the straight path) is the centre of the circle, every point of the circle is allowed as the circular-path point. The radial rays of the circle therefore form a family of allowed rod configurations, together making up one ellipse. The other ellipse corresponds to the configurations in which the straight-path point moves from -2 to $+2$, pushing the circular-path point ahead of it or pulling it behind on the upper or lower arc. Two rod configurations belong to both families: those with the circle's centre as the straight-path point and $(0, 1)$ or $(0, -1)$ as the circular-path point. In such a configuration, the mechanical system can not only move forwards and backwards but also change direction in an essential way.

What do the trajectories of a point on the moving rod look like? The total trajectory is the union of the two trajectories corresponding to the two irreducible components of the system. How many self-intersection points are there?

For a point

$$P = P_1 + u(P_2 - P_1)$$

on the coupling rod, its coordinates are

$$(z_1, z_2) = (x_1 + u(x_2 - x_1), uy_2).$$

For $u = 0$, the trajectory is the real interval $[-2, 2]$, and for $u = 1$, it is the unit circle. So now let

$$u \neq 0, 1.$$

The projection of the radial components of the system is simply a circle of radius u . The projection of the other ellipse is again an ellipse, which can intersect the circle in different ways.

Edition note. The radius is $|u|$ if negative u is allowed. There is also a degenerate case omitted in the source: on the second component, $x_1 = 2x_2$, so $(z_1, z_2) = ((2 - u)x_2, uy_2)$ with $x_2^2 + y_2^2 = 1$. For $u = 2$, its image is the segment $\{0\} \times [-2, 2]$, not an ellipse. For $u \neq 0, 2$, that component projects to a nondegenerate ellipse. See also [Exercise 8.23](#).

Worksheet 8

Practice exercises

Exercise 8.1

Determine the possible intersections of two perpendicular cylinders.

Exercise 8.2

Find an equation as simple as possible for the mechanical system given by the X -axis, the translated parabola

$$Y = X^2 + 1,$$

and distance 2.

Exercise 8.3

Let

$$H_1, H_2 \in K[X]$$

be polynomials in one variable, and let

$$C_1 = V(Y - H_1), \quad C_2 = V(Y - H_2)$$

be their corresponding graphs in $\mathbb{A}_K^2$. Show that the associated mechanical system, for a specified distance d , can be described using two variables.

Exercise 8.4

Determine equations for the mechanical system given by the unit circle, the x -axis, and distance 1. What are its irreducible components?

Exercise 8.5

Let

$$M \subseteq \mathbb{A}_{\mathbb{R}}^4$$

be the mechanical system given by two paths B_1 and B_2 and distance d . Show that there is a natural injective map

$$M \longrightarrow B_1 \times B_2.$$

Exercise 8.6

Let

$$B = V(G) \subseteq \mathbb{A}_{\mathbb{R}}^2$$

be a path. Regard it as a mechanical system M in the sense that the two points at distance d must lie on this one path. Show that there is a natural fixed-point-free bijection

$$M \longrightarrow M.$$

Exercise 8.7

Determine equations for the mechanical system given by the x -axis as the common path and distance 1.

Exercise 8.8

Determine equations for the mechanical system given by the unit circle as the common path and distance 2.

Exercise 8.9 ★

Consider the mechanical system defined by the unit circle and its tangent line through $(0, 1)$, with coupling distance

$$d = 2.$$

Show that this system can be described using two variables.

Exercise 8.10

Consider the mechanical system defined by the unit circle and its tangent line through $(0, 1)$, with coupling distance $d = 2$; see Example 8.5. Show that this system is irreducible. To do so, apply [Exercise 7.24](#) to the ring extension

$$R = \mathbb{R}[X_2, Y_2]/(X_2^2 + Y_2^2 - 1) \subseteq \mathbb{R}[X_1]/(X_1^2 - 2X_1X_2 - 2Y_2 - 2).$$

Exercise 8.11

Determine equations for the mechanical system given by the x -axis, the y -axis, and distance 1.

Exercise 8.12

Determine equations for the mechanical system given by the union of the coordinate axes as the common path and distance 1.

Exercise 8.13

Let

$$B_1 = V(F_1), \quad B_2 = V(F_2) \subseteq \mathbb{A}_{\mathbb{R}}^2$$

be two paths, and fix a distance d . Compare the mechanical system M for these paths with the system N associated with the single path $B_1 \cup B_2$. Show that there are two natural injective maps

$$M \longrightarrow N.$$

Let L_i be the mechanical system associated with B_i as the sole path. Show that there is a natural surjective map

$$L_1 \uplus L_2 \uplus M \uplus M \longrightarrow N.$$

Exercise 8.14

Let

$$M \subseteq \mathbb{A}_{\mathbb{R}}^4$$

be a mechanical system. Show that the rule

$$\begin{aligned} M &\longrightarrow S^1(d), \\ (P_1, P_2) &\longmapsto P_1 - P_2 \end{aligned}$$

defines a map from the system to the circle with centre $(0, 0)$ and radius d . What does surjectivity of this map mean? Can its image consist of only finitely many points?

Exercise 8.15

Consider the mechanical system

$$M \subseteq \mathbb{A}_{\mathbb{R}}^4$$

for two intersecting lines. Show that the map in [Exercise 8.14](#) is bijective.

Exercise 8.16

Consider the mechanical system

$$M \subseteq \mathbb{A}_{\mathbb{R}}^4$$

for the unit circle, the circle with centre $(0, 2)$ and radius 4, and coupling distance 5. Show that the map in [Exercise 8.14](#) is not surjective. What is its image?

Preparation for regularity

In the following exercises, we consider the zero locus

$$V = V(f_1, \dots, f_k) \subseteq \mathbb{A}_{\mathbb{R}}^n$$

from the viewpoint of differential geometry. Recall the definition of a regular point of a differentiable map from Analysis 2. In the present context, this notion is applied to the map

$$\varphi = (f_1, \dots, f_k) : \mathbb{A}_{\mathbb{R}}^n \longrightarrow \mathbb{A}_{\mathbb{R}}^k$$

at a point $P \in V$.

Let V_0 and W be finite-dimensional real vector spaces, let $G \subseteq V_0$ be open, let $P \in G$, and let

$$\varphi : G \longrightarrow W$$

be differentiable at P . The point P is called a *regular point* of φ if

$$\text{rank}(D\varphi)_P = \min(\dim V_0, \dim W).$$

Otherwise, P is called a *critical point* or a *singular point*.

Exercise 8.17 ★

Consider the mechanical system given by the x -axis and the circle with centre $(0, 2)$ and radius 1. Let the coupling distance be $d > 0$.

1. Set up equations describing this system.
2. Determine for which values of d the system is regular at every point.
3. Determine the critical points as a function of d . How can these points be explained as a property of the mechanical system?

Exercise 8.18

Consider the mechanical system given by the x -axis and the circle with centre $(0, 2)$ and radius 1. Let the coupling distance be $d > 0$.

1. Give a geometric explanation, then prove, that for $d > 3$ this mechanical system is not path-connected in the real topology.
2. Give a geometric explanation, then prove, that for $1 \leq d \leq 3$ this mechanical system is path-connected.

Exercises for submission

Exercise 8.19 - 6 points (2+2+2)

Consider the mechanical system given by the x -axis, the parabola $V(Y - X^2)$, and coupling distance 1.

1. Determine equations for this mechanical system using as few variables as possible.
2. Does the system have critical points?
3. Determine an equation for the trajectory of the midpoint of the connecting rod.

Exercise 8.20 - 2 points

Determine equations for the mechanical system given by the unit circle as the common path and distance 1.

Exercise 8.21 - 6 points (2+2+2)

Consider the mechanical system given by the parabola $V(Y - X^2)$, the circle with centre $(0, 2)$ and radius 1, and coupling distance 1.

1. Determine equations for this mechanical system using as few variables as possible.
2. Determine its connected components in the metric topology.
3. Determine its connected components in the Zariski topology.

Exercise 8.22 - 6 points (2+4)

Consider the mechanical system given by the x -axis and the circle with centre $(0, 2)$ and radius 1. Let the coupling distance be $d > 0$. Continue [Exercise 8.17](#).

1. Eliminate y_1 from the system's equations.
2. Using the single equation in x_1 and x_2 describing the system, determine for which values of d the system is regular at every point.

Exercise 8.23 - 6 points

Let

$$C \subseteq \mathbb{A}_{\mathbb{R}}^3$$

be the intersection of two cylinders of radius 1, about the x -axis and the y -axis respectively; thus C is the union of two ellipses. Consider the perpendicular projection

$$p_v : \mathbb{A}_{\mathbb{R}}^3 \longrightarrow \mathbb{A}_{\mathbb{R}}^2$$

specified by a vector

$$v = (a, b, c) \neq 0.$$

Characterise the possible images under these projections as a function of a, b, c .

Exercise 8.24 - 3 points

Consider the map

$$\begin{aligned} \mathbb{A}_K^2 &\longrightarrow \mathbb{A}_K^2, \\ (x, y) &\longmapsto (x^2, y^2) = (u, v). \end{aligned}$$

What are the images of the plane and the unit circle under this map for $K = \mathbb{R}$, and what are they for $K = \mathbb{C}$? In the real case, if the circle is traversed once, how many times is its image traversed?

Source navigation: [Lecture 8](#) - [public solutions for Unit 8](#) - [Worksheet 7](#) - [Worksheet 9](#) ([source](#))

Public Solutions to Worksheet 8

At the frozen revision boundary, the source provides public solutions only for Exercises 8.9 and 8.17. No additional solutions have been created for this edition.

Solution to Exercise 8.9

With

$$P_1 = (x_1, 1) \quad \text{and} \quad P_2 = (x_2, y_2),$$

we obtain the two conditions

$$x_2^2 + y_2^2 = 1$$

and

$$(x_2 - x_1)^2 + (y_2 - 1)^2 = 4.$$

Subtracting the first equation from the second gives

$$\begin{aligned} 0 &= (x_2 - x_1)^2 + (y_2 - 1)^2 - 4 - (x_2^2 + y_2^2 - 1) \\ &= x_2^2 + x_1^2 - 2x_1x_2 + y_2^2 - 2y_2 + 1 - 4 - x_2^2 - y_2^2 + 1 \\ &= x_1^2 - 2x_1x_2 - 2y_2 - 2. \end{aligned}$$

Together with the unit-circle equation, this equation is equivalent to the original system. From the new second equation, we can eliminate y_2 using

$$y_2 = \frac{1}{2}x_1^2 - x_1x_2 - 1.$$

Thus the system can be described in x_1 and x_2 alone, as the zero locus of the polynomial

$$\begin{aligned} x_2^2 + y_2^2 - 1 &= x_2^2 + \left(\frac{1}{2}x_1^2 - x_1x_2 - 1\right)^2 - 1 \\ &= x_2^2 + \frac{1}{4}x_1^4 + x_1^2x_2^2 + 1 - x_1^3x_2 - x_1^2 + 2x_1x_2 - 1 \\ &= \frac{1}{4}x_1^4 - x_1^3x_2 + x_1^2x_2^2 - x_1^2 + 2x_1x_2 + x_2^2. \end{aligned}$$

[Back to Exercise 8.9.](#)

Solution to Exercise 8.17

1. Let

$$P_1 = (x_1, y_1)$$

be the point on the circle and

$$P_2 = (x_2, 0)$$

the point on the x -axis. The equations are

$$x_1^2 + (y_1 - 2)^2 = 1$$

and

$$(x_1 - x_2)^2 + y_1^2 = d^2.$$

2. Write

$$f_1 = x_1^2 + (y_1 - 2)^2 - 1 = x_1^2 + y_1^2 - 4y_1 + 3$$

and

$$f_2 = (x_1 - x_2)^2 + y_1^2 - d^2 = x_1^2 + x_2^2 - 2x_1x_2 + y_1^2 - d^2.$$

The Jacobian matrix with respect to (x_1, x_2, y_1) is

$$\begin{pmatrix} 2x_1 & 0 & 2y_1 - 4 \\ 2x_1 - 2x_2 & 2x_2 - 2x_1 & 2y_1 \end{pmatrix}.$$

Depending on d , we must determine at which points the linear map $\mathbb{R}^3 \rightarrow \mathbb{R}^2$ given by this matrix is surjective, that is, has rank 2. Its rank is not 2 exactly when every pair of columns is linearly dependent, or equivalently when all 2×2 minors vanish. After removing the common factor 4, the three polynomials are

$$x_1(x_2 - x_1), \quad (y_1 - 2)(x_2 - x_1), \quad x_1y_1 - (x_1 - x_2)(y_1 - 2).$$

If $x_1 \neq 0$, we must have $x_1 = x_2$ and $y_1 = 0$. But this is not a point of the system. Thus $x_1 = 0$. If $x_2 \neq 0$, we must have $y_1 = 2$, which does not satisfy the first equation of the system. Hence $x_2 = 0$. The system's equations then give

$$(y_1 - 2)^2 = 1 \quad \text{and} \quad y_1^2 = d^2.$$

The first equation forces $y_1 = 1$ or $y_1 = 3$, so $d = 1$ or $d = 3$. Thus the system is regular at every point exactly when $d \neq 1, 3$.

3. For $d = 1$, the calculation above shows that $(0, 0, 1)$ is the only critical point; indeed it is the only point of the system, which explains the singularity.

For $d = 3$, the point $(0, 0, 3)$ is the only critical point of the system. It is a crossing point, since the rod can move in four directions there: both coordinates (x_1, x_2) in the positive direction, both in the negative direction, or in either of the two mixed directions.

[Back to Exercise 8.17.](#)

Source navigation: [Worksheet 8 - Lecture 8](#)

Lecture 9: Noetherian Rings, Hilbert's Basis Theorem, and Modules

Noetherian rings

In the next few lectures, we will further develop the algebraic side of algebraic geometry. Our first aim is to show that if R is a Noetherian ring, then the polynomial ring $R[X]$ is also Noetherian (Hilbert's basis theorem). This also holds when adjoining several (finitely many) variables, in particular for polynomial rings in finitely many variables over a field. We recall the notion of a Noetherian ring.

Definition: Noetherian ring

A commutative ring R is called *Noetherian* if every ideal in it is finitely generated.

Proposition: characterisation of Noetherian rings

For a commutative ring R , the following statements are equivalent.

1. R is Noetherian.
2. Every ascending chain of ideals

$$\mathfrak{a}_1 \subseteq \mathfrak{a}_2 \subseteq \mathfrak{a}_3 \subseteq \cdots$$

becomes *stationary*; that is, there is an n such that

$$\mathfrak{a}_n = \mathfrak{a}_{n+1} = \cdots.$$

Proof (1) $\Rightarrow$ (2). Let

$$\mathfrak{a}_1 \subseteq \mathfrak{a}_2 \subseteq \mathfrak{a}_3 \subseteq \cdots$$

be an ascending chain of ideals in R . Consider its union

$$\mathfrak{a} = \bigcup_{n \in \mathbb{N}} \mathfrak{a}_n,$$

which is again an ideal in R . Since R is Noetherian, $\mathfrak{a}$ is finitely generated, say

$$\mathfrak{a} = (f_1, \dots, f_k).$$

All the f_i lie in the union of the ideals $\mathfrak{a}_n$. Since these ideals form an ascending chain, there is an n such that

$$f_1, \dots, f_k \in \mathfrak{a}_n.$$

Then, for every $m \geq 0$,

$$(f_1, \dots, f_k) \subseteq \mathfrak{a}_n \subseteq \mathfrak{a}_{n+m} \subseteq \bigcup_{n \in \mathbb{N}} \mathfrak{a}_n \subseteq (f_1, \dots, f_k).$$

All these inclusions must be equalities, so the chain is stationary from n onwards.

(2) $\Rightarrow$ (1). Let $\mathfrak{a}$ be an ideal in R . Suppose $\mathfrak{a}$ is not finitely generated. We can successively construct an infinite strictly ascending chain of ideals

$$\mathfrak{a}_1 \subset \mathfrak{a}_2 \subset \cdots \subseteq \mathfrak{a},$$

where each $\mathfrak{a}_n$ is finitely generated. Suppose we have already constructed

$$\mathfrak{a}_1 \subset \mathfrak{a}_2 \subset \cdots \subset \mathfrak{a}_n \subseteq \mathfrak{a}.$$

Since $\mathfrak{a}_n$ is finitely generated but $\mathfrak{a}$ is not, the inclusion $\mathfrak{a}_n \subseteq \mathfrak{a}$ is strict. Thus there is an element

$$f_{n+1} \in \mathfrak{a}, \quad f_{n+1} \notin \mathfrak{a}_n.$$

The ideal

$$\mathfrak{a}_{n+1} := \mathfrak{a}_n + (f_{n+1})$$

strictly extends the chain. This contradicts (2).

Lemma: quotients of Noetherian rings

If R is Noetherian, every quotient ring $R/\mathfrak{b}$ is also Noetherian.

Proof Let $\mathfrak{a} \subseteq R/\mathfrak{b}$ be an ideal and let $\tilde{\mathfrak{a}} \subseteq R$ be its preimage ideal. By assumption this is finitely generated, say

$$\tilde{\mathfrak{a}} = (f_1, \dots, f_n).$$

The residue classes of these generators, $\bar{f}_1, \dots, \bar{f}_n$, form a generating set for the ideal $\mathfrak{a}$. Indeed, for $\bar{g} \in \mathfrak{a}$, we have in R

$$g = \sum_{i=1}^n r_i f_i,$$

and hence in $R/\mathfrak{b}$

$$\bar{g} = \sum_{i=1}^n \bar{r}_i \bar{f}_i.$$

Hilbert's basis theorem

Like many fundamental results in commutative algebra, Hilbert's basis theorem, to which we now turn, goes back to David Hilbert, specifically his 1890 paper *Ueber die Theorie der algebraischen Formen* ("On the Theory of Algebraic Forms").

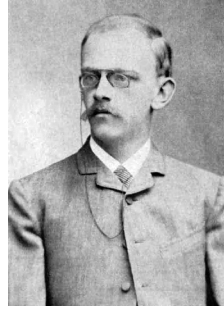


Figure 60: David Hilbert (1862–1943)

David Hilbert (1862–1943); unknown creator (1886), Commons, public domain.

Hilbert's basis theorem

If R is Noetherian, the polynomial ring $R[X]$ is also Noetherian.

Proof Let $\mathfrak{b}$ be an ideal in the polynomial ring $R[X]$. For $n \in \mathbb{N}$, define an ideal $\mathfrak{a}_n$ in R by

$$\mathfrak{a}_n = \{c \in R \mid \text{there is } F \in \mathfrak{b} \text{ with } F = cX^n + c_{n-1}X^{n-1} + \dots + c_1X + c_0\}.$$

Thus $\mathfrak{a}_n$ consists of all leading coefficients of degree- n polynomials in $\mathfrak{b}$. Clearly $\mathfrak{a}_n$ is an ideal in R (here we allow 0 as a leading coefficient). Moreover,

$$\mathfrak{a}_n \subseteq \mathfrak{a}_{n+1},$$

since a polynomial F of degree n with leading coefficient c can be multiplied by X to give a polynomial of degree $n+1$ with the same leading coefficient. Since R is Noetherian, this ascending chain of ideals becomes stationary; choose n such that

$$\mathfrak{a}_n = \mathfrak{a}_{n+1} = \cdots.$$

For each $i \leq n$, choose a finite generating set

$$\mathfrak{a}_i = (c_{i1}, \dots, c_{ik_i}),$$

and choose corresponding polynomials

$$F_{ij} = c_{ij}X^i + \text{terms of lower degree}$$

in $\mathfrak{b}$ (which exist by the definition of $\mathfrak{a}_i$).

We claim that $\mathfrak{b}$ is generated by all the polynomials

$$\{F_{ij} \mid 0 \leq i \leq n, 1 \leq j \leq k_i\}.$$

For each $G \in \mathfrak{b}$, we prove by induction on its degree that it can be written as an $R[X]$ -linear combination of these F_{ij} .

Edition note. The source says “ R -linear combination” here. The factors X^{d-i} in the final induction step show that $R[X]$ is intended; the assertion is finite ideal generation, not finite generation as an R -module.

If G is constant, that is, $G \in R$, this is clear. Let G have degree d , and suppose the statement has been proved for smaller degrees. Write

$$G = cX^d + c_{d-1}X^{d-1} + \cdots + c_1X + c_0.$$

We have $c \in \mathfrak{a}_d$, so c is an R -linear combination of the c_{ij} with $0 \leq i \leq n$ and $1 \leq j \leq k_i$. If $d \leq n$, then c can be written as an R -linear combination of the c_{dj} , say

$$c = \sum_{j=1}^{k_d} r_j c_{dj}.$$

Thus

$$G - \sum_{j=1}^{k_d} r_j F_{dj} \in \mathfrak{b}$$

has smaller degree, so the induction hypothesis applies. If $d > n$, then

$$c = \sum_{i=0, \dots, n, j=1, \dots, k_i} r_{ij} c_{ij}.$$

Therefore

$$G - \sum_{i=0, \dots, n, j=1, \dots, k_i} r_{ij} X^{d-i} F_{ij}$$

also belongs to $\mathfrak{b}$ and has smaller degree. This completes the induction, so $\mathfrak{b}$ is finitely generated.

Corollary: finitely many variables

If R is Noetherian, then

$$R[X_1, \dots, X_n]$$

is also Noetherian.

Proof Apply Hilbert's basis theorem inductively along the chain

$$R \subset R[X_1] \subset (R[X_1])[X_2] = R[X_1, X_2] \subset (R[X_1, X_2])[X_3] = R[X_1, X_2, X_3] \subset \dots \subset R[X_1, \dots, X_n].$$

Corollary: polynomial rings over fields

If K is a field, then $K[X_1, \dots, X_n]$ is Noetherian.

Proof This is a special case of Corollary 9.5.

In particular, Hilbert's basis theorem means that every closed subvariety

$$V \subseteq \mathbb{A}_K^n$$

of affine space can be described by finitely many polynomials. Thus every algebraic zero locus is already the zero locus of finitely many polynomials.

Corollary: a fibre over the origin

Let $V \subseteq \mathbb{A}_K^n$ be an affine algebraic set. Then there is a map

$$\varphi : \mathbb{A}_K^n \longrightarrow \mathbb{A}_K^m$$

whose components are given by polynomials

$$F_i \in K[X_1, \dots, X_n], \quad \varphi = (F_1, \dots, F_m),$$

such that V is the preimage of the origin

$$0 \in \mathbb{A}_K^m.$$

Proof Let $\mathfrak{a}$ be an ideal describing V , so

$$V = V(\mathfrak{a}).$$

By Hilbert's basis theorem, there are

$$F_1, \dots, F_m \in K[X_1, \dots, X_n]$$

with $\mathfrak{a} = (F_1, \dots, F_m)$. Then

$$V = V(\mathfrak{a}) = V(F_1) \cap \dots \cap V(F_m).$$

Combine these polynomials into a map

$$\varphi = (F_1, \dots, F_m) : \mathbb{A}_K^n \longrightarrow \mathbb{A}_K^m.$$

We have $\varphi(P) = 0$ exactly when all its component functions vanish, which happens exactly when $P \in V(F_i)$ for every i ; thus $V = \varphi^{-1}(0)$.

Definition: algebra of finite type

Let R be a commutative ring. An R -algebra A is called *of finite type* (or *finitely generated*) if it has the form

$$A = R[X_1, \dots, X_n]/\mathfrak{a}.$$

Thus a finitely generated R -algebra has a presentation as a quotient ring of a polynomial algebra over R in finitely many variables. Such a presentation is by no means unique.

Corollary: finite-type algebras over a Noetherian ring

If R is Noetherian, every R -algebra of finite type is also Noetherian. In particular, for a field K , every K -algebra of finite type is Noetherian.

Proof This follows from Corollary 9.5 and Lemma 9.3.

Decomposition into irreducible components

Hilbert's basis theorem implies that every ascending chain of ideals

$$\mathfrak{a}_1 \subseteq \mathfrak{a}_2 \subseteq \mathfrak{a}_3 \subseteq \dots$$

in $K[X_1, \dots, X_n]$ becomes stationary. For descending chains of affine algebraic subsets of affine space, this has the following consequence.

Theorem: the Zariski topology is Noetherian

In affine space $\mathbb{A}_K^n$, every descending sequence of closed sets

$$V_1 \supseteq V_2 \supseteq \dots$$

becomes stationary.

Proof Let

$$V_1 \supseteq V_2 \supseteq \dots$$

be a descending chain of affine algebraic subsets of $\mathbb{A}_K^n$. By Lemma 3.7, their corresponding vanishing ideals satisfy

$$\text{Id}(V_i) \subseteq \text{Id}(V_{i+1}).$$

By Corollary 9.6, this chain of ideals becomes stationary, say for $i \geq i_0$. By Lemma 3.8(3),

$$V_i = V(\text{Id}(V_i)).$$

Hence, for $i \geq i_0$,

$$V_i = V(\text{Id}(V_i)) = V(\text{Id}(V_{i+1})) = V_{i+1},$$

so the descending chain becomes stationary.

Taking complements, it follows that every ascending chain of Zariski-open sets in affine space also becomes stationary. Such a topology is called *Noetherian* (more generally, a partial order in which every ascending chain becomes stationary is called Noetherian). In a Noetherian space, every nonempty collection of open sets (or closed sets) has a maximal (or minimal) element. This is useful as a proof principle called *Noetherian induction*: to prove that a property E holds for all closed subsets, consider the collection of closed subsets that do not satisfy E . We want to show that this collection is empty; if it were nonempty, it would have a minimal element, which we then lead to a contradiction. The validity of the principle rests on the fact that a nonempty set with no minimal element allows an infinite descending chain to be constructed. A typical example of this principle is the following theorem.

Theorem: decomposition into irreducible components

Every affine algebraic set $V \subseteq \mathbb{A}_K^n$ has a unique decomposition

$$V = V_1 \cup \cdots \cup V_k$$

into irreducible sets V_i such that

$$V_i \not\subseteq V_j \quad \text{for } i \neq j.$$

Edition clarification. The components in this statement are closed affine algebraic subsets, as required by the source's proof using closed decompositions. Uniqueness is up to reordering. For the empty set, the decomposition is the empty union.

Proof of existence (Noetherian induction) Suppose not every affine algebraic set has such a decomposition. Then there is a minimal set, say V , without such a decomposition. The set V cannot be irreducible, so it has a nontrivial decomposition

$$V = V_1 \cup V_2.$$

Since V_1 and V_2 are proper subsets of V , each has a finite expression as a union of irreducible sets. Combining these expressions gives a finite expression for V , a contradiction.

Proof of uniqueness Let

$$V = V_1 \cup \cdots \cup V_k = W_1 \cup \cdots \cup W_m$$

be two decompositions into irreducible sets (with no inclusions within either decomposition). We have

$$V_1 = V_1 \cap V = V_1 \cap (W_1 \cup \cdots \cup W_m) = (V_1 \cap W_1) \cup \cdots \cup (V_1 \cap W_m).$$

Since V_1 is irreducible, $V_1 \subseteq W_j$ for some j . By the same argument, $W_j \subseteq V_i$ for some i , whence $i = 1$ and $V_1 = W_j$. Similarly, V_2 and so on reappear in the decomposition on the right, so the decomposition is unique.

The sets $V_1, \dots, V_k$ in this theorem are called the *irreducible components* of V .

Modules

Definition: module

Let R be a commutative ring and

$$M = (M, +, 0)$$

an additively written commutative group. We call M an R -module if an operation

$$R \times M \longrightarrow M, \quad (r, v) \longmapsto rv = r \cdot v,$$

called *scalar multiplication*, is specified and satisfies the following axioms (for arbitrary $r, s \in R$ and $u, v \in M$):

$$\begin{aligned} r(su) &= (rs)u, \\ r(u + v) &= (ru) + (rv), \\ (r + s)u &= (ru) + (su), \\ 1u &= u. \end{aligned}$$

Definition: submodule

Let R be a commutative ring and M an R -module. A subset

$$U \subseteq M$$

is called an *R -submodule* if it is a subgroup of $(M, 0, +)$ and $ru \in U$ for every $u \in U$ and $r \in R$.

Definition: a generating set for a module

Let R be a commutative ring and M an R -module. A family

$$v_i \in M \quad (i \in I)$$

is called a *generating set* for M if every $v \in M$ has an expression

$$v = \sum_{i \in J} r_i v_i,$$

where $J \subseteq I$ is finite and $r_i \in R$.

Definition: finitely generated module

Let R be a commutative ring and M an R -module. The module M is called *finitely generated* (or *finite*) if it has a finite generating set v_i ($i \in I$), that is, one with a finite index set.

A commutative ring R itself is naturally an R -module if ring multiplication is interpreted as scalar multiplication. Its ideals are exactly the R -submodules of R . For ideals, the notions of an *ideal generating set* and a *module generating set* coincide. A vector space is simply a module over a field.

Worksheet 9

Practice exercises

Exercise 9.1

Explain why the ring

$$\mathbb{Z}[X, Y, Z, W] / (XY - ZW, 5X^8 - YZ^3 + 2WXY)$$

is Noetherian.

Exercise 9.2

Let R be a commutative ring and

$$\mathfrak{a}_1 \subseteq \mathfrak{a}_2 \subseteq \mathfrak{a}_3 \subseteq \cdots$$

an ascending chain of ideals. Show that the union

$$\bigcup_{n \in \mathbb{N}} \mathfrak{a}_n$$

is also an ideal. Give a simple example showing that a union of ideals need not be an ideal in general.

Exercise 9.3

Show that the product $R \times S$ of Noetherian rings R and S is again Noetherian.

Exercise 9.4

Let K be a field. Show that in $K[X, Y]$ there is no upper bound on the number of generators in a minimal generating set of an ideal.

Hint: Consider the powers $(X, Y)^m$.

Exercise 9.5

Let K be a field and

$$K[X_n, n \in \mathbb{N}]$$

the polynomial ring over K in infinitely many variables. Describe an ideal that is not finitely generated and an infinite strictly ascending chain of ideals in it.

Exercise 9.6 ★

Show that a subring

$$R \subseteq S$$

of a Noetherian ring need not be Noetherian.

Exercise 9.7

Give an example of a non-Noetherian ring whose reduction is a field.

Exercise 9.8

Let R be a commutative ring and $\mathfrak{a} \subset R$ a proper ideal with quotient ring $R/\mathfrak{a}$. Give an example showing that $\mathfrak{a}$ can be finitely generated and $R/\mathfrak{a}$ Noetherian even though R itself is not Noetherian.

Exercise 9.9

Let R be a commutative ring and $\mathfrak{b} \subseteq R[X]$ an ideal containing at least one monic polynomial. What does this imply for the chain of ideals in R constructed in the proof of Hilbert's basis theorem?

Exercise 9.10

Let R be a commutative ring. Characterise the ideals

$$\mathfrak{b} \subseteq R[X]$$

for which the chain of ideals in R constructed in the proof of Hilbert's basis theorem is constant.

Exercise 9.11

Let K be a field and $R = K[X]$ the polynomial ring over K . For the ideals

$$\mathfrak{b}_m = (X, Y)^m \subseteq R[Y] = K[X, Y],$$

determine the chain of ideals in R constructed in the proof of Hilbert's basis theorem. When does it become stationary?

Exercise 9.12

For the ideal

$$I = (6, 6x^2 + 2x + 3, 3x^3 + 5, 2x^5 + x - 4, 4x^7 - 3x)$$

in $\mathbb{Z}[x]$, determine the chain of ideals constructed in the proof of Hilbert's basis theorem and the corresponding generating set for I . Write the generators above as linear combinations of the constructed generating set.

Exercise 9.13 ★

Let R be a commutative ring and A a commutative R -algebra. Suppose A is generated over R by the family $a_i \in A$ ($i \in I$). Prove that if A is finitely generated, then it is also generated by a finite subfamily of the a_i .

Exercise 9.14

Consider ascending and descending chains of affine algebraic sets in $\mathbb{A}_K^n$ and of ideals in $K[X_1, \dots, X_n]$. Show the following.

1. For a finite field, every ascending chain

$$V_0 \subseteq V_1 \subseteq V_2 \subseteq \dots$$

of affine algebraic sets becomes stationary.

2. For an infinite field and $n \geq 1$, not every ascending chain of affine algebraic sets

$$V_0 \subseteq V_1 \subseteq V_2 \subseteq \dots$$

becomes stationary.

3. For any field and $n \geq 1$, not every descending chain of ideals

$$\mathfrak{a}_0 \supseteq \mathfrak{a}_1 \supseteq \mathfrak{a}_2 \supseteq \dots$$

becomes stationary.

4. For an infinite field and $n \geq 1$, there are strictly descending chains of affine algebraic sets of arbitrary length.

Exercise 9.15

Show that the set $\mathbb{R}$ of real numbers with its metric topology is not a Noetherian topological space.

Exercise 9.16

Let G be a commutative group. Show that there is exactly one way to give G the structure of a $\mathbb{Z}$ -module. Thus commutative groups and $\mathbb{Z}$ -modules are equivalent objects.

Exercise 9.17

Let R and A be commutative rings. Show that A is an R -algebra if and only if A is an R -module additionally satisfying

$$r(ab) = (ra)b \quad \text{for all } r \in R, a, b \in A.$$

Exercise 9.18 ★

Let V be a module over the commutative ring R . Let

$$s_1, \dots, s_k \in R \quad \text{and} \quad v_1, \dots, v_n \in V.$$

Prove that

$$\left(\sum_{i=1}^k s_i \right) \cdot \left(\sum_{j=1}^n v_j \right) = \sum_{1 \leq i \leq k, 1 \leq j \leq n} s_i \cdot v_j.$$

Exercise 9.19

Let R be a commutative ring, M and N two R -modules, and $\varphi : M \rightarrow N$ a module homomorphism. Prove the following statements.

1. If $S \subseteq M$ is an R -submodule, then its image $\varphi(S)$ is a submodule of N .
2. In particular, the image of the map

$$\text{bild } \varphi = \varphi(M)$$

is a submodule of N .

3. If $T \subseteq N$ is a submodule, then the preimage

$$\varphi^{-1}(T)$$

is a submodule of M .

4. In particular, the kernel

$$\varphi^{-1}(0)$$

is a submodule of M .

Exercises for submission**Exercise 9.20 (3 points)**

Let R be a commutative ring and $\mathfrak{a}$ an ideal with quotient ring

$$S = R/\mathfrak{a}.$$

Show that the ideals of S correspond uniquely to the ideals of R containing $\mathfrak{a}$. Show that the same holds for prime ideals, radical ideals, and maximal ideals.

Exercise 9.21 (4 points)

Let R be a Noetherian integral domain. Show that every nonzero nonunit of R can be written as a product of irreducible elements.

Edition note. The source says “every element”. Zero and units must be excluded from that formulation; equivalently, every nonzero element is a unit times a finite product of irreducibles, with the empty product allowed.

Exercise 9.22 (4 points)

Show that $\mathbb{Q}$ is not an algebra of finite type over $\mathbb{Z}$.

Exercise 9.23 (4 points)

Let K be a field and $A = K[X, Y]$. Find a K -subalgebra of A that is not finitely generated.

Exercise 9.24 (4 points)

For the ideal

$$I = (10, 6x^2 + 8, 4x^3 - 12)$$

in $\mathbb{Z}[x]$, determine the chain of ideals constructed in the proof of Hilbert’s basis theorem and the corresponding generating set for I . Write the original generators as linear combinations of the constructed generating set.

Public Solutions to Worksheet 9

At the frozen revision boundary, the source provides public solutions only for Exercises 9.6, 9.13, and 9.18. No additional solutions have been created for this edition.

Solution to Exercise 9.6

Consider

$$S = K[X, Y]$$

as the polynomial ring in two variables over a field K . By Corollary 9.6, this ring is Noetherian. Within it, consider the subring

$$R = \{Xg(X, Y) + c \mid g \in K[X, Y], c \in K\}$$

and the chain of ideals in that subring

$$\mathfrak{a}_n = (X, XY, \dots, XY^n).$$

For every $n \in \mathbb{N}$, we have

$$XY^{n+1} \in \mathfrak{a}_{n+1} \setminus \mathfrak{a}_n,$$

so the chain does not become stationary. By Proposition 9.2, R is therefore not Noetherian.

[Back to Exercise 9.6.](#)

Solution to Exercise 9.13

We have

$$A = R[f_1, \dots, f_n] \subseteq R[a_i, i \in I].$$

Each f_j can be written as a polynomial expression in the elements of the family a_i , with coefficients in R . For each j , only finitely many a_i occur. Hence all the generators f_j belong to

$$R[a_i, i \in I']$$

for some finite subfamily $I' \subseteq I$. Thus

$$A = R[f_1, \dots, f_n] \subseteq R[a_i, i \in I'] \subseteq A,$$

and therefore $A = R[a_i, i \in I']$.

[Back to Exercise 9.13.](#)

Solution to Exercise 9.18

We prove the statement by double induction on $k, n \geq 1$. The cases

$$(k, n) = (1, 1), \quad (1, 2), \quad (2, 1)$$

are immediately clear or follow directly from the module axioms.

For $k = 1$ and arbitrary n , we prove the statement by induction on n , with the base case supplied by the preceding observation. Suppose the statement has been proved for some n , and let $n + 1$ vectors $v_1, \dots, v_n, v_{n+1} \in V$ be given. Using the case $(1, 2)$ and the induction hypothesis, we obtain

$$\begin{aligned} s \cdot \left(\sum_{j=1}^{n+1} v_j \right) &= s \cdot \left(\sum_{j=1}^n v_j + v_{n+1} \right) \\ &= s \cdot \left(\sum_{j=1}^n v_j \right) + s v_{n+1} \\ &= \sum_{1 \leq j \leq n} s \cdot v_j + s v_{n+1} \\ &= \sum_{1 \leq j \leq n+1} s \cdot v_j. \end{aligned}$$

Now consider the statement for fixed k and arbitrary n . For $k = 1$, it has already been proved. Suppose it has been proved for some fixed k . Let scalars

$$s_1, \dots, s_k, s_{k+1} \in R$$

and vectors

$$v_1, \dots, v_n \in V$$

be given. Using the cases $(2, 1)$ and $(1, n)$ and the induction hypothesis, we obtain

$$\begin{aligned}
\left(\sum_{i=1}^{k+1} s_i\right) \cdot \left(\sum_{j=1}^n v_j\right) &= \left(\sum_{i=1}^k s_i + s_{k+1}\right) \cdot \left(\sum_{j=1}^n v_j\right) \\
&= \left(\sum_{i=1}^k s_i\right) \cdot \left(\sum_{j=1}^n v_j\right) + s_{k+1} \cdot \left(\sum_{j=1}^n v_j\right) \\
&= \sum_{1 \leq i \leq k, 1 \leq j \leq n} s_i \cdot v_j + \sum_{j=1}^n s_{k+1} \cdot v_j \\
&= \sum_{1 \leq i \leq k+1, 1 \leq j \leq n} s_i \cdot v_j.
\end{aligned}$$

Back to Exercise 9.18.

Lecture 10: Noetherian modules and Hilbert's Nullstellensatz

Noetherian modules

We want to show that, for a Noetherian ring R and a finitely generated R -module, every R -submodule is again finitely generated. Modules with this property are called Noetherian.

Definition: Noetherian module

Let R be a commutative ring and M an R -module. The module M is called *Noetherian* if every R -submodule of M is finitely generated.

For $M = R$, this agrees with the definition of a Noetherian ring, since the R -submodules of R are precisely its ideals.

In the following statements we use the following terminology and notation.

Definition: short exact sequence

Let R be a commutative ring and let M_1, M_2, M_3 be R -modules. A diagram of the form

$$0 \longrightarrow M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

is called a *short exact sequence* of R -modules if M_1 is an R -submodule of M_2 and M_3 is a quotient module of M_2 isomorphic to M_2/M_1 .

Exactness means that at every position we have

$$\ker \varphi_{i+1} = \operatorname{bild} \varphi_i,$$

where the φ_i denote the R -module homomorphisms in the sequence.

Lemma: the Noetherian property in a short exact sequence

Let R be a commutative ring and let

$$0 \longrightarrow M_1 \longrightarrow M \longrightarrow M_3 \longrightarrow 0$$

be a short exact sequence of R -modules. Then M is Noetherian if and only if both M_1 and M_3 are Noetherian.

Proof First suppose that M is Noetherian, and let $U \subseteq M_1$ be a submodule. Then U is also a submodule of M , so it is finitely generated by assumption. Now let $V \subseteq M_3$ be a submodule of the quotient module. Let $\widetilde{V}$ be the inverse image of V in M under the quotient map. By assumption, $\widetilde{V}$ is finitely generated, and the images of a generating set also generate the image module V .

Conversely, suppose that the two outer modules M_1 and M_3 are Noetherian, and let $U \subseteq M$ be a submodule. Let $U_3 \subseteq M_3$ be its image submodule. The module U_3 is generated by finitely many elements $s_1, \dots, s_n$, and we may assume that

$$s_i = \overline{r}_i$$

are the images of elements $r_i \in U$. Consider $U \cap M_1$. This is a submodule of M_1 , and hence finitely generated, say by $t_1, \dots, t_k$, which we regard as elements of U . We claim that

$$r_1, \dots, r_n, t_1, \dots, t_k$$

form a generating set for U . To see this, take an arbitrary $m \in U$. Then

$$\overline{m} = \sum_{i=1}^n a_i s_i.$$

Thus the element $m - \sum_{i=1}^n a_i r_i$ maps to 0 on the right. It belongs to the kernel of the quotient map, and hence to M_1 . On the other hand, this element also belongs to U , and therefore to the intersection $M_1 \cap U$, which is generated by $t_1, \dots, t_k$. We can therefore write

$$m - \sum_{i=1}^n a_i r_i = \sum_{j=1}^k b_j t_j,$$

or, equivalently,

$$m = \sum_{i=1}^n a_i r_i + \sum_{j=1}^k b_j t_j.$$

Theorem: finite modules over a Noetherian ring

Let R be a Noetherian commutative ring and M a finitely generated R -module. Then M is a Noetherian module.

Proof We prove the statement by induction on the number n of module generators of M . For $n = 0$, we have the zero module. Let $n = 1$. Then there is a surjective map

$$R \longrightarrow M \cong R/\mathfrak{a}.$$

By Lemma 10.3, a quotient module of a Noetherian module is again Noetherian. Since the ring R itself is Noetherian by assumption, M is also Noetherian.

Now let $n \geq 2$ and suppose the statement has been proved for smaller values. Let $m_1, \dots, m_n$ be a generating set for M . Denote by M_1 the R -submodule generated by $m_1, \dots, m_{n-1}$. This submodule gives rise to a short exact sequence

$$0 \longrightarrow M_1 \longrightarrow M \longrightarrow M/M_1 =: M_3 \longrightarrow 0.$$

The module on the left is generated by $n - 1$ elements and is Noetherian by the induction hypothesis. The module on the right is generated by the residue class of m_n , hence by one element, and is therefore also Noetherian. By Lemma 10.3, M is Noetherian as well.

Hilbert's Nullstellensatz — algebraic version

For an R -algebra A , both terms *finite* and *finitely generated* will be important. The former means that A , regarded as an R -module, is finitely generated; the latter means that A is finitely generated as an algebra. The polynomial ring $R[X_1, \dots, X_n]$ is finitely generated in the second sense: the variables form a finite set of algebra generators. However, the polynomial ring is not finitely generated as a module; the simplest set of module generators consists of all monomials.

We want to prove the algebraic version of Hilbert's Nullstellensatz. For this we need the following two lemmas.

Lemma: subalgebras beneath a finite extension

Let R be a Noetherian commutative ring and A a finitely generated R -algebra. Let $B \subseteq A$ be an R -subalgebra over which A is finite (as a B -module). Then B is also a finitely generated R -algebra.

Proof We write

$$A = R[x_1, \dots, x_n]$$

and

$$A = Ba_1 + \dots + Ba_m,$$

with $a_i \in A$. *Edition supplement:* adjoin 1 to this finite module-generating family if necessary and take $a_1 = 1$. This ensures that the module $\tilde{A}$ constructed below contains 1; the source's claim that it is an S -algebra requires this step. Set

$$x_i = \sum_{j=1}^m b_{ij}a_j \quad \text{and} \quad a_i a_j = \sum_{k=1}^m b_{ijk}a_k,$$

with coefficients $b_{ij}, b_{ijk} \in B$. Consider the R -subalgebra S of B generated by these coefficients and the S -submodule

$$\tilde{A} = Sa_1 + \dots + Sa_m \subseteq A.$$

The products $a_i a_j$ again belong to this module, so $\tilde{A}$ is even an S -algebra. Since all the x_i also belong to $\tilde{A}$, we obtain $A = \tilde{A}$. This means that A is a finite S -module. By Corollary 9.9, S is a Noetherian ring; by Theorem 10.4, the S -submodule

$$B \subseteq A$$

is also a finite S -module. Finally, the chain

$$R \subseteq S \subseteq B$$

shows that B is a finitely generated R -algebra.

Lemma: the rational function field is not finitely generated

Let K be a field and $R = K(X)$ the associated rational function field. Then R is not a finitely generated K -algebra.

Proof Suppose that the rational functions

$$F_i = \frac{P_i}{Q_i}, \quad i = 1, \dots, n,$$

with $P_i, Q_i \in K[X]$ and $Q_i \neq 0$, form a finite generating set for $K(X)$. By passing to a common denominator, we may assume that all the denominators agree, namely $Q_i = Q$. In particular, the assumption says that the rational function field could be obtained by localising at just one element. Since Q is not constant (otherwise $K[X] = K(X)$, which is false), we have $Q - 1 \neq 0$, and hence

$$\frac{1}{Q-1} \in K(X).$$

There is therefore a representation

$$\frac{1}{Q-1} = \frac{P}{Q^s}$$

for a suitable s . Consequently,

$$Q^s = (Q-1)P.$$

Since Q^s and $Q-1$ generate the unit ideal in $K[X]$, this equality implies that $Q-1$ alone already generates the unit ideal, so $Q-1$ is a unit. But this would mean that Q is constant, a contradiction.

The following statement is the algebraic version of Hilbert's Nullstellensatz.

Theorem: Hilbert's Nullstellensatz, algebraic version

Let K be a field and $K \subseteq L$ a field extension that is finitely generated as a K -algebra. Then L is finite over K .

Proof Set

$$L = K[x_1, \dots, x_n].$$

Let K_i be the field of fractions of $K[x_1, \dots, x_i]$ inside L . Thus we have a chain of fields

$$K = K_0 \subseteq K_1 \subseteq \dots \subseteq K_n = L.$$

We want to show that L is finite over K . By Theorem 2.8 of the course Field and Galois Theory (Osnabrück 2018–2019), it suffices to show that each step in this chain of fields is finite. Suppose that $K_i \subseteq K_{i+1}$ is not finite, but all subsequent steps are finite. Apply Lemma 10.5 to

$$K \subseteq K_{i+1} \subset L.$$

We obtain that K_{i+1} is finitely generated over K . In particular, K_{i+1} is also finitely generated over K_i . On the other hand, K_{i+1} is the field of fractions of $K_i[x_{i+1}]$. Thus we have a chain

$$K_i \subseteq K_i[x_{i+1}] \subseteq Q(K_i[x_{i+1}]) = K_{i+1},$$

where K_{i+1} is finitely generated over K_i , but not finite. If x_{i+1} were algebraic over K_i , it would also be finite, and by Exercise 10.1, $K_i[x_{i+1}]$ would already be a field. The whole of the last chain would then be finite, contrary to the choice of i . Thus x_{i+1} is transcendental over K_i . But then $K_i[x_{i+1}]$ is isomorphic to a polynomial ring in one variable, and $Q(K_i[x_{i+1}])$ is isomorphic to the rational function field over K_i . By Lemma 10.6, this field is not finitely generated, again a contradiction.

Theorem: inverse images of maximal ideals

Let K be a field and A, B two K -algebras of finite type. Let

$$\varphi : A \longrightarrow B$$

be a K -algebra homomorphism. Then, for every maximal ideal $\mathfrak{m}$ of B , its inverse image $\varphi^{-1}(\mathfrak{m})$ is also a maximal ideal.

Proof Let $\mathfrak{m}$ be a maximal ideal of B . From Exercise 4.19 we know that the inverse image of a prime ideal under any ring homomorphism is again prime. Thus $\varphi^{-1}(\mathfrak{m})$ is initially a prime ideal; call it $\mathfrak{p}$. We obtain induced ring homomorphisms

$$K \longrightarrow A/\mathfrak{p} \longrightarrow B/\mathfrak{m} = L,$$

where L is a field and both homomorphisms are injective and of finite type. Since the composite map is of finite type and K, L are both fields, Theorem 10.7 says that this map is finite. We want to show that the intermediate ring $A/\mathfrak{p}$ is a field. This follows from Exercise 10.2.

Theorem: radicals as intersections of maximal ideals

Let K be a field and A a K -algebra of finite type. Then every radical ideal in A is an intersection of maximal ideals.

Proof By Exercise 10.17, every radical ideal is an intersection of prime ideals. It therefore suffices to show that every prime ideal in a finitely generated algebra is an intersection of maximal ideals. Let $\mathfrak{p}$ be a prime ideal and $f \notin \mathfrak{p}$. The ideal $\mathfrak{p}$ remains a prime ideal in the localisation

$$B := A_f.$$

In A_f there is a maximal ideal $\mathfrak{m} \subset A_f$ containing $\mathfrak{p}A_f$. Regard A_f as a finitely generated K -algebra and consider

$$\varphi : A \longrightarrow A_f.$$

We have

$$\mathfrak{p} \subseteq \varphi^{-1}(\mathfrak{m}) \quad \text{and} \quad f \notin \varphi^{-1}(\mathfrak{m}).$$

By Theorem 10.8, $\varphi^{-1}(\mathfrak{m})$ is maximal.

Theorem: maximal ideals are point ideals

Let K be an algebraically closed field and A a finitely generated K -algebra. Then every quotient of A by a maximal ideal is isomorphic to K . In other words, every maximal ideal of A is a point ideal.

Proof Let $\mathfrak{m}$ be a maximal ideal of the finitely generated K -algebra A , and consider

$$K \longrightarrow A \longrightarrow A/\mathfrak{m} =: L.$$

Here L is both a field and a finitely generated K -algebra. By Theorem 10.7, L must be a finite K -algebra. Since K is algebraically closed, we must have $K = L$.

Worksheet 10

Practice exercises

Exercise 10.1 ★

Let K be a field and A a commutative K -algebra that is finite as a K -module. Show that an element $f \in A$ is a unit if and only if it is a non-zero-divisor.

Exercise 10.2

Let K and L be fields, let $K \subseteq L$ be a finite field extension, and let A be an intermediate ring,

$$K \subseteq A \subseteq L.$$

Show that A is also a field.

Exercise 10.3

Let $R \subseteq S$ be a finite ring extension and $f \in R$. Show that if f , regarded as an element of S , is a unit, then f is a unit in R .

Exercise 10.4

Let R be a commutative ring and M an R -module. Show that M is Noetherian if and only if every ascending chain of R -submodules

$$M_0 \subseteq M_1 \subseteq M_2 \subseteq \dots$$

becomes stationary.

Exercise 10.5

Let $f \in R$ be a non-zero-divisor in a commutative ring R . Show that this gives a short exact sequence of R -modules

$$0 \longrightarrow R \xrightarrow{f} R \longrightarrow R/(f) \longrightarrow 0.$$

Exercise 10.6 ★

Let R be a commutative ring and $I, J \subseteq R$ ideals. Show that the sequence

$$0 \longrightarrow R/(I \cap J) \longrightarrow R/I \times R/J \longrightarrow R/(I + J) \longrightarrow 0$$

with maps $r \mapsto (r, r)$ and $(s, t) \mapsto s - t$ is exact.

Exercise 10.7

Let R be a commutative ring and N an R -module with R -submodules

$$L \subseteq M \subseteq N.$$

Show that the quotient modules are related by the short exact sequence

$$0 \longrightarrow M/L \longrightarrow N/L \longrightarrow N/M \longrightarrow 0.$$

Exercise 10.8

Let R be a commutative ring and

$$\varphi : M \longrightarrow N$$

an R -module homomorphism between R -modules M and N . Show that this gives a short exact sequence

$$0 \longrightarrow \ker \varphi \longrightarrow M \longrightarrow \operatorname{bild} \varphi \longrightarrow 0.$$

Let R be a commutative ring and M an R -module. The R -module

$$M^* = \operatorname{Hom}_R(M, R)$$

is called the *dual module* of M .

Exercise 10.9 ★

Let R be a commutative ring and let

$$0 \longrightarrow L \longrightarrow M \longrightarrow N \longrightarrow 0$$

be a short exact sequence of R -modules L, M, N . Show that this gives an exact sequence of dual modules

$$0 \longrightarrow N^* \longrightarrow M^* \longrightarrow L^*.$$

Exercise 10.10

Let K be a field and let

$$0 \longrightarrow L \longrightarrow M \longrightarrow N \longrightarrow 0$$

be a short exact sequence of K -vector spaces L, M, N . Show that this gives a short exact sequence of dual spaces

$$0 \longrightarrow N^* \longrightarrow M^* \longrightarrow L^* \longrightarrow 0.$$

Exercise 10.11

Let $a \neq 0$ be an integer. We consider the short exact sequence of $\mathbb{Z}$ -modules

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot a} \mathbb{Z} \longrightarrow \mathbb{Z}/(a) \longrightarrow 0.$$

Show that, for $a \geq 2$, the sequence that is exact by Exercise 10.9,

$$0 \longrightarrow (\mathbb{Z}/(a))^* \longrightarrow \mathbb{Z}^* \longrightarrow \mathbb{Z}^*,$$

cannot be extended exactly to the right by $\longrightarrow 0$.

Exercise 10.12

Let R be a commutative ring and let

$$0 \longrightarrow L \longrightarrow M \longrightarrow N \longrightarrow 0$$

be a short exact sequence of R -modules. Suppose that L has a set of R -module generators with k elements and N has a set of R -module generators with n elements. Show that M has a set of R -module generators with $k + n$ elements.

Exercise 10.13

Let R be a commutative ring, A a commutative finite R -algebra, and M a finite A -module. Show that M is also a finite R -module.

The following exercises use the notion of an Artinian module, which is “dual” to the notion of a Noetherian module.

Let R be a commutative ring. An R -module M is called *Artinian* if every descending chain of R -submodules

$$M_1 \supseteq M_2 \supseteq M_3 \supseteq \cdots$$

becomes stationary. A commutative ring R is called *Artinian* if it is Artinian as an R -module.

Exercise 10.14

Let A be an Artinian integral domain. Show that A is a field. Give an example of an Artinian commutative ring that is not a field.

Exercise 10.15

Let R be a commutative ring and M an R -module. Show that if M is Artinian and

$$\phi : M \longrightarrow M$$

is R -linear and injective, then ϕ is an isomorphism. Also formulate and prove an analogous statement for the case where M is Noetherian.

Exercise 10.16 ★

Let R be a commutative ring and let $f \in R$ be non-nilpotent. Show that there is a prime ideal $\mathfrak{p}$ with $f \notin \mathfrak{p}$.

Exercise 10.17 ★

Let $\mathfrak{a}$ be a radical ideal in a commutative ring. Show that $\mathfrak{a}$ is an intersection of prime ideals.

One approach follows from the preceding exercise; another follows from Exercise 13.5 below.

Exercise 10.18

Let K be a field and $P \in K[X]$ a nonconstant polynomial. Show that P is not algebraic over K .

Exercise 10.19

Let K be a field and $L = K(X)$ the field of fractions of the polynomial ring $K[X]$. Let M be an intermediate field with

$$K \subseteq M \subseteq L, \quad M \neq K.$$

Show that $M \subseteq L$ is a finite field extension.

Exercise 10.20 ★

Let A be a finitely generated $\mathbb{Z}$ -algebra and $\mathfrak{m} \subseteq A$ a maximal ideal. Show that the quotient ring $A/\mathfrak{m}$ is a finite field.

Let K be a field and A a commutative K -algebra. Elements $f_1, \dots, f_n \in A$ are called *algebraically dependent* if there is a nonzero polynomial $P \in K[X_1, \dots, X_n]$ such that

$$P(f_1, \dots, f_n) = 0.$$

Exercise 10.21

Let $K[X_1, \dots, X_n]$ be the polynomial ring over a field K . Show that the variables $X_1, \dots, X_n$ are algebraically independent.

Exercise 10.22

Let $K[X_1, \dots, X_n]$ be the polynomial ring over a field K , and let $n + 1$ polynomials

$$f_1, \dots, f_{n+1} \in K[X_1, \dots, X_n]$$

be given. Show that these polynomials are algebraically dependent.

Exercise 10.23

Let

$$\varphi : \mathbb{A}_K^m \longrightarrow \mathbb{A}_K^n$$

be a polynomial map between affine spaces with $m < n$. Show that φ is not surjective.

Exercise 10.24

Let A be a commutative K -algebra over a field K , and let n elements $f_1, \dots, f_n \in A$ be given. Show that these elements are algebraically independent if and only if the K -algebra they generate, $K[f_1, \dots, f_n]$, is isomorphic to the polynomial ring $K[X_1, \dots, X_n]$.

Exercises for submission**Exercise 10.25 - 3 points**

Let K be an algebraically closed field and $F \in K[X, Y]$ a nonconstant polynomial. Show that the quotient ring

$$K[X, Y]/(F)$$

can be regarded as a finite $K[T]$ -algebra.

Exercise 10.26 - 3 points

Let R, S, T be commutative rings, and let $\varphi : R \rightarrow S$ and $\psi : S \rightarrow T$ be ring homomorphisms such that S is finite over R and T is finite over S . Show that T is also finite over R .

Exercise 10.27 - 5 points

Let A be a commutative ring and let

$$0 \longrightarrow M \longrightarrow N \longrightarrow P \longrightarrow 0$$

be a short exact sequence of A -modules. Show that N is Artinian if and only if both M and P are Artinian.

Exercise 10.28 - 4 points (1+3)

Let R be a commutative ring, and let $M_i, i \in \mathbb{N}$, be R -modules with fixed R -module homomorphisms

$$\varphi_i : M_i \longrightarrow M_{i+1}.$$

The sequence

$$\cdots \longrightarrow M_i \longrightarrow M_{i+1} \longrightarrow M_{i+2} \longrightarrow M_{i+3} \longrightarrow \cdots$$

is called *exact* if, for every i ,

$$\text{Kern}(\varphi_i) = \text{Bild}(\varphi_{i-1}).$$

1. Show that, in the case of a short exact sequence, this definition agrees with Definition 10.2 in the lecture.
2. Now suppose that the sequence is exact, $R = K$ is a field, all the M_i are finitely generated, $M_0 = 0$, and $M_i = 0$ for all $i \geq n$ for some n . Show that

$$\sum_{i=0}^n (-1)^i \dim_K M_i = 0.$$

Edition note. Part 2 requires the sequence to be exact. The source defines this property immediately beforehand but does not repeat it as a hypothesis; it is made explicit here.

Exercise 10.29 - 3 points

Let K be a field and A a finite K -algebra. Show that A is Artinian.

Public Solutions to Worksheet 10

At the frozen revision boundary, the source provides public solutions only to Exercises 10.1, 10.6, 10.9, 10.16, 10.17, and 10.20. No additional solutions have been created for this edition.

Solution to Exercise 10.1

If f is a unit, there is a $g \in A$ with $gf = 1$. From $fh = 0$ we immediately obtain

$$h = gfh = g0 = 0.$$

Thus f is a non-zero-divisor.

Conversely, if f is a non-zero-divisor, consider the K -linear multiplication map

$$\mu_f : A \longrightarrow A, \quad h \longmapsto fh.$$

This map is injective. Since A is finite as a module over the field K , it is a finite-dimensional K -vector space. An injective endomorphism of a finite-dimensional vector space is also surjective. In particular, there is a $g \in A$ with $fg = 1$. This means that f is a unit.

[Back to Exercise 10.1.](#)

Solution to Exercise 10.6

Choose a representative $r \in R$ of a class in $R/(I \cap J)$ that maps to $(r, r) = 0$ in $R/I \times R/J$. Both components are 0, so $r \in I$ and $r \in J$. Thus $r \in I \cap J$, and hence the class of r on the left is 0. The map on the left is therefore injective.

The composite map is given by

$$r \longmapsto (r, r) \longmapsto r - r,$$

so it is the zero map. Conversely, if (s, t) maps to 0 on the right, then $s - t \in I + J$. Say

$$s - t = a + b, \quad a \in I, \quad b \in J.$$

Then

$$s - a = t + b$$

in R . This element also represents (s, t) , so (s, t) comes from the left. Surjectivity of the last map follows immediately by choosing $t = 0$.

[Back to Exercise 10.6.](#)

Solution to Exercise 10.9

Surjectivity of the map $M \rightarrow N$ immediately implies that the map

$$N^* \longrightarrow M^*$$

is injective: an R -linear map $N \rightarrow R$ whose composite with $M \rightarrow N$ is the zero map must itself be the zero map.

Since the composite $L \rightarrow M \rightarrow N$ is the zero map, the same holds for the corresponding dual map.

It remains to show that a linear form $f \in M^*$ that maps to 0 in L^* comes from a dual form in N^* . This condition says that the restriction of f to the submodule $L \subseteq M$ is the zero map. In other words, L is contained in the kernel of f . By the homomorphism theorem, there is an induced homomorphism

$$\tilde{f} : M/L \longrightarrow R$$

whose composite with $M \rightarrow M/L$ equals f . Since $M/L \cong N$, this is the desired statement.

[Back to Exercise 10.9.](#)

Solution to Exercise 10.16

Consider the set of ideals

$$\mathcal{M} = \{\mathfrak{a} \text{ an ideal} \mid f^r \notin \mathfrak{a} \text{ for every } r\}.$$

This set is nonempty because it contains the zero ideal. Moreover, $\mathcal{M}$ is inductively ordered by inclusion. Indeed, if the $\mathfrak{a}_i, i \in I$, form a totally ordered subset of $\mathcal{M}$, then their union is also an ideal containing no power of f . By Zorn's lemma, $\mathcal{M}$ therefore has a maximal element.

We claim that every such maximal element, say $\mathfrak{p}$, is a prime ideal. Take $g, h \in R$ with $gh \in \mathfrak{p}$, and suppose that $g, h \notin \mathfrak{p}$. Then there are strict inclusions

$$\mathfrak{p} \subsetneq \mathfrak{p} + (g), \quad \mathfrak{p} \subsetneq \mathfrak{p} + (h).$$

Since $\mathfrak{p}$ is maximal in $\mathcal{M}$, neither ideal on the right belongs to $\mathcal{M}$. Thus there are $r, s \in \mathbb{N}$ such that

$$f^r \in \mathfrak{p} + (g) \quad \text{and} \quad f^s \in \mathfrak{p} + (h).$$

But multiplying these two relations gives the contradiction

$$f^{r+s} \in \mathfrak{p} + (gh) \subseteq \mathfrak{p}.$$

Thus $\mathfrak{p}$ is prime and, by the definition of $\mathcal{M}$, $f \notin \mathfrak{p}$.

[Back to Exercise 10.16.](#)

Solution to Exercise 10.17

Edition note. The source directly equates ideals in R with ideals in $R/\mathfrak{a}$. The argument below makes the required inverse-image correspondence explicit.

Let $\mathfrak{a} \subseteq R$ be a radical ideal. Then $\mathfrak{a} = \sqrt{\mathfrak{a}}$. The nilradical of $R/\mathfrak{a}$ is the intersection of all prime ideals in this quotient ring. Under the correspondence between ideals of $R/\mathfrak{a}$ and ideals of R containing $\mathfrak{a}$, this gives

$$\sqrt{\mathfrak{a}} = \bigcap_{\substack{\mathfrak{p} \supseteq \mathfrak{a} \\ \mathfrak{p} \text{ prime}}} \mathfrak{p}.$$

Since $\mathfrak{a} = \sqrt{\mathfrak{a}}$, we obtain

$$\mathfrak{a} = \bigcap_{\substack{\mathfrak{p} \supseteq \mathfrak{a} \\ \mathfrak{p} \text{ prime}}} \mathfrak{p}.$$

[Back to Exercise 10.17.](#)

Solution to Exercise 10.20

Consider the composite map

$$\mathbb{Z} \xrightarrow{\varphi} A \longrightarrow A/\mathfrak{m} = L,$$

which is also of finite type. The inverse image $\varphi^{-1}(\mathfrak{m})$ is a prime ideal in $\mathbb{Z}$, so it is either (0) or (p) for a prime number p .

In the first case there is a factorisation

$$\mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow L.$$

By Hilbert's Nullstellensatz, L is finite over $\mathbb{Q}$, and by Lemma 10.5, $\mathbb{Q}$ would then have to be finitely generated over $\mathbb{Z}$, which is not the case. Thus the first case is impossible.

Consequently, the second case holds and there is a factorisation

$$\mathbb{Z} \longrightarrow \mathbb{Z}/(p) \longrightarrow L.$$

By Hilbert's Nullstellensatz, L is finite over the finite field $\mathbb{Z}/(p)$, so L itself is finite.

[Back to Exercise 10.20.](#)

Lecture 11: Hilbert's Nullstellensatz and coordinate rings

Hilbert's Nullstellensatz - geometric version

We now prove the geometric version of Hilbert's Nullstellensatz. For an algebraically closed field, this theorem establishes a one-to-one relationship between affine algebraic sets in affine space $\mathbb{A}_K^n$ and radical ideals in the polynomial ring.

Theorem: the geometric Hilbert Nullstellensatz

Let K be an algebraically closed field and let

$$V = V(\mathfrak{a}) \subseteq \mathbb{A}_K^n$$

be an affine algebraic set described by the ideal $\mathfrak{a}$. Let

$$F \in K[X_1, \dots, X_n]$$

be a polynomial vanishing on V . Then F belongs to the radical of $\mathfrak{a}$; that is, there is an $r \in \mathbb{N}$ with

$$F^r \in \mathfrak{a}.$$

Proof Suppose that F does not belong to the radical of $\mathfrak{a}$. By Theorem 10.9, there is a maximal ideal

$$\mathfrak{m} \subset K[X_1, \dots, X_n]$$

with $\mathfrak{a} \subseteq \mathfrak{m}$ and $F \notin \mathfrak{m}$. By Theorem 10.10,

$$K[X_1, \dots, X_n]/\mathfrak{m} = K,$$

and hence

$$\mathfrak{m} = (X_1 - a_1, \dots, X_n - a_n)$$

for some $a_1, \dots, a_n \in K$. The property $F \notin \mathfrak{m}$ means that F is nonzero in the corresponding residue field, namely

$$F(a_1, \dots, a_n) \neq 0.$$

But since $\mathfrak{a} \subseteq \mathfrak{m}$, the point $(a_1, \dots, a_n)$ belongs to V . By assumption, F must vanish at that point. This is a contradiction.

Theorem: correspondence between algebraic sets and radical ideals

Let K be an algebraically closed field, with polynomial ring $K[X_1, \dots, X_n]$ and affine space $\mathbb{A}_K^n$. There is a natural correspondence between affine algebraic sets in $\mathbb{A}_K^n$ and radical ideals in $K[X_1, \dots, X_n]$.

In this correspondence, a radical ideal is sent to its zero locus, and an affine algebraic set is sent to its vanishing ideal.

Proof Let $V \subseteq \mathbb{A}_K^n$ be affine algebraic. By part (3) of Lemma 3.8,

$$V = V(\text{Id}(V)).$$

For a radical ideal $I \subseteq K[X_1, \dots, X_n]$, part (2) of the same lemma gives the inclusion

$$I \subseteq \text{Id}(V(I)).$$

The reverse inclusion,

$$\text{Id}(V(I)) \subseteq I,$$

is the content of Hilbert's Nullstellensatz.

Corollary: covers by principal open sets

Let K be an algebraically closed field and let

$$F_i \in K[X_1, \dots, X_n], \quad i \in I,$$

be polynomials such that

$$\mathbb{A}_K^n = \bigcup_{i \in I} D(F_i).$$

Then the F_i generate the unit ideal in $K[X_1, \dots, X_n]$.

Proof Let $\mathfrak{b}$ be the ideal generated by all the F_i . The assumption says that

$$\bigcap_{i \in I} V(F_i) = V(\mathfrak{b})$$

is empty. Thus $V(\mathfrak{b}) \subseteq V(1)$. By Hilbert's Nullstellensatz, a power of 1, hence 1 itself, belongs to $\mathfrak{b}$. Therefore $\mathfrak{b}$ is the unit ideal.

Corollary: the vanishing ideal of an intersection

Let K be an algebraically closed field and V_1, V_2 two affine algebraic sets in $\mathbb{A}_K^n$. Then

$$\text{Id}(V_1 \cap V_2) = \text{rad}(\text{Id}(V_1) + \text{Id}(V_2)).$$

Proof Let

$$\mathfrak{a}_1 = \text{Id}(V_1), \quad \mathfrak{a}_2 = \text{Id}(V_2).$$

The statement follows from

$$\begin{aligned} \text{rad}(\mathfrak{a}_1 + \mathfrak{a}_2) &= \text{Id}(V(\text{rad}(\mathfrak{a}_1 + \mathfrak{a}_2))) \\ &= \text{Id}(V(\mathfrak{a}_1 + \mathfrak{a}_2)) \\ &= \text{Id}(V(\mathfrak{a}_1) \cap V(\mathfrak{a}_2)), \end{aligned}$$

where the first equality comes from Theorem 11.1.

These properties also fail without the assumption that the field is algebraically closed, as the following example shows.

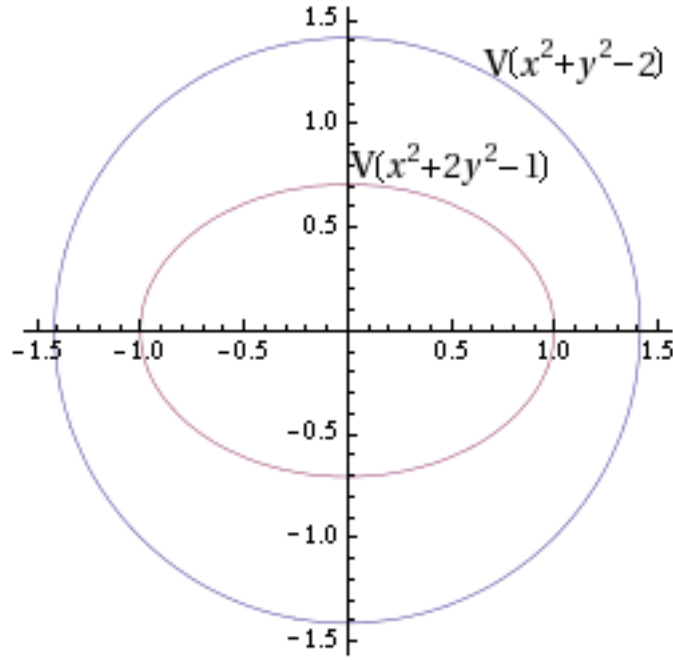


Figure 61: Two disjoint ellipses

Two disjoint ellipses; Pmidden, created with Mathematica; public domain.

Example: two disjoint real quadrics

We consider the two algebraic curves

$$V_1 = V(X^2 + Y^2 - 2), \quad V_2 = V(X^2 + 2Y^2 - 1) \subseteq \mathbb{A}_K^2.$$

For $K = \mathbb{R}$, both are irreducible quadrics. Their intersection is described by the ideal

$$(X^2 + Y^2 - 2, X^2 + 2Y^2 - 1) = (Y^2 + 1, X^2 - 3).$$

Since the polynomial $Y^2 + 1$ has no real zero,

$$V_1 \cap V_2 = \emptyset.$$

The vanishing ideal of the empty intersection is of course the unit ideal, whereas the sum of the two vanishing ideals is not the unit ideal.

The coordinate ring of an affine algebraic set

Let $V \subseteq \mathbb{A}_K^n$ be an affine algebraic set with vanishing ideal $\text{Id}(V)$. Every polynomial

$$F \in K[X_1, \dots, X_n]$$

defines a function on affine space and thus induces a function on the subset V :

$$\begin{array}{ccc} \mathbb{A}_K^n & \xrightarrow{F} & K \\ \uparrow & \nearrow & \\ V & & \end{array}$$

By Definition 3.4, an element of the vanishing ideal induces the zero function on V . Two polynomials $G, H \in K[X_1, \dots, X_n]$ whose difference belongs to the vanishing ideal induce the same function on V . It is therefore natural to regard the quotient ring

$$K[X_1, \dots, X_n]/\text{Id}(V)$$

as the ring of polynomial (or algebraic) functions on V .

Definition: coordinate ring

For an affine algebraic set

$$V \subseteq \mathbb{A}_K^n$$

with vanishing ideal $\text{Id}(V)$, the ring

$$R(V) = K[X_1, \dots, X_n]/\text{Id}(V)$$

is called the *coordinate ring* of V .

This notion is not entirely unproblematic, especially when K is not algebraically closed; see the examples below. We first record some elementary properties.

Proposition: basic properties of coordinate rings

Let $V \subseteq \mathbb{A}_K^n$ be an affine algebraic set and

$$R = K[X_1, \dots, X_n]/\text{Id}(V)$$

its coordinate ring. Then the following statements hold.

1. R is reduced.
2. $V = \emptyset$ if and only if R is the zero ring.
3. V is irreducible if and only if R is an integral domain.
4. V consists of a single point if and only if $R = K$.
5. If K is algebraically closed and $V = V(\mathfrak{a})$, then

$$R = K[X_1, \dots, X_n]/\text{rad}(\mathfrak{a}).$$

Proof Let $I = \text{Id}(V)$ be the vanishing ideal of V .

1. This follows from Lemma 3.14 and Exercise 3.13.
2. The equality $V = \emptyset$ is equivalent to $1 \in I$, which in turn is equivalent to $R = 0$.
3. This follows from Lemma 4.3 and Exercise 4.17.
4. Let $V = \{P\}$ with $P = (a_1, \dots, a_n)$. Then

$$I = (X_1 - a_1, \dots, X_n - a_n)$$

and its coordinate ring is

$$K[X_1, \dots, X_n]/(X_1 - a_1, \dots, X_n - a_n) \cong K.$$

Conversely, if the coordinate ring is K , the corresponding quotient homomorphism must be an evaluation homomorphism $X_i \mapsto a_i$. The vanishing ideal of V must be a point ideal, and $P = (a_1, \dots, a_n) \in V$. If

there were another point $Q \in V$, $Q \neq P$, we would obtain a contradiction, because not all the $X_i - a_i$ vanish at Q .

5. If K is algebraically closed, Hilbert's Nullstellensatz gives

$$\text{Id}(V) = \text{rad}(\mathfrak{a}).$$

Theorem: the coordinate ring of affine space over an infinite field

Let K be an infinite field. The vanishing ideal of affine space $\mathbb{A}_K^n$ is the zero ideal, and its coordinate ring is the polynomial ring $K[X_1, \dots, X_n]$.

Proof We prove the statement by induction on the number of variables. For $n = 1$, it follows from the fact that a polynomial of degree d has at most d roots.

For the induction step, let

$$F \in K[X_1, \dots, X_n]$$

be a polynomial vanishing at every point of $\mathbb{A}_K^n = K^n$. Write F as

$$F = P_d X_n^d + P_{d-1} X_n^{d-1} + \dots + P_1 X_n + P_0,$$

with

$$P_d, \dots, P_0 \in K[X_1, \dots, X_{n-1}].$$

We must show that $F = 0$, which is equivalent to $P_i = 0$ for every $i = 0, \dots, d$. Suppose, without loss of generality, that P_d is not the zero polynomial. By the induction hypothesis, P_d is not the zero function either. Thus there is a point $(a_1, \dots, a_{n-1})$ with

$$P_d(a_1, \dots, a_{n-1}) \neq 0.$$

Consequently, $F(a_1, \dots, a_{n-1})$ is a nonzero polynomial of degree d in the single variable X_n . By the one-variable case, it cannot be the zero function, a contradiction.

Edition note: The source TeX witness writes the linear term as $P_1 X_0$. The context of expansion in the variable X_n requires $P_1 X_n$; this edition uses the consistent index and preserves the source misprint in this note.

Example: finite fields

Theorem 11.8 is false for finite fields. Over a finite field, affine space consists of only finitely many points, and many polynomials vanish at all these points. Typical examples are the polynomials

$$X_i^q - X_i,$$

where q is the number of elements of the field.

Example: quotient rings and coordinate rings can differ

Let

$$R = \mathbb{R}[X, Y]/(X^2 + Y^2).$$

Since squares of real numbers are never negative, the zero locus of $X^2 + Y^2$ consists only of the origin:

$$V(X^2 + Y^2) = \{(0, 0)\}.$$

Its vanishing ideal is the maximal ideal (X, Y) , so the corresponding coordinate ring is

$$\mathbb{R}[X, Y]/(X, Y) \cong \mathbb{R}.$$

Thus the coordinate ring can be very different from the initial quotient ring whose ideal was used to define the zero locus.

Hilbert's Nullstellensatz for affine algebraic sets

Hilbert's Nullstellensatz, as formulated for affine space and the polynomial ring, holds correspondingly for every $V(\mathfrak{a})$ and the quotient ring $K[X_1, \dots, X_n]/\mathfrak{a}$.

Corollary: the Nullstellensatz on an affine algebraic set

Let K be an algebraically closed field and let

$$R = K[X_1, \dots, X_n]/\mathfrak{a}$$

be a K -algebra of finite type with zero locus

$$V = V(\mathfrak{a}) \subseteq \mathbb{A}_K^n.$$

Let $\mathfrak{b}$ be an ideal in R , and $F \in R$ an element vanishing on $V(\mathfrak{b}) \subseteq V$. Then there is an $r \in \mathbb{N}$ with

$$F^r \in \mathfrak{b}$$

in R .

Proof The vanishing condition $V(\mathfrak{b}) \subseteq V(F)$ inside $V = V(\mathfrak{a})$, translated back into affine space, says that

$$V(\mathfrak{a} + \mathfrak{b}) = V(\mathfrak{a}) \cap V(\mathfrak{b}) \subseteq V(F),$$

where F is now a representative polynomial in $K[X_1, \dots, X_n]$ and $\mathfrak{b}$ is the inverse-image ideal in that ring. By Hilbert's Nullstellensatz for affine space, there is an $r \in \mathbb{N}$ with

$$F^r \in \mathfrak{a} + \mathfrak{b}.$$

Modulo $\mathfrak{a}$, this says precisely that $F^r \in \mathfrak{b}$ in R .

Corollary: principal open covers of an affine algebraic set

Let K be an algebraically closed field and $V = V(\mathfrak{a}) \subseteq \mathbb{A}_K^n$ an affine algebraic set described by the ideal $\mathfrak{a}$. Let

$$F_i \in K[X_1, \dots, X_n], \quad i \in I,$$

be such that

$$V = \bigcup_{i \in I} D(F_i).$$

Then the classes of all the F_i generate the unit ideal in $K[X_1, \dots, X_n]/\mathfrak{a}$.

Proof Let $\mathfrak{b}$ be the ideal in $K[X_1, \dots, X_n]/\mathfrak{a}$ generated by all the $F_i, i \in I$. The assumption says that

$$V(\mathfrak{b}) = \bigcap_{i \in I} V(F_i)$$

is empty on V . Since $V(1)$ is also empty, $V(1) \subseteq V(\mathfrak{b})$. By Hilbert's Nullstellensatz, a power of 1, hence 1 itself, belongs to $\mathfrak{b}$.

Corollary: a polynomial with no zeros is a unit

Let K be an algebraically closed field and $V = V(\mathfrak{a}) \subseteq \mathbb{A}_K^n$ an affine algebraic set described by the ideal $\mathfrak{a}$. If $F \in K[X_1, \dots, X_n]$ has no zero on V , then its class is a unit in the quotient ring

$$K[X_1, \dots, X_n]/\mathfrak{a}.$$

Proof This is a special case of Corollary 11.12.

In Example 11.5, the function $X^2 + 2Y^2 - 1$ has no zero on the real zero locus $V(X^2 + Y^2 - 2)$, so its value is a unit at every point of that locus. However, it is not a unit in the coordinate ring

$$\mathbb{R}[X, Y]/(X^2 + Y^2 - 2).$$

Worksheet 11

Practice exercises

Exercise 11.1

Let K be an algebraically closed field. Prove Hilbert's Nullstellensatz directly for the polynomial ring in one variable.

Exercise 11.2

Let K be an algebraically closed field and

$$f, g \in K[X_1, \dots, X_n].$$

Show that

$$V(f) \subseteq V(g)$$

holds if and only if there are a natural number r and $h \in K[X_1, \dots, X_n]$ with

$$fh = g^r.$$

Also consider the special cases where f , or where g , is a constant polynomial.

Exercise 11.3

Show that, in the correspondence given by Hilbert's Nullstellensatz, points correspond to maximal ideals.

Exercise 11.4

Show that, in the correspondence given by Hilbert's Nullstellensatz, irreducible varieties correspond to prime ideals.

Exercise 11.5

Let K be an algebraically closed field. Prove directly the following special case of Hilbert's Nullstellensatz: if

$$f \in K[X_1, \dots, X_n]$$

has no zero in K^n , then f is a nonzero constant polynomial.

Exercise 11.6 ★

Consider the two polynomials $X^2 + Y^2$ and $X^2 - Y^3$ and the corresponding algebraic curves over the fields $\mathbb{R}$ and $\mathbb{C}$.

1. Does

$$V(X^2 + Y^2) \subseteq V(X^2 - Y^3)$$

hold in $\mathbb{A}_{\mathbb{R}}^2$?

2. Does the same inclusion hold in $\mathbb{A}_{\mathbb{C}}^2$?
3. Does $X^2 - Y^3$ belong to the radical of $(X^2 + Y^2)$ in $\mathbb{R}[X, Y]$?
4. Does $X^2 - Y^3$ belong to the radical of $(X^2 + Y^2)$ in $\mathbb{C}[X, Y]$?

Exercise 11.7 ★

Let polynomials

$$f_1, \dots, f_k \in \mathbb{C}[X_1, \dots, X_n]$$

be given, regarded as functions

$$f_i : \mathbb{C}^n \longrightarrow \mathbb{C}.$$

Let $f \in \mathbb{C}[X_1, \dots, X_n]$ be another polynomial, and let

$$g_1, \dots, g_k : \mathbb{C}^n \longrightarrow \mathbb{C}$$

be functions that need not be polynomial. Suppose that the following equality of functions holds:

$$f = g_1 f_1 + \dots + g_k f_k.$$

Show that f belongs to the radical of $(f_1, \dots, f_k)$.

Exercise 11.8

Let K be a field and $n \in \mathbb{N}_+$. Show that all functions $\varphi : K^n \rightarrow K$ of the form

$$\varphi = \frac{P}{Q},$$

where $P, Q \in K[X_1, \dots, X_n]$ and Q has no zero on K^n , form a commutative ring. Show that if K is algebraically closed, this ring coincides with the polynomial ring.

Exercise 11.9

Let K be a finite field. Show that there are only finitely many zero loci in $\mathbb{A}_K^n$, but infinitely many radical ideals in $K[X_1, \dots, X_n]$.

Exercise 11.10

Let R be a commutative ring and let $f_j, j \in J$, be a family of elements of R . Suppose that the f_j together generate the unit ideal. Show that there is a finite subfamily

$$f_j, \quad j \in J_0 \subseteq J,$$

that also generates the unit ideal.

Exercise 11.11

Let K be an algebraically closed field and

$$\mathfrak{a}, \mathfrak{b} \subseteq K[X_1, \dots, X_n]$$

radical ideals. Show that the zero loci $V(\mathfrak{a})$ and $V(\mathfrak{b})$ are affine-linearly equivalent if and only if there is an affine-linear change of variables taking one ideal to the other.

Aside: extension ideals

Let

$$\varphi : A \longrightarrow B$$

be a ring homomorphism between commutative rings A and B . For an ideal $\mathfrak{a} \subseteq A$, the ideal in B generated by $\varphi(\mathfrak{a})$ is called the *extension ideal* of $\mathfrak{a}$ under φ . It is denoted by $\mathfrak{a}B$. If φ is surjective, this is simply the image ideal.

Exercise 11.12

Let

$$\mathfrak{a} \subseteq \mathbb{R}[X_1, \dots, X_n]$$

be an ideal and $f \in \mathbb{R}[X_1, \dots, X_n]$. Show that $f \in \mathfrak{a}$ if and only if

$$f \in \mathfrak{a}\mathbb{C}[X_1, \dots, X_n]$$

for this extension ideal.

Exercise 11.13

Sketch the graphs of the functions x and y on $V(xy)$. Convince yourself that the product xy is the zero function.

Exercise 11.14

Determine the coordinate ring of an affine algebraic set $V \subseteq \mathbb{A}_K^n$ consisting of d points.

Exercise 11.15

Determine the coordinate ring of the affine algebraic set

$$V = V(5X - 8Y + 3) \subseteq \mathbb{A}_K^2.$$

Exercise 11.16

Consider the hyperbola $V(xy - 1)$ over the field $K = \mathbb{Z}/(11)$. Determine the inverse of $4x^3$ in the corresponding coordinate ring.

Exercise 11.17

Let K be a field and

$$V, W \subseteq \mathbb{A}_K^n$$

affine algebraic sets with $V \subseteq W$. Define a K -algebra homomorphism between the two coordinate rings $R(V)$ and $R(W)$, and describe its main properties. Give an example of two affine algebraic sets neither of which contains the other, but whose coordinate rings are isomorphic.

Exercise 11.18

Let K be a field and

$$\mathfrak{a}, \mathfrak{b} \subseteq K[X_1, \dots, X_n]$$

ideals with the same radical. Show that there is a natural bijection between the radical ideals of the quotient rings

$$K[X_1, \dots, X_n]/\mathfrak{a} \quad \text{and} \quad K[X_1, \dots, X_n]/\mathfrak{b}.$$

Exercise 11.19

Let K be a field with q elements and $V = V(\mathfrak{a}) \subseteq \mathbb{A}_K^n$ an affine algebraic set. Show that the coordinate ring of V need not equal

$$K[x_1, \dots, x_n] / ((x_1^q - x_1, \dots, x_n^q - x_n) + \mathfrak{a}).$$

Edition note: The source typographically places $+\mathfrak{a}$ outside the denominator of the quotient ring. The parentheses above make the intended mathematical reading explicit. With this reading, however, the source assertion is false: the displayed quotient always is the coordinate ring of V . Indeed, the quotient by the Frobenius equations is the ring of all functions $K^n \rightarrow K$, a finite product of copies of K ; imposing $\mathfrak{a}$ leaves precisely the factors indexed by $V(\mathfrak{a})$. This is an editorial correction, not a public source solution.

Exercise 11.20

Let K be a field of characteristic 0. Consider the intersection of a cylinder and a sphere,

$$C = V(X^2 + Y^2 - 1) \cap V((X - 3)^2 + Y^2 + Z^2 - 7) \subseteq \mathbb{A}_K^3.$$

Show that the coordinate ring of C can be written as a quotient ring of a polynomial ring in two variables.

Exercises for submission**Exercise 11.21 (4 points)**

Let $F \in \mathbb{C}[X_1, \dots, X_n]$ and let $U \subseteq \mathbb{A}_{\mathbb{C}}^n$ be a nonempty subset open in the metric topology. If $F|_U = 0$ is the zero function, show that F is the zero polynomial.

Exercise 11.22 (3 points)

Prove Corollary 11.3 directly from Theorem 10.10.

Exercise 11.23 (7 points)

Let K be an algebraically closed field and R the polynomial ring in n variables over K . We want to understand an alternative proof, based on Corollary 11.3, that

$$\text{Id}(V(J)) = \text{rad}(J)$$

for every ideal J in R . Let $f \in \text{Id}(V(J))$. Consider the ring $R[T]$ and show that the ideal

$$J' = (J, 1 - f \cdot T)$$

is the unit ideal. Deduce that f belongs to the radical of J .

Exercise 11.24 (3 points)

Let $F \in K[X_1, \dots, X_n]$ and consider the polynomial map

$$\begin{aligned} \varphi : \mathbb{A}_K^n &\longrightarrow \mathbb{A}_K^{n+1}, \\ (x_1, \dots, x_n) &\longmapsto (x_1, \dots, x_n, F(x_1, \dots, x_n)), \end{aligned}$$

which defines a bijection between affine space and the graph of F . For an affine algebraic set $V(\mathfrak{a}) \subseteq \mathbb{A}_K^n$, consider the image $V' = \varphi(V)$. Show that V' is also affine algebraic and give an ideal describing it. Show that V is irreducible if and only if V' is irreducible.

Exercise 11.25 (5 points)

Consider the two algebraic curves

$$V(x^2 + y^2 - 2) \quad \text{and} \quad V(x^2 + 2y^2 - 1)$$

over the field $\mathbb{Z}/(7)$. Show that their intersection is empty, then find an extension field $K \supseteq \mathbb{Z}/(7)$ over which it is nonempty. Calculate all intersection points over K and over every other extension field. Also describe the coordinate ring of the intersection.

Exercise 11.26 (4 points)

Let K be a field, and $P_1, \dots, P_n$ finitely many points in the affine plane $\mathbb{A}_K^2$. Let $a_1, \dots, a_n \in K$ be arbitrarily prescribed values. Show that there is a polynomial $F \in K[X, Y]$ with

$$F(P_i) = a_i \quad \text{for every } i = 1, \dots, n.$$

Edition note: The points must be pairwise distinct, as is implicit in the source's reference to finitely many points. If repetitions are allowed, the prescribed values must agree whenever $P_i = P_j$.

Public Solutions to Worksheet 11

At the frozen revision boundary, the source provides public solutions only to Exercises 11.6 and 11.7. No additional solutions have been created for this edition.

Solution to Exercise 11.6

1. The only real point of $V(X^2 + Y^2)$ is the origin $(0, 0)$, and this point lies on $V(X^2 - Y^3)$. Thus

$$V(X^2 + Y^2) \subseteq V(X^2 - Y^3) \subseteq \mathbb{A}_{\mathbb{R}}^2.$$

2. The corresponding inclusion does not hold over the complex numbers. For example,

$$(1, i) \in V(X^2 + Y^2),$$

but since $1 \neq i^3$, this point does not lie on $V(X^2 - Y^3)$.

3. Suppose that $X^2 - Y^3$ belonged to the radical of $(X^2 + Y^2)$ in $\mathbb{R}[X, Y]$. After extending scalars, the same would immediately hold in $\mathbb{C}[X, Y]$. But the next part shows that this is not the case.
4. From part (2) and the easy direction of Hilbert's Nullstellensatz, it follows that $X^2 - Y^3$ does not belong to the radical of $(X^2 + Y^2)$ in $\mathbb{C}[X, Y]$.

[Back to Exercise 11.6.](#)

Solution to Exercise 11.7

We claim that

$$V(f_1, \dots, f_k) \subseteq V(f).$$

Once this claim is proved, Hilbert's Nullstellensatz says that f belongs to the radical of $(f_1, \dots, f_k)$.

Let

$$P = (x_1, \dots, x_n) \in \mathbb{C}^n$$

and suppose that $P \in V(f_1, \dots, f_k)$. This means that $f_i(P) = 0$ for every i . Then

$$\begin{aligned} f(P) &= (g_1 f_1 + \dots + g_k f_k)(P) \\ &= g_1(P) f_1(P) + \dots + g_k(P) f_k(P) \\ &= 0. \end{aligned}$$

Thus $P \in V(f)$, proving the claim.

[Back to Exercise 11.7.](#)

Lecture 12: The K -spectrum and its functoriality

"Born to see, appointed to behold."

— Johann Wolfgang von Goethe

The K -spectrum



Figure 62: Black-and-white portrait of Alexander Grothendieck seated

Alexander Grothendieck (1928–2014); photograph: Konrad Jacobs, Oberwolfach Photo Collection/MFO; [CC BY-SA 2.0 de](#).

How are affine algebraic sets and their coordinate rings related? Meaningful answers can be expected only for infinite ground fields, because in the finite case there are too few points. A satisfactory theory even requires us to restrict ourselves to algebraically closed fields, or else—and this is the viewpoint of scheme theory developed by Alexander Grothendieck—to consider not only K -points, but also maximal ideals and prime ideals as points.

A first important question is the following. A K -algebra R of finite type has several representations, generally on an equal footing, as a quotient ring of a polynomial algebra; say,

$$K[X_1, \dots, X_n]/\mathfrak{a} \cong R \cong K[X_1, \dots, X_m]/\mathfrak{b}.$$

These two representations give the zero loci

$$V(\mathfrak{a}) \subseteq \mathbb{A}_K^n \quad \text{and} \quad V(\mathfrak{b}) \subseteq \mathbb{A}_K^m.$$

How are these two zero loci related?

Example: three representations of the affine line

Consider the polynomial ring in one variable

$$R = K[T].$$

The first object corresponding to it is the affine line $\mathbb{A}_K^1$. But R can also be obtained in quite different ways as a quotient ring of a polynomial algebra in several variables. For example, let

$$a \in K, \quad a \neq 0,$$

and consider the quotient ring $K[X, Y]/(aY + bX)$. As a K -algebra, this ring is isomorphic to R , as shown by the map

$$\begin{aligned} K[X, Y]/(aY + bX) &\longrightarrow K[T], \\ X &\longmapsto T, \\ Y &\longmapsto -\frac{b}{a}T. \end{aligned}$$

The corresponding zero locus,

$$V(aY + bX) \subset \mathbb{A}_K^2,$$

is simply the line in the affine plane described by the equation

$$Y = -\frac{b}{a}X.$$

Another way to represent the polynomial ring in one variable as a quotient ring is

$$K[X, Y]/(Y - P(X)),$$

where $P(X)$ is an arbitrary polynomial in the single variable X . The ring homomorphism

$$\begin{aligned} K[X, Y]/(Y - P(X)) &\longrightarrow K[T], \\ X &\longmapsto T, \\ Y &\longmapsto P(T) \end{aligned}$$

again shows that there is an isomorphism with the polynomial ring in one variable. The corresponding zero locus is simply the graph of $P(X)$.



Figure 63: Black horizontal line

Horizontal line; Astur1, public domain.

lineaire functie, cartesiaans assenstelsel

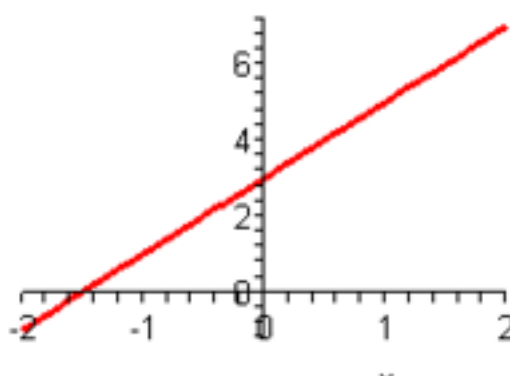


Figure 64: Graph of a red straight line in a Cartesian coordinate system

Graph of a linear function in Cartesian coordinates; MADe, CC BY-SA 3.0.

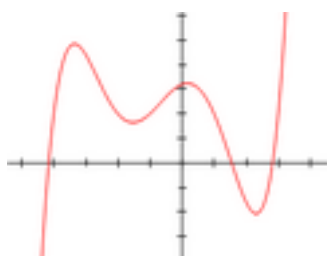


Figure 65: Red graph of a polynomial of degree five

Graph of a polynomial of degree five; Derbeth, CC BY-SA 3.0.

The point of this example is that all three geometric objects are zero loci for different quotient-ring presentations of $K[T]$. From the viewpoint of algebraic geometry, they are three equally valid representations of the affine line, even though they “look” different. In algebraic geometry we must look at them in a way that makes them look the same. What we see are merely different embeddings of the “actual, true” geometric object intrinsically associated with a K -algebra: the K -spectrum.

Definition: the K -spectrum

For a commutative K -algebra R of finite type, the set of all K -algebra homomorphisms

$$\text{Hom}_K(R, K)$$

is called the K -spectrum of R and is denoted by

$$K\text{-Spek}(R).$$

We regard the elements of the K -spectrum $K\text{-Spek}(R)$ as points and usually denote them by P , although by definition they are maps, namely K -algebra homomorphisms from R to K . For a ring element $f \in R$, we then write $f(P)$ rather than $P(f)$ for the value of f under the ring homomorphism denoted by P . Indeed, it is not unusual to regard a point as an evaluation of functions defined in some neighbourhood of that point.

The K -spectrum is again equipped with a Zariski topology. For an ideal $\mathfrak{a} \subseteq R$ —or even an arbitrary subset of R —we declare the subset

$$V(\mathfrak{a}) = \{P \in K\text{-Spek}(R) \mid f(P) = 0 \text{ for all } f \in \mathfrak{a}\}$$

to be closed. This does indeed define a topology; see Exercise 12.8. Its open complement is denoted by $D(\mathfrak{a})$.

Lemma: points of affine space as homomorphisms

Let K be a field and $K[X_1, \dots, X_n]$ the polynomial ring in n variables. The K -algebra homomorphisms from $K[X_1, \dots, X_n]$ to K are naturally in bijection with the points of affine space

$$\mathbb{A}_K^n = K^n.$$

The point $(a_1, \dots, a_n)$ corresponds to the substitution homomorphism $X_i \mapsto a_i$. In other words,

$$K\text{-Spek}(K[X_1, \dots, X_n]) = \mathbb{A}_K^n.$$

Proof A K -algebra homomorphism is always determined by a set of K -algebra generators. Thus the values on the variables X_i determine a K -algebra homomorphism from $K[X_1, \dots, X_n]$ to K . Such a substitution homomorphism is defined by $X_i \mapsto a_i$, and every choice of values $(a_1, \dots, a_n)$ is allowed here.

Example: the K -spectrum of K

The K -spectrum of the K -algebra K consists of a single point: the identity

$$\text{id} : K \longrightarrow K$$

is the only K -algebra homomorphism from K to K . In general there may be other field automorphisms of K , but these are not K -algebra homomorphisms.

The following theorem is crucial: it establishes a bijective relationship between the K -spectrum of R and the zero locus arising from a quotient-ring presentation of R .

Theorem: the K -spectrum and zero loci

Let K be a field and R a finitely generated commutative K -algebra with K -spectrum $K\text{-Spek}(R)$. Let

$$R = K[X_1, \dots, X_n]/\mathfrak{a}$$

be a quotient-ring presentation of R , with quotient homomorphism

$$\varphi : K[X_1, \dots, X_n] \longrightarrow R$$

and corresponding zero locus $V(\mathfrak{a}) \subseteq \mathbb{A}_K^n$. The map

$$\begin{aligned} K\text{-Spek}(R) &\longrightarrow \mathbb{A}_K^n, \\ P &\longmapsto P \circ \varphi \end{aligned}$$

gives a bijection between $K\text{-Spek}(R)$ and $V(\mathfrak{a})$, and this bijection is a homeomorphism for the Zariski topology.

Proof First, the map above is well-defined because the composite

$$P \circ \varphi : K[X_1, \dots, X_n] \xrightarrow{\varphi} K[X_1, \dots, X_n]/\mathfrak{a} \cong R \xrightarrow{P} K$$

defines a K -algebra homomorphism from the polynomial ring to K . By Lemma 12.3, this is the substitution homomorphism at some $(a_1, \dots, a_n)$ and can be identified with the corresponding point of affine space; explicitly,

$$a_i = P(\varphi(X_i)).$$

Since $P \circ \varphi$ factors through R , the ideal $\mathfrak{a}$ maps to 0. Thus the image point

$$P \circ \varphi = (a_1, \dots, a_n)$$

lies in $V(\mathfrak{a})$. We therefore obtain a map

$$\begin{aligned} K\text{-Spek}(R) &\longrightarrow V(\mathfrak{a}) \subseteq \mathbb{A}_K^n, \\ P &\longmapsto P \circ \varphi, \end{aligned}$$

which remains to be proved bijective.

Let $P_1, P_2 \in K\text{-Spek}(R)$ be two distinct points. They are distinct K -algebra homomorphisms. Since a K -algebra homomorphism is determined by its values on a set of K -algebra generators, they must differ on at least one image of a variable. The corresponding coordinate values therefore also differ, so $P_1 \circ \varphi \neq P_2 \circ \varphi$. Thus the map is injective.

For surjectivity, let $(a_1, \dots, a_n) \in V(\mathfrak{a})$. The corresponding K -algebra homomorphism,

$$\begin{aligned} K[X_1, \dots, X_n] &\longrightarrow K, \\ X_i &\longmapsto a_i, \end{aligned}$$

annihilates every $F \in \mathfrak{a}$. Hence this ring homomorphism factors through $K[X_1, \dots, X_n]/\mathfrak{a}$. The resulting homomorphism is the required preimage in $K\text{-Spek}(R)$.

For the topological assertion, take $G \in R$, a preimage $\tilde{G} \in K[X_1, \dots, X_n]$, and a point $P \in K\text{-Spek}(R)$ with image point $\tilde{P} = P \circ \varphi \in V(\mathfrak{a})$. Then

$$G(P) = P(G) = P(\varphi(\tilde{G})) = (P \circ \varphi)(\tilde{G}) = \tilde{G}(\tilde{P}),$$

so zero loci on the two sides also correspond. The bijection is therefore a homeomorphism.

This theorem says that every K -spectrum of a K -algebra R of finite type can be identified with a Zariski-closed subset of some $\mathbb{A}_K^n$. Such an identification is called a *closed embedding*.

Corollary: topological independence of the presentation

Let K be a field and R a finitely generated commutative K -algebra with two quotient-ring presentations

$$R \cong K[X_1, \dots, X_n]/\mathfrak{a} \quad \text{and} \quad R \cong K[X_1, \dots, X_m]/\mathfrak{b},$$

and corresponding zero loci

$$V(\mathfrak{a}) \subseteq \mathbb{A}_K^n \quad \text{and} \quad V(\mathfrak{b}) \subseteq \mathbb{A}_K^m.$$

With their induced Zariski topologies, these two zero loci are homeomorphic.

Proof By Theorem 12.5, both zero loci are homeomorphic to $K\text{-Spek}(R)$, and hence also to each other.

If R is the zero ring, its K -spectrum is empty. If K is not algebraically closed, the spectrum of other rings can also be empty. However, if K is algebraically closed and $R \neq 0$, the spectrum is nonempty. Under this assumption there is again a Hilbert Nullstellensatz; see Exercise 12.10.

The K -spectrum as a functor

Theorem: the spectrum map

Let K be a field, let R and S be commutative K -algebras of finite type, and let

$$\varphi : R \longrightarrow S$$

be a K -algebra homomorphism. It induces a map

$$\begin{aligned} \varphi^* : K\text{-Spek}(S) &\longrightarrow K\text{-Spek}(R), \\ P &\longmapsto P \circ \varphi. \end{aligned}$$

This map is continuous for the Zariski topology.

Proof The existence of the map is clear: to the K -algebra homomorphism $P : S \rightarrow K$ we assign the composite

$$R \xrightarrow{\varphi} S \xrightarrow{P} K.$$

The inverse image of the open set $D(f) \subseteq K\text{-Spek}(R)$ is

$$\begin{aligned} (\varphi^*)^{-1}(D(f)) &= \{P \in K\text{-Spek}(S) \mid \varphi^*(P) \in D(f)\} \\ &= \{P \in K\text{-Spek}(S) \mid P \circ \varphi \in D(f)\} \\ &= \{P \in K\text{-Spek}(S) \mid (P \circ \varphi)(f) \neq 0\} \\ &= \{P \in K\text{-Spek}(S) \mid P(\varphi(f)) \neq 0\} \\ &= D(\varphi(f)). \end{aligned}$$

Thus inverse images of open sets are again open, and the map is continuous.

The map φ^* introduced in Theorem 12.7 is called the *spectrum map* associated with φ .

Proposition: some forms of spectrum maps

Let K be a field and $\varphi : R \rightarrow S$ a K -algebra homomorphism between K -algebras of finite type, with associated spectrum map φ^* . The following statements hold.

1. For a K -algebra homomorphism $P : R \rightarrow K$, the induced spectrum map P^* is the map taking the unique point

$$\{\text{id}\} = K\text{-Spek}(K)$$

to the point $P \in K\text{-Spek}(R)$.

2. The substitution homomorphism defined by $F \in R$,

$$\begin{aligned} \varphi : K[T] &\longrightarrow R, \\ T &\longmapsto F, \end{aligned}$$

induces the spectrum map

$$\begin{aligned} \varphi^* : K\text{-Spek}(R) &\longrightarrow K\text{-Spek}(K[T]) = \mathbb{A}_K^1, \\ P &\longmapsto F(P). \end{aligned}$$

3. If $\varphi : R \rightarrow S$ is surjective, then the spectrum map

$$\varphi^* : K\text{-Spek}(S) \longrightarrow K\text{-Spek}(R)$$

is a closed embedding with image $V(\ker \varphi)$.

4. The spectrum map associated with a surjective map

$$K[X_1, \dots, X_n] \longrightarrow S$$

agrees with the map

$$\varphi^* : K\text{-Spek}(S) \longrightarrow K\text{-Spek}(K[X_1, \dots, X_n]) \cong \mathbb{A}_K^n$$

defined in Theorem 12.5.

5. Let $F_i \in K[X_1, \dots, X_n]$ for $i = 1, \dots, m$, and let

$$\begin{aligned} \varphi : K[Y_1, \dots, Y_m] &\longrightarrow K[X_1, \dots, X_n], \\ Y_i &\longmapsto F_i \end{aligned}$$

be the associated substitution homomorphism. Under the identification in Lemma 12.3, the spectrum map

$$\varphi^* : \mathbb{A}_K^n = K\text{-Spek}(K[X_1, \dots, X_n]) \longrightarrow \mathbb{A}_K^m = K\text{-Spek}(K[Y_1, \dots, Y_m])$$

agrees with the direct polynomial map

$$(x_1, \dots, x_n) \longmapsto (F_1(x_1, \dots, x_n), \dots, F_m(x_1, \dots, x_n)).$$

Proof

(1) follows from $\text{id} \circ P = P$.

For (2), under the composite

$$K[T] \xrightarrow{\varphi} R \xrightarrow{P} K,$$

T is sent to $P(F) = F(P)$.

(3) rests on considerations similar to those in the proof of Theorem 12.5; these also prove (4). For (5), see Exercise 12.18.

Statement (2) says in particular that the elements of the ring R can be regarded as functions from the K -spectrum $K\text{-Spek}(R)$ to $\mathbb{A}_K^1$. Thus we have introduced a geometric object that realises ring elements as functions.

Further properties of the K -spectrum

Lemma: adjoining one variable

Let K be a field and R a finitely generated commutative K -algebra. Then there is a natural bijection

$$K\text{-Spek}(R[X]) \cong K\text{-Spek}(R) \times \mathbb{A}_K^1.$$

Proof A K -algebra homomorphism $R[T] \rightarrow K$ induces a K -algebra homomorphism $R \rightarrow K$, while T maps to a particular element $a \in K$. Conversely, these two data uniquely determine a K -algebra homomorphism $R[T] \rightarrow K$.

Warning: the statement above gives only a natural bijection at the level of points. If the product set on the right is equipped with the product topology, this bijection need not be a homeomorphism with the Zariski topology on the left. In particular, for an infinite field K ,

$$\mathbb{A}_K^2 = \mathbb{A}_K^1 \times \mathbb{A}_K^1,$$

but the Zariski topology on the affine plane is not the product of the Zariski topology on the affine line with itself.

Edition note: The source states the failure of homeomorphism without qualification. The general statement is “need not”: for example, when $R = K$ the bijection is a homeomorphism. The affine-plane counterexample requires K infinite; over a finite field all these finite K -spectra are discrete.

Remark: products via tensor products

If

$$X = K\text{-Spek}(R) \quad \text{and} \quad Y = K\text{-Spek}(S),$$

then the product set $X \times Y$ can also be represented as the K -spectrum of a K -algebra, namely

$$X \times Y \cong K\text{-Spek}(R \otimes_K S),$$

where $\otimes$ denotes the tensor product. We shall not discuss this in detail. To give some intuition, however, take

$$R = K[X_1, \dots, X_n]/\mathfrak{a} \quad \text{and} \quad S = K[Y_1, \dots, Y_m]/\mathfrak{b}.$$

Then

$$R \otimes_K S \cong K[X_1, \dots, X_n, Y_1, \dots, Y_m]/(\mathfrak{a} + \mathfrak{b}).$$

With this *ad hoc* definition, it is not yet clear that the result is independent of the chosen quotient-ring presentations.

Worksheet 12

Practice exercises

Exercise 12.1

Explain the concept of a point in Euclidean (coordinate-free) geometry and in Cartesian geometry. In which situations is it useful to introduce coordinates?

Exercise 12.2

What is a point in algebraic geometry? Which notions of a point have you encountered in the course on algebraic curves, and how are they related?

Exercise 12.3

Determine the K -spectrum of K^d .

Exercise 12.4

Determine the $\mathbb{R}$ -spectrum of the $\mathbb{R}$ -algebra $\mathbb{C}$.

Exercise 12.5

Determine the $\mathbb{R}$ -spectrum of

$$\mathbb{R}[X, Y]/(X^2 + Y^2 - 1).$$

Exercise 12.6 ★

Let K be an algebraically closed field and R a commutative K -algebra of finite type. Show that the points of

$$K\text{-Spek}(R)$$

correspond to the maximal ideals of R .

Exercise 12.7

Let R be a commutative K -algebra of finite type. Show that, for every ideal $\mathfrak{a} \subseteq R$, we have

$$V(\mathfrak{a}) = V(\text{rad}(\mathfrak{a}))$$

inside $K\text{-Spek}(R)$.

Exercise 12.8

Show that the Zariski topology on the K -spectrum of a commutative K -algebra R of finite type is indeed a topology.

Exercise 12.9

Let K be a field and

$$V \subseteq \mathbb{A}_K^n$$

an affine algebraic set with vanishing ideal $\text{Id}(V)$ and coordinate ring

$$R := K[X_1, \dots, X_n]/\text{Id}(V).$$

Show that the K -spectrum of R is homeomorphic to V .

All versions of Hilbert's Nullstellensatz, such as Corollary 11.11, carry over to the K -spectrum of a K -algebra of finite type over an algebraically closed field K .

Exercise 12.10

Let K be an algebraically closed field and R a K -algebra of finite type. Prove that there is a bijective correspondence between the closed subsets of the K -spectrum

$$K\text{-Spek}(R)$$

and the radical ideals of R .

Exercise 12.11

Let K be an algebraically closed field and R a reduced K -algebra of finite type. Prove the *identity theorem* in the following form: if $f, g \in R$ satisfy

$$f(P) = g(P)$$

for all $P \in K\text{-Spek}(R)$, then $f = g$.

Exercise 12.12 ★

Let K be a field, R a finitely generated K -algebra, $\mathfrak{a} \subseteq R$ an ideal, and

$$X = K\text{-Spek}(R).$$

What is the relationship between the two statements

$$V(\mathfrak{a}) = \emptyset \quad \text{and} \quad \mathfrak{a} \text{ is the unit ideal,}$$

and between the two statements

$$V(\mathfrak{a}) = X \quad \text{and} \quad \mathfrak{a} \text{ is nilpotent?}$$

Show that the answer depends on whether K is algebraically closed.

Exercise 12.13

For a commutative K -algebra R of finite type, describe the spectrum map associated with the algebra structure homomorphism

$$K \longrightarrow R.$$

Exercise 12.14

Let R, S, T be commutative K -algebras of finite type, and let

$$\varphi : R \longrightarrow S \quad \text{and} \quad \psi : S \longrightarrow T$$

be K -algebra homomorphisms. Show that the corresponding spectrum maps satisfy

$$(\psi \circ \varphi)^* = \varphi^* \circ \psi^*.$$

Also show that the map Id^* associated with the identity $\text{Id} : R \rightarrow R$ is itself the identity.

Exercise 12.15

Give an example of two commutative K -algebras R, S of finite type and a continuous map between their K -spectra that cannot arise from a K -algebra homomorphism.

Exercise 12.16

Let K be a field, R a commutative K -algebra of finite type, and $F \in R$. Let

$$\varphi^* : K\text{-Spek}(R) \longrightarrow \mathbb{A}_K^1$$

be the spectrum map associated with the substitution homomorphism. Show that

$$(\varphi^*)^{-1}(0) = V(F).$$

Exercise 12.17

Let K be a field and R a commutative K -algebra of finite type, with reduction

$$S = R_{\text{red}}.$$

Show that there is a natural homeomorphism

$$K\text{-Spek}(R) \cong K\text{-Spek}(S).$$

Exercise 12.18

Let K be a field and let

$$F_i \in K[X_1, \dots, X_n], \quad i = 1, \dots, m,$$

be polynomials. Let

$$\begin{aligned} \varphi : K[Y_1, \dots, Y_m] &\longrightarrow K[X_1, \dots, X_n], \\ Y_i &\longmapsto F_i \end{aligned}$$

be the corresponding substitution homomorphism. Show that, under the identification in Lemma 12.3, the spectrum map

$$\varphi^* : \mathbb{A}_K^n = K\text{-Spek}(K[X_1, \dots, X_n]) \longrightarrow \mathbb{A}_K^m = K\text{-Spek}(K[Y_1, \dots, Y_m])$$

agrees with the direct polynomial map

$$(x_1, \dots, x_n) \longmapsto (F_1(x_1, \dots, x_n), \dots, F_m(x_1, \dots, x_n)).$$

Exercise 12.19

Which “functors” in mathematics do you know?

In the following exercises, for an arbitrary topological space—for example a manifold, a subset of $\mathbb{R}^n$, or a real interval—we consider the ring of continuous real-valued functions on it. The spaces should be viewed as analogous to K -spectra, and their function rings as analogous to coordinate rings.

Exercise 12.20

Let X be a topological space and

$$R = C(X, \mathbb{R}) = \{f : X \longrightarrow \mathbb{R} \mid f \text{ is continuous}\}.$$

Show that R is a commutative ring.

Exercise 12.21

Consider the ring of continuous functions

$$R = C(\mathbb{R}, \mathbb{R})$$

from $\mathbb{R}$ to $\mathbb{R}$. Is this ring an integral domain?

Exercise 12.22

Let $T \subseteq \mathbb{R}$ be a subset. Show that, in the ring of continuous functions

$$R = C(\mathbb{R}, \mathbb{R}),$$

the subset

$$I = \{f \in R \mid f(x) = 0 \text{ for all } x \in T\}$$

is an ideal of R .

Exercise 12.23

Consider the ideal associated with

$$T = \{0\} \subseteq \mathbb{R}$$

in the sense of Exercise 12.22. Is it a principal ideal?

Exercise 12.24

Let X be a topological space and

$$R = C^0(X, \mathbb{R})$$

the ring of continuous functions on X . For a subset $T \subseteq X$, show that

$$I = \{f \in R \mid f|_T = 0\}$$

is an ideal of R . Define a ring homomorphism

$$R/I \longrightarrow C^0(T, \mathbb{R}).$$

Is this homomorphism always injective? Is it always surjective?

Exercise 12.25

Let X and Y be topological spaces and

$$\varphi : X \longrightarrow Y$$

a continuous map. Show that it induces a ring homomorphism

$$\begin{aligned} C(Y, \mathbb{R}) &\longrightarrow C(X, \mathbb{R}), \\ f &\longmapsto f \circ \varphi. \end{aligned}$$

Exercise 12.26

Let $X \subseteq \mathbb{R}$ and let $C(X, \mathbb{R})$ be the ring of continuous functions from X to $\mathbb{R}$. Restriction of functions gives a ring homomorphism

$$\begin{aligned} \varphi : C(\mathbb{R}, \mathbb{R}) &\longrightarrow C(X, \mathbb{R}), \\ f &\longmapsto f|_X. \end{aligned}$$

1. Show that φ is surjective if and only if X is closed.
2. For which sets X is φ injective?

Exercises for submission**Exercise 12.27 (4 points: 2+2)**

Let K be an infinite field, R a commutative K -algebra of finite type, and $F \in R$. Let

$$\varphi^* : K\text{-Spek}(R) \longrightarrow \mathbb{A}_K^1$$

be the spectrum map associated with the substitution homomorphism.

1. Show that F is constant if and only if φ^* is constant.
2. Show that this statement need not hold for a finite field.

Hint: Also note the different meanings of “constant” in these two contexts.

Edition note: Part (1), as stated in the source, needs an additional hypothesis: infinitude of K alone is insufficient. For example, in $R = K[\varepsilon]/(\varepsilon^2)$ the nonconstant element $F = \varepsilon$ induces the constant zero function, even when K is infinite. A sufficient correction is to assume that K is algebraically closed and R is reduced, as in Exercise 12.11. More generally, it suffices that evaluation on K -points separates elements of R . This is an editorial clarification of the hypothesis, not a public source solution.

Exercise 12.28 (4 points)

Give an example of two $\mathbb{C}$ -algebras R and S of finite type that are integral domains, and a $\mathbb{C}$ -algebra homomorphism

$$\varphi : R \longrightarrow S,$$

which is not a ring isomorphism, but whose induced spectrum map

$$\varphi^* : \mathbb{C}\text{-Spek}(S) \longrightarrow \mathbb{C}\text{-Spek}(R)$$

is a homeomorphism.

Exercise 12.29 (3 points)

Let K be an algebraically closed field and R a K -algebra of finite type. Consider the finite extension

$$\varphi : R \longrightarrow S = R[X] / (X^n + r_{n-1}X^{n-1} + \cdots + r_2X^2 + r_1X + r_0).$$

Show that

$$\varphi^* : K\text{-Spek}(S) \longrightarrow K\text{-Spek}(R)$$

is surjective.

Exercise 12.30 (5 points)

Consider the ideal

$$\mathfrak{a} = (U^5 - V^3, U^{11} - W^3, V^{11} - W^5) \subseteq K[U, V, W]$$

and its zero locus

$$Z = V(\mathfrak{a}) \subseteq \mathbb{A}_K^3.$$

Show that $W - U^2V$ belongs to the radical of $\mathfrak{a}$. Use this to show that Z is isomorphic to a plane algebraic curve.

Hint: Use the fact that a radical is the intersection of all prime ideals containing it, or reduce to the case where K is algebraically closed.

Public Solutions to Worksheet 12

At the frozen revision boundary, the source provides public solutions only to Exercises 12.6 and 12.12. No additional solutions have been created for this edition.

Solution to Exercise 12.6

A K -point is a K -algebra homomorphism

$$\varphi : A \longrightarrow K.$$

Since A is a K -algebra, this homomorphism is surjective. Its kernel is a maximal ideal of A . Since A is of finite type over an algebraically closed field, the theorem that maximal ideals in an algebra of finite type over an algebraically closed field are point ideals applies. Thus the residue field at every maximal ideal equals K .

Edition note: The exercise names the algebra R , whereas the source solution uses A . The source solution's notation A is retained here.

[Back to Exercise 12.6.](#)

Solution to Exercise 12.12

If $\mathfrak{a}$ is the unit ideal, then $V(\mathfrak{a}) = \emptyset$, because 1 vanishes at no point. The converse holds if K is algebraically closed. Indeed, from

$$V(1) \subseteq V(\mathfrak{a})$$

—since both sets are empty—Hilbert's Nullstellensatz immediately gives

$$1 = 1^n \in \mathfrak{a}.$$

For $K = \mathbb{R}$, the converse fails. The polynomial

$$F = X^2 + 1$$

is not a unit, but its zero locus is empty.

If $\mathfrak{a}$ is nilpotent, then every element of it is nilpotent and therefore vanishes under every ring homomorphism to a field, since fields are reduced. For an algebraically closed ground field, the converse again holds. If $V(\mathfrak{a}) = X$, then for each $f \in \mathfrak{a}$ we have

$$V(\mathfrak{a}) \subseteq V(f) = X = V(0).$$

By Hilbert's Nullstellensatz, this implies

$$f^n = 0,$$

so f is nilpotent. In a Noetherian ring, this also implies that the ideal $\mathfrak{a}$ itself is nilpotent.

Over a finite field, this converse fails. For $K = \mathbb{F}_2$, the polynomial

$$X^2 - X \in K[X]$$

is not nilpotent, but vanishes at both points—that is, at all points—of K .

[Back to Exercise 12.12.](#)

Lecture 13: The open sets $D(f)$, connectedness, and idempotent elements

The open sets $D(f)$

We shall show that the Zariski-open subsets

$$D(f) \subseteq K\text{-Spek}(R)$$

are themselves homeomorphic to the K -spectrum of a K -algebra of finite type. For this we need the notions of a multiplicative system and localisation.

Definition: multiplicative system

Let R be a commutative ring. A subset $S \subseteq R$ is called a *multiplicative system* if it satisfies the following two properties:

1. $1 \in S$;
2. if $f, g \in S$, then $fg \in S$.

Example: powers of an element

Let R be a commutative ring and $f \in R$. The powers

$$\{f^n \mid n \in \mathbb{N}\}$$

form a multiplicative system.

Definition: localisation inside the field of fractions

Let R be an integral domain and $S \subseteq R$ a multiplicative system with $0 \notin S$. The subring

$$R_S := \left\{ \frac{f}{g} \mid f \in R, g \in S \right\} \subseteq Q(R)$$

is called the *localisation* of R at S .

For localisation at a single element f , we simply write R_f instead of $R_{\{f^n \mid n \in \mathbb{N}\}}$. For the definition of localisation for arbitrary commutative rings, see Exercise 13.1.

Theorem: $D(f)$ as the K -spectrum of R_f

Let K be a field, R a K -algebra of finite type, and $f \in R$. The Zariski-open set

$$D(f) \subseteq K\text{-Spek}(R)$$

is naturally homeomorphic to $K\text{-Spek}(R_f)$.

Proof Consider the canonical K -algebra homomorphism

$$\varphi : R \longrightarrow R_f$$

and its spectrum map

$$\begin{aligned} \varphi^* : K\text{-Spek}(R_f) &\longrightarrow K\text{-Spek}(R), \\ P &\longmapsto P \circ \varphi. \end{aligned}$$

By Theorem 12.7 this map is continuous. Since f becomes a unit in R_f , for every P we have

$$f(P \circ \varphi) = P(\varphi(f)) \neq 0.$$

Thus the image of φ^* lies in $D(f)$.

Conversely, take $Q \in D(f)$. Thus $Q : R \rightarrow K$ is a K -algebra homomorphism with $Q(f) \neq 0$. The element $Q(f)$ is a unit in K . By the universal property of localisation (see Exercise 13.6), Q extends to a homomorphism $R_f \rightarrow K$. This extension is the required preimage, so φ^* is surjective as a map to $D(f)$.

To prove injectivity, let $P_1, P_2 : R_f \rightarrow K$ be two K -algebra homomorphisms whose composites with $R \rightarrow R_f$ agree. For $r \in R$ and $s \in \mathbb{N}$ we have

$$P_1\left(\frac{r}{f^s}\right) = P_1(rf^{-s}) = P_1(r)P_1(f^s)^{-1},$$

and the same formula holds for P_2 . Since their values on R agree, we obtain $P_1 = P_2$.

Finally, the Zariski-open sets of $K - \text{Spek}(R_f)$ are covered by sets $D(g)$ with $g \in R_f$. Since f is a unit in R_f , we may take $g \in R$. This set $D(g)$ equals

$$(\varphi^*)^{-1}(D(gf)),$$

where $D(gf)$ on the right is an open set in $K - \text{Spek}(R)$. Thus the bijection above is a homeomorphism.

Remark: a closed realisation of $D(f)$

Theorem 13.4 says in particular that the open set

$$D(f) \subseteq K - \text{Spek}(R)$$

is itself the K -spectrum of a K -algebra of finite type, namely R_f , which is generated over R by $1/f$. Since

$$R_f \cong R[T]/(Tf - 1)$$

(see Exercise 13.4), it can be realised as a closed set in an affine space. If

$$R = K[X_1, \dots, X_n]/\mathfrak{a},$$

then the surjective ring homomorphism

$$K[X_1, \dots, X_n, T] \longrightarrow (K[X_1, \dots, X_n]/\mathfrak{a})[T] \longrightarrow \frac{(K[X_1, \dots, X_n]/\mathfrak{a})[T]}{(Tf - 1)} \cong R_f$$

gives a closed embedding of $D(f)$ into $\mathbb{A}_K^{n+1}$ by Proposition 12.8(3). If ψ is the composite inclusion

$$D(f) \subseteq K - \text{Spek}(R) \subseteq \mathbb{A}_K^n,$$

this closed embedding can also be viewed as

$$\psi \times \frac{1}{f} : D(f) \longrightarrow \mathbb{A}_K^n \times \mathbb{A}_K^1.$$

Here the product of varieties appears again.

Example: the punctured affine line as a hyperbola

Continuing Remark 13.5, consider the open set

$$D(X) = \{P \in \mathbb{A}_K^1 \mid P \neq 0\} \subset \mathbb{A}_K^1.$$

This set is called the *punctured affine line*. On it, X is invertible, so the rational function $1/X$ is defined. Together with the open inclusion $D(X) \subseteq \mathbb{A}_K^1$, this function gives a closed inclusion

$$\begin{aligned} D(X) &\longrightarrow V(XY - 1) \subseteq \mathbb{A}_K^2, \\ x &\longmapsto \left(x, \frac{1}{x}\right). \end{aligned}$$

Its image is a hyperbola closed in the affine plane. Thus the punctured affine line and this hyperbola are homeomorphic; the corresponding rings,

$$K[X]_X = K[X, X^{-1}] \quad \text{and} \quad K[X, Y]/(XY - 1),$$

are also isomorphic.

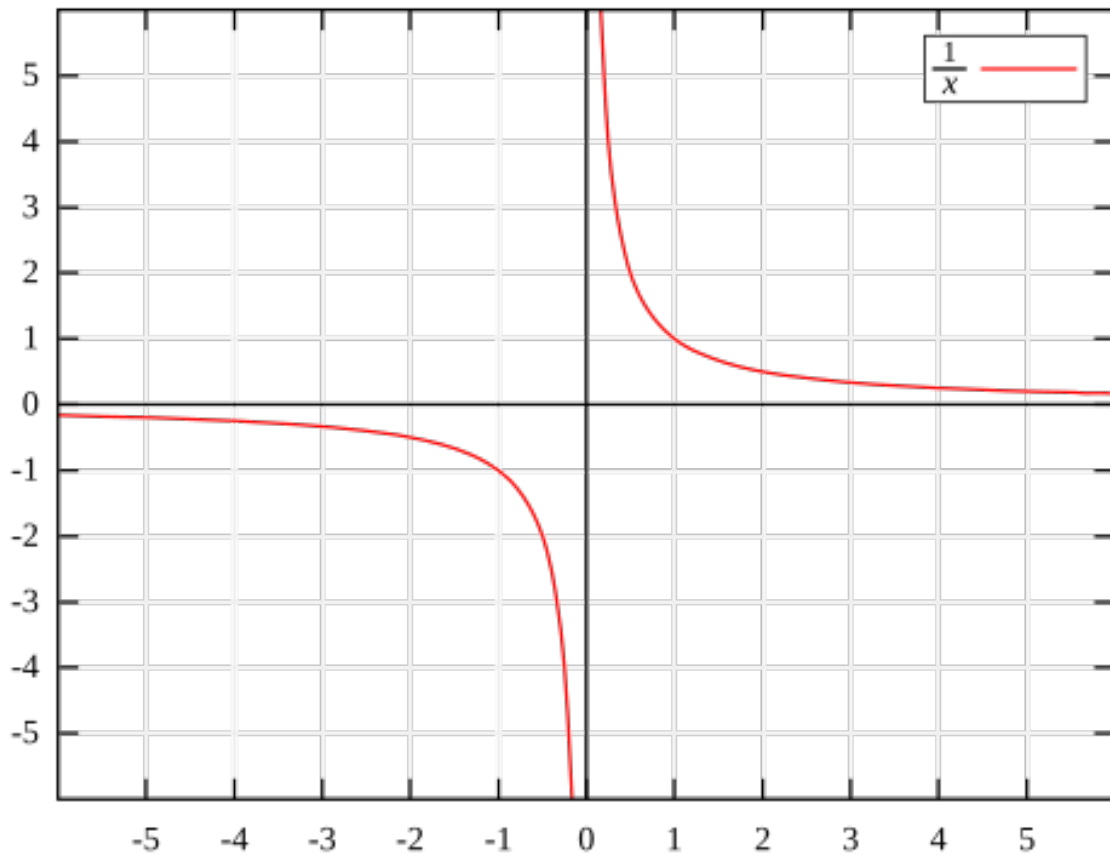


Figure 66: Two branches of the hyperbola y equals one over x in the coordinate plane

Graph of the hyperbola $y = 1/x$; Ktims, [CC BY-SA 3.0](#).

Connectedness and idempotent elements

We want to understand how connectedness of an affine algebraic set is reflected in its coordinate ring, and how its connected components can be characterised. The following example shows that a satisfactory theory cannot be expected over a field that is not algebraically closed.

Example: connectedness can change after a field extension

As in Example 11.5, consider the two algebraic curves

$$V_1 = V(X^2 + Y^2 - 2) \quad \text{and} \quad V_2 = V(X^2 + 2Y^2 - 1) \subseteq \mathbb{A}_K^2.$$

Their intersection is described by the ideal

$$(X^2 + Y^2 - 2, X^2 + 2Y^2 - 1) = (Y^2 + 1, X^2 - 3).$$

For $K = \mathbb{R}$ we have $V_1 \cap V_2 = \emptyset$. Consequently,

$$V = V_1 \cup V_2$$

is disconnected; V_1 and V_2 are both its irreducible components and its connected components. The coordinate ring of V is

$$\mathbb{R}[X, Y]/((X^2 + Y^2 - 2)(X^2 + 2Y^2 - 1)).$$

One might expect the function on V that is constantly 1 on V_1 and constantly 0 on V_2 to occur in the coordinate ring. This is not so. The reason is that, after extending scalars to the complex numbers, $V_{\mathbb{C}}$ is connected. Hence the complex coordinate ring has only trivial idempotents, and this property passes to the real coordinate ring.

Definition: idempotent element

An element e of a commutative ring is called *idempotent* if

$$e^2 = e.$$

The elements 0 and 1 are idempotent.

Definition: product ring

Let $R_1, \dots, R_n$ be commutative rings. The product

$$R_1 \times \cdots \times R_n,$$

with componentwise addition and multiplication, is called the *product ring* of the R_i , $i = 1, \dots, n$.

A product ring has many idempotents, namely elements each of whose components is either 0 or 1.

Definition: connected ring

A commutative ring R is called *connected* if it has exactly two idempotent elements, namely $0 \neq 1$.

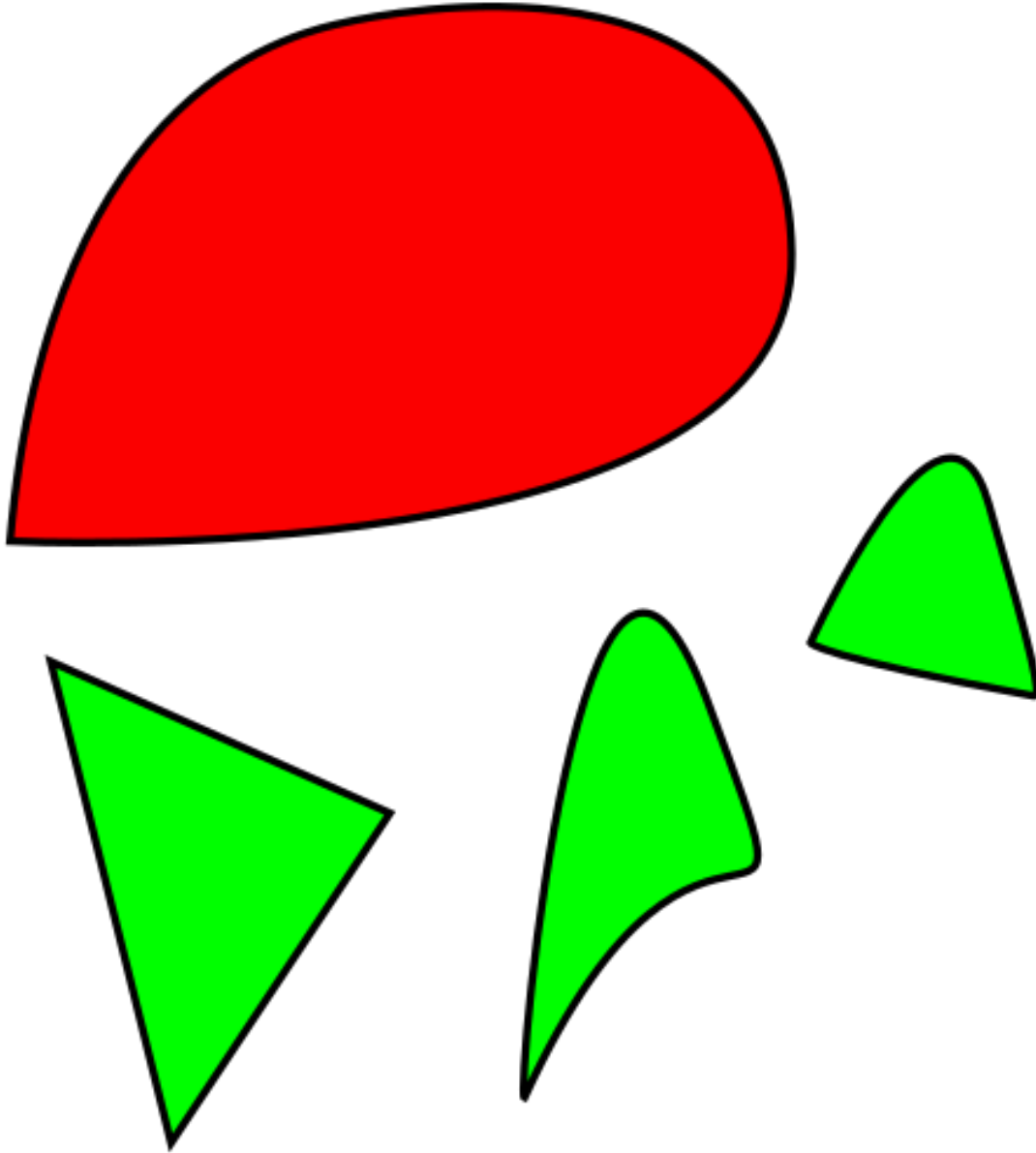


Figure 67: One unbroken red shape and two separate green shapes

A connected topological space (red) and a disconnected space (green); Dbc334, public domain.

Definition: connected topological space

A topological space X is called *connected* if exactly two subsets of X —namely $\emptyset$ and the whole space $X \neq \emptyset$ —are both open and closed.

The empty set and the whole space are always both open and closed. Such sets are also called *boundaryless* or *clopen*. The empty topological space is not considered connected, since it has only one subset that is both open and closed.

Lemma: the K -spectrum of a product ring

Let K be a field and R_1, R_2 K -algebras of finite type. For $R = R_1 \times R_2$ there is a natural homeomorphism

$$K\text{-Spek}(R_1 \times R_2) \cong K\text{-Spek}(R_1) \uplus K\text{-Spek}(R_2).$$

The embeddings from right to left are induced by the projections $R \rightarrow R_i, i = 1, 2$.

Proof The projection $R_1 \times R_2 \rightarrow R_1$ is a K -algebra homomorphism and, by Proposition 12.8(3), induces a continuous map—indeed, a closed embedding—

$$K\text{-Spek}(R_1) \longrightarrow K\text{-Spek}(R_1 \times R_2).$$

The same holds for R_2 . Together, the two maps give a continuous map from the disjoint union on the right to the left-hand side.

Take $P \in K\text{-Spek}(R_1 \times R_2)$, that is, a K -algebra homomorphism $P : R_1 \times R_2 \rightarrow K$. Let

$$e_1 = (1, 0), \quad e_2 = (0, 1).$$

Since $e_1 + e_2 = 1$ and $e_1 e_2 = 0$, exactly one of these elements maps under P to 0, and the other to 1. If, say, e_1 maps to 0, then $R_1 \times 0$ maps to 0 as well. Thus P factors through one of the projections. This proves surjectivity.

For injectivity, take two distinct points in the disjoint union. If they lie in the same component, their images remain distinct because the map on that component is a closed embedding. If they lie in different components, their values on e_1 are 0 and 1, respectively, so they are also distinct as points of the spectrum of the product.

This bijective map is a homeomorphism because the two closed embeddings combine to form a closed map.

Theorem: idempotent elements and clopen subsets

Let K be an algebraically closed field and R a reduced commutative K -algebra of finite type. The map

$$e \mapsto D(e)$$

gives a bijection between the idempotent elements of R and the subsets of $K\text{-Spek}(R)$ that are both open and closed.

Proof First,

$$D(e) = V(1 - e)$$

is both open and closed. This follows from

$$D(e) \cup D(1 - e) = D(1) = K\text{-Spek}(R)$$

and

$$D(e) \cap D(1 - e) = D(e(1 - e)) = D(e - e^2) = D(0) = \emptyset.$$

Thus the map is well-defined.

Let e_1, e_2 be idempotents with

$$U = D(e_1) = D(e_2).$$

An idempotent in a field can take only the values 0 and 1. Thus both e_1 and e_2 take the value 1 on U and 0 outside U . They have the same value at every point. The identity theorem for reduced algebras over an algebraically closed field gives $e_1 = e_2$. This proves injectivity.

Now let $U = D(\mathfrak{a})$ be both open and closed. There is another ideal $\mathfrak{b}$ with

$$D(\mathfrak{a}) \cup D(\mathfrak{b}) = K\text{-Spek}(R), \quad D(\mathfrak{a}) \cap D(\mathfrak{b}) = \emptyset.$$

By Corollary 11.12, $\mathfrak{a}$ and $\mathfrak{b}$ together generate the unit ideal. Thus there are $a \in \mathfrak{a}$ and $b \in \mathfrak{b}$ with $a + b = 1$. Since

$$D(a) \cap D(b) = D(ab) = \emptyset,$$

Exercise 12.11 says that ab is nilpotent. The ring R is reduced, so $ab = 0$. Consequently,

$$a = a \cdot 1 = a(a + b) = a^2 + ab = a^2,$$

so a is idempotent. Since $D(a) \subseteq D(\mathfrak{a})$, $D(b) \subseteq D(\mathfrak{b})$, and $D(a) \cup D(b) = K - \text{Spek}(R)$, we obtain

$$U = D(\mathfrak{a}) = D(a).$$

This proves surjectivity.

It follows that, over an algebraically closed field, a reduced K -algebra R of finite type is connected if and only if $K - \text{Spek}(R)$ is connected.

The last statement also holds without the reducedness assumption, since idempotent elements correspond bijectively after passing to the reduction; see Exercises 13.27 and 13.30.

Worksheet 13

Practice exercises

Exercise 13.1

Let R be a commutative ring and $S \subseteq R$ a multiplicative system. The localisation R_S is defined step by step as follows. First, let M be the set of formal fractions with denominator in S , namely

$$M = \left\{ \frac{r}{s} \mid r \in R, s \in S \right\}.$$

Show that

$$\frac{r}{s} \sim \frac{r'}{s'} \iff \text{there is } t \in S \text{ with } trs' = tr's$$

defines an equivalence relation on M . Denote its set of equivalence classes by R_S . Define a ring structure on R_S and a ring homomorphism $R \rightarrow R_S$.

Exercise 13.2

Let R be an integral domain and $S \subseteq R$ a multiplicative system with $0 \notin S$.

1. Show that the localisation

$$R_S := \left\{ \frac{f}{g} \mid f \in R, g \in S \right\} \subseteq Q(R)$$

is a subring of $Q(R)$.

2. Show that not every subring of $Q(R)$ is a localisation.

Exercise 13.3 ★

Show that the field of rational numbers $\mathbb{Q}$ has uncountably many subrings.

Exercise 13.4

Let R be a commutative ring and $f \in R$, with localisation R_f . Prove the R -algebra isomorphism

$$R_f \cong R[T]/(Tf - 1).$$

Exercise 13.5

Let R be a commutative ring, $f \in R$, and R_f the corresponding localisation. Show that f is nilpotent if and only if R_f is the zero ring.

In the following exercises on localisation, you may assume, if you wish, that the rings involved are integral domains.

Exercise 13.6 ★

Let R, A be commutative rings, $S \subseteq R$ a multiplicative system, and

$$\varphi : R \longrightarrow A$$

a ring homomorphism such that $\varphi(s)$ is a unit in A for every $s \in S$. Show that there is a unique ring homomorphism

$$\tilde{\varphi} : R_S \longrightarrow A$$

extending φ .

Exercise 13.7

Let R be a commutative ring and $S \subseteq R$ a multiplicative system. Show that the prime ideals of R_S correspond precisely to the prime ideals of R disjoint from S .

Exercise 13.8 ★

Let K be a field, $R = K[X, Y]$, $S \subseteq R$ a multiplicative system, and $F \in R$. Show that there is a unique R -algebra isomorphism

$$(R/(F))_S \cong (R_S)/(F),$$

where the localisation on the left is taken at the image of S in $R/(F)$.

Exercise 13.9 ★

Let R be a commutative ring, $\mathfrak{a} \subseteq R$ an ideal, and $S \subseteq R$ a multiplicative system. Show that there is a natural ring isomorphism

$$(R/\mathfrak{a})_S \cong R_S/\mathfrak{a}R_S,$$

where the localisation on the left is taken at the image of S in $R/\mathfrak{a}$.

Exercise 13.10

Let K be an algebraically closed field, R, S commutative K -algebras of finite type, $f \in R$, and

$$\varphi : R \longrightarrow S$$

a K -algebra homomorphism. Show that the spectrum map φ^* factors through $D(f)$ if and only if $\varphi(f)$ is a unit in S .

Exercise 13.11 ★

Let K be an algebraically closed field and R a K -algebra of finite type that is an integral domain. For $f, g \in R$, show that the following statements are equivalent:

1. $D(f) \subseteq D(g)$;
2. there is an R -algebra homomorphism $R_g \rightarrow R_f$.

Also show that this equivalence fails for $K = \mathbb{R}$.

The following exercise uses the notion of a saturated multiplicative system. A multiplicative system S in a commutative ring R is called *saturated* if the following holds: whenever $g \in R$ divides some $f \in S$, we also have $g \in S$.

Exercise 13.12

Let A, B be commutative rings and $\varphi : A \rightarrow B$ a ring homomorphism. Show that the inverse image

$$\varphi^{-1}(B^\times)$$

of the unit group is a saturated multiplicative system in A .

Exercise 13.13

Let R be a commutative ring. Show that the set of all non-zero-divisors in R forms a saturated multiplicative system.

Exercise 13.14 ★

Give an example of a $\mathbb{C}$ -algebra R of finite type that is an integral domain, and a multiplicative system $S \subseteq R$, $0 \notin S$, such that R_S is not a field, but every maximal ideal of R becomes the unit ideal in R_S .

Exercise 13.15 ★

Show that every integral domain is a connected ring.

Exercise 13.16

Let R be a commutative ring and $f \in R$. If f is both nilpotent and idempotent, show that $f = 0$.

Exercise 13.17 ★

For every $n \geq 2$, give a commutative ring R and an element $x \in R$, $x \neq 0$, satisfying

$$nx = 0 \quad \text{and} \quad x^n = 0.$$

Exercise 13.18

Let R be a commutative ring and $e \in R$ an idempotent element. Show that there is a natural ring isomorphism

$$R_e \cong R/(1 - e).$$

This shows once again that $D(e)$ is both open and closed.

Exercise 13.19

Let R, S be commutative rings. Show that the subset $R \times 0$ of the product ring $R \times S$ is a principal ideal.

Exercise 13.20 ★

Let $p \in \mathbb{Z}$ be a prime number and $n \in \mathbb{N}$. Show that the residue class ring $\mathbb{Z}/(p^n)$ has only the two trivial idempotents, 0 and 1.

Edition note: To have two distinct idempotents one needs $n \geq 1$. The source allows $n \in \mathbb{N}$; at $n = 0$ the quotient is the zero ring, in which $0 = 1$ is its only element.

Exercise 13.21 ★

Write the residue class ring

$$\mathbb{Q}[X]/(X^4 - 1)$$

as a product of fields involving only $\mathbb{Q}$ and $\mathbb{Q}[i]$. Write the residue class of $X^3 + X$ as a tuple in this product decomposition.

Exercise 13.22

Let K be a field, $a_1, \dots, a_n \in K$ distinct elements, and

$$F = (X - a_1) \cdots (X - a_n) \in K[X].$$

Show that the residue class ring $K[X]/(F)$ is isomorphic to the product ring K^n .

Exercise 13.23

Let K be an algebraically closed field. Show that, for every nonzero polynomial $F \in K[X]$, its residue class ring has the structure

$$K[X]/(F) \cong K[T]/(T^{n_1}) \times \cdots \times K[T]/(T^{n_r}).$$

Also show that

$$\deg(F) = n_1 + \cdots + n_r.$$

Exercise 13.24 ★

Let K be a field and

$$A = K[X_1, \dots, X_m]/\mathfrak{a}, \quad B = K[Y_1, \dots, Y_n]/\mathfrak{b}$$

K -algebras of finite type. Set

$$\ell = \max(m, n).$$

Show that the K -spectrum of the product ring $A \times B$ can be realised as a closed subset of $\mathbb{A}_K^{\ell+1}$.

Exercise 13.25

Let X be a nonempty disconnected topological space. Show that there is a continuous function

$$f : X \longrightarrow \mathbb{R}, \quad f \neq 0, 1,$$

where $\mathbb{R}$ has the metric topology, that is idempotent in the ring of continuous functions on X .

Exercise 13.26

Let X be a topological space with a disjoint decomposition

$$X = U \sqcup V$$

into open subsets $U, V \subseteq X$. Show that the natural map

$$\begin{aligned} C(X, \mathbb{R}) &\longrightarrow C(U, \mathbb{R}) \times C(V, \mathbb{R}), \\ f &\longmapsto (f|_U, f|_V) \end{aligned}$$

is bijective.

Exercise 13.27 ★

Let R be a commutative ring with reduction S . Show that the map sending each idempotent of R to its residue class in S is injective.

Exercise 13.28 ★

Let R be a commutative ring with an element $n \in R$ such that $n^2 = 0$, and let

$$S = R/(n).$$

Show that every idempotent element e of S has an idempotent preimage in R .

Exercise 13.29

Let R be a Noetherian commutative ring with reduction S . Show that there is a sequence of commutative rings R_i , $1 \leq i \leq n$, and surjective ring homomorphisms

$$\varphi_i : R_i \longrightarrow R_{i+1}$$

such that the composite map

$$R = R_0 \longrightarrow R_1 \longrightarrow \cdots \longrightarrow R_{n-1} \longrightarrow R_n = S$$

is the reduction map, and each φ_i is the quotient homomorphism $R_i \rightarrow R_i/(x_i)$ for some $x_i \in R_i$ with $x_i^2 = 0$.

Edition note: The source writes the domain and codomain of the last quotient homomorphism as $R \rightarrow R/(x_i)$; the notation $R_i \rightarrow R_i/(x_i)$ above follows the context of the sequence and the element $x_i \in R_i$.

Exercise 13.30

Let R be a commutative ring with reduction S . Show that the map sending each idempotent of R to its residue class in S is surjective.

The following statement is a version of the Chinese remainder theorem.

Exercise 13.31 ★

Let R be a commutative ring and let $\mathfrak{a}_j$, $j = 1, \dots, n$, be ideals satisfying

$$\mathfrak{a}_i + \mathfrak{a}_j = R$$

for all $i \neq j$. Show that

$$R/(\mathfrak{a}_1 \cdots \mathfrak{a}_n) \cong R/\mathfrak{a}_1 \times \cdots \times R/\mathfrak{a}_n.$$

Exercises for submission

Exercise 13.32 (4 points)

Let R be a principal ideal domain with field of fractions $Q = Q(R)$. Show that every intermediate ring

$$R \subseteq S \subseteq Q$$

is a localisation.

Exercise 13.33 (5 points: 1+2+1+1)

Consider the curve C given by

$$Y^2 = X^3 + X^2$$

(see Example 6.3) and the open set $U = D(X) \subseteq C$.

1. Find a closed realisation of U in $\mathbb{A}_K^3$.
2. Show that there is also a closed realisation in $\mathbb{A}_K^2$.
3. Is U isomorphic to an open subset of the affine line?
4. Sketch the image curve under the map

$$\begin{aligned} U &\longrightarrow \mathbb{A}_{\mathbb{R}}^2, \\ (x, y) &\longmapsto \left(\frac{1}{x}, y\right). \end{aligned}$$

Exercise 13.34 (4 points)

Consider the union V of two parallel lines and the union W of the coordinate axes. Describe a surjective map between V and W that is as natural as possible—decide in which direction—both geometrically and algebraically. Is there also a surjective polynomial map in the opposite direction?

Exercise 13.35 (3 points)

Determine all nilpotent elements and all idempotent elements of $\mathbb{Z}/(175)$.

Exercise 13.36 (4 points)

Let K be an algebraically closed field. Consider the intersection of the two algebraic curves

$$V(X^2 + Y^2 - 1) \quad \text{and} \quad V(Y - X^2).$$

Identify the residue class ring

$$R = K[X, Y]/(X^2 + Y^2 - 1, Y - X^2)$$

with a product ring, and describe the quotient map $K[X, Y] \rightarrow R$ using this identification. Determine preimages in $K[X, Y]$ for all the idempotents of that product ring.

Exercise 13.37 (6 points)

Let K be a field and A a finite-dimensional reduced K -algebra. Show that A is a finite direct product of finite field extensions of K .

Hint. You may use without proof that A has only finitely many prime ideals.

Public Solutions to Worksheet 13

At the frozen revision boundary, the source provides public solutions only to Exercises 13.3, 13.6, 13.8, 13.9, 13.11, 13.14, 13.15, 13.17, 13.20, 13.21, 13.24, 13.27, 13.28, and 13.31. No additional solutions have been created for this edition.

Solution to Exercise 13.3

Let T be a subset of the set of prime numbers. Since there are infinitely many prime numbers, there are uncountably many choices of T . Associate with T the multiplicative system $M(T)$ consisting of all integers whose prime factorisations contain only primes from T . The localisation

$$\mathbb{Z}_{M(T)} \subseteq \mathbb{Q}$$

consists of all rational numbers that can be written with a denominator whose prime factorisation uses only primes from T . Uniqueness of prime factorisation in $\mathbb{Z}$ shows that these subrings are distinct for distinct choices of T .

[Back to Exercise 13.3.](#)

Solution to Exercise 13.6

For the diagram of ring homomorphisms to commute, we must have

$$\tilde{\varphi}(1/s) = \varphi(s)^{-1}$$

for $s \in S$, and hence

$$\tilde{\varphi}(a/s) = \varphi(a)\varphi(s)^{-1}.$$

Thus there is at most one such ring homomorphism, and it must be given by the last formula.

We need to show that this formula is well-defined. Let $a/s = b/t$ with $s, t \in S$. This means that there is an $r \in S$ such that $rta = rsb$. Then

$$\varphi(r)\varphi(t)\varphi(a) = \varphi(r)\varphi(s)\varphi(b).$$

Multiplying both sides by the unit $\varphi(r)^{-1}\varphi(t)^{-1}\varphi(s)^{-1}$ gives

$$\varphi(a)\varphi(s)^{-1} = \varphi(b)\varphi(t)^{-1}.$$

As an example of verifying the homomorphism properties, for addition we obtain

$$\begin{aligned} \tilde{\varphi}\left(\frac{a}{s} + \frac{b}{t}\right) &= \tilde{\varphi}\left(\frac{at + bs}{st}\right) \\ &= \varphi(at + bs)\varphi(st)^{-1} \\ &= (\varphi(a)\varphi(t) + \varphi(s)\varphi(b))\varphi(s)^{-1}\varphi(t)^{-1} \\ &= \varphi(a)\varphi(s)^{-1} + \varphi(b)\varphi(t)^{-1} \\ &= \tilde{\varphi}\left(\frac{a}{s}\right) + \tilde{\varphi}\left(\frac{b}{t}\right). \end{aligned}$$

[Back to Exercise 13.6.](#)

Solution to Exercise 13.8

All homomorphisms below are R -algebra homomorphisms and are uniquely determined by the stated properties. The homomorphism $R \rightarrow R_S$ first induces

$$R/(F) \longrightarrow R_S/(F).$$

Since the image of S in $R/(F)$ becomes units in $R_S/(F)$, the universal property of localisation gives a homomorphism

$$(R/(F))_S \longrightarrow R_S/(F), \quad \frac{\bar{r}}{\bar{s}} \mapsto \overline{\left(\frac{r}{s}\right)}.$$

This map is surjective: every element on the right is represented by r/s with $s \in S$ and comes from $\bar{r}/\bar{s}$.

For injectivity, suppose that $\bar{r}/\bar{s}$ maps to 0. Then $r/s \in (F)R_S$, so $r/s = Fa/t$ for some $a \in R$ and $t \in S$. Translating this equality back into R gives

$$tr = sFa.$$

Thus $tr = 0$ in $R/(F)$. Since $t \in S$, this gives $\bar{r}/\bar{s} = 0$ in $(R/(F))_S$.

Edition note: The source's cancellation in $R = K[X, Y]$ assumes $0 \notin S$. If $0 \in S$, both localised rings are zero and the asserted isomorphism is immediate; the displayed cross-multiplication argument is used only in the other case.

[Back to Exercise 13.8.](#)

Solution to Exercise 13.9

The ring homomorphism

$$R \longrightarrow R_S/\mathfrak{a}R_S$$

sends $\mathfrak{a}$ to 0 and therefore induces a homomorphism

$$R/\mathfrak{a} \longrightarrow R_S/\mathfrak{a}R_S.$$

The universal property of localisation then induces

$$(R/\mathfrak{a})_S \longrightarrow R_S/\mathfrak{a}R_S, \quad \frac{[r]}{s} \mapsto \left[\frac{r}{s}\right].$$

This formula immediately shows surjectivity. If the image $[r/s]$ is zero, then $r/s \in \mathfrak{a}R_S$, and hence $r \in \mathfrak{a}R_S$. Thus there is a $t \in S$ with $tr \in \mathfrak{a}$. Hence $[tr] = 0$ in $R/\mathfrak{a}$, and consequently $[r]/s = 0$ in $(R/\mathfrak{a})_S$. The map is therefore also injective.

Edition note: The source writes $[r/s] \in \mathfrak{a}R_S$ in the kernel argument. The ideal-membership statement concerns the representative r/s in R_S , as made explicit above.

[Back to Exercise 13.9.](#)

Solution to Exercise 13.11

If (2) holds, we can in particular write

$$\frac{1}{g} = \frac{r}{f^n}, \quad \text{or equivalently} \quad f^n = rg.$$

Thus g divides a power of f , that is, $f \in \text{rad}(g)$. Conversely, if $f \in \text{rad}(g)$, then g is a unit in R_f , and the universal property of localisation gives an R -algebra homomorphism $R_g \rightarrow R_f$.

From $f \in \text{rad}(g)$ it follows immediately that f vanishes on $V(g)$, so $V(g) \subseteq V(f)$. If K is algebraically closed, the reverse implication follows from Hilbert's Nullstellensatz. Since $D(f) \subseteq D(g)$ is equivalent to $V(g) \subseteq V(f)$, the two statements in the exercise are equivalent.

Edition note: The source's fraction argument applies directly when $f, g \neq 0$. If $f = 0$, both conditions hold because $D(0) = \emptyset$ and R_0 is the zero ring. If $f \neq 0$ and $g = 0$, neither holds: $D(f) \neq \emptyset$ by the Nullstellensatz, and there is no unital homomorphism from the zero ring to the nonzero ring R_f .

For $K = \mathbb{R}$, take $R = K[X]$, $f = 1$, and $g = X^2 + 1$. The polynomial g has no real zero, so

$$V(g) = \emptyset = V(1)$$

and $D(f) \subseteq D(g)$, but g is not a unit in $R_f = R$.

Edition note: In the last example, the source displays $V(g) = \mathbb{A}_{\mathbb{R}}^1 = V(1)$. Both zero loci in question are empty; the relation between the open sets remains as stated above.

[Back to Exercise 13.11.](#)

Solution to Exercise 13.14

Take

$$R = \mathbb{C}[X, Y]$$

and let S be the multiplicative system consisting of all products of elements of the form $X - a$, $a \in \mathbb{C}$. The maximal ideals of R have the form

$$(X - a, Y - b).$$

Thus every maximal ideal contains an element of S and becomes the unit ideal in R_S . However, R_S is not a field: precisely the prime elements $X - a$ are made into units, whereas other prime elements, such as Y , are not.

[Back to Exercise 13.14.](#)

Solution to Exercise 13.15

An idempotent element e satisfies

$$e(1 - e) = e - e^2 = e - e = 0.$$

In a ring without zero divisors, this implies $e = 1$ or $e = 0$.

[Back to Exercise 13.15.](#)

Solution to Exercise 13.17

Consider the residue class ring

$$R = (\mathbb{Z}/n\mathbb{Z})[X]/(X^n)$$

and write x for the residue class of X . The element x is nonzero: in the polynomial ring, X cannot be a multiple of X^n when $n \geq 2$ for degree reasons. In R we have $x^n = 0$, so $ny = 0$ for every $y \in R$, in particular $nx = 0$. Moreover, $x^n = 0$ because the entire ideal (X^n) is made zero when forming the quotient ring.

[Back to Exercise 13.17.](#)

Solution to Exercise 13.20

Let $e \in \mathbb{Z}/(p^n)$ be idempotent. Choosing an integer representative, the equation $e^2 = e$ means that

$$p^n \mid e(e - 1).$$

The integers e and $e - 1$ are coprime, so they cannot both be divisible by p . Since p^n divides their product, the whole factor p^n must divide either e or $e - 1$. Thus

$$e = 0 \quad \text{or} \quad e = 1$$

in $\mathbb{Z}/(p^n)$.

Edition note: The source writes factorisations $e = bp^i$ and $e - 1 = cp^j$ with $i + j = n$. These exponents may be chosen as divisibility exponents; they need not be the exact p -adic valuations. The coprimality argument above expresses the same step directly. For $n = 0$, the conclusion still holds, but $0 = 1$; see the note to Exercise 13.20.

[Back to Exercise 13.20.](#)

Solution to Exercise 13.21

We have

$$X^4 - 1 = (X^2 - 1)(X^2 + 1) = (X + 1)(X - 1)(X^2 + 1).$$

The polynomial $X^2 + 1 \in \mathbb{Q}[X]$ is irreducible because it has no rational root. Thus this is the prime factorisation, and the monic factors above are pairwise nonassociate. The Chinese remainder theorem for principal ideal domains gives

$$\begin{aligned} \mathbb{Q}[X]/(X^4 - 1) &\cong \mathbb{Q}[X]/(X + 1) \times \mathbb{Q}[X]/(X - 1) \times \mathbb{Q}[X]/(X^2 + 1) \\ &\cong \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q}[i]. \end{aligned}$$

The last isomorphism uses the substitutions $X \mapsto -1$, $X \mapsto 1$, and $\mathbb{Q}[X]/(X^2 + 1) \cong \mathbb{Q}[i]$. The element $X^3 + X = X(X^2 + 1)$ maps under the three projections to -2 , 2 , and 0 . Its tuple is therefore

$$(-2, 2, 0).$$

[Back to Exercise 13.21.](#)

Solution to Exercise 13.24

Without loss of generality, suppose $m \geq n$. We can write

$$B \cong K[X_1, \dots, X_n]/\mathfrak{b} \cong K[X_1, \dots, X_m]/(\mathfrak{b} + (X_{n+1}, \dots, X_m)).$$

Denote this extended ideal by $\mathfrak{b}'$. The two K -spectra have thus been realised as closed subsets of the same affine space. Use one additional variable Z to separate them, and consider

$$C = K[X_1, \dots, X_m, Z]/(Z(Z-1), Z\mathfrak{a}, (Z-1)\mathfrak{b}').$$

The K -spectrum of C is the disjoint union of the two given spectra. Indeed, the set

$$V = V(Z(Z-1), Z\mathfrak{a}, (Z-1)\mathfrak{b}') \subseteq \mathbb{A}_K^{m+1}$$

satisfies $Z = 0$ or $Z = 1$. The part with $Z = 0$ is

$$\begin{aligned} V_0 &= V(Z, Z(Z-1), Z\mathfrak{a}, (Z-1)\mathfrak{b}') \\ &= V(Z, -\mathfrak{b}') \\ &\cong K\text{-Spek}(K[X_1, \dots, X_m, Z]/(Z, \mathfrak{b}')) \\ &\cong K\text{-Spek}(K[X_1, \dots, X_n]/\mathfrak{b}) \\ &= K\text{-Spek}(B), \end{aligned}$$

whereas the part with $Z = 1$ is

$$\begin{aligned} V_1 &= V(Z-1, Z(Z-1), Z\mathfrak{a}, (Z-1)\mathfrak{b}') \\ &= V(Z-1, Z\mathfrak{a}) \\ &\cong K\text{-Spek}(K[X_1, \dots, X_m, Z]/(Z-1, \mathfrak{a})) \\ &\cong K\text{-Spek}(K[X_1, \dots, X_m]/\mathfrak{a}) \\ &= K\text{-Spek}(A). \end{aligned}$$

[Back to Exercise 13.24.](#)

Solution to Exercise 13.27

Let $e, f \in R$ be idempotent and suppose their images in the reduction are equal. Then $e - f$ is nilpotent in R . Thus there is an $n \in \mathbb{N}$ with

$$(e - f)^n = 0.$$

We may take n to be odd. By the binomial theorem, symmetry of binomial coefficients, and idempotence, we obtain

$$\begin{aligned}
0 &= (e - f)^n = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} e^k f^{n-k} \\
&= e^n - f^n + \sum_{k=1}^{n-1} (-1)^{n-k} \binom{n}{k} e^k f^{n-k} \\
&= e - f + \sum_{k=1}^{n-1} (-1)^{n-k} \binom{n}{k} e^k f^{n-k} \\
&= e - f + \sum_{k=1}^{(n-1)/2} (-1)^{n-k} \binom{n}{k} (e^k f^{n-k} - e^{n-k} f^k) \\
&= e - f + \sum_{k=1}^{(n-1)/2} (-1)^{n-k} \binom{n}{k} (ef - ef) \\
&= e - f.
\end{aligned}$$

Hence $e = f$.

Edition note: The source omits the alternating signs in its binomial sums. The factors $(-1)^{n-k}$ above correct that omission. Since n is odd, the terms with indices k and $n - k$ have opposite signs, which justifies the displayed pairing and cancellation.

[Back to Exercise 13.27.](#)

Solution to Exercise 13.28

Take a preimage $f \in R$ of e . Since e is idempotent, the element

$$c = f^2 - f$$

lies in (n) , so $c^2 = 0$. Consider

$$g = f + c - 2cf.$$

This element also maps to e . Moreover,

$$\begin{aligned}
g^2 &= (f + c - 2cf)^2 \\
&= f^2 + c^2 + 4c^2f^2 + 2cf - 4cf^2 - 4c^2f \\
&= f^2 + 2cf - 4cf^2 \\
&= f + c + 2cf - 4c(f + c) \\
&= f + c + 2cf - 4cf \\
&= f + c - 2cf \\
&= g.
\end{aligned}$$

Thus g is an idempotent preimage of e .

[Back to Exercise 13.28.](#)

Solution to Exercise 13.31

The general case follows from the case $n = 2$, so it suffices to consider two ideals $\mathfrak{a}$ and $\mathfrak{b}$. The natural map

$$R \longrightarrow R/\mathfrak{a} \times R/\mathfrak{b}$$

has kernel $\mathfrak{a} \cap \mathfrak{b}$. For comaximal ideals this intersection equals the product $\mathfrak{a}\mathfrak{b}$. We therefore obtain an injective ring homomorphism

$$R/(\mathfrak{a}\mathfrak{b}) \longrightarrow R/\mathfrak{a} \times R/\mathfrak{b}.$$

To prove surjectivity, take (r, s) on the right. Choose $a \in \mathfrak{a}$ and $b \in \mathfrak{b}$ with $a + b = 1$. The element

$$r - ar + s - sb$$

is a preimage of (r, s) . Modulo $\mathfrak{a}$, it becomes

$$r - ar + s - sb = r + s - s(1 - a) = r + s - s = r,$$

and similarly modulo $\mathfrak{b}$ it becomes s .

[Back to Exercise 13.31.](#)

Lecture 14: Algebraic Functions on Varieties

Algebraic functions

What is a morphism between two affine algebraic sets V and W ? We first consider the case in which

$$W = \mathbb{A}_K^1$$

is the affine line. Suppose that

$$V = V(\mathfrak{a}) \subseteq \mathbb{A}_K^n$$

is given as a closed subset of an affine space. Every polynomial

$$F \in K[X_1, \dots, X_n]$$

then gives a map

$$F : \mathbb{A}_K^n \longrightarrow \mathbb{A}_K^1 = K$$

and hence, by restriction, a map on V . We already considered this when defining the coordinate ring. Likewise, an element F of a finitely generated K -algebra R gives a function

$$\begin{aligned} K\text{-Spek}(R) &\longrightarrow \mathbb{A}_K^1, \\ P &\longmapsto F(P). \end{aligned}$$

This is also the map of spectra that, by Proposition 12.8(2), corresponds to the substitution homomorphism

$$K[T] \longrightarrow R, \quad T \longmapsto F.$$

On the open set

$$D(F) \cong K\text{-Spek}(R_F),$$

the function $1/F$ is well defined by Theorem 13.4. We shall now explain what an algebraic function on an arbitrary Zariski-open set $U \subseteq V$ is. The following definition is arranged so that being “algebraic” is a local property.

Definition: algebraic functions on an open set

Let K be an algebraically closed field, let R be a K -algebra of finite type, and let

$$V = K\text{-Spek}(R).$$

Let $P \in V$, let $U \subseteq V$ be a Zariski-open set with $P \in U$, and let

$$f : U \longrightarrow \mathbb{A}_K^1 = K$$

be a function. We call f *algebraic* (also *regular* or *polynomial*) at P if there are elements $G, H \in R$ such that

$$P \in D(H) \subseteq U$$

and

$$f(Q) = \frac{G(Q)}{H(Q)} \quad \text{for every } Q \in D(H).$$

We call f *algebraic on U* if it is algebraic at every point of U .

Of course, every element $f \in R$ defines an algebraic function on every open subset of the K -spectrum. In general, however, it is rather difficult to give a concise description of all algebraic functions.

Remark: locality and fractional representations

In Definition 14.1, the condition $D(H) \subseteq U$ is not essential. If there is a representation $f = G/H$ on $D(H)$ with $P \in D(H)$, choose H' such that

$$P \in D(H') \subseteq U.$$

On

$$D(H) \cap D(H') = D(HH')$$

we may use the representation

$$f = \frac{GH'}{HH'}.$$

If $f = G/H$ is a fractional representation at P , the same representation works for every point of $D(H)$. Thus f is algebraic on the whole open set $D(H)$. In particular, we need not work with infinitely many different representations: finitely many fractions G_i/H_i for a cover

$$U = \bigcup_{i \in I} D(H_i)$$

suffice.

For $K = \mathbb{C}$, an algebraic function is also continuous in the metric topology; when $R = \mathbb{C}[X_1, \dots, X_n]$, it is holomorphic.

Example: a function glued from two fractions

Let

$$V = V(WX - ZY) \subseteq \mathbb{A}_K^4$$

and let

$$U = D(X, Y) = D(X) \cup D(Y) \subset V$$

be the Zariski-open set defined by X and Y . The function on U defined by

$$f = \frac{Z}{X} = \frac{W}{Y}$$

is algebraic. The two fractions clearly give algebraic functions on $D(X)$ and $D(Y)$ respectively. To define a single function on U , their values must agree on the intersection $D(X) \cap D(Y) = D(XY)$. Take

$$Q = (w, x, y, z) \in D(XY) \cap V.$$

We have $x, y \neq 0$ and $wx = zy$, so

$$\frac{Z}{X}(Q) = \frac{z}{x} = \frac{w}{y} = \frac{W}{Y}(Q).$$

Lemma: algebraic functions form an algebra

Let K be an algebraically closed field, let R be a K -algebra of finite type, let $V = K\text{-Spek}(R)$, and let $U \subseteq V$ be Zariski-open. The algebraic functions on U form a subring—in fact, a K -subalgebra—of the ring of all functions $U \rightarrow K$, with operations performed in K .

Proof We must check that the constant zero and one functions, the negative of an algebraic function, and the sum and product of two algebraic functions are again algebraic. We restrict ourselves to the sum. Let f_1, f_2 be algebraic and let $P \in U$. There are $G_1, H_1, G_2, H_2 \in R$ such that

$$f_1(Q) = \frac{G_1(Q)}{H_1(Q)} \quad (Q \in D(H_1) \subseteq U), \quad P \in D(H_1),$$

and

$$f_2(Q) = \frac{G_2(Q)}{H_2(Q)} \quad (Q \in D(H_2) \subseteq U), \quad P \in D(H_2).$$

Set $H = H_1 H_2$. Then

$$P \in D(H) = D(H_1) \cap D(H_2) \subseteq U.$$

For $Q \in D(H)$ we have

$$\begin{aligned} (f_1 + f_2)(Q) &= f_1(Q) + f_2(Q) \\ &= \frac{G_1(Q)}{H_1(Q)} + \frac{G_2(Q)}{H_2(Q)} \\ &= \frac{G_1(Q)H_2(Q) + G_2(Q)H_1(Q)}{H_1(Q)H_2(Q)} \\ &= \frac{(G_1H_2 + G_2H_1)(Q)}{(H_1H_2)(Q)}. \end{aligned}$$

Thus the sum has a fractional representation on the Zariski-open neighbourhood $D(H)$ of P . The other cases are similar.

Definition: the ring of algebraic sections

In the situation above, the ring

$$\Gamma(U, \mathcal{O}) = \{f : U \longrightarrow K \mid f \text{ is algebraic}\}$$

is called the *ring of algebraic functions* on U . It is also called the *structure ring* or *ring of sections* on U . By Lemma 14.4 this set is indeed a ring. The symbol $\mathcal{O}$ (pronounced “O”) denotes the so-called *structure sheaf*.

Lemma: restriction maps

Let K be an algebraically closed field and R a K -algebra of finite type. Let $U_1 \subseteq U_2$ be open subsets of $V = K - \text{Spek}(R)$. There is a natural K -algebra homomorphism

$$\Gamma(U_2, \mathcal{O}) \longrightarrow \Gamma(U_1, \mathcal{O}).$$

Proof A function $f : U_2 \rightarrow K$ immediately gives a function on U_1 by restriction. The local algebraic description of f at each point $P \in U_2$ also applies on the smaller subset U_1 .

The map in this lemma is called the *restriction map*.

Lemma: independence of a principal open ambient space

Let K be an algebraically closed field and R a K -algebra of finite type. Let $F \in R$ and let

$$U \subseteq D(F) \subseteq V = K - \text{Spek}(R)$$

be open. The definition of $\Gamma(U, \mathcal{O})$ gives the same ring whether we take the ambient space to be V or

$$D(F) = K - \text{Spek}(R_F).$$

Proof Functions on U clearly depend only on U , not on an ambient space. It remains to show that the local algebraic condition also depends only on U .

Take $P \in U$. A representation

$$\varphi = \frac{G}{H} \text{ on } D(H), \quad P \in D(H), \quad G, H \in R,$$

immediately gives a fractional representation on $D(HF)$ by regarding G, H as elements of R_F .

Conversely, suppose that there is a representation over R_F

$$\varphi = \frac{\tilde{G}}{\tilde{H}} \text{ on } D(\tilde{H}), \quad P \in D(\tilde{H}),$$

where

$$\tilde{G} = \frac{G}{F^r}, \quad \tilde{H} = \frac{H}{F^s}.$$

For $Q \in D(HF)$ we have

$$\varphi(Q) = \frac{\tilde{G}(Q)}{\tilde{H}(Q)} = \frac{G(Q)/F^r(Q)}{H(Q)/F^s(Q)} = \frac{G(Q)F^s(Q)}{H(Q)F^r(Q)}.$$

In the last step we multiplied numerator and denominator by F^{r+s} . The final numerator and denominator lie in R , and $HF^r(P) \neq 0$. Thus $D(HF^r)$ is an open neighbourhood of P giving a representation relative to V .

Edition note: In the source's last neighbourhood argument, take $r \geq 1$, which is always possible by multiplying the numerator and denominator of $\tilde{G} = G/F^r$ by F . Then $D(HF^r) = D(HF) \subseteq U$. If $r = 0$ were retained, $D(HF^r) = D(H)$ could extend outside the original domain.

Lemma: the relation between two rational representations

Let K be an algebraically closed field and R a K -algebra of finite type. Let $f : U \rightarrow K$ be an algebraic function on a Zariski-open set $U \subseteq V = K\text{-Spek}(R)$. Suppose that near $P \in U$ it has two representations

$$\frac{G_1}{H_1} \quad \text{and} \quad \frac{G_2}{H_2},$$

with $G_1, H_1, G_2, H_2 \in R$ and $P \in D(H_1), D(H_2) \subseteq U$. Then there is an $r \in \mathbb{N}$ such that

$$H_1^r H_2^r (G_1 H_2 - G_2 H_1)^r = 0 \quad \text{in } R.$$

If R is reduced, we even have

$$H_1 H_2 (G_1 H_2 - G_2 H_1) = 0.$$

Proof Consider the element

$$F = H_1 H_2 (G_1 H_2 - G_2 H_1)$$

on V . We show that it induces the zero function. Take $Q \in V$. If $H_1(Q) = 0$ or $H_2(Q) = 0$, then $F(Q) = 0$ immediately. If neither is zero, then $Q \in D(H_1) \cap D(H_2)$ and

$$\frac{G_1(Q)}{H_1(Q)} = f(Q) = \frac{G_2(Q)}{H_2(Q)}.$$

Hence $G_1(Q)H_2(Q) = G_2(Q)H_1(Q)$ and again $F(Q) = 0$. By Hilbert's Nullstellensatz there is an r such that $F^r = 0$ in R . If R is reduced, $F = 0$.

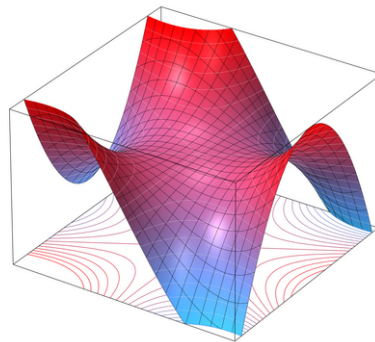


Figure 68: Copper-coloured monkey saddle surface with three valleys and three ridges

The graph of a global function on two-dimensional affine space; Inductiveload, public domain. Source details are given in the Unit 14 media credits.

Theorem: global sections on an affine spectrum

Let K be an algebraically closed field, let R be a reduced K -algebra of finite type, and let $V = K\text{-Spek}(R)$. Then

$$\Gamma(V, \mathcal{O}) = R.$$

Proof Every $F \in R$ directly gives an algebraic function on all of V , so there is a K -algebra homomorphism

$$R \longrightarrow \Gamma(V, \mathcal{O}).$$

If F induces the zero function at every point, Theorem 11.1 and the reducedness of R imply $F = 0$. Thus the map is injective.

Now let $f : V \rightarrow K$ be an algebraic function. For every $P \in V$ there are $G_P, H_P \in R$ with $P \in D(H_P)$ and

$$f = \frac{G_P}{H_P} \quad \text{on } D(H_P).$$

The sets $D(H_P)$ cover V . By Corollary 11.12, the elements H_P generate the unit ideal, so finitely many of them already generate the unit ideal. Denote them by

$$H_i = H_{P_i}, \quad i = 1, \dots, m.$$

Then the $D(H_i)$ cover all of V . On each intersection $D(H_i H_j) = D(H_i) \cap D(H_j)$ we have

$$f(Q) = \frac{G_i(Q)}{H_i(Q)} = \frac{G_j(Q)}{H_j(Q)}.$$

By Lemma 14.8 and reducedness,

$$H_i H_j G_i H_j = H_i H_j G_j H_i$$

in R . Replace H_i by H_i^2 and G_i by $G_i H_i$. The representation G_i/H_i remains unchanged, while the last relation simplifies to

$$H_i G_j = H_j G_i.$$

Since the H_i generate the unit ideal, there are $A_i \in R$ with

$$\sum_{i=1}^m A_i H_i = 1.$$

Set

$$F = \sum_{i=1}^m A_i G_i.$$

We claim that F induces f on all of V . Take $Q \in V$; without loss of generality, suppose $Q \in D(H_1)$. Then

$$\begin{aligned}
f(Q) &= \frac{G_1(Q)}{H_1(Q)} \\
&= \frac{G_1(Q)}{H_1(Q)} \left(\sum_{i=1}^m A_i H_i \right) (Q) \\
&= \sum_{i=1}^m A_i(Q) \frac{G_1(Q) H_i(Q)}{H_1(Q)} \\
&= \sum_{i=1}^m A_i(Q) G_i(Q) \\
&= F(Q).
\end{aligned}$$

The homomorphism above is therefore also surjective.

Corollary: sections on a principal open set

Let $F \in R$ in the situation of the preceding theorem. Then

$$\Gamma(D(F), \mathcal{O}) = R_F.$$

Proof This follows directly from Lemma 14.7 and Theorem 14.9.

Remark: Hilbert's fourteenth problem

One variant of Hilbert's fourteenth problem asks whether the ring of algebraic functions $\Gamma(U, \mathcal{O})$ is finitely generated for every open set U . This is true for open sets of the form $U = D(f)$, also when R is regular or factorial, and in small dimensions. In general, however, it is false.

Edition note: "Small dimensions" is the source's informal wording; it specifies no dimension bound or additional hypotheses. No precise low-dimensional theorem is being asserted by that phrase here.

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Worksheet 14

Practice exercises

Exercise 14.1

Let K be an algebraically closed field and let $R = K[X]$ be the polynomial ring over K . Show that every algebraic function f on an open set

$$U = D(F) \subseteq K\text{-}\mathrm{Spek}(R) = \mathbb{A}_K^1$$

has the form

$$f = \frac{G}{H},$$

where $G, H \in R$ have no common nonunit factor and $D(F) \subseteq D(H)$.

Exercise 14.2 ★

Let K be an algebraically closed field and let R be a K -algebra of finite type that is a unique factorisation domain. Show that every algebraic function f on an open set

$$U \subseteq K\text{-Spek}(R)$$

has the form $f = G/H$, where $G, H \in R$ have no common nonunit factor and $U \subseteq D(H)$.

Exercise 14.3

Complete the proof of Lemma 14.4.

Exercise 14.4

Let $U \subseteq K\text{-Spek}(R)$ be an open subset of the K -spectrum of a K -algebra R over an algebraically closed field K , and let

$$f \in \Gamma(U, \mathcal{O})$$

be an algebraic function. Show that f defines a continuous map to K .

Exercise 14.5

Show that the ring $\Gamma(U, \mathcal{O})$ is reduced.

Exercise 14.6

Let K be an algebraically closed field. Consider the point

$$P = (0, 1) \in V(X^3 - Y^3 + 1) = C \subseteq \mathbb{A}_K^2$$

and set $U = C \setminus \{P\}$. Describe an algebraic function on U that cannot be extended to an algebraic function on all of C .

Exercise 14.7 ★

Consider Neil's parabola

$$C = V(Y^2 - X^3) \subseteq \mathbb{A}_K^2$$

and the point $P = (1, 1) \in C$. Find an algebraic function defined on $C \setminus \{P\}$ but not on all of C .

Hint. Find two different factorisations of $X^3 - X^2$.

Edition note: This exercise uses the standing assumption of this lecture that K is algebraically closed. The source does not repeat it here; the notion of algebraic function used in Definition 14.1 is stated under that assumption.

Exercise 14.8

Let K be an algebraically closed field, let R be a K -algebra of finite type that is an integral domain, and let

$$U = D(\mathfrak{a}) \subseteq K\text{-Spek}(R), \quad \mathfrak{a} = (f_1, \dots, f_n).$$

Show that

$$\Gamma(U, \mathcal{O}) = \bigcap_{i=1}^n R_{f_i},$$

where the intersection is taken inside the fraction field $Q(R)$.

Edition note: For this fraction-field interpretation, assume $U \neq \emptyset$ and omit any zero generators f_i . The source does not state these qualifications. Localisation at zero is the zero ring and cannot be viewed as a subring of $Q(R)$; for $U = \emptyset$ the ring of functions is the zero ring instead.

Exercise 14.9

Let R be a commutative K -algebra of finite type over an algebraically closed field, let $U \subseteq K\text{-Spek}(R)$ be open, and let $f : U \rightarrow K$ be a function. Suppose that

$$U = \bigcup_{i \in I} U_i$$

is an open cover and that every restriction $f_i = f|_{U_i}$ is an algebraic function. Show that f itself is algebraic.

Exercise 14.10

Let K be an algebraically closed field and let R be a K -algebra of finite type that is an integral domain. Show that for open sets $U \subseteq V$, the restriction map

$$\Gamma(V, \mathcal{O}) \longrightarrow \Gamma(U, \mathcal{O})$$

is injective.

Edition note: The source needs the hypothesis $U \neq \emptyset$ (unless V is also empty). Restriction from a nonempty V to the empty set sends all functions to the sole element of the zero ring and is not injective.

Exercise 14.11

Consider

$$V = V(XW - YZ) \subseteq \mathbb{A}_K^4.$$

Describe an open set $U \subseteq \mathbb{A}_K^4$ such that the ring homomorphism corresponding to $U \cap V \subseteq U$,

$$\Gamma(U, \mathcal{O}) \longrightarrow \Gamma(U \cap V, \mathcal{O}),$$

is not surjective.

The next exercise uses the concept of the limit of a map. Exercise 1.9 may be helpful.

Exercise 14.12

Consider the curve

$$C = V(Y^2 - X^2 - X^3) \subseteq \mathbb{A}_{\mathbb{C}}^2,$$

the point $P = (0, 0) \in C$, and the open complement $U = C \setminus \{P\}$.

1. Show that Y/X is an algebraic function on U that cannot be extended algebraically to all of C .
2. Show that the limit at P of the map $\varphi = Y/X : U \rightarrow \mathbb{C}$ does not exist.
3. Show that there are sequences $(w_n)_{n \in \mathbb{N}}$ and $(z_n)_{n \in \mathbb{N}}$ in U , both converging to P , whose image sequences under φ converge to different values.

The following concepts are important in many areas of mathematics and concisely capture essential properties of the structure sheaf on a K -spectrum.

Supporting definition: presheaves

Let X be a topological space. A *presheaf* $\mathcal{F}$ on X is an assignment associating:

- to every open set $U \subseteq X$, a set $\mathcal{F}(U)$;
- to every inclusion of open sets $U \subseteq V$, a map

$$\rho_{V,U} : \mathcal{F}(V) \longrightarrow \mathcal{F}(U),$$

such that

$$\rho_{U,U} = \text{Id}_{\mathcal{F}(U)}$$

and, for $U \subseteq V \subseteq W$,

$$\rho_{W,U} = \rho_{V,U} \circ \rho_{W,V}.$$

The maps $\rho_{V,U}$ are called *restriction maps*. The presheaf is a *presheaf of groups* if every $\mathcal{F}(U)$ is a group and every restriction map is a group homomorphism. It is a *presheaf of commutative rings* if every $\mathcal{F}(U)$ is a commutative ring and every restriction map is a ring homomorphism.

Exercise 14.13

Show that the assignment associating to every open set

$$U \subseteq K\text{-}\text{Spek}(R)$$

the ring of algebraic functions $\Gamma(U, \mathcal{O})$, and to every inclusion $U_1 \subseteq U_2$ the restriction map (see Lemma 14.6)

$$\Gamma(U_2, \mathcal{O}) \longrightarrow \Gamma(U_1, \mathcal{O}),$$

is a presheaf of K -algebras.

Supporting definition: sheaves

A *sheaf* $\mathcal{F}$ on a topological space X is a presheaf satisfying the following two properties.

1. **Local uniqueness.** For every open cover $U = \bigcup_{i \in I} U_i$ and $s, t \in \mathcal{F}(U)$, if

$$\rho_{U, U_i}(s) = \rho_{U, U_i}(t) \quad \text{for all } i \in I,$$

then $s = t$.

2. **Gluing.** For every open cover $U = \bigcup_{i \in I} U_i$ and sections $s_i \in \mathcal{F}(U_i)$ that are compatible on each intersection, meaning

$$\rho_{U_i, U_i \cap U_j}(s_i) = \rho_{U_j, U_i \cap U_j}(s_j) \quad (i, j \in I),$$

there is an $s \in \mathcal{F}(U)$ with $s_i = \rho_{U, U_i}(s)$ for all $i \in I$.

Exercise 14.14

Let X, Y be topological spaces. For every open set $U \subseteq X$, set

$$\mathcal{C}(U) = C^0(U, Y) = \{\varphi : U \rightarrow Y \mid \varphi \text{ is continuous}\}.$$

Show that this assignment is a sheaf on X .

Exercise 14.15

Let X be a topological space. For every open set $U \subseteq X$, set

$$\mathcal{C}(U) = C^0(U, \mathbb{R}) = \{\varphi : U \rightarrow \mathbb{R} \mid \varphi \text{ is continuous}\}.$$

Show that this assignment is a sheaf of commutative $\mathbb{R}$ -algebras on X .

Exercise 14.16

Let M be a differentiable manifold. For every open set $U \subseteq M$, consider the set $C^1(U, \mathbb{R})$ of differentiable functions on U . Let

$$M = \bigcup_{i \in I} U_i$$

be an open cover.

1. Show that if $V \subseteq U$ is open and $f \in C^1(U, \mathbb{R})$, then $f|_V \in C^1(V, \mathbb{R})$.
2. Let $f \in C^1(M, \mathbb{R})$. Show that $f = 0$ if and only if $f|_{U_i} = 0$ for every i .
3. Suppose that functions $f_i \in C^1(U_i, \mathbb{R})$ are given satisfying the compatibility condition

$$f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$$

for all i, j . Show that there is an $f \in C^1(M, \mathbb{R})$ with $f|_{U_i} = f_i$ for all i .

Exercise 14.17

Show that the assignment associating to every open set $U \subseteq K\text{-Spek}(R)$ the ring of algebraic functions $\Gamma(U, \mathcal{O})$, and to every inclusion $U_1 \subseteq U_2$ the restriction map (see Lemma 14.6)

$$\Gamma(U_2, \mathcal{O}) \longrightarrow \Gamma(U_1, \mathcal{O}),$$

is a sheaf of K -algebras.

The following exercises concern ultrafilters and minimal prime ideals. We give the definitions.

A prime ideal $\mathfrak{p}$ in a commutative ring is called a *minimal prime ideal* if there is no prime ideal $\mathfrak{q}$ with $\mathfrak{q} \subsetneq \mathfrak{p}$.

Let R be a commutative ring. A multiplicative system $F \subseteq R$ is called an *ultrafilter* if $0 \notin F$ and F is maximal among the multiplicative systems not containing 0.

Exercise 14.18

Let R be a commutative ring and let $F \subseteq R$ be a multiplicative system with $0 \notin F$. Show that F is an ultrafilter if and only if for every $g \in R$ with $g \notin F$ there are $f \in F$ and $n \in \mathbb{N}$ such that

$$fg^n = 0.$$

Exercise 14.19

Let R be a commutative ring and let $F \subseteq R$ be an ultrafilter. Show that the complement $R \setminus F$ is a minimal prime ideal in R .

Exercise 14.20

Let R be a commutative ring and let S be a multiplicative system with $0 \notin S$. Show that S is contained in an ultrafilter.

Hint. Use Zorn's lemma.

Exercise 14.21

Let R be a reduced commutative ring. Show that every zero divisor is contained in a minimal prime ideal.

Exercise 14.22

Let K be an algebraically closed field and let $\mathfrak{a} \subseteq K[X_1, \dots, X_n]$ be a radical ideal. Set

$$R = K[X_1, \dots, X_n]/\mathfrak{a}, \quad V = V(\mathfrak{a}).$$

Show that the irreducible components of V correspond to the minimal prime ideals of its coordinate ring R .

Exercise 14.23

Let K be an algebraically closed field and let R be a commutative K -algebra of finite type. Show that the minimal prime ideals of R correspond to the irreducible components of $K\text{-Spek}(R)$.

Exercises for submission**Exercise 14.24 (3 points)**

Let K be an algebraically closed field, let $P \in \mathbb{A}_K^2$, and let

$$U = \mathbb{A}_K^2 \setminus \{P\}.$$

Show that

$$\Gamma(U, \mathcal{O}) = K[X, Y].$$

In other words, every algebraic function defined away from a single point of the affine plane extends to that point.

Exercise 14.25 (5 points: 1+2+2)

Consider Neil's parabola

$$C = V(X^2 - Y^3) \subseteq \mathbb{A}_{\mathbb{C}}^2.$$

1. Show that $D(X) = D(Y)$ on C .
2. Show that on $U = D(Y) \subseteq C$, the fraction X/Y defines an algebraic function that cannot be extended algebraically to all of C .
3. Show that the continuous function $X/Y : D(Y) \rightarrow \mathbb{C}$ has a continuous extension to all of C .

Exercise 14.26 (4 points)

Let R be a K -algebra of finite type that is an integral domain over an algebraically closed field K , and let

$$q \in Q = Q(R)$$

be an element of the fraction field of R . Show that

$$\mathfrak{a} = \left\{ f \in R \mid \text{there is } n \in \mathbb{N} \text{ with } f^n q \in R \right\}$$

is an ideal in R . Show also that

$$D(\mathfrak{a}) \subseteq K\text{-Spek}(R)$$

is the maximal domain of definition of the algebraic function q .

Exercise 14.27 (4 points)

Let R be a commutative ring and let $f_1, \dots, f_n \in R$ generate the unit ideal. Suppose that every localisation R_{f_i} , $i = 1, \dots, n$, is Noetherian. Show that R is also Noetherian.

Public Solutions to Worksheet 14

At the frozen revision boundary, the source provides public solutions only for Exercises 14.2 and 14.7. No additional solutions have been created for this edition.

Solution to Exercise 14.2

Edition note: If $U = \emptyset$, the fraction $0/1$ represents its unique function. The source's cover-combining argument below concerns the nonempty case.

Let

$$U = \bigcup_{i \in I} D(H_i)$$

with I finite, and suppose that on each $D(H_i)$ the function f has a representation

$$f = \frac{F_i}{H_i},$$

meaning that

$$f(Q) = \frac{F_i(Q)}{H_i(Q)} \quad \text{for every } Q \in D(H_i).$$

On the intersection $D(H_i) \cap D(H_j)$ we have

$$\frac{F_i(Q)}{H_i(Q)} = f(Q) = \frac{F_j(Q)}{H_j(Q)}.$$

Thus

$$H_j(Q)F_i(Q) - H_i(Q)F_j(Q) = 0$$

for every $Q \in D(H_i) \cap D(H_j)$. Consequently the element

$$H_i H_j (H_j F_i - H_i F_j)$$

induces the zero function on all of $K - \text{Spek}(R)$. Since R is an integral domain and K is algebraically closed, the identity theorem gives

$$H_i H_j (H_j F_i - H_i F_j) = 0$$

in R . After discarding empty members of the cover, H_i and H_j are nonzero; since R is an integral domain, it follows that

$$H_j F_i - H_i F_j = 0,$$

and hence

$$H_j F_i = H_i F_j.$$

By unique prime factorisation, there are elements A, B, C, D and a unit u such that

$$H_j F_i = (AB)(CD) = (u^{-1}AD)(uBC) = H_i F_j.$$

Hence

$$\frac{F_i}{H_i} = \frac{uCD}{AD} = \frac{uC}{A} = \frac{uBC}{AB} = \frac{F_j}{H_j}.$$

The fractional representation uC/A holds on $D(H_i) \cup D(H_j)$. In this way we can combine two members of the cover and reduce the index set I . Since I is finite, repetition eventually produces a single fraction G/H valid on all of U . By cancelling common factors, we may choose G and H with no common nonunit factor, and of course $U \subseteq D(H)$.

[Back to Exercise 14.2.](#)

Solution to Exercise 14.7

The maximal ideal corresponding to P is $(X - 1, Y - 1)$. In the coordinate ring

$$R = K[X, Y]/(Y^2 - X^3)$$

we have

$$X^2(X - 1) = X^3 - X^2 = Y^2 - X^2 = (Y - X)(Y + X).$$

We may therefore set, in the fraction field,

$$f := \frac{X^2}{Y - X} = \frac{X + Y}{X - 1}.$$

These representations define an algebraic function on

$$D(Y - X, X - 1) = D(Y - 1, X - 1) = C \setminus \{P\}.$$

To show that this function is not defined on all of C , consider the map

$$\begin{aligned} \varphi : \mathbb{A}_K^1 &\longrightarrow C, \\ t &\longmapsto (t^2, t^3). \end{aligned}$$

We have

$$\varphi^{-1}(C \setminus \{P\}) = \mathbb{A}_K^1 \setminus \{1\}.$$

The pullback of f under this map is

$$\frac{t^2 + t^3}{t^2 - 1} = \frac{t^2(1 + t)}{(t + 1)(t - 1)} = \frac{t^2}{t - 1}.$$

This function has a pole at $t = 1$ and cannot be extended to an algebraic function on the whole affine line. Hence f cannot be extended algebraically to all of C either.

Edition note: in the final cancellation step, the source displays $-t^2/(t - 1)$. The factorisation on the preceding line gives $t^2/(t - 1)$ without a minus sign. The pole at $t = 1$ and the conclusion of the proof are unchanged.

[Back to Exercise 14.7.](#)

Lecture 15: Affine and Quasi-affine Varieties, Local Rings, and Stalks

Affine and quasi-affine varieties

Definition: affine varieties

Let K be an algebraically closed field and let R be a K -algebra of finite type. The K -spectrum

$$V = K\text{-Spek}(R),$$

with every Zariski-open set $U \subseteq V$ equipped with the ring of algebraic functions $\Gamma(U, \mathcal{O})$, is called an *affine variety*.

An open subset of an affine variety, again with every open set equipped with its structure ring, is called a *quasi-affine variety*. A quasi-affine variety is covered by finitely many open sets of the form $D(f)$, each of which is itself an affine variety. Some authors reserve the term variety for irreducible K -spectra.

When R is reduced, Theorem 14.9 ensures that no information is lost in

$$V = K\text{-Spek}(R),$$

since the ring R can be recovered as $\Gamma(V, \mathcal{O})$. For arbitrary R , the pointwise algebraic functions instead recover the reduction R_{red} ; nilpotents are invisible on K -points. This is not possible from the topological space alone.

Edition note. The source states the recovery claim without reducedness. The qualification above records the hypothesis needed for the pointwise definition used here.

Local rings

For a given point P in a K -spectrum, we are interested in all algebraic functions defined at P and admitting a rational representation on some neighbourhood of P . These functions are defined on different neighbourhoods, and in general there is no smallest neighbourhood on which all algebraic functions defined at P are simultaneously defined. We have a system of rings

$$(\Gamma(U, \mathcal{O}))_{P \in U}$$

that we want to understand geometrically and algebraically. It turns out that this system has a meaningful limit—called a *direct limit* or *colimit*—and that it agrees with the localisation $R_{\mathfrak{m}}$ at the maximal ideal $\mathfrak{m}$ corresponding to P . We begin with the algebraic terminology.

Edition note. The source says unconditionally that there is no smallest neighbourhood. The qualification “in general” allows for isolated points, for which $\{P\}$ is a smallest neighbourhood of P .

Definition: local rings

A commutative ring R is called *local* if it has exactly one maximal ideal.

For a nonzero commutative ring, equivalently, the complement of the group of units of R is closed under addition. The simplest local rings are fields. Every local ring R with maximal ideal $\mathfrak{m}$ has a quotient $R/\mathfrak{m}$ that is a field, called its *residue field*. We shall soon see that every point of a K -spectrum has an associated local ring describing the “local appearance” of the variety at that point algebraically.

Edition note. The source omits the nonzero-ring qualification in the additive characterisation. For the zero ring the nonunit set is empty and hence additively closed, although there is no maximal ideal.

Definition: localisation at a prime ideal

Let R be a commutative ring and let $\mathfrak{p}$ be a prime ideal. The localisation at the multiplicative system

$$S = R \setminus \mathfrak{p}$$

is called the *localisation* of R at $\mathfrak{p}$ and is denoted by $R_{\mathfrak{p}}$. Thus

$$R_{\mathfrak{p}} = \left\{ \frac{f}{g} \mid f \in R, g \notin \mathfrak{p} \right\}.$$

The following theorem explains this terminology.

Theorem: localisation at a prime ideal is local

Let R be a commutative ring and let $\mathfrak{p}$ be a prime ideal in R . Then $R_{\mathfrak{p}}$ is a local ring with maximal ideal

$$\mathfrak{p}R_{\mathfrak{p}} = \left\{ \frac{f}{g} \mid f \in \mathfrak{p}, g \notin \mathfrak{p} \right\}.$$

Proof The displayed set is indeed an ideal in

$$R_{\mathfrak{p}} = \left\{ \frac{f}{g} \mid f \in R, g \notin \mathfrak{p} \right\}.$$

We show that the complement of $\mathfrak{p}R_{\mathfrak{p}}$ consists only of units, so this ideal must be maximal. Let

$$q = \frac{f}{g} \in R_{\mathfrak{p}}$$

but suppose $q \notin \mathfrak{p}R_{\mathfrak{p}}$. Then $f, g \notin \mathfrak{p}$, so the reciprocal fraction g/f also belongs to the localisation.

Fraction fields and function fields

If R is an integral domain, its fraction field is the localisation at the zero prime ideal. We now show that for the associated irreducible affine variety $K\text{-Spek}(R)$, every algebraic function naturally belongs to the fraction field.

Lemma: algebraic functions as elements of the fraction field

Let K be an algebraically closed field, let R be a K -algebra of finite type that is an integral domain, and let

$$\emptyset \neq U \subseteq K\text{-Spek}(R)$$

be an open subset. There is a uniquely determined injective R -algebra homomorphism

$$\Gamma(U, \mathcal{O}) \longrightarrow Q(R).$$

In particular, every algebraic function defined on a nonempty open set U is an element of the fraction field $Q(R)$.

Proof Take $P \in U$ and suppose that the algebraic function f is given on a neighbourhood of P by

$$f = G/H, \quad G, H \in R, \quad H \neq 0.$$

The fraction G/H can immediately be regarded as an element of the fraction field. Let $Q \in U$ be another point with a representation

$$f = G'/H'.$$

By Lemma 14.8, since R is an integral domain,

$$GH' = G'H$$

in R . Thus the element of the fraction field is well defined. The resulting map is clearly a ring homomorphism and makes the diagram

$$\begin{array}{ccc} R & & \\ \downarrow & \searrow & \\ \Gamma(U, \mathcal{O}) & \longrightarrow & Q(R) \end{array}$$

commute. These properties also determine the map uniquely: algebraic functions arising from elements of R must map to those same elements in the fraction field, so the image of every fraction is determined.

For injectivity, if $G/H = 0$ in the fraction field, then $G = 0$, and the corresponding function is zero on $D(H)$. For another representation G'/H' of the same function, the relation above again gives $G' = 0$; the function is therefore zero on all of U .

Uniqueness also immediately shows that for two open sets $U \subseteq U'$ the diagram

$$\begin{array}{ccc} \Gamma(U', \mathcal{O}) & & \\ \downarrow & \searrow & \\ \Gamma(U, \mathcal{O}) & \longrightarrow & Q(R) \end{array}$$

commutes, with the restriction homomorphism on the left. From now on, in the integral-domain case we identify an algebraic function with its corresponding element of the fraction field.

Topological filters and their stalks

The result of the preceding section says that the fraction field can be obtained as an ordered union of all rings of sections $\Gamma(U, \mathcal{O})$ as U ranges over the nonempty open sets. A similar construction can be made for suitably structured systems of open sets in general. For this we need the concept of a filter.

Definition: topological filters

Let X be a topological space. A system F of open subsets of X is called a *topological filter* if, for open sets U, V , the following hold:

1. $X \in F$;
2. if $U \in F$ and $U \subseteq V$, then $V \in F$;
3. if $U, V \in F$, then $U \cap V \in F$.

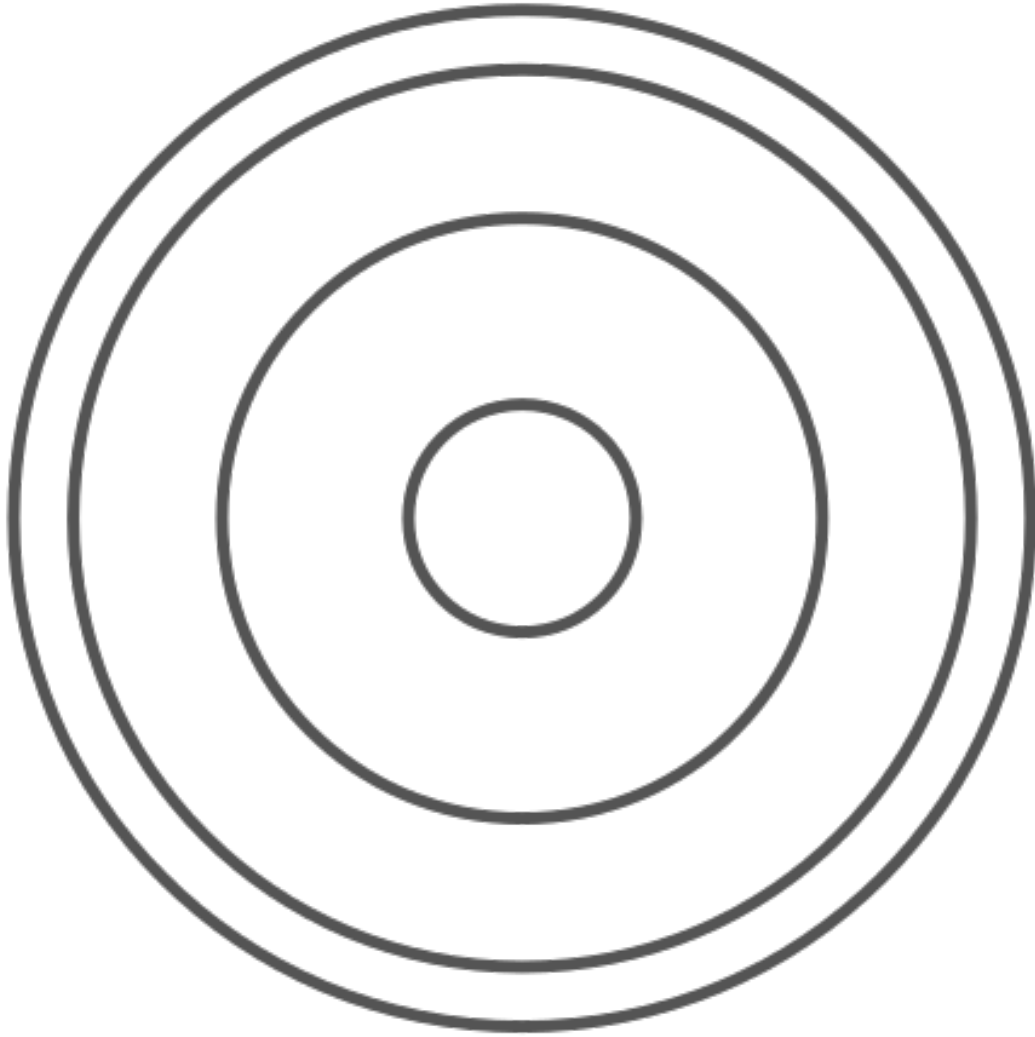


Figure 69: Four concentric grey circles on a transparent background

Schematic depiction of a neighbourhood filter; Andreas Pietzowski, [CC BY-SA 4.0](#). Source details are given in the Unit 15 media credits.

Definition: neighbourhood filters

Let X be a topological space and let $M \subseteq X$. The system

$$\mathcal{U}(M) = \{U \subseteq X \mid U \text{ is open and } M \subseteq U\}$$

is called the *neighbourhood filter* of M .

This is clearly a topological filter. In particular, a point $P \in X$ has a neighbourhood filter $\mathcal{U}(P)$ consisting of all its open neighbourhoods.

Suppose that two open neighbourhoods U_1, U_2 of P and two algebraic functions

$$f_1 \in \Gamma(U_1, \mathcal{O}), \quad f_2 \in \Gamma(U_2, \mathcal{O})$$

are given. Initially their sum $f_1 + f_2$ —and likewise their product—makes no sense because their domains differ. In the integral-domain case we can regard both as elements of the fraction field and add them there. Alternatively, we can pass to the intersection $U_1 \cap U_2$, which is also an open neighbourhood of P , and add the restrictions of the two functions there. The important property of a filter is that together with any two of its open sets it contains their intersection, with the inclusions

$$U_1 \cap U_2 \subseteq U_1, U_2$$

and the corresponding restriction maps

$$\Gamma(U_1, \mathcal{O}), \Gamma(U_2, \mathcal{O}) \longrightarrow \Gamma(U_1 \cap U_2, \mathcal{O}).$$

This observation is made precise by the concepts of a directed set and a directed system.

Definition: directed sets

A nonempty ordered set $(I, \preceq)$ is called *directedly ordered*, or simply *directed*, if for every $i, j \in I$ there is a $k \in I$ such that

$$i, j \preceq k.$$

Edition note. The source does not require I to be nonempty. The standard convention is used here; it is also needed below for a colimit of groups to carry a group structure.

We regard a topological filter as a set ordered by inclusion. Its intersection property makes it directed; the direction of the order is

$$\preceq = \supseteq.$$

Definition: ordered and directed systems

Let $(I, \preceq)$ be an ordered index set. A family of sets

$$M_i, \quad i \in I,$$

is called an *ordered system of sets* if:

1. $\varphi_{ii} = \text{id}_{M_i}$ for every $i \in I$;
2. for $i \preceq j$ there is a map $\varphi_{ij} : M_i \rightarrow M_j$;
3. for $i \preceq j \preceq k$ we have

$$\varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}.$$

Edition note. The identity-map axiom is part of the usual definition and is added explicitly; the source states only the transition maps and their composition law.

If the index set is also directed, the family is called a *directed system of sets*.

If all the M_i are groups (or rings) and all maps between them are group homomorphisms (or ring homomorphisms), we speak of an ordered or directed system of groups (or rings).

Definition: colimits

Let $(M_i)_{i \in I}$ be a directed system of sets. The set

$$\operatorname{colim}_{i \in I} M_i = \left(\bigcup_{i \in I} M_i \right) / \sim$$

is called the *colimit* (also the *direct limit* or *inductive limit*) of the system. Here $\sim$ is the equivalence relation declaring two elements $m \in M_i$ and $n \in M_j$ equivalent if there is a $k \in I$ with $i, j \preceq k$ and

$$\varphi_{ik}(m) = \varphi_{jk}(n).$$

In particular, $s_i \in M_i$ is equivalent to its image $\varphi_{ik}(s_i) \in M_k$ for every $i \preceq k$.

Edition note: in the last sentence the source writes $s_i \in M$, although the system defines only the sets M_i . This edition supplies the required index, $s_i \in M_i$.

For a directed system of groups (or rings), the colimit of sets above can also be given a group (or ring) structure. Two elements of the colimit represented by $s_i \in M_i$ and $s_j \in M_j$ can be replaced by their images in some M_k with $i, j \preceq k$, and the operation is then defined in M_k ; see Exercise 15.23.

Our principal example is the directed system of rings

$$\Gamma(U, \mathcal{O}), \quad U \in F,$$

directed by a topological filter. Its colimit has a name of its own.

Definition: the stalk at a filter

Let $(V, \mathcal{O})$ be a quasi-affine variety and let F be a topological filter in V . The colimit

$$\mathcal{O}_F = \operatorname{colim}_{U \in F} \Gamma(U, \mathcal{O})$$

is called the *stalk* of $\mathcal{O}$ at F .

The stalk at the neighbourhood filter of a point P is also called the stalk at P and is denoted by $\mathcal{O}_P$.

Theorem: the stalk at a point is a localisation

Let R be a reduced commutative algebra of finite type over an algebraically closed field K . Let

$$P \in K - \operatorname{Spek}(R)$$

be a point with corresponding maximal ideal $\mathfrak{m} \subseteq R$. There is a natural isomorphism of R -algebras

$$R_{\mathfrak{m}} \longrightarrow \mathcal{O}_P.$$

Proof The stalk $\mathcal{O}_P$ has a unique R -algebra structure because the whole space belongs to the filter. If $F \in R$ and $F \notin \mathfrak{m}$, then $1/F$ is defined on the open neighbourhood $D(F)$ of P . There we have $F \cdot (1/F) = 1$, so F becomes a unit in the colimit. By the universal property of localisation, there is an R -algebra homomorphism

$$R_{\mathfrak{m}} \longrightarrow \mathcal{O}_P.$$

We prove that this map is bijective. First take $f \in \mathcal{O}_P$. This element is represented by an algebraic function

$$f \in \Gamma(U, \mathcal{O}), \quad P \in U.$$

In particular, f has a rational representation at P : on $D(H)$ we have

$$f = G/H, \quad P \in D(H).$$

The last condition means $H(P) \neq 0$, or equivalently $H \notin \mathfrak{m}$. Thus $G/H \in R_{\mathfrak{m}}$ and maps to f . This proves surjectivity.

For injectivity, take G/H with $H \notin \mathfrak{m}$ and suppose its image in the stalk is zero. This means that G/H is the zero function on some open neighbourhood U of P . We may choose

$$P \in D(H') \subseteq U \cap D(H)$$

and, by Corollary 14.10, write on that set, explicitly taking $G' = 0$,

$$G/H = G'/H' = 0.$$

By Lemma 14.8,

$$H(H')^2G = 0$$

in R . Since H and H' become units in $R_{\mathfrak{m}}$, we obtain $G/H = 0$ in the localisation.

Lemma: sections as an intersection of local rings

Let K be an algebraically closed field, let R be a K -algebra of finite type that is an integral domain, and let

$$U \subseteq K\text{-}\text{Spek}(R)$$

be a nonempty open set. Then

$$\Gamma(U, \mathcal{O}) = \bigcap_{P \in U} \mathcal{O}_P,$$

where the intersection is taken inside the fraction field $Q(R)$.

Edition note. The source allows $U = \emptyset$, but then the section ring is not represented by an intersection of subrings of $Q(R)$. The nonempty hypothesis is therefore necessary for this formulation.

Proof For every $P \in U$ there are injective ring homomorphisms

$$\Gamma(U, \mathcal{O}) \longrightarrow \mathcal{O}_P \longrightarrow Q(R).$$

Consequently there is an injective ring homomorphism

$$\Gamma(U, \mathcal{O}) \longrightarrow \bigcap_{P \in U} \mathcal{O}_P.$$

Conversely, let $f \in Q(R)$ belong to the intersection on the right. For every $P \in U$ there is a representation $f = G/H$ with

$$P \in D(H) \subseteq U.$$

This says precisely that f is an algebraic function on U .

Definition: function fields

Let V be an irreducible quasi-affine variety. The stalk $\mathcal{O}_V$ at the filter of all nonempty open sets in V is a field, called the *function field* of V .

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Worksheet 15

Practice exercises

Exercise 15.1

Let R be a commutative ring. Show that the following statements are equivalent.

1. R has exactly one maximal ideal.
2. The set of nonunits $R \setminus R^\times$ is an ideal in R .

Exercise 15.2

Let R be a nonzero commutative ring. Show that R is local if and only if, whenever $a + b$ is a unit, at least one of a and b is a unit.

Edition note. The source omits the nonzero-ring hypothesis. The zero ring satisfies the displayed unit-sum criterion but has no maximal ideal, so it is not local under the definition used in the lecture.

Exercise 15.3

Determine all subrings of the rational numbers $\mathbb{Q}$ that are local rings.

Exercise 15.4

Let R be a commutative local ring. Show that R is connected.

Exercise 15.5

Let $\mathfrak{n}$ be a maximal ideal in a commutative ring R . Let $R_{\mathfrak{n}}$ be the localisation of R at $\mathfrak{n}$, and let

$$\mathfrak{m} = \mathfrak{n}R_{\mathfrak{n}}$$

be the maximal ideal of $R_{\mathfrak{n}}$. Show that

$$R/\mathfrak{n} = R_{\mathfrak{n}}/\mathfrak{m}.$$

Exercise 15.6 ★

Let R be a commutative ring and let $\mathfrak{p}$ be a prime ideal. The quotient ring

$$S = R/\mathfrak{p}$$

is an integral domain with fraction field $Q = Q(S)$, while $R_{\mathfrak{p}}$ is a local ring with maximal ideal $\mathfrak{p}R_{\mathfrak{p}}$. Show that there is a natural isomorphism

$$Q(S) \cong R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}.$$

This field is also called the residue field at $\mathfrak{p}$.

Exercise 15.7

For $a \in \mathbb{C}$, show that the substitution homomorphism

$$\begin{aligned}\mathbb{C}[X] &\longrightarrow \mathbb{C}, \\ X &\longmapsto a,\end{aligned}$$

agrees with the evaluation map to the residue field at the prime ideal $(X - a)$, namely

$$\mathbb{C}[X]_{(X-a)}/(X-a)\mathbb{C}[X]_{(X-a)}.$$

Exercise 15.8

Let R be a local ring with residue field K . Show that R and K have the same characteristic if and only if R contains a field.

Exercise 15.9 ★

Let R be a local ring and let $\mathfrak{a}$ be an ideal in R . Show that the map

$$R^\times \longrightarrow (R/\mathfrak{a})^\times$$

is surjective.

Exercise 15.10

Let K be an algebraically closed field and let

$$R = K[X_1, \dots, X_n].$$

Show that all localisations of R at maximal ideals are mutually isomorphic.

Exercise 15.11

Let K be a field and consider the coordinate cross

$$V = K\text{-Spek}(K[X, Y]/(XY)).$$

For each point $P \in V$, determine whether the local ring at P is an integral domain.

Exercise 15.12

Consider Neil's parabola

$$C = V(X^2 - Y^3) \subseteq \mathbb{A}_K^2$$

over an algebraically closed field K . Show that all localisations of C at points $P \neq (0, 0)$ are mutually isomorphic, but are not isomorphic to the localisation at the origin.

Exercise 15.13

Let R be the localisation at the origin of the curve

$$C = V(Y^2 - X^2 - X^3) \subseteq \mathbb{A}_K^2,$$

and let S be the localisation of the coordinate cross at the origin. Are these two local rings isomorphic?

Exercise 15.14

Let R be a commutative ring and let $\mathfrak{p}$ be a prime ideal. Show that $\mathfrak{p}$ is a minimal prime ideal if and only if the reduction of the localisation $R_{\mathfrak{p}}$ is a field.

Exercise 15.15

Let R be a commutative ring, let $\mathfrak{m}$ be a maximal ideal with localisation $R_{\mathfrak{m}}$, and let $\mathfrak{a}$ be an ideal contained in the kernel of the localisation map. Show that $R_{\mathfrak{m}}$ is also a localisation of $R/\mathfrak{a}$.

Exercise 15.16

Let K be a field and let R be a finitely generated K -algebra. Let

$$S = R_{\mathfrak{m}}$$

be the localisation of R at a maximal ideal $\mathfrak{m}$. Show that the residue field of S is a finite extension of K .

Exercise 15.17

Let R be a commutative ring, let $f \in R$, and let $\mathfrak{a}$ be an ideal. Show that

$$f \in \mathfrak{a}$$

if and only if for every prime ideal $\mathfrak{p}$ we have

$$f \in \mathfrak{a}R_{\mathfrak{p}}.$$

Remark. For this reason, ideal membership is called a local property.

Exercise 15.18

Let R be a commutative ring. Prove that the following statements are equivalent.

1. R is reduced.
2. For every prime ideal $\mathfrak{p}$, the ring $R_{\mathfrak{p}}$ is reduced.
3. For every maximal ideal $\mathfrak{m}$, the ring $R_{\mathfrak{m}}$ is reduced.

Remark. For this reason, reducedness is called a local property. Also give an example of a commutative ring that is not an integral domain but whose localisations at prime ideals are all integral domains.

Exercise 15.19 ★

Let K be a field and let R, S be finitely generated K -algebras that are integral domains. Let

$$\varphi : R \longrightarrow S$$

be a K -algebra homomorphism, and let $\mathfrak{n}$ be a maximal ideal in S with

$$\varphi^{-1}(\mathfrak{n}) = \mathfrak{m}.$$

Suppose the induced map is an isomorphism

$$R_{\mathfrak{m}} \longrightarrow S_{\mathfrak{n}}.$$

Show that there is an $f \in R$ with $f \notin \mathfrak{m}$ such that

$$R_f \longrightarrow S_{\varphi(f)}$$

is an isomorphism.

Exercise 15.20

Let K be an algebraically closed field, let R be a finitely generated commutative K -algebra, and let F_1, F_2 be topological filters in $K\text{-Spek}(R)$ with $F_1 \subseteq F_2$. Show that there is a ring homomorphism

$$\mathcal{O}_{F_1} \longrightarrow \mathcal{O}_{F_2}.$$

Exercise 15.21

Let K be an algebraically closed field, let R be a finitely generated commutative K -algebra, and let

$$P \in K\text{-Spek}(R).$$

Without using Theorem 15.12, show that the stalk $\mathcal{O}_P$ is a local ring.

Exercise 15.22 ★

Let K be a field, let R be a finitely generated K -algebra that is an integral domain with fraction field $Q(R)$, and let $q \in Q(R)$. Show that the set

$$\left\{ P \in K\text{-Spek}(R) \mid q \in \mathcal{O}_P \right\}$$

is open in $K\text{-Spek}(R)$, where $\mathcal{O}_P$ denotes the local ring at P .

Exercise 15.23

Let I be a directed index set and let $(G_i)_{i \in I}$ be a directed system of abelian groups. Show that its colimit is an abelian group.

Exercise 15.24

Let I be a directed index set and let $(M_i)_{i \in I}$ be a directed system of sets. Let N be another set. Suppose that for every $i \in I$ a map

$$\psi_i : M_i \longrightarrow N$$

is given such that for every $i \preccurlyeq j$ we have

$$\psi_i = \psi_j \circ \varphi_{ij},$$

where φ_{ij} are the maps of the system. Prove the universal property of the colimit: there is exactly one map

$$\psi : \text{colim}_{i \in I} M_i \longrightarrow N$$

such that

$$\psi_i = \psi \circ j_i,$$

where $j_i : M_i \rightarrow \text{colim}_{i \in I} M_i$ are the natural maps.

Show also that if $(M_i)_{i \in I}$ is a directed system of groups, N is a group, and all the ψ_i are group homomorphisms, then ψ is a group homomorphism as well.

Exercises for submission

Exercise 15.25 (4 points)

Describe the set M of all 2×3 matrices of rank at most one over a field K as the K -spectrum of a suitable K -algebra. Show that there is an isomorphism between a nonempty Zariski-open subset of M and an open set in $\mathbb{A}_K^4$.

Exercise 15.26 (4 points)

Let K be a field, let R be a K -algebra of finite type, and let $P_1, \dots, P_n$ be finitely many points in

$$X = K\text{-Spek}(R).$$

Show that the neighbourhood filter of these points is generated by open sets of the form $D(f)$. In other words, for every open set U containing $P_1, \dots, P_n$, show that there is an $F \in R$ with

$$P_1, \dots, P_n \in D(F) \subseteq U.$$

Exercise 15.27 (5 points: 1+2+2)

Let K be a field, let R be a commutative K -algebra of finite type, and let S be a multiplicative system in R . Define

$$F(S) = \left\{ U \subseteq K\text{-Spek}(R) \mid U \text{ is open and there is } f \in S \text{ with } D(f) \subseteq U \right\}.$$

1. Show that $F = F(S)$ is a topological filter in $K\text{-Spek}(R)$.
2. Show that there is a ring homomorphism

$$R_S \longrightarrow \mathcal{O}_F.$$

3. Show that the homomorphism in part 2 is an isomorphism if K is algebraically closed and R is reduced.

Edition note. Although the lecture defines the pointwise structure rings over an algebraically closed field, the same local-fraction and colimit construction is used verbatim in parts 1 and 2 over an arbitrary field. The reconstruction assertion in part 3 retains its stated algebraically closed and reduced hypotheses.

Exercise 15.28 (4 points)

Let

$$X = K\text{-Spek}(R)$$

be an affine variety, let $P_1, \dots, P_n \in X$ be finitely many distinct points, let F be their neighbourhood filter, and let $\mathcal{O}_F$ be the associated stalk. Show that $\mathcal{O}_F$ is a local ring if and only if $n = 1$.

Edition note. The source's n points are understood to form a finite set of distinct points; allowing repetitions would make the criterion false.

Exercise 15.29 (4 points)

Let R be a commutative ring and let $S \subseteq R$ be a multiplicative system. Consider the following partial order on S : set $f \preceq g$ if f divides a power of g , identifying two elements when this relation holds in both directions. Show that the commutative rings

$$R_f, \quad f \in S,$$

form a directed system and that

$$\operatorname{colim}_{f \in S} R_f = R_S.$$

Public Solutions to Worksheet 15

At the frozen revision boundary, the source provides public solutions only for Exercises 15.6, 15.9, 15.19, and 15.22. No additional solutions have been created for this edition.

Solution to Exercise 15.6

Consider the commutative diagram of ring homomorphisms

$$\begin{array}{ccccc} R & \longrightarrow & R/\mathfrak{p} & \longrightarrow & Q(R/\mathfrak{p}) \\ \downarrow & & \downarrow \varphi & & \downarrow \psi \\ R_{\mathfrak{p}} & \longrightarrow & R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}} & = & R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}. \end{array}$$

The maps φ and ψ are to be constructed. Under the ring homomorphism

$$R \longrightarrow R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}},$$

the prime ideal $\mathfrak{p}$ maps to zero, giving an induced homomorphism φ . The map φ sends every nonzero element

$$[r] \in R/\mathfrak{p}, \quad [r] \neq 0,$$

represented by $r \notin \mathfrak{p}$, to a unit. By the universal property of localisation, φ therefore extends to the fraction field:

$$\psi : Q(R/\mathfrak{p}) \longrightarrow R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}.$$

As a ring homomorphism between fields, ψ is injective. Every element of the residue field on the right can be represented by a fraction r/s in $R_{\mathfrak{p}}$ with $s \notin \mathfrak{p}$. It is the image of

$$\frac{[r]}{[s]} \in Q(R/\mathfrak{p}),$$

since $[s] \neq 0$. Thus ψ is also surjective, and is an isomorphism.

[Back to Exercise 15.6.](#)

Solution to Exercise 15.9

If $\mathfrak{a} = R$, the quotient is the zero ring and the statement is clear. We therefore suppose that

$$\mathfrak{a} \subseteq \mathfrak{m},$$

where $\mathfrak{m}$ is the unique maximal ideal of R .

Let $r \in R$ represent a unit in $R/\mathfrak{a}$, and choose $s \in R$ such that

$$rs = 1 \quad \text{in } R/\mathfrak{a}.$$

This means that

$$rs - 1 \in \mathfrak{a} \subseteq \mathfrak{m}.$$

If r were not a unit, then $r \in \mathfrak{m}$ and hence $rs \in \mathfrak{m}$. But this would give the contradiction

$$1 = (1 - rs) + rs \in \mathfrak{m}.$$

Thus r itself is a unit. Every unit in $R/\mathfrak{a}$ therefore has a unit preimage in R .

[Back to Exercise 15.9.](#)

Solution to Exercise 15.19

We first show that the map

$$R_f \longrightarrow S_{\varphi(f)}$$

is surjective for a suitable $f \in R$. Choose a set of K -algebra generators $x_1, \dots, x_n$ for S . By surjectivity of the local map, there are elements

$$y_i = \frac{r_i}{g_i} \in R_{\mathfrak{m}}, \quad g_i \notin \mathfrak{m},$$

so that $y_i g_i = r_i$ in $R_{\mathfrak{m}}$, and with $\varphi(y_i) = x_i$ in $S_{\mathfrak{n}}$. The last equality means that for some $h_i \notin \mathfrak{n}$ we have

$$h_i(\varphi(r_i) - \varphi(g_i)x_i) = 0$$

in S . Since S is an integral domain and $h_i \neq 0$, it follows that

$$\varphi(r_i) = \varphi(g_i)x_i$$

in S .

Set

$$f = g_1 \cdots g_n.$$

Since every $g_i \notin \mathfrak{m}$ and $\mathfrak{m}$ is prime, we have $f \notin \mathfrak{m}$. All the y_i can be written over the common denominator f , so $y_i \in R_f$. Thus every generator x_i belongs to the image of $R_f \rightarrow S_{\varphi(f)}$. The inverse denominator powers $\varphi(f)^{-k}$ are also images of the inverse powers f^{-k} in R_f . The map is therefore surjective.

Edition note: the source immediately deduces surjectivity from the equalities in $S_{\mathfrak{n}}$ without writing out the cancellation of h_i above. That step is valid precisely because S is assumed to be an integral domain; this edition makes the dependence explicit without changing the argument.

We now prove injectivity. Suppose $q \in R_f$ maps to zero. Its image is then also zero in $S_{\mathfrak{n}}$, and q comes from an element of $R_{\mathfrak{m}}$. Since the local map is an isomorphism, $q = 0$ in $R_{\mathfrak{m}}$. Since R is an integral domain, this also implies $q = 0$ in R_f . The map is therefore injective and, together with surjectivity, an isomorphism.

[Back to Exercise 15.19.](#)

Solution to Exercise 15.22

We show that every point P with $q \in \mathcal{O}_P$ has an open neighbourhood on which the same property holds at every point. The set in the exercise is then a union of these open neighbourhoods and hence open.

The local ring at P has the form

$$\mathcal{O}_P = R_{\mathfrak{m}}$$

for a maximal ideal $\mathfrak{m}$ in R . Membership $q \in R_{\mathfrak{m}}$ means that

$$q = \frac{r}{f}$$

with $f \notin \mathfrak{m}$. Hence $P \in D(f)$, so $D(f)$ is an open neighbourhood of P . For every $P' \in D(f)$, the element f is again an allowable denominator. Thus $q \in \mathcal{O}_{P'}$ for every $P' \in D(f)$, as required.

[Back to Exercise 15.22.](#)

Lecture 16: Irreducible Filters, Morphisms, and Fibres

Irreducible filters



Figure 70: Coffee grounds retained in a paper filter in a white ceramic holder

A filter can be identified with what it retains. Elke Wetzig (Elya), CC BY-SA 3.0. Source details are given in the Unit 16 media credits.

In the preceding lecture we saw that a point P in the K -spectrum

$$K\text{-}\mathrm{Spek}(R)$$

determines its neighbourhood filter, and that the stalk at this filter is the localisation of R at the corresponding maximal ideal

$$\mathfrak{m}_P \subseteq R.$$

We also saw that if R is an integral domain, the stalk at the filter of all nonempty open sets gives the fraction field of R , which in turn is the localisation at the zero ideal. The concept of an irreducible filter generalises this relationship.

Definition: irreducible filters

A topological filter F is called *irreducible* if

$$\emptyset \notin F$$

and the following condition holds: if U, V are two open sets with

$$U \cup V \in F,$$

then $U \in F$ or $V \in F$.

For Zariski filters, that is, topological filters in the Zariski topology, the following correspondence holds.

Theorem: prime ideals, irreducible closed sets, and irreducible filters

Let K be an algebraically closed field, let R be a commutative K -algebra of finite type, and let

$$X = K\text{-}\text{Spek}(R).$$

The following objects correspond to one another:

1. prime ideals in R ;
2. irreducible closed subsets of X ;
3. irreducible filters in X .

An irreducible closed subset $Y \subseteq X$ corresponds to the filter

$$F(Y) = \{U \subseteq X \mid U \text{ is open and } U \cap Y \neq \emptyset\}.$$

The stalk of the structure sheaf at this filter is the localisation $R_{\mathfrak{p}}$, where $\mathfrak{p}$ is the corresponding prime ideal.

Edition note: for the structure sheaf of algebraic functions used here, the stalk formula presupposes that R is reduced. For a general R , replace it in that formula by $R/\sqrt{(0)}$. Nilpotent elements vanish as functions. The prime ideal/filter correspondence itself is unchanged.

Proof The correspondence between prime ideals and irreducible closed subsets is already known: a prime ideal $\mathfrak{p}$ corresponds to the irreducible closed subset $V(\mathfrak{p})$; see Lemma 4.3 and Proposition 11.7.

The stated construction for an irreducible closed set Y does indeed give an irreducible filter. Irreducibility follows immediately from the definition; only the intersection property of a filter needs checking. Let

$$U, V \in F(Y),$$

so that $Y \cap U$ and $Y \cap V$ are nonempty. Since Y is irreducible,

$$(Y \cap U) \cap (Y \cap V) = Y \cap (U \cap V)$$

is also nonempty. Thus $U \cap V \in F(Y)$.

Now let F be an irreducible topological filter. We claim that the complement of

$$S = \{f \in R \mid D(f) \in F\}$$

is a prime ideal. The set S is immediately a saturated multiplicative system. It remains to show that its complement is closed under addition. Let $g, h \in R$ with $g + h \in S$. Then $D(g + h) \in F$. Since

$$D(g) \cup D(h) = D(g, h) \supseteq D(g + h),$$

the set $D(g) \cup D(h)$ also belongs to F . Irreducibility of F gives $D(g) \in F$ or $D(h) \in F$, hence $g \in S$ or $h \in S$. Thus the complement of S is closed under addition and is a prime ideal.

Composing the three correspondences always returns the original object. To see this, it suffices to note that an irreducible Zariski filter is generated by open sets of the form $D(f)$; see Exercise 16.1. The assertion about the stalk is a special case of Exercise 15.27.

The filter associated with an irreducible closed set Y is also called the *generic filter* of Y , and its stalk the *generic stalk* of Y . The correspondence in Theorem 16.2 gives, as a special case, the relationship between minimal prime ideals, irreducible components, and ultrafilters. At the other extreme, maximal ideals, points, and neighbourhood filters correspond.

Morphisms between varieties

Definition: morphisms

Let X and Y be quasi-affine varieties, and let

$$\psi : Y \longrightarrow X$$

be a continuous map. We call ψ a *morphism* of quasi-affine varieties if, for every open set $U \subseteq X$ and every algebraic function

$$f \in \Gamma(U, \mathcal{O}),$$

the composite function

$$f \circ \psi : \psi^{-1}(U) \longrightarrow U \xrightarrow{f} \mathbb{A}_K^1$$

belongs to $\Gamma(\psi^{-1}(U), \mathcal{O})$.

Remark: pullback homomorphisms

By definition, a morphism $\psi : Y \rightarrow X$ induces, for every open set $U \subseteq X$, a ring homomorphism

$$\tilde{\psi} : \Gamma(U, \mathcal{O}) \longrightarrow \Gamma(\psi^{-1}(U), \mathcal{O}).$$

In particular, there is a global ring homomorphism

$$\tilde{\psi} : \Gamma(X, \mathcal{O}) \longrightarrow \Gamma(Y, \mathcal{O}).$$

If $U_1 \subseteq U_2$ are open sets in X , we have a commutative diagram of continuous maps

$$\begin{array}{ccc} \psi^{-1}(U_1) & \longrightarrow & U_1 \\ \downarrow & & \downarrow \\ \psi^{-1}(U_2) & \longrightarrow & U_2, \end{array}$$

whose vertical arrows are open inclusions. This gives a commutative diagram of ring homomorphisms

$$\begin{array}{ccc} \Gamma(\psi^{-1}(U_1), \mathcal{O}) & \longleftarrow & \Gamma(U_1, \mathcal{O}) \\ \uparrow & & \uparrow \\ \Gamma(\psi^{-1}(U_2), \mathcal{O}) & \longleftarrow & \Gamma(U_2, \mathcal{O}). \end{array}$$

Edition note: the source calls U_1, U_2 open sets in Y . However, the preceding definition, the expressions $\psi^{-1}(U_i)$, and both diagrams require $U_i \subseteq X$. This edition displays the ambient space determined by that context.

We collect some elementary properties of morphisms.

Proposition: open inclusions and composition

Let K be an algebraically closed field and let U, X, Y, Z be quasi-affine varieties. Then:

1. an open inclusion $U \subseteq X$ is a morphism;
2. if $\theta : Z \rightarrow Y$ and $\psi : Y \rightarrow X$ are morphisms, then $\psi \circ \theta$ is also a morphism.

The following properties are more important.

Theorem: algebra homomorphisms induce morphisms

Let K be an algebraically closed field and let R, S be commutative K -algebras of finite type, with K -spectra

$$X = K\text{-Spek}(R), \quad Y = K\text{-Spek}(S).$$

Every K -algebra homomorphism

$$\varphi : R \longrightarrow S$$

induces a map of spectra

$$\varphi^* : Y \longrightarrow X$$

that is a morphism.

Proof By Theorem 12.7, the map $\varphi^* : Y \rightarrow X$ is already known to be continuous. Let $U \subseteq X$ be open, set

$$V = (\varphi^*)^{-1}(U),$$

and take an algebraic function $f : U \rightarrow K$. We must show that $f \circ \varphi^* : V \rightarrow K$ is also algebraic. Take $P \in V$, set $Q = \varphi^*(P)$, and choose

$$Q \in D(H) \subseteq U, \quad f = G/H \quad \text{on } D(H), \quad G, H \in R.$$

By Theorem 12.7,

$$P \in (\varphi^*)^{-1}(D(H)) = D(\varphi(H)).$$

On this open set we have

$$f \circ \varphi^* = \frac{\varphi(G)}{\varphi(H)}.$$

Indeed, for $\tilde{P} \in D(\varphi(H))$,

$$\begin{aligned} (f \circ \varphi^*)(\tilde{P}) &= f(\varphi^*(\tilde{P})) \\ &= \frac{G(\varphi^*(\tilde{P}))}{H(\varphi^*(\tilde{P}))} \\ &= \frac{(\varphi(G))(\tilde{P})}{(\varphi(H))(\tilde{P})}. \end{aligned}$$

Thus the composite is locally a rational function with nonzero denominator, as required.

Remark: pullback on principal open sets

In the situation of Theorem 16.6, the ring homomorphism associated with $U = D(f)$ is the natural map

$$\Gamma(D(f), \mathcal{O}) \cong R_f \longrightarrow \Gamma((\varphi^*)^{-1}(D(f)), \mathcal{O}) = \Gamma(D(\varphi(f)), \mathcal{O}) = S_{\varphi(f)}.$$

Edition note: the identifications with these localisations use reduced R and S ; otherwise use their reductions, as in the stalk formula above.

Lemma: a global algebraic function is a morphism to the affine line

Let U be a quasi-affine variety over an algebraically closed field K , and let

$$f \in \Gamma(U, \mathcal{O})$$

be an algebraic function. Then f defines a morphism

$$f : U \longrightarrow \mathbb{A}_K^1 \cong K.$$

Proof By Exercise 14.4, the map is continuous. Let $U \subseteq K - \text{Spek}(R)$, take an open set

$$V = D(s) \subseteq \mathbb{A}_K^1, \quad W = f^{-1}(V) \subseteq U,$$

and an algebraic function on V ,

$$q = \frac{r}{s^n} \in \Gamma(V, \mathcal{O}) = K[T]_s.$$

We need to show that $q \circ f$ is algebraic on W . Take $P \in W$ and a local representation

$$f = G/H$$

on a neighbourhood $D(H) \ni P$. Then

$$(q \circ f)(P) = q(f(P)) = \frac{r}{s^n} \left(\frac{G}{H}(P) \right) = \frac{r(G(P)/H(P))}{(s(G(P)/H(P)))^n}.$$

The denominator is nonzero because $f(P) \in D(s)$; this is therefore the required rational representation.

Theorem: morphisms to the affine line are global sections

Let $U \subseteq K - \text{Spek}(R)$ be a quasi-affine variety, where R is a commutative K -algebra of finite type over an algebraically closed field K . There is a natural bijection

$$\begin{aligned} \text{Mor}(U, \mathbb{A}_K^1) &\longrightarrow \Gamma(U, \mathcal{O}), \\ \psi &\longmapsto \tilde{\psi}(T), \end{aligned}$$

where T is the variable in

$$K[T] = \Gamma(\mathbb{A}_K^1, \mathcal{O}).$$

In particular, a morphism from U to the affine line is uniquely determined by the global ring homomorphism

$$\tilde{\psi} : K[T] \longrightarrow \Gamma(U, \mathcal{O}).$$

Proof The displayed map is well defined and surjective. Given a global algebraic function $f \in \Gamma(U, \mathcal{O})$, we first have a map $f : U \rightarrow K$. The variable T , which corresponds to the identity map on $K = \mathbb{A}_K^1$, pulls back along f to the element f itself. By Lemma 16.8, f is a morphism.

Injectivity follows because both the morphism and the algebraic function are uniquely determined by their underlying continuous map.

Theorem: morphisms to an affine variety as algebra homomorphisms

Let $U \subseteq K - \text{Spek}(R)$ be a quasi-affine variety, where R is a commutative K -algebra of finite type over an algebraically closed field K . Let S be another commutative K -algebra of finite type. There is a natural bijection

$$\begin{aligned} \text{Mor}(U, K - \text{Spek}(S)) &\longrightarrow \text{Hom}_K^{\text{alg}}(S, \Gamma(U, \mathcal{O})), \\ \psi &\longmapsto \tilde{\psi}. \end{aligned}$$

Here $\tilde{\psi}$ denotes the global ring homomorphism associated with ψ .

Proof The map is well defined. Theorem 16.9 proves the assertion for $S = K[T]$. Since a morphism to affine space $\mathbb{A}_K^n$ is determined by its components, and a K -algebra homomorphism from $K[T_1, \dots, T_n]$ by the substitutions for the T_i , the assertion also holds for every polynomial ring $K[T_1, \dots, T_n]$.

Now write

$$S = K[T_1, \dots, T_n]/\mathfrak{a}, \quad K - \text{Spek}(S) \cong V(\mathfrak{a}) = V \subseteq \mathbb{A}_K^n.$$

Composing a morphism $U \rightarrow K - \text{Spek}(S)$ with the closed inclusion into affine space again gives a morphism. Thus there is a commutative diagram

$$\begin{array}{ccc} \text{Mor}(U, K - \text{Spek}(S)) & \longrightarrow & \text{Hom}_K^{\text{alg}}(S, \Gamma(U, \mathcal{O})) \\ \downarrow & & \downarrow \\ \text{Mor}(U, \mathbb{A}_K^n) & \longrightarrow & \text{Hom}_K^{\text{alg}}(K[T_1, \dots, T_n], \Gamma(U, \mathcal{O})). \end{array}$$

The lower map is already known to be bijective, and both vertical maps are injective. We need only check that the lower map identifies the two upper subsets.

A morphism $U \rightarrow \mathbb{A}_K^n$ that factors through V as a map is also a morphism to V . It suffices to check the morphism property on principal open sets $D(H)$ with $H \in S$. If $\tilde{H} \in K[T_1, \dots, T_n]$ represents H , the map

$$K[T_1, \dots, T_n]_{\tilde{H}} \longrightarrow S_H$$

is surjective. Thus every element of S_H pulls back to an algebraic function. On the right of the diagram, an algebra homomorphism belongs to the upper subset exactly when $\mathfrak{a}$ is contained in its kernel. The assertion now follows from Exercise 16.8.

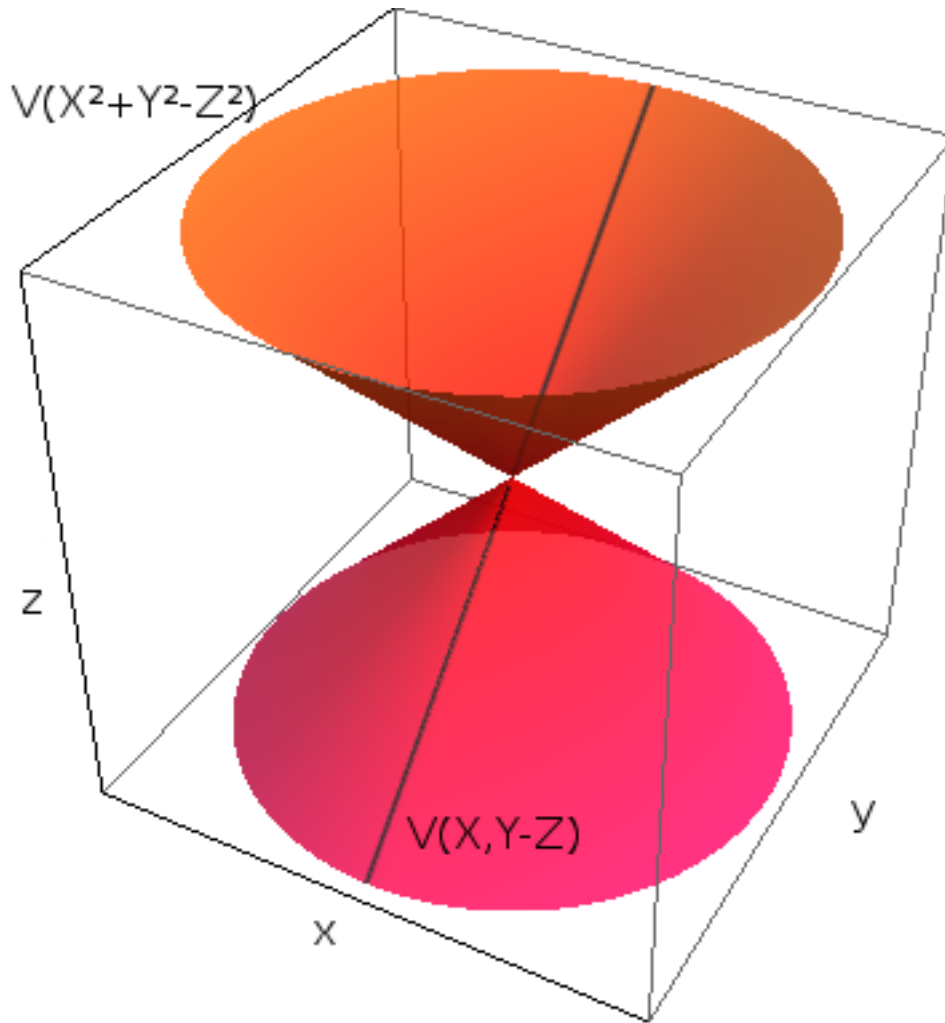


Figure 71: A double cone intersected by a sloping line, with x, y, and z coordinate axes

Not every function defined on the cone away from the line extends to affine space with that line removed. Pmidden, public domain. Source details are given in the Unit 16 media credits.

Example: a restriction map that is not surjective

Consider the standard cone as the closed subset

$$V = V(X^2 + Y^2 - Z^2) \subseteq \mathbb{A}_K^3.$$

Let

$$U = D(X, Z - Y) \subseteq \mathbb{A}_K^3.$$

The intersection $U \cap V = D(X, Z - Y)$, now regarded inside V , is open in V . The associated ring homomorphism

$$\Gamma(U, \mathcal{O}) \longrightarrow \Gamma(U \cap V, \mathcal{O})$$

is not surjective. On the left there is simply the polynomial ring in three variables; compare Exercise 14.24. On the other hand, the equation

$$X^2 = Z^2 - Y^2 = (Z - Y)(Z + Y)$$

gives an algebraic function on $U \cap V$,

$$\frac{X}{Z-Y} = \frac{Z+Y}{X}.$$

This function does not belong to the image of the map, because it does not extend to a function on the whole cone.

Edition note: this example requires $\text{char}(K) \neq 2$. In characteristic 2 the zero locus is the plane $X + Y - Z = 0$, and the displayed function is 1 on $U \cap V$, so it does extend. The source does not state the characteristic restriction.

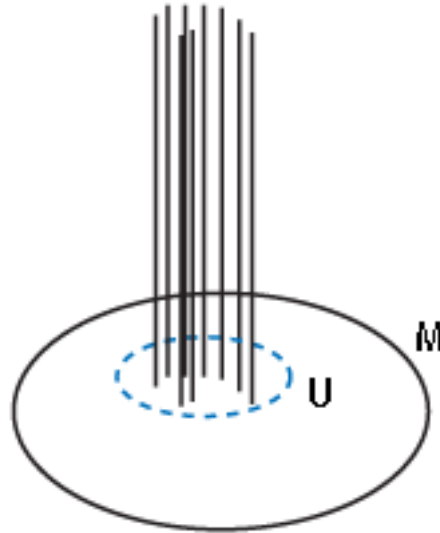


Figure 72: Vertical fibre lines above an oval region M , with a dashed open region U inside M

The fibres of a map: M is the codomain, the domain is the union of all fibres, and the map goes from top to bottom. 132ninme', CC BY-SA 3.0. Source details are given in the Unit 16 media credits.

Definition: fibres

Let

$$\psi : Y \longrightarrow X$$

be a morphism between affine varieties. For a point $P \in X$, the preimage

$$\psi^{-1}(P) \subseteq Y$$

is called the *fibre* over P . As a closed subset of Y , it is itself an affine variety.

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Worksheet 16

Practice exercises

Exercise 16.1 ★

Let

$$X = K\text{-}\mathrm{Spek}(R)$$

be the K -spectrum of a finitely generated commutative K -algebra. Show that an irreducible filter is generated by open sets of the form $D(f)$.

Exercise 16.2

Let $X = K\text{-}\mathrm{Spek}(R)$ be an affine variety and let $Z \subseteq X$ be a closed subset. Show that the neighbourhood filter $\mathcal{U}(Z)$ is generated by open sets of the form $D(f)$.

Exercise 16.3

Let X be a topological space. Show that every ultrafilter is irreducible.

Exercise 16.4

Show that in the correspondence of Theorem 16.2, maximal ideals, points of the K -spectrum, and neighbourhood filters of points correspond to one another.

Exercise 16.5

Show that in the correspondence of Theorem 16.2, minimal prime ideals, irreducible components of the K -spectrum, and ultrafilters correspond to one another.

Exercise 16.6

Let K be an algebraically closed field. Consider the affine plane $\mathbb{A}_K^2$ together with the x -axis

$$V = V(y).$$

Show that the following set is a saturated multiplicative system:

$$S = \left\{ f \in K[X, Y] \mid \text{the homogeneous component } f_{\deg(f)} \text{ contains } x^{\deg(f)} \right\}.$$

Sketch the zero loci of some polynomials belonging to S and some not belonging to S .

Let F be the associated topological filter. Compare F with the neighbourhood filter of V and the generic filter of V .

Exercise 16.7

Let U be a quasi-affine variety over an algebraically closed field K . Show that the units in $\Gamma(U, \mathcal{O}_U)$ correspond to morphisms from U to

$$\mathbb{A}_K^\times = \mathbb{A}_K^1 \setminus \{0\}.$$

Exercise 16.8

Let U be a quasi-affine variety over an algebraically closed field K and let

$$\psi : U \longrightarrow \mathbb{A}_K^n$$

be a morphism. Show that ψ factors through the closed subset

$$V(\mathfrak{a}) \subseteq \mathbb{A}_K^n$$

if and only if $\mathfrak{a}$ is contained in the kernel of the global ring homomorphism

$$\tilde{\psi} : K[T_1, \dots, T_n] \longrightarrow \Gamma(U, \mathcal{O}_U).$$

Exercise 16.9

Let U and V be quasi-affine varieties over an algebraically closed field K , and let

$$W = U \uplus V$$

be their disjoint union. Show that for every quasi-affine variety Z , a morphism $W \rightarrow Z$ is the same as a pair of morphisms, one from U to Z and one from V to Z .

Exercise 16.10 ★

Give an example of two affine varieties V_1, V_2 and a bijective morphism

$$\psi : V_1 \longrightarrow V_2$$

whose inverse map is not continuous.

Exercise 16.11 ★

Let

$$V = V(x^2 + y^2 - 1) \subseteq \mathbb{A}_K^2$$

be the unit circle over a field K , and let $P = (a, b)$ and $Q = (c, d)$ be points on V . Show that there is an automorphism

$$\varphi : V \longrightarrow V$$

with $\varphi(P) = Q$.

Exercise 16.12 ★

1. In the real case, sketch the zero loci

$$V = V(XY, XZ, YZ) \subseteq \mathbb{A}_K^3$$

and

$$W = V(ST(S - T)) \subseteq \mathbb{A}_K^2.$$

2. Construct a bijective morphism

$$\varphi : V \longrightarrow W.$$

3. Show that if the characteristic of K is not 2, the morphism φ is an isomorphism away from the origin.

Exercise 16.13 ★

Show that the map

$$\begin{aligned}\varphi : V(Z^2 + W^2 - 1) &\longrightarrow V(X^2 + Y^2 - 1), \\ (Z, W) &\longmapsto (Z^2 - W^2, 2ZW) = (X, Y)\end{aligned}$$

is a morphism from the unit circle to itself. Show that the preimage of every point

$$P \in V(X^2 + Y^2 - 1)$$

consists of two points.

Edition note: the two-point assertion needs additional hypotheses, for example that K is algebraically closed and $\text{char}(K) \neq 2$. In characteristic 2 the map is constant; over $\mathbb{Q}$, the point $(0, 1)$ has no preimage. The source does not state these restrictions.

Exercise 16.14

Let X and Z be quasi-affine varieties over an algebraically closed field K . For every open set $U \subseteq X$, consider

$$\mathcal{F}(U) := \{\varphi : U \rightarrow Z \mid \varphi \text{ is a morphism}\}.$$

Show that $\mathcal{F}$ is a sheaf on X .

Exercise 16.15 ★

Describe a K -algebra homomorphism whose induced map of K -spectra describes addition on K .

Exercise 16.16

Let K be an algebraically closed field. Show that addition, multiplication, negation, inversion, and division on K can be realised as morphisms on their respective natural domains.

Exercise 16.17

Let R be a commutative ring, let

$$\mathfrak{a} = (f_1, \dots, f_n)$$

be a finitely generated ideal, and let $f \in R$. The R -algebra

$$A = R[T_1, \dots, T_n]/(f_1 T_1 + \dots + f_n T_n + f)$$

is called the *forcing algebra* for $f_1, \dots, f_n, f$. Show that A has the following property: for every ring homomorphism $\varphi : R \rightarrow S$ to a commutative ring S satisfying

$$\varphi(f) \in \mathfrak{a}S,$$

there is an R -algebra homomorphism

$$\vartheta : A \longrightarrow S.$$

Show also that this homomorphism is *not* uniquely determined in general.

Edition note: the source's non-uniqueness statement is understood in this general sense. Particular data can force uniqueness, for example $n = 1$ and $f_1 = 1$.

Exercise 16.18

Let R be a commutative K -algebra of finite type over an algebraically closed field. For $f_1, \dots, f_n, f \in R$, let

$$A = R[T_1, \dots, T_n]/(f_1 T_1 + \dots + f_n T_n + f)$$

be the associated forcing algebra. Characterise the fibres of the morphism

$$K\text{-Spek}(A) \longrightarrow K\text{-Spek}(R).$$

Exercises for submission**Exercise 16.19 (4 points)**

Let K be an algebraically closed field, let R, S be K -algebras of finite type that are integral domains, and let

$$\varphi : R \longrightarrow S$$

be a K -algebra homomorphism, with associated morphism

$$\varphi^* : K\text{-Spek}(S) \longrightarrow K\text{-Spek}(R).$$

Show that the following statements are equivalent.

1. φ is injective.
2. The image of φ^* is dense in $K\text{-Spek}(R)$.
3. φ induces a ring homomorphism $Q(R) \rightarrow Q(S)$.

Exercise 16.20 (4 points)

Let K be an algebraically closed field, let R, S be K -algebras of finite type that are integral domains, and suppose a K -algebra homomorphism

$$\varphi : Q(R) \longrightarrow Q(S)$$

between their fraction fields is given. Show that there is an open set

$$U \subseteq K\text{-Spek}(S)$$

and a morphism

$$U \longrightarrow K\text{-Spek}(R)$$

that induces φ .

Exercise 16.21 (5 points)

Give an example of two affine algebraic curves C_1, C_2 over $\mathbb{C}$ and a bijective morphism

$$\psi : C_1 \longrightarrow C_2$$

whose inverse map is not continuous in the metric topology.

Exercise 16.22 (6 points)

Give an example of two irreducible affine varieties V_1, V_2 and a bijective morphism

$$\psi : V_1 \longrightarrow V_2$$

whose inverse map is not continuous in the Zariski topology and hence is not a morphism either.

Exercise 16.23 (3 points)

Let

$$U \subseteq K\text{-}\mathrm{Spek}(R)$$

be a quasi-affine variety and let $f \in \Gamma(U, \mathcal{O})$ be an algebraic function. Let

$$f = g_i/h_i, \quad i = 1, \dots, n,$$

be local representations of f on open sets $D(h_i) \subseteq U$ covering U . Show that its zero fibre is the closed subset

$$f^{-1}(0) = V(h_1g_1, \dots, h_ng_n) \cap U.$$

Edition note: on the left of the local representation, the source writes $q = g_i/h_i$, although the function defined and being represented is f . This edition displays the variable determined by the context of the exercise.

Public Solutions to Worksheet 16

At the frozen revision boundary, the source provides public solutions only for Exercises 16.1, 16.10, 16.11, 16.12, 16.13, and 16.15. No additional solutions have been created for this edition. Solution 16.13 is retained only as far as it is actually available in the source; that boundary is explained where it occurs.

Solution to Exercise 16.1

Let F be an irreducible filter. For every $U \in F$, write

$$U = D(f_1) \cup \cdots \cup D(f_k).$$

Since F is irreducible, at least one $D(f_i)$ belongs to F . Since

$$D(f_i) \subseteq U,$$

the open sets of the form $D(f)$ belonging to F generate the filter.

[Back to Exercise 16.1.](#)

Solution to Exercise 16.10

Consider the affine line $\mathbb{A}_K^1$ and the punctured line

$$Y = \mathbb{A}_K^1 \setminus \{0\},$$

which is affine because it can be realised as a hyperbola. Consider the disjoint union

$$Z = Y \sqcup \{P\}$$

with one additional point. There is a natural morphism

$$Z \longrightarrow \mathbb{A}_K^1$$

that is the open inclusion on Y and sends P to the origin. This map is bijective. However, $\{P\}$ is open on the left but not on the right. Thus the inverse map is not continuous.

[Back to Exercise 16.10.](#)

Solution to Exercise 16.11

It suffices to give, for $(1, 0)$ and $P = (a, b) \in V$, an automorphism of the circle taking $(1, 0)$ to P . The required automorphism from P to Q is then obtained by composing maps of this kind and, where necessary, their inverses.

Consider the bijective linear map

$$\varphi : K^2 \longrightarrow K^2$$

given by the matrix

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

It sends $(1, 0)$ to (a, b) . A point $(x, y) \in V$ maps to

$$(ax - by, bx + ay).$$

For the image point we have

$$\begin{aligned}
 (ax - by)^2 + (bx + ay)^2 &= a^2x^2 - 2abxy + b^2y^2 + b^2x^2 + 2abxy + a^2y^2 \\
 &= (a^2 + b^2)x^2 + (a^2 + b^2)y^2 \\
 &= x^2 + y^2 \\
 &= 1.
 \end{aligned}$$

Thus the image point again lies on the circle, and φ induces an algebraic map $V \rightarrow V$. The linear map with matrix

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

gives the inverse morphism. We therefore obtain an automorphism.

Edition note: in the second term on the first line of the calculation, the source writes $(bx + ax)^2$. Both the map just defined and the expansion on the following line require $(bx + ay)^2$, which is displayed here.

[Back to Exercise 16.11.](#)

Solution to Exercise 16.12

1. We have

$$V = V(XY, XZ, YZ) = V(X, Y) \cup V(X, Z) \cup V(Y, Z),$$

the union of the three coordinate axes in three-dimensional affine space.

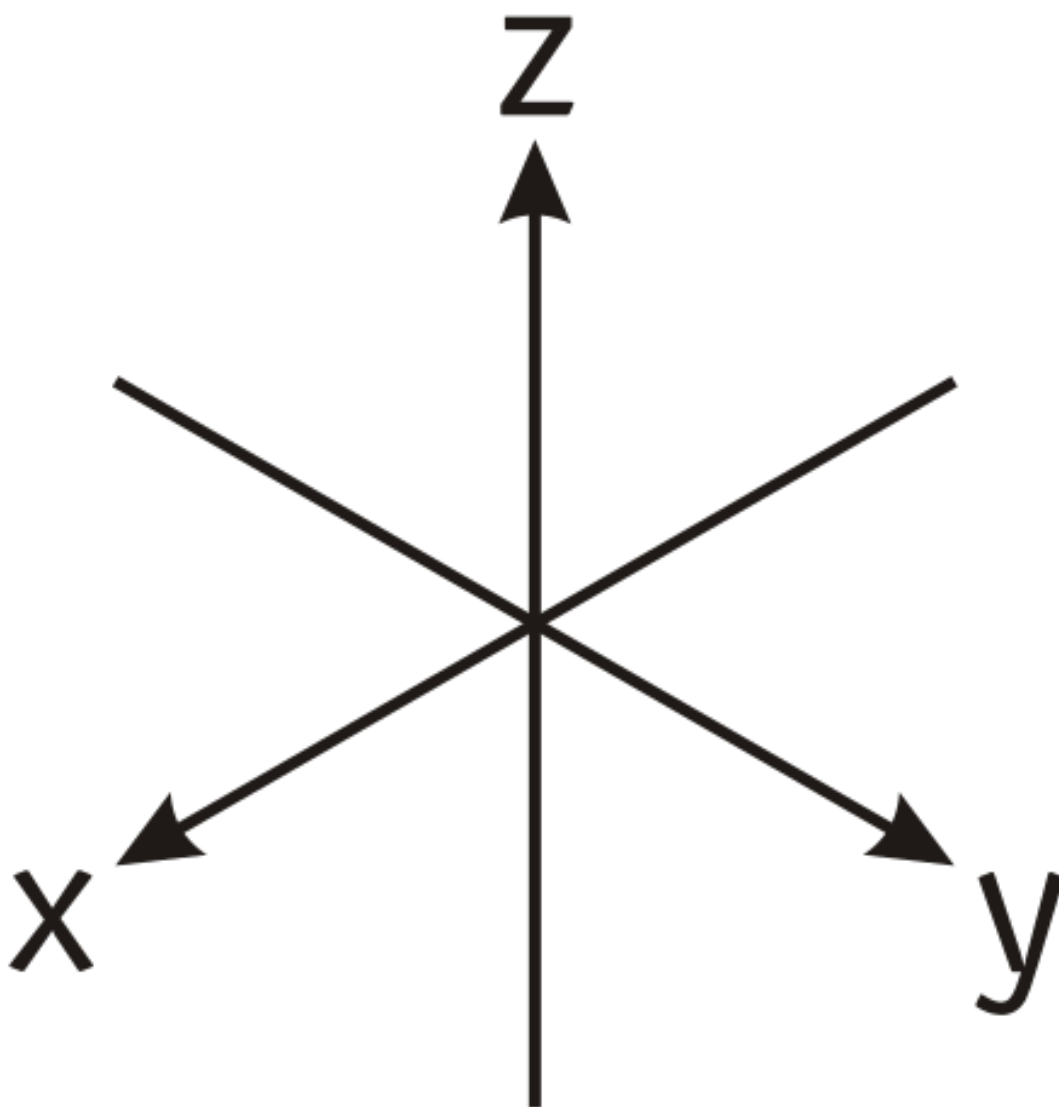


Figure 73: Three coordinate axes meeting at the origin

Sketch of the union of the three coordinate axes. Kalan, [CC BY-SA 3.0](#). Source details are given in the Unit 16 media credits.

Edition note: the source displays $V(XY) \cup V(XZ) \cup V(YZ)$ on the right. That union is not the common zero locus of the three polynomials and does not consist only of the three axes. This edition displays the component decomposition matching the left-hand side and the source's following sentence.

2. The linear map

$$K^3 \longrightarrow K^2$$

given, with respect to the standard bases, by the matrix

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

is the identity on the XY -plane and sends the Z -axis to the main diagonal in that plane. Thus the image of the union of the axes lies entirely in

$$W = V(ST(S - T)),$$

giving a morphism

$$\varphi : V \longrightarrow W.$$

This morphism is bijective because each of the lines involved is mapped bijectively to one of the lines.

3. Algebraically, there is a K -algebra homomorphism

$$K[S, T]/(ST(S - T)) \longrightarrow K[X, Y, Z]/(XY, XZ, YZ),$$

with

$$S \mapsto X + Z, \quad T \mapsto Y + Z.$$

It induces a homomorphism of localisations

$$(K[S, T]/(ST(S - T)))_{S+T} \longrightarrow (K[X, Y, Z]/(XY, XZ, YZ))_{X+Y+2Z}.$$

The intersection of $V(S + T)$, respectively $V(X + Y + 2Z)$, with each of the three lines consists only of the origin, using $\text{char}(K) \neq 2$. Thus both localisations describe the complement of the origin.

In the variables

$$A = X + Z, \quad B = Y + Z,$$

the ring on the right can be written as

$$K[A, B, Z, (A + B)^{-1}] / ((A - Z)Z, (B - Z)Z, (A - Z)(B - Z)).$$

In this ring,

$$\begin{aligned} 2AB &= 2Z(A + B) - 2Z^2 \\ &= 2Z(A + B) - AZ - BZ \\ &= Z(A + B), \end{aligned}$$

so

$$Z = \frac{2AB}{A + B}.$$

Thus Z can be eliminated. Since

$$A - Z = A - \frac{2AB}{A + B} = \frac{A^2 + AB - 2AB}{A + B} = \frac{A^2 - AB}{A + B},$$

the ideal generators become

$$(A - Z)Z = \frac{A^2 - AB}{A + B} \frac{2AB}{A + B} = \frac{2A^2B(A - B)}{(A + B)^2},$$

$$(B - Z)Z = -\frac{2AB^2(A - B)}{(A + B)^2},$$

and

$$\begin{aligned}(A - Z)(B - Z) &= \frac{A^2 - AB}{A + B} \frac{B^2 - AB}{A + B} \\ &= \frac{2A^2B^2 - AB^3 - A^3B}{(A + B)^2}.\end{aligned}$$

Edition note: the source prints a positive sign before the fraction for $(B - Z)Z$. The displayed substitution gives a negative sign. Changing the sign does not change the generated ideal, but the algebraic equality is displayed here with the correct sign.

Since $A + B$ and 2 are units, the first two generators give

$$A^3B = A^2B^2 = AB^3,$$

so the third generator is redundant. Moreover,

$$AB(A - B)(A + B) = AB(A^2 - B^2)$$

belongs to the ideal. Since $A + B$ is a unit, $AB(A - B)$ also belongs to the ideal; conversely, it generates the same ideal. Thus the map given by $S \mapsto A$ and $T \mapsto B$ is an isomorphism on the complement of the origin.

[Back to Exercise 16.12.](#)

Solution to Exercise 16.13

There is a morphism

$$V(Z^2 + W^2 - 1) \longrightarrow \mathbb{A}_K^2.$$

It therefore suffices to check that its image satisfies the circle equation. Indeed,

$$\begin{aligned}X^2 + Y^2 &= (Z^2 - W^2)^2 + 4Z^2W^2 \\ &= (Z^2 - (1 - Z^2))^2 + 4Z^2(1 - Z^2) \\ &= (2Z^2 - 1)^2 + 4Z^2(1 - Z^2) \\ &= 4Z^4 - 4Z^2 + 1 + 4Z^2 - 4Z^4 \\ &= 1.\end{aligned}$$

Source-solution boundary: the frozen public solution stops after proving that the image satisfies the circle equation. It does not prove the second assertion of the exercise, that every fibre consists of two points. This edition does not invent a continuation absent from the source.

[Back to Exercise 16.13.](#)

Solution to Exercise 16.15

Consider the substitution homomorphism

$$\begin{aligned}K[X] &\longrightarrow K[Y, Z], \\ X &\longmapsto Y + Z.\end{aligned}$$

The induced map of spectra is

$$\begin{aligned}\mathbb{A}_K^2 &\cong K^2 \longrightarrow \mathbb{A}_K^1 \cong K, \\ (y, z) &\longmapsto y + z.\end{aligned}$$

This is exactly addition on K .

[Back to Exercise 16.15.](#)

Lecture 17: Monoid Rings and Groups of Differences

Having developed the theory sufficiently far, we now turn to a broad class of examples: monoid rings.

Monoid rings

Definition: monoid rings

Let M be a commutative monoid written additively and let R be a commutative ring. The *monoid ring* $R[M]$ is constructed as follows. As an R -module,

$$R[M] = \bigoplus_{m \in M} R e_m,$$

that is, $R[M]$ is the free module with basis

$$(e_m)_{m \in M}.$$

Multiplication on basis elements is defined by

$$e_m \cdot e_k := e_{m+k}$$

and extended distributively to all of $R[M]$. The identity $0 \in M$ determines the multiplicative identity

$$1 = e_0.$$

Remark: the form of elements and multiplication

Every element of a monoid ring has a unique expression

$$f = \sum_{m \in \widetilde{M}} a_m e_m,$$

where $\widetilde{M} \subseteq M$ is finite and $a_m \in R$. Addition is componentwise, while multiplication is explicitly given by

$$\begin{aligned}fg &= \left(\sum_{m \in \widetilde{M}} a_m e_m \right) \left(\sum_{k \in \widetilde{M}} b_k e_k \right) \\ &= \sum_{\ell \in M} \left(\sum_{\substack{m+k=\ell \\ m \in \widetilde{M}, k \in \widetilde{M}}} a_m b_k \right) e_\ell.\end{aligned}$$

Only finitely many ℓ occur, and each inner sum is also finite. This is what distributive extension means in the definition above.

It is customary to write X^m in place of e_m , where X is a suggestive symbol reminiscent of a variable. The rule

$$X^m X^k = X^{m+k}$$

resembles the corresponding rule for polynomial rings. Indeed, polynomial rings are special cases of monoid rings, and this notation comes from that case. A full proof that the construction really gives a ring with associative and distributive multiplication also works as in the polynomial-ring case. Usually we simply write

$$\sum_{m \in M} a_m X^m,$$

with almost all $a_m = 0$. Elements of the form X^m are called *monomials*. The map

$$\begin{aligned} M &\longrightarrow R[M], \\ m &\longmapsto X^m \end{aligned}$$

is a monoid homomorphism, using the multiplicative monoid structure on the right.

A monoid ring is naturally an R -algebra: an element $f \in R$, regarded in $R[M]$, is

$$f = f \cdot 1 = fX^0.$$

Thus R is also called the *base ring* of the monoid ring. Monoid rings are already interesting when the base ring is a field.

Example: polynomial rings

Let n be a natural number and let

$$M = \mathbb{N}^n.$$

Thus M is the direct product of n copies of the natural numbers.

Every $k \in \mathbb{N}^n$ is an n -tuple $k = (k_1, \dots, k_n)$ with $k_i \in \mathbb{N}$, and can be written as

$$(k_1, \dots, k_n) = k_1(1, 0, \dots, 0) + \dots + k_n(0, \dots, 0, 1).$$

Writing

$$X_i = X^{e_i} = X^{(0, \dots, 0, 1, 0, \dots, 0)}$$

for the monomial corresponding to the i th basis element, we obtain

$$X^k = X_1^{k_1} X_2^{k_2} \dots X_n^{k_n}.$$

Thus the monoid ring of $\mathbb{N}^n$ over R is precisely the polynomial ring in n variables. In particular,

$$R[\mathbb{N}] = R[X].$$

The monoid ring of the trivial monoid $\{0\}$ is the base ring itself.

Example: Laurent rings

Let n be a natural number and let

$$M = \mathbb{Z}^n.$$

Thus M is the direct product of n copies of the integers.

The monoid M is the free abelian group of rank n . Every $k \in \mathbb{Z}^n$ is an n -tuple $k = (k_1, \dots, k_n)$ with $k_i \in \mathbb{Z}$, which can be written as

$$(k_1, \dots, k_n) = k_1(1, 0, \dots, 0) + \dots + k_n(0, \dots, 0, 1).$$

As in Example 17.3, the corresponding monomial can be written uniquely as

$$X^k = X_1^{k_1} X_2^{k_2} \dots X_n^{k_n}, \quad X_i = X^{e_i}.$$

Hence

$$R[M] = R[X_1, \dots, X_n, X_1^{-1}, \dots, X_n^{-1}].$$

This ring is isomorphic to the localisation of the polynomial ring at the product of all the variables:

$$R[M] = R[X_1, \dots, X_n, X_1^{-1}, \dots, X_n^{-1}] = R[X_1, \dots, X_n]_{X_1 \dots X_n}.$$

It is called the *Laurent ring* in n variables over R .

The universal property of monoid rings**Theorem: the universal property**

Let R be a commutative ring, let M be a commutative monoid, let B be a commutative R -algebra, and let

$$\varphi : M \longrightarrow (B, \cdot, 1)$$

be a monoid homomorphism. Then there is exactly one R -algebra homomorphism

$$\tilde{\varphi} : R[M] \longrightarrow B$$

making the following diagram commute:

$$\begin{array}{ccc} M & \longrightarrow & R[M] \\ & \searrow & \downarrow \tilde{\varphi} \\ & & B. \end{array}$$

Proof An R -module homomorphism

$$\tilde{\varphi} : R[M] \longrightarrow B$$

is determined by the images of the basis elements $(X^m)_{m \in M}$. The diagram commutes exactly when

$$\tilde{\varphi}(X^m) = \varphi(m).$$

This condition determines the map uniquely and immediately makes it an R -module homomorphism. We need only check multiplication. First,

$$\tilde{\varphi}(1) = \tilde{\varphi}(X^0) = \varphi(0) = 1.$$

Moreover,

$$\begin{aligned}\tilde{\varphi}(X^m X^k) &= \tilde{\varphi}(X^{m+k}) \\ &= \varphi(m+k) \\ &= \varphi(m)\varphi(k) \\ &= \tilde{\varphi}(X^m)\tilde{\varphi}(X^k).\end{aligned}$$

Thus the map respects multiplication on monomials. For

$$f = \sum_{m \in M} a_m X^m, \quad g = \sum_{k \in M} b_k X^k,$$

with finite support, we obtain

$$\begin{aligned}\tilde{\varphi}(fg) &= \tilde{\varphi}\left(\sum_{\ell \in M} \left(\sum_{m+k=\ell} a_m b_k\right) X^\ell\right) \\ &= \sum_{\ell \in M} \left(\sum_{m+k=\ell} a_m b_k\right) \varphi(\ell) \\ &= \sum_{m,k \in M} a_m b_k \varphi(m)\varphi(k) \\ &= \left(\sum_{m \in M} a_m \varphi(m)\right) \left(\sum_{k \in M} b_k \varphi(k)\right) \\ &= \tilde{\varphi}(f)\tilde{\varphi}(g).\end{aligned}$$

Consequently $\tilde{\varphi}$ is a ring homomorphism.

Corollary: functoriality in the monoid

Let R be a commutative ring, let M, N be commutative monoids, and let

$$\varphi : M \longrightarrow N$$

be a monoid homomorphism. It induces an R -algebra homomorphism

$$\begin{aligned}\tilde{\varphi} : R[M] &\longrightarrow R[N], \\ X^m &\longmapsto X^{\varphi(m)}.\end{aligned}$$

Proof Apply Theorem 17.5 to the R -algebra $B = R[N]$ and the composite monoid homomorphism

$$M \xrightarrow{\varphi} N \longrightarrow R[N].$$

Remark: substitution from a polynomial algebra

A family $(m_i)_{i \in I}$ in a monoid M determines a monoid homomorphism

$$\mathbb{N}^{(I)} \longrightarrow M$$

sending the i th basis element e_i to m_i . When $I = \{1, \dots, n\}$ is finite, Corollary 17.6 gives an R -algebra homomorphism

$$R[\mathbb{N}^n] = R[X_1, \dots, X_n] \longrightarrow R[M].$$

This is the substitution homomorphism given by

$$X_i \mapsto X^{m_i}.$$

Definition: ring-valued points

For a commutative monoid M and a commutative ring R , a monoid homomorphism

$$M \longrightarrow (R, \cdot, 1)$$

is called an R -valued point of M .

Remark: monoid points and the K -spectrum

By Theorem 17.5, an R -valued point of M is equivalent to an R -algebra homomorphism from $R[M]$ to R . This terminology is especially common when the base ring is a field K . In that case,

$$\begin{aligned} K\text{-Spek}(K[M]) &= \text{Hom}_K^{\text{alg}}(K[M], K) \\ &= \text{Mor}_{\text{mon}}(M, K) \\ &= \{K\text{-valued points of } M\}. \end{aligned}$$

Thus the K -spectrum already has a simple, purely multiplicative description at the monoid level. As we shall see, this means that the K -spectra of monoid rings generally have much clearer descriptions than spectra of rings in general. Nevertheless, the monoid ring remains indispensable for defining the Zariski topology and the sheaf of algebraic functions on $K\text{-Spek}(K[M])$.

Remark: generators and binomial relations

A commutative monoid is often described by finitely many generators $e_1, \dots, e_r$ together with binomial relations of the form

$$n_1 e_1 + \dots + n_r e_r = m_1 e_1 + \dots + m_r e_r, \quad n_i, m_i \in \mathbb{N}.$$

A K -valued point

$$\varphi : M \longrightarrow K$$

is uniquely determined by $a_i = \varphi(e_i)$. For every binomial relation holding in M , these values must satisfy

$$a_1^{n_1} \dots a_r^{n_r} = a_1^{m_1} \dots a_r^{m_r}.$$

Lemma: injectivity and surjectivity

Let R be a nonzero commutative ring, let M, N be commutative monoids, and let

$$\varphi : M \longrightarrow N$$

be a monoid homomorphism. The map φ is injective (respectively, surjective) if and only if the associated R -algebra homomorphism

$$\tilde{\varphi} : R[M] \longrightarrow R[N]$$

is injective (respectively, surjective).

Proof Suppose φ is injective and

$$\tilde{\varphi}\left(\sum_{m \in M} a_m X^m\right) = \sum_{m \in M} a_m X^{\varphi(m)} = 0.$$

Since all the $\varphi(m)$ are distinct, every $a_m = 0$. Conversely, if φ is not injective, take $m \neq k$ with $\varphi(m) = \varphi(k)$. Then

$$\tilde{\varphi}(X^m) = \tilde{\varphi}(X^k), \quad X^m \neq X^k,$$

so $\tilde{\varphi}$ is not injective.

If φ is surjective, then for any element $\sum_{n \in N} a_n X^n \in R[N]$, choose a preimage $m_n \in M$ of n . The element $\sum_{n \in N} a_n X^{m_n}$ is a preimage of it. Conversely, if $n \in N$ is not in the image of φ , the nonzero monomial X^n cannot be in the image of $\tilde{\varphi}$.

Corollary: generating sets

Let R be a nonzero commutative ring, let M be a commutative monoid, and let $(m_i)_{i \in I}$ be a family of elements of M . The family $(m_i)_{i \in I}$ generates M as a monoid if and only if $(X^{m_i})_{i \in I}$ generates $R[M]$ as an R -algebra.

Proof The family $(m_i)_{i \in I}$ generates M exactly when the monoid homomorphism

$$\mathbb{N}^{(I)} \longrightarrow M$$

is surjective. By Lemma 17.11, this is equivalent to surjectivity of

$$\begin{aligned} R[X_i \mid i \in I] &\longrightarrow R[M], \\ X_i &\longmapsto X^{m_i}, \end{aligned}$$

which says precisely that the X^{m_i} generate $R[M]$ as an R -algebra.

Corollary: functoriality in the base ring

Let R be a commutative ring, let S be an R -algebra, and let M be a commutative monoid. There is a natural R -algebra homomorphism

$$\begin{aligned} R[M] &\longrightarrow S[M], \\ \sum_{m \in M} a_m X^m &\longmapsto \sum_{m \in M} a_m X^m, \end{aligned}$$

where coefficients from R are viewed through the structure map $R \rightarrow S$.

Proof Apply Theorem 17.5 to the R -algebra $S[M]$ and the natural monoid homomorphism $M \rightarrow S[M]$.

The group of differences of a monoid

We want to know when a monoid ring is an integral domain (which is possible only when the base ring is an integral domain) and how its fraction field can then be described. In the fraction field, every nonzero element must be invertible, in particular the monomials X^m . It is therefore natural to look for an additive group containing M .

Edition note: after consistently using X^m for monomials, the source prints T^m in the last sentence. This edition retains the notation X^m established in this lecture.

Definition: the group of differences

Let M be a commutative monoid. The set of *formal differences*

$$\Gamma(M) = \{m - n \mid m, n \in M\}$$

is equipped with addition

$$(m_1 - n_1) + (m_2 - n_2) := (m_1 + m_2) - (n_1 + n_2)$$

and the identification

$$m_1 - n_1 = m_2 - n_2$$

whenever there is a $u \in M$ such that

$$u + m_1 + n_2 = u + m_2 + n_1.$$

The resulting object $\Gamma(M)$ is called the *group of differences* of M .

Exercise 17.13 asks the reader to show that it really is a group. The construction is modelled on the construction of fraction fields, with multiplicative notation replaced by additive notation. The construction of the group of differences is actually more elementary. For example,

$$\Gamma(\mathbb{N}) = \mathbb{Z}.$$

There is a natural monoid homomorphism

$$\begin{aligned} M &\longrightarrow \Gamma(M), \\ m &\longmapsto m - 0. \end{aligned}$$

We usually write simply m for $m - 0$. This map need not be injective, because the extra element u may occur in the identification above, and this cannot be avoided. We now characterise the monoids for which that extra element is unnecessary.

Definition: the cancellation law

A commutative monoid M is said to satisfy the *cancellation law* (or to be a *cancellative monoid*) if

$$m + n = m + k, \quad m, n, k \in M,$$

always implies $n = k$.

For such a monoid, the map $M \rightarrow \Gamma(M)$ is injective; see Exercise 17.16.

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Worksheet 17

Practice exercises

Exercise 17.1

Calculate

$$(4T^{(-1,3)} - 6T^{(-2,5)} + 5T^{(0,-2)} + 3T^{(-1,1)}) (-7T^{(1,0)} + 8T^{(4,5)} - 4T^{(-3,5)} + 6T^{(3,-1)})$$

in the monoid ring $K[\mathbb{Z}^2]$.

Exercise 17.2

Calculate

$$(4T^0 - 9T^1 + 9T^2 + 7T^3 - 4T^4) (4T^0 - T^1 + 5T^2 + 7T^3 - T^4)$$

in the monoid ring $K[\mathbb{Z}/(5)]$.

Exercise 17.3 ★

Calculate

$$(3T^0 - 2T^1 + 5T^2) (4T^0 - 6T^1 + 5T^2)$$

in the monoid ring

$$\mathbb{Z}/(7)[\mathbb{Z}/(3)]$$

over the field $\mathbb{Z}/(7)$.

Exercise 17.4

Show that multiplication in the monoid ring of a commutative monoid satisfies the associative and distributive laws.

Exercise 17.5

Let K be a field. Find a commutative monoid M such that there is an isomorphism

$$K[M] \cong K[X, Y, U, V]/(UX - VY).$$

Exercise 17.6

Let R be a commutative ring. Prove the R -algebra isomorphism

$$R[\mathbb{Z}^n] \cong R[X_1, \dots, X_n]_{X_1 \cdots X_n}$$

using the universal properties of monoid rings and localisation.

Exercise 17.7

Show that the coordinate ring of the standard cone

$$V(Z^2 - X^2 - Y^2)$$

over $\mathbb{C}$ can be realised as a monoid ring.

Exercise 17.8

Let R be a commutative ring and let

$$A = R[X_1, \dots, X_n].$$

For a family of monomials

$$\{X^\nu \mid \nu \in J \subseteq \mathbb{N}^n\},$$

show that the R -subalgebra of A generated by these monomials is the monoid ring $R[M]$, where M is the submonoid of $\mathbb{N}^n$ generated by $(\nu)_{\nu \in J}$.

Exercise 17.9

Let M be a commutative monoid and let $e \in M$ satisfy

$$2e = e.$$

For a field K , show that T^e is an idempotent element of $K[M]$.

Exercise 17.10

Let $G \neq \{0\}$ be a finite abelian group. Show that the monoid ring $\mathbb{C}[G]$ is not connected, although there is no element $e \neq 0$ in G satisfying

$$2e = e.$$

Edition note: the source entity title and the context of commutative monoid rings specify that G is abelian, but the sentence displayed in the source mentions only a finite group. This edition displays the commutativity hypothesis stated in the exercise entity.

Exercise 17.11

Give an example of a submonoid

$$M \subseteq \mathbb{N}^2$$

that is not finitely generated.

Invertible elements of a monoid are also called its *units*. They form the *group of units* of the monoid.

Exercise 17.12 ★

Let M be a commutative monoid, let K be a field, and let $m \in M$. Show that m is a unit in M if and only if T^m is a unit in $K[M]$.

Exercise 17.13

Show that the group of differences of a commutative monoid is indeed a group.

Exercise 17.14

Let M be a commutative monoid. Show that its group of differences

$$\Gamma = \Gamma(M)$$

is an abelian group and has the following universal property: for every monoid homomorphism

$$\varphi : M \longrightarrow G$$

to a group G , there is exactly one group homomorphism

$$\tilde{\varphi} : \Gamma \longrightarrow G$$

extending φ .

Exercise 17.15

Let $M \subseteq N$ be commutative monoids. Show that

$$\widetilde{M} = \{n \in N \mid \text{there is } k \in \mathbb{N}_+ \text{ with } kn \in M\}$$

is a submonoid of N containing M .

Exercise 17.16

Let M be a commutative monoid with group of differences $\Gamma = \Gamma(M)$. Show that the following statements are equivalent.

1. M satisfies the cancellation law.
2. The canonical map $M \rightarrow \Gamma(M)$ is injective.
3. M can be realised as a submonoid of a group.

Exercise 17.17

Let M be a commutative monoid and let R be a commutative ring. Characterise the subsets $I \subseteq M$ for which

$$R[I] = \bigoplus_{m \in I} RT^m \subseteq R[M]$$

is an ideal in $R[M]$.

Edition note: the source prints $\bigoplus_{m \in I} T^m$ without the coefficient module R . Since $R[I]$ is intended as an R -submodule of $R[M]$, this edition displays $\bigoplus_{m \in I} RT^m$.

Exercise 17.18

Consider the monoid homomorphism

$$\mathbb{N}^2 \longrightarrow \mathbb{Z}, \quad e_1 \longmapsto 1, \quad e_2 \longmapsto -1.$$

Describe the associated maps between the monoid rings (over a field K) and between the associated K -spectra.

Exercise 17.19

Consider the commutative monoids

$$M = \mathbb{N}^r, \quad N = \mathbb{N}^s.$$

Show that a monoid homomorphism from M to N is uniquely determined by a matrix with entries in $\mathbb{N}$, with r columns and s rows. What does the associated map of spectra look like?

Exercise 17.20

Let M be a square $r \times r$ matrix with entries in $\mathbb{N}$, with associated monoid map and map of spectra

$$\varphi : K\text{-Spek}(K[\mathbb{N}^r]) \longrightarrow K\text{-Spek}(K[\mathbb{N}^r]),$$

where K is an infinite field. Show that

$$\det M \neq 0$$

if and only if the image of φ contains a nonempty open subset of $K\text{-Spek}(K[\mathbb{N}^r])$.

Edition note: this is the intended meaning of the source's phrase "maps surjectively onto an open set". The stated assumption that K is infinite is insufficient: for $K = \mathbb{R}$ and $M = (2)$, the image of $x \mapsto x^2$ contains no nonempty Zariski-open subset of the line. The assertion holds if K is algebraically closed.

The next exercise uses the following definition. A *filter* F in a commutative monoid M is a submonoid that is also *closed under taking divisors*: if $f \in F$ and $g \mid f$, then $g \in F$.

Exercise 17.21

Let M be a commutative monoid. Show that M has a smallest filter and that this filter forms a group.

Exercise 17.22

Let M be a commutative monoid and let $f \in M$. Consider the set

$$M_f = \{m - nf \mid m \in M, n \in \mathbb{N}\} / \sim,$$

where

$$m - nf \sim m' - n'f$$

if and only if there is a $k \in \mathbb{N}$ such that

$$m + n'f + kf = m' + nf + kf$$

in M .

1. Show that $\sim$ is an equivalence relation.
2. Define a monoid structure on M_f .
3. Let R be a commutative ring and let $T^f \in R[M]$ be the monomial corresponding to f . Show that

$$R[M_f] \cong R[M]_{T^f}.$$

Exercise 17.23

Let M, N be finitely generated commutative monoids, with

$$K\text{-Spek}(K[M]) = \text{Mor}_{\text{mon}}(M, K)$$

and

$$K\text{-Spek}(K[N]) = \text{Mor}_{\text{mon}}(N, K).$$

For a monoid homomorphism $\varphi : M \rightarrow N$, show that the associated map of spectra can be defined in two different ways that agree in substance.

Exercise 17.24

Consider the commutative monoid M with three generators e, f, g and the single relation

$$e + f = 5g.$$

Determine the K -spectrum of M for various fields K .

Exercise 17.25

Let M be a finite commutative monoid. Show that its K -spectrum is also finite.

Exercise 17.26

In

$$R = \mathbb{R}[\mathbb{Q}_{\geq 0}],$$

calculate the product

$$(X^2 + 4X^{3/2} - 5X + X^{1/2})(2X^{3/2} + 4X - 7X^{1/2}).$$

Exercise 17.27

Let K be a field. Show that every element

$$F \in R = K[\mathbb{Q}_{\geq 0}]$$

can be written as a polynomial in $X^{1/b}$ for some $b \in \mathbb{N}_+$. In other words, there is a $P \in K[Y]$ such that

$$F = P(X^{1/b}).$$

Which polynomial can be chosen for

$$F = X^{1/2} + X^{1/3} + X^{1/5}?$$

Exercise 17.28

Show that in $R = K[\mathbb{Q}_{\geq 0}]$, the element X has no factorisation into irreducible elements.

Exercise 17.29

Show that in $R = \mathbb{R}[\mathbb{Q}_{\geq 0}]$, the element $X^2 + 1$ is not irreducible.

Exercise 17.30

Show that there are no irreducible elements in $R = \mathbb{C}[\mathbb{Q}_{\geq 0}]$.

Exercise 17.31 ★

Determine all divisors of X in the ring

$$R = K[\mathbb{Q}_{\geq 0}],$$

where K is a field.

Exercise 17.32 ★

Determine the units in the ring

$$R = K[\mathbb{Q}],$$

where K is a field.

Exercises for submission**Exercise 17.33 (6 points)**

Let $M \subseteq N$ be finitely generated commutative cancellative monoids. Show that, for a field K , the ring homomorphism

$$K[M] \subseteq K[N]$$

is finite if and only if for every $n \in N$ there is a $k \in \mathbb{N}_+$ with $kn \in M$.

Exercise 17.34 (4 points)

Let

$$M = (\mathbb{Q}, +)$$

be the additive group of rational numbers. Determine

$$\mathbb{Q}\text{-Spek}(\mathbb{Q}[M]).$$

What happens if $\mathbb{Q}$ is replaced by $\mathbb{R}$?

Exercise 17.35 (4 points)

Let

$$\varphi : M \longrightarrow N$$

be a homomorphism of commutative monoids. Show that the set of all points in $K\text{-Spek}(K[N])$ sent by the map of spectra to the point

$$1 \in K\text{-Spek}(K[M])$$

(the point corresponding to the constant map $M \rightarrow 1$) itself has the structure of the K -spectrum of a suitable monoid.

Exercise 17.36 (4 points)

Consider monoids of the form

$$M = (\mathbb{Z}/(m), +).$$

Describe $K\text{-Spek}(K[M])$ in general and specifically for

$$K = \mathbb{R}, \mathbb{C}, \mathbb{Z}/(5).$$

Find the idempotent elements in

$$\mathbb{C}[\mathbb{Z}/(3)].$$

Exercise 17.37 (4 points)

Let M be a commutative monoid. Define bijections between the following objects.

1. Filters in M .
2. $\text{Mor}_{\text{mon}}(M, (\{0, 1\}, 1, \cdot))$.
3. $\mathbb{F}_2\text{-Spek}(M)$.
4. The set

$$\left\{ \varphi \in K\text{-Spek}(K[M]) \mid \varphi(M) \subseteq \{0, 1\} \right\},$$

where K is a field.

Exercise 17.38 (3 points)

Let M, N be commutative monoids and let K be a field. What is the relationship between

$$K\text{-Spek}(K[M \times N])$$

and

$$K\text{-Spek}(K[M]), \quad K\text{-Spek}(K[N])?$$

Exercise 17.39 (4 points)

Let K be a field and let G be a group. Consider the group ring $K[G]$, and let M be a $K[G]$ -module. Show that:

1. M is nothing other than a K -vector space V together with a group homomorphism

$$\rho : G \longrightarrow \text{Aut}_K(V);$$

2. a $K[G]$ -module homomorphism

$$\varphi : M \longrightarrow M$$

is a K -linear map also satisfying

$$\varphi \circ \rho(g) = \rho(g) \circ \varphi$$

for every $g \in G$.

The map ρ is called a *representation* of G . Such representations are often easier to handle than G itself and frequently give useful insights into G .

Edition note: the right-hand side of the source equation prints $\rho \circ \varphi$ without the argument g . This edition displays $\rho(g) \circ \varphi$, the correctly typed endomorphism expressing the intertwining condition.

Public Solutions to Worksheet 17

At the frozen revision boundary, the source provides public solutions only for Exercises 17.3, 17.12, 17.31, and 17.32. No additional solutions have been created for this edition.

Solution to Exercise 17.3

In $\mathbb{Z}/(7)[\mathbb{Z}/(3)]$, coefficients are calculated modulo 7 and exponents of T modulo 3. Thus

$$\begin{aligned} & (3T^0 - 2T^1 + 5T^2)(4T^0 - 6T^1 + 5T^2) \\ &= (3T^0 + 5T^1 + 5T^2)(4T^0 + T^1 + 5T^2) \\ &= 5T^0 + 3T^1 + T^2 + 6T^1 + 5T^2 + 4T^0 + 6T^2 + 5T^0 + 4T^1 \\ &= 6T + 5T^2. \end{aligned}$$

[Back to Exercise 17.3.](#)

Solution to Exercise 17.12

If m is a unit in M , there is an $n \in M$ with

$$m + n = 0.$$

Then

$$T^m T^n = T^{m+n} = T^0 = 1,$$

so T^m is a unit in the monoid ring.

Conversely, suppose T^m is a unit in the monoid ring. There is an element

$$P = \sum_{n \in E} a_n T^n$$

with finite support $E \subseteq M$, all displayed coefficients $a_n \in K$ nonzero, and

$$T^m P = \sum_{n \in E} a_n T^{m+n} = 1.$$

The coefficient of T^0 on the left is 1, so there is at least one $n \in E$ such that

$$m + n = 0.$$

Thus m has an inverse in M .

Edition note: the source passes directly to the assertion that all exponents $m + n$ equal zero. The argument above first uses the coefficient of T^0 to obtain one such n , which already proves that m is a unit. Translation by m is then bijective, so the source's stronger assertion also follows. This ordering avoids assuming cancellation prematurely.

[Back to Exercise 17.12.](#)

Solution to Exercise 17.31

The divisors of X are exactly the elements of the form

$$aX^q, \quad a \neq 0, \quad q \in \mathbb{Q}_{\geq 0}, \quad q \leq 1.$$

Every such element is indeed a divisor, since

$$(aX^q)(a^{-1}X^{1-q}) = X^qX^{1-q} = X^{q+1-q} = X.$$

Conversely, let

$$P = a_{q_1}X^{q_1} + a_{q_2}X^{q_2} + \cdots + a_{q_n}X^{q_n}, \quad 0 \leq q_1 < q_2 < \cdots < q_n,$$

be a divisor of X . There is then a

$$Q = b_{r_1}X^{r_1} + b_{r_2}X^{r_2} + \cdots + b_{r_m}X^{r_m}, \quad 0 \leq r_1 < r_2 < \cdots < r_m,$$

with all displayed coefficients nonzero and $PQ = X$. The lowest- and highest-exponent terms in the product are

$$a_{q_1}b_{r_1}X^{q_1+r_1} \quad \text{and} \quad a_{q_n}b_{r_m}X^{q_n+r_m};$$

both are nonzero. For $PQ = X$, we must have

$$q_1 + r_1 = q_n + r_m = 1.$$

Since the exponents are strictly ordered, this is possible only if $n = m = 1$. Thus $P = aX^q$ with $0 \leq q \leq 1$, as claimed.

Edition note: in the final extreme term, the source prints $a_{q_n}b_{r_n}$, although the support of Q has m terms and the next line itself uses r_m . This edition displays the terminal index $a_{q_n}b_{r_m}$.

[Back to Exercise 17.31.](#)

Solution to Exercise 17.32

The units are exactly the elements of the form

$$aX^q, \quad a \neq 0, \quad q \in \mathbb{Q}.$$

Such an element is a unit because

$$(aX^q)(a^{-1}X^{-q}) = X^qX^{-q} = X^{q-q} = X^0 = 1.$$

Conversely, let

$$P = a_{q_1}X^{q_1} + a_{q_2}X^{q_2} + \cdots + a_{q_n}X^{q_n}, \quad q_1 < q_2 < \cdots < q_n,$$

be a unit. There is a

$$Q = b_{r_1}X^{r_1} + b_{r_2}X^{r_2} + \cdots + b_{r_m}X^{r_m}, \quad r_1 < r_2 < \cdots < r_m,$$

with all displayed coefficients nonzero and $PQ = 1$. The lowest- and highest-exponent terms in the product are

$$a_{q_1}b_{r_1}X^{q_1+r_1} \quad \text{and} \quad a_{q_n}b_{r_m}X^{q_n+r_m},$$

and both are nonzero. For the product to equal 1, we must have

$$q_1 + r_1 = q_n + r_m = 0.$$

The strict ordering of the exponents forces $n = m = 1$. Thus $P = aX^q$.

[Back to Exercise 17.32.](#)

Lecture 18: Monomial Curves and Numerical Monoids

Monomial curves

We now specialise the theory of monoid rings to the one-dimensional case, obtaining the rings that describe monomial curves.

Definition: monomial curve

A *monomial curve* is the image of the affine line $\mathbb{A}_K^1$ under a map of the form

$$\begin{aligned} \mathbb{A}_K^1 &\longrightarrow \mathbb{A}_K^n, \\ t &\longmapsto (t^{e_1}, \dots, t^{e_n}), \end{aligned}$$

where $e_i \geq 1$ for every i .

We shall shortly see that the image of such a monomial map is Zariski closed. Thus a monomial curve really is an algebraic curve. A monomial curve is parametrised and hence rational, although it is not generally a plane curve. Sometimes the map itself is also called a monomial curve.

Edition note: the source's closed-image assertion holds for relatively prime exponents over any field, and for arbitrary exponents over an algebraically closed field. Without such a hypothesis it can fail: $t \mapsto t^2$ over $\mathbb{R}$ has a non-Zariski-closed image.

We often restrict attention to exponents e_i whose greatest common divisor is 1. This is no essential restriction. If m is the greatest common divisor of all the e_i , we can write

$$e_i = mf_i$$

with $f_1, \dots, f_n$ relatively prime, and factor the map as

$$\mathbb{A}_K^1 \longrightarrow \mathbb{A}_K^1 \longrightarrow \mathbb{A}_K^n, \quad t \longmapsto t^m = s, \quad s \longmapsto (s^{f_1}, \dots, s^{f_n}).$$

The first map is simply exponentiation, while the second is a monomial curve map with relatively prime exponents.

Edition note: although the domain of the second map is the affine line with a single coordinate s , the source prints the image coordinates as $(s_1^{f_1}, \dots, s_n^{f_n})$. This edition displays powers of the defined scalar input, namely $(s^{f_1}, \dots, s^{f_n})$.

Remark: the underlying monoid map

The monomial map

$$t \longmapsto (t^{e_1}, \dots, t^{e_n})$$

is precisely the map of K -spectra associated with the monoid homomorphism

$$\mathbb{N}^n \longrightarrow \mathbb{N}$$

that sends the i th basis vector to e_i ; compare Remark 17.9.

Edition note: the source prints the coordinates as $(t_1^{e_1}, \dots, t_n^{e_n})$, although the input is a single scalar t . This edition displays coordinates of the correct type.

This monoid homomorphism factors as

$$\mathbb{N}^n \longrightarrow M \longrightarrow \mathbb{N},$$

where M is the submonoid of $\mathbb{N}$ generated by $e_1, \dots, e_n$. Such a submonoid is called a *numerical monoid*. The first map is surjective. At the ring level we obtain

$$K[\mathbb{N}^n] = K[X_1, \dots, X_n] \longrightarrow K[M] \longrightarrow K[\mathbb{N}] = K[T],$$

while geometrically we obtain the spectrum maps

$$\mathbb{A}_K^1 \longrightarrow K\text{-Spek}(K[M]) \subseteq \mathbb{A}_K^n.$$

Thus the image of the affine line lies in the K -spectrum of the monoid ring $K[M]$. Theorem 18.10 will show that

$$\mathbb{A}_K^1 \longrightarrow K\text{-Spek}(K[M])$$

is always surjective and, when the exponents are relatively prime, also injective.

Edition note: “always surjective” requires, for example, algebraic closure of K . For a general field, Theorem 18.10 proves bijectivity under the relatively prime hypothesis. If $M = 2\mathbb{N}$ and $K = \mathbb{R}$, this spectrum map is $t \mapsto t^2$ and is not surjective.

Example: Neil's parabola

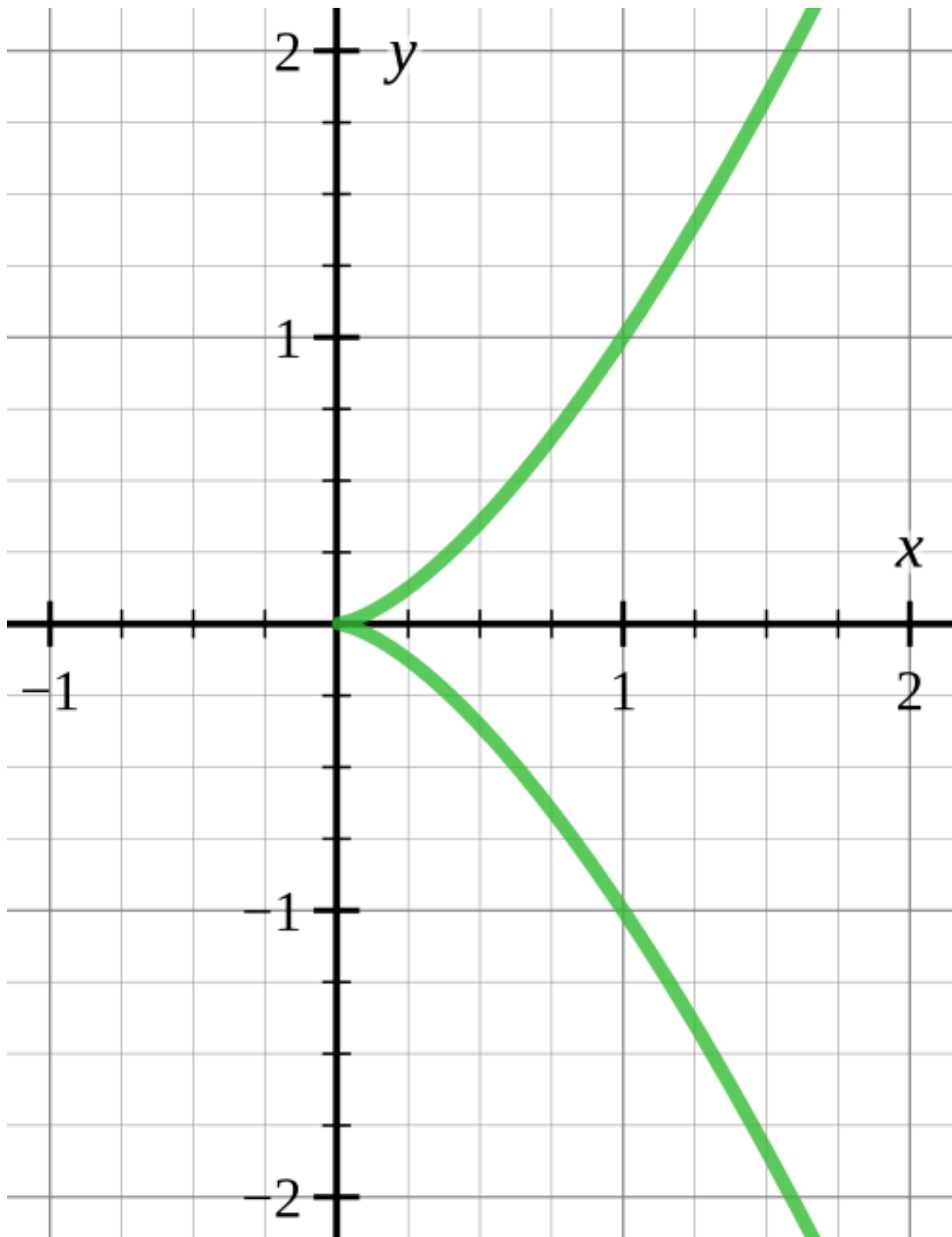


Figure 74: A cuspidal curve with two smooth arcs meeting in a sharp point at the origin

Neil's parabola with a cuspidal singularity at the origin. Georg-Johann, [CC BY-SA 3.0](#). Source details and the discrepancy between rights labels are recorded in the Unit 18 media credits.

Neil's parabola C is the image of the monomial map

$$\begin{aligned}\mathbb{A}_K^1 &\longrightarrow \mathbb{A}_K^2, \\ t &\longmapsto (t^2, t^3) = (x, y).\end{aligned}$$

The corresponding equation is

$$y^2 = x^3,$$

so that

$$C = V(Y^2 - X^3).$$

A monomial curve is determined solely by its tuple of exponents $(e_1, \dots, e_n)$, or equivalently by the numerical monoid they generate. Thus very little combinatorial information is required. Nevertheless, these curves provide a rich collection of examples in the theory of algebraic curves. The same phenomenon holds more generally for monoid rings and the algebraic varieties they define.

Invariants of numerical monoids

Lemma: all sufficiently large numbers belong to the monoid

Let $M \subseteq \mathbb{N}$ be a numerical monoid generated by relatively prime natural numbers $e_1, \dots, e_n$. Every $m \in \mathbb{N}$ has a representation

$$m = a_1 e_1 + \dots + a_n e_n$$

with

$$0 \leq a_i < e_{i+1}, \quad 1 \leq i \leq n-1.$$

If m is sufficiently large, we can also arrange that $a_n \geq 0$, so that all coefficients in the representation are nonnegative.

Proof Since $e_1, \dots, e_n$ are relatively prime, there is initially a representation

$$m = b_1 e_1 + \dots + b_n e_n$$

with integer coefficients. We put it into the required form step by step. By division with remainder, write

$$b_1 = c_1 e_2 + a_1, \quad 0 \leq a_1 < e_2.$$

Substitute this into the representation of m and combine the term $c_1 e_2 e_1$ with $b_2 e_2$. Treat the new second coefficient in the same way, using the third generator. Continuing this process puts the first $n-1$ coefficients into the required form.

In this form, the sum of the first $n-1$ terms is bounded by a constant depending only on the generators. If m exceeds this bound, the last term, and therefore its coefficient a_n , must be nonnegative.

Thus, beyond some threshold, every natural number belongs to the monoid generated by the relatively prime exponents. The least such threshold has its own name.

Definition: conductor

Let $M \subseteq \mathbb{N}$ be a numerical monoid generated by relatively prime elements. The least number f satisfying

$$\mathbb{N}_{\geq f} \subseteq M$$

is called the *conductor* of M .

The following invariants can also be computed discretely at the level of the numerical monoid. Later we shall define them for arbitrary algebraic curves; in that generality, they are usually harder to compute.

Definition: embedding dimension

Let $M \subseteq \mathbb{N}$ be a numerical monoid with relatively prime generators. The smallest number of elements in a generating system for M is called its *embedding dimension*.

Definition: multiplicity

Let $M \subseteq \mathbb{N}$ be a numerical monoid with relatively prime generators. The least positive element

$$e \in M, \quad e \geq 1,$$

is called the *multiplicity* of M and is denoted by $e(M)$.

Definition: degree of singularity

Let $M \subseteq \mathbb{N}$ be a numerical monoid with relatively prime generators. The number of gaps, that is, the number of elements of

$$\mathbb{N} \setminus M,$$

is called the *degree of singularity* of M and is denoted by $\delta(M)$.

Example: the monoid generated by 5, 8, and 11

Consider the numerical monoid M generated by 5, 8, 11. Thus M consists of all sums

$$5a_1 + 8a_2 + 11a_3, \quad a_1, a_2, a_3 \geq 0.$$

Its elements include

$$0, 5, 8, 10, 11, 13, 15, 16, 18, 19, 20, 21, 22, 23, 24, 25, \dots$$

Since 18, 19, 20, 21, 22 are five consecutive numbers in M and $5 \in M$, all subsequent numbers also belong to M . Hence its conductor is 18, its multiplicity is 5, its degree of singularity is 10, and its embedding dimension is 3.

Parametrisation and canonical generators**Theorem: the monomial parametrisation is a bijection**

Let $M \subseteq \mathbb{N}$ be a submonoid generated by relatively prime natural numbers $e_1, \dots, e_n$. The monomial map

$$\mathbb{A}_K^1 \longrightarrow K\text{-Spek}(K[M])$$

is a bijection.

Proof By Remark 17.9, this can be viewed as the natural map

$$\mathbb{A}_K^1 = \text{Mor}_{\text{mon}}(\mathbb{N}, K) \longrightarrow \text{Mor}_{\text{mon}}(M, K)$$

induced by the inclusion $M \subseteq \mathbb{N}$.

To prove injectivity, suppose that $a, b \in K$ and

$$a^m = b^m$$

for every $m \in M$. If $b = 0$, then immediately $a = 0$. We may therefore assume $b \neq 0$, and likewise $a \neq 0$. By Lemma 18.4, all natural numbers from some f onwards belong to M . In particular,

$$a^f = b^f, \quad a^{f+1} = b^{f+1}.$$

It follows that

$$a = \frac{a^{f+1}}{a^f} = \frac{b^{f+1}}{b^f} = b.$$

For surjectivity, take a monoid homomorphism

$$\varphi : M \longrightarrow K.$$

We must extend it to all of $\mathbb{N}$. Write

$$\varphi(e_i) = a_i \in K.$$

These values satisfy

$$a_j^{e_i} = \varphi(e_j)^{e_i} = \varphi(e_i e_j) = \varphi(e_i)^{e_j} = a_i^{e_j}.$$

If one a_i is 0, then all the a_i are 0, and the map sending every positive number to 0 (and 0 to 1) gives an extension. Thus we may assume that all the a_i are units.

Since the e_i are relatively prime, there are integers $m_1, \dots, m_n \in \mathbb{Z}$ such that

$$m_1 e_1 + \dots + m_n e_n = 1.$$

Set

$$a = a_1^{m_1} \dots a_n^{m_n}.$$

We claim that the homomorphism $\mathbb{N} \rightarrow K$ determined by $1 \mapsto a$, namely $k \mapsto a^k$, extends φ . It suffices to check this on the e_i . For e_1 ,

$$\begin{aligned} a^{e_1} &= (a_1^{m_1} \dots a_n^{m_n})^{e_1} \\ &= a_1^{e_1 m_1} a_2^{e_1 m_2} \dots a_n^{e_1 m_n} \\ &= a_1^{1 - \sum_{i=2}^n m_i e_i} (a_2^{e_1 m_2} \dots a_n^{e_1 m_n}) \\ &= a_1 (a_1^{-m_2 e_2} a_2^{e_1 m_2}) \dots (a_1^{-m_n e_n} a_n^{e_1 m_n}) \\ &= a_1 (a_1^{-e_2} a_2^{e_1})^{m_2} \dots (a_1^{-e_n} a_n^{e_1})^{m_n} \\ &= a_1, \end{aligned}$$

because each factor in the penultimate line equals 1 by the relations $a_j^{e_i} = a_i^{e_j}$. The same argument applies to every e_i .

Remark: proof using the group of differences

The surjectivity in the preceding theorem can also be seen using the universal property of the group of differences. The group of differences of a numerical monoid with relatively prime generators is $\mathbb{Z}$. The case in which a generator maps to zero is handled as in the proof. If all images are nonzero, we obtain a monoid homomorphism

$$\varphi : M \longrightarrow K^\times.$$

The universal property of the group of differences gives a unique extension

$$\tilde{\varphi} : \mathbb{Z} \longrightarrow K^\times,$$

which supplies the required preimage.

Edition note: at this step the source refers to the universal property of monoid rings. The construction actually used extends from M to its group of differences $\mathbb{Z}$, so this edition states the universal property of the group of differences.

The next two statements show that a numerical monoid has a canonical generating system. This gives the embedding dimension an interpretation that can later be transferred to arbitrary Noetherian local rings. The monoid ring itself is of course not local, but its localisation at the singularity of a numerical monoid is local.

Lemma: the canonical minimal generating system

Let $M \subseteq \mathbb{N}$ be a numerical monoid with relatively prime generators. Set

$$M_+ = \{m \in M \mid m \geq 1\}$$

and

$$M_+ + M_+ = \{m + m' \mid m, m' \in M_+\}.$$

Then

$$M_+ \setminus (M_+ + M_+)$$

is a generating system for M , and every other generating system contains this set.

Proof An element

$$m \in M_+ \setminus (M_+ + M_+)$$

cannot be written as a sum of two other positive elements of M . It therefore belongs to every generating system.

Conversely, this set already generates M . Otherwise, choose the least element $x \in M$ that it does not generate. Since x does not belong to $M_+ \setminus (M_+ + M_+)$, we have

$$x = x_1 + x_2, \quad x_1, x_2 \in M_+.$$

Both summands are smaller than x , so each can be written as a sum of elements of the canonical set. This immediately gives a contradiction.

Edition note: the source prints $x_1, x_2 \in M_+ + M_+$. From $x \in M_+ + M_+$, the definition of the sumset gives only $x_1, x_2 \in M_+$; this is the membership needed for the minimal-descent argument.

Corollary: embedding dimension from the canonical generators

In the situation of the preceding lemma, the embedding dimension of M equals the number of elements of

$$M_+ \setminus (M_+ + M_+).$$

Proof This follows directly from Lemma 18.12.

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Worksheet 18

Practice exercises

Exercise 18.1

A counterfeiter produces 3-euro and 7-euro notes.

1. Show that there are only finitely many amounts that he cannot pay exactly. What is the largest amount he cannot pay?
2. What is the smallest amount he can pay in two different ways?
3. Describe the set M of whole-euro amounts he can pay with his notes.

Exercise 18.2

A counterfeiter produces 4-euro, 9-euro, and 11-euro notes.

1. Show that there are only finitely many amounts that he cannot pay exactly. What is the largest amount he cannot pay?
2. What is the smallest amount he can pay in two different ways?
3. Describe explicitly the set M of whole-euro amounts he can pay with his notes.

Exercise 18.3 ★

A counterfeiter produces 7-euro, 11-euro, 13-euro, and 37-euro notes. How many whole-euro amounts can he not pay with his notes, and what is the largest amount he cannot pay? Determine the multiplicity and embedding dimension of the associated numerical monoid.

Exercise 18.4 ★

For the numerical monoid

$$M \subseteq \mathbb{N}$$

generated by 4, 7, and 17, determine its embedding dimension, multiplicity, conductor, and degree of singularity.

Exercise 18.5

For the numerical monoid M generated by 5, 7, and 9, determine its embedding dimension, multiplicity, conductor, and degree of singularity.

Exercise 18.6

Let $M \subseteq \mathbb{N}$ be a numerical monoid generated by relatively prime natural numbers. Show that its embedding dimension is at most its multiplicity.

Exercise 18.7

Let $M \subseteq \mathbb{N}$ be a numerical monoid generated by relatively prime numbers, whose embedding dimension equals its multiplicity. Show that the largest generator in a minimal generating system is greater than or equal to the conductor.

Exercise 18.8

Give an example of a numerical monoid M with multiplicity 3 and embedding dimension 3 whose conductor is prime and does not belong to the minimal generating system.

Exercise 18.9

Let $M \subseteq \mathbb{N}$ be a numerical monoid generated by relatively prime elements. Suppose that its multiplicity equals its conductor. Determine a minimal generating system and the embedding dimension of M .

Exercise 18.10 ★

Consider Neil's parabola

$$C = V(Y^2 - X^3) \subseteq \mathbb{A}_{\mathbb{C}}^2$$

and the point $P = (1, 1) \in C$. Show that the maximal ideal associated with P , namely the ideal $(X - 1, Y - 1)$ in the coordinate ring

$$R = \mathbb{C}[X, Y]/(Y^2 - X^3),$$

cannot be described as the radical of an ideal generated by a single element $f \in R$.

The preceding exercise shows that every curve D meeting Neil's parabola at $(1, 1)$ also meets it at at least one other point. This also generalises Exercise 1.13.

Exercise 18.11 ★

Consider the submonoid of the natural numbers generated by 2 and 3, namely

$$M = \{0, 2, 3, 4, 5, \dots\} \subset \mathbb{N},$$

and the quotient ring

$$R = \mathbb{Z}/(9).$$

Also consider the sets of R -valued points of M and $\mathbb{N}$, namely the sets of monoid homomorphisms $\text{Mor}_{\text{mon}}(M, R)$ and $\text{Mor}_{\text{mon}}(\mathbb{N}, R)$, and the restriction map

$$\begin{aligned} \psi : \text{Mor}_{\text{mon}}(\mathbb{N}, R) &\longrightarrow \text{Mor}_{\text{mon}}(M, R), \\ \pi &\longmapsto \pi|_M. \end{aligned}$$

1. Determine the units and nilpotent elements of R .
2. Determine the number of elements of $\text{Mor}_{\text{mon}}(\mathbb{N}, R)$.
3. Show that every $\rho \in \text{Mor}_{\text{mon}}(M, R)$ for which $\rho(2)$ is a unit in R has an extension to $\mathbb{N}$.
4. Determine all $\pi \in \text{Mor}_{\text{mon}}(\mathbb{N}, R)$ whose restriction to $M \setminus \{0\}$ is the zero function.
5. Determine all $\rho \in \text{Mor}_{\text{mon}}(M, R)$ that have no extension to $\mathbb{N}$.
6. Determine the number of elements of $\text{Mor}_{\text{mon}}(M, R)$.

Exercise 18.12

Let M be a numerical monoid. Determine all filters in M .

Exercise 18.13

Let M be a numerical monoid and K a field. Show that there is an $f \in K[M]$ such that

$$K[M]_f = K[X]_X.$$

Edition note: this equality inside $K(X)$ requires that the generators of M have greatest common divisor 1, a hypothesis omitted here in the source. For $M = 2\mathbb{N}$, every such localisation remains inside $K(X^2)$ and cannot contain X .

Exercise 18.14

Let M be a numerical monoid, K a field, and $K[M]$ the associated monoid ring. Show that

$$K[M] \cong K[\mathbb{N}]$$

if and only if $M = \mathbb{N}$.

Edition note: assume here that the generators of M have greatest common divisor 1. With the broader definition of numerical monoid used at the start of the lecture, the source's assertion is false: $K[2\mathbb{N}] = K[X^2] \cong K[X]$, although $2\mathbb{N} \neq \mathbb{N}$.

Exercise 18.15 ★

Give an example of an injective monoid homomorphism

$$M \longrightarrow N$$

between commutative monoids for which the associated spectrum map is not surjective.

Exercise 18.16

Let M be a commutative monoid and K a field.

1. Show that the diagonal map

$$\begin{aligned} \Delta : M &\longrightarrow M \times M, \\ m &\longmapsto (m, m) \end{aligned}$$

is a monoid homomorphism.

2. Describe the associated K -algebra homomorphism

$$K[M] \longrightarrow K[M \times M] \cong K[M] \otimes K[M].$$

3. Describe the associated spectrum map

$$K\text{-Spek}(K[M]) \times K\text{-Spek}(K[M]) \longrightarrow K\text{-Spek}(K[M]).$$

In particular, show that this makes $K\text{-Spek}(K[M])$ itself a commutative monoid.

Exercise 18.17

Specialise Exercise 18.16 to the monoids

$$M = \mathbb{N}, \mathbb{Z}, \mathbb{N}^r, \mathbb{Z}^r.$$

Exercise 18.18

Let M be a commutative monoid, K a field, and

$$P \in K\text{-Spek}(M)$$

a fixed point.

1. Show that there is a continuous map

$$\begin{aligned}\Psi_P : K\text{-Spek}(M) &\longrightarrow K\text{-Spek}(M), \\ Q &\longmapsto Q \cdot P.\end{aligned}$$

2. Describe the associated K -algebra homomorphism $K[M] \longrightarrow K[M]$.
3. Characterise the points P for which this map is bijective.

Exercise 18.19

Let M, N be commutative monoids, $\varphi : M \longrightarrow N$ a monoid homomorphism, and K a field. Show that the spectrum map

$$\varphi^* : K\text{-Spek}(N) \longrightarrow K\text{-Spek}(M)$$

is also a monoid homomorphism.

Exercise 18.20

Let $a, b, c, d, r, s \geq 1$ be natural numbers. Consider the continuously differentiable curve

$$\begin{aligned}[0, 1] &\longrightarrow \mathbb{R}^2, \\ t &\longmapsto (t^r, t^s).\end{aligned}$$

Compute the line integral along this path for the vector field

$$F(x, y) = (x^a y^b, x^c y^d).$$

Exercises to submit**Exercise 18.21 (6 points)**

Let M be a numerical monoid generated by two relatively prime elements $d > e$. Determine its embedding dimension, multiplicity, conductor, and degree of singularity.

Exercise 18.22 (3 points)

For the numerical monoid M generated by 3, 7, 9, and 11, determine its embedding dimension, multiplicity, conductor, and degree of singularity.

Exercise 18.23 (4 points)

Classify all numerical monoids M with relatively prime generators satisfying

$$f(M) \leq 6.$$

For each monoid, give its embedding dimension, multiplicity, and degree of singularity.

Exercise 18.24 (8 points: 1+3+1+2+1)

Consider the submonoid of the natural numbers generated by 2 and 5, namely

$$M = \{0, 2, 4, 5, 6, \dots\} \subset \mathbb{N},$$

and the quotient ring $R = \mathbb{Z}/(4)$. Also consider the sets of R -valued points of M and $\mathbb{N}$, namely $\text{Mor}_{\text{mon}}(M, R)$ and $\text{Mor}_{\text{mon}}(\mathbb{N}, R)$, and the restriction map

$$\begin{aligned} \psi : \text{Mor}_{\text{mon}}(\mathbb{N}, R) &\longrightarrow \text{Mor}_{\text{mon}}(M, R), \\ \pi &\longmapsto \pi|_M. \end{aligned}$$

1. Determine the units and nilpotent elements of R , and the number of elements of $\text{Mor}_{\text{mon}}(\mathbb{N}, R)$.
2. Show that every $\rho \in \text{Mor}_{\text{mon}}(M, R)$ for which $\rho(2)$ is a unit in R has an extension to $\mathbb{N}$.
3. Determine all $\pi \in \text{Mor}_{\text{mon}}(\mathbb{N}, R)$ whose restriction to $M \setminus \{0\}$ is the zero function.
4. Determine all $\rho \in \text{Mor}_{\text{mon}}(M, R)$ that have no extension to $\mathbb{N}$.
5. Determine the number of elements of $\text{Mor}_{\text{mon}}(M, R)$.

Exercise 18.25 (3 points)

Let $M \subseteq \mathbb{N}$ be a numerical monoid and K a field. Define

$$M_+ = M \cap \mathbb{N}_+$$

and

$$nM_+ = \left\{ m \in M \mid \text{there is a representation } m = m_1 + \dots + m_n \text{ with } m_i \in M_+ \right\}.$$

Show that the nM_+ are “ideals” in M , that M_+ determines a maximal ideal $\mathfrak{m}$ in $K[M]$, and that the ideal associated with nM_+ equals $\mathfrak{m}^n$.

Exercise 18.26 (4 points)

Let K be an algebraically closed field and M, N numerical monoids with $M \subseteq N$. Show that the associated spectrum map is surjective.

Source hint: Use Theorem 18.10.

Exercise 18.27 (3 points)

Let M, N be numerical monoids. For which numerical invariants ν among multiplicity, conductor, degree of singularity, and embedding dimension does the inclusion

$$M \subseteq N$$

imply the inequality

$$\nu(M) \geq \nu(N)?$$

Give a proof or a counterexample.

Exercise 18.28 (3 points)

Let M be a numerical monoid not isomorphic to $\mathbb{N}$, and let K be a field. Show that the monoid ring $K[M]$ has irreducible elements that are not prime. Give elements of $K[M]$ with two essentially different factorisations into irreducible elements.

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Public Solutions to Worksheet 18

At the frozen revision boundary, the source provides public solutions only for Exercises 18.3, 18.4, 18.10, 18.11, and 18.15. No additional solutions have been created for this edition.

Solution to Exercise 18.3

We compute the sums that can be formed from the four numbers. We do this by adding multiples of 7 to sums formed from the larger numbers. The multiples of 7 are

$$7, 14, 21, 28, 35, 42, \dots$$

Starting from 11 gives

$$11, 18, 25, 32, 39, 46, \dots$$

Starting from 13 gives

$$13, 20, 27, 34, 41, \dots$$

Starting from $22 = 11 + 11$ gives

$$22, 29, 36, 43, \dots,$$

and starting from $24 = 11 + 13$ gives

$$24, 31, 38, 45, \dots$$

We also add

$$26 = 13 + 13, \quad 33 = 11 + 11 + 11, \quad 35 = 11 + 11 + 13, \quad 37 = 11 + 13 + 13.$$

The last equality also shows that the generator 37 is redundant. We now have a gap-free sequence from 31 to 37, of length 7, so every larger number also belongs to the monoid. The number 30 does not belong to it. Hence the conductor is 31, and 30 is the largest amount that cannot be paid.

The multiplicity is the least positive number, namely 7, and the embedding dimension is 3 because 37 is redundant. The gaps are

$$1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, 16, 17, 19, 23, 30.$$

Thus the degree of singularity is 16; this is exactly the number of amounts that cannot be paid.

[Back to Exercise 18.3.](#)

Solution to Exercise 18.4

The monoid contains all multiples of 4, namely

$$4, 8, 12, 16, 20, 24, \dots$$

It also contains all sums of 7 and a multiple of 4,

$$7, 11, 15, 19, 23, \dots,$$

all sums of $2 \cdot 7 = 14$ and a multiple of 4,

$$14, 18, 22, 26, \dots,$$

and all sums of $3 \cdot 7 = 21$ and a multiple of 4,

$$21, 25, \dots$$

Thus all numbers from 18 onwards are covered, since every residue class modulo 4 has a representative in the monoid. Since the generator 17 is also available, all numbers from 14 onwards belong to the monoid. Hence the conductor is 14.

The multiplicity is the least positive number in the monoid, namely 4. The embedding dimension is 3, since the generator 17 cannot be omitted. The gaps are

$$1, 2, 3, 5, 6, 9, 10, 13,$$

so the degree of singularity is 8.

[Back to Exercise 18.4.](#)

Solution to Exercise 18.10

Consider the injective ring homomorphism

$$\begin{aligned}\varphi : R &\longrightarrow \mathbb{C}[T], \\ X &\longmapsto T^2, \\ Y &\longmapsto T^3.\end{aligned}$$

The extension of the ideal $(X - 1, Y - 1)$ is

$$(T^2 - 1, T^3 - 1) = ((T - 1)(T + 1), (T - 1)(T^2 + T + 1)).$$

The radical of this ideal is $(T - 1)$, since 1 is the only common root of the two polynomials. This also follows from the Nullstellensatz; in this case the corresponding ideals are in fact equal.

Suppose that $(X - 1, Y - 1) = \sqrt{(f)}$ for some $f \in R$. After extending to $\mathbb{C}[T]$, we must have

$$\varphi(f) = c(T - 1)^n$$

for some $n \in \mathbb{N}_+$ and $c \in \mathbb{C}^\times$. The coefficient of T in this polynomial is $cn(-1)^{n-1} \neq 0$. However, every element in the image of R has the form

$$\varphi(f) = a_0 + a_2T^2 + a_3T^3 + \dots + a_dT^d,$$

and therefore has no linear term. This is a contradiction.

Edition note: the source writes $\varphi(f) = (T - 1)^n$ directly. Equality of radicals gives only a nonzero scalar multiple $c(T - 1)^n$. This edition retains the argument with the necessary factor c ; the linear coefficient remains nonzero.

[Back to Exercise 18.10.](#)

Solution to Exercise 18.11

1. The elements

$$1, 2, 4, 5, 7, 8$$

are units, since they are all relatively prime to 9. The elements 0, 3, 6 are nilpotent, since their squares are 0 in R .

2. For a monoid homomorphism

$$\pi : \mathbb{N} \longrightarrow R$$

we must have $\pi(0) = 1$. The homomorphism is uniquely determined by $\pi(1)$, and any element of R can be chosen as this value. Thus

$$|\text{Mor}_{\text{mon}}(\mathbb{N}, R)| = 9.$$

3. Suppose that $\rho(2)$ is a unit. The only possible choice is

$$\pi(1) = \rho(3)\rho(2)^{-1},$$

since we must have

$$\rho(3) = \pi(3) = \pi(2)\pi(1) = \rho(2)\pi(1).$$

We check that this really defines a homomorphism from $\mathbb{N}$ to R . On all other numbers, π is already fixed by ρ . We must check $\pi(i + j) = \pi(i)\pi(j)$ for every $i, j \in \mathbb{N}$. The cases in which one summand is 0, or both summands are at least 2, follow immediately because ρ is a homomorphism. For $i = j = 1$,

$$\begin{aligned} \pi(1)\pi(1) &= \rho(3)\rho(2)^{-1}\rho(3)\rho(2)^{-1} \\ &= \rho(3+3)\rho(2+2)^{-1}\rho(2)^{-1}\rho(2) \\ &= \rho(6)\rho(6)^{-1}\rho(2) \\ &= \rho(2) = \pi(2). \end{aligned}$$

For $j \geq 2$,

$$\begin{aligned} \pi(1)\pi(j) &= \pi(1)\rho(j) \\ &= \rho(3)\rho(2)^{-1}\rho(j) \\ &= \rho(3+j)\rho(2)^{-1} \\ &= \rho(1+j)\rho(2)\rho(2)^{-1} \\ &= \rho(1+j) = \pi(1+j). \end{aligned}$$

4. If $\pi(1)$ is a unit, its entire image consists of units, so the restriction to $M \setminus \{0\}$ is not the zero function. If

$$\pi(1) \in \{0, 3, 6\},$$

then $\pi(2) = 0$, and hence $\pi(i) = 0$ for every $i \geq 2$. Thus exactly these three choices give the required zero restriction.

5. By part 3, only homomorphisms ρ with $\rho(2)$ nilpotent can fail to have an extension. In that case $\rho(3)$ is also nilpotent. If $\rho(3)$ were a unit, then

$$\rho(3 + 3) = \rho(6) = \rho(2 + 2 + 2)$$

would be a unit, and hence $\rho(2)$ would also be a unit, a contradiction.

Conversely, choose arbitrary nilpotent elements as $\rho(2)$ and $\rho(3)$. We must then have $\rho(n) = 0$ for every $n \geq 4$, since every such number can be written as $n = 2i + 3j$ with $i + j \geq 2$. Every such choice does indeed give a monoid homomorphism $M \rightarrow R$. It has an extension to $\mathbb{N}$ only when

$$\rho(2) = \rho(3) = 0.$$

The other eight pairs of nilpotent values have no extension.

6. The six homomorphisms with $\rho(2)$ a unit, the eight nonextendible homomorphisms from part 5, and the one homomorphism that is zero on $M \setminus \{0\}$ give

$$6 + 8 + 1 = 15$$

elements of $\text{Mor}_{\text{mon}}(M, R)$.

[Back to Exercise 18.11.](#)

Solution to Exercise 18.15

Consider the inclusion

$$\mathbb{N} \subset \mathbb{Z}.$$

For any field K , the map

$$\varphi : \mathbb{N} \longrightarrow (K, \cdot, 1)$$

that sends 0 to 1 and every positive number to 0 is a monoid homomorphism, and hence a point of $K\text{-Spek}(K[\mathbb{N}])$. This homomorphism cannot be extended to a monoid homomorphism on all of $\mathbb{Z}$, since 1 is invertible in $\mathbb{Z}$ and must therefore map to a unit. Thus the spectrum map induced by the inclusion above is not surjective.

[Back to Exercise 18.15.](#)

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Lecture 19: Quotient Ring Presentations and Integrality

Quotient ring presentations for monomial curves

Let

$$M \subseteq \mathbb{N}$$

be a numerical monoid generated by natural numbers $e_1, \dots, e_n$ with $\gcd(e_1, \dots, e_n) = 1$. The associated surjection

$$\mathbb{N}^n \longrightarrow M \subseteq \mathbb{N}$$

gives a surjection

$$\begin{aligned} K[X_1, \dots, X_n] &\longrightarrow K[M], \\ X_i &\longmapsto T^{e_i}, \end{aligned}$$

and a closed embedding

$$C = K\text{-}\mathrm{Spek}(K[M]) \hookrightarrow \mathbb{A}_K^n.$$

Which equations describe the curve C ?

Theorem: binomial equations defining the curve

Let $M \subseteq \mathbb{N}$ be a submonoid generated by $e_1, \dots, e_n$ with $\gcd(e_1, \dots, e_n) = 1$, and let

$$\mathbb{N}^n \longrightarrow M$$

be the associated surjective map, with quotient homomorphism

$$\varphi : K[X_1, \dots, X_n] \longrightarrow K[M].$$

Then the kernel ideal is described by

$$\ker \varphi = \left(\prod_{i \in I_1} X_i^{r_i} - \prod_{i \in I_2} X_i^{s_i} \mid \begin{array}{l} I_1, I_2 \subseteq \{1, \dots, n\} \text{ disjoint,} \\ \sum_{i \in I_1} r_i e_i = \sum_{i \in I_2} s_i e_i, \\ r_i, s_i \geq 1 \end{array} \right).$$

Proof The displayed elements belong to the kernel ideal, as follows directly from

$$\begin{aligned} \varphi \left(\prod_{i \in I_1} X_i^{r_i} \right) &= \prod_{i \in I_1} (T^{e_i})^{r_i} \\ &= T^{\sum_{i \in I_1} r_i e_i}. \end{aligned}$$

Edition note: in this line the source switches from the variable T specified in the homomorphism to lowercase t . This edition consistently uses the single variable T ; no mathematical content is changed.

Conversely, let

$$F \in K[X_1, \dots, X_n]$$

be a polynomial with $\varphi(F) = 0$. Write

$$F = \sum_{\nu} a_{\nu} X^{\nu}, \quad \nu = (\nu_1, \dots, \nu_n).$$

Then

$$\begin{aligned} \varphi(F) &= \sum_{\nu} a_{\nu} T^{\sum_{i=1}^n \nu_i e_i} \\ &= \sum_{k \geq 0} \left(\sum_{\nu: \sum_{i=1}^n \nu_i e_i = k} a_{\nu} \right) T^k. \end{aligned}$$

Edition note: the source writes the summation bound $\sum_{k=0}$ without an upper bound. Since F is a polynomial and this line groups terms by nonnegative weight k , this edition writes $\sum_{k \geq 0}$; only the notation is completed.

Since this polynomial is zero, all its coefficients vanish. Hence, for every k , the polynomial

$$F_k = \sum_{\nu: \sum_{i=1}^n \nu_i e_i = k} a_\nu X^\nu$$

also lies in the kernel. We may therefore assume that F contains only monomials X^ν with the same value

$$\sum_{i=1}^n \nu_i e_i = k.$$

Take a monomial X^ν appearing in F with $a_\nu \neq 0$. At least one other monomial, say X^μ , must also appear, since a single monomial does not map to zero. Write

$$F = a_\nu (X^\nu - X^\mu) + (F - a_\nu X^\nu + a_\nu X^\mu).$$

The monomial X^ν no longer appears in the summand on the right, and no new monomial is introduced. In $X^\nu - X^\mu$, factor out the variables occurring on both sides as far as possible. This gives

$$X^\nu - X^\mu = X_1^{b_1} \cdots X_n^{b_n} \left(\prod_{i \in I_1} X_i^{r_i} - \prod_{i \in I_2} X_i^{s_i} \right),$$

with I_1 and I_2 disjoint and

$$\sum_{i \in I_1} e_i r_i = \sum_{i \in I_2} e_i s_i.$$

Thus the first summand in the above expression for F belongs to the ideal generated by the stated binomials. We can continue with the second summand, which contains one fewer monomial. This process terminates and proves the claim.

The equations in the preceding theorem are called *binomial equations*. The simplest binomial equations have the form

$$X_i^{e_j / \gcd(e_i, e_j)} = X_j^{e_i / \gcd(e_i, e_j)}, \quad i \neq j.$$

For a plane monomial curve, this is also the only equation.

Corollary: the equation of a plane monomial curve

Let C be the plane monomial curve given by

$$t \mapsto (t^{e_1}, t^{e_2}) = (x, y),$$

with e_1 and e_2 relatively prime. Then

$$C = V(X^{e_2} - Y^{e_1}).$$

Proof This follows immediately from Theorem 19.1.

For monomial space curves, the defining equations can still be determined relatively easily, since one can always isolate a single variable.

Example: the twisted cubic

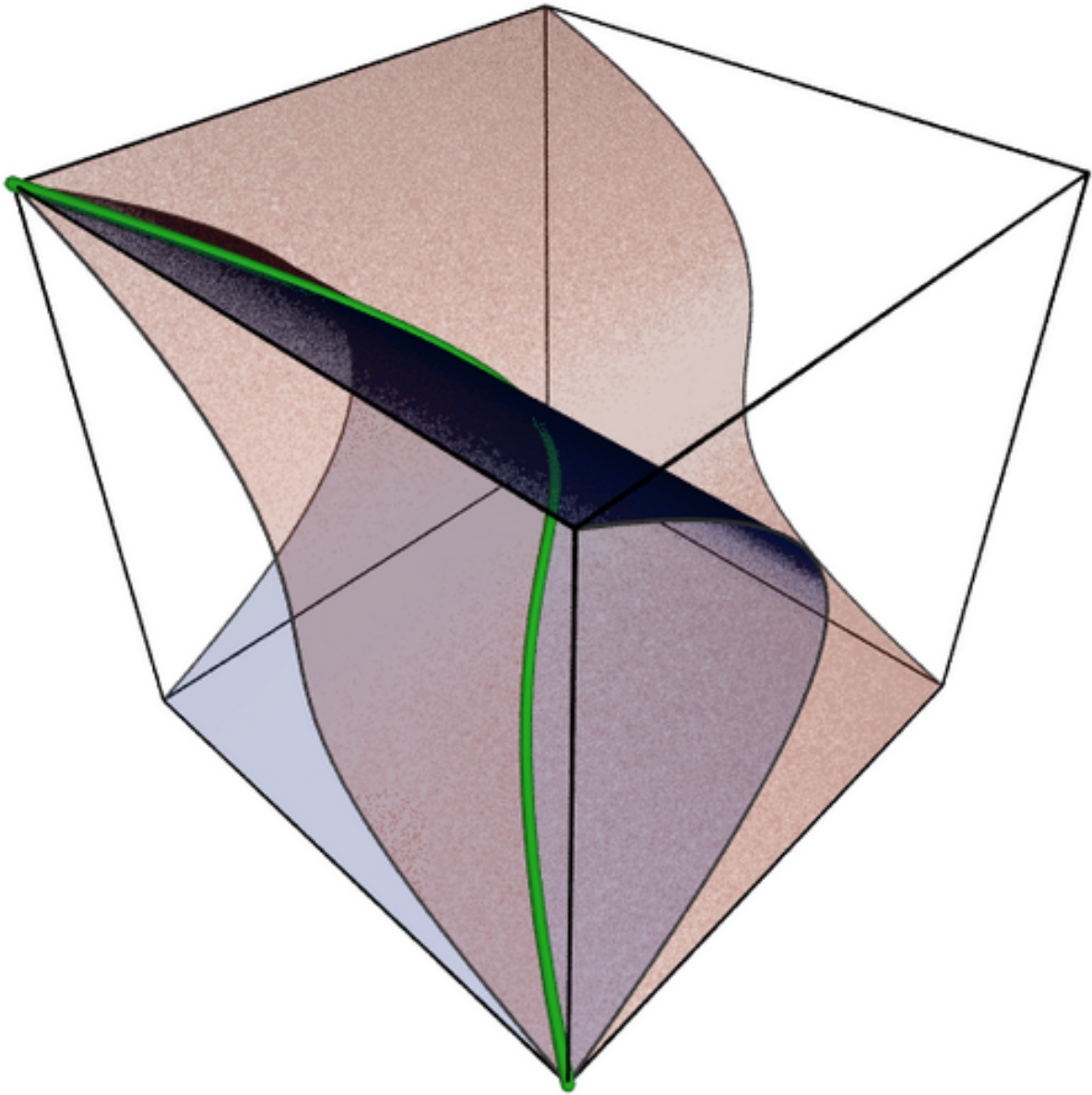


Figure 75: A green space curve twisting along two translucent surfaces inside a three-dimensional box

The twisted cubic. Claudio Rocchini, [CC BY 3.0](#). The lecture source has an inline CC BY-SA 3.0 label, whereas the frozen Commons metadata offers the CC BY 3.0 option used for this component.

Let

$$C \subset \mathbb{A}_K^3$$

be the *twisted cubic*, that is, the image of the monomial map

$$t \mapsto (t, t^2, t^3).$$

This curve is isomorphic to the affine line and is therefore smooth. By Theorem 19.1, its defining ideal is

$$\begin{aligned} \mathfrak{a} &= (Y - X^2, Z - X^3, Y^3 - Z^2, Z - XY) \\ &= (Y - X^2, Z - X^3). \end{aligned}$$

The last two ideal generators are redundant, since they can be expressed in terms of the first two. Thus

$$C = V(Y - X^2, Z - X^3).$$

The images of C under the three different projections are

$$C_1 = V(Z^2 - Y^3), \quad C_2 = V(Z - X^3), \quad C_3 = V(Y - X^2).$$

The curves C_2 and C_3 are isomorphic to the affine line, being graphs of maps, whereas C_1 is the singular Neil's parabola.

Example: the monomial curve with exponents 3, 4, and 5

Let C be the monomial curve given by

$$t \mapsto (t^3, t^4, t^5) = (x, y, z).$$

For each of the three variables, Theorem 19.1 requires us to determine which powers, after substituting the powers of t , can also be expressed as monomials in the other two variables.

First, the equations involving only two variables are

$$Y^3 = X^4, \quad Z^3 = X^5, \quad Z^4 = Y^5.$$

As in the plane case, there can be only one basic relation for each pair of variables.

In a relation involving all three variables, one variable occurs alone on one side. Start with X . Neither X nor X^2 can be expressed using the other variables, but

$$X^3 = T^9 = YZ.$$

No other combination independent of this relation is possible. In general, a double representation

$$X^k = Y^i Z^j = Y^a Z^b$$

gives a relation between powers of Y and powers of Z , after cancelling the smaller powers. Since all relations involving only two variables have already been listed, each power of X gives at most one new relation.

We have already finished with relations in which X stands alone. Indeed, if

$$X^k = Y^i Z^j,$$

then $k \geq 3$. If $i = 0$ or $j = 0$, the relation is already listed. Thus assume $i, j \geq 1$. Using $X^3 = YZ$, we can reduce the exponents in the equation: subtract 3 from the exponent of X and 1 from each exponent of Y and Z .

For Y we immediately obtain

$$Y^2 = ZX,$$

which again reduces all other equations. For Z we obtain

$$Z^2 = X^2 Y$$

and

$$Z^3 = XY^3.$$

There are no smaller monomials in X and Y that can be expressed as a power of Z . Hence every other relation can be reduced to an earlier one using one of these relations.

Altogether, the curve C has the presentation

$$C = V(Y^3 - X^4, Z^3 - X^5, Z^4 - Y^5, X^3 - YZ, Y^2 - XZ, Z^2 - X^2Y, Z^3 - XY^3).$$

Edition note: in this final list the source invokes the unbraced fraction macro `\mathdisplaybruch` immediately before the relation $X^3 - YZ$. This edition displays the relation stated and proved immediately beforehand, $X^3 - YZ$.

Integrality

Definition: equation of integral dependence

Let R and S be commutative rings and

$$R \subseteq S$$

a ring extension. For $x \in S$, an equation of the form

$$x^n + r_{n-1}x^{n-1} + r_{n-2}x^{n-2} + \cdots + r_1x + r_0 = 0,$$

with $r_i \in R$ for $i = 0, \dots, n-1$ is called an *equation of integral dependence* for x .

Definition: integral element

Let $R \subseteq S$ be an extension of commutative rings. An element $x \in S$ is called *integral over R* if it satisfies an equation of integral dependence with coefficients in R .

Definition: integral closure

Let $R \subseteq S$ be an extension of commutative rings. The set of all elements $x \in S$ integral over R is called the *integral closure* of R in S .

Definition: integral extension

Let $R \subseteq S$ be an extension of commutative rings. The ring S is called *integral over R* , and $R \subseteq S$ an *integral ring extension*, if every $x \in S$ is integral over R .

The ring S is integral over R if and only if the integral closure of R in S equals S .

Lemma: characterisation of integral elements

Let $R \subseteq S$ be an extension of commutative rings. For an element $x \in S$, the following statements are equivalent.

1. The element x is integral over R .
2. There is an R -subalgebra T of S containing x that is finitely generated as an R -module.
3. There is a finitely generated R -submodule M of S containing a non-zero-divisor of S and satisfying $xM \subseteq M$.

Proof (1) $\Rightarrow$ (2). Consider the R -subalgebra of S generated by the powers of x ,

$$R[x].$$

This subalgebra consists of all polynomial expressions in x with coefficients in R . An equation of integral dependence

$$x^n + r_{n-1}x^{n-1} + r_{n-2}x^{n-2} + \cdots + r_1x + r_0 = 0$$

gives

$$x^n = -r_{n-1}x^{n-1} - r_{n-2}x^{n-2} - \cdots - r_1x - r_0.$$

Thus x^n can be expressed as a polynomial expression of smaller degree. Multiplying this last equation by x^i allows every power of x with exponent at least n to be replaced by a polynomial expression of smaller degree. Eventually all such powers can be expressed with degree at most $n-1$. Hence

$$R[x] = R + Rx + Rx^2 + \cdots + Rx^{n-2} + Rx^{n-1},$$

and $x^0 = 1, x, x^2, \dots, x^{n-1}$ form a finite generating system for the R -module $T = R[x]$.

(2) $\Rightarrow$ (3). Suppose

$$x \in T \subseteq S,$$

where T is an R -subalgebra finitely generated as an R -module. Then $xT \subseteq T$, and T contains the non-zero-divisor 1.

(3) $\Rightarrow$ (1). Let $M \subseteq S$ be a finitely generated R -submodule satisfying $xM \subseteq M$, and choose generators $y_1, \dots, y_n$ for M . For every i , the element xy_i is an R -linear combination of the y_j , so

$$xy_i = \sum_{j=1}^n r_{ij}y_j, \quad r_{ij} \in R.$$

In matrix form,

$$x \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} r_{1,1} & r_{1,2} & \cdots & r_{1,n} \\ r_{2,1} & r_{2,2} & \cdots & r_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{n,1} & r_{n,2} & \cdots & r_{n,n} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}.$$

Thus

$$0 = \underbrace{\begin{pmatrix} x - r_{1,1} & -r_{1,2} & \cdots & -r_{1,n} \\ -r_{2,1} & x - r_{2,2} & \cdots & -r_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ -r_{n,1} & -r_{n,2} & \cdots & x - r_{n,n} \end{pmatrix}}_A \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}.$$

The entries of the matrix A lie in S . If A^{adj} is the adjugate (classical adjoint) matrix of A , then

$$A^{\text{adj}}Ay = 0,$$

where $y = (y_1, \dots, y_n)^T$. By the adjugate identity,

$$A^{\text{adj}}A = (\det A)I_n,$$

so

$$((\det A)I_n)y = 0.$$

Thus $(\det A)y_j = 0$ for every j , and hence

$$(\det A)z = 0$$

for every $z \in M$. By hypothesis, M contains a non-zero-divisor of S , so $\det A = 0$. But this determinant is a monic polynomial expression in x of degree n . Thus x satisfies an equation of integral dependence.

Corollary: integral closure is a subalgebra

Let $R \subseteq S$ be an extension of commutative rings. The integral closure of R in S is an R -subalgebra of S .

Proof The equations of integral dependence $X - r = 0$, for $r \in R$, show that every element of R is integral over R . Let $x_1, x_2 \in S$ be integral over R . By the characterisation of integrality, there are R -subalgebras

$$T_1, T_2 \subseteq S$$

with $x_1 \in T_1$ and $x_2 \in T_2$, each finitely generated as an R -module. Let $y_1, \dots, y_n$ be an R -generating system for T_1 and $z_1, \dots, z_m$ an R -generating system for T_2 . We may assume $y_1 = z_1 = 1$.

Consider the finitely generated R -module

$$T = T_1 T_2 = \langle y_i z_j \mid i = 1, \dots, n, j = 1, \dots, m \rangle.$$

This module plainly contains $x_1 + x_2, x_1 x_2$, and 1. The R -module T is also an R -algebra. Indeed, for two arbitrary elements,

$$\left(\sum r_{ij} y_i z_j \right) \left(\sum s_{k\ell} y_k z_\ell \right) = \sum r_{ij} s_{k\ell} y_i y_k z_j z_\ell,$$

and $y_i y_k \in T_1$ and $z_j z_\ell \in T_2$, so this linear combination belongs to T .

Hence the sum and product of two integral elements are again integral. Thus the integral closure is a subring of S containing R , that is, an R -subalgebra.

Definition: integrally closed

Let $R \subseteq S$ be an extension of commutative rings. The ring R is called *integrally closed in S* if the integral closure of R in S equals R .

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Worksheet 19

Practice exercises

Exercise 19.1

Determine ideal generators for the monomial curve given by

$$\begin{aligned} \mathbb{A}_K^1 &\longrightarrow \mathbb{A}_K^3, \\ t &\longmapsto (t^2, t^3, t^4). \end{aligned}$$

Exercise 19.2

Determine ideal generators for the monomial curve given by

$$\begin{aligned} \mathbb{A}_K^1 &\longrightarrow \mathbb{A}_K^3, \\ t &\longmapsto (t^4, t^5, t^6). \end{aligned}$$

Exercise 19.3

Let

$$R \subseteq S$$

be an integral ring extension of integral domains, and let $F \subseteq R$ be a multiplicative system. Show that the localised extension

$$R_F \subseteq S_F$$

is also integral.

Exercise 19.4 ★

Let R and S be integral domains and

$$R \subseteq S$$

an integral ring extension. Let $f \in R$ be an element that is a unit in S . Show that f is already a unit in R .

Exercise 19.5

Let $R \subseteq S$ be an integral ring extension and $f \in R$. Show that if f , regarded as an element of S , is a unit, then f is a unit in R .

Exercise 19.6

Give an example of an integral ring extension

$$R \subseteq S$$

having a non-zero-divisor $f \in R$ that becomes a zero divisor in S .

Exercise 19.7

Let

$$P = X^2 - 3X + 7$$

and

$$Q = Y^3 - Y^2 + 4Y - 5.$$

Explain why the ring extension

$$\mathbb{Z} \subseteq \mathbb{Z}[X, Y]/(P, Q)$$

is integral, and find equations of integral dependence for $x + y$ and xy . Lowercase letters denote the residue classes of the variables.

Exercise 19.8

Let $R \subseteq S$ be a ring extension between finite commutative rings R and S . Show that this ring extension is integral.

Exercise 19.9

Let R be a commutative ring and

$$S = R[X_1, \dots, X_n]/\mathfrak{a}$$

a finitely generated R -algebra that is integral over R . Show that S is a finitely generated R -module.

Exercise 19.10

Let

$$\varphi : R \longrightarrow S$$

be an integral homomorphism of commutative rings, and let $R \longrightarrow R'$ be another ring homomorphism. Show that the base-changed homomorphism

$$\begin{aligned} \varphi' : R' &\longrightarrow R' \otimes_R S, \\ f &\longmapsto f \otimes 1 \end{aligned}$$

is also integral.

Exercise 19.11

For every field K of characteristic $\neq 2$, show that the ring homomorphism

$$\begin{aligned} K[X, Y]/(X^2 + Y^2 - 1) &\longrightarrow K[U, V]/(U^2 + V^2 - 1), \\ (X, Y) &\longmapsto (U^2 - V^2, 2UV) \end{aligned}$$

is integral.

Exercise 19.12 ★

Let K be a field and suppose a ring extension finite as a $K[X]$ -module is given in the form

$$K[X] \subseteq K[X][Y]/(Y^n + P_{n-1}(X)Y^{n-1} + \dots + P_2(X)Y^2 + P_1(X)Y + P_0(X)) = R,$$

with $P_i(X) \in K[X]$, where the extension ring R is an integral domain. Choose k such that the degree of P_i is at most $k(n-i)$ for all

$$i = 0, 1, \dots, n-1.$$

Show that YX^{-k} satisfies an equation of integral dependence of degree n over $K[X^{-1}]$, and that the extension algebra

$$K[X^{-1}, YX^{-k}]$$

has the same field of fractions as R .

Exercise 19.13

1. Let R be an integral domain. Show that R is integrally closed in the polynomial ring $R[X]$.
2. Give an example of a commutative ring R that is not integrally closed in its polynomial ring.

Exercises to submit

Exercise 19.14 (4 points)

Let $M \subseteq \mathbb{N}$ be the numerical submonoid generated by 3, 5, 7. Determine a quotient ring presentation of the associated monoid ring.

Exercise 19.15 (3 points)

Let R, S, T be commutative rings and

$$\varphi : R \longrightarrow S, \quad \psi : S \longrightarrow T$$

ring homomorphisms such that S is integral over R and T is integral over S . Show that T is also integral over R .

Source hint: Compare [Exercise 10.26](#).

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Public Solutions to Worksheet 19

At the frozen revision boundary, the source provides public solutions only for Exercises 19.4 and 19.12. No additional solutions have been created for this edition.

Solution to Exercise 19.4

Let $s \in S$ be the inverse of f , so that $fs = 1$. Since S is integral over R , there is an equation of integral dependence for s , say

$$s^n + a_{n-1}s^{n-1} + \cdots + a_1s + a_0 = 0, \quad a_i \in R.$$

Multiplying this equation by f^n gives

$$(fs)^n + a_{n-1}f(fs)^{n-1} + \cdots + a_1f^{n-1}(fs) + a_0f^n = 0,$$

or, since $fs = 1$,

$$1 + a_{n-1}f + \cdots + a_1f^{n-1} + a_0f^n = 0.$$

Factoring out f gives

$$1 + f(a_{n-1} + \cdots + a_1f^{n-2} + a_0f^{n-1}) = 0,$$

and hence

$$f(-a_{n-1} - \cdots - a_1f^{n-2} - a_0f^{n-1}) = 1.$$

The expression in parentheses belongs to R . Thus f also has an inverse in R .

[Back to Exercise 19.4.](#)

Solution to Exercise 19.12

Multiply the given equation of integral dependence by X^{-kn} . In the field of fractions Q of the quotient ring defining R , we obtain

$$Y^n X^{-kn} + \sum_{i=0}^{n-1} P_i Y^i X^{-kn} = 0.$$

Here

$$Y^n X^{-kn} = (YX^{-k})^n$$

and

$$\begin{aligned} P_i Y^i X^{-kn} &= P_i Y^i X^{-ki} X^{-k(n-i)} \\ &= (YX^{-k})^i P_i X^{-k(n-i)}. \end{aligned}$$

The condition on k ensures that

$$P_i X^{-k(n-i)}$$

is a polynomial in X^{-1} . Thus the resulting equation is a monic equation of integral dependence of degree n for YX^{-k} over $K[X^{-1}]$.

Since R is an integral domain, the original defining polynomial is irreducible. After the invertible change of variable over $K(X)$, the resulting monic polynomial is irreducible in $K[X^{-1}][Z]$, where Z is a formal variable subsequently mapped to YX^{-k} .

Edition note: the source calls this equation irreducible “in $K[X^{-1}, YX^{-k}]$ ”. In that quotient algebra the relation is zero; the appropriate polynomial ring in which to state irreducibility is $K[X^{-1}][Z]$. This edition clarifies the ambient polynomial ring and formal variable; the claim is not needed for either of the two requested conclusions.

We have

$$K[X^{-1}][YX^{-k}] \subseteq Q.$$

The field of fractions of the left-hand side contains X as the inverse of X^{-1} , and then contains

$$Y = (YX^{-k})X^k.$$

Its field of fractions therefore contains $K(X, Y) = Q$. The reverse inclusion is clear from the inclusion above, so the two fields of fractions are equal.

[Back to Exercise 19.12.](#)

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Lecture 20: Normal Rings and Normalisation

Normal rings and normalisation

Definition: normal integral domain

An integral domain is called *normal* if it is integrally closed in its field of fractions.

Unique factorisation domains provide important examples of normal rings.

Theorem: unique factorisation domains are normal

Let R be a unique factorisation domain. Then R is normal.

Proof Let

$$K = Q(R)$$

be the field of fractions of R , and let $q \in K$ satisfy an equation of integral dependence

$$q^n + r_{n-1}q^{n-1} + r_{n-2}q^{n-2} + \cdots + r_1q + r_0 = 0, \quad r_i \in R.$$

Write

$$q = \frac{a}{b}, \quad a, b \in R, \quad b \neq 0,$$

in lowest terms, so that a and b have no common prime divisor. We must show that b is a unit in R , since then

$$q = ab^{-1} \in R.$$

Multiplying the equation of integral dependence above by b^n gives, in R ,

$$a^n + (r_{n-1}b)a^{n-1} + (r_{n-2}b^2)a^{n-2} + \cdots + (r_1b^{n-1})a + r_0b^n = 0.$$

If b is not a unit, it has a prime divisor p . This element p divides all the terms

$$(r_{n-i}b^i)a^{n-i}, \quad i \geq 1,$$

and hence also divides the first term a^n . Thus p divides a , contradicting the assumption that a and b have no common prime divisor.

Lemma: localisations of normal integral domains are normal

Let R be a normal integral domain and $S \subseteq R$ a multiplicative system. Then the localisation R_S is also normal.

Proof See Exercise 20.6.

Definition: normalisation of an integral domain

Let R be an integral domain with field of fractions $Q(R)$. The integral closure of R in $Q(R)$ is called the *normalisation* of R .

By Corollary 19.10, the normalisation is a subring of the field of fractions. It is a nontrivial fact that if R is of finite type over a field, then its normalisation is also of finite type.

Normalisation of monoid rings

We shall discuss when monoid rings are normal and how their normalisations can be described. First we need conditions ensuring that a monoid ring over an integral domain is again an integral domain.

Definition: torsion-free monoid

A commutative monoid M is called *torsion-free* if, for $m, n \in M$ and a positive integer $r \in \mathbb{N}_+$, the equality

$$rm = rn$$

always implies

$$m = n.$$

Theorem: torsion-free monoid rings are integral domains

Let R be an integral domain and M a torsion-free commutative monoid satisfying the cancellation law. Then the monoid ring $R[M]$ is an integral domain.

Proof First,

$$M \subseteq \Gamma(M),$$

where $\Gamma(M)$ is the group of differences of M . Hence

$$R[M] \subseteq R[\Gamma(M)]$$

is a subring, so it suffices to prove the statement for $R[\Gamma(M)]$. Since M is torsion-free, Exercise 20.10 shows that $\Gamma(M)$ is also torsion-free. Thus we may assume that M itself is a torsion-free commutative group.

Suppose

$$\left(\sum_{m \in M} a_m X^m \right) \left(\sum_{m \in M} b_m X^m \right) = 0.$$

All but finitely many coefficients in these two sums are zero. Hence the entire calculation takes place in a finitely generated subgroup U of the torsion-free group M . By the structure theorem for finitely generated torsion-free commutative groups,

$$U \cong \mathbb{Z}^n.$$

Thus we may even assume that $M = \mathbb{Z}^n$. In this case $R[M]$ is a localisation of a polynomial ring over an integral domain, and is therefore an integral domain.

Without the cancellation law, a monoid ring over an integral domain can have zero divisors.

Example: zero divisors without cancellation

Let M be a monoid containing two distinct elements m and n with

$$m + n = n + n.$$

Without cancellation, this equation does not imply $m = n$. In the monoid ring over any integral domain R we have

$$X^m - X^n \neq 0 \quad \text{and} \quad X^n \neq 0,$$

but

$$(X^m - X^n)X^n = X^{m+n} - X^{n+n} = X^{2n} - X^{2n} = 0.$$

Definition: normalisation of a monoid

Let M be a torsion-free commutative monoid satisfying the cancellation law, with group of differences $\Gamma(M)$. The submonoid

$$\widetilde{M} = \{m \in \Gamma(M) \mid \text{there is } r \in \mathbb{N}_+ \text{ with } rm \in M\}$$

is called the *normalisation* of M .

Theorem: normalisation of a monoid ring

Let M be a torsion-free commutative monoid satisfying the cancellation law, with group of differences $\Gamma(M)$ and normalisation

$$M \subseteq \widetilde{M} \subseteq \Gamma(M).$$

Let R be a normal integral domain. Then the normalisation of the monoid ring $R[M]$ is the monoid ring

$$R[\widetilde{M}].$$

In particular, the monoid ring of a normal monoid over a normal ring is itself normal.

Proof First,

$$R[M] \subseteq R[\widetilde{M}] \subseteq R[\Gamma(M)] \subseteq Q(R)[\Gamma(M)] \subseteq Q(R[M]).$$

Take $m \in \widetilde{M}$ with

$$m = n - k, \quad n, k \in M,$$

and with

$$rm = \underbrace{m + \cdots + m}_{r \text{ times}} \in M.$$

Then

$$T^m = \frac{T^n}{T^k}$$

is an element of the field of fractions, while

$$(T^m)^r \in R[M].$$

Thus T^m satisfies a pure equation of integral dependence over $R[M]$ and belongs to the normalisation of $R[M]$. Hence

$$R[\widetilde{M}] \subseteq R[M]^{\text{norm}}.$$

Edition note: in the last two formulas the source changes the base ring from R to K , although the theorem and the entire argument specify R . This edition consistently retains R . The source also prints $rm = M$ in the defining condition; by the definition of $\widetilde{M}$, the required relation, displayed here, is $rm \in M$.

For the reverse inclusion, we can replace M by $\widetilde{M}$ and thus restrict attention to the case in which M is normal. First one proves that, for a torsion-free commutative group G , the group ring $R[G]$ is normal. This follows from the fact that a polynomial ring over a normal domain is again normal. It remains to show that $R[M]$ is integrally closed in $R[\Gamma(M)]$.

An element

$$q \in R[\Gamma(M)]$$

Edition note — brackets in the group ring. The frozen source writes $R(\Gamma(M))$ in this line. In keeping with the specified ambient ring, this edition uses $R[\Gamma(M)]$; the argument is unchanged.

and its equation of integral dependence lie in the monoid ring of a finitely generated subgroup

$$U \subseteq \Gamma(M).$$

We may therefore assume

$$\Gamma(M) = \mathbb{Z}^n.$$

At this point some convex geometry enters, which we shall not develop. In any case, a normal submonoid

$$M \subseteq \mathbb{Z}^n$$

can be expressed as the intersection of $\mathbb{Z}^n$ with a polyhedral cone in $\mathbb{Q}^n$ or $\mathbb{R}^n$. By Gordan's lemma, this cone is in turn a finite intersection of half-spaces H_i .

Edition note: the finite polyhedral-cone argument in this paragraph requires M to be finitely generated, a hypothesis not stated in the source theorem. For the stated generality, take an integral element $q = a/b$ and let $N \subseteq M$ be the finitely generated submonoid containing the supports of a , b , and of the coefficients of an integral equation for q . The affine case gives $q \in R[\tilde{N}]$, while normality of M gives $\tilde{N} \subseteq M$. Thus the theorem's conclusion is unchanged, but the displayed finite intersection is justified only after this finite reduction.

A half-space H is specified by a linear map

$$p : V = \mathbb{R}^n \longrightarrow \mathbb{R}$$

through

$$H = p^{-1}(\mathbb{R}_+).$$

Thus M is a finite intersection

$$M = \bigcap_{i \in I} M_i, \quad M_i = p_i^{-1}(\mathbb{N}),$$

with

$$M_i \cong \mathbb{N} \times \mathbb{Z}^{n-1}.$$

Consequently,

$$R[M] = \bigcap_{i \in I} R[M_i]$$

is normal by Exercise 20.7, since each

$$R[M_i] \cong R[\mathbb{N} \times \mathbb{Z}^{n-1}]$$

is normal.

Example: the Whitney umbrella

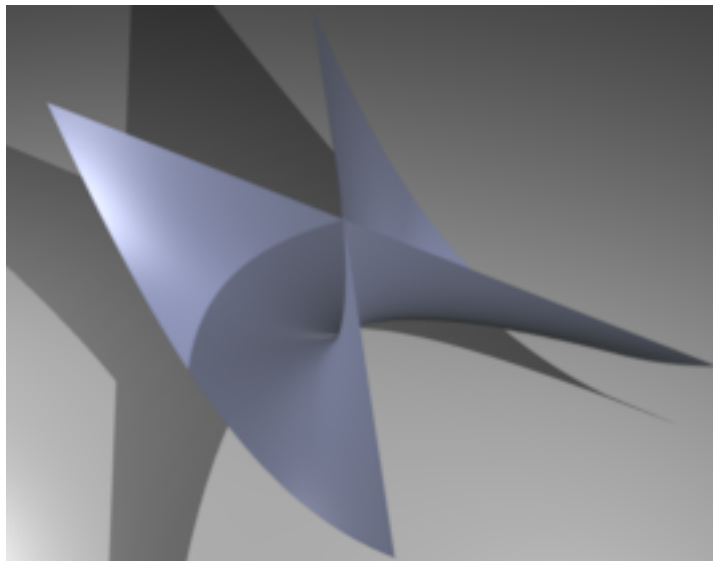


Figure 76: A bluish-grey Whitney umbrella surface intersecting itself in three-dimensional space

The Whitney umbrella. Claudio Rocchini, CC BY 2.5. The source's inline licence label differs from the options available in the Commons metadata; the frozen rights details are recorded in the Unit 20 media credits.

Consider the algebraic surface given by the equation

$$X^2Z = Y^2.$$

We shall view it as the surface associated with a monoid ring. Set

$$M = \langle (1, 0), (1, 1), (0, 2) \rangle \subseteq \mathbb{N}^2.$$

Since

$$(1, 1) - (1, 0) = (0, 1),$$

its group of differences is $\mathbb{Z}^2$. Moreover,

$$2(0, 1) = (0, 2) \in M,$$

so $\mathbb{N}^2$ is the normalisation of M . The three generators give a surjective monoid homomorphism

$$\begin{aligned} \mathbb{N}^3 &\longrightarrow M, \\ e_i &\longmapsto m_i. \end{aligned}$$

Geometrically, the monomial map

$$\mathbb{N}^3 \longrightarrow M \subseteq \mathbb{N}^2$$

corresponds to the map

$$\begin{aligned} \mathbb{A}_K^2 &\longrightarrow K\text{-}\mathrm{Spek}(K[M]) \hookrightarrow \mathbb{A}_K^3, \\ (s, t) &\longmapsto (s, st, t^2). \end{aligned}$$

Under this monoid homomorphism,

$$2e_1 + e_3 \longmapsto (2, 2) \quad \text{and} \quad 2e_2 \longmapsto (2, 2).$$

This gives the equation

$$X^2Z = Y^2,$$

which can of course also be read directly from the parametrisation.

Edition note: the source prints that both elements map to $(1, 1)$. With the displayed generators, both map to $(2, 2)$. This edition transparently corrects these coordinates; the relation $X^2Z = Y^2$ is unchanged.

The defining equation can also be written as

$$Z = \left(\frac{Y}{X} \right)^2.$$

Thus, starting from $K[X, Y]$, we adjoin the square of Y/X .

Example: the standard monomial cone

Consider the submonoid

$$M = \langle (1, 0), (-1, 2), (0, 1) \rangle \subseteq \mathbb{Z}^2.$$

Its associated monoid ring satisfies

$$K[M] \cong K[X, Y, Z]/(Z^2 - XY).$$

We claim that the monoid is normal, that is, equal to its normalisation. The two generators $(1, 0)$ and $(-1, 2)$ each determine a line in $\mathbb{R}^2$, and the monoid consists of all lattice points inside the cone determined by these lines. The lattice points in this cone are given by the two conditions

$$\{(s, t) \in \mathbb{Z}^2 \mid t \geq 0 \text{ and } t \geq -2s\}.$$

A point in this set with $s \geq 0$ plainly belongs to M . Now let (s, t) be a point of the set with $s < 0$. By the second linear condition, we can write

$$(s, t) = -s(-1, 2) + (t + 2s)(0, 1),$$

and this point belongs to M because $t + 2s \geq 0$.

The two lines also immediately describe M as

$$M = M_1 \cap M_2,$$

with

$$M_1 = \{(s, t) \in \mathbb{Z}^2 \mid t \geq 0\} \cong \mathbb{Z} \times \mathbb{N}$$

and

$$M_2 = \{(s, t) \in \mathbb{Z}^2 \mid t \geq -2s\} \cong \mathbb{Z} \times \mathbb{N}.$$

The second identification comes from the $\mathbb{Z}$ -basis $(-1, 2), (0, 1)$. This explicit description shows that the associated monoid ring is normal.

Monomial curves and normalisation

Later we shall see that an algebraic curve is normal if and only if it is nonsingular. For monomial curves, the normalisation is easy to describe.

Theorem: normalisation of a monomial curve

Let

$$M \subseteq \mathbb{N}$$

be a submonoid generated by relatively prime numbers $e_1, \dots, e_n$, and let

$$K[M] \subseteq K[T]$$

be the associated extension of monoid rings. Then $K[T]$ is the normalisation of $K[M]$.

In other words, the monomial map

$$\mathbb{A}_K^1 \longrightarrow K\text{-Spek}(K[M])$$

is a normalisation.

Proof We have

$$K[M] = K[T^{e_1}, \dots, T^{e_n}] \subseteq K[T].$$

Since the exponents are relatively prime, they generate 1. Multiplicatively, this means that there is a monomial in these powers, allowing negative exponents, that equals T . Thus T is a quotient of elements of $K[M]$, and the two fields of fractions are equal.

On the other hand, T satisfies an equation of integral dependence over $K[M]$, for example

$$X^{e_1} - T^{e_1} = 0.$$

Here X is the polynomial variable, whereas T^{e_1} is a coefficient in $K[M]$. Since $K[T]$ is normal—indeed, it is a unique factorisation domain because it is a principal ideal domain—it is the normalisation of $K[M]$.

Edition note: the source uses T both for the polynomial variable and for the element being tested, adding the qualification “read correctly”. This edition distinguishes them by naming the polynomial variable X ; the mathematical content is unchanged.

Monomial curves thus provide many examples in which normalisation is a bijection at the level of K -spectra. The map is also a homeomorphism for the Zariski topology, which is very simple in the curve case. Nevertheless, it would be wrong to regard the two curves as identical. If $e_i \neq 1$ for every i , normalisation is not a bijection at the ring level. In algebraic geometry, we must not look only at the set-theoretic or topological shape of the zero locus; we must not forget the underlying rings and equations. The difference is also visible in the embedded situation, where Neil’s parabola has a cusp.

Normalisation gives a new interpretation of the degree of singularity of a monomial curve.

Lemma: degree of singularity as a dimension of the normalisation quotient

Let $M \subseteq \mathbb{N}$ be a numerical monoid defined by relatively prime generators. Let

$$R = K[M]$$

be its associated monoid ring and

$$R^{\text{norm}} = K[T]$$

its normalisation. Then

$$\delta(M) = \dim_K(R^{\text{norm}}/R).$$

Proof The normalisation has the K -basis

$$\{T^m \mid m \in \mathbb{N}\},$$

whereas the monoid ring $K[M]$ has the K -basis

$$\{T^m \mid m \in M\}.$$

Thus the quotient vector space

$$K[T]/K[M]$$

has the K -basis

$$\{T^m \mid m \in \mathbb{N} \setminus M\}.$$

The dimension of the quotient vector space is the number of elements in a basis, namely the number of gaps in M . This is precisely the degree of singularity of M .

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Worksheet 20

Practice exercises

Prove each of the following four exercises in two ways: directly and using the normality of $\mathbb{Z}$.

Exercise 20.1 ★

Show that $\sqrt{2}$ is irrational.

Exercise 20.2

Let p be a prime number. Using unique prime factorisation of natural numbers, show that the real number $\sqrt{p}$ is irrational.

Exercise 20.3 ★

Use unique prime factorisation in $\mathbb{Z}$ to prove that $9^{1/3}$ is irrational.

Exercise 20.4 ★

Let

$$n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$$

be the canonical prime factorisation of the natural number n . Let k be a positive natural number, and suppose that not all exponents α_i are multiples of k . Show that the real number

$$n^{1/k}$$

is irrational.

Exercise 20.5 ★

Let R be a normal integral domain and $f \in R$ with $f \neq 0$. Show that the localisation R_f is also normal.

Exercise 20.6

Let R be a normal integral domain and $S \subseteq R$ a multiplicative system. Show that the localisation R_S is also normal.

Exercise 20.7

Let K be a field and $(R_i \subseteq K)_{i \in I}$ a family of normal subrings. Show that the intersection

$$\bigcap_{i \in I} R_i$$

is also normal.

Exercise 20.8

Let R be an integral domain. Show that R is normal if and only if it equals its normalisation.

Exercise 20.9

Let R be an integral domain. Suppose that its normalisation equals its field of fractions $Q(R)$. Show that R itself is already a field.

Exercise 20.10

Let M be a torsion-free monoid. Show that its group of differences $\Gamma(M)$ is also torsion-free.

Exercise 20.11

Let M be a commutative group. Show that torsion-freeness of M is equivalent to the following property: if $m \in M$ and

$$rm = 0$$

for some positive natural number $r \in \mathbb{N}$, then $m = 0$. Also show that this equivalence need not hold for a monoid.

Exercise 20.12 ★

Consider the zero locus

$$C = V(Y^2 - X^4) \subseteq \mathbb{A}_{\mathbb{C}}^2.$$

1. Is C irreducible?
2. Can the coordinate ring $\mathbb{C}[X, Y]/(Y^2 - X^4)$ be obtained as a monoid ring?
3. Can the coordinate ring $\mathbb{C}[X, Y]/(Y^2 - X^4)$ be obtained as the monoid ring of a submonoid

$$M \subseteq \mathbb{N} \times \mathbb{Z}/(2)?$$

Exercise 20.13 ★

Let $n \in \mathbb{N}_+$ and let

$$M \subseteq \mathbb{N} \times \mathbb{Z}/(n)$$

be a submonoid. Show that $m \in M$ is a unit if and only if it is a unit when regarded as an element of $\mathbb{N} \times \mathbb{Z}/(n)$.

Exercise 20.14 ★

Let $n \in \mathbb{N}_+$. Show that the $\mathbb{C}$ -spectrum of the commutative monoid

$$M = \mathbb{N} \times \mathbb{Z}/(n)$$

consists of n irreducible components, all isomorphic to the affine line $\mathbb{A}_{\mathbb{C}}^1$.

Exercise 20.15

Let R be an integral domain with normalisation R^{norm} . Show that

$$\mathfrak{f} = \left\{ g \in R \mid gR^{\text{norm}} \subseteq R \right\}$$

is an ideal of R .

Exercise 20.16

Let $M \subseteq \mathbb{N}$ be a numerical monoid generated by relatively prime natural numbers. Show that the conductor ideal of the associated monoid ring $K[M]$ satisfies

$$\mathfrak{f} = (M_{\geq f}),$$

where f denotes the conductor of the monoid.

Exercise 20.17 ★

Let $M \subseteq \mathbb{N}$ be a numerical monoid. Show that its degree of singularity equals each of the following three numbers.

1. The maximum length of a chain of monoids

$$M = M_0 \subset M_1 \subset M_2 \subset \cdots \subset M_n = \mathbb{N}.$$

2. The maximum length of a chain of K -algebras

$$K[M] = R_0 \subset R_1 \subset R_2 \subset \cdots \subset R_n = K[T].$$

3. The maximum length of a chain—a flag—of K -vector subspaces

$$K[M] = V_0 \subset V_1 \subset V_2 \subset \cdots \subset V_n = K[T].$$

Exercise 20.18

Consider [Example 20.11](#). What are the values of the three generators under the monoid homomorphisms φ_1, φ_2 to $\mathbb{Z}$ mentioned there, which describe the monoid? Determine the cokernel of the group homomorphism

$$\begin{aligned} \Gamma(M) &\longrightarrow \mathbb{Z}^2, \\ m &\longmapsto (\varphi_1(m), \varphi_2(m)). \end{aligned}$$

This cokernel is called the **divisor class group** of the monoid ring.

Edition note on the source: The cited example states the inequalities $t \geq 0$ and $t \geq -2s$, but does not label or explicitly define φ_1, φ_2 . Following the order of these inequalities, the intended functionals can be inferred as $\varphi_1(s, t) = t$ and $\varphi_2(s, t) = t + 2s$. These formulas are an explicitly disclosed editorial inference, not an unstated assertion attributed to the source.

Exercises to submit**Exercise 20.19 (3 points)**

Let R be a normal integral domain and

$$R \subseteq S$$

an integral ring extension. Let $f \in R$. Show that the principal ideal generated by f satisfies

$$R \cap (f)S = (f)R.$$

Exercise 20.20 (6 points)

Let R be a normal integral domain. Show that the polynomial ring $R[X]$ is also normal.

Exercise 20.21 (5 points)

Let R be a normal integral domain and $a \in R$. Suppose that a has no square root in R . Show that the polynomial $X^2 - a$ is a prime element of $R[X]$.

Source hint: Use the field of fractions $Q(R)$.

Source warning: In this setting, being prime need not be equivalent to being irreducible.

Exercise 20.22 (4 points)

Let R be an integral domain. Show that the following three properties are equivalent.

1. R is normal.
2. For every prime ideal $\mathfrak{p}$, the localisation $R_{\mathfrak{p}}$ is normal.
3. For every maximal ideal $\mathfrak{m}$, the localisation $R_{\mathfrak{m}}$ is normal.

Normality is therefore called a **local property**.

Exercise 20.23 (2 points)

Let

$$M \subseteq \Gamma(M) \cong \mathbb{Z}^n$$

be a monoid, and consider the set

$$M^* = \left\{ \varphi : \Gamma(M) \longrightarrow \mathbb{Z} \mid \varphi(M) \subseteq \mathbb{N} \right\}.$$

Show that M^* is a normal submonoid of $\text{Hom}(\mathbb{Z}^n, \mathbb{Z})$.

This monoid is called the **dual monoid** of M .

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Public Solutions to Worksheet 20

At the frozen revision boundary, the source provides public solutions only for Exercises 20.1, 20.3, 20.4, 20.5, 20.12, 20.13, 20.14, and 20.17. The solutions to Exercises 20.1 and 20.4 are wrapper pages transcluding separate proof bodies; this edition includes those frozen proof bodies in full. No additional solutions have been created for this edition.

Solution to Exercise 20.1

We assume that there is a rational number whose square is 2 and derive a contradiction. Thus suppose that

$$x \in \mathbb{Q}$$

and

$$x^2 = 2.$$

Every rational number can be written as a fraction with integer numerator and denominator. We may therefore write

$$x = \frac{a}{b}.$$

We may also assume that the fraction is in lowest terms, so that a and b have no nontrivial common divisor. This choice merely simplifies the representation and is not the assumption we intend to refute. In fact, we only need at least one of a and b to be odd: if both are even, we can divide both by 2 and continue as necessary.

The equation $x^2 = 2$ means

$$x^2 = \left(\frac{a}{b}\right)^2 = \frac{a^2}{b^2} = 2.$$

Multiplying by b^2 gives an equation in $\mathbb{Z}$ (indeed, in $\mathbb{N}$),

$$2b^2 = a^2.$$

Thus a^2 is even, being a multiple of 2. Consequently a itself is even, since the square of an odd number is odd. We can therefore write

$$a = 2c$$

for some $c \in \mathbb{Z}$. Substituting into the equation above gives

$$2b^2 = (2c)^2 = 2^2c^2.$$

Dividing by 2, we obtain

$$b^2 = 2c^2.$$

For the same reason, b^2 , and hence b , is also even. This contradicts our choice that a and b are not both even.

[Back to Exercise 20.1.](#)

Solution to Exercise 20.3

Suppose that there is a representation

$$9^{1/3} = \frac{a}{b}$$

with $a, b \in \mathbb{N}_+$. If a and b have a common divisor at least 2, we can cancel it. Thus we may assume that a and b are relatively prime. Cubing the original equation gives

$$9 = \frac{a^3}{b^3},$$

or

$$3^2 b^3 = a^3.$$

This number has a unique prime factorisation. Since 3 occurs in it, $3 \mid a^3$; as 3 is prime, also $3 \mid a$. Hence the exponent of 3 in the prime factorisation on the right is at least 3. On the other hand, because a and b are relatively prime, b is not divisible by 3, so the exponent of 3 on the left is exactly 2. This is a contradiction.

[Back to Exercise 20.3.](#)

Solution to Exercise 20.4

The number

$$n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$$

cannot have a k th root in $\mathbb{Z}$, since in a k th power all prime-factor exponents are multiples of k , whereas by hypothesis this does not hold for all the α_i .

Since $\mathbb{Z}$ is a unique factorisation domain, it is normal. Therefore there cannot be an

$$x \in Q(\mathbb{Z}) = \mathbb{Q}$$

with

$$x^k = n.$$

Thus the real number $n^{1/k}$ is irrational.

[Back to Exercise 20.4.](#)

Solution to Exercise 20.5

Let

$$q \in Q(R) = Q(R_f)$$

be an element of the field of fractions satisfying an equation of integral dependence over R_f . Thus there is an equation

$$q^n + g_{n-1}q^{n-1} + \cdots + g_1q + g_0 = 0, \quad g_i \in R_f.$$

Each g_i can be written as a fraction whose denominator is a power of f . We can choose one fixed power f^k as a common denominator. Increasing k if necessary, we may also assume that k is a multiple of n .

Multiplying the equation by f^{kn} gives

$$(f^k q)^n + g_{n-1} f^k (f^k q)^{n-1} + \dots + g_1 f^{k(n-1)} (f^k q) + f^{kn} g_0 = 0.$$

All coefficients in this equation lie in R . It is therefore an equation of integral dependence for $f^k q$ over R . Since R is normal, we obtain $f^k q \in R$, and hence

$$q = \frac{b}{f^k} \in R_f$$

for some $b \in R$. Thus the localisation R_f is also normal.

[Back to Exercise 20.5.](#)

Solution to Exercise 20.12

1. We have

$$Y^2 - X^4 = (Y - X^2)(Y + X^2),$$

so the curve is reducible.

2. Consider the monoid M with two generators e, f and the single relation

$$2f = 4e.$$

Then

$$\mathbb{C}[M] \cong \mathbb{C}[X, Y]/(Y^2 - X^4).$$

3. Take

$$e = (1, 1) \in \mathbb{N} \times \mathbb{Z}/(2)$$

and

$$f = (2, 1) \in \mathbb{N} \times \mathbb{Z}/(2).$$

We have $2f = 4e$. All other relations are multiples of this one. Indeed, if

$$af = be, \quad a, b \in \mathbb{N},$$

then comparing the first coordinates gives $b = 2a$, while comparing the second coordinates requires a to be even.

[Back to Exercise 20.12.](#)

Solution to Exercise 20.13

Let

$$m = (r, s) \in M.$$

If m is a unit in M , it is certainly also a unit in $\mathbb{N} \times \mathbb{Z}/(n)$, since its inverse in M belongs to that larger monoid.

Conversely, suppose that m is a unit in $\mathbb{N} \times \mathbb{Z}/(n)$. We must first have $r = 0$. For

$$m = (0, s), \quad 0 \leq s < n,$$

its inverse in $\mathbb{N} \times \mathbb{Z}/(n)$ is

$$(0, n - s) = (0, -s).$$

But

$$(n - 1)(0, s) = (0, (n - 1)s) = (0, -s).$$

Since $(0, s) \in M$ and M is closed under addition, this inverse also belongs to M . Hence m is a unit in M .

[Back to Exercise 20.13.](#)

Solution to Exercise 20.14

We have

$$\begin{aligned} \mathbb{C}[M] &\cong \mathbb{C}[S, T]/(S^n - 1) \\ &\cong \mathbb{C}[T][S]/(S^n - 1) \\ &\cong \mathbb{C}[T][S]/\left(\prod_{\zeta^n=1} (S - \zeta)\right) \\ &\cong \left(\mathbb{C}[S]/\left(\prod_{\zeta^n=1} (S - \zeta)\right)\right)[T], \end{aligned}$$

where ζ runs through all n complex roots of unity. Furthermore,

$$\mathbb{C}[S]/\left(\prod_{\zeta^n=1} (S - \zeta)\right) \cong \mathbb{C}^n,$$

with the isomorphism given by

$$S \mapsto (\zeta_0, \zeta_1, \dots, \zeta_{n-1}).$$

Consequently,

$$\left(\mathbb{C}[S]/\left(\prod_{\zeta^n=1} (S - \zeta)\right)\right)[T] \cong \mathbb{C}^n[T] \cong (\mathbb{C}[T])^n$$

is a product ring of n polynomial rings $\mathbb{C}[T]$. Therefore its $\mathbb{C}$ -spectrum is the disjoint union of n copies of

$$\mathrm{Spec}_{\mathbb{C}}(\mathbb{C}[T]) \cong \mathbb{A}_{\mathbb{C}}^1.$$

Each of these affine lines is irreducible.

Edition note: in the three products, the source uses the dummy index η in $\prod_{\eta} (S - \zeta)$, whereas the factors and explanatory prose use ζ . This edition transparently makes the indexing consistent as $\prod_{\zeta^n=1} (S - \zeta)$.

[Back to Exercise 20.14.](#)

Solution to Exercise 20.17

The degree of singularity δ is the number of gaps of M in $\mathbb{N}$. This equals

$$\dim_K(R^{\text{norm}}/R) = \dim_K(K[T]/K[M]).$$

1. In a chain of monoids

$$M = M_0 \subsetneq M_1 \subsetneq M_2 \subsetneq \cdots \subsetneq M_n = \mathbb{N},$$

at least one element must be added at each step. Hence $n \leq \delta$. Conversely, define M_{i+1} successively by adjoining to M_i the largest element not yet in M_i . The result is still a monoid and has exactly one more element than M_i . This procedure produces a chain of length δ , as required.

2. This chain of length δ gives a chain of K -algebras

$$K[M] = K[M_0] \subsetneq K[M_1] \subsetneq K[M_2] \subsetneq \cdots \subsetneq K[M_\delta] = K[\mathbb{N}].$$

All these inclusions are strict: if $m \in M_{i+1} \setminus M_i$, then

$$T^m \in K[M_{i+1}] \setminus K[M_i].$$

The next part gives the general reason that no longer chain exists.

3. The algebra chain in part 2 is, in particular, a chain of vector subspaces over K . Since

$$\dim_K(K[\mathbb{N}]/K[M]) = \delta,$$

there can be no longer chain of vector subspaces: these chains correspond to chains in the quotient vector space $K[\mathbb{N}]/K[M]$, and in a vector space of dimension δ , the maximum length of a chain of strict inclusions is δ .

Edition note: in step 2 the source writes $T^m \in M_{i+1} \setminus M_i$. Since the M_i consist of exponents, the correctly typed relation is $m \in M_{i+1} \setminus M_i$, which then gives $T^m \in K[M_{i+1}] \setminus K[M_i]$. This edition states that implication explicitly.

[Back to Exercise 20.17.](#)

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Lecture 21: Discrete Valuation Rings and Nakayama's Lemma

Discrete valuation rings

We now continue the local study of algebraic curves. In what follows, we shall obtain various characterisations of when a point on a curve is nonsingular—or smooth. Associated with a point P on a curve is the local ring at P , obtained by localising the affine coordinate ring of the curve at the maximal ideal corresponding to P . If

$$P = (a, b) \in C = V(F) \subseteq \mathbb{A}_K^2,$$

this local ring can be described in two ways:

$$K[X, Y]_{(X-a, Y-b)} / (F) \cong (K[X, Y] / (F))_{\mathfrak{m}},$$

where $\mathfrak{m}$ is the maximal ideal now regarded in the quotient ring. This ring captures the essential algebraic properties of the point. The first important concept is that of a discrete valuation ring.

Definition: discrete valuation ring

A *discrete valuation ring* is a principal ideal domain R having exactly one prime element up to associates.

A generator of the maximal ideal of a discrete valuation ring is also called a *local uniformiser*.

Lemma: first properties of discrete valuation rings

A discrete valuation ring is a local Noetherian principal ideal domain with exactly two prime ideals, namely

$$(0) \subset \mathfrak{m},$$

where $\mathfrak{m}$ is its maximal ideal.

Proof A discrete valuation ring is not a field. In a principal ideal domain that is not a field, every maximal ideal is generated by a prime element, and prime generators of distinct maximal ideals cannot be associates. Since a discrete valuation ring has only one prime element up to associates, it has only one maximal ideal.

Likewise, every nonzero prime ideal in a principal ideal domain is generated by a prime element. Hence its only nonzero prime ideal is $\mathfrak{m}$; together with the zero ideal, this gives exactly two prime ideals.

Example: localisation of $K[X]$ at (X)

Let K be a field, $K[X]$ the polynomial ring, and

$$R = K[X]_{(X)}$$

the localisation at the maximal ideal

$$\mathfrak{m} = (X).$$

Then R is a discrete valuation ring. Its two prime ideals are

$$(0) \subset (X).$$

The ring R is a principal ideal domain because $K[X]$ is a principal ideal domain. Since there is only one maximal ideal, there is only one prime element up to associates, namely X .

Example: localisation of $\mathbb{Z}$ at (p)

Let p be a prime number and

$$R = \mathbb{Z}_{(p)}$$

the localisation at the maximal ideal

$$\mathfrak{m} = (p).$$

Then R is a discrete valuation ring. Its two prime ideals are

$$(0) \subset (p).$$

The ring R is a principal ideal domain because $\mathbb{Z}$ is a principal ideal domain. Since there is only one maximal ideal, there is only one prime element up to associates, namely p .

Definition: order in a discrete valuation ring

Let R be a discrete valuation ring with prime element p . For every nonzero element $f \in R$, there are an $n \in \mathbb{N}$ and a unit $u \in R^\times$ such that

$$f = up^n.$$

The number n is called the *order* of f , written

$$\text{ord}(f) = n.$$

Thus the order is simply the exponent of the unique prime element, up to associates, in the prime factorisation of f .

Lemma: properties of the order

Let R be a discrete valuation ring with maximal ideal

$$\mathfrak{m} = (p).$$

The order map

$$\begin{aligned} R \setminus \{0\} &\longrightarrow \mathbb{N}, \\ f &\longmapsto \text{ord}(f) \end{aligned}$$

has the following properties. For $f, g \in R \setminus \{0\}$,

1. we have

$$\text{ord}(fg) = \text{ord}(f) + \text{ord}(g);$$

2. if $f + g \neq 0$, then

$$\text{ord}(f + g) \geq \min\{\text{ord}(f), \text{ord}(g)\};$$

3. $f \in \mathfrak{m}$ if and only if

$$\text{ord}(f) \geq 1;$$

4. $f \in R^\times$ if and only if

$$\text{ord}(f) = 0.$$

Proof. See Exercise 21.2.

Edition note — source correction. The source defines ord only on $R \setminus \{0\}$ but prints the inequality for sums without excluding $f + g = 0$. This edition adds the condition $f + g \neq 0$, so that every occurrence of ord lies within its domain.

We shall now prove an important characterisation of discrete valuation rings. In particular, it shows that a normal Noetherian local integral domain with exactly two prime ideals is already a discrete valuation ring. First we need a lemma.

Lemma 21.7: nilpotence of the maximal ideal

Let S be a Noetherian local commutative ring. Suppose that its maximal ideal $\mathfrak{m}$ is its only prime ideal. Then there is an $n \in \mathbb{N}$ such that

$$\mathfrak{m}^n = 0.$$

Proof First we claim that every element of S is either a unit or nilpotent. Take a nonunit $f \in S$. Since S is local,

$$f \in \mathfrak{m}.$$

Suppose that f is not nilpotent.

Edition bridge 21.A — the prime ideal lemma. If an element f of a commutative ring is not nilpotent, there is a prime ideal $\mathfrak{p}$ that does not contain f . Indeed, the multiplicative set $T = \{1, f, f^2, \dots\}$ does not contain zero. By Zorn's lemma, choose an ideal $\mathfrak{p}$ maximal among ideals disjoint from T . If $ab \in \mathfrak{p}$ but $a, b \notin \mathfrak{p}$, then both $\mathfrak{p} + (a)$ and $\mathfrak{p} + (b)$ meet T . Multiplying one element from each intersection gives an element of $T \cap \mathfrak{p}$, a contradiction. Thus $\mathfrak{p}$ is prime and, since $\mathfrak{p} \cap T = \emptyset$, we have $f \notin \mathfrak{p}$. The source cites this fact using the numbering of another course; this edition supplies its statement and proof to make the argument self-contained.

This prime ideal $\mathfrak{p}$ differs from $\mathfrak{m}$, since $f \in \mathfrak{m}$ but $f \notin \mathfrak{p}$. This contradicts the assumption that $\mathfrak{m}$ is the only prime ideal. Thus every element of the maximal ideal is nilpotent.

Since S is Noetherian, the ideal $\mathfrak{m}$ has a finite generating system

$$\mathfrak{m} = (f_1, \dots, f_k).$$

If $\mathfrak{m} = 0$, take $n = 1$. Hence we may now assume $k \geq 1$.

Choose $m \in \mathbb{N}$ such that

$$f_i^m = 0 \quad \text{for all } i = 1, \dots, k,$$

and set $n = km$. Every element of $\mathfrak{m}^n$ is a linear combination of products of the form

$$\left(\sum_{i=1}^k a_{i1} f_i \right) \left(\sum_{i=1}^k a_{i2} f_i \right) \cdots \left(\sum_{i=1}^k a_{in} f_i \right).$$

Upon expansion, each term is a monomial

$$f_1^{r_1} \cdots f_k^{r_k} \quad \text{with} \quad \sum_{i=1}^k r_i = n.$$

For at least one index i ,

$$r_i \geq \frac{n}{k} = m.$$

Since $f_i^m = 0$, every such monomial is zero. Hence $\mathfrak{m}^n = 0$.

Theorem: characterisation of discrete valuation rings

Let R be a Noetherian local integral domain having exactly two prime ideals

$$(0) \subset \mathfrak{m}.$$

The following statements are equivalent.

1. R is a discrete valuation ring.
2. R is a principal ideal domain.
3. R is a unique factorisation domain.
4. R is normal.
5. The maximal ideal $\mathfrak{m}$ is principal.

Proof The implication $(1) \Rightarrow (2)$ follows directly from Definition 21.1.

The implication $(2) \Rightarrow (3)$ follows from Theorem 9.3 in Commutative Algebra.

The implication $(3) \Rightarrow (4)$ follows from Theorem 20.2.

To prove $(4) \Rightarrow (5)$, take

$$f \in \mathfrak{m}, \quad f \neq 0.$$

Edition bridge 21.B – application to $R/(f)$. Set $S = R/(f)$. The ring S remains Noetherian and local, with maximal ideal $\widetilde{\mathfrak{m}} = \mathfrak{m}/(f)$. Prime ideals of S correspond to prime ideals of R containing (f) . Since the only prime ideals of R are (0) and $\mathfrak{m}$, and $f \neq 0$, the only prime ideal containing (f) is $\mathfrak{m}$. Thus $\widetilde{\mathfrak{m}}$ is the only prime ideal of S . Lemma 21.7 gives $\widetilde{\mathfrak{m}}^n = 0$ for some n , which, pulled back to R , means exactly that $\mathfrak{m}^n \subseteq (f)$.

Choose n minimal such that

$$\mathfrak{m}^n \subseteq (f) \quad \text{and} \quad \mathfrak{m}^{n-1} \not\subseteq (f).$$

Choose

$$g \in \mathfrak{m}^{n-1} \setminus (f)$$

and consider

$$h := \frac{f}{g} \in Q(R).$$

Here $g \neq 0$. Its inverse,

$$h^{-1} = \frac{g}{f},$$

does not belong to R , since otherwise $g \in (f)$. Because R is normal, h^{-1} is not integral over R either. By the module criterion for integrality in Lemma 19.9, applied in particular to the maximal ideal $\mathfrak{m} \subset R$, we have

$$h^{-1}\mathfrak{m} \not\subseteq \mathfrak{m}.$$

On the other hand, by the choice of g ,

$$h^{-1}\mathfrak{m} = \frac{g}{f}\mathfrak{m} \subseteq \frac{\mathfrak{m}^n}{f} \subseteq R.$$

Thus $h^{-1}\mathfrak{m}$ is an ideal of R not contained in the maximal ideal. Hence

$$h^{-1}\mathfrak{m} = R.$$

This equality gives $h \in \mathfrak{m}$. Moreover, for every $x \in \mathfrak{m}$, we have $h^{-1}x \in R$, so

$$x = h(h^{-1}x) \in (h).$$

Therefore

$$(h) = \mathfrak{m},$$

and the maximal ideal is principal.

Now prove (5) $\Rightarrow$ (1). Suppose

$$\mathfrak{m} = (\pi).$$

The element π is prime and is the only prime element up to associates. Take a nonzero nonunit $f \in R$. Then

$$f \in \mathfrak{m},$$

so $f = \pi g_1$. The element g_1 is either a unit or again belongs to $\mathfrak{m}$. In the latter case,

$$g_1 = \pi g_2 \quad \text{and} \quad f = \pi^2 g_2.$$

We claim that this process terminates, giving

$$f = \pi^k u$$

for some $k \in \mathbb{N}$ and unit u . Otherwise, for arbitrarily large n , we could write

$$f = \pi^n g_n.$$

Applying Lemma 21.7 to $R/(f)$ as in Bridge 21.B, there is an $m \in \mathbb{N}$ such that

$$(\pi^m) = \mathfrak{m}^m \subseteq (f).$$

If $n \geq m + 1$, then for some $a, b \in R$ we obtain

$$\pi^m = af = a\pi^{m+1}b,$$

and, since R is an integral domain, cancelling π^m gives the contradiction

$$1 = ab\pi.$$

Thus every nonzero nonunit is a product of a power of π and a unit. In particular, R is a unique factorisation domain. Since R is Noetherian, every ideal can be written as

$$\mathfrak{a} = (f_1, \dots, f_s).$$

The zero ideal is already principal. If $\mathfrak{a} \neq 0$, remove zero generators and write each remaining generator as

$$f_i = \pi^{n_i} u_i$$

with u_i a unit. If

$$n = \min_i n_i,$$

then

$$\mathfrak{a} = (\pi^n).$$

Thus R is a principal ideal domain with exactly one prime element up to associates; by definition, it is a discrete valuation ring.

Nakayama's lemma

Within the class of Noetherian local integral domains having exactly two prime ideals $(0) \subset \mathfrak{m}$ —and hence not fields—the preceding theorem shows that R is a discrete valuation ring if and only if the maximal ideal $\mathfrak{m}$ is generated by one element. It is therefore natural to ask, more generally, how many generators are needed for the maximal ideal of the local ring at a point on an algebraic curve. This leads to the embedding dimension, which we have already encountered for monomial curves. It is also the dimension of the vector space over $R/\mathfrak{m}$

$$\mathfrak{m}/\mathfrak{m}^2.$$

To explain this connection, we need some preparation, in particular Nakayama's lemma.

Edition note — correction to the source's scope. The source's transition sentence states the equivalence between “discrete valuation ring” and “principal maximal ideal” without repeating the hypotheses. This edition explicitly retains the scope of the theorem: Noetherian local integral domains with exactly two prime ideals.

The following construction is used in Nakayama's lemma. Let V be an R -module, $U \subseteq V$ a submodule, and $I \subseteq R$ an ideal. The notation IU denotes the R -submodule of V generated by all elements

$$fv, \quad f \in I, \quad v \in U.$$

This is also a submodule of U . If U itself is an ideal—that is, an R -submodule of R —the construction agrees with the product of ideals. The quotient module V/IU is naturally not only an R -module but also an (R/I) -module. If I is maximal, this quotient module is even a vector space over the residue field R/I .

Nakayama's lemma

Let $(R, \mathfrak{m})$ be a local ring and V a finitely generated R -module. If

$$\mathfrak{m}V = V,$$

then

$$V = 0.$$

Proof Let $v_1, \dots, v_n$ be a generating system for V . Since $v_i \in \mathfrak{m}V$, for each v_i there is a representation

$$v_i = a_{i1}v_1 + \dots + a_{in}v_n, \quad a_{ij} \in \mathfrak{m}.$$

Thus, for every i ,

$$(1 - a_{ii})v_i = a_{i1}v_1 + \dots + a_{i,i-1}v_{i-1} + a_{i,i+1}v_{i+1} + \dots + a_{in}v_n.$$

Since $a_{ii} \in \mathfrak{m}$, the coefficient $1 - a_{ii}$ is a unit in the local ring R . We can solve for v_i , so v_i is redundant in the generating system. Removing generators one by one eventually leaves no generators. Hence V is the zero module.

Edition bridge 21.C – minimal generators corollary. Let $(R, \mathfrak{m})$ be a local ring, $k = R/\mathfrak{m}$ its residue field, and V a finitely generated R -module. Elements $v_1, \dots, v_r$ generate V if and only if their classes generate the k -vector space $V/\mathfrak{m}V$. The forward implication is immediate. Conversely, set $N = Rv_1 + \dots + Rv_r$. The hypothesis on the classes gives $V = N + \mathfrak{m}V$, so $V/N = \mathfrak{m}(V/N)$. Nakayama's lemma gives $V/N = 0$, hence $V = N$. Consequently the minimal number of generators of V is $\dim_k(V/\mathfrak{m}V)$. If $\mathfrak{m}$ is finitely generated, in particular in the Noetherian setting above, then $\mu_R(\mathfrak{m}) = \dim_k(\mathfrak{m}/\mathfrak{m}^2)$, the embedding dimension. This corollary supplies the tool needed to assess the number of ideal generators in Exercises 21.25–21.26, without providing solutions to those exercises.

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Worksheet 21

Practice exercises

Exercise 21.1

Let

$$M \subseteq \mathbb{N}$$

be a numerical monoid generated by relatively prime generators, and let $K[M]$ be its monoid ring over a field K . Write $M_+ = M \setminus \{0\}$ and let

$$R = K[M]_{\mathfrak{m}}$$

be the localisation at the maximal ideal

$$\mathfrak{m} = K[M_+] = \langle T^m \mid m \in M_+ \rangle.$$

Show that R is a discrete valuation ring only in the case $M = \mathbb{N}$.

Source correction note: In the last expression for $\mathfrak{m}$, the source indexes the generators by all $m \in M$, which would include $T^0 = 1$ and give the entire ring. The index is corrected to $m \in M_+$ to describe the intended ideal of positive monomials.

Exercise 21.2

Let R be a discrete valuation ring with maximal ideal $\mathfrak{m} = (p)$. Show that the order function

$$\text{ord} : R \setminus \{0\} \longrightarrow \mathbb{N}$$

has the following properties.

1. For all $f, g \in R \setminus \{0\}$,

$$\text{ord}(fg) = \text{ord}(f) + \text{ord}(g).$$

2. For all $f, g \in R \setminus \{0\}$ with $f + g \neq 0$,

$$\text{ord}(f + g) \geq \min\{\text{ord}(f), \text{ord}(g)\}.$$

3. For every $f \in R \setminus \{0\}$, we have $f \in \mathfrak{m}$ if and only if $\text{ord}(f) \geq 1$.
4. For every $f \in R \setminus \{0\}$, we have $f \in R^\times$ if and only if $\text{ord}(f) = 0$.

Source correction note: The condition $f + g \neq 0$ is added in part 2 because the stated domain of ord does not contain 0.

Exercise 21.3 ★

Let R be a discrete valuation ring with field of fractions Q . Show that there is no proper intermediate ring between R and Q .

Exercise 21.4

Let R be a discrete valuation ring with field of fractions Q . Characterise the finitely generated R -submodules of Q . Into what form can a generating system be put?

Exercise 21.5

Let R be a discrete valuation ring. For every $q \in Q(R) \setminus \{0\}$, define

$$\text{ord}(q) \in \mathbb{Z}$$

so that this definition agrees with the order of elements of R and defines a group homomorphism

$$Q(R) \setminus \{0\} \longrightarrow \mathbb{Z}.$$

What is the kernel of this homomorphism?

Exercise 21.6

Let K be a field with $\text{char}(K) \neq 2$, and let

$$V = V(x^2 + y^2 - 1) \subseteq \mathbb{A}_K^2$$

be the unit circle over K . Let $P = (a, b) \in V$ be a point.

1. Show that the local ring R of V at P is a discrete valuation ring.
2. Deduce that the coordinate ring

$$K[X, Y]/(X^2 + Y^2 - 1)$$

is normal.

3. For this part, assume in addition that -1 is not a square in K . Show that

$$K[X, Y]/(X^2 + Y^2 - 1)$$

is not a unique factorisation domain.

4. Determine the orders of X and $Y - 1$ in the local ring at $(0, 1)$.

Source correction note: The source places no restriction on the characteristic and claims that the ring in part 3 is always nonfactorial. In characteristic 2, the curve equation is a square and the coordinate ring is nonreduced. If $\text{char}(K) \neq 2$, the ring is factorial exactly when -1 is a square in K ; the statement above adds the minimal hypotheses making all four parts true. The source's permission to assume K algebraically closed is also removed because it conflicts with part 3.

Exercise 21.7

Let R be a discrete valuation ring with maximal ideal $\mathfrak{m} = (\pi)$, and let $K = R/(\pi)$ be its residue field. Show that for every $n \in \mathbb{N}$ there is an R -module isomorphism

$$(\pi^n)/(\pi^{n+1}) \longrightarrow K.$$

Exercise 21.8 ★

Let K be a field and

$$\nu : (K^\times, \cdot, 1) \longrightarrow (\mathbb{Z}, +, 0)$$

a surjective group homomorphism satisfying

$$\nu(f + g) \geq \min\{\nu(f), \nu(g)\}$$

for all $f, g \in K^\times$ with $f + g \neq 0$. Show that

$$R = \{f \in K^\times \mid \nu(f) \geq 0\} \cup \{0\}$$

is a discrete valuation ring.

Source correction note: The condition $f + g \neq 0$ is added because ν is defined only on $K^\times$.

Exercise 21.9

Let $f \in \mathbb{C}[X]$ with $f \neq 0$ and $a \in \mathbb{C}$. Show that the following three “orders” of f at a coincide.

1. The order of vanishing of f at a , namely the least order of a derivative satisfying

$$f^{(k)}(a) \neq 0.$$

2. The exponent of the linear factor $X - a$ in the factorisation of f into irreducible polynomials.
3. The order of f in the localisation $\mathbb{C}[X]_{(X-a)}$ of $\mathbb{C}[X]$ at the maximal ideal $(X - a)$.

The preceding exercise can be extended to other base fields using formal differentiation. Let K be a field and $K[X]$ the polynomial ring over K . For a polynomial

$$F = \sum_{i=0}^n a_i X^i \in K[X],$$

the polynomial

$$F' = na_n X^{n-1} + (n-1)a_{n-1} X^{n-2} + \cdots + 3a_3 X^2 + 2a_2 X + a_1$$

is called the **formal derivative** of F .

Exercise 21.10

Determine the formal derivative of

$$2X^7 + X^6 + 2X^5 + X^4 + X^3 + X^2 + 2 \in (\mathbb{Z}/(3))[X].$$

Exercise 21.11

Let K be a field and $K[X]$ the polynomial ring over K . Prove the following rules for formal differentiation $F \mapsto F'$.

1. The derivative of a constant polynomial is 0.
2. Differentiation is K -linear.
3. The product rule holds:

$$(FG)' = FG' + F'G.$$

Exercise 21.12

Let K be a field, $F \in K[X]$, and let $a \in K$ already be a root of F . Show that a is a multiple root of F if and only if

$$F'(a) = 0,$$

where F' denotes the formal derivative of F .

Source correction note: The hypothesis $F(a) = 0$ is added. Without it, $F'(a) = 0$ can also hold when a is not a root of F at all.

Exercise 21.13

Let K be a field of positive characteristic $p > 0$. Determine the set of all polynomials $F \in K[T]$ whose formal derivative satisfies $F' = 0$.

Exercise 21.14

Show that over a field K of characteristic 0, the following identity holds for $i \leq n$:

$$\frac{1}{i!} (X^n)^{(i)} = \binom{n}{i} X^{n-i}.$$

Exercise 21.15

Let K be a field of characteristic 0, $f \in K[X]$ with $f \neq 0$, and $a \in K$. Show that the following three “orders” of f at a coincide.

1. The order of vanishing of f at a , namely the least order of a formal derivative satisfying

$$f^{(k)}(a) \neq 0.$$

2. The exponent of the linear factor $X - a$ in the factorisation of f .
3. The order of f in the localisation $K[X]_{(X-a)}$ of $K[X]$ at the maximal ideal $(X - a)$.

Use the following definition for the next exercises. Let R be a commutative ring, $\mathfrak{a} \subseteq R$ an ideal, and $U \subseteq V$ a submodule of an R -module V . By $\mathfrak{a}U$ we mean the submodule generated by all products

$$fv \quad \text{with } f \in \mathfrak{a} \text{ and } v \in U.$$

Exercise 21.16

Let $\mathfrak{a}, \mathfrak{b} \subseteq R$ be ideals in a commutative ring. Show that the ideal product $\mathfrak{a}\mathfrak{b}$ agrees with the product $\mathfrak{a}\mathfrak{b}$ formed from the ideal $\mathfrak{a}$ and the R -submodule $\mathfrak{b} \subseteq R$ according to the definition above.

Exercise 21.17

Let $\mathfrak{a}, \mathfrak{b} \subseteq R$ be ideals in a commutative ring and $U \subseteq V$ a submodule of an R -module V . Show that

$$(\mathfrak{a} \cdot \mathfrak{b}) \cdot U = \mathfrak{a} \cdot (\mathfrak{b} \cdot U).$$

Exercise 21.18

Let $\mathfrak{a}, \mathfrak{b} \subseteq R$ be ideals in a commutative ring and $U \subseteq V$ a submodule of an R -module V . Show that

$$(\mathfrak{a} + \mathfrak{b}) \cdot U = \mathfrak{a} \cdot U + \mathfrak{b} \cdot U.$$

Exercise 21.19

Let $\mathfrak{a} \subseteq R$ be an ideal in a commutative ring and $U, W \subseteq V$ submodules of an R -module V . Show that

$$\mathfrak{a} \cdot (U + W) = \mathfrak{a} \cdot U + \mathfrak{a} \cdot W.$$

Exercise 21.20

Let $(R, \mathfrak{m})$ be a Noetherian local ring and $n \in \mathbb{N}$. Show that

$$\mathfrak{m}^{n+1} = \mathfrak{m}^n$$

implies

$$\mathfrak{m}^n = 0.$$

Source correction note: The quantification $n \in \mathbb{N}$ is added explicitly; the source uses n without introducing it.

Exercise 21.21

Let $(R, \mathfrak{m})$ be a local integral domain that is not a field, and let Q be its field of fractions. Show that

$$\mathfrak{m}Q = Q.$$

Exercises to submit**Exercise 21.22 (4 points)**

Let K be a field and $K(T)$ the field of rational functions over K . Find a discrete valuation ring

$$R \subseteq K(T)$$

satisfying

$$Q(R) = K(T) \quad \text{and} \quad R \cap K[T] = K.$$

Exercise 21.23 (4 points)

Let K be a field. A **power series in one variable** over K is a formal expression of the form

$$a_0 + a_1T + a_2T^2 + a_3T^3 + \cdots, \quad a_i \in K.$$

Thus there may be infinitely many nonzero coefficients a_i . Define a ring structure on the set of all power series extending the ring structure on the polynomial ring in one variable. Show that this ring is a discrete valuation ring.

Exercise 21.24 (4 points)

Let R be an integral domain with the following property: for any two elements $f, g \in R$, either f divides g or g divides f . Suppose that R is Noetherian but not a field. Show that R is a discrete valuation ring.

Exercise 21.25 (3 points)

Show that every ideal in

$$K[X, Y]_{(X, Y)} / (X^2 - Y^3)$$

can be generated by at most two elements.

Exercise 21.26 (3 points)

Give an example of a plane monomial curve and an ideal in the associated local ring at its singularity that cannot be generated by two elements.

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Public Solutions to Worksheet 21

At the frozen revision boundary, the source provides public solutions only for Exercises 21.3 and 21.8. Both are complete source bodies without wrapper transclusions. No additional solutions have been created for this edition.

Solution to Exercise 21.3

Let the maximal ideal of R be

$$\mathfrak{m} = (\pi).$$

The field of fractions of R is

$$Q(R) = R_\pi,$$

and every nonzero element of this field has the form

$$u\pi^n, \quad u \in R^\times, \quad n \in \mathbb{Z}.$$

Suppose

$$R \subsetneq T \subseteq Q(R).$$

Since the first inclusion is strict, there is an element

$$u\pi^n \in T$$

with $n < 0$. Since u is a unit in $R \subseteq T$, we also have $\pi^n \in T$. Moreover, $-n - 1 \geq 0$, so

$$\pi^{-1} = \pi^{-n-1}\pi^n \in T.$$

Thus $R_\pi \subseteq T$. Together with $T \subseteq Q(R) = R_\pi$, this gives

$$T = Q(R).$$

Hence there is no proper intermediate ring between R and its field of fractions.

[Back to Exercise 21.3.](#)

Solution to Exercise 21.8

First we show that R is a subring of the field K . By definition, $0 \in R$. Since ν is a group homomorphism,

$$\nu(1) = 0,$$

so $1 \in R$. For $f, g \in R$, closure under multiplication is immediate if either element is zero. If f and g are nonzero, then

$$\nu(fg) = \nu(f) + \nu(g) \geq 0,$$

so $fg \in R$. For addition, if either f or g is zero, or if $f + g = 0$, closure is again immediate. In the remaining case, $f, g, f + g$ are all nonzero and the hypothesis gives

$$\nu(f + g) \geq \min\{\nu(f), \nu(g)\} \geq 0.$$

Thus $f + g \in R$. Moreover,

$$\nu(-1) + \nu(-1) = \nu((-1)^2) = \nu(1) = 0,$$

so $\nu(-1) = 0$ and $-1 \in R$. Hence R is also closed under negation and is a commutative ring.

Next we show that R is local. Set

$$\mathfrak{m} := \{f \in K^\times \mid \nu(f) \geq 1\} \cup \{0\} \subseteq R.$$

This set contains 0. For $f, g \in \mathfrak{m}$, the cases involving zero or satisfying $f + g = 0$ are immediate. Otherwise,

$$\nu(f + g) \geq \min\{\nu(f), \nu(g)\} \geq 1,$$

so $f + g \in \mathfrak{m}$. For $f \in \mathfrak{m}$ and $g \in R$, the cases $f = 0$ or $g = 0$ are also immediate. If both are nonzero, then

$$\nu(gf) = \nu(g) + \nu(f) \geq 1,$$

so $gf \in \mathfrak{m}$. Thus $\mathfrak{m}$ is an ideal.

The complement $R \setminus \mathfrak{m}$ consists exactly of the elements $h \in K^\times$ with

$$\nu(h) = 0.$$

For such an element,

$$\nu(h^{-1}) = -\nu(h) = 0,$$

so $h^{-1} \in R$. Thus every element of $R \setminus \mathfrak{m}$ is a unit. Consequently $\mathfrak{m}$ is the unique maximal ideal and R is local.

It remains to show that R is a discrete valuation ring. Since ν is surjective, there is a $p \in K^\times$ with

$$\nu(p) = 1.$$

In particular, $p \in R$. We show that p is prime. For nonzero elements $x, y \in R$, the element y is a multiple of x exactly when

$$\nu(y) \geq \nu(x),$$

since this condition is equivalent to $y/x \in R$. Now suppose that $p \mid xy$ for $x, y \in R$. If $xy = 0$, one factor is zero and is certainly a multiple of p . If $xy \neq 0$, then

$$1 = \nu(p) \leq \nu(xy) = \nu(x) + \nu(y).$$

Since $\nu(x)$ and $\nu(y)$ are nonnegative integers, either $\nu(x) \geq 1$ or $\nu(y) \geq 1$. By the divisibility criterion above, p divides either x or y . Thus p is prime.

By the same argument, every nonzero element $x \in R$ with

$$n = \nu(x)$$

is associated to p^n . Indeed, $\nu(x/p^n) = 0$, so x/p^n is a unit in R . Thus R is a principal ideal domain whose ideals are exactly 0 and

$$(p^n), \quad n \in \mathbb{N}.$$

Hence R is a discrete valuation ring.

Edition note: the source defines ν only on $K^\times$ but states the inequality for $\nu(f + g)$ without the condition $f + g \neq 0$. Since $\nu(0)$ is undefined, this edition uses the inequality only when $f + g \neq 0$ and handles zero sums separately; it does not extend ν to 0.

[Back to Exercise 21.8.](#)

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Lecture 22: Embedding Dimension and Smooth and Singular Points

Embedding dimension

Definition: embedding dimension

Let R be a Noetherian local commutative ring with maximal ideal $\mathfrak{m}$. The minimum number of generators of the ideal $\mathfrak{m}$ is called the *embedding dimension* of R , written

$$\text{embdim}(R).$$

A Noetherian local integral domain of dimension one—that is, whose only prime ideals are the zero ideal and the maximal ideal—is a discrete valuation ring if and only if its embedding dimension is 1, by Theorem 21.8. The embedding dimension is always at least as large as the dimension of a local ring. Rings for which equality holds play a special role and are called *regular rings*. We have already encountered embedding dimension for monomial curves and must show that the definition given there (Definition 18.6) agrees with this new one.

We first prove another characterisation, which follows from Nakayama's lemma.

Lemma: module generators and generators modulo $\mathfrak{m}$

Let $(R, \mathfrak{m}, K)$ be a local ring and V a finitely generated R -module. Then the minimum number of generators

$$\mu(V)$$

equals the dimension of the K -vector space

$$V/\mathfrak{m}V.$$

Proof We prove the slightly more general statement that elements

$$v_1, \dots, v_n \in V$$

generate V as an R -module if and only if their residue classes in $V/\mathfrak{m}V$ generate it over $R/\mathfrak{m}$. One direction is immediate. Thus let $v_1, \dots, v_n \in V$ be elements whose classes modulo $\mathfrak{m}$ form a generating set. Let

$$U \subseteq V$$

be the R -submodule generated by the v_i . The hypothesis means that

$$V = U + \mathfrak{m}V.$$

Consider the quotient module V/U . In it we have

$$(V/U)\mathfrak{m} = V/U.$$

Nakayama's lemma gives

$$V/U = 0,$$

and hence

$$V = U.$$

Corollary: embedding dimension as the dimension of the cotangent space

Let $(R, \mathfrak{m}, K)$ be a Noetherian local ring. Its embedding dimension equals

$$\mu(\mathfrak{m}) = \dim_K (\mathfrak{m}/\mathfrak{m}^2).$$

Proof This follows immediately from Nakayama's lemma applied to the ideal $\mathfrak{m}$ and the finitely generated R -module $\mathfrak{m}$.

The R -module

$$\mathfrak{m}/\mathfrak{m}^2,$$

which occurs above and is a vector space over $R/\mathfrak{m}$, is also called the *cotangent space* of the local ring.

Lemma: the cotangent space before and after localisation

Let R be a Noetherian commutative ring and $\mathfrak{n}$ a maximal ideal. Let

$$S = R_{\mathfrak{n}}$$

be the localisation at $\mathfrak{n}$, with maximal ideal

$$\mathfrak{m} = \mathfrak{n}R_{\mathfrak{n}}.$$

Then

$$\mathfrak{m}/\mathfrak{m}^2 \cong \mathfrak{n}/\mathfrak{n}^2.$$

In particular, the embedding dimension of the localisation equals

$$\dim_{R/\mathfrak{n}} (\mathfrak{n}/\mathfrak{n}^2).$$

Proof By Exercise 15.5,

$$R/\mathfrak{n} \cong R_{\mathfrak{n}}/\mathfrak{m},$$

so the residue fields agree; denote this common residue field by K . The natural R -module homomorphism

$$\mathfrak{n} \longrightarrow \mathfrak{m}$$

induces a homomorphism of K -vector spaces

$$\mathfrak{n}/\mathfrak{n}^2 \longrightarrow \mathfrak{m}/\mathfrak{m}^2.$$

This map is surjective, since R -module generators of $\mathfrak{n}$ map to $R_{\mathfrak{n}}$ -module generators of $\mathfrak{m}$, whose classes modulo $\mathfrak{m}^2$ generate the K -vector space.

To prove injectivity, take

$$f \in \mathfrak{n}$$

that maps to 0 on the right. This means that

$$f \in \mathfrak{m}^2$$

in the localisation $R_{\mathfrak{n}}$. There are therefore elements

$$g_1, \dots, g_n \in \mathfrak{n}, \quad h_1, \dots, h_n \in \mathfrak{n},$$

and elements

$$q_i = \frac{a_i}{s_i} \in R_{\mathfrak{n}}, \quad s_i \notin \mathfrak{n}, \quad a_i \in R,$$

such that

$$f = \frac{a_1}{s_1} g_1 h_1 + \dots + \frac{a_n}{s_n} g_n h_n.$$

Translated back into R , this means that there is an element

$$s \notin \mathfrak{n}$$

with

$$sf = b_1 g_1 h_1 + \dots + b_n g_n h_n$$

for some $b_i \in R$. Since s does not belong to the maximal ideal $\mathfrak{n}$, there are $r \in R$ and $g \in \mathfrak{n}$ such that

$$g + rs = 1.$$

Multiplying the preceding equation by r , we obtain

$$(1 - g)f = r(b_1 g_1 h_1 + \dots + b_n g_n h_n),$$

or, equivalently,

$$f = r(b_1 g_1 h_1 + \dots + b_n g_n h_n) + gf.$$

The right-hand side plainly belongs to $\mathfrak{n}^2$. Thus f represents zero in $\mathfrak{n}/\mathfrak{n}^2$, proving injectivity.

Lemma: equality of numerical and algebraic embedding dimension

Let K be a field and M a numerical monoid generated by coprime natural numbers. Let

$$R = K[M]$$

be the corresponding monoid ring, with maximal ideal

$$\mathfrak{n} = (M_+),$$

and let $R_{\mathfrak{n}}$ be its localisation. Then the numerical embedding dimension of M —or of $K[M]$ —equals the embedding dimension of the local ring $R_{\mathfrak{n}}$.

Proof We have

$$\mathfrak{n} = (M_+) = \bigoplus_{m \in M_+} K T^m$$

and

$$\mathfrak{n}^2 = (2M_+) = \bigoplus_{m \in 2M_+} K T^m.$$

The quotient space is therefore

$$\mathfrak{n}/\mathfrak{n}^2 = \bigoplus_{m \in M_+ \setminus 2M_+} K T^m.$$

Its K -dimension equals the number of elements of $M_+ \setminus 2M_+$. By Corollary 18.13, $M_+ \setminus 2M_+$ is the minimal monoid generating set of M . Thus this K -dimension equals the numerical embedding dimension.

On the other hand, by Lemma 22.4, the K -dimension of $\mathfrak{n}/\mathfrak{n}^2$ equals the embedding dimension of the corresponding local ring $R_{\mathfrak{n}}$.

Smooth and singular points

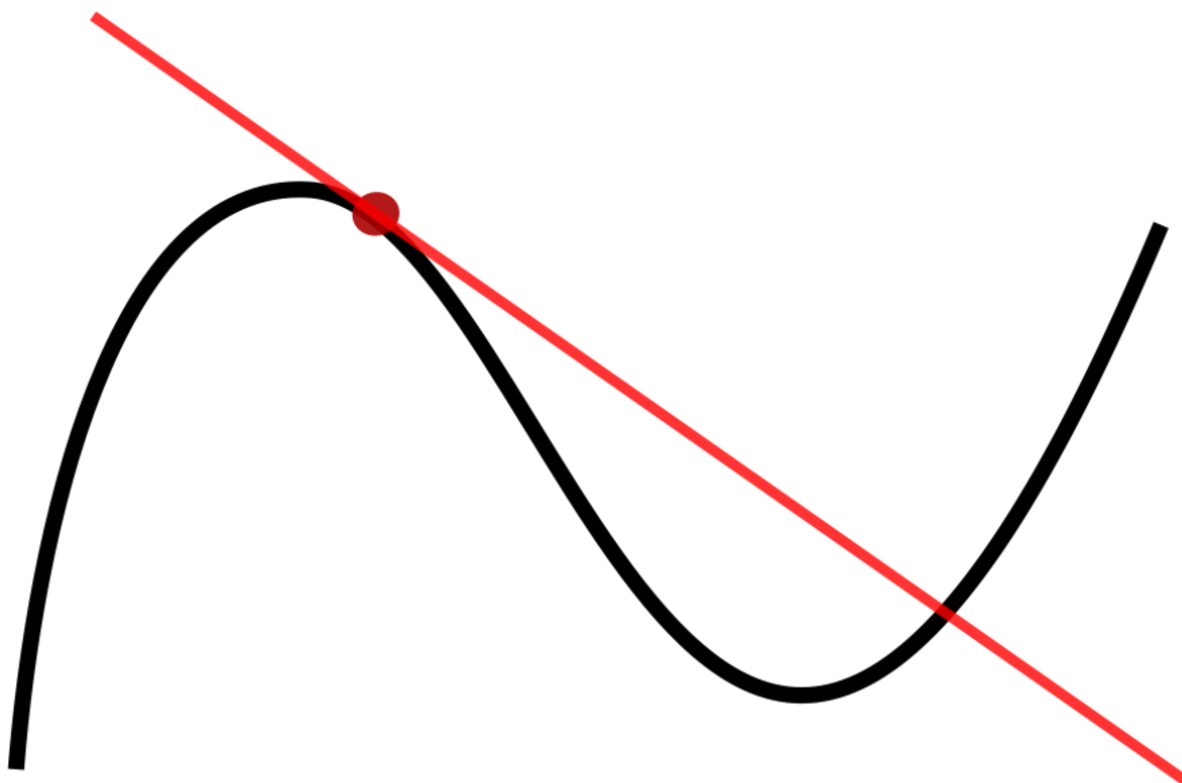


Figure 77: A black curve with a red point and a red tangent line touching the curve at that point

A tangent line to a curve. Work by Jacj on English Wikipedia; later versions uploaded by Oleg Alexandrov. Public domain; local file: `authority/assets/Tangent_to_a_curve.svg`.

Edition note — media credit. The source credit names AxelBoldt and CC BY-SA 3.0. The frozen Commons metadata for this file states public domain and records the Jacj/Oleg Alexandrov history; that Commons metadata governs the component credit in this edition.

Let K be a field and

$$F \in K[X, Y], \quad F \neq 0,$$

a polynomial without repeated factors. Since we are interested only in the corresponding curve, over an algebraically closed field this is no restriction, by Hilbert's Nullstellensatz. For every point

$$(a, b) \in \mathbb{A}_K^2,$$

we may pass to the variables $X - a$ and $Y - b$. This shifts the point to the origin. Thus, when studying a polynomial's behaviour at a point, we may always restrict attention to the origin.

Now let

$$P = (0, 0).$$

Write F in terms of its homogeneous components as

$$F = F_d + F_{d-1} + \cdots + F_1 + F_0,$$

where F_i is homogeneous of degree i . What can we read off from the individual homogeneous components? First, we immediately have

$$P \in V(F) \iff F_0 = 0.$$

Substituting the coordinates of P , namely $(0, 0)$, into F makes all the higher-degree components vanish, leaving only the constant component F_0 . Since our main interest is the behaviour of a curve at one of its points, we shall often restrict attention to the situation

$$F_0 = 0.$$

Which is the first nonzero homogeneous component F_i ? What role does this index i play, and what role do its linear factors play?

Suppose first that

$$F_0 = 0 \quad \text{and} \quad F_1 = aX + bY.$$

This linear form, which may be zero, can also be characterised by partial derivatives:

$$\frac{\partial F}{\partial X}(P) = a \quad \text{and} \quad \frac{\partial F}{\partial Y}(P) = b.$$

Here and below, polynomials are differentiated formally. Thus

$$F_1 = 0 \iff \frac{\partial F}{\partial X}(P) = \frac{\partial F}{\partial Y}(P) = 0.$$

If this does not hold, it is natural to regard the line defined by

$$F_1(X, Y) = 0$$

as the tangent to the curve at P . An initial indication is that in the linear case

$$F = F_1,$$

the line should coincide with its tangent.

Definition: smooth and singular points

Let K be a field and $F \in K[X, Y]$ a nonzero polynomial. Let

$$P \in C = V(F) \subseteq \mathbb{A}_K^2$$

be a point on the corresponding affine plane curve. The point P is called a *smooth point* of C if

$$\frac{\partial F}{\partial X}(P) \neq 0 \quad \text{or} \quad \frac{\partial F}{\partial Y}(P) \neq 0.$$

Otherwise it is called *singular*.

The curve is called *smooth* if it is smooth at every one of its points.

Edition bridge 22.A — scope of the terminology. This lecture works with plane curves presented by a polynomial F without repeated factors, as stipulated above. In this definition, “smooth” means the formal derivative criterion just stated for that presentation. The edition does not extend this to an unconditional identification with scheme-theoretic smoothness over arbitrary fields, particularly imperfect fields.

Definition: multiplicity and tangent lines

Let K be an algebraically closed field and $F \in K[X, Y]$ a nonzero polynomial. Let

$$P \in C = V(F) \subseteq \mathbb{A}_K^2$$

be a point on the corresponding affine plane curve, and suppose that P is the origin after an affine change of coordinates by translation. Let

$$F = F_d + F_{d-1} + \cdots + F_m$$

be the homogeneous decomposition of F , with

$$F_d \neq 0, \quad F_m \neq 0, \quad d \geq m.$$

The number m is called the *multiplicity* of the curve at P . Let

$$F_m = G_1 \cdots G_m$$

be its decomposition into linear factors. Each line

$$V(G_i), \quad i = 1, \dots, m,$$

is called a *tangent line* to C at P . The multiplicity of G_i in F_m is also called the *multiplicity* of that tangent line.

Edition note – source correction. The source calls moving a general point P to the origin a “linear transformation of variables”. Translation $(X, Y) \mapsto (X - a, Y - b)$ is an affine change of coordinates, not a linear transformation unless $P = (0, 0)$. The edition uses the correct term without changing the homogeneous decomposition in the new coordinates.

The point is smooth if and only if its multiplicity is 1. In that case there is exactly one tangent line through the point, and its slope can be computed from the partial derivatives.

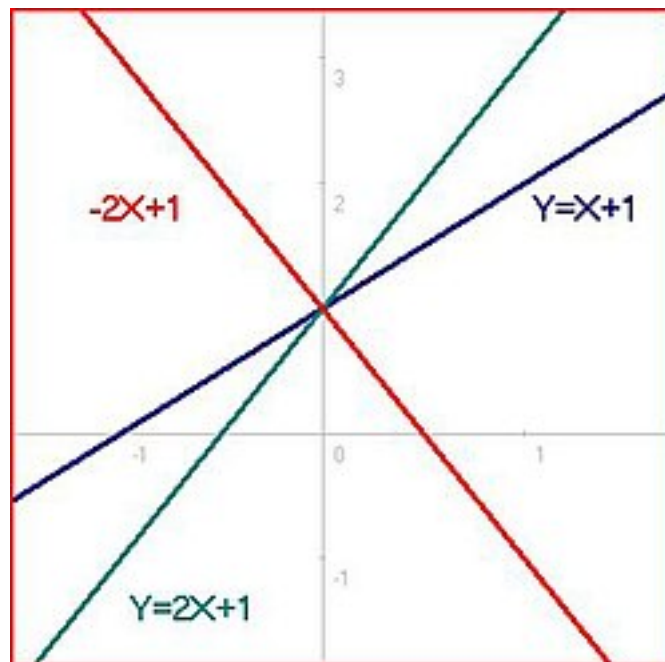


Figure 78: Three straight lines, coloured red, green, and blue, meeting at the point with coordinates zero, one

Lines intersecting at $(0, 1)$. Cronholm144, public domain; local file: authority/assets/250px-3_equations_-5.JPG.

Example: several lines through the origin

Suppose that d distinct lines

$$L_1, \dots, L_d$$

in the affine plane are given, all passing through the origin. Let

$$a_i X + b_i Y = 0, \quad i = 1, \dots, d,$$

be their equations, which are determined only up to a scalar. Their union is described by the product

$$F = (a_1 X + b_1 Y) \cdots (a_d X + b_d Y).$$

In particular,

$$F = F_d$$

is homogeneous of degree d . Here each linear factor defines a tangent line through the origin. The multiplicity is d .

Example: the folium of Descartes

Figure 79: Portrait of René Descartes with shoulder-length hair, a white collar, and black clothing against a dark background

René Descartes (1596–1650). Portrait after Frans Hals; uploaded to Commons by Dedden; public domain; local file: authority/assets/250px-Frans_Hals_-_Portret_van_René_Descartes.jpg.

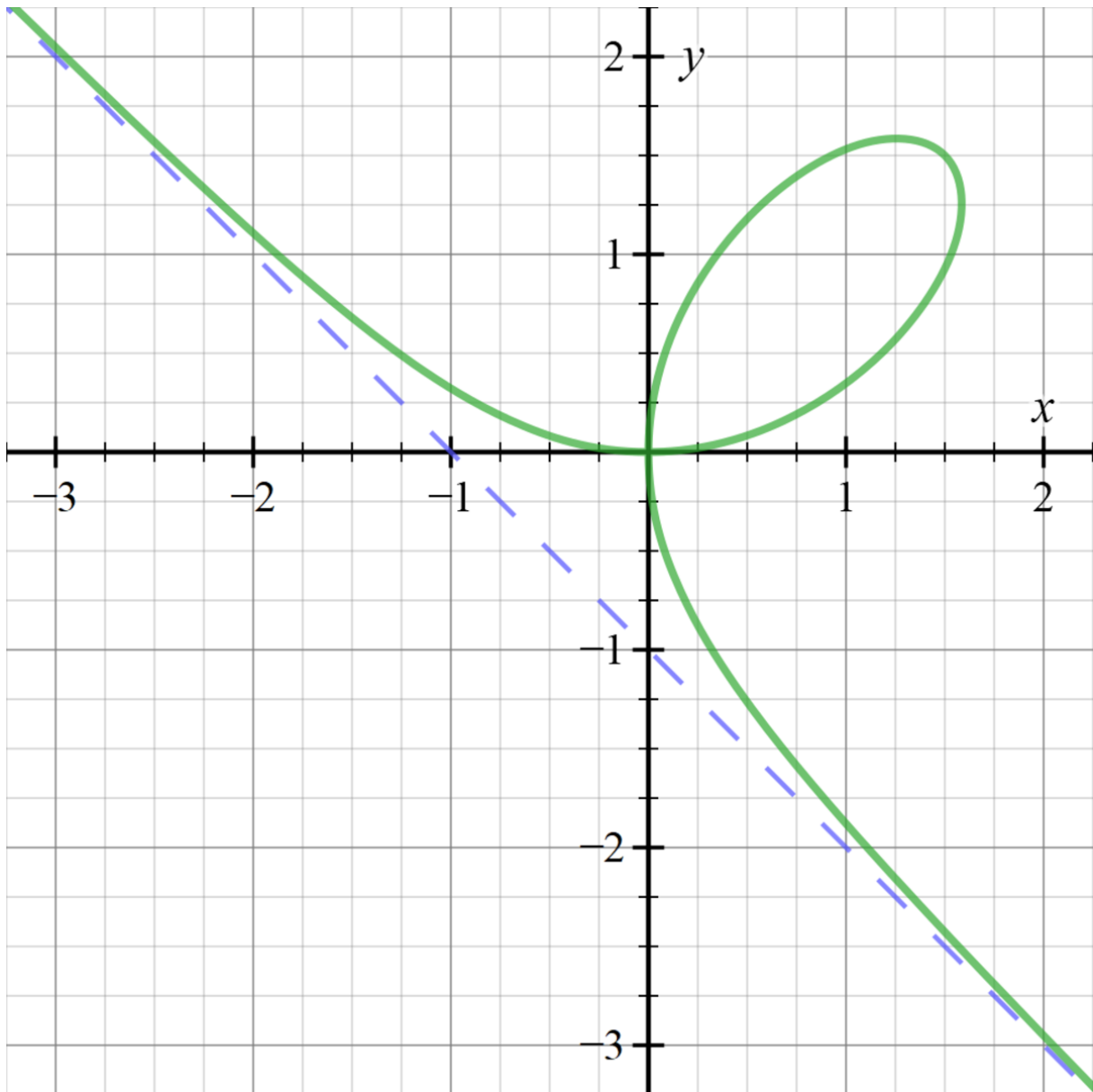


Figure 80: Green graph of the folium of Descartes with a loop in the first quadrant, a node at the origin, and a dashed blue asymptote

The folium of Descartes $x^3 + y^3 - 3xy = 0$, with axes, grid, and asymptote. Georg-Johann, [CC BY-SA 3.0](#); local file: *authority/assets/Kartesisches-Blatt.svg*.

Edition note – characteristic. The assumption $\text{char}(K) \neq 3$ applies throughout this example. The source states it below when discussing the other points, but it is already needed for the multiplicity-two assertion at the origin: in characteristic 3, $F_2 = 0$ and $F = (X + Y)^3$, contrary to the standing squarefree premise.

The folium of Descartes is described by the equation

$$F = X^3 + Y^3 - 3XY = 0.$$

The number 3 is not essential here and can be replaced by any other nonzero number. The homogeneous components of the curve equation are

$$F_3 = X^3 + Y^3$$

and

$$F_2 = -3XY.$$

Thus the origin on the folium of Descartes has multiplicity two and is singular. Both the X -axis and the Y -axis are tangent lines, each with multiplicity one. At every other point the curve is smooth, provided the base field does not have characteristic 3. From

$$\frac{\partial F}{\partial X} = 3X^2 - 3Y = 0 \quad \text{and} \quad \frac{\partial F}{\partial Y} = 3Y^2 - 3X = 0$$

we obtain

$$Y = X^2 \quad \text{and} \quad X = Y^2,$$

and hence also

$$Y = Y^4,$$

and similarly for X . Thus

$$Y = X = 0,$$

or X and Y are both cube roots of unity: either both are 1, or they are the other two cube roots of unity, in either order. However, at these other common zeros of the partial derivatives, F takes the value -1 . These are therefore not points of the curve.

Remark: the tangent-line equation at a smooth point

At a smooth point

$$P \in C = V(F)$$

of a plane algebraic curve, the multiplicity is

$$m = 1.$$

If $P = (0, 0)$, the linear term of the curve equation is

$$F_1 = uX + vY = 0,$$

and

$$\frac{\partial F}{\partial X}(P) = u \quad \text{and} \quad \frac{\partial F}{\partial Y}(P) = v,$$

since the higher-degree homogeneous components of F contribute nothing to the partial derivatives at the origin. Thus this linear equation is the tangent-line equation.

For an arbitrary smooth point

$$P = (a, b) \in C,$$

the tangent-line equation can likewise be read off directly from the partial derivatives of F at P . The tangent line is given by

$$\frac{\partial F}{\partial X}(P)(X - a) + \frac{\partial F}{\partial Y}(P)(Y - b) = 0.$$

Remark: the tangent map and the direction of the tangent line

Let $F \in K[X, Y]$ have corresponding plane algebraic curve C , and let

$$P \in C = V(F)$$

be a smooth point of the curve. The map

$$F : \mathbb{A}_K^2 \longrightarrow \mathbb{A}_K^1$$

and the point P determine a linear *tangent map*, also called the *total differential*, between the corresponding tangent spaces:

$$T_P F = \left(\frac{\partial F}{\partial X}(P), \frac{\partial F}{\partial Y}(P) \right) : T_P \mathbb{A}_K^2 \cong \mathbb{A}_K^2 \longrightarrow T_{F(P)} \mathbb{A}_K^1 = T_0 \mathbb{A}_K^1 \cong \mathbb{A}_K^1,$$

$$(s, t) \longmapsto \frac{\partial F}{\partial X}(P)s + \frac{\partial F}{\partial Y}(P)t.$$

Since P is smooth, this linear map is not the zero map. The direction of the tangent line to C at P is the kernel of this tangent map. When identifying the tangent plane at P with the surrounding affine plane, we must identify P with the origin: the tangent line must pass through that point, whereas the kernel specifies only a linear direction.

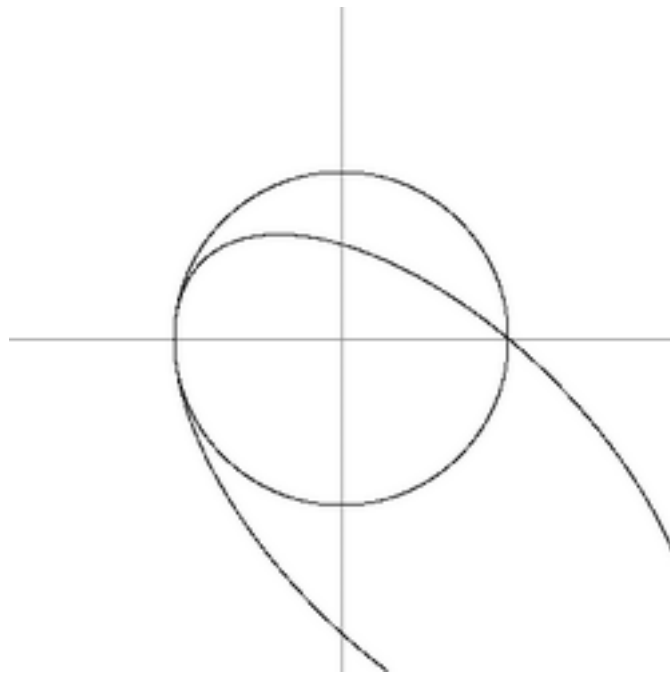


Figure 81: Two black curves—a circle and a curved branch—intersecting at two points, shown against grey coordinate axes

On an algebraic curve, intersection points of irreducible components are never smooth. The lecture source credits Michael Larsen; the Commons metadata records uploader Maksim and the licence [CC BY-SA 3.0](#); local file: `authority/assets/250px-Intersect3.png`.

The following statement shows that an intersection point of two irreducible components can never be smooth.

Lemma: a smooth point lies on only one component

Let

$$C = V(F)$$

be a plane algebraic curve and

$$F = F_1 \cdots F_n$$

its decomposition into distinct prime factors. Let $P \in C$ be a smooth point of the curve. Then P lies on only one component

$$C_i = V(F_i)$$

of the curve.

Proof. See Exercise 22.9.

Corollary: a smooth connected curve is irreducible

Let

$$C \subseteq \mathbb{A}_K^2$$

be a smooth plane algebraic curve, connected in the Zariski topology, over an algebraically closed field K . Then C is irreducible.

Proof By Lemma 22.12, the irreducible components of the curve are disjoint. They are then also its connected components. Thus there is only one irreducible component, so the curve is irreducible.

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Worksheet 22

Practice exercises

Exercise 22.1

Let R be the local ring at the intersection of the three coordinate axes in three-dimensional space. Determine its embedding dimension.

Exercise 22.2

Give an example of a curve

$$C \subseteq \mathbb{A}_K^n$$

with points $P_1, P_2, P_3 \in C$ whose embedding dimensions are respectively 1, 2, 3.

Exercise 22.3

Let $H(X) \in K[X]$, $F = Y - H$, and let

$$C = V(F) \subseteq \mathbb{A}_K^2$$

be the graph of H , regarded as a plane algebraic curve. Let

$$P = (a, b) = (a, H(a))$$

be a point on this graph.

1. Show that the multiplicity of C at P is 1.
2. Show that the tangent to C at P agrees with the usual tangent to the graph at a .

Exercise 22.4

Let K be a field and let

$$F_1, \dots, F_m \in K[X_1, \dots, X_\ell]$$

and

$$G_1, \dots, G_n \in K[X_1, \dots, X_m]$$

be polynomials giving rise to polynomial maps

$$\mathbb{A}_K^\ell \xrightarrow{F} \mathbb{A}_K^m \xrightarrow{G} \mathbb{A}_K^n.$$

Let $J(F)_P$ and $J(G)_Q$ be the Jacobian matrices defined by formal partial differentiation. Prove the formal chain rule

$$J(G \circ F)_P = J(G)_{F(P)} \circ J(F)_P.$$

Exercise 22.5 ★

1. Show that formal partial differentiation with respect to one variable in the polynomial ring $K[X_1, \dots, X_n]$ commutes with dehomogenisation with respect to another variable.
2. Show that this does not hold when both operations concern the same variable.

Exercise 22.6 ★

Let

$$H \in K[X_1, \dots, X_n]$$

be a homogeneous polynomial of degree e in the standard grading. Show that

$$eH = X_1 \frac{\partial H}{\partial X_1} + \dots + X_n \frac{\partial H}{\partial X_n}.$$

Exercise 22.7

Consider the curve given by

$$y = 2x^4 + 3x^2 - x + 1$$

with the point

$$P = (1, 5).$$

Find a coordinate transformation taking P to $(0, 0)$ and the tangent at P to the x -axis.

Exercise 22.8

Let K be a field and $F \in K[X, Y]$ a nonconstant polynomial all of whose prime factors have multiplicity one. Let

$$C = V(F)$$

be the corresponding plane curve. Assume in addition that F remains squarefree after extending scalars to an algebraic closure of K . Show that C has only finitely many singular points.

Edition note — correction to the source hypotheses: Requiring the prime factors of F to have multiplicity one only in $K[X, Y]$ is insufficient when K is imperfect. For example, in characteristic $p > 0$, a reduced polynomial in $K[X, Y]$ may become a p th power after extending scalars, so that both partial derivatives vanish and its geometric singular locus has positive dimension. This edition states the precise condition needed in the argument: F is geometrically reduced. This is automatic, for example, if K is perfect and F is squarefree.

Exercise 22.9 ★

Prove Lemma 22.12.

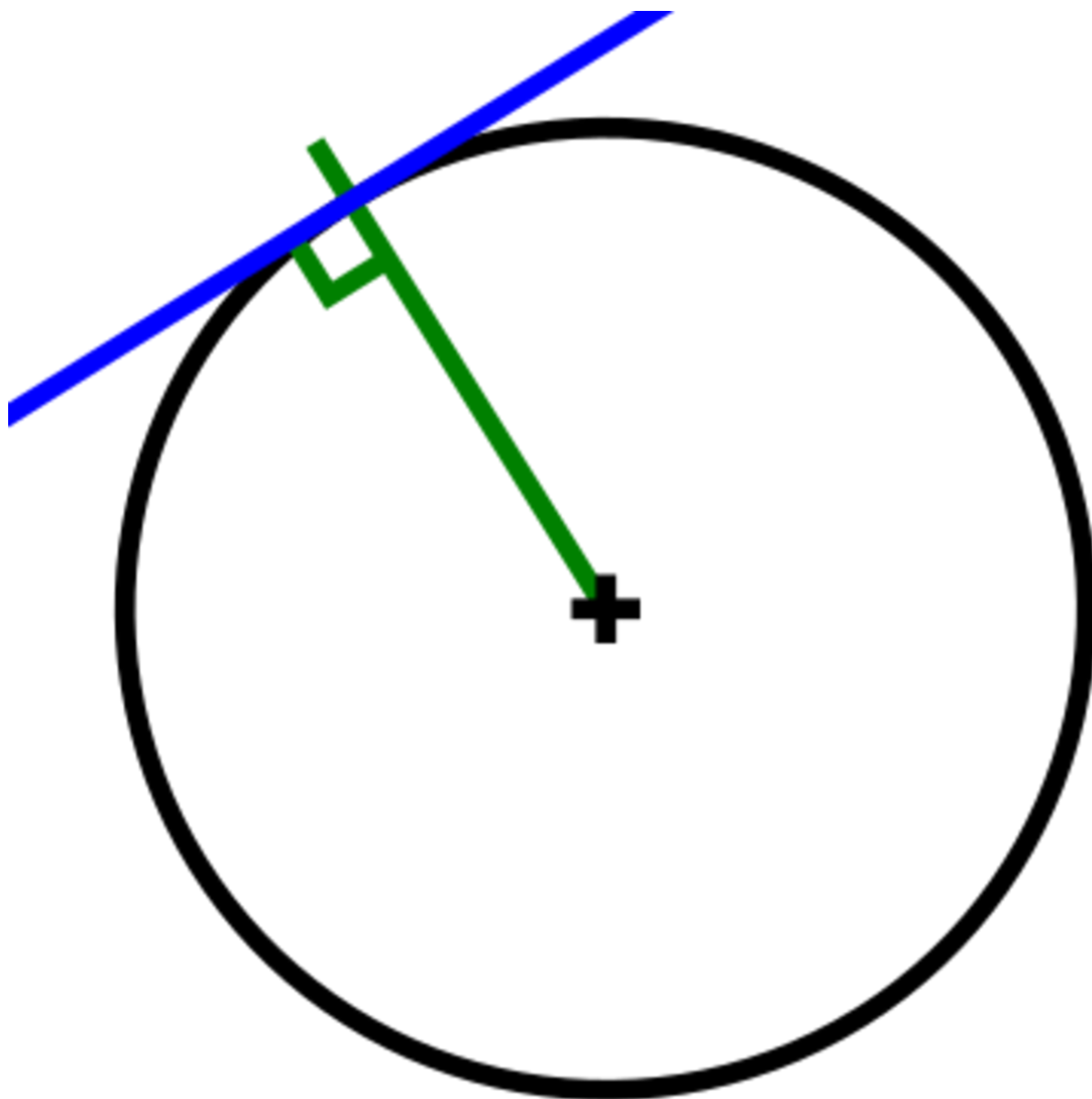


Figure 82: Diagram of a circle with a radius to the point of tangency and a tangent line perpendicular to that radius

Figure: A tangent to a circle at a point is perpendicular to the radius ending at that point. Work by Christophe Dang Ngoc Chan (Cdang), derivative SVG version by Hagman; CC BY-SA 3.0.

Exercise 22.10 ★

Show that the unit circle over a field of characteristic $\neq 2$ is smooth, and determine the tangent-line equation at each of its points.

Exercise 22.11

Let K be a field.

1. Show that the graph of a polynomial $F \in K[X]$ is a smooth algebraic curve.
2. Let $F, G \in K[X]$ be polynomials with no common zero. Show that the graph of the rational function F/G is also a smooth algebraic curve.

Exercise 22.12 ★

Determine the singular points of the plane algebraic curve

$$V\left(-2X^3 + 3X^2Y - Y + \frac{2}{3}\sqrt{\frac{1}{3}}\right) \subseteq \mathbb{A}_{\mathbb{C}}^2.$$

Exercise 22.13

For the trajectory computed in Example 8.5, determine the coordinates of the points where the curve is singular.

Exercise 22.14 ★

Determine the prime factorisation of the polynomial

$$X^3 + XY^2 \in \mathbb{C}[X, Y],$$

and determine the singularities of the corresponding affine curve, together with their multiplicities and tangent lines.

Exercise 22.15 ★

Determine the multiplicity and tangent lines at the origin $(0, 0)$ of the plane algebraic curve

$$C = V(Y^4 + X^3 + 3XY^2 + 2X^2Y) \subseteq \mathbb{A}_{\mathbb{C}}^2.$$

Exercise 22.16 ★

For the zero locus given by the polynomial

$$V^3 + U^2V - 2UV + 2U^2 - 4U - 2V,$$

use partial derivatives to determine a singular point. Perform a coordinate transformation taking this point to the origin. Determine the multiplicity and tangent lines at that point.

Edition note — correction to the source's scope: The exercise does not specify the base field, while the source solution divides by 2 and 3 and then factors using $\sqrt{3}$. To follow that solution, assume $\text{char}(K) \notin \{2, 3\}$ and $\sqrt{3} \in K$ (for example, $K = \mathbb{R}$ or $\mathbb{C}$). Over other fields, the point found can still be checked directly, but the factorisation and tangent multiplicities must be interpreted over the relevant base field; in characteristic 2, the tangent cone has one double tangent line.

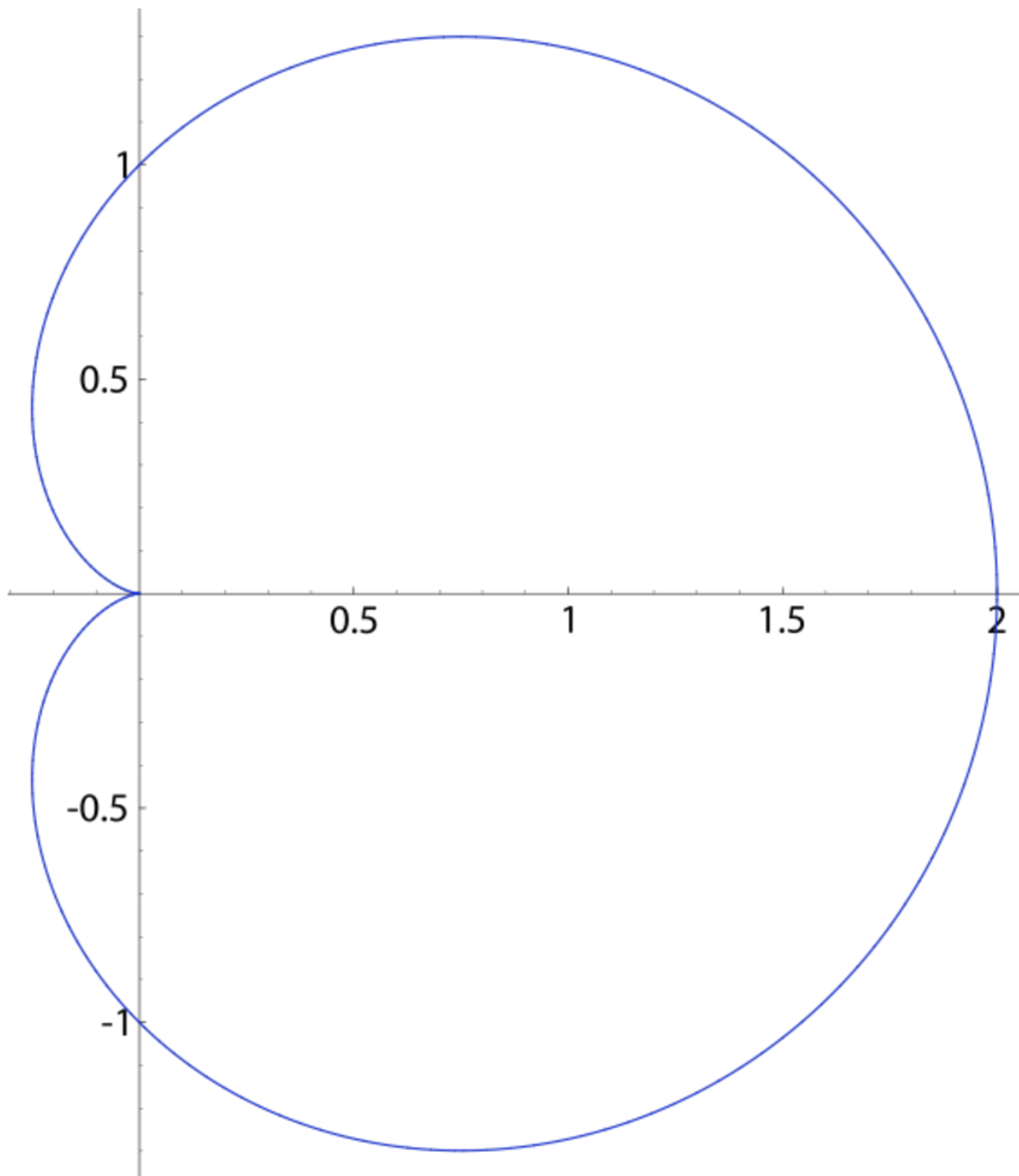


Figure 83: Symmetric cardioid with a cusp on the left and a rounded lobe on the right

Figure: The cardioid with polar equation $r = a(1 + \cos \theta)$ for $a = 1$. Work by D.328; CC BY-SA 3.0.

Exercise 22.17

Determine the singularities, including their multiplicities and tangent lines, of the cardioid given by

$$V\left((X^2 + Y^2)^2 - 2X(X^2 + Y^2) - Y^2\right).$$

Exercise 22.18 ★

Consider the two real curves

$$V(X^5 - X^3 + 2XY + 7Y^2 - 9)$$

at $(1, 1)$ and

$$V(X^4 + Y^4 - 3X^2Y^2 + 5X + 7Y)$$

at the origin. Are these curves locally diffeomorphic to one another at the specified points?

Exercises for submission

Exercise 22.19 (3 points)

Let K be a field of characteristic $p \geq 0$. Characterise the polynomials $F \in K[X, Y]$ satisfying each of the following three conditions:

1. the first partial derivative is 0;
2. the second partial derivative is 0;
3. both partial derivatives are 0.

Exercise 22.20 (4 points)

For the curve

$$V(X^3 + Y^3 - 3XY + 1),$$

determine its singular points over $\mathbb{R}$ and over $\mathbb{C}$. In each case give the multiplicities and tangent lines.

Exercise 22.21 (3 points)

Let K be an algebraically closed field and $G, H \in K[X, Y]$ polynomials satisfying

$$G(P) = H(P) = 0$$

at a specified point $P \in \mathbb{A}_K^2$. Let $F = GH$. Show that every tangent to G at P and every tangent to H at P is also a tangent to F at P .

Exercise 22.22 (6 points)

Let K be an algebraically closed field. Consider the curve

$$C = V(x^3 + 5x^2y - 6xy^2 - x^2 - xy + 4y^2).$$

1. Determine the tangent lines at the origin.
2. Show that

$$P = (1, 2)$$

is a point on the curve, and compute the tangent line or lines to C at P using derivatives.

3. Transform the variables so that P is the origin in the new variables, and determine the tangent line or lines at P from the transformed curve equation.

Exercise 22.23 (4 points)

For the algebraic curve

$$C = V(9y^4 + 10x^2y^2 + x^4 - 12y^3 - 12x^2y + 4y^2),$$

determine its singularities together with their multiplicities and tangent lines.

Source hint: Compare Example 8.5.

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Public Solutions to Worksheet 22

At the frozen revision boundary, the source provides public solutions only to Exercises 22.5, 22.6, 22.9, 22.10, 22.12, 22.14, 22.15, 22.16, and 22.18. No additional solutions have been created for this edition. The solution to Exercise 22.9 transcludes a proof page; the text below translates the actual proof body from the frozen recursive transclusion closure.

Solution to Exercise 22.5

1. Since both processes are linear, it suffices to consider a monomial

$$X_1^{\nu_1} X_2^{\nu_2} X_3^{\nu_3} \cdots X_n^{\nu_n},$$

differentiating with respect to X_1 and dehomogenising with respect to X_2 . Its partial derivative is

$$\nu_1 X_1^{\nu_1-1} X_2^{\nu_2} X_3^{\nu_3} \cdots X_n^{\nu_n},$$

and dehomogenisation gives

$$\nu_1 X_1^{\nu_1-1} X_3^{\nu_3} \cdots X_n^{\nu_n}.$$

If we dehomogenise first, we obtain

$$X_1^{\nu_1} X_3^{\nu_3} \cdots X_n^{\nu_n},$$

whose partial derivative likewise gives

$$\nu_1 X_1^{\nu_1-1} X_3^{\nu_3} \cdots X_n^{\nu_n}.$$

2. Consider X_1 . Differentiating with respect to X_1 gives 1, which is unchanged by dehomogenisation. If we first dehomogenise with respect to X_1 , we obtain 1, whose derivative is 0.

[Back to Exercise 22.5.](#)

Solution to Exercise 22.6

Since partial differentiation and multiplication by a variable are linear, it suffices to prove the statement for a monomial. Thus let

$$H = X_1^{\nu_1} \cdot X_2^{\nu_2} \cdots X_n^{\nu_n}.$$

Then

$$\begin{aligned} X_1 \frac{\partial H}{\partial X_1} + \cdots + X_n \frac{\partial H}{\partial X_n} &= \nu_1 X_1 \cdot X_1^{\nu_1-1} \cdot X_2^{\nu_2} \cdots X_n^{\nu_n} \\ &\quad + \nu_2 X_2 \cdot X_1^{\nu_1} \cdot X_2^{\nu_2-1} \cdots X_n^{\nu_n} + \cdots \\ &\quad + \nu_n X_n \cdot X_1^{\nu_1} \cdot X_2^{\nu_2} \cdots X_n^{\nu_n-1} \\ &= \nu_1 X_1^{\nu_1} \cdot X_2^{\nu_2} \cdots X_n^{\nu_n} + \nu_2 X_1^{\nu_1} \cdot X_2^{\nu_2} \cdots X_n^{\nu_n} + \cdots \\ &\quad + \nu_n X_1^{\nu_1} \cdot X_2^{\nu_2} \cdots X_n^{\nu_n} \\ &= (\nu_1 + \nu_2 + \cdots + \nu_n) X_1^{\nu_1} \cdot X_2^{\nu_2} \cdots X_n^{\nu_n}. \end{aligned}$$

This is the required statement, since the sum of the exponents is the degree of the monomial.

[Back to Exercise 22.6.](#)

Solution to Exercise 22.9

Since P is a smooth point of the curve, we may assume without loss of generality that

$$\frac{\partial F}{\partial X}(P) \neq 0.$$

By the product rule,

$$\begin{aligned}\frac{\partial F}{\partial X}(P) &= \frac{\partial(F_1 \cdots F_n)}{\partial X}(P) \\ &= \sum_{k=1}^n F_1(P) \cdots F_{k-1}(P) \cdot \frac{\partial F_k}{\partial X}(P) \cdot F_{k+1}(P) \cdots F_n(P).\end{aligned}$$

Now suppose that

$$P \in C_i \cap C_j$$

for $i \neq j$, that is,

$$F_i(P) = F_j(P) = 0.$$

Every product in the sum above would then have a zero factor, so

$$\frac{\partial F}{\partial X}(P) = 0,$$

contrary to the smoothness of P .

[Back to Exercise 22.9.](#)

Solution to Exercise 22.10

The defining polynomial is

$$X^2 + Y^2 - 1,$$

and its partial derivatives are $2X$ and $2Y$. Since the characteristic is assumed not to be 2 and the origin is not on the circle, the curve is smooth. By the remark on the tangent-line equation at a smooth point in Lecture 22, at a point (a, b) on the circle the tangent-line equation is

$$2a(X - a) + 2b(Y - b) = 0.$$

After dividing by 2, this can be written as

$$aX + bY - a^2 - b^2 = aX + bY - 1 = 0.$$

Edition note — source correction. The source prints only the left-hand expressions while calling them “tangent-line equations”. This edition restores the equality to zero that makes them actual equations.

[Back to Exercise 22.10.](#)

Solution to Exercise 22.12

Let F be the defining polynomial. Its partial derivatives are

$$\frac{\partial F}{\partial X} = -6X^2 + 6XY \quad \text{and} \quad \frac{\partial F}{\partial Y} = 3X^2 - 1.$$

Set both polynomials equal to zero. The second equation gives $x^2 = \frac{1}{3}$, hence

$$x = \pm \sqrt{\frac{1}{3}}.$$

In the first equation we can factor out the nonzero factor $6X$, so we must have $x = y$. Thus

$$y = \pm \sqrt{\frac{1}{3}}.$$

For $x = y$, the curve equation becomes

$$x^3 - x + \frac{2}{3}\sqrt{\frac{1}{3}} = 0.$$

At $x = \sqrt{\frac{1}{3}}$, the left-hand side has value

$$\frac{1}{3}\sqrt{\frac{1}{3}} - \sqrt{\frac{1}{3}} + \frac{2}{3}\sqrt{\frac{1}{3}} = 0,$$

so

$$\left(\sqrt{\frac{1}{3}}, \sqrt{\frac{1}{3}} \right)$$

is a point on the curve. By contrast, at $x = y = -\sqrt{\frac{1}{3}}$ we obtain

$$-\frac{1}{3}\sqrt{\frac{1}{3}} + \sqrt{\frac{1}{3}} + \frac{2}{3}\sqrt{\frac{1}{3}} = \frac{4}{3}\sqrt{\frac{1}{3}} \neq 0,$$

so this is not a point on the curve. The curve's only singularity is therefore

$$\left(\sqrt{\frac{1}{3}}, \sqrt{\frac{1}{3}} \right).$$

[Back to Exercise 22.12.](#)

Solution to Exercise 22.14

Clearly,

$$X(X + iY)(X - iY)$$

is a factorisation of the polynomial into prime factors. To determine the singular points, we examine the partial derivatives:

$$\frac{\partial F}{\partial X} = 3X^2 + Y^2 \quad \text{and} \quad \frac{\partial F}{\partial Y} = 2XY.$$

These both vanish precisely when $(x, y) = (0, 0)$. Since this point also satisfies the curve equation, it is the singular point of the curve. The defining polynomial is already homogeneous of degree 3, so the multiplicity is 3. The tangent lines are therefore given by

$$V(X), \quad V(X + iY), \quad V(X - iY).$$

[Back to Exercise 22.14.](#)

Solution to Exercise 22.15

The multiplicity is the degree of the lowest-degree homogeneous component, namely 3. To determine the tangent lines, we must factor $X^3 + 3XY^2 + 2X^2Y$ into linear factors. We have

$$X^3 + 3XY^2 + 2X^2Y = X(X^2 + 3Y^2 + 2XY).$$

Furthermore,

$$\begin{aligned} X^2 + 3Y^2 + 2XY &= (X + Y)^2 - Y^2 + 3Y^2 \\ &= (X + Y)^2 + 2Y^2 \\ &= (X + Y + \sqrt{2}iY)(X + Y - \sqrt{2}iY). \end{aligned}$$

The tangent lines are thus

$$X = 0$$

(the Y -axis), and

$$X = -(1 + \sqrt{2}i)Y \quad \text{and} \quad X = (-1 + \sqrt{2}i)Y.$$

[Back to Exercise 22.15.](#)

Solution to Exercise 22.16

The following source computation uses the edition's hypotheses in Exercise 22.16, namely $\text{char}(K) \notin \{2, 3\}$ and $\sqrt{3} \in K$.

Let

$$F = V^3 + U^2V - 2UV + 2U^2 - 4U - 2V.$$

Then

$$\frac{\partial F}{\partial U} = 2UV - 2V + 4U - 4 \quad \text{and} \quad \frac{\partial F}{\partial V} = 3V^2 + U^2 - 2U - 2.$$

The first equation gives the following condition for a singular point:

$$V(U - 1) = -2U + 2, \quad \text{or} \quad V = \frac{-2U + 2}{U - 1},$$

where the latter form requires $U \neq 1$. We therefore first consider the case $U = 1$. The first partial derivative is then zero regardless of V , while the second gives the condition

$$3V^2 + 1 - 2 - 2 = 0, \quad V^2 = 1, \quad V = \pm 1.$$

The curve equation gives

$$V^3 + V - 2V - 2V - 2 = V^3 - 3V - 2 = 0,$$

which is satisfied by $V = -1$. Therefore

$$P = (1, -1)$$

is a singular point of the curve.

In the new variables $X = U - 1$ and $Y = V + 1$, the point P becomes the origin. Substituting $U = X + 1$ and $V = Y - 1$ transforms the curve equation into

$$\begin{aligned} & (Y - 1)^3 + (X + 1)^2(Y - 1) - 2(X + 1)(Y - 1) \\ & \quad + 2(X + 1)^2 - 4(X + 1) - 2(Y - 1) \\ & = Y^3 - 3Y^2 + 3Y - 1 + (X^2 + 2X + 1)(Y - 1) \\ & \quad - 2XY + 2X - 2Y + 2 + 2X^2 + 4X + 2 - 4X - 4 - 2Y + 2 \\ & = Y^3 - 3Y^2 + 3Y - 1 + X^2Y + 2XY + Y - X^2 - 2X - 1 \\ & \quad - 2XY + 2X - 2Y + 2 + 2X^2 + 4X + 2 - 4X - 4 - 2Y + 2 \\ & = Y^3 + X^2Y - 3Y^2 + X^2. \end{aligned}$$

Thus the lowest-degree homogeneous component is

$$X^2 - 3Y^2 = (X - \sqrt{3}Y)(X + \sqrt{3}Y).$$

The multiplicity is therefore two, and the two tangent lines through the singular point are described by

$$X = \pm\sqrt{3}Y.$$

[Back to Exercise 22.16.](#)

Solution to Exercise 22.18

The partial derivative of the first polynomial with respect to X is

$$5X^4 - 3X^2 + 2Y,$$

whose value at the specified point is

$$4 \neq 0.$$

The partial derivative of the second polynomial with respect to X is

$$4X^3 - 6XY^2 + 5,$$

whose value at the specified point is

$$5 \neq 0.$$

Both curves are therefore smooth at these points. By the implicit function theorem, each is locally diffeomorphic to an open real interval, so they are locally diffeomorphic to one another.

[Back to Exercise 22.18.](#)

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Lecture 23: Cotangent Spaces and Hilbert–Samuel Multiplicity

Interpretation as a cotangent space

We give another indication that calling $\mathfrak{m}/\mathfrak{m}^2$ the *cotangent space* is well justified. From analysis, we know that for a point $P \in M$ on a manifold M and a differentiable function $f : M \rightarrow \mathbb{R}$, the differential

$$df : T_P M \rightarrow \mathbb{R}$$

is linear (see Lemma 9.10 (Differential Geometry (Osnabrück 2023)), part (3)). Thus df is an element of the cotangent space $T_P^* M$. The overall assignment

$$C^1(M, \mathbb{R}) \rightarrow T_P^* M, \quad f \mapsto df,$$

is a derivation: it satisfies the Leibniz rule

$$d(fg) = f dg + g df.$$

We now introduce the general algebraic concept.

Definition: algebraic derivation

Let R be a commutative ring, A a commutative R -algebra, and M an A -module. An R -linear map

$$\delta : A \rightarrow M$$

is called an R -*derivation* with values in M if

$$\delta(ab) = a\delta(b) + b\delta(a)$$

for all $a, b \in A$.

Theorem: the canonical derivation into the cotangent space

Let K be a field, R a K -algebra of finite type, and

$$P \in K\text{-}\mathrm{Spek}(R)$$

a point with corresponding maximal ideal $\mathfrak{m}$. Then the map

$$\begin{aligned} d : R &\rightarrow \mathfrak{m}/\mathfrak{m}^2, \\ f &\mapsto df := \overline{f - f(P)} \end{aligned}$$

is a K -derivation.

Proof

There is a canonical isomorphism $K \rightarrow R/\mathfrak{m}$ between the base field and the residue field. The map d is well defined because

$$(f - f(P))(P) = 0,$$

so $f - f(P) \in \mathfrak{m}$. Its K -linearity is immediate. For the product rule, all the following equalities are understood in $\mathfrak{m}/\mathfrak{m}^2$:

$$\begin{aligned}
d(fg) &= \overline{fg - (fg)(P)} \\
&= \overline{fg - f(P)g(P)} \\
&= \overline{fg - f(P)g(P) + (f - f(P))(g - g(P))} \\
&= \overline{2fg - f \cdot g(P) - g \cdot f(P)} \\
&= \overline{f(g - g(P)) + g(f - f(P))} \\
&= f dg + g df.
\end{aligned}$$

In the third step we add an element of $\mathfrak{m}^2$, whose class in the quotient is zero. This proves the Leibniz rule. $\square$

Edition note – clarification of the source notation: The source writes the chain above without a residue-class bar on each term. These are equalities modulo $\mathfrak{m}^2$, not polynomial equalities in R .

Smooth points and normal points

We shall show that a point on a plane algebraic curve is smooth precisely when the corresponding local ring is a discrete valuation ring. Smoothness at a point was initially defined extrinsically, with reference to the ambient plane, whereas being a discrete valuation ring depends only on the curve's coordinate ring. The following lemma handles one direction. For the other, we must first develop an intrinsic multiplicity for a local ring.

Lemma: smooth points give discrete valuation rings

Let K be a field, $F \in K[X, Y]$ a nonzero polynomial without repeated factors, and

$$P \in C = V(F)$$

a smooth point of the curve. If R is the local ring of the curve at P , then R is a discrete valuation ring.

Proof

First, R is a Noetherian local ring and, by Lemma 22.12, an integral domain. Its only prime ideals are therefore the zero ideal and the maximal ideal $\mathfrak{m}_P$. We shall show that this maximal ideal is principal.

We may assume that P is the origin and write

$$F = F_d + \cdots + F_1, \quad F_1 \neq 0,$$

where each F_i is homogeneous of degree i . Since P is smooth, this form has a nonzero linear term. A linear change of variables allows us to arrange that $F_1 = Y$. Collect all the pure powers of X , that is, the monomials not involving Y , and factor Y out of the remaining terms. The equation $F = 0$ can then be written as

$$Y(1 + G) = XH(X), \quad G \in (X, Y).$$

The element $1 + G$ is a unit in $K[X, Y]_{(X, Y)}$, and hence also in the local ring of the curve at the origin,

$$R = K[X, Y]_{(X, Y)} / (F).$$

In R we have

$$Y = \frac{H}{1 + G} X.$$

Thus the maximal ideal of R is generated by X alone. By Theorem 21.8, R is a discrete valuation ring. $\square$

Hilbert–Samuel multiplicity

Lemma: quotients by powers of the maximal ideal are finite-dimensional

Let R be a Noetherian local ring with maximal ideal $\mathfrak{m}$ and residue field

$$K = R/\mathfrak{m}.$$

Then the quotient modules $\mathfrak{m}^n/\mathfrak{m}^{n+1}$ are finite-dimensional over K . If R contains a field K mapping isomorphically onto the residue field, the quotient rings $R/\mathfrak{m}^n$ are also finite-dimensional over K .

Proof

We write

$$\mathfrak{m}^n/\mathfrak{m}^{n+1} \cong \mathfrak{m}^n/(\mathfrak{m}^n)\mathfrak{m}.$$

This is the situation of Lemma 22.2. Since $\mathfrak{m}^n$ is a finitely generated ideal, the quotient module is finite-dimensional over the residue field.

For the quotient rings, consider the short exact sequence of R -modules

$$0 \longrightarrow \mathfrak{m}^n/\mathfrak{m}^{n+1} \longrightarrow R/\mathfrak{m}^{n+1} \longrightarrow R/\mathfrak{m}^n \longrightarrow 0.$$

Under the additional hypothesis, this is also a short exact sequence of K -vector spaces, so their dimensions add. The space on the left is finite-dimensional by the part just proved. Induction on n now gives the desired result, with initial case $R/\mathfrak{m} = K$. $\square$

For a plane algebraic curve

$$V = V(F) \subseteq \mathbb{A}_K^2$$

and a point $P = (a, b) \in V$, the local ring is

$$K[X, Y]_{(X-a, Y-b)}/(F).$$

Its residue field is K itself. Thus all the hypotheses of Lemma 23.4 hold, and all the following dimensions are over the base field.

Theorem: multiplicity via the Hilbert–Samuel function

Let

$$P \in V = V(F) \subseteq \mathbb{A}_K^2$$

be a point on an affine plane curve. Let

$$R = \mathcal{O}_{V,P}$$

be its local ring, with maximal ideal $\mathfrak{m}$. Then the multiplicity m_P of P satisfies

$$m_P = \dim_K (\mathfrak{m}^n/\mathfrak{m}^{n+1})$$

for all sufficiently large n .

Proof

Consider the short exact sequence of K -vector spaces

$$0 \longrightarrow \mathfrak{m}^n/\mathfrak{m}^{n+1} \longrightarrow R/\mathfrak{m}^{n+1} \longrightarrow R/\mathfrak{m}^n \longrightarrow 0.$$

By Lemma 23.4, all dimensions are finite. The assertion that the dimension of $\mathfrak{m}^n/\mathfrak{m}^{n+1}$ is eventually constant and equal to the multiplicity is equivalent to the assertion that the difference

$$\dim_K(R/\mathfrak{m}^{n+1}) - \dim_K(R/\mathfrak{m}^n)$$

is eventually constant and equal to the multiplicity. By induction, this is equivalent to the existence of a constant c such that

$$\dim_K(R/\mathfrak{m}^n) = m_P n + c$$

for sufficiently large n .

After a translation, we may assume that P is the origin. Set

$$\mathfrak{a} = (X, Y) \subseteq S = K[X, Y].$$

Then

$$K[X, Y]/(\mathfrak{a}^n + (F)) = R/\mathfrak{m}^n,$$

so it suffices to prove the statement for the quotient on the left. By hypothesis, F has the form

$$F = F_m + F_{m+1} + \cdots, \quad m = m_P,$$

and in particular $F \in \mathfrak{a}^m$. If $G \in \mathfrak{a}^{n-m}$ with $n \geq m$, then $GF \in \mathfrak{a}^n$. There is therefore a short exact sequence

$$0 \longrightarrow S/\mathfrak{a}^{n-m} \xrightarrow{\cdot F} S/\mathfrak{a}^n \longrightarrow S/(\mathfrak{a}^n, F) = R/\mathfrak{m}^n \longrightarrow 0.$$

Injectivity on the left follows from a direct degree argument; see Exercise 23.4. We know that

$$\dim_K(S/\mathfrak{a}^n) = \frac{n(n+1)}{2}.$$

Hence, for $n \geq m$,

$$\begin{aligned} \dim_K(R/\mathfrak{m}^n) &= \frac{n(n+1)}{2} - \frac{(n-m)(n-m+1)}{2} \\ &= \frac{n^2 + n - (n-m)^2 - n + m}{2} \\ &= \frac{2nm - m^2 + m}{2} \\ &= mn - \frac{m(m-1)}{2}. \end{aligned}$$

This is the required linear form. $\square$

Remark: multiplicity as an intrinsic invariant

Theorem 23.5 says in particular that the multiplicity of a point on a plane curve is an invariant of the local ring of the curve at that point. It therefore depends only on intrinsic properties of the curve, not on its realisation in an ambient plane.

Every Noetherian local ring has a *Hilbert–Samuel multiplicity*, defined in terms of the dimensions over $R/\mathfrak{m}$ of the quotient modules $\mathfrak{m}^n/\mathfrak{m}^{n+1}$. In the one-dimensional case it is

$$\lim_{n \rightarrow \infty} \dim_{R/\mathfrak{m}} (\mathfrak{m}^n/\mathfrak{m}^{n+1}),$$

since this function eventually becomes constant—a nontrivial fact. If R contains a field K isomorphic to its residue field, as is the case for the local rings of curves considered here, the same number is also given by

$$\lim_{n \rightarrow \infty} \frac{\dim_K (R/\mathfrak{m}^n)}{n}.$$

Theorem: smoothness, multiplicity, discrete valuation, and normality

Let K be a field and $F \in K[X, Y]$ a nonconstant polynomial without repeated factors, with corresponding algebraic curve

$$C = V(F).$$

Let $P = (a, b) \in C$, with maximal ideal

$$\mathfrak{m} = (X - a, Y - b)$$

and local ring

$$R = K[X, Y]_{\mathfrak{m}}/(F).$$

The following statements are equivalent.

1. P is a smooth point of the curve.
2. The multiplicity of P is one.
3. R is a discrete valuation ring.
4. R is a normal integral domain.

Proof

The equivalence (1) $\Leftrightarrow$ (2) follows from Definition 22.7 of multiplicity. The equivalence (3) $\Leftrightarrow$ (4) was proved in Theorem 21.8, and the implication (1) $\Rightarrow$ (3) in Lemma 23.3. It remains to prove (3) $\Rightarrow$ (2); by Theorem 23.5 we may work with Hilbert–Samuel multiplicity.

It suffices to show that, for the local ring of a plane curve that is a discrete valuation ring, all the quotient modules

$$\mathfrak{m}^n/\mathfrak{m}^{n+1} \cong \mathfrak{m}^n/\mathfrak{m}^n\mathfrak{m}$$

are one-dimensional over the residue field $R/\mathfrak{m} \cong K$. Since $\mathfrak{m}^n = (\pi^n)$, this follows immediately from Nakayama’s lemma. $\square$

Monomial curves and multiplicity

Let $M \subseteq \mathbb{N}$ be a numerical monoid generated by coprime natural numbers

$$e_1 < e_2 < \cdots < e_r.$$

The least generator e_1 is also called the *numerical multiplicity* of M . We shall show that this does indeed give the correct ring-theoretic multiplicity. Set

$$M_+ = \{m \in M \mid m \geq 1\}$$

and, for $n \geq 1$,

$$nM_+ = \{m \in M \mid m = m_1 + \cdots + m_n \text{ for some } m_i \in M_+\}.$$

Both are monoid ideals of M . Thus the monomial spaces

$$K[nM_+] = \bigoplus_{m \in nM_+} KT^m$$

are ideals in the monoid ring. In particular,

$$\mathfrak{m} = K[M_+]$$

is a maximal ideal, and its powers are

$$\mathfrak{m}^n = K[nM_+].$$

Lemma: bounds for monoid difference sets

Let $M \subseteq \mathbb{N}$ be a numerical monoid of numerical multiplicity e_1 . Choose $\ell \geq 1$ such that $\mathbb{N}_{\geq \ell} \subseteq M$. Then, for each $n \geq 1$,

$$ne_1 - \ell \leq \#(M \setminus nM_+) \leq (n-1)e_1 + \ell.$$

Proof

The lower bound follows from the fact that the smallest number in nM_+ is ne_1 . Thus $0, 1, \dots, ne_1 - 1$ lie outside it. All numbers at least ℓ belong to M , so at least $ne_1 - \ell$ of these ne_1 numbers belong to M but not to nM_+ .

For the upper bound, we claim that every number at least $(n-1)e_1 + \ell$ belongs to nM_+ . Let

$$x \geq (n-1)e_1 + \ell.$$

Write

$$x = (n-1)e_1 + \ell', \quad \ell' \geq \ell.$$

Since $\ell' \in M_+$, the right-hand side is a sum of n elements of M_+ : $n-1$ summands equal to e_1 and one equal to ℓ' . Thus $x \in nM_+$, proving the upper bound. $\square$

Edition note – correction to the source’s bound: The source describes ℓ only as “a number” and says in the final step that the summands belong to M . The hypothesis $\ell \geq 1$, which can always be achieved by increasing the threshold, ensures that all n summands really lie in M_+ , as required by the definition of nM_+ . The source also leaves the range of n implicit. Here $n \geq 1$: the displayed definition by sums of positive elements does not give the zeroth ideal power when $n = 0$.

Corollary: numerical multiplicity equals Hilbert–Samuel multiplicity

Let $M \subseteq \mathbb{N}$ be a numerical monoid generated by coprime numbers, with numerical multiplicity e_1 . Let

$$\mathfrak{m} = K[M_+]$$

be the maximal ideal of the monoid ring $K[M]$ corresponding to the origin. Then

$$\lim_{n \rightarrow \infty} \frac{\dim_K (K[M]/\mathfrak{m}^n)}{n} = e_1.$$

In other words, numerical multiplicity equals Hilbert–Samuel multiplicity.

Proof

Since $\mathfrak{m}^n = K[nM_+]$, the quotient ring

$$K[M]/\mathfrak{m}^n = K[M]/K[nM_+]$$

has the monomials T^m with $m \in M \setminus nM_+$ as a basis over K . Its dimension therefore equals $\#(M \setminus nM_+)$. By the bounds in Lemma 23.8,

$$\frac{\#(M \setminus nM_+)}{n} \rightarrow e_1.$$

The same convergence holds for these dimensions. $\square$

Edition note – clarification of the source notation: The source notation $K[M]/(nM_+)$ means the quotient by the monomial ideal $K[nM_+]$; the parentheses do not denote scalar multiplication of a set by n .

Worksheet 23**Practice exercises****Exercise 23.1**

Let K be a field, $K[X_1, \dots, X_n]$ the polynomial ring over K , and

$$\frac{\partial}{\partial X_1}$$

formal partial differentiation with respect to X_1 , that is, the map

$$\begin{aligned} K[X_1, \dots, X_n] &\longrightarrow K[X_1, \dots, X_n], \\ f &\longmapsto \frac{\partial f}{\partial X_1}. \end{aligned}$$

Show that this map is a K -derivation.

Exercise 23.2

Consider the maximal ideal

$$\mathfrak{m} = (X_1, \dots, X_n) \subseteq K[X_1, \dots, X_n]$$

in the polynomial ring over a field K , together with its powers $\mathfrak{m}^d$. Show that the monomials

$$X_1^{\nu_1} \cdots X_n^{\nu_n}, \quad \sum_{i=1}^n \nu_i < d,$$

form a K -basis of the quotient ring

$$K[X_1, \dots, X_n]/\mathfrak{m}^d.$$

Exercise 23.3

Consider the union of the coordinate axes

$$V(xy) \subseteq \mathbb{A}_K^2$$

and the local ring R at the origin, with maximal ideal $\mathfrak{m}$. Describe explicitly a K -basis of the quotient rings $R/\mathfrak{m}^n$ and determine their dimensions.

Exercise 23.4 ★

Let

$$F = F_m + \cdots + F_d \in K[X, Y]$$

be the homogeneous decomposition of a polynomial, with $m \leq d$ and $F_m \neq 0$, and let $\mathfrak{m} = (X, Y)$. Show that, for every $n \geq m$, the multiplication map

$$\begin{aligned} K[X, Y] &\longrightarrow K[X, Y], \\ G &\longmapsto FG \end{aligned}$$

induces a well-defined injective homomorphism of $K[X, Y]$ -modules

$$K[X, Y]/\mathfrak{m}^{n-m} \longrightarrow K[X, Y]/\mathfrak{m}^n.$$

Edition note – clarification of the source hypothesis: Here F_m is the lowest nonzero homogeneous component. The source leaves $F_m \neq 0$ implicit, but it is required for injectivity and is used in the source argument and the corrected solution below.

Exercise 23.5 ★

Let $e \in \mathbb{N}_+$ and

$$M := \{0\} \cup \mathbb{N}_{\geq e} \subseteq \mathbb{N}.$$

1. Determine nM_+ for $n \in \mathbb{N}_+$.
2. Determine $\#(M \setminus nM_+)$.
3. Let K be a field and set

$$R = K[M]_{\mathfrak{m}}, \quad \mathfrak{m} = K[M_+] \subseteq K[M].$$

Determine $\dim_K(R/\mathfrak{m}^n)$.

Exercise 23.6

For the numerical monoid $M \subseteq \mathbb{N}$ generated by 4 and 9, compute the quantities appearing in the bounds of Lemma 23.8 for $n \leq 6$.

Krull dimension

Some of the following exercises use the Krull dimension of a commutative ring. Since our main interest is in curves, which correspond to one-dimensional rings, we shall not develop a systematic dimension theory.

Definition: prime ideal chains and Krull dimension

Let R be a commutative ring. A chain of prime ideals

$$\mathfrak{p}_0 \subset \mathfrak{p}_1 \subset \cdots \subset \mathfrak{p}_n$$

is called a *prime ideal chain of length n* . Thus we count the inclusions, not the prime ideals in the chain. The *dimension*, or *Krull dimension*, of R is the supremum of all lengths of prime ideal chains, denoted by

$$\dim(R).$$

Exercise 23.7

Let R be a principal ideal domain that is not a field. Show that its Krull dimension is one.

Exercises for submission

Exercise 23.8 (4 points)

For the monoid $M \subseteq \mathbb{N}$ generated by 5, 8, 11, compute the quantities appearing in the bounds of Lemma 23.8 for $n \leq 5$.

Exercise 23.9 (3 points)

Let $\mathfrak{a} \subseteq R$ be an ideal in a commutative ring, and suppose that the only prime ideal containing $\mathfrak{a}$ is a maximal ideal $\mathfrak{m}$. Show that

$$R/\mathfrak{a} \cong R_{\mathfrak{m}}/\mathfrak{a}R_{\mathfrak{m}}.$$

Deduce that, for a maximal ideal $\mathfrak{m}$ in a Noetherian commutative ring, there is an isomorphism

$$R/\mathfrak{m}^n \cong R_{\mathfrak{m}}/\mathfrak{m}^n R_{\mathfrak{m}}$$

for every n .

Exercise 23.10 (5 points)

Let K be an algebraically closed field and $R = K[X, Y]$ the polynomial ring in two variables. Show that R has Krull dimension two.

Exercise 23.11 (5 points)

Let R be a Noetherian commutative ring. Show that the following statements are equivalent.

1. R has Krull dimension 0.
2. R is an Artinian ring.
3. R has finitely many prime ideals, all of which are maximal.
4. There is $n \in \mathbb{N}$ such that

$$(\mathfrak{m}R_{\mathfrak{m}})^n = 0$$

for every maximal ideal $\mathfrak{m}$.

5. The reduction $R_{\text{red}} = R/\sqrt{(0)}$ is a finite product of fields.

Edition note – correction to the source statement: In item (4), the source writes $\mathfrak{m}^n = 0$ in R for every maximal ideal. This is not equivalent to the other four items: for example, in $R = K \times K$ both maximal ideals are idempotent rather than nilpotent, although R is zero-dimensional and Artinian. The edition replaces this with the uniform local formulation above, which correctly characterises zero-dimensional Noetherian rings. The “product” in item (5) is also explicitly stated to be *finite*, as forced by the Noetherian/Artinian condition.

Exercise 23.12 (3 points)

Let R be a commutative ring of finite Krull dimension d . Show that the Krull dimension of the polynomial ring $R[X]$ is at least $d + 1$.

Source remark: Over a Noetherian base ring, passing to the polynomial ring increases the dimension by exactly one; this stronger result is harder to prove.

Public Solutions to Worksheet 23

At the frozen revision boundary, the source provides public solutions only to Exercises 23.4 and 23.5. No additional solutions have been created for this edition.

Solution to Exercise 23.4

Write $S = K[X, Y]$. Since the lowest homogeneous term of F has degree m , we have $F \in \mathfrak{m}^m$. To prove well-definedness, let $G \in \mathfrak{m}^{n-m}$. Then

$$FG \in \mathfrak{m}^m \mathfrak{m}^{n-m} = \mathfrak{m}^n.$$

Thus the S -module map

$$S \xrightarrow{\cdot F} S \longrightarrow S/\mathfrak{m}^n$$

vanishes on $\mathfrak{m}^{n-m}$. By the universal property of quotient modules, it induces an S -module homomorphism

$$S/\mathfrak{m}^{n-m} \xrightarrow{\cdot F} S/\mathfrak{m}^n.$$

To prove injectivity, suppose that the class of G maps to zero, so that $FG \in \mathfrak{m}^n$. Assume $G \notin \mathfrak{m}^{n-m}$. If q is the degree of the lowest nonzero homogeneous term G_q of G , then

$$q < n - m.$$

The lowest homogeneous term of FG is $F_m G_q$. Since $K[X, Y]$ is an integral domain and $F_m, G_q \neq 0$, we have

$$F_m G_q \neq 0, \quad \deg(F_m G_q) = m + q < n.$$

Hence $FG \notin \mathfrak{m}^n$, a contradiction. Thus $G \in \mathfrak{m}^{n-m}$ and its class is zero. The induced homomorphism is injective. $\square$

Edition note – correction to the source solution: The source says that FG has a monomial of degree “less than m ”; the required bound is less than n . The argument using the lowest homogeneous term $F_m G_q$ above also rules out cancellation. Since multiplication by F is generally not a ring homomorphism, the factorisation uses the universal property of quotient *modules*.

[Back to Exercise 23.4](#)

Solution to Exercise 23.5

1. We claim that

$$nM_+ = \mathbb{N}_{\geq ne}.$$

Membership $k \in nM_+$ means that there are $m_1, \dots, m_n \in M_+ = \mathbb{N}_{\geq e}$ with

$$k = m_1 + \dots + m_n.$$

Since $m_j \geq e$ for every j , we obtain $k \geq ne$. Conversely, if $k \geq ne$, then

$$k = (n-1)e + m, \quad m := k - (n-1)e \geq e.$$

The right-hand side is a sum of n elements of M_+ , so $k \in nM_+$.

2. We obtain

$$M \setminus nM_+ = M \setminus \mathbb{N}_{\geq ne} = \{0, e, e+1, \dots, ne-1\}.$$

Therefore

$$\#(M \setminus nM_+) = ne - e + 1 = (n-1)e + 1.$$

3. The ideal $\mathfrak{m}^n$ is the monomial ideal $K[nM_+]$. Localisation does not change this finite-dimensional quotient, and there are isomorphisms of K -vector spaces

$$\begin{aligned} R/\mathfrak{m}^n &\cong K[M]/K[nM_+] \\ &\cong \text{span}_K\{T^m \mid m \in M \setminus nM_+\}. \end{aligned}$$

Hence

$$\dim_K(R/\mathfrak{m}^n) = (n-1)e + 1.$$

Edition note – correction to the source solution: In the first part, the source writes $n_j \geq e$ after introducing $m_1, \dots, m_n$; the correct notation is $m_j \geq e$. In the third part, the source notation identifies the quotient with $K[M \setminus nM_+]$, as though the complement defined a monoid ring. The edition states the correct objects: the quotient by the monomial ideal $K[nM_+]$ and the vector space with monomial basis indexed by its complement.

[Back to Exercise 23.5](#)

Lecture 24: Tangent Lines and Formal Power Series Rings

Tangent lines from parametrisations

Theorem 24.1: the derivative vector of a parametrisation

Let K be an infinite field and

$$\varphi: \mathbb{A}_K^1 \longrightarrow \mathbb{A}_K^n$$

a map given by n polynomials in one variable,

$$\varphi = (\varphi_1(t), \dots, \varphi_n(t)),$$

whose image is contained in the curve

$$C = V(F_1, \dots, F_m).$$

Take $Q \in \mathbb{A}_K^1$ and

$$P = \varphi(Q) \in C.$$

Then the derivative vector

$$\left(\frac{\partial \varphi_1}{\partial t}(Q), \dots, \frac{\partial \varphi_n}{\partial t}(Q) \right)$$

lies in the kernel of the linear tangent map

$$(TF)_P: \mathbb{A}_K^n \longrightarrow \mathbb{A}_K^m$$

defined by the Jacobian matrix

$$\left(\frac{\partial F_i}{\partial X_j}(P) \right)_{ij}.$$

If $n = 2$, the derivatives $\varphi'_1(Q)$ and $\varphi'_2(Q)$ do not both vanish, and P is a smooth point of C , then

$$\left(\frac{\partial \varphi_1}{\partial t}(Q), \frac{\partial \varphi_2}{\partial t}(Q) \right)$$

determines the direction of the tangent line to C at P .

Proof Write $F = (F_1, \dots, F_m)$. Since

$$\varphi(\mathbb{A}_K^1) \subseteq V(F_1, \dots, F_m),$$

the composite $F \circ \varphi$ is the constant map to the origin. Since K is infinite, each component polynomial representing $F_i \circ \varphi$ is the zero polynomial. The formal chain rule for polynomials therefore gives

$$0 = (T(F \circ \varphi))_Q = (TF)_P \circ (T\varphi)_Q.$$

Thus the image of $(T\varphi)_Q$, spanned by the derivative vector above, is contained in the kernel of $(TF)_P$.

In the plane case stated in the final part of the theorem, choose a reduced defining polynomial H for C . The same chain-rule argument with H in place of the tuple F puts the derivative vector in $\ker(dH_P)$. This kernel is one-dimensional because P is smooth. The image of $(T\varphi)_Q$ is also one-dimensional because the derivative vector is nonzero. The inclusion is therefore an equality, so that vector determines the tangent direction. $\square$

Edition note — defining equations. The source treats the kernel of the original Jacobian as necessarily one-dimensional in the plane case. This requires equations generating the reduced curve's ideal near P ; arbitrary equations with the same zero locus need not do so. For example, $Y^2 = 0$ defines the line $Y = 0$ set-theoretically but has zero differential on that line. The argument above uses a reduced equation for the final assertion; the initial chain-rule inclusion for all F_i is unchanged.

Example 24.2: why the field must be infinite

Let K be a finite field with

$$q = p^e$$

elements, where p is prime and $e \geq 1$. The map

$$\begin{aligned} \mathbb{A}_K^1 &\longrightarrow \mathbb{A}_K^2, \\ t &\longmapsto (t^q - t, t^q - t) \end{aligned}$$

sends every K -rational point to the single image point $(0, 0)$, since $t^q = t$ for every $t \in K$. However, the formal derivative vector of this polynomial parametrisation is

$$(-1, -1).$$

Thus a map on K -rational points can be constant in positive characteristic even though the formal derivative of its defining polynomials is nonzero. The origin is a smooth point on every line

$$C = V(aX + bY),$$

with $(a, b) \neq (0, 0)$.

Its tangent direction is the kernel of the linear form $aX + bY$, but this form annihilates $(-1, -1)$ only when $a = -b$. The assumption that K is infinite in Theorem 24.1 therefore cannot be omitted.

Edition note. Constancy here refers explicitly to the function on K -rational points. The morphism defined by the pair of polynomials $(t^q - t, t^q - t)$ is not constant. This distinction prevents “constant” from being mistaken for a statement about the polynomials or the morphism itself.

Example 24.3: a tangent line from a parametrisation

In this example we work over a field K of characteristic zero. Returning to Example 6.3, consider the curve

$$V(y^2 - x^2 - x^3)$$

with parametrisation

$$(\varphi(t), \psi(t)) = (t^2 - 1, t(t^2 - 1)) = (x, y).$$

For

$$F = y^2 - x^2 - x^3,$$

the partial derivatives are

$$\frac{\partial F}{\partial x} = -2x - 3x^2 \quad \text{and} \quad \frac{\partial F}{\partial y} = 2y.$$

The Jacobian matrix of the parametrisation, viewed as a row vector, is

$$\left(\frac{\partial \varphi}{\partial t}, \frac{\partial \psi}{\partial t} \right) = (2t, 3t^2 - 1).$$

With $P = (\varphi(t), \psi(t))$, the formal polynomial chain-rule computation indeed gives

$$\begin{aligned} & \left(\frac{\partial F}{\partial x}(P), \frac{\partial F}{\partial y}(P) \right) \begin{pmatrix} 2t \\ 3t^2 - 1 \end{pmatrix} \\ &= (-2(t^2 - 1) - 3(t^2 - 1)^2, 2(t^3 - t)) \begin{pmatrix} 2t \\ 3t^2 - 1 \end{pmatrix} \\ &= -4t(t^2 - 1) - 6t(t^2 - 1)^2 + 2(t^3 - t)(3t^2 - 1) \\ &= -4t^3 + 4t - 6t^5 + 12t^3 - 6t + 6t^5 - 2t^3 - 6t^3 + 2t \\ &= 0. \end{aligned}$$

For $t = 2$, for example, the image point is

$$P = (3, 6).$$

The derivative vector is $(4, 11)$, while the partial derivatives at P give the gradient $(-33, 12)$, which is perpendicular to the tangent direction vector. The tangent line can be written as

$$\{(3, 6) + s(4, 11) \mid s \in K\}$$

or as

$$V(-11x + 4y + 9).$$

Edition note. The source does not specify a characteristic restriction for this numerical example. This edition restricts it to characteristic zero because the coefficients 2, 3, and 11, as well as the particular value $t = 2$, can change or vanish upon reduction in small positive characteristic. In positive characteristic, smoothness and the tangent direction must be checked again over the field in question.

Tangent lines to space curves

Although our discussion is mainly restricted to plane curves, derivatives can also be used to define smooth and singular points on curves in higher-dimensional spaces, and indeed on arbitrary varieties. As an illustration, suppose that a space curve is given by two polynomials with no common component,

$$F, G \in K[X, Y, Z].$$

Not every space curve can be described in this way. For $P \in C = V(F, G)$, assume also that (F, G) generates the ideal of the reduced curve locally at P . Consider again the map given by the Jacobian matrix

$$\begin{pmatrix} \frac{\partial F}{\partial x} & \frac{\partial F}{\partial y} & \frac{\partial F}{\partial z} \\ \frac{\partial G}{\partial x} & \frac{\partial G}{\partial y} & \frac{\partial G}{\partial z} \end{pmatrix}_P : \mathbb{A}_K^3 \longrightarrow \mathbb{A}_K^2.$$

The point P is smooth on the curve precisely when this matrix has rank two. Its kernel is then one-dimensional and defines the tangent line.

Edition note — reduced-curve hypothesis. The source assumes only that F and G have no common component. For the intrinsic smoothness of the reduced curve, this alone is insufficient: $F = X^2, G = Y$ have no common component and their zero locus is the smooth Z -axis, but their Jacobian has rank one there. The added local ideal-generation hypothesis makes the stated rank criterion applicable to the reduced curve.

Example 24.4: the intersection of two cylinders

Assume $\text{char}(K) \neq 2$. Returning to Example 4.6, consider the intersection C of the two cylinders

$$F = x^2 + y^2 - 1 \quad \text{and} \quad G = y^2 + z^2 - 1.$$

Their partial derivative vectors are

$$\partial F = (2x, 2y, 0) \quad \text{and} \quad \partial G = (0, 2y, 2z).$$

A singular point occurs when the map defined by this Jacobian matrix has rank at most one, that is, when the two partial derivative vectors are linearly dependent and the point actually lies on the corresponding variety. Linear dependence requires

$$xy = xz = yz = 0.$$

On the curve with both parameters equal to 1, the curve equations exclude the cases $x = y = 0$ and $y = z = 0$. Thus the remaining candidates satisfy

$$x = z = 0,$$

and at these values the vectors are indeed linearly dependent for every y . When $x = z = 0$, only

$$y = \pm 1$$

give points on the curve. These are therefore exactly the two singular points of C . They are also the two intersection points of the two circles that form the irreducible components of C , as in Example 4.6.

For the version with unequal radii, write $r_1, r_2 \in K^\times$ for the *nonzero squared radii* and use

$$F = x^2 + y^2 - r_1, \quad G = y^2 + z^2 - r_2.$$

If the given quantities are the radii ρ_1 and ρ_2 themselves, then $r_i = \rho_i^2$. The dependence condition $xy = xz = yz = 0$ must now be solved together with the two curve equations. In the case $x = z = 0$, these equations become

$$y^2 = r_1 \quad \text{and} \quad y^2 = r_2.$$

Thus this case forces $r_1 = r_2$. The case $x = y = 0$ forces $r_1 = 0$, while $y = z = 0$ forces $r_2 = 0$. All three are impossible when $r_1, r_2 \neq 0$ and $r_1 \neq r_2$. The intersection curve is therefore smooth when the squared radii are nonzero and distinct. For real cylinders with positive radii, nonvanishing is automatic.

Edition note. The frozen live semantic text uses $G = y^2 + z^2 - 1$; the historical official PDF incorrectly prints $G = x^2 + z^2 - 1$. This edition follows the live semantic text. The source also calls r_1, r_2 “radii” but then uses the equations $y^2 = r_i$; here these parameters are clarified as squared radii. Their nonvanishing is also made explicit, since if $r_1 = 0$ or $r_2 = 0$, the Jacobian rank can drop even when $r_1 \neq r_2$. The assumption $\text{char}(K) \neq 2$ is made visible because all the displayed derivatives vanish in characteristic 2.

Power series rings

Definition 24.5: formal power series

Let R be a commutative ring and $T_1, \dots, T_n$ a set of variables. A *formal power series* is an expression of the form

$$F = \sum_{\nu} a_{\nu} T^{\nu} = \sum_{\nu} a_{\nu} T_1^{\nu_1} \cdots T_n^{\nu_n},$$

where

$$a_{\nu} \in R$$

for every multi-index

$$\nu = (\nu_1, \dots, \nu_n) \in \mathbb{N}^n.$$

Two power series are added coefficientwise and multiplied in the same way as polynomials. In one variable,

$$\begin{aligned} F \cdot G &= \left(\sum_{i=0}^{\infty} a_i T^i \right) \left(\sum_{j=0}^{\infty} b_j T^j \right) \\ &= \sum_{k=0}^{\infty} c_k T^k, \end{aligned}$$

where

$$c_k = \sum_{i=0}^k a_i b_{k-i}.$$

Definition 24.6: the power series ring

Let R be a commutative ring. The notation

$$R[[X_1, \dots, X_n]]$$

denotes the *power series ring in n variables*, also called the *ring of formal power series in n variables*.

We shall mainly use the one-variable power series ring $K[[T]]$ over a field K . Under suitable hypotheses, power series rings allow us to find “formal parametrisations” of branches of algebraic curves at a point; this will be treated in the next lecture. First we need to understand some basic properties of power series rings.

Edition note — scope of existence. The source states this for arbitrary algebraic curves at every point over a general field. Some hypothesis, or a suitable scalar extension, is necessary. For example, over $\mathbb{R}$ the curve $V(X^2 + Y^2)$ admits no nonconstant formal parametrisation through the origin by series in $\mathbb{R}[[T]]$. Writing the two series as $G = \sum a_i T^i$ and $H = \sum b_i T^i$, if n were the first degree occurring in either one, the coefficient of T^{2n} would be $a_n^2 + b_n^2 = 0$, forcing $a_n = b_n = 0$. The next lecture proves existence for a tangent of multiplicity one and also records an obstruction in a multiple-tangent example.

Theorem 24.7: the unit criterion

Let K be a field. A formal power series

$$F = \sum_{n=0}^{\infty} a_n T^n \in K[[T]]$$

is a unit if and only if its constant term satisfies $a_0 \neq 0$.

Proof The condition is necessary because formal evaluation at $T = 0$,

$$\begin{aligned} \text{ev}_0: K[[T]] &\longrightarrow K, \\ F &\longmapsto F(0) = a_0, \end{aligned}$$

is a ring homomorphism. A unit must therefore map to a nonzero element of K .

Conversely, assume $a_0 \neq 0$. We shall construct

$$G = \sum_{j=0}^{\infty} b_j T^j$$

such that

$$FG = \left(\sum_{i=0}^{\infty} a_i T^i \right) \left(\sum_{j=0}^{\infty} b_j T^j \right) = 1.$$

For the constant coefficient we require

$$a_0 b_0 = 1,$$

which has the unique solution $b_0 = a_0^{-1}$. Inductively, suppose that b_j has been constructed for $j < n$ so that all coefficients c_k of FG with $1 \leq k < n$ are zero. The condition for the n th coefficient is

$$0 = c_n = a_0 b_n + a_1 b_{n-1} + \dots + a_{n-1} b_1 + a_n b_0.$$

All values except b_n have already been determined. Since $a_0 \neq 0$, this equation has exactly one solution for b_n . This induction constructs an inverse G for F . $\square$

Edition note. The live semantic text ends its explanation of the constant-term homomorphism with an unfilled exercise placeholder. This edition does not retain that dangling reference; evaluation at $T = 0$ is stated and used directly.

Corollary 24.8: a discrete valuation ring

If K is a field, then the one-variable power series ring

$$R = K[[T]]$$

is a discrete valuation ring.

Proof First, R is a local ring with maximal ideal

$$\mathfrak{m} = (T).$$

Indeed, if a power series F is not a unit, Theorem 24.7 says that its constant term is zero. Consequently,

$$F = T\tilde{F}$$

for the power series $\tilde{F}$ obtained by shifting the indices.

The absence of zero divisors follows by considering initial terms. If F and G are nonzero power series, write

$$F = a_k T^k + a_{k+1} T^{k+1} + \dots$$

and

$$G = b_\ell T^\ell + b_{\ell+1} T^{\ell+1} + \dots,$$

where $a_k \neq 0$ and $b_\ell \neq 0$. Since all earlier coefficients vanish, the coefficient of degree $k + \ell$ in their product is

$$c_{k+\ell} = a_k b_\ell \neq 0.$$

It remains to show that R is Noetherian; in fact, it is a principal ideal domain. For a nonzero ideal $I \subseteq R$, let j be the smallest index of a nonzero coefficient among all series in I . Choose $H \in I$ with initial term of degree j . Then $H = T^j U$, where U is a unit by Theorem 24.7, so $T^j \in I$. The minimality of j also gives $I \subseteq (T^j)$, hence

$$I = (T^j).$$

Thus R is a local principal ideal domain with maximal ideal (T) , and is therefore a discrete valuation ring. $\square$

Edition note. Both the live semantic text and the historical PDF write the second term of G as $a_{\ell+1} T^{\ell+1}$. The correct coefficient family is $b_{\ell+1}$, as displayed above.

Power series can not only be added and multiplied. Under certain additional conditions, one power series can also be substituted into another. This operation corresponds to composition of maps.

Definition 24.9: substitution of power series

Let K be a field and

$$F = \sum_{i=0}^{\infty} a_i T^i \in K[[T]].$$

Let

$$G = \sum_{j=0}^{\infty} b_j T^j$$

be another power series with constant term $b_0 = 0$. The series

$$\begin{aligned} F(G) &= a_0 + a_1 \left(\sum_{j=0}^{\infty} b_j T^j \right) + a_2 \left(\sum_{j=0}^{\infty} b_j T^j \right)^2 \\ &\quad + a_3 \left(\sum_{j=0}^{\infty} b_j T^j \right)^3 + \cdots \\ &= \sum_{k=0}^{\infty} c_k T^k \end{aligned}$$

is called the *composite power series*. Its coefficients are determined by

$$c_0 = a_0$$

and, for $k \geq 1$,

$$c_k = \sum_{s=0}^k a_s \left(\sum_{j_1 + \cdots + j_s = k} b_{j_1} \cdots b_{j_s} \right),$$

where the inner sum runs over all ordered s -tuples

$$(j_1, \dots, j_s) \in \mathbb{N}_+^s.$$

Since $b_0 = 0$, only indices $j \geq 1$ occur, so every sum determining a coefficient is finite. These formulas agree with ordinary polynomial substitution when F and G are polynomials. Substituting power series into power series produces substitution homomorphisms between power series rings.

Lemma 24.10: substitution is a homomorphism

Let K be a field and $G \in K[[S]]$ a power series with constant term zero. Substitution of G defines a K -algebra homomorphism

$$\begin{aligned} K[[T]] &\longrightarrow K[[S]], \\ F &\longmapsto F(G). \end{aligned}$$

Proof The map is well defined. To show that it is a ring homomorphism, we need only compare the relevant coefficients. Each depends on only finitely many coefficients of the series involved. The required identities therefore follow from the polynomial case. The map also preserves scalars in K , so it is a K -algebra homomorphism. $\square$

Lemma 24.11: a formal change of parameter

Let K be a field and

$$G = \sum_{j=0}^{\infty} b_j T^j \in K[[T]]$$

with $b_0 = 0$ and $b_1 \neq 0$. Then the substitution homomorphism determined by

$$T \mapsto G$$

is a K -algebra automorphism of $K[[T]]$.

Proof We first construct a power series

$$F = \sum_{i=0}^{\infty} a_i T^i$$

with

$$F(G) = T.$$

We must have $a_0 = 0$ and $a_1 = b_1^{-1}$. For $k \geq 2$, suppose inductively that the coefficients of F through a_{k-1} have been constructed to give the required coefficients. By Definition 24.9, the condition on c_k is

$$\begin{aligned} 0 = c_k &= \sum_{s=0}^k a_s \left(\sum_{j_1+\dots+j_s=k} b_{j_1} \cdots b_{j_s} \right) \\ &= \sum_{s=0}^{k-1} a_s \left(\sum_{j_1+\dots+j_s=k} b_{j_1} \cdots b_{j_s} \right) + a_k b_1^k. \end{aligned}$$

Since $b_1 \neq 0$, this equation determines a_k uniquely.

Now consider the composite

$$K[[T]] \xrightarrow{T \mapsto F} K[[T]] \xrightarrow{T \mapsto G} K[[T]].$$

The composite map is substitution $T \mapsto T$, which is the identity. Therefore the second map, determined by $T \mapsto G$, is surjective. By Corollary 24.8, $K[[T]]$ is a discrete valuation ring, and its ideals are known. If the kernel of the second map were nonzero, it would contain some T^j , but its image is $G^j \neq 0$. Only the zero ideal can therefore be the kernel. The map is also injective, hence bijective and a K -algebra automorphism. $\square$

Worksheet 24

Warm-up exercises

Exercise 24.1

Give an example of a smooth curve

$$C \subseteq \mathbb{A}_K^2$$

with a parametrisation whose differential vanishes at at least one point.

Exercise 24.2

Consider the union of the coordinate axes

$$V(xy) \subseteq \mathbb{A}_K^2$$

and the local ring R at the origin, with maximal ideal $\mathfrak{m}$. Describe explicitly a K -basis of the quotient rings $R/\mathfrak{m}^n$ and determine their dimensions.

Exercise 24.3

Let K be a field and $K[[T]]$ the formal power series ring. Determine the power series inverse of $1 - T$.

Exercise 24.4 ★

Let K be a field and

$$\mathfrak{m} = (T) \subseteq K[T]$$

the maximal ideal corresponding to the origin, with localisation

$$R = K[T]_{\mathfrak{m}}.$$

Define a K -algebra homomorphism

$$\varphi : R \longrightarrow K[[T]]$$

satisfying $\varphi(T) = T$, where $K[[T]]$ denotes the formal power series ring.

Exercise 24.5

Compute the first five coefficients, up to and including c_4 , of the composite power series $F(G)$ in the sense of Definition 24.9.

Exercises for submission**Exercise 24.6 (5 points)**

Give an example of an irreducible real polynomial

$$F \in \mathbb{R}[X, Y]$$

whose two partial derivatives agree and are nonconstant. Show that this is impossible over $\mathbb{C}$.

Exercise 24.7 (3 points)

Consider the curve

$$C = V(Y^2 - X^2 - X^3)$$

with the parametrisation discussed in Example 24.3. Determine the singular points of the curve, together with their multiplicities and tangent lines. Also compute the image points and tangent lines for the parameter values

$$t = -1, 0, 1.$$

For the geometric conclusions about tangent lines and the parameter values above, take the base field to be $\mathbb{R}$; in particular, it has characteristic zero.

Edition note - correction to the source equation: The source displays $C = V(X^2 - Y^2 - Y^3)$, but the referenced parametrisation is

$$(x, y) = (t^2 - 1, t(t^2 - 1)).$$

Direct substitution gives

$$y^2 - x^2 - x^3 = (t^2 - 1)^2 (t^2 - 1 - (t^2 - 1)) = 0.$$

The edition therefore uses $C = V(Y^2 - X^2 - X^3)$, in agreement with the parametrisation and the source's object category.

Exercise 24.8 (3 points)

Describe a formal power series over $\mathbb{C}$ that converges in no neighbourhood of the origin.

Exercise 24.9 (3 points)

Let K be a field. Compare the two rings

$$(K[X])[[Y]] \quad \text{and} \quad (K[[Y]])[X].$$

In particular, determine whether one is contained in the other and, if so, in which direction the inclusion holds.

Exercise 24.10 (6 points)

Let R be a Noetherian commutative ring. Show that

$$R[[T_1, \dots, T_n]]$$

is Noetherian.

Hint. Take inspiration from the proof of Hilbert's basis theorem.

Public Solutions to Worksheet 24

At the frozen revision boundary, the source provides a public solution only to Exercise 24.4. No additional solutions have been created for this edition.

Solution to Exercise 24.4

We start with the K -algebra homomorphism

$$\begin{aligned} K[T] &\longrightarrow K[[T]], \\ T &\longmapsto T. \end{aligned}$$

Every polynomial $P \in K[T] \setminus (T)$ has constant term

$$P(0) \neq 0.$$

By the unit criterion for the formal power series ring, P is a unit in $K[[T]]$. Since every element of the denominator set $K[T] \setminus (T)$ maps to a unit, the universal property of localisation gives a unique K -algebra homomorphism

$$K[T]_{(T)} \longrightarrow K[[T]].$$

Explicitly, it is given by

$$\frac{f}{P} \longmapsto fP^{-1}.$$

Edition note – correction to the source solution: After stating $P \notin (T)$, the source displays only the symbol $\neq 0$ without a left-hand side. The edition restores the required statement $P(0) \neq 0$; German spelling errors with no mathematical effect are also naturally corrected in translation.

[Back to Exercise 24.4](#)

Lecture 25: Power Series Solutions for Algebraic Curves

Power series solutions for algebraic curves

Let $F \neq 0$ be a polynomial describing a plane algebraic curve C , and assume that

$$P = (0, 0) \in C.$$

This is no restriction, since it can always be achieved by a translation. How can we describe the curve near the origin using power series? In other words, when is there a ring homomorphism determined by nonconstant power series G and H with constant term zero,

$$\begin{aligned} K[X, Y] &\longrightarrow K[[T]], \\ X &\longmapsto G, \\ Y &\longmapsto H, \end{aligned}$$

such that

$$F(G, H) = 0?$$

Equivalently, we seek a ring homomorphism

$$K[X, Y]/(F) \longrightarrow K[[T]].$$

Thus the problem is to find power series solutions of the equation

$$F(X, Y) = 0$$

that describe more precisely the behaviour of the curve around the point solution $(0, 0)$.

The basic approach is a power series ansatz, as also used in the theory of differential equations. We start with

$$G = \sum_{k=0}^{\infty} a_k T^k \quad \text{and} \quad H = \sum_{\ell=0}^{\infty} b_{\ell} T^{\ell},$$

where the coefficients a_k and b_{ℓ} are initially unknown. Direct substitution into $F = 0$, followed by expansion of the products, produces an expression that is in principle infinite. For each power T^k , however, the expression for its coefficient is determined by only finitely many data: the finitely many coefficients of F , and only the coefficients of G and H through degree k , are needed.

Edition note — coefficient range. The source says only coefficients below degree k are relevant. A degree- k coefficient can be required: for $F = X$, the coefficient of T^k in $F(G, H)$ is a_k . The finite-dependence claim is retained with the corrected bound “through degree k ”.

Since we require

$$F(G, H) = 0,$$

the coefficients of F , G , and H must make the coefficient of every T^k vanish.

We then seek conditions for the existence of solutions, their form, and their uniqueness. The condition

$$a_0 = b_0 = 0$$

is an initial condition expressing that the power series solution passes through the origin.

A condition on the linear terms of the power series, namely a_1 and b_1 , emerges immediately. This further justifies interpreting the linear factors of the lowest-degree homogeneous component F_m in the homogeneous decomposition of F as tangent-line equations.

Lemma 25.1: the linear term lies on a tangent line

Let K be an algebraically closed field and

$$F \in K[X, Y]$$

a polynomial with homogeneous decomposition

$$F = F_m + \cdots + F_d, \quad d \geq m \geq 1, \quad F_m \neq 0.$$

Let

$$F_m = \prod_{\lambda=1}^m (u_{\lambda} X + v_{\lambda} Y)$$

be the factorisation of F_m into linear factors. These linear factors define the tangent lines to the curve

$$C = V(F)$$

at $P = (0, 0)$. Let

$$G = \sum_{n=0}^{\infty} a_n T^n \quad \text{and} \quad H = \sum_{\ell=0}^{\infty} b_\ell T^\ell$$

be elements of $K[[T]]$ giving a solution of the curve equation through the origin, that is,

$$a_0 = b_0 = 0 \quad \text{and} \quad F(G, H) = 0.$$

Then, for some λ ,

$$u_\lambda a_1 + v_\lambda b_1 = 0.$$

In other words, the pair of linear terms of the two power series is constrained by one of the tangent lines.

Proof Substitute

$$G = a_1 T + a_2 T^2 + \dots \quad \text{and} \quad H = b_1 T + b_2 T^2 + \dots$$

into F . A homogeneous component F_k is a sum of terms $c_{ij} X^i Y^j$ with $i + j = k$. We can immediately factor out T^k and obtain an expression of the form

$$\begin{aligned} F_k(G, H) &= \left(\sum_{i+j=k} c_{ij} a_1^i b_1^j \right) T^k \\ &\quad + \left(\sum_{i+j=k} c_{ij} (i a_1^{i-1} a_2 b_1^j + j a_1^i b_1^{j-1} b_2) \right) T^{k+1} + \dots \end{aligned}$$

Thus a_1 and b_1 enter the coefficient of T^k directly through F_k ; in general that coefficient also receives more complicated contributions from F_ℓ with $\ell < k$. For F_m , there are no lower homogeneous components. The decisive equation for a_1 and b_1 is therefore

$$\sum_{i+j=m} c_{ij} a_1^i b_1^j = 0,$$

or, equivalently,

$$F_m(a_1, b_1) = 0.$$

Since F_m is a product of linear factors, the row vector (a_1, b_1) must annihilate one of them. This is the assertion. $\square$

Note that Lemma 25.1 does not exclude the possibility

$$a_1 = b_1 = 0.$$

Indeed, realising a curve by power series along a prescribed tangent line is possible only under additional conditions; see [Theorem 25.2](#) and the examples below.

The computational effort needed to determine a power series solution can be greatly reduced by restricting to “graph solutions”, where one power series is simply a linear polynomial prescribed by a tangent line and the other is a power series to be determined. This is often no essential restriction, as follows from [Lemma 24.11](#). That lemma lets us easily reparametrise

$$G, H \in K[[T]]$$

when their linear terms do not both vanish. Assume

$$G = a_1 T + \cdots, \quad a_1 \neq 0.$$

Choose a power series $U(T)$ that is the compositional inverse of G . Then

$$G(U(T)) = T \quad \text{and} \quad H(U(T)) = \widetilde{H}(T).$$

Following the original map by a power series ring automorphism gives the composite

$$K[X, Y] \xrightarrow{X \mapsto G, Y \mapsto H} K[[T]] \xrightarrow{T \mapsto U(T)} K[[T]],$$

which has the particularly simple form

$$X \mapsto T, \quad Y \mapsto \widetilde{H}.$$

That is, we seek to realise the curve as the graph of a formal function in one variable.

Edition note - correction to the source's composition order. The source writes $U(G(T)) = T$ and $U(H(T)) = \widetilde{H}(T)$. However, the displayed arrow substitutes $T \mapsto U(T)$ after the first map, so its resulting images are $G(U(T))$ and $H(U(T))$. The edition uses the composition order corresponding to that arrow.

Theorem 25.2: a tangent of multiplicity one gives a graph solution

Let K be a field and

$$F \in K[X, Y]$$

a nonzero polynomial with

$$(0, 0) \in C = V(F).$$

Let

$$F = F_d + \cdots + F_m, \quad d \geq m, \quad F_m \neq 0,$$

be the homogeneous decomposition of F , and let $uX + vY$ be a simple linear factor of F_m , that is, a linear polynomial defining a tangent line of multiplicity 1. Then there are power series

$$G = \sum_{n=0}^{\infty} a_n T^n, \quad H = \sum_{\ell=0}^{\infty} b_\ell T^\ell \in K[[T]]$$

such that

$$F(G, H) = 0, \quad a_0 = b_0 = 0, \quad a_1 u + b_1 v = 0.$$

Moreover, one of the two power series can be chosen to be a linear polynomial.

Edition note - clarification of the initial condition. The source writes “ $a_0, b_0 = 0$ ”. The edition states the intended equality unambiguously as $a_0 = b_0 = 0$.

Proof By a linear change of variables, we may assume that

$$uX + vY = Y.$$

We shall construct a power series solution with

$$G = T$$

and

$$H = b_2T^2 + b_3T^3 + \cdots.$$

Since $a_1 = 1$ and $b_1 = 0$, this solution satisfies the linear condition specified by the tangent line.

Write

$$F = \sum_{i,j} c_{ij} X^i Y^j.$$

We have

$$c_{m,0} = 0,$$

since otherwise Y could not be a linear factor of F_m . Moreover,

$$c_{m-1,1} \neq 0,$$

since if this coefficient were zero, Y would be a linear factor with multiplicity at least 2.

We now show that these initial data determine a unique power series

$$H = b_2T^2 + b_3T^3 + \cdots.$$

Substituting G and H into F gives one condition for each k , since the resulting coefficient of T^k must be zero. The k th coefficient is a sum of expressions of the form

$$c_{ij} b_{\ell_1} \cdots b_{\ell_j}, \quad i + \sum_{\rho=1}^j \ell_\rho = k.$$

These expressions may occur repeatedly, with a multinomial coefficient. Since $\ell_\rho \geq 2$, the term b_ℓ does not yet occur when

$$k < m + \ell - 1.$$

It first occurs in the coefficient with

$$k = m + \ell - 1,$$

and its only occurrence there is

$$c_{m-1,1} b_\ell.$$

The other terms in that coefficient involve only the c_{ij} and b_r with $r < \ell$. Since $c_{m-1,1} \neq 0$, this determines b_ℓ uniquely. The coefficients b_ℓ can therefore be constructed inductively, each value being uniquely determined by the corresponding coefficient equation. $\square$

Example 25.3: the graph of a rational function

Consider the affine plane curve of degree three given by

$$F = X^3 + XY + Y = 0.$$

Its partial derivatives are

$$\frac{\partial F}{\partial X} = 3X^2 + Y \quad \text{and} \quad \frac{\partial F}{\partial Y} = X + 1.$$

The second partial derivative vanishes only when $X = -1$, but on that line F has value -1 . Thus the curve is smooth. At the origin the partial derivatives have values $(0, 1)$. The tangent line is therefore the X -axis, in agreement with the fact that the linear term of the curve equation is Y .

We compute the power series

$$Y = H(T) = \sum_{\ell=0}^{\infty} b_{\ell} T^{\ell}$$

that describes the curve as a graph at the origin, with $X = T$. The initial conditions are

$$b_0 = b_1 = 0.$$

The subsequent coefficients must satisfy

$$F(T, H) = T^3 + TH + H = 0,$$

or

$$T^3 + T(b_2T^2 + b_3T^3 + \cdots) + (b_2T^2 + b_3T^3 + \cdots) = 0.$$

For b_2 , the second coefficient of the equation immediately gives

$$b_2 = 0.$$

For b_3 , the third coefficient gives

$$1 + b_3 = 0,$$

hence $b_3 = -1$. The subsequent coefficients give the relation

$$b_{\ell-1} + b_{\ell} = 0.$$

Thus the later coefficients alternate between 1 and -1 , giving a simple recurrence, and

$$H = -T^3 + T^4 - T^5 + T^6 - T^7 + \cdots.$$

Rewriting the curve equation as

$$Y = \frac{-X^3}{1 + X}$$

shows that this is the graph of a rational function with a pole at $X = -1$. The power series above describes that rational function's graph as the graph of a formal-analytic function.

Example 25.4: the folium of Descartes as a formal graph

Consider the folium of Descartes

$$X^3 + Y^3 - 3XY = 0$$

at the origin, with tangent line $Y = 0$. We seek the power series describing as a graph the branch of the curve corresponding to this tangent line. Set

$$X = T$$

and

$$H = b_2T^2 + b_3T^3 + b_4T^4 + \dots,$$

assuming that the characteristic of K is not 3. The coefficients b_ℓ are determined by

$$\begin{aligned} 0 &= T^3 + H^3 - 3TH \\ &= T^3 + (b_2T^2 + b_3T^3 + \dots)^3 - 3T(b_2T^2 + b_3T^3 + \dots). \end{aligned}$$

This substitution and expansion first gives a condition at $k = 3$. The term X^3 , or T^3 , needs to be considered only once, when $k = 3$. The term Y^3 contributes only from $k \geq 6$ onwards, since $Y = H$ is a multiple of T^2 . The term XY must be considered from $k = 3$ onwards.

For b_2 we obtain

$$1 - 3b_2 = 0,$$

hence

$$b_2 = \frac{1}{3}.$$

The coefficient b_3 first occurs in the condition for the fourth coefficient, where it stands alone, so

$$b_3 = 0.$$

For the same reason,

$$b_4 = 0.$$

For b_5 , the sixth coefficient is decisive, and now the term Y^3 must also be included. The condition is

$$b_2^3 - 3b_5 = 0,$$

so

$$b_5 = \frac{1}{81}.$$

For b_6, b_7, b_8 , note that the term

$$Y^3 = (b_2T^2 + b_5T^5 + \dots)(b_2T^2 + b_5T^5 + \dots)(b_2T^2 + b_5T^5 + \dots)$$

next contributes at the ninth coefficient, with contribution $3b_2^2b_5$. Thus b_6 and b_7 stand alone and must be zero. For b_8 we obtain

$$3b_2^2b_5 - 3b_8 = 0,$$

hence

$$b_8 = \frac{1}{729}.$$

The beginning of the power series describing this branch of the curve as a graph is therefore

$$H = \frac{1}{3}T^2 + \frac{1}{81}T^5 + \frac{1}{729}T^8 + \dots$$

Example 25.5: Neil's parabola without a nonzero linear term

Consider Neil's parabola given by

$$X^3 - Y^2 = 0.$$

The origin is singular and has only one tangent line, namely

$$Y = 0.$$

However, this tangent has multiplicity two, so [Theorem 25.2](#) does not apply. In fact, there is no power series solution at the origin with a nonzero linear term.

To see this, suppose that

$$X = G = a_1T + a_2T^2 + \dots$$

and

$$Y = H = b_1T + b_2T^2 + \dots$$

satisfy the curve equation. After substitution, the second coefficient gives

$$-b_1^2 = 0,$$

so $b_1 = 0$. The third coefficient then gives

$$a_1^3 = 0,$$

so $a_1 = 0$ as well.

Nevertheless, there are power series solutions for Neil's parabola through the origin. We may take the monomial solution

$$G = T^2 \quad \text{and} \quad H = T^3.$$

This even gives a bijection between the affine line and Neil's parabola, but its linear term is indeed zero.

Remark 25.6: comparison with the implicit function theorem

Let

$$K = \mathbb{R} \quad \text{or} \quad K = \mathbb{C},$$

and $F \in K[X, Y]$. If

$$\left(\frac{\partial F}{\partial x}(P), \frac{\partial F}{\partial y}(P) \right) \neq (0, 0),$$

that is, if P is a regular point of the function F , or equivalently a smooth point of

$$C = V(F - F(P)),$$

then the implicit function theorem guarantees that, in a metric neighbourhood of P , the curve can be expressed as the graph of a differentiable function.

Edition note - correction to the source wording. The source writes “ $K = \mathbb{R}$ or $= \mathbb{C}$ ”. The edition supplies the subject in the second alternative, writing $K = \mathbb{R}$ or $K = \mathbb{C}$.

Worksheet 25**Warm-up exercises****Exercise 25.1 ★**

For the plane algebraic curve

$$V(X^3 + Y^2 - XY + X),$$

determine a nonconstant power series solution

$$X = F(Y)$$

at the origin through degree six.

Exercise 25.2 ★

For the plane algebraic curve

$$V(X^2Y + X^2 + Y^2 - 5XY + Y),$$

determine a nonconstant power series solution

$$Y = F(X)$$

at the origin through fifth order.

The following exercises concern the completion of a local ring.

Exercise 25.3

Let R be a local ring with maximal ideal $\mathfrak{m}$. Consider the diagram

$$\longrightarrow R/\mathfrak{m}^4 \longrightarrow R/\mathfrak{m}^3 \longrightarrow R/\mathfrak{m}^2 \longrightarrow R/\mathfrak{m}.$$

The maps are the canonical projections

$$\varphi_n : R/\mathfrak{m}^{n+1} \longrightarrow R/\mathfrak{m}^n$$

induced by the ideal inclusions $\mathfrak{m}^{n+1} \subseteq \mathfrak{m}^n$. A sequence of elements

$$a_n \in R/\mathfrak{m}^n$$

is called *compatible* if

$$\varphi_n(a_{n+1}) = a_n$$

for every n . Define a ring structure on the set of all such compatible sequences. This ring is called the *completion* of R . Also show that there is a canonical ring homomorphism from R to its completion.

Exercise 25.4

Let R be a one-dimensional Noetherian local commutative ring. Show that the canonical map from R to its completion is injective.

Remark. This injectivity holds for every Noetherian local ring, but the proof is more difficult.

Exercise 25.5

Let R be a commutative ring and I an ideal. Show that, for each $x \in R$, the family

$$\{x + I^n \mid n \in \mathbb{N}\}$$

defines a neighbourhood basis at x . These families define the I -adic topology on R . Show also that this topology is Hausdorff if and only if

$$\bigcap_n I^n = \{0\}.$$

Remark. The completion of a local ring with respect to its maximal ideal is precisely its topological completion for this topology.

Exercises for submission**Exercise 25.6 (4 points)**

Consider the cardioid

$$V((X^2 + Y^2)^2 - 2X(X^2 + Y^2) - Y^2)$$

at $(2, 0)$. Determine a formal parametrisation of the curve at this point, through the fifth term, in terms of a tangent parameter.

Edition note - base field. The source does not specify the base field. For the geometric interpretation of the cardioid and tangent parameter in this exercise, the edition uses $\mathbb{R}$ as the base field; in particular, it has characteristic zero.

Exercise 25.7 (4 points)

Let K be a field with $\text{char}(K) \neq 2$. Consider the unit circle

$$X^2 + Y^2 = 1$$

at $(1, 0)$. Determine power series

$$G, H \in K[[T]]$$

with initial conditions

$$a_0 = 1, \quad a_1 = 0, \quad b_0 = 0, \quad b_1 = 1,$$

and satisfying

$$G(T)^2 + H(T)^2 = 1.$$

Edition note - characteristic. The source places no restriction on the characteristic of K . The condition $\text{char}(K) \neq 2$ is added because, in characteristic 2, the coefficient of T^2 in the required equation would force $1 = 0$, so series with these initial conditions could not exist.

Exercise 25.8 (4 points)

Consider Neil's parabola

$$C = V(Y^3 - X^2)$$

at $(1, 1)$. Find a parametrisation of the curve at this point by power series through the fifth term, such that one of the series is a linear polynomial.

Exercise 25.9 (3 points)

Let K be a field. A *formal Laurent series with finite principal part* is an infinite sum of the form

$$F = \sum_{n=k}^{\infty} a_n T^n, \quad a_n \in K, \quad k \in \mathbb{Z}.$$

Show that the ring of these formal series, with suitable ring operations, is isomorphic to the field of fractions of the power series ring $K[[T]]$.

Exercise 25.10 (4 points)

Let K be a field and $K[T]$ the polynomial ring in one variable. Let R be the localisation of $K[T]$ at the maximal ideal

$$\mathfrak{m} = (T).$$

Show that the completion of R is isomorphic to the power series ring $K[[T]]$.

Exercise 25.11 (4 points)

Let K be a field and

$$R = K[[T]]$$

the power series ring. Show that T has no square root in R . Show also that, when $K = \mathbb{Z}/(7)$, the element $T + 2$ has a square root in R , and determine the first five coefficients of one such square root.

Exercise 25.12 (5 points)

Let

$$F \in K[X, Y]$$

be an irreducible polynomial and

$$R = K[X, Y]/(F)$$

the integral coordinate ring of the plane curve

$$C = V(F).$$

Let

$$R \longrightarrow S = R^{\text{norm}}$$

be the normalisation of R , and let

$$R \longrightarrow K[[T]]$$

be the ring homomorphism corresponding to a nonconstant formal power series solution of the curve. Show that there is a unique ring homomorphism

$$S \longrightarrow K[[T]]$$

making the diagram

$$\begin{array}{ccc} R & \longrightarrow & S \\ & \searrow & \downarrow \\ & & K[[T]] \end{array}$$

commute.

Exercise for upload**Exercise 25.13 (4 points)**

Using suitable software, plot one of the example curves from the lecture, together with the various polynomial approximations computed there.

Edition note - discrepancy in source points. The official worksheet displays 4 points for this exercise, whereas the transcluded semantic exercise page records 3 points. The edition retains the displayed value of 4 and records the exercise page's value of 3 without silently reconciling them.

Public Solutions to Worksheet 25

At the frozen revision boundary, the source provides public solutions only to Exercises 25.1 and 25.2. The frozen authority query records the other eleven candidate solution pages as absent. No additional solutions have been created for this edition.

Solution to Exercise 25.1

We make the ansatz

$$X = F(Y) = \sum_{i=0}^{\infty} a_i Y^i$$

and determine the coefficients $a_0, \dots, a_6$ from the condition

$$\left(\sum_{i=0}^{\infty} a_i Y^i \right)^3 + Y^2 - \left(\sum_{i=0}^{\infty} a_i Y^i \right) Y + \left(\sum_{i=0}^{\infty} a_i Y^i \right) = 0.$$

Since the power series must approximate the curve at the origin, we must have

$$a_0 = 0.$$

For Y^1 , the coefficient condition is

$$a_1 = 0,$$

since the first three summands in the equation contribute nothing. For Y^2 , we obtain

$$1 + a_2 = 0, \quad \text{hence} \quad a_2 = -1.$$

This deals with the second summand Y^2 . For Y^3 , we obtain

$$-a_2 + a_3 = 0, \quad \text{hence} \quad a_3 = a_2 = -1.$$

For Y^4 , we obtain

$$-a_3 + a_4 = 0, \quad \text{hence} \quad a_4 = a_3 = -1,$$

and for Y^5 ,

$$-a_4 + a_5 = 0, \quad \text{hence} \quad a_5 = a_4 = -1.$$

At Y^6 , the first summand must be included for the first time. We obtain

$$a_2^3 - a_5 + a_6 = 0, \quad \text{hence} \quad a_6 = a_5 - a_2^3 = -1 - (-1)^3 = 0.$$

[Back to Exercise 25.1](#)

Solution to Exercise 25.2

We make the ansatz

$$Y = F(X) = \sum_{n=0}^{\infty} a_n X^n$$

(and $X = X$), and determine the coefficients successively by comparing coefficients of powers of X . Since the solution must pass through the origin, we require $a_0 = 0$.

$$X^1 : \quad a_1 = 0.$$

$$X^2 : \quad 1 + a_2 = 0, \quad \text{hence} \quad a_2 = -1.$$

$$X^3 : \quad -5a_2 + a_3 = 0, \quad \text{hence} \quad a_3 = -5.$$

$$X^4 : \quad a_2 + a_2^2 - 5a_3 + a_4 = 0, \quad \text{hence} \quad a_4 = 5a_3 = -25.$$

$$\begin{aligned} X^5 : \quad a_3 + 2a_2a_3 - 5a_4 + a_5 &= 0, \\ a_5 &= -a_3 - 2a_2a_3 + 5a_4 \\ &= 5 - 10 - 125 \\ &= -130. \end{aligned}$$

The initial terms of the power series are therefore

$$F = -X^2 - 5X^3 - 25X^4 - 130X^5 + \dots$$

[Back to Exercise 25.2](#)

Lecture 26: Intersection Multiplicity

Intersection multiplicity

Let two plane algebraic curves

$$C, D \subseteq \mathbb{A}_K^2$$

be given, with no common component. By Theorem 4.8, the intersection $C \cap D$ consists of only finitely many points. We want to describe quantitatively how the two curves intersect at a point

$$P \in C \cap D.$$

For this purpose, it is useful to consider a slightly more general situation. We write

$$C = V(F) \quad \text{and} \quad D = V(G),$$

and allow prime factors to occur repeatedly in both F and G . In other words, from now on we distinguish $V(F)$ from $V(F^n)$, although they are the same geometric object.

Lemma 26.1: finite dimension of the quotient

Let K be a field and let

$$F, G \in K[X, Y]$$

be two polynomials with no common prime divisor. Let

$$P \in V(F, G)$$

and let

$$R = K[X, Y]_{\mathfrak{m}_P}$$

be the corresponding localisation. Then the quotient ring

$$R/(F, G)$$

is finite-dimensional as a vector space over K .

Proof Let $\mathfrak{m}$ be the maximal ideal of R . Since F and G have no common divisor, there is no other prime ideal between (F, G) and $\mathfrak{m}$ in R . Consequently, every nonunit in $R/(F, G)$ is nilpotent. Thus, for some s ,

$$\mathfrak{m}^s \subseteq (F, G) \subseteq \mathfrak{m}.$$

Hence there is a surjection

$$R/\mathfrak{m}^s \longrightarrow R/(F, G).$$

By Lemma 23.3 in the source numbering, the ring on the left is finite-dimensional over K . The ring on the right is therefore also finite-dimensional over K . $\square$

This lemma makes the following definition meaningful.

Definition 26.2: intersection multiplicity

Let K be a field and let

$$F, G \in K[X, Y]$$

be two nonconstant polynomials with no common component, and let

$$P \in V(F) \cap V(G) = V(F, G).$$

The dimension

$$\dim_K (K[X, Y]_{\mathfrak{m}_P}/(F, G))$$

is called the *intersection multiplicity* of the curves $V(F)$ and $V(G)$ at P . It is denoted by

$$\text{mult}_P(F, G) \quad \text{or} \quad \text{mult}_P(V(F), V(G)).$$

Example 26.3: intersection of a curve with a line

Let

$$C = V(F)$$

and a line

$$L = V(cX + dY)$$

in the affine plane $\mathbb{A}_K^2$ be given, where L is not a component of C . Let

$$P = (a, b) \in C \cap L.$$

The quotient ring

$$K[X, Y]_{\mathfrak{m}_P} / (F, cX + dY)$$

can be computed by solving the linear equation for one of the variables. If $d \neq 0$, substitute

$$Y = -\frac{c}{d}X$$

into F to obtain the one-variable polynomial

$$\tilde{F}(X) = F\left(X, -\frac{c}{d}X\right).$$

Thus

$$K[X, Y]_{\mathfrak{m}_P} / (F, cX + dY) \cong K[X]_{(X-a)} / (\tilde{F}).$$

If $d = 0$, then $c \neq 0$, so we solve for X and obtain the analogous statement in Y , localised at $(Y - b)$. Equivalently, we may first form the one-variable quotient ring and then localise at the corresponding point.

Now suppose that K is algebraically closed. In the case $d \neq 0$, we have a factorisation

$$\tilde{F} = u(X - \lambda_1)^{\nu_1} \cdots (X - \lambda_k)^{\nu_k}, \quad u \in K^\times.$$

Since P is a zero, we must have $a = \lambda_i$ for some i . On localising at $(X - a)$, all the other linear factors become units. The remaining factor gives a ring isomorphic to

$$K[X] / (X - \lambda_i)^{\nu_i},$$

which has dimension ν_i over K .

Editorial note - elimination cases and localisation. The source immediately replaces Y by $-(c/d)X$ without stating the condition $d \neq 0$, and then writes $K[X]_P$. This edition separates the cases $d \neq 0$ and $c \neq 0$ and specifies the correct one-variable localisation ideal, namely $(X - a)$ or $(Y - b)$. The source also suppresses the nonzero leading coefficient in the factorisation of $\tilde{F}$; the scalar u is displayed here and, being a unit, does not affect the quotient or its dimension.

Lemma 26.4: intersection with a line

Let K be an algebraically closed field, let

$$F = F_m + \cdots + F_d \in K[X, Y], \quad m \leq d,$$

be the homogeneous decomposition of a polynomial, and let

$$L = V(aX + bY)$$

be a line through the origin P which is not a component of $V(F)$. Then

$$\text{mult}_P(L, V(F)) \geq m_P(F) = m.$$

In other words, the intersection multiplicity of a curve with a line is at least the multiplicity of the curve at the intersection point. If L is not a tangent to the curve, equality holds.

Proof Set

$$R = K[X, Y]_{(X, Y)} \quad \text{and} \quad H = aX + bY.$$

Without loss of generality, suppose $b \neq 0$, so that $H = 0$ can be written as $Y = cX$ for some $c \in K$. If $b = 0$, then $a \neq 0$, and the same argument applies after interchanging X and Y .

First suppose that L is not a tangent to $V(F)$ at P , and hence L is not a component of $V(F_m)$. Then

$$R/(F, H) \cong K[X]_{(X)} / (F_m(X, cX) + \cdots + F_d(X, cX)).$$

Since $F_m(X, cX) \neq 0$, the polynomial generating the ideal can be written as $X^m u$ with u a unit. The quotient ring therefore has dimension m over K .

In the general case there is a least index i , with $m \leq i \leq d$, such that

$$F_i(X, cX) \neq 0.$$

Such an index must exist, since otherwise L would be a component of $V(F)$. By the same argument, the dimension of the quotient ring is $i \geq m$. $\square$

Editorial note - choice of coordinates in the proof. The source assumes $b \neq 0$ without explaining the other case. This edition states that the choice is without loss of generality, since for $b = 0$ the variables X and Y can be interchanged.

Lemma 26.5: basic properties

Let K be an algebraically closed field, let

$$F, G \in K[X, Y]$$

be two polynomials with no common component, and let $P \in \mathbb{A}_K^2$. Then the following hold:

1. $\text{mult}_P(F, G) = 0$ if and only if $P \notin V(F, G)$.
2. $\text{mult}_P(F, G) = \text{mult}_P(G, F)$.
3. Intersection multiplicity is unchanged by an affine change of variables.
4. If $F = F_1 F_2$ and $F_2(P) \neq 0$, then

$$\text{mult}_P(F, G) = \text{mult}_P(F_1, G).$$

5. For every $H \in K[X, Y]$,

$$\text{mult}_P(F, G) = \text{mult}_P(F, G + HF).$$

Proof. This is immediate.

The fourth assertion can also be phrased as follows: intersection multiplicity depends only on those components of F and G that pass through P .

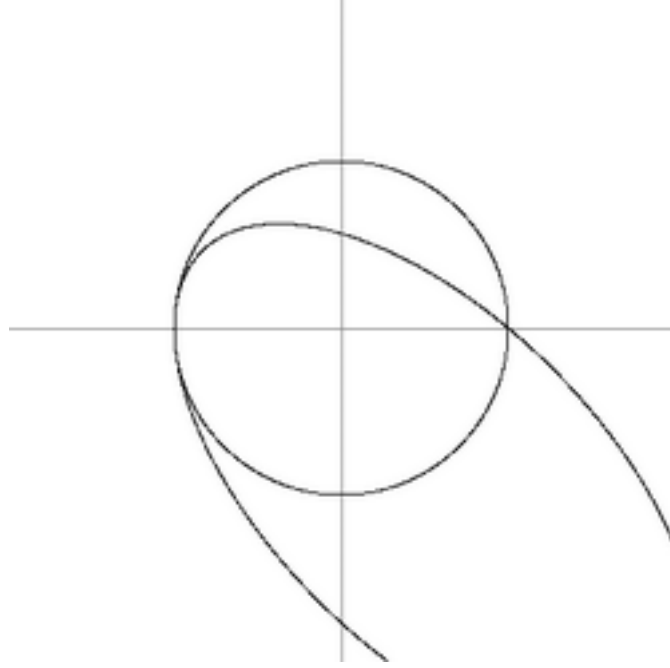


Figure 84: A circle and a curve tangent on the left and crossing transversely on the right

One transverse and one nontransverse intersection. Image created by Michael Larsen and uploaded to Commons by Maksim; CC BY-SA 3.0; local file: authority/assets/250px-Intersect3.png.

Definition 26.6: transverse intersection

Let

$$F, G \in K[X, Y] \quad \text{and} \quad P \in V(F, G).$$

The curves $V(F)$ and $V(G)$ are said to *intersect transversely* at P if P is a smooth point of both curves and their tangent lines at P are distinct.

Lemma 26.7: characterisation of transverse intersection

Let K be a field and let

$$F, G \in K[X, Y]$$

be two polynomials with no common component. Let

$$P \in V(F, G) \subseteq \mathbb{A}_K^2$$

be an intersection point. Then $V(F)$ and $V(G)$ intersect transversely at P if and only if

$$\text{mult}_P(V(F), V(G)) = 1.$$

Proof Let

$$R = K[X, Y]_{\mathfrak{m}_P}$$

be the local ring of the plane at P . First suppose that the intersection is transverse. Both curves are smooth at P , and by Lemma 23.2 in the source numbering,

$$B = R/(F)$$

is a discrete valuation ring. Since the tangent lines are distinct, after a change of coordinates we may assume that the tangent to $V(F)$ is $V(Y)$ and the tangent to $V(G)$ is $V(X)$. In B , the element X is a local uniformiser. Since

$$G = X + H, \quad H \in \mathfrak{m}_P^2,$$

the element G is also a local uniformiser in B . Hence

$$B/(G) = K,$$

and the intersection multiplicity is one.

Conversely, suppose that

$$\dim_K R/(F, G) = 1.$$

Since P is a K -rational point, this quotient is the residue field K . Its maximal ideal therefore vanishes, or equivalently,

$$(F, G) = \mathfrak{m}_P$$

in R . Passing to the quotient modulo $\mathfrak{m}_P^2$, the linear terms of F and G generate the two-dimensional cotangent space $\mathfrak{m}_P/\mathfrak{m}_P^2$. Both linear terms are therefore nonzero and linearly independent. Thus both curves are smooth at P , and the kernels of the two linear forms, namely their tangent lines, are distinct. The intersection is therefore transverse. $\square$

Editorial note - correction to the converse proof. The source deduces smoothness of both curves by citing Lemma 26.4. However, that lemma concerns only the intersection of a curve with a line, and so does not justify the asserted conclusion for two arbitrary polynomials. Moreover, Lemma 26.4 is stated over an algebraically closed field, whereas Lemma 26.7 is stated over an arbitrary field. This edition replaces that step with a direct argument in $\mathfrak{m}_P/\mathfrak{m}_P^2$, valid for the K -rational point in the statement.

Theorem 26.8: additivity formula

Let

$$F, G \in K[X, Y]$$

be two polynomials with no common prime divisor, with factorisations

$$F = \prod_{i=1}^m F_i^{\nu_i} \quad \text{and} \quad G = \prod_{j=1}^n G_j^{\mu_j}.$$

Then, for every $P \in \mathbb{A}_K^2$,

$$\text{mult}_P(F, G) = \sum_{i,j} \nu_i \mu_j \text{mult}_P(F_i, G_j).$$

Away from intersection points, the multiplicities on both sides are understood to be zero, as in Lemma 26.5.

Editorial note - quantification of the point. The source displays P in the formula without introducing or quantifying it. This edition states that the identity holds for every $P \in \mathbb{A}_K^2$, with the zero convention outside the intersection.

Proof By induction, it suffices to prove the special case $F = F_1 F_2$. Set

$$R = K[X, Y]_{\mathfrak{m}_P}.$$

Since

$$(F_1 F_2, G) \subseteq (F_2, G),$$

there is a surjective map

$$R/(F_1 F_2, G) \longrightarrow R/(F_2, G).$$

On the other hand, multiplication by F_2 induces an R -module homomorphism

$$R/(F_1, G) \longrightarrow R/(F_1 F_2, G).$$

We claim that there is a short exact sequence

$$0 \longrightarrow R/(F_1, G) \xrightarrow{\cdot F_2} R/(F_1 F_2, G) \longrightarrow R/(F_2, G) \longrightarrow 0.$$

Surjectivity of the right-hand map is clear, as is the fact that the composition of the two maps is zero. Suppose a class $z \in R/(F_1 F_2, G)$ maps to zero on the right. In R we can write

$$z = AF_2 + BG.$$

Thus AF_2 represents the same class in $R/(F_1 F_2, G)$, and that class comes from the left.

Now suppose a class $w \in R/(F_1, G)$ maps to zero under multiplication by F_2 . In R this means

$$wF_2 = CF_1 F_2 + DG,$$

or

$$(w - CF_1)F_2 = DG.$$

Since F and G have no common prime divisor, neither do F_2 and G . Hence F_2 divides D , giving

$$w - CF_1 = \widetilde{D}G.$$

Thus $w = 0$ in $R/(F_1, G)$, and the left-hand map is injective.

Additivity of dimension in short exact sequences now gives

$$\begin{aligned} \text{mult}_P(F_1 F_2, G) &= \dim_K R/(F_1 F_2, G) \\ &= \dim_K R/(F_1, G) + \dim_K R/(F_2, G) \\ &= \text{mult}_P(F_1, G) + \text{mult}_P(F_2, G). \end{aligned}$$

Induction over all factors of F and G proves the formula. $\square$

Theorem 26.9: product decomposition of a zero-dimensional Noetherian ring

Let R be a commutative Noetherian ring with only finitely many prime ideals

$$\mathfrak{m}_1, \dots, \mathfrak{m}_n,$$

all of which are maximal. Then there is a canonical isomorphism

$$R \cong R_{\mathfrak{m}_1} \times \cdots \times R_{\mathfrak{m}_n}.$$

Proof These maximal ideals are also the minimal prime ideals. Their intersection,

$$\mathfrak{a} = \bigcap_i \mathfrak{m}_i,$$

therefore consists entirely of nilpotent elements. Since R is Noetherian, there is an s such that

$$\mathfrak{a}^s = 0.$$

For each i , consider the localisation

$$R \longrightarrow R_{\mathfrak{m}_i}.$$

We claim that this localisation is isomorphic to

$$R/\mathfrak{a}_i, \quad \mathfrak{a}_i := \mathfrak{m}_i^s.$$

Since

$$\prod_i \mathfrak{m}_i \subseteq \bigcap_i \mathfrak{m}_i,$$

we obtain

$$\left(\prod_i \mathfrak{m}_i \right)^s \subseteq \left(\bigcap_i \mathfrak{m}_i \right)^s,$$

and hence

$$\mathfrak{a}_1 \cdots \mathfrak{a}_n = 0.$$

Take $i = 1$. For every $j \neq 1$, there is an element

$$g_j \in \mathfrak{m}_j \quad \text{with} \quad g_j \notin \mathfrak{m}_1.$$

For every $f \in \mathfrak{a}_1$, we have

$$f g_2^s \cdots g_n^s = 0.$$

Since $g_2^s \cdots g_n^s \notin \mathfrak{m}_1$, this element becomes a unit after localisation. Thus f maps to zero, and we obtain a ring homomorphism

$$R/\mathfrak{a}_1 \longrightarrow R_{\mathfrak{m}_1}.$$

The right-hand side is also a localisation of the quotient ring on the left. Distinct maximal ideals are pairwise comaximal, and this remains true of their powers. Hence $\mathfrak{a}_1$ is contained only in $\mathfrak{m}_1$. Thus $R/\mathfrak{a}_1$ is itself a zero-dimensional local ring, so the map above is an isomorphism. The same argument applies for every i .

The original map can therefore be written as

$$R \longrightarrow \prod_{i=1}^n R/\mathfrak{a}_i.$$

Since the ideals $\mathfrak{a}_i$ are pairwise comaximal, the Chinese Remainder Theorem says that this map is an isomorphism. $\square$

Corollary 26.10: the global quotient as a product of local rings

Let K be an algebraically closed field and let

$$F, G \in K[X, Y]$$

be two polynomials with no common prime divisor. Let

$$P_1, \dots, P_n \in \mathbb{A}_K^2$$

be all the points of $V(F, G)$, with corresponding maximal ideals $\mathfrak{m}_1, \dots, \mathfrak{m}_n$ in $K[X, Y]$. Then there is a canonical isomorphism

$$K[X, Y]/(F, G) \cong \prod_{i=1}^n \left(K[X, Y]_{\mathfrak{m}_i}/(F, G) \right).$$

Proof Since F and G have no common prime divisor, the ideal (F, G) is contained in only finitely many prime ideals, all of them maximal. Consequently the quotient ring

$$K[X, Y]/(F, G)$$

satisfies the hypotheses of Theorem 26.9. Since K is algebraically closed, these maximal ideals correspond bijectively to the intersection points of $V(F)$ and $V(G)$. This gives the asserted isomorphism. $\square$

Editorial note - ring symbol in the proof. The source writes $R/(F, G)$ in this proof without defining R . From the corollary's statement and the application of Theorem 26.9, the intended ring is $K[X, Y]/(F, G)$; this edition writes it explicitly.

Theorem 26.11: sum of intersection multiplicities

Let K be an algebraically closed field and let

$$F, G \in K[X, Y]$$

be two polynomials with no common prime divisor. Then

$$\dim_K(K[X, Y]/(F, G)) = \sum_P \text{mult}_P(F, G),$$

where the sum runs over all points $P \in V(F, G)$.

Proof This follows directly from the isomorphism proved in Corollary 26.10, since the dimension of a finite product of vector spaces is the sum of the dimensions of its factors. $\square$

Finally, we record without proof the following theorem, which gives a bound relating intersection multiplicity to the multiplicities of the two curves.

Theorem 26.12: lower bound for intersection multiplicity

Let

$$F, G \in K[X, Y]$$

be two polynomials with no common component and let

$$P \in V(F, G).$$

Then

$$\text{mult}_P(F, G) \geq m_P(F) m_P(G).$$

Editorial note - finiteness hypothesis. The source does not state that F and G must have no common component. Without this condition, the local quotient in the definition of intersection multiplicity can be infinite-dimensional. This edition adds the hypothesis already governing the entire discussion in this lecture.

Proof reference See Fulton, *Algebraic Curves*, Chapter III.3.

Worksheet 26**Warm-up exercises****Exercise 26.1**

For each n , give an example of two plane algebraic curves familiar from school which intersect at exactly one point with intersection multiplicity n .

Exercise 26.2

Consider the curve given by

$$y = 2x^4 + 3x^2 - x + 1$$

with the point

$$P = (1, 5).$$

Find a change of coordinates that takes P to $(0, 0)$ and the tangent line at P to the x -axis.

Exercise 26.3 (3 points)

Let the monomial plane curve

$$C = V(X^d - Y^e),$$

be given, with d and e coprime. Compute the intersection multiplicity of this curve with every line G through the origin which is not a component of C .

Editorial note - the common-component case. The source asks for every line through the origin. If $d = e = 1$, the curve $C = V(X - Y)$ itself is one of these lines, and the finite intersection multiplicity defined in the lecture is not available for a curve intersecting itself. This edition excludes lines which are components of C .

Exercise 26.4 ★

Determine the intersection multiplicity at the origin of the folium of Descartes

$$C = V(X^3 + Y^3 - 3XY)$$

with every affine line in the affine plane. Assume that the characteristic of the field is not 3.

Exercises to submit**Exercise 26.5 (4 points)**

Compute the intersection multiplicity of the two monomial curves

$$C = V(X^5 - Y^2) \quad \text{and} \quad D = V(X^7 - Y^3)$$

at the origin.

Exercise 26.6 (4 points)

Let K be a field, and let

$$C = V(F) \quad \text{and} \quad D = V(G)$$

be two plane algebraic curves with no common component. Let

$$P \in C$$

be a smooth point, so that the local ring

$$R = K[X, Y]_{\mathfrak{m}_P} / (F)$$

is a discrete valuation ring. Show that

$$\text{mult}_P(F, G) = \text{ord}(G),$$

where ord denotes the order of the nonzero image of G in the valuation ring R .

Editorial note - finiteness condition. The source does not state that the two curves must have no common component. This condition, or locally the condition that the image of G in R is nonzero, is necessary for both sides to be finite numbers. This edition states it explicitly.

Exercise 26.7 (4 points)

In the real plane, let $r > 0$. Consider the parabola

$$C = V(Y - X^2)$$

and the circle D with centre $(0, r)$ and radius r , namely

$$D = V(X^2 + (Y - r)^2 - r^2).$$

Determine the intersection points of C and D and their respective intersection multiplicities.

Editorial note - geometric scope. The source specifies a centre and radius without fixing a base field or a condition on r . To give “a circle of radius r ” its usual geometric meaning and prevent degeneration to a point, this edition interprets the exercise over $\mathbb{R}$ with $r > 0$ and writes out the circle’s equation.

Exercise 26.8 (4 points)

For each $a \in \mathbb{C}$, describe the quotient ring

$$\mathbb{C}[X, Y] / (XY - 1, X^2 + Y^2 - a)$$

as a product of local rings. Also give the dimension of each factor ring as a vector space over $\mathbb{C}$.

Exercise 26.9 (4 points)

For the curve

$$V(X^3 + Y^3 - 3XY + 1),$$

determine its singular points over $\mathbb{R}$ and over $\mathbb{C}$. For each point, give its multiplicity and tangent lines.

Exercise 26.10 (3 points)

Consider the curve

$$y = 2x^4 + 3x^2 - x + 1$$

at the point

$$P = (1, 5),$$

in the coordinates found in Exercise 26.2. Determine the power series for the curve at P along the tangent line.

The following exercise is probably more difficult.

Exercise 26.11 (8 points)

Let two distinct monomial plane curves

$$C = V(X^d - Y^e) \quad \text{and} \quad D = V(X^r - Y^s),$$

be given, with d, e coprime and r, s coprime. Compute the intersection multiplicity of the two curves at the origin.

Editorial note - zero-locus symbol. In the second equation, the source writes $D = (X^r - Y^s)$ and omits the symbol V , although the text calls D a curve. This edition restores the intended expression, $D = V(X^r - Y^s)$.

Public Solutions to Worksheet 26

At the frozen revision boundary, the source provides a public solution only for Exercise 26.4. The frozen authority query reports the other ten candidate solution pages as absent. No additional solutions have been created for this edition.

Solution to Exercise 26.4

If a line does not pass through the origin, its intersection multiplicity with the folium at the origin is zero. Thus it suffices to consider lines through the origin. These can all be written as

$$V(Y - aX), \quad a \in K,$$

or as the vertical line $V(X)$.

For $a = 0$, that is, the line $V(Y)$, the quotient ring is

$$K[X, Y]_{(X, Y)} / (Y, X^3 + Y^3 - 3XY) \cong K[X]_{(X)} / (X^3).$$

This ring has dimension 3 over K , so the intersection multiplicity is 3. Since the equation of the folium is symmetric in X and Y , the same result holds for the line $V(X)$.

Now take a line $V(Y - aX)$ with $a \neq 0$. Then

$$\begin{aligned} K[X, Y]_{(X, Y)} / (Y - aX, X^3 + Y^3 - 3XY) &\cong K[X]_{(X)} / (X^3 + a^3X^3 - 3aX^2) \\ &= K[X]_{(X)} / (X^2(-3a + (1 + a^3)X)). \end{aligned}$$

Since $\text{char}(K) \neq 3$ and $a \neq 0$, the factor

$$-3a + (1 + a^3)X$$

is a unit in $K[X]_{(X)}$. Hence the quotient ring is isomorphic to

$$K[X]_{(X)} / (X^2),$$

which has dimension 2. Thus the lines $V(X)$ and $V(Y)$ have intersection multiplicity 3 at the origin, whereas every other line through the origin has intersection multiplicity 2.

Editorial note - list of lines through the origin. The source states that these lines have the form $V(Y - aX)$ or $V(Y)$, thereby listing $V(Y)$ twice and omitting the vertical line $V(X)$. The source's next step itself treats $V(X)$ by symmetry. This edition restores the intended list: $V(Y - aX)$ for $a \in K$, together with $V(X)$. The source also leaves the number after “ K -dimension” blank in the $V(Y)$ case; the displayed quotient has basis $1, X, X^2$, so the dimension 3 supplied here agrees with the source's stated multiplicity.

[Back to Exercise 26.4](#)

Lecture 27: Projective Space

Projective space



Figure 85: A dandelion with many lines and stalks radiating from a single centre

Lines through a point. Waugsberg; historical course label: CC BY-SA 2.5; frozen Commons description: CC BY-SA 3.0. Both notices are retained; local file: authority/assets/Loewenzahn_20.jpg.

Definition 27.1: projective space

Let K be a field. The n -dimensional projective space

$$\mathbb{P}_K^n$$

consists of all lines through the origin in the affine space $\mathbb{A}_K^{n+1}$, each such line being regarded as a point. Such a projective point is represented by *homogeneous coordinates*

$$(a_0, a_1, \dots, a_n),$$

where the a_i are not all zero. Two such coordinate tuples represent the same point precisely when one is obtained from the other by multiplication by a scalar

$$\lambda \in K^\times.$$

We shall gradually equip projective space with additional structures.

Theorem 27.2: the standard open cover by affine spaces

Let K be a field, let $\mathbb{P}_K^n$ be a projective space, and let

$$i \in \{0, 1, \dots, n\}.$$

Indexing the source coordinates by all $j \neq i$, there is a natural map

$$\begin{aligned} \varphi_i : \mathbb{A}_K^n &\longrightarrow \mathbb{P}_K^n, \\ (u_0, \dots, \widehat{u_i}, \dots, u_n) &\longmapsto [u_0 : \dots : u_{i-1} : 1 : u_{i+1} : \dots : u_n]. \end{aligned}$$

This map is injective and induces a bijection onto the set of projective points whose i th homogeneous coordinate is nonzero, namely

$$D_+(X_i) := \{[x_0 : \dots : x_n] \in \mathbb{P}_K^n \mid x_i \neq 0\}.$$

Its inverse is given by

$$[x_0 : \dots : x_n] \longmapsto \left(\frac{x_0}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_n}{x_i} \right).$$

Projective space is covered by these $n + 1$ affine spaces. The complement of the affine chart

$$\mathbb{A}_K^n \cong D_+(X_i) \subseteq \mathbb{P}_K^n$$

is a projective space of dimension $n - 1$.

Editorial note - chart insertion indices. The source writes the source coordinates as $(u_1, \dots, u_n)$ and then inserts 1 “in position i ” for $i \in \{0, \dots, n\}$. This notation is ambiguous at the endpoints and does not display a coordinate u_0 . This edition indexes coordinates by $\{0, \dots, n\} \setminus \{i\}$ and uses a hat to mark the omitted coordinate.

Proof The map is well defined because the coordinate 1 ensures that at least one homogeneous coordinate is nonzero. If

$$[u_0 : \dots : u_{i-1} : 1 : u_{i+1} : \dots : u_n] = [v_0 : \dots : v_{i-1} : 1 : v_{i+1} : \dots : v_n],$$

then some $\lambda \in K^\times$ multiplies every coordinate on the right. Comparing the i th coordinates gives $1 = \lambda$, so all the other coordinates agree and the map is injective.

On $D_+(X_i)$, every point has exactly one representative with i th coordinate equal to 1, obtained by dividing all coordinates by x_i . This proves the formula for the inverse. Every projective point has at least one nonzero coordinate, so the charts $D_+(X_i)$ cover $\mathbb{P}_K^n$.

The complement of $D_+(X_i)$ is

$$V_+(X_i) = \{[x_0 : \dots : x_{i-1} : 0 : x_{i+1} : \dots : x_n] \mid \text{at least one } x_j \neq 0\}.$$

Retaining the identification of tuples that differ by a scalar factor, this set is $\mathbb{P}_K^{n-1}$. $\square$

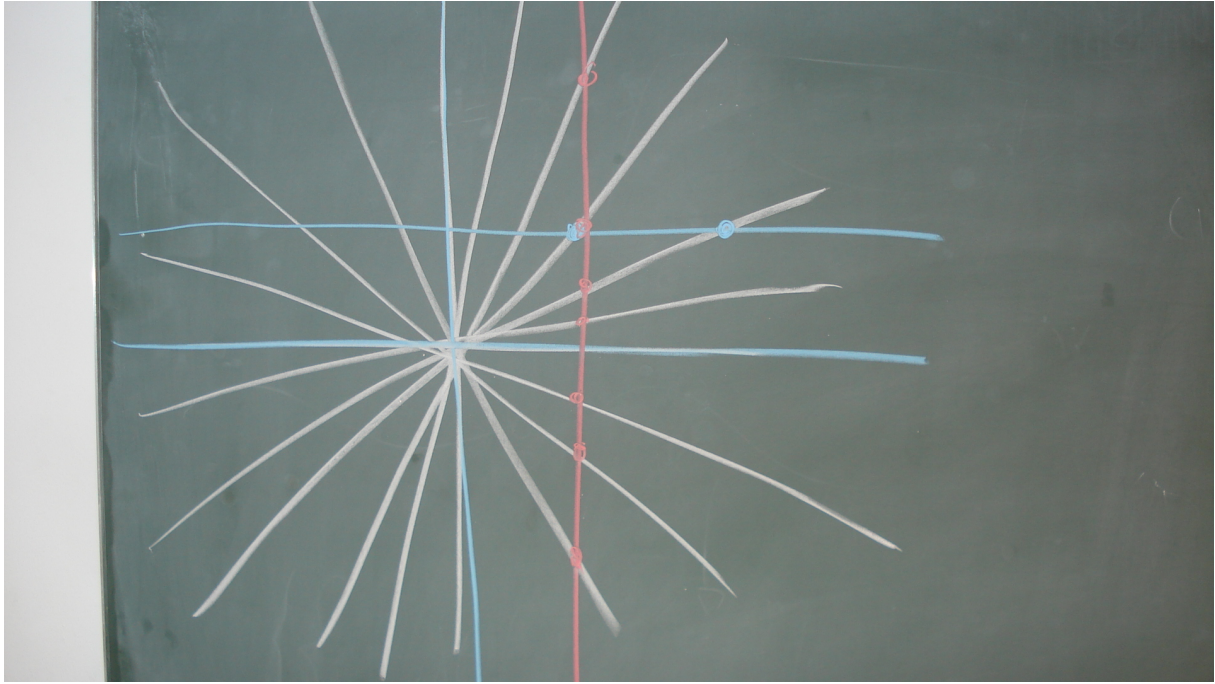


Figure 86: First of three diagrams of affine charts on the projective line

Illustration of the projective line, part 1. Darapti, CC BY-SA 3.0; local file: authority/assets/Projektiveline1 bb.jpg.

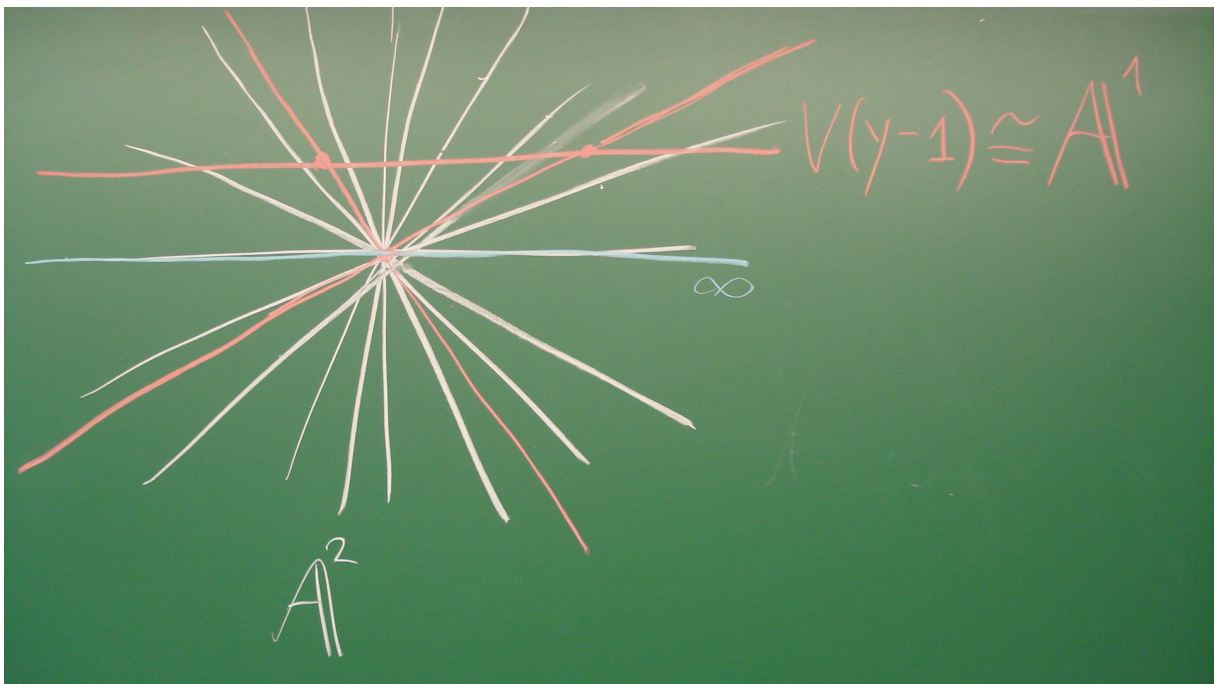


Figure 87: Second of three diagrams of affine charts on the projective line

Illustration of the projective line, part 2. Darapti, CC BY-SA 3.0; local file: authority/assets/Projektiveline2 bb.jpg.

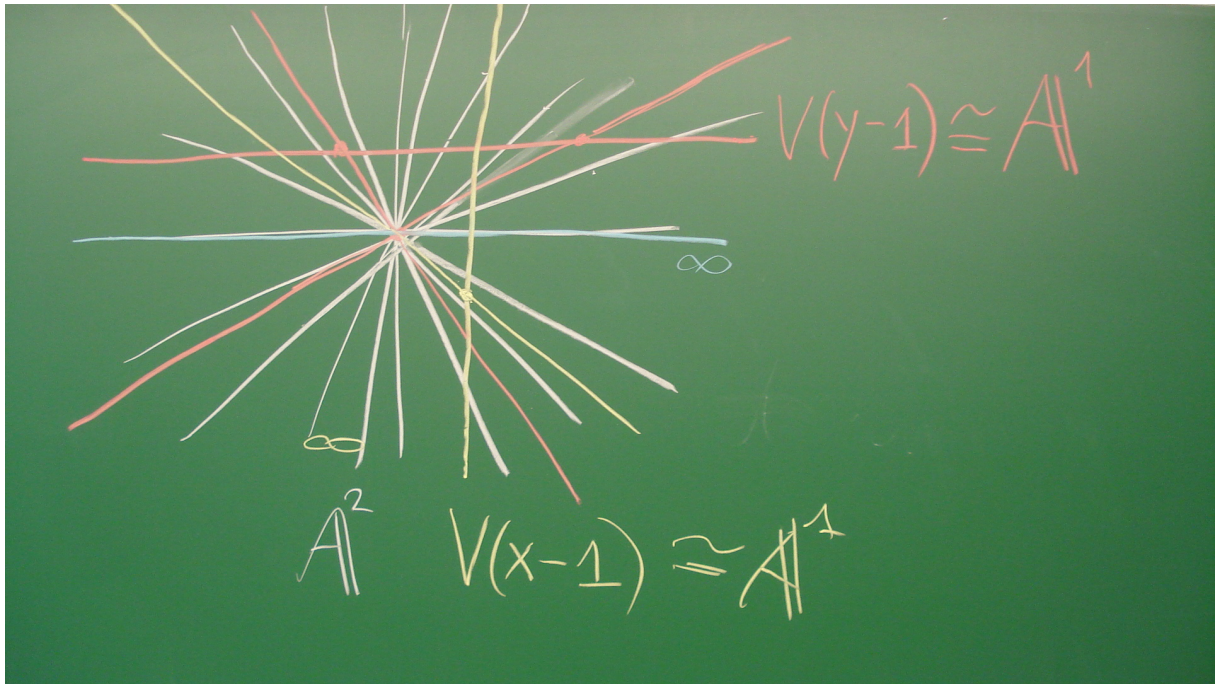


Figure 88: Third of three diagrams of affine charts on the projective line

Illustration of the projective line, part 3. Darapti, CC BY-SA 3.0; local file: authority/assets/Projektiveline3 bb.jpg.

Example 27.3: the projective line

The projective line $\mathbb{P}_K^1$ is the set of lines through the origin in the affine plane $\mathbb{A}_K^2$. Such a line is either the x -axis or intersects the line

$$V(y - 1)$$

at exactly one point. The line $V(y - 1)$ is parallel to the x -axis and passes through $(0, 1)$. Conversely, every point

$$P \in V(y - 1) \cong \mathbb{A}_K^1$$

uniquely determines a line through the origin. Thus the projective line consists of an affine line and one additional point, called the point “at infinity”.

This point is not intrinsically different from the other projective points. Take any line G through the origin and a parallel line $L \neq G$. The line L can play the role of the affine line, while G represents the point at infinity as seen from that affine chart.

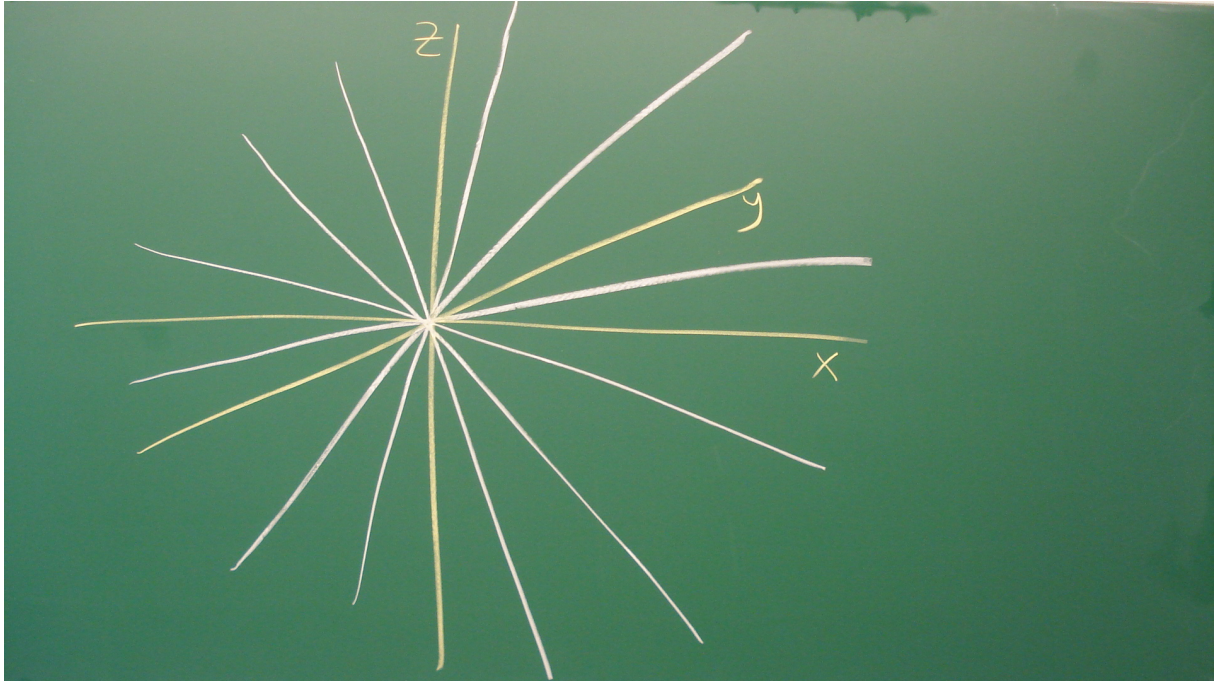


Figure 89: First of four diagrams of affine charts and points at infinity in the projective plane

Illustration of the projective plane, part 1. Darapti, CC BY-SA 3.0; local file: authority/assets/Projektiveplane1bb.jpg.

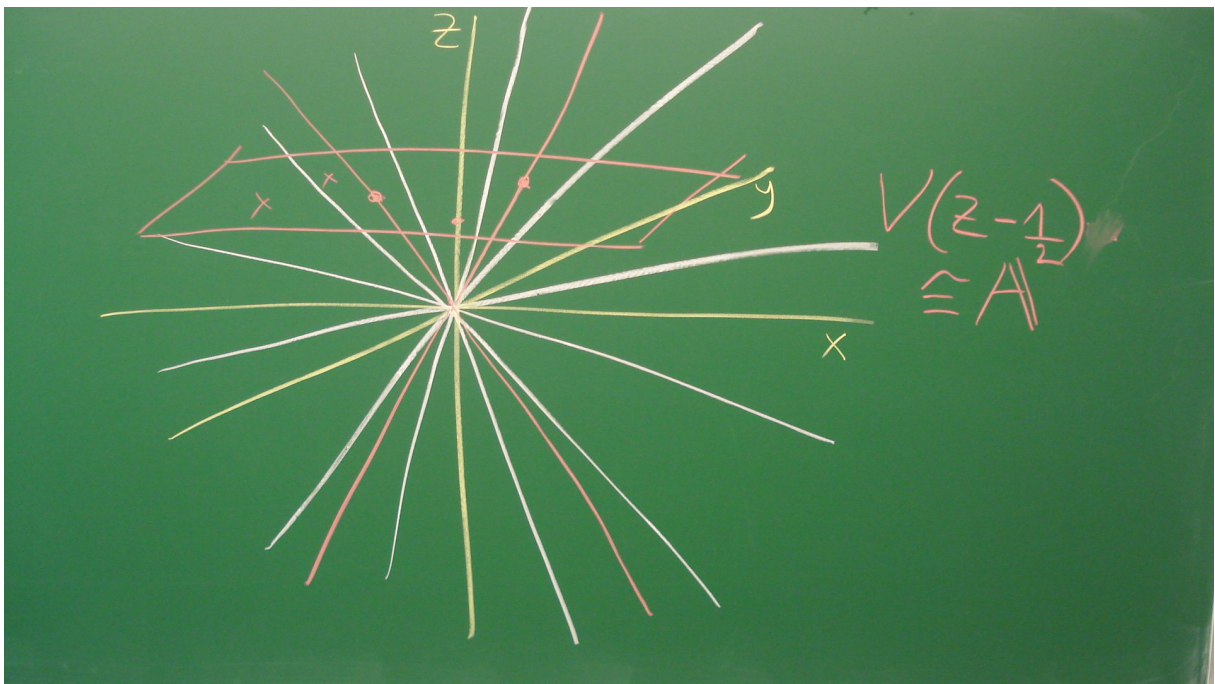


Figure 90: Second of four diagrams of affine charts and points at infinity in the projective plane

Illustration of the projective plane, part 2. Darapti, CC BY-SA 3.0; local file: authority/assets/Projektiveplane2bb.jpg.

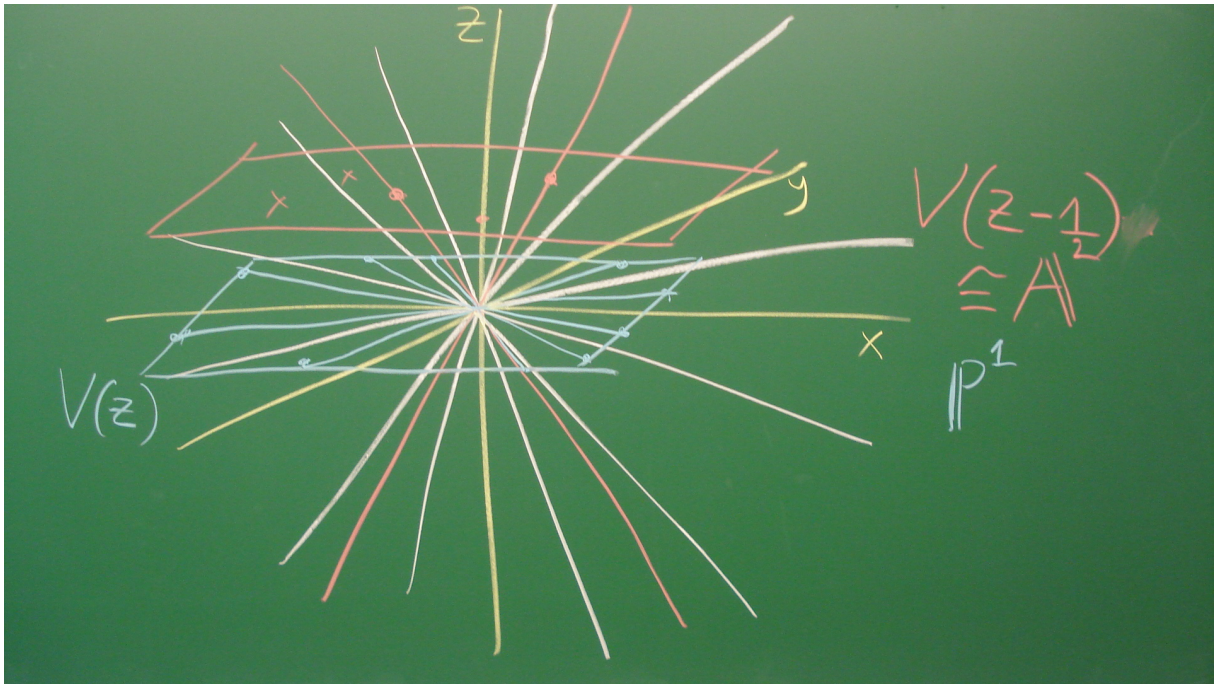


Figure 91: Third of four diagrams of affine charts and points at infinity in the projective plane

Illustration of the projective plane, part 3. Darapti, CC BY-SA 3.0; local file: *authority/assets/Projektiveplan e3bb.jpg*.

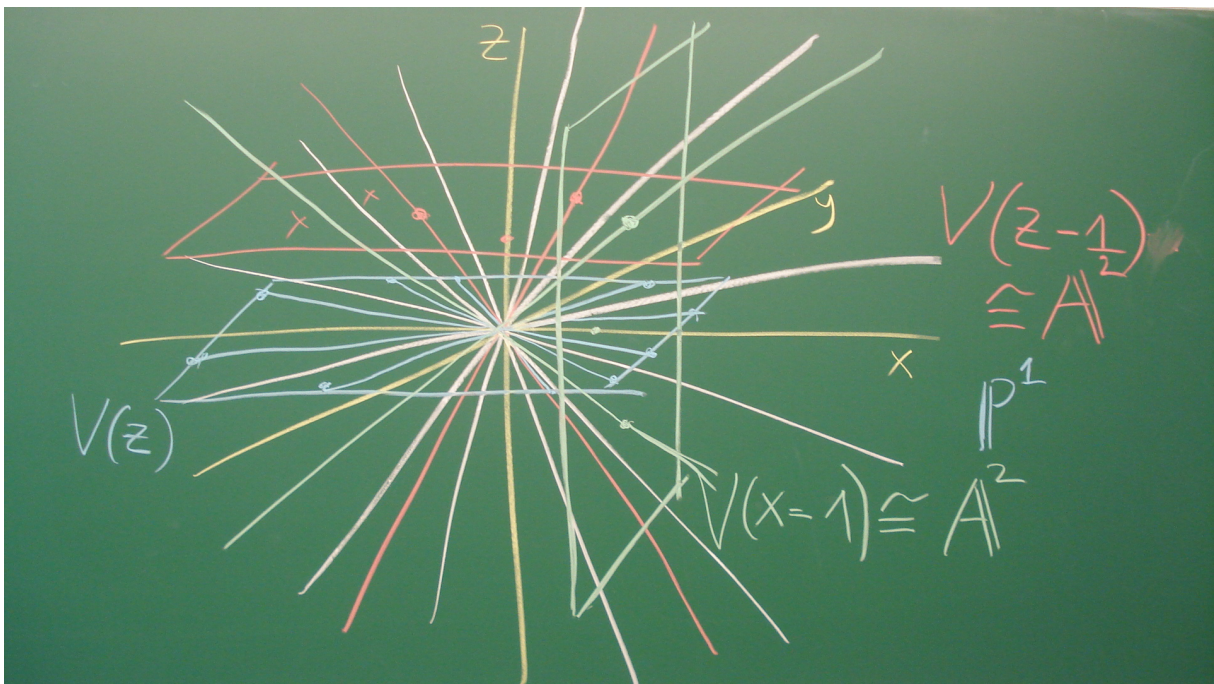


Figure 92: Fourth of four diagrams of affine charts and points at infinity in the projective plane

Illustration of the projective plane, part 4. Darapti, CC BY-SA 3.0; local file: *authority/assets/Projektiveplan e4bb.jpg*.

Example 27.4: the projective plane

Points in the projective plane $\mathbb{P}_K^2$ correspond to lines through the origin in the affine space $\mathbb{A}_K^3$. Each point of the projective plane is represented by a tuple

$$(x, y, z),$$

where x, y, z are not all zero. Two tuples are identified if one is obtained from the other by multiplication by a nonzero scalar. The projective plane is covered by three affine planes

$$D_+(X), \quad D_+(Y), \quad D_+(Z).$$

The chart $D_+(Z)$ consists of all points with nonzero third coordinate. Multiplying the coordinates by z^{-1} gives the representative

$$\left(\frac{x}{z}, \frac{y}{z}, \frac{z}{z}\right) = (u, v, 1),$$

so this chart is indeed an affine plane. Its complement is $V_+(Z)$, the set of points with third coordinate zero. Retaining scalar identification makes $V_+(Z)$ a projective line.

A point $(x, y, 0)$ on this line, together with the origin $(0, 0, 1)$ of $D_+(Z)$, determines the direction (x, y) of a line through the origin in the affine plane. The homogeneous equation of this line is

$$yX - xY = 0,$$

or $V_+(yX - xY)$. Thus we can picture the projective plane as an affine plane with one additional point at infinity for every direction of a line through the origin.

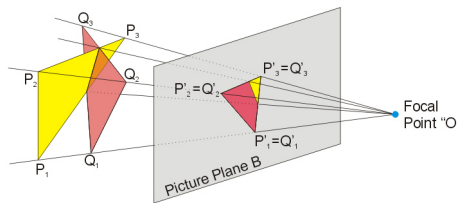


Figure 93: Perspective diagram with projection rays joining an object, a projection centre, and an image plane

Principle of perspective projection. Historical course credit: Fantagu; the frozen Commons description credits the drawing to Joachim Baecker and identifies Fantagu as the uploader. CC BY-SA 3.0; local file: authority/assets/Perspective_Projection_Principle.jpg.

Zeros of homogeneous polynomials

For an arbitrary polynomial

$$F \in K[X_0, \dots, X_n],$$

the assertion that a point $P \in \mathbb{P}_K^n$ is a zero of F is generally not well defined. The value can change when a coordinate representative of P is multiplied by a scalar. The situation is different for homogeneous polynomials.

Lemma 27.5: vanishing of a homogeneous polynomial is well defined

Let K be a field and let

$$F \in K[X_0, \dots, X_n]$$

be a homogeneous polynomial of degree d . For every $(x_0, \dots, x_n) \in K^{n+1}$ and $\lambda \in K$,

$$F(\lambda x_0, \dots, \lambda x_n) = \lambda^d F(x_0, \dots, x_n).$$

In particular, F vanishes at $(x_0, \dots, x_n)$ if and only if it vanishes at $\lambda(x_0, \dots, x_n)$ for every $\lambda \neq 0$.

Proof It suffices to check each homogeneous monomial. For

$$X_0^{d_0} \dots X_n^{d_n}, \quad \sum_{i=0}^n d_i = d,$$

we obtain

$$\begin{aligned} (\lambda X_0)^{d_0} \dots (\lambda X_n)^{d_n} &= (\lambda^{d_0} X_0^{d_0}) \dots (\lambda^{d_n} X_n^{d_n}) \\ &= \lambda^d X_0^{d_0} \dots X_n^{d_n}. \end{aligned}$$

Linearity then gives the assertion for F . $\square$

The lemma makes the property “vanishes or does not vanish” well defined at a projective point. However, the numerical value of a homogeneous polynomial at a projective point need not be intrinsically defined. In general, a homogeneous polynomial does not define a function on projective space by evaluation of representatives.

Editorial note - scope of the evaluation claim. The source makes the last assertion without qualification. Constant polynomials, for example, do give well-defined values, so this edition states it as a general warning rather than an assertion about every homogeneous polynomial.

Definition 27.6: projective zero locus

Let K be a field and let

$$F \in K[X_0, \dots, X_n]$$

be a homogeneous polynomial. The set

$$V_+(F) = \{P = [x_0 : \dots : x_n] \in \mathbb{P}_K^n \mid F(x_0, \dots, x_n) = 0\}$$

is called the *projective zero locus* of F .

To determine $V_+(F)$, we can use the disjoint decomposition

$$\mathbb{P}_K^n = D_+(X_0) \uplus V_+(X_0),$$

and likewise for any other variable. In the chart $D_+(X_0) \cong \mathbb{A}_K^n$, we set $X_0 = 1$ and solve

$$F(1, X_1, \dots, X_n) = 0.$$

The polynomial may become inhomogeneous, one variable is eliminated, and the ambient dimension stays the same, but the problem becomes affine. On $V_+(X_0) \cong \mathbb{P}_K^{n-1}$, we set $X_0 = 0$ and solve

$$F(0, X_1, \dots, X_n) = 0.$$

Here one variable is again eliminated, the polynomial remains homogeneous, and the dimension of the projective space decreases by one.

Editorial note - dehomogenisation. The source displays the malformed expressions $F\{1/X_0\}$ and $F\{0/X_0\}$. The immediately preceding prose specifies the substitutions $X_0 = 1$ and $X_0 = 0$. This edition writes the intended polynomials explicitly as $F(1, X_1, \dots, X_n)$ and $F(0, X_1, \dots, X_n)$.

Example 27.7: homogeneous linear polynomials

The simplest homogeneous polynomials in $K[X_0, \dots, X_n]$ are those of degree one,

$$F = a_0X_0 + a_1X_1 + \dots + a_nX_n,$$

with coefficients not all zero. The affine zero locus $V(F)$ in $\mathbb{A}_K^{n+1}$ is an n -dimensional affine space through the origin. The projective zero locus $V_+(F)$ in $\mathbb{P}_K^n$ is isomorphic to a projective space of dimension $n - 1$.

Editorial note - index of the linear term. The source writes $a_0X_0 + a_1X_0 + \dots + a_nX_n$. The second term must use X_1 for the expression to be a general linear form in the variables $X_0, \dots, X_n$.

Definition 27.8: homogeneous ideal

Let K be a field and let

$$\mathfrak{a} \subseteq K[X_1, \dots, X_n]$$

be an ideal. The ideal $\mathfrak{a}$ is called *homogeneous* if, for every $H \in \mathfrak{a}$ with homogeneous decomposition

$$H = \sum_i H_i,$$

every homogeneous component H_i also belongs to $\mathfrak{a}$.

Definition 27.9: projective variety

For a homogeneous ideal

$$\mathfrak{a} \subseteq K[X_0, \dots, X_n],$$

the set

$$V_+(\mathfrak{a}) = \{P \in \mathbb{P}_K^n \mid F(P) = 0 \text{ for every homogeneous } F \in \mathfrak{a}\}$$

is called the *projective zero locus* or *projective variety* of $\mathfrak{a}$.

Definition 27.10: the Zariski topology on projective space

Projective space $\mathbb{P}_K^n$ is equipped with the *Zariski topology* by declaring the sets

$$V_+(\mathfrak{a}) \subseteq \mathbb{P}_K^n,$$

for every homogeneous ideal

$$\mathfrak{a} \subseteq K[X_0, \dots, X_n],$$

to be the closed sets.

Thus the open sets of projective space have the form

$$D_+(\mathfrak{a}) := \mathbb{P}_K^n \setminus V_+(\mathfrak{a}).$$

In particular, each standard open set $D_+(X_i)$ is isomorphic to an affine space of dimension n .

Remark 27.11: projective points are closed

Let

$$P = [a_0 : \cdots : a_n] \in \mathbb{P}_K^n.$$

The point P is closed. More precisely,

$$P = V_+(\mathfrak{a}_P), \quad \mathfrak{a}_P = (a_i X_j - a_j X_i \mid 0 \leq i, j \leq n).$$

If $a_0 \neq 0$, this ideal can also be written as

$$\mathfrak{a}_P = \left(X_j - \frac{a_j}{a_0} X_0 \mid j \neq 0 \right);$$

the generators $a_i X_j - a_j X_i$ with $i \neq 0$ are then redundant. This ideal is clearly homogeneous, and $P \in V_+(\mathfrak{a}_P)$. If

$$Q = [b_0 : \cdots : b_n] \in V_+(\mathfrak{a}_P),$$

then, since $a_0 \neq 0$,

$$b_j - \frac{a_j}{a_0} b_0 = 0$$

for every j . Hence

$$(b_0, \dots, b_n) = \frac{b_0}{a_0} (a_0, \dots, a_n),$$

so $Q = P$ as projective points.

The ideal $\mathfrak{a}_P$ is not a maximal ideal in the polynomial ring. It is a homogeneous prime ideal: the quotient by it is isomorphic to a polynomial ring in one variable. In $\mathbb{A}_K^{n+1}$, this ideal defines the line through the origin corresponding to the projective point P .

Editorial note - maximality claim. The source states that $\mathfrak{a}_P$ is maximal among all homogeneous ideals other than the irrelevant ideal $(X_0, \dots, X_n)$. This claim is false without further conditions. For example, after choosing $a_0 \neq 0$, the homogeneous ideal $\mathfrak{a}_P + (X_0^2)$ lies strictly between $\mathfrak{a}_P$ and the irrelevant ideal. This edition retains the correct and necessary statement: $\mathfrak{a}_P$ is a homogeneous prime ideal and defines exactly the projective point P .

There is no natural map from all of $\mathbb{A}_K^{n+1}$ to $\mathbb{P}_K^n$, since the origin does not determine a line. There is, however, a natural map

$$\begin{aligned} \mathbb{A}_K^{n+1} \setminus \{0\} &\longrightarrow \mathbb{P}_K^n, \\ (x_0, \dots, x_n) &\longmapsto [x_0 : \cdots : x_n]. \end{aligned}$$

This map sends a nonzero point to the line through that point and the origin. It is called the *canonical map* or *cone map*. The inverse image of $D_+(X_i)$ under this map is $D(X_i)$.

Projective space over $\mathbb{R}$ and $\mathbb{C}$

We now develop a topological picture of projective space for $\mathbb{K} = \mathbb{R}$ and $\mathbb{K} = \mathbb{C}$. The real n -dimensional sphere is

$$S^n = \{x \in \mathbb{R}^{n+1} \mid \|x\| = 1\},$$

where

$$\|x\| = \sqrt{x_0^2 + \cdots + x_n^2}$$

is the Euclidean norm.

Theorem 27.12: representation by spheres

Real projective space $\mathbb{P}_{\mathbb{R}}^n$ can be represented by the sphere $S^n \subseteq \mathbb{R}^{n+1}$ modulo the equivalence relation identifying each pair of antipodal points.

Complex projective space $\mathbb{P}_{\mathbb{C}}^n$ can be represented by the sphere

$$S^{2n+1} \subseteq \mathbb{R}^{2n+2} \cong \mathbb{C}^{n+1}$$

modulo the equivalence relation identifying $z, w \in S^{2n+1}$ if

$$z = \lambda w$$

for some $\lambda \in S^1 \subseteq \mathbb{C}$.

Proof We treat the real and complex cases together. Each point of the sphere S determines a real or complex line through the origin in the ambient space, and hence a projective point. Two points $z, w \in S$ determine the same line precisely when

$$z = \lambda w$$

for some $\lambda \in \mathbb{K}$. Multiplicativity of the norm gives

$$\|z\| = |\lambda| \|w\|.$$

Since both norms equal one, $|\lambda| = 1$. In the real case this means $\lambda = \pm 1$, so the identified points form antipodal pairs. In the complex case it means $\lambda \in S^1 \subseteq \mathbb{C}$. $\square$

Altogether, we have surjective maps

$$S^n \subseteq \mathbb{R}^{n+1} \setminus \{0\} \longrightarrow \mathbb{P}_{\mathbb{R}}^n$$

in the real case, and

$$S^{2n+1} \subseteq \mathbb{R}^{2n+2} \setminus \{0\} \cong \mathbb{C}^{n+1} \setminus \{0\} \longrightarrow \mathbb{P}_{\mathbb{C}}^n$$

in the complex case. Real and complex projective spaces are equipped with the quotient topology of the metric topology of the real vector space. Thus $U \subseteq \mathbb{P}_{\mathbb{K}}^n$ is declared open if its inverse image in $\mathbb{A}_{\mathbb{K}}^{n+1} \setminus \{0\}$ is open. Equivalently, its inverse image on the corresponding sphere is open. With this metric or natural topology, the maps above are continuous.

Lemma 27.13: open charts and manifold structure

For real and complex projective spaces, the sets $D_+(X_i)$ are open in the natural topology and homeomorphic to $\mathbb{R}^n$ and $\mathbb{C}^n$, respectively. In particular, real and complex projective spaces are topological manifolds.

Proof The inverse image of $D_+(X_i)$ under the canonical map

$$\mathbb{A}_{\mathbb{K}}^{n+1} \setminus \{0\} \longrightarrow \mathbb{P}_{\mathbb{K}}^n$$

is $D(X_i)$, the complement of an n -dimensional vector subspace, and is therefore open in the natural topology. Consider the continuous map

$$\mathbb{K}^n \cong V(X_i - 1) \subset D(X_i) \longrightarrow D_+(X_i).$$

This map is bijective. To show that it is a homeomorphism, it suffices to show that it is open. Let

$$U \subseteq V(X_i - 1) \cong \mathbb{K}^n$$

be open and let U' be its image in $D_+(X_i)$. The inverse image of U' in $D(X_i)$ is the cone

$$U'' = \{\lambda P \mid \lambda \in \mathbb{K}^\times, P \in U\}.$$

The map

$$\begin{aligned} \mathbb{K}^\times \times V(X_i - 1) &\longrightarrow D(X_i), \\ (\lambda, P) &\longmapsto \lambda P \end{aligned}$$

is a homeomorphism, with inverse

$$Q \longmapsto \left(Q_i, \frac{Q}{Q_i} \right).$$

Consequently,

$$U'' \cong \mathbb{K}^\times \times U$$

is open in $D(X_i)$. By the definition of the quotient topology, U' is open. Thus the bijection is a homeomorphism. $\square$

Editorial note - neighbourhood in the cone. The source chooses an open ball B around P and then asserts without qualification that its cone lies in U'' . For this conclusion to hold, the ball must be chosen with $P \in B \subseteq U$. This edition gives an equivalent global argument using the homeomorphism $\mathbb{K}^\times \times V(X_i - 1) \cong D(X_i)$, which also closes this gap.



Figure 94: A blue sphere shown in three dimensions, representing the complex projective line

The projective line over $\mathbb{C}$ is a sphere. Historical course credit: Kieff; the frozen Commons description credits Lucas Vieira (LucasVB). Public domain; local file: `authority/assets/Blue-sphere.png`.

Corollary 27.14: compactness and the Hausdorff property

Real and complex projective spaces are compact and Hausdorff in their natural topology.

Proof For each such projective space, there is a continuous surjective map from the corresponding sphere. The sphere is closed and bounded in a finite-dimensional real vector space, and is therefore compact by the Heine–Borel Theorem. A continuous image of a compact space is compact. Hence real and complex projective spaces are compact.

Now take two distinct points

$$P, Q \in \mathbb{P}_{\mathbb{K}}^n, \quad \mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}.$$

Since $\mathbb{K}$ is infinite, there is a homogeneous linear form L vanishing at neither P nor Q . Indeed, in the dual space

the forms vanishing at P and those vanishing at Q each form a proper hyperplane, and the union of these two hyperplanes does not fill the entire dual space.

By a linear change of coordinates, L can be made one of the homogeneous coordinates. Then

$$P, Q \in D_+(L) \cong \mathbb{K}^n.$$

By Lemma 27.13, this chart is homeomorphic to a real or complex Euclidean space and is therefore Hausdorff. Hence P and Q have disjoint open neighbourhoods. Thus the whole projective space is Hausdorff. $\square$

Editorial note - two points in one chart. The source assumes that any two projective points lie together in one of the standard charts $D_+(X_i)$. This is false, for example for $[1 : 0]$ and $[0 : 1]$ in $\mathbb{P}^1$. This edition chooses a linear form L vanishing at neither point and uses a change of coordinates to obtain an affine chart $D_+(L)$ containing both.

Editorial note - broken reference. In the compactness argument, the source refers to “Fakt *****”. This edition replaces the broken placeholder with the standard result actually used: a continuous image of a compact space is compact.

Worksheet 27

Warm-up exercises

Exercise 27.1

Define an equivalence relation on the set

$$\mathbb{A}_K^{n+1} \setminus \{0\}$$

such that its quotient by this equivalence relation is n -dimensional projective space.

Exercise 27.2

Define the notion of a *projective linear subspace* of a projective space $\mathbb{P}_K^n$.

Exercise 27.3

Let

$$\mathfrak{a} \subseteq K[X_1, \dots, X_n]$$

be an ideal. Show that $\mathfrak{a}$ is homogeneous if and only if $\mathfrak{a}$ is generated by homogeneous elements.

Exercise 27.4

Let

$$K = \mathbb{F}_q$$

be a finite field with q elements. Compute the number of elements of projective space $\mathbb{P}_K^n$ in two different ways.

The following three exercises concern the Zariski topology on projective spaces.

Exercise 27.5

Show that the Zariski topology on projective space is indeed a topology.

Exercise 27.6

Let K be an infinite field and let $\mathbb{P}_K^n$ be projective space. Characterise the homogeneous ideals $\mathfrak{a}$ satisfying

$$D_+(\mathfrak{a}) = \emptyset.$$

Exercise 27.7

Let K be an infinite field. Show that projective space $\mathbb{P}_K^n$ is irreducible.

Exercises to submit**Exercise 27.8 (3 points)**

Show that two distinct points P and Q in the projective plane uniquely determine a projective line containing both. How is the equation of this line computed from the coordinates of the two points?

Determine the homogeneous equation of the line through the points

$$(2, 3, 7) \quad \text{and} \quad (1, 5, -2).$$

Exercise 27.9 (3 points)

Let $\mathbb{P}_K^n$ be a projective space of dimension n , and let

$$X, Y \subseteq \mathbb{P}_K^n$$

be projective linear subspaces of dimensions r and s , respectively. Suppose that

$$r + s \geq n.$$

Show that

$$X \cap Y \neq \emptyset.$$

Exercise 27.10 (3 points)

Let K be an infinite field and let

$$P_1, \dots, P_m$$

be a finite collection of points in projective space $\mathbb{P}_K^n$. Show that there is a homogeneous linear form

$$L \in K[X_0, \dots, X_n]$$

such that all these points belong to the open set $D_+(L)$.

The next exercise requires an additional definition.

For a homogeneous ideal I in

$$R = A[X_0, \dots, X_n]$$

with the standard grading, the *saturation* of I is defined as

$$\{r \in R \mid \text{there exists } k \text{ such that } r(R_+)^k \subseteq I\}.$$

Here R_+ is the irrelevant ideal

$$\bigoplus_{d \geq 1} R_d = (X_0, \dots, X_n).$$

Exercise 27.11 (3 points)

Let A be a commutative ring and let

$$R = A[X_0, \dots, X_n]$$

be the polynomial ring with the standard grading. Show that the saturation of a homogeneous ideal I is again a homogeneous ideal.

Public Solutions to Worksheet 27

At the frozen revision boundary, the source provides no public solutions for any of the eleven exercises in Worksheet 27. The frozen authority query checked all eleven exact candidate solution-page titles and reported all of them absent. No additional solutions have been created for this edition.

The complete negative evidence is bound in `authority/wikiversity/unit-27/ORDERED_EXERCISE_MAP.json` and the API witness `authority/wikiversity/unit-27/worksheet-solution-candidates-api.json`.

Lecture 28: Projective Varieties and Projective Plane Curves

Projective varieties

Definition 28.1: projective variety

A *projective variety* is a Zariski-closed subset

$$V_+(\mathfrak{a}) \subseteq \mathbb{P}_K^n,$$

where $\mathfrak{a}$ is a homogeneous ideal in $K[X_0, X_1, \dots, X_n]$. Thus a projective variety Y is the zero locus in projective space of a finite collection of homogeneous polynomials.

With the induced topology, a projective variety again carries a Zariski topology. Its open sets have the form $D_+(\mathfrak{b})$ for a homogeneous ideal $\mathfrak{b}$, either in $K[X_0, \dots, X_n]$ or in the quotient ring

$$K[X_0, \dots, X_n]/\mathfrak{a},$$

which is also called the *homogeneous coordinate ring* of $V_+(\mathfrak{a})$. In particular, every homogeneous element $F \in K[X_0, \dots, X_n]$ defines an open set

$$D_+(F) \subseteq Y.$$

Lemma 28.2: covering by affine varieties

Let $Y \subseteq \mathbb{P}_K^n$ be a projective variety. The affine spaces

$$D_+(X_i) \cong \mathbb{A}_K^n \subset \mathbb{P}_K^n$$

give affine varieties

$$D_+(X_i) \cap Y$$

which cover Y . In particular, for every point $P \in Y$ and every open neighbourhood $P \in U$, there is an affine open neighbourhood of P contained in U .

Proof Within Y , we have

$$D_+^Y(X_i) := Y \cap D_+^{\mathbb{P}_K^n}(X_i) \cong Y \cap \mathbb{A}_K^n,$$

where $D_+^Y(X_i)$ denotes the relative open set in Y , whereas $D_+^{\mathbb{P}_K^n}(X_i)$ is the standard chart in the ambient projective space. Thus $D_+^Y(X_i)$ is a closed subset (see Exercise 28.2) of the affine space $D_+^{\mathbb{P}_K^n}(X_i) \cong \mathbb{A}_K^n$, and is therefore an affine variety. Since the sets $D_+^{\mathbb{P}_K^n}(X_i)$ cover projective space, the sets $D_+^Y(X_i)$ cover Y .

Editorial bridge. To obtain the claim about an arbitrary open neighbourhood U , choose one of the affine charts above containing P , then shrink the intersection to a principal open set containing P and contained in U . That principal open set is affine, giving the required affine neighbourhood. $\square$

Editorial note - relative and ambient notation. The source uses $D_+(X_i)$ both for the relative open set in Y and for the standard chart in $\mathbb{P}_K^n$. This edition displays the intersection with Y to distinguish the two meanings.

An immediate consequence is that local concepts developed for affine varieties also apply to projective varieties. To check a property at a point, we can pass straight to an affine open neighbourhood of that point. This applies, for example, to smoothness, normality, and regular functions.

Algebraic functions and morphisms

Using the result just proved, we can again define what is meant by a regular or algebraic function on a projective variety.

Definition 28.3: regular function

Let K be an algebraically closed field, let $Y \subseteq \mathbb{P}_K^n$ be a projective variety, let $U \subseteq Y$ be an open set, and let $P \in U$. A function

$$f : U \longrightarrow \mathbb{A}_K^1 = K$$

is called *algebraic*, *regular*, or *polynomial* at P if there is an affine open neighbourhood

$$P \in V \subseteq U$$

such that $f|_V$ is algebraic at P . The function f is called algebraic on U if it is algebraic at every point of U .

For an open set U , the set of all regular functions on U again forms a commutative K -algebra, denoted $\Gamma(U, \mathcal{O})$. From now on, a projective variety means a projective zero locus equipped with the induced Zariski topology and the *structure sheaf* $\mathcal{O}$ of regular functions.

These concepts extend immediately to open subsets, leading to the notion of a quasiprojective variety.

Definition 28.4: quasiprojective variety

An open subset of a projective variety, equipped with the induced Zariski topology and the structure sheaf of algebraic functions, is called a *quasiprojective variety*.

In particular, both projective varieties and affine varieties are quasiprojective. For the latter, an affine variety $Y \subseteq \mathbb{A}_K^n$ can be extended to a projective variety $\tilde{Y} \subseteq \mathbb{P}_K^n$ containing Y as an open subset.

The definition of a morphism also applies word for word in this more general situation.

Definition 28.5: morphism of quasiprojective varieties

Let X and Y be quasiprojective varieties over an algebraically closed field, and let

$$\psi : Y \longrightarrow X$$

be a continuous map. The map ψ is called a *morphism* if, for every open set $U \subseteq X$ and every algebraic function $f \in \Gamma(U, \mathcal{O}_X)$, the composition

$$f \circ \psi : \psi^{-1}(U) \longrightarrow U \xrightarrow{f} \mathbb{A}_K^1$$

belongs to $\Gamma(\psi^{-1}(U), \mathcal{O}_Y)$.

Homogenisation and projective closure

Let K be algebraically closed, and consider the hyperbola

$$V(XY - 1) \subset \mathbb{A}_K^2 \subset \mathbb{P}_K^2.$$

The hyperbola is closed in the affine plane but not in the projective plane. Embed the affine plane as $V(Z - 1)$ in three-dimensional space, and consider the lines through the origin and the points of the hyperbola. Geometrically these lines tilt increasingly and, in the real or complex picture, approach the x -axis and the y -axis. The following algebraic computation gives a statement valid over this base field.

Source-condition note REVIEW-AK-26-30-C08 - base field. The source does not state a field hypothesis in this introductory paragraph. This edition uses the algebraically closed-field setting of Definition 28.3 and Theorem 28.8; over a finite field, the stated non-closedness fails for the Zariski topology on K -points. The source's expression "approach" remains Euclidean intuition for $\mathbb{R}$ or $\mathbb{C}$.

Definition 28.6: homogenisation of an ideal

For an ideal

$$\mathfrak{a} \subseteq K[X_1, \dots, X_n],$$

the ideal in $K[X_1, \dots, X_n, Z]$ generated by the homogenisations of all elements of $\mathfrak{a}$ is called the *homogenisation* $\mathfrak{a}^h$ of $\mathfrak{a}$.

In general, homogenising only a generating set of the ideal $\mathfrak{a}$ is not enough.

Definition 28.7: projective closure

For an affine variety

$$V(\mathfrak{a}) \subseteq \mathbb{A}_K^n \subseteq \mathbb{P}_K^n,$$

the Zariski closure of $V(\mathfrak{a})$ in $\mathbb{P}_K^n$ is called the *projective closure* of $V(\mathfrak{a})$.

Theorem 28.8: projective closure by homogenisation

Let K be an algebraically closed field and let

$$V = V(\mathfrak{a}) \subseteq \mathbb{A}_K^n \cong D_+(X_0)$$

be an affine variety. The projective closure of $V(\mathfrak{a})$ in $\mathbb{P}_K^n$ is $V_+(\mathfrak{b})$, where $\mathfrak{b}$ is the homogenisation of $\mathfrak{a}$ in $K[X_0, X_1, \dots, X_n]$.

Proof A point $P = (x_1, \dots, x_n)$ in $\mathbb{A}_K^n$ determines the point $\widehat{P} = (1, x_1, \dots, x_n)$ in $\mathbb{P}_K^n$. For $F \in K[X_1, \dots, X_n]$ and its homogenisation $\widehat{F}$, we have

$$F(P) = \widehat{F}(\widehat{P}).$$

Consequently, all homogeneous polynomials in $\mathfrak{b}$ vanish on $V(\mathfrak{a})$, so

$$V(\mathfrak{a}) \subseteq V_+(\mathfrak{b}).$$

We obtain a commutative diagram in which every arrow is injective,

$$\begin{array}{ccc} V(\mathfrak{a}) & \longrightarrow & V_+(\mathfrak{b}) \\ \downarrow & & \downarrow \\ \mathbb{A}_K^n & \longrightarrow & \mathbb{P}_K^n. \end{array}$$

Write the projective closure of $V(\mathfrak{a})$ as $V_+(\mathfrak{c})$ for a homogeneous ideal $\mathfrak{c}$. Minimality of the closure gives $V_+(\mathfrak{c}) \subseteq V_+(\mathfrak{b})$. To prove the reverse inclusion, it suffices to show

$$\mathfrak{c} \subseteq \text{rad}(\mathfrak{b}).$$

Take a nonzero homogeneous polynomial $F \in \mathfrak{c}$ and write

$$F = X_0^r G,$$

where G is not a multiple of X_0 . Since F vanishes on $V(\mathfrak{a})$ and X_0 does not vanish on $V(\mathfrak{a}) \subseteq D_+(X_0)$, the polynomial G also vanishes there. Its dehomogenisation

$$g = G(1, X_1, \dots, X_n)$$

therefore vanishes on $V(\mathfrak{a})$. Hilbert's Nullstellensatz gives an integer $N \geq 1$ such that $g^N \in \mathfrak{a}$. Because X_0 does not divide G , the degree of g equals the degree of G , so homogenising g^N gives G^N . Hence $G^N \in \mathfrak{b}$, and consequently

$$F^N = X_0^{rN} G^N \in \mathfrak{b}.$$

Thus $F \in \text{rad}(\mathfrak{b})$. This proves the required inclusion and shows that the closure is exactly $V_+(\mathfrak{b})$. $\square$

Source correction REVIEW-AK-26-30-C06 - the radical step. The source replaces $\mathfrak{a}$ by its radical and then concludes membership in the original homogenised ideal $\mathfrak{b}$, although replacing $\mathfrak{a}$ can change that ideal. The argument above keeps $\mathfrak{a}$ and $\mathfrak{b}$ fixed and proves the required power-membership $F^N \in \mathfrak{b}$.

Projective plane curves

Definition 28.9: projective plane curve

A *projective plane curve* is the zero locus

$$C = V_+(F) \subset \mathbb{P}_K^2$$

of a nonconstant homogeneous polynomial $F \in K[X, Y, Z]$.

For an affine plane curve $V = V(G) \subset \mathbb{A}_K^2 \subset \mathbb{P}_K^2$, the Zariski closure of V in $\mathbb{P}_K^2$ is called the projective closure of the curve.

Corollary 28.10: an equation for the projective closure of a curve

Let K be an algebraically closed field and let

$$V = V(G) \subseteq \mathbb{A}_K^2 \subseteq \mathbb{P}_K^2, \quad G \in K[X, Y].$$

The Zariski closure of V in $\mathbb{P}_K^2$ is

$$C = V_+(H),$$

where H is the homogenisation of G in $K[X, Y, Z]$.

Proof This follows directly from Theorem 28.8 and the fact that the homogenisation of a principal ideal is generated by the homogenisation of its generator. $\square$

Editorial note - two unresolved source references. In the preceding step, the source prints `nach Aufgabe *****\nolinkurl{` while linking to the [exercise on homogenising a principal ideal] (<https://de.wikiversity.org/wiki/Hauptideal/Homogenisierung/Aufgabe>). After the corollary, it also prints `}siehe Aufgabe *****` while linking to the [exercise on a homogeneous equation for the projective closure over a field that is not algebraically closed](#). Neither exercise number is supplied; this edition preserves the exact linked identities without guessing their numbers.

Without the assumption that the field is algebraically closed, the assertion need not hold.

Remark 28.11: affine charts and points at infinity

Let $G \in K[X, Y]$ and let $F \in K[X, Y, Z]$ be its homogenisation. We recover G from F by setting $Z = 1$. The polynomial G describes the intersection $D_+(Z) \cap V_+(F)$. The other two affine pieces,

$$D_+(X) \cap V_+(F) \quad \text{and} \quad D_+(Y) \cap V_+(F),$$

play equal roles and give affine neighbourhoods of the points of $C = V_+(F)$ not lying in $D_+(Z)$.

To check smoothness at $P \in C$, choose an affine open neighbourhood, preferably one of $D_+(L) \cap C$ with $L = X, Y, Z$, and apply the derivative criterion to the affine equation in that chart. The result does not depend on the chart chosen, although one chart may be computationally more convenient.

From the viewpoint of the affine curve $V(G)$, the points at infinity are

$$V_+(F) \cap V_+(Z).$$

This is the intersection of the projective curve with a projective line.

The intersection is finite unless the line $V_+(Z)$ is a component of the curve. This cannot occur when we start with an affine curve, since Z does not divide the homogenisation F . Write the homogeneous decomposition

$$G = G_d + \cdots + G_m, \quad m \leq d,$$

so that

$$F = G_d + G_{d-1}Z + \cdots + G_mZ^{d-m}.$$

Setting $Z = 0$ shows that the points at infinity are given by the projective zeros of the homogeneous polynomial $G_d(X, Y)$. Thus the degree d immediately bounds the number of points at infinity on the curve.

Example 28.12: conic sections as affine charts

Assume $\text{char}(K) \neq 2$, and consider the standard cone

$$V(X^2 + Y^2 - Z^2) \subset \mathbb{A}_K^3.$$

Since its equation is homogeneous, the cone can also be regarded as the projective plane curve of degree two

$$V_+(X^2 + Y^2 - Z^2) \subset \mathbb{P}_K^2.$$

Intersections of the cone with arbitrary planes $E \subset \mathbb{A}_K^3$ are called *conic sections*. If E does not pass through the origin, it can naturally be identified with an open affine plane $D_+(L) \subseteq \mathbb{P}_K^2$, where L is a homogeneous linear form describing the vector subspace parallel to E . The intersections of the cone with E are different affine pieces of the same projective curve. In particular, circles, hyperbolas, and parabolas are such affine pieces.

By contrast, intersections with planes through the origin, viewed projectively, are the finite sets

$$V_+(X^2 + Y^2 - Z^2) \cap V_+(L).$$

Editorial note - characteristic two. The source imposes no restriction on the characteristic. In characteristic 2, the polynomial above becomes $(X + Y + Z)^2$; if $L = X + Y + Z$, the projective intersection is the whole line, not a finite set. The characteristic restriction above ensures that the conic is nonsingular and has no line as a component.

Definition 28.13: Fermat curve

Let K be a field and let $d \geq 1$. The projective plane curve

$$V_+(X^d + Y^d + Z^d) \subseteq \mathbb{P}_K^2$$

is called the *Fermat curve* of degree d . For $d = 1$, it is simply a projective line.

Editorial note - projective zero-locus operator. In this definition the source writes $V(X^d + Y^d + Z^d)$ although the object lies in $\mathbb{P}_K^2$; the next lemma correctly writes V_+ . This edition consistently uses V_+ .

Lemma 28.14: smoothness of Fermat curves

Let K be an algebraically closed field of characteristic $p \geq 0$, and let

$$C = V_+(X^d + Y^d + Z^d) \subset \mathbb{P}_K^2$$

be the Fermat curve of degree d . If the characteristic of K does not divide d , then C is smooth.

Proof Smoothness is a local property, so it suffices to work on any affine piece. By symmetry, consider

$$V(X^d + Y^d + 1) \subset \mathbb{A}_K^2.$$

The partial derivatives are dX^{d-1} and dY^{d-1} . The characteristic hypothesis gives $d \neq 0$. If $d = 1$, both derivatives are nonzero constants and never vanish. If $d > 1$, both vanish simultaneously only at $x = y = 0$, which does not lie on the curve. $\square$

Editorial note - degree-one case. The source immediately states that both derivatives vanish simultaneously only at $(0, 0)$; this does not hold for $d = 1$, when both derivatives are nonzero constants. The case distinction above preserves the smoothness conclusion for all permitted d .



Figure 95: Football pattern illustrating a surface of genus zero

Sphere, or surface of genus zero. OpenClipart file, currently uploaded to Commons by MapGrid, CC0 1.0; the historical course label records Ranveig/PD.

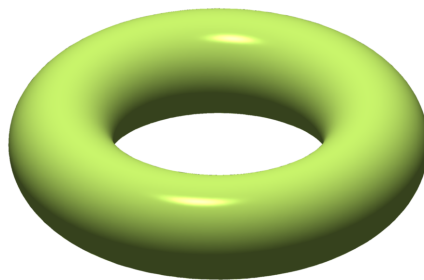


Figure 96: A torus as a surface with one handle

Torus, a surface of genus one. Oleg Alexandrov, public domain.



Figure 97: A double torus as a surface with two handles

Double torus, a surface of genus two. Oleg Alexandrov, public domain.

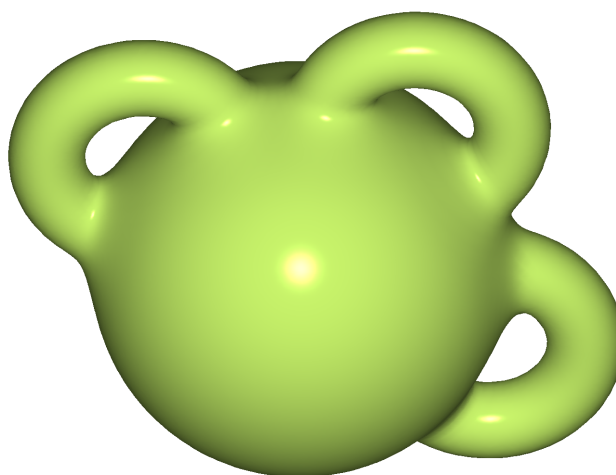


Figure 98: A sphere with three handles

Sphere with three handles, a surface of genus three. Oleg Alexandrov, public domain.

Remark 28.15: topological shape and genus

Over the base field $\mathbb{C}$, a smooth connected projective curve can be viewed as a compact oriented real two-dimensional manifold. Topologically, such a manifold is homeomorphic to a sphere with g handles attached. The number g is called the *genus* of the real surface, and also the genus of the curve.

Source-condition note REVIEW-AK-26-30-C 10 - connectedness. The source says “a smooth projective curve” here, but the description by one sphere with g handles and one genus presupposes that the curve is connected. This edition states that convention explicitly.

The complex projective line is a two-dimensional sphere with no handles, so its genus is 0. A surface of genus 1 is a torus (like a car tyre), homeomorphic to $S^1 \times S^1$. In the source’s exposition, projective curves whose underlying topological manifolds have genus one are called elliptic curves.

Editorial note - convention for elliptic curves. In modern terminology, an elliptic curve usually means a smooth projective curve of genus one together with a base point. The source’s statement describes a genus-one curve without a chosen base point; this edition retains the exposition and notes the difference in convention.

Genus also has algebraic definitions and is therefore defined for smooth connected projective curves over every algebraically closed field. It equals the K -dimension of the first cohomology group of the structure sheaf, and also the K -dimension of the space of global differential forms on the curve.

For every g , there is a projective curve of genus g . In particular, every compact oriented real two-dimensional surface can be realised as a complex projective curve. Such objects are also called *Riemann surfaces*.

For a smooth plane curve

$$C = V_+(F) \subset \mathbb{P}_K^2$$

of degree $d = \deg(F)$, the genus is

$$g = \frac{(d-1)(d-2)}{2}.$$

Smooth projective plane curves of degree one or two, namely lines and conics, have genus 0 and are isomorphic to the projective line. For $d = 3$ we obtain genus 1, giving elliptic curves in the source's convention, whereas $d = 4$ gives genus 3. Thus not every genus can be realised by a smooth plane curve. For example, giving explicit equations for a curve of genus 2 is by no means easy.

Worksheet 28

Warm-up exercises

Exercise 28.1

Let

$$P = (a_0, \dots, a_n) \in \mathbb{P}_K^n$$

be a point in projective space. Show that there is an affine open neighbourhood

$$U \cong \mathbb{A}_K^n \subset \mathbb{P}_K^n$$

such that P corresponds to the origin in this affine space.

Exercise 28.2

Let

$$D_+(L) \cong \mathbb{A}_K^n \subseteq \mathbb{P}_K^n,$$

where L is a homogeneous linear form in $K[X_0, \dots, X_n]$. Show that the Zariski topology on projective space induces the Zariski topology on this affine space.

Exercise 28.3

For each $n \in \mathbb{Z}$, define the power map

$$x \mapsto x^n$$

as a morphism from the projective line to itself. What are the fibres of this morphism?

Editorial note - all integer exponents. The source quantifier $n \in \mathbb{Z}$ is retained. The cases $n = 0$, $n < 0$, and characteristic dividing $|n|$ must be distinguished; this edition does not silently substitute an assumption of $n > 0$ or characteristic zero.

Exercise 28.4

Determine the projective closure of the complex cardioid

$$V((X^2 + Y^2)^2 - 2X(X^2 + Y^2) - Y^2),$$

and in particular its points at infinity.

Exercise 28.5

Show that the cone map

$$\mathbb{A}_K^{n+1} \setminus \{0\} \longrightarrow \mathbb{P}_K^n$$

is a morphism of quasiprojective varieties.

Exercise 28.6

Give an example showing that the cone map

$$\mathbb{A}_K^{n+1} \setminus \{0\} \longrightarrow \mathbb{P}_K^n$$

need not be a closed map.

Exercise 28.7

Let X and Y be quasiprojective varieties and let $\varphi : X \rightarrow Y$ be a continuous map. Let

$$Y = \bigcup_{i \in I} U_i$$

be an open cover. Show that φ is a morphism if and only if, for every i , the restriction

$$\varphi_i : \varphi^{-1}(U_i) \longrightarrow U_i$$

is a morphism.

Exercise 28.8

Let K be an algebraically closed field. Determine the ring of global sections

$$\Gamma(\mathbb{P}_K^1, \mathcal{O}_{\mathbb{P}_K^1}).$$

What does this imply for a morphism

$$\mathbb{P}_K^1 \longrightarrow \mathbb{A}_K^1?$$

Source-condition note REVIEW-AK-26-30-C08 - classical base-field convention. The source says only “field” in Exercises 28.8, 28.10 and 28.14. The structure sheaf used in this chapter was defined in Definition 28.3 over an algebraically closed field; with the source’s topology on K -points, the asserted closure and global-function conclusions can fail over other fields. This edition keeps all three exercises in that defined setting.

Exercise 28.9

Define and characterise when an irreducible quasiprojective variety is *normal*.

Exercise 28.10 *

Let K be an algebraically closed field. Consider the affine plane curve

$$C = V(Y - X^3 + X + 2).$$

Define an isomorphism between C and the affine line $\mathbb{A}_K^1$. Can such an isomorphism be extended to an isomorphism between $\mathbb{P}_K^1$ and the projective closure

$$\overline{C} \subset \mathbb{P}_K^2?$$

Exercises to submit**Exercise 28.11 (3 points)**

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$. Let $H \subset \mathbb{K}^{n+1}$ be an n -dimensional affine subspace not containing the origin, and let $\widetilde{H}$ be the subspace parallel to H through the origin. Let $U \subseteq H$ be open in the metric topology on $H \cong \mathbb{K}^n$, and let V be the union of all lines through the origin and a point of U . Show that

$$V \cap (\mathbb{K}^{n+1} \setminus \widetilde{H})$$

is open.

Exercise 28.12 (4 points)

For the cone map

$$\mathbb{A}_K^{n+1} \setminus \{0\} \longrightarrow \mathbb{P}_K^n,$$

determine the Zariski closure in $\mathbb{P}_K^n$ of the image of the closed set

$$V(\mathfrak{a}) \cap (\mathbb{A}_K^{n+1} \setminus \{0\}).$$

Exercise 28.13 (3 points)

Let X be an irreducible quasiprojective variety with function field $L = K(X)$. Let U be a nonempty open set, and let $(U_i)_{i \in I}$ be a cover of U by nonempty open sets:

$$U = \bigcup_{i \in I} U_i.$$

Show that

$$\Gamma(U, \mathcal{O}) = \bigcap_{i \in I} \Gamma(U_i, \mathcal{O}),$$

where the intersection is taken inside L .

Source-condition note REVIEW-AK-26-30-C09 - nonempty opens. The source allows arbitrary open sets. Identifying a section ring with a subring of the function field requires the open set to be nonempty; empty members of a cover must therefore be omitted before taking this intersection.

Exercise 28.14 (3 points)

Let K be an algebraically closed field and let $\mathbb{P}_K^n$ be projective space over K . Show that the only global algebraic functions are constants, that is,

$$\Gamma(\mathbb{P}_K^n, \mathcal{O}_{\mathbb{P}_K^n}) = K.$$

Source remark. This statement holds for every connected projective variety over an algebraically closed field.

Public Solutions to Worksheet 28

At the frozen revision boundary, the source provides a public solution only for Exercise 28.10. The frozen authority query reports the other thirteen candidate solution pages as absent. No additional solutions have been created for this edition.

Solution to Exercise 28.10

Under the algebraically closed-field hypothesis stated in the reviewed exercise, an isomorphism is given by

$$x \mapsto (x, x^3 - x - 2) = (x, y).$$

At the ring level, this map corresponds to the substitution homomorphism

$$\begin{aligned} K[X, Y]/(Y - X^3 + X + 2) &\longrightarrow K[X], \\ X &\mapsto X, \\ Y &\mapsto X^3 - X - 2. \end{aligned}$$

This homomorphism is well defined and surjective. Since Y can be eliminated directly on the left, the source ring is isomorphic to $K[X]$. Thus the map is indeed an isomorphism of affine curves.

This isomorphism cannot be extended to an isomorphism with the projective line. The projective closure of the curve is

$$\overline{C} = V_+(YZ^2 - X^3 + XZ^2 + 2Z^3).$$

Exactly one point at infinity is added, namely $(0, 1, 0)$. In the affine neighbourhood $D_+(Y)$, this point becomes the origin on the affine curve

$$V(Z^2 - X^3 + XZ^2 + 2Z^3).$$

The origin has multiplicity two and is therefore not smooth.

Editorial bridge. The lowest-degree term Z^2 explains this multiplicity two. Since the projective line is smooth, $\overline{C}$ is not isomorphic to $\mathbb{P}_K^1$.

Editorial note - singular/plural agreement. The source uses a grammatical form referring to “points” but then gives exactly one point, $(0, 1, 0)$. The homogeneous equation also gives only that point. This edition translates the consistent mathematical meaning: exactly one point at infinity.

Lecture 29: Projections and Parametrised Projective Curves

Projection away from a point

Definition 29.1: projection away from a point

The map

$$\begin{aligned} \mathbb{P}_K^n \setminus \{(1, 0, \dots, 0)\} &\longrightarrow \mathbb{P}_K^{n-1}, \\ (x_0, x_1, \dots, x_n) &\mapsto (x_1, \dots, x_n) \end{aligned}$$

is called the *projection away from the point* $(1, 0, \dots, 0)$.

This map is a well-defined morphism outside the *centre* of projection $(1, 0, \dots, 0)$. Each other point is sent to the point of $\mathbb{P}_K^{n-1}$ corresponding to the line through that point and the centre. Hence the map is surjective, and each fibre is a projective line with the centre removed, and thus an affine line. In other words, we have a so-called *line bundle* over $\mathbb{P}_K^{n-1}$.

This map extends the cone map

$$\mathbb{A}_K^n \setminus \{0\} \longrightarrow \mathbb{P}_K^{n-1}$$

to punctured projective space. The corresponding map can be defined for any centre; see Exercise 29.7.

Maps to $\mathbb{P}_K^1$

The following theorem gives a new version of Noether normalisation.

Theorem 29.2: projecting a plane curve to $\mathbb{P}_K^1$

Let K be an algebraically closed field and let

$$C \subseteq \mathbb{P}_K^2$$

be a projective plane curve of degree d . Then there is a surjective morphism

$$C \longrightarrow \mathbb{P}_K^1$$

such that each fibre consists of at most d points.

Proof Choose a point

$$P \in \mathbb{P}_K^2$$

not lying on the curve. Such a point exists because, in particular, K is infinite. Consider projection away from P ; its restriction induces a morphism

$$C \hookrightarrow \mathbb{P}_K^2 \setminus \{P\} \longrightarrow \mathbb{P}_K^1.$$

The fibre of this morphism over a point

$$Q \in \mathbb{P}_K^1,$$

representing a direction at $P \in \mathbb{P}_K^2$ consists exactly of those points of the curve lying on the line determined by Q ,

$$G = V_+(aX + bY + cZ) \cong \mathbb{P}_K^1 \subseteq \mathbb{P}_K^2.$$

Thus the fibre over Q can be described on G by eliminating a variable from the curve equation

$$C = V_+(F)$$

using the line equation. The result is a nonzero homogeneous polynomial $\overline{F}$ in two variables of degree d ; it cannot be zero, since then P would lie on the curve. As we work over an algebraically closed field, $\overline{F}$ has at least one and at most d zeros, all distinct from P . This proves surjectivity and the bound on the number of points in each fibre. $\square$

Theorem 29.3: a rational function as a morphism to $\mathbb{P}_K^1$

Let K be a field and let

$$C \subseteq \mathbb{P}_K^2$$

be a smooth irreducible projective plane curve. Let

$$D = C \cap D_+(Z) \cong K\text{-Spek}(R)$$

be an affine piece of this curve, and let

$$q = \frac{g}{h} \in Q(R)$$

be a rational function, with $g, h \in R$ and $h \neq 0$. Then there is exactly one morphism

$$\varphi : C \longrightarrow \mathbb{P}_K^1$$

such that the diagram

$$\begin{array}{ccc} D(h) & \xrightarrow{g/h} & \mathbb{A}_K^1 \cong D_+(s) \\ \downarrow & & \downarrow \\ C & \xrightarrow{\varphi} & \mathbb{P}_K^1 \end{array}$$

commutes.

Moreover, every genuine pole $P \in D$, meaning a point with $h(P) = 0$ and $g(P) \neq 0$, is mapped to the point at infinity $\infty \in \mathbb{P}_K^1$.

Proof First we define on D an extension

$$\varphi : D \longrightarrow \mathbb{P}_K^1$$

of the rational function g/h . If $q = 0$, the constant map with value 0 is the required extension, so assume from now on that $q \neq 0$. Take a point $P \in D$ on the curve. If $P \in D(h)$, there is nothing to do. Thus suppose $h(P) = 0$.

Source correction REVIEW-AK-26-30-C12 - the zero rational function. The source immediately writes $g/h = u\pi^n$ with u a unit, which is possible only for a nonzero element of the function field. The theorem also permits $q = 0$; the separate constant-map case above closes that gap without changing the nonzero case.

Since the curve is smooth, Theorem 23.6 shows that its local ring B at P is a discrete valuation ring. The quotient g/h can therefore be written there as

$$\frac{g}{h} = u\pi^n,$$

with $u \in B^\times$, $n \in \mathbb{Z}$, and π a uniformiser (a generator of the maximal ideal). There is an open neighbourhood

$$P \in D(\psi) \subseteq D$$

such that π and u are defined on $D(\psi)$ and u is a unit there. If $n \geq 0$, then

$$\frac{g}{h} \in R_\psi,$$

so the point of indeterminacy is removable even for a map to $\mathbb{A}_K^1$. If $n \leq 0$, the reciprocal quotient

$$\frac{h}{g} = u^{-1}\pi^{-n}$$

is defined on $D(\psi)$ as a map to $\mathbb{A}_K^1$. Using the “embedding with coordinates interchanged”

$$\mathbb{A}_K^1 \cong D_+(t) \hookrightarrow \mathbb{P}_K^1,$$

we obtain a map to $\mathbb{P}_K^1$.

We must show that these two morphisms to the projective line agree wherever both are defined. These are the points P where g/h has neither a zero nor a pole. Compatibility follows because on an open neighbourhood

$$P \in U$$

there is a map

$$\frac{g}{h} : U \longrightarrow (\mathbb{A}_K^1)^\times = \mathbb{A}_K^1 \setminus \{0\},$$

and the diagram

$$\begin{array}{ccc} (\mathbb{A}_K^1)^\times & \xrightarrow{i^{-1}} & \mathbb{A}_K^1 \cong D_+(t) \\ \downarrow & & \downarrow \\ \mathbb{A}_K^1 \cong D_+(s) & \longrightarrow & \mathbb{P}_K^1 \end{array}$$

commutes. This gives a well-defined morphism on the affine piece D .

Source correction AGC-CORR-0127 - the object with zeros and poles. The source refers to φ in this sentence, although zeros and poles are properties of the rational function g/h . This edition displays the correct object without changing the overlap condition or the gluing diagram.

For an arbitrary point on the projective curve C and an affine neighbourhood

$$P \in D' \subseteq C,$$

we are in the same situation, because

$$D_+(h) \cap D' \neq \emptyset,$$

so the rational function is defined on a nonempty open set, possibly with different numerator and denominator. The preceding argument therefore applies in the same way.

Uniqueness follows because, on every affine open set

$$P \in U$$

the intersection $U \cap D_+(h)$ is nonempty. A morphism from an integral variety to the affine line $\mathbb{A}_K^1$ is uniquely determined by its rational function. $\square$

Example 29.4: inversion on the projective line

The inversion map

$$\begin{aligned} \mathbb{A}_K^1 \supset D(z) &\longrightarrow \mathbb{A}_K^1, \\ z &\longmapsto z^{-1} \end{aligned}$$

extends to a bijective morphism

$$\begin{aligned} \mathbb{P}_K^1 &\longrightarrow \mathbb{P}_K^1, \\ (x, y) &\longmapsto (y, x). \end{aligned}$$

This follows directly from Theorem 29.3. Every point $z \neq 0$ is sent to $1/z$, while zero is sent to the point at infinity ∞ .

Parametrised projective plane curves

Suppose a curve with rational parametrisation

$$s \longmapsto \left(\frac{\varphi_1(s)}{\psi(s)}, \frac{\varphi_2(s)}{\psi(s)} \right).$$

is given. In Theorem 6.11 we saw that its image satisfies an algebraic equation. In that theorem's proof we already used the homogenised parametrisation; it now reappears as a projective extension.

Theorem 29.5: projective extension of a rational parametrisation

Let K be an infinite field, and let

$$\begin{aligned} \mathbb{A}_K^1 \supseteq D(\psi) &\longrightarrow \mathbb{A}_K^2, \\ s &\longmapsto \left(\frac{\varphi_1(s)}{\psi(s)}, \frac{\varphi_2(s)}{\psi(s)} \right) \end{aligned}$$

be a rational parametrisation in reduced form, meaning that $\varphi_1, \varphi_2, \psi$ have no common divisor. Let d be the largest degree of the polynomials involved, and let

$$\widehat{\varphi_1}, \quad \widehat{\varphi_2}, \quad \widehat{\psi}$$

be their homogenisations with respect to the new variable t . Each of H_1, H_2, H_3 is obtained from the corresponding homogenisation by multiplication by a suitable power of t , so that all three have degree d .

Then H_1, H_2, H_3 define a morphism

$$\begin{aligned} H : \mathbb{P}_K^1 &\longrightarrow \mathbb{P}_K^2, \\ (s, t) &\longmapsto (H_1(s, t), H_2(s, t), H_3(s, t)), \end{aligned}$$

such that the diagram

$$\begin{array}{ccc} \mathbb{A}_K^1 \supseteq D(\psi) & \longrightarrow & \mathbb{A}_K^2 \cong D_+(Z) \\ \downarrow & & \downarrow \\ \mathbb{P}_K^1 & \xrightarrow{H} & \mathbb{P}_K^2 \end{array}$$

commutes. Moreover, the image of H lies on the projective closure of the affine image curve.

Source-condition note REVIEW-AK-26-30-C13 - infinitude of the base field. The source does not state this hypothesis, but its proof uses the assertion that a finite open subset of $\mathbb{P}_K^1$ is empty. For the source's topology on K -points this requires K to be infinite; it fails over a finite field, where every subset of $\mathbb{P}_K^1$ is finite.

Proof By Exercise 29.6, the map H is well defined on all of $\mathbb{P}_K^1$, since $\varphi_1, \varphi_2, \psi$ have no common divisor. For commutativity, it suffices to note that a point

$$s \in D(\psi) \subseteq \mathbb{A}_K^1$$

is sent, on the one hand, through $(s, 1)$ to

$$(H_1(s, 1), H_2(s, 1), H_3(s, 1)) = (\varphi_1(s), \varphi_2(s), \psi(s)),$$

and, on the other hand, to

$$\left(\frac{\varphi_1(s)}{\psi(s)}, \frac{\varphi_2(s)}{\psi(s)}, 1 \right) = (\varphi_1(s), \varphi_2(s), \psi(s))$$

as a projective point.

For the additional assertion, let C be the affine closure of the image and let

$$\overline{C} \subseteq \mathbb{P}_K^2$$

be its projective closure. Consider the open complement

$$U = \mathbb{P}_K^2 \setminus \overline{C}.$$

Since the map is continuous, the inverse image $H^{-1}(U)$ is open in $\mathbb{P}_K^1$ and can contain only points of $\mathbb{P}_K^1 \setminus D(\psi)$. But a finite open subset of the projective line must be empty. $\square$

Theorem 29.6: projective closure of the graph of a polynomial

Let K be an algebraically closed field and let

$$F \in K[X]$$

be a polynomial in one variable of degree $d \geq 1$. The projective closure C of the graph

$$V(Y - F(X))$$

is described by

$$V_+(YZ^{d-1} - \widehat{F}(X, Z)),$$

where $\widehat{F}(X, Z)$ is the homogenisation of F . If $d = 1$ and $F = aX + b$, then C has one additional point, the smooth point $(1, a, 0)$. If $d \geq 2$, the additional point is $(0, 1, 0)$, which is singular when $d \geq 3$. For $d \geq 2$, this point at infinity has multiplicity $d - 1$.

Source correction AGC-CORR-0128 - projective zero-locus operator. The source prints V for a homogeneous equation in $\mathbb{P}_K^2$. This edition uses V_+ , in keeping with the projective ambient space and the notation of the subsequent proof.

Proof The equation of the projective closure follows directly from Corollary 28.10. To determine the intersection of C with the projective line $V_+(Z)$ at infinity, set $Z = 0$ in the equation.

If $d = 1$, the curve equation is the line equation

$$V_+(Y - aX - bZ),$$

and its intersection with $V_+(Z)$ gives the unique point $(1, a, 0)$. If $d \geq 2$, the curve equation is

$$V_+(YZ^{d-1} - s_d X^d - s_{d-1} X^{d-1} Z - \cdots - s_0 Z^d),$$

with $s_d \neq 0$. Setting $Z = 0$ leaves

$$V_+(-s_d X^d),$$

so $X = 0$. This gives the unique point at infinity $(0, 1, 0)$.

To compute the multiplicity, consider the affine equation of the curve on $D_+(Y)$. Setting $Y = 1$ gives the affine equation

$$V(Z^{d-1} - s_d X^d - s_{d-1} X^{d-1} Z - \cdots - s_0 Z^d),$$

and in these coordinates $(0, 1, 0)$ becomes the origin. Hence its multiplicity is $d - 1$, with the unique tangent line given by $Z = 0$. If $d \geq 3$, the multiplicity is at least 2, so the point is singular. $\square$

Source correction AGC-U29-SRC-001 / AGC-CORR-0126 - undefined symbol in the singularity bound. In the last sentence of the proof, the source prints $g \geq 3$, although no quantity g is defined in this theorem. The degree defined and used throughout the calculation is d ; this edition therefore displays the intended bound as $d \geq 3$ and explicitly retains the multiplicity argument $d - 1 \geq 2$.

The theorem can be understood as follows. If $d \geq 2$, the point $(0, 1, 0)$ is the unique point at infinity and represents the direction of the y -axis. The line at infinity $V_+(Z)$ is the unique tangent line at this point.

Source correction REVIEW-AK-26-30-C14 - direction versus asymptote. The source calls the y -axis the graph's only asymptote. The projective point $(0, 1, 0)$ records the direction of that axis, not an affine asymptotic line; in the usual affine sense a polynomial graph of degree at least two has no linear asymptote. The source's valid tangent-line statement is retained.

The normalisation of C is $\mathbb{P}_K^1$. By Theorem 29.5, applied to the affine parametrisation of the graph

$$\begin{aligned} \mathbb{A}_K^1 &\longrightarrow \mathbb{A}_K^1 \times \mathbb{A}_K^1 = \mathbb{A}_K^2 \cong D_+(Z) \subset \mathbb{P}_K^2, \\ x &\longmapsto (x, F(x)) = (x, F(x), 1), \end{aligned}$$

the normalisation map is given by

$$\begin{aligned} \mathbb{P}_K^1 &\longrightarrow C \subset \mathbb{P}_K^2, \\ (x, t) &\longmapsto (xt^{d-1}, \widehat{F}(x, t), t^d). \end{aligned}$$

The point at infinity $(1, 0)$ is sent to

$$(0, s_d, 0) = (0, 1, 0).$$

Theorem 29.7: projective closure of the graph of a rational function

Let K be an algebraically closed field, and let

$$G, H \in K[X]$$

be polynomials in one variable of degrees $d, e \geq 1$, respectively, with no common root. Let $H \neq 0$ and let

$$F(X) = \frac{G(X)}{H(X)}$$

be the corresponding rational function. Let $\widehat{G}(X, Z)$ and $\widehat{H}(X, Z)$ be their respective homogenisations. If $d > e$, the projective closure C of the graph of $F(X)$ is described by

$$V_+(\widehat{H}(X, Z)YZ^{d-e-1} - \widehat{G}(X, Z)),$$

whereas if $d \leq e$, it is described by

$$V_+(\widehat{H}(X, Z)Y - \widehat{G}(X, Z)Z^{e-d+1}).$$

Proof The affine description of the curve is

$$V(YH - G).$$

By Corollary 28.10, the projective closure is described by the homogenisation of $YH - G$. This is determined by the larger of the degrees of YH and G ; the summand of smaller degree must be “filled out” with a suitable power of Z . This yields the two equations above. $\square$

Monomial projective curves

For the monomial plane curve

$$s \mapsto (s^e, s^d) = (x, y)$$

with coprime exponents $e > d$, Theorem 29.5 gives the monomial projective curve

$$(s, t) \mapsto (s^e, s^d t^{e-d}, t^e).$$

On the open set $D_+(t)$ this is the original map, whereas on $D_+(s)$ it becomes the affine map

$$t \mapsto (t^{e-d}, t^e).$$

Theorem 29.8: singularities of monomial projective curves

Let $e > d$ be coprime. For the monomial projective plane curve of degree e

$$C : (s, t) \mapsto (s^e, s^d t^{e-d}, t^e),$$

the following statements hold.

1. The curve is described by the homogeneous equation of degree e

$$Y^e = X^d Z^{e-d}.$$

2. The curve is smooth at all points other than $(0, 0, 1)$ and $(1, 0, 0)$.

3. The curve has multiplicity d at $(0, 0, 1)$ and multiplicity $e - d$ at $(1, 0, 0)$.
4. If $e \geq 3$, the curve is not smooth.

Proof

1. The affine equation is $X^d - Y^e$. By Corollary 28.10, the projective closure is described by its homogenisation, namely

$$V_+(X^d Z^{e-d} - Y^e).$$

Source correction AGC-CORR-0129 - projective zero-locus operator. The source prints V for the homogenisation defining the closure in $\mathbb{P}_K^2$; this edition uses V_+ . The affine loci below retain V .

2. On the affine curve

$$V(X^d - Y^e) \subseteq \mathbb{A}_K^2 \subseteq \mathbb{P}_K^2,$$

by the current source's [normalisation result for affine monomial curves](#), only the origin—corresponding to the projective point $(0, 0, 1)$ —can fail to be smooth. Points of the curve outside $D_+(Z)$ are obtained by setting $Z = 0$ in the equation. This forces $Y = 0$, leaving only the point $(1, 0, 0)$.

Semantic-source note REVIEW-AK-26-30-C15. The expanded 2012 witness cites Theorem 20.12 here; the frozen current proof instead links to the named normalisation result above. This edition preserves the current source target and the unchanged smoothness conclusion.

3. Multiplicity at a point is a local property. The point $(0, 0, 1)$ corresponds to the origin on the affine monomial curve

$$V(X^d - Y^e),$$

which, by Corollary 23.8, has multiplicity equal to the smaller exponent, namely d . The point $(1, 0, 0)$ lies in $D_+(X)$, where the affine equation is

$$V(Y^e - Z^{e-d}).$$

Its multiplicity is again the smaller exponent, namely $e - d$.

4. This follows from item 3. $\square$

Worksheet 29

Warm-up exercises

Exercise 29.1

Let K be an algebraically closed field. Show that every projective plane curve has nonempty intersection with every projective line in the projective plane.

Exercise 29.2 *

Let

$$K = \mathbb{Z}/(5),$$

and consider the two affine plane algebraic curves

$$C = V(X^2 + Y^2 - 1) \quad \text{and} \quad D = V(X^3 - 2Y^2 + 3).$$

- Determine the intersection $C \cap D$.
- Determine the points in

$$V_+(X^2 + Y^2 - Z^2) \setminus V(X^2 + Y^2 - 1).$$

- Determine the points in

$$V_+(X^3 - 2Y^2Z + 3Z^3) \setminus V(X^3 - 2Y^2 + 3).$$

- Is $V_+(X^2 + Y^2 - Z^2)$ the projective closure of $V(X^2 + Y^2 - 1)$?

Exercise 29.3 *

Let K be a field. Show that the local rings of the projective line $\mathbb{P}_K^1$ at all its K -rational points are isomorphic to one another. Give the simplest possible description of this ring.

Source-convention note REVIEW-AK-26-30-C17 - K -rational points. The frozen source-page identity explicitly says “ K -points”, although the exercise body says only “all local rings”. The restriction is essential: scheme-theoretic points of $\mathbb{P}_K^1$ include the generic point, whose local ring is not isomorphic to the local ring at a K -rational closed point.

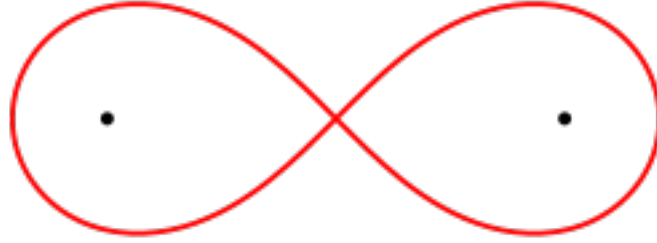


Figure 99: Bernoulli's lemniscate. By Zorgit, public domain.

Exercise 29.4

For Bernoulli's lemniscate given by

$$V((X^2 + Y^2)^2 - X^2 + Y^2),$$

determine its singularities and its points at infinity in $\mathbb{P}_{\mathbb{C}}^2$. Compute the multiplicity and tangent lines at all these points.

Exercise 29.5

Consider the projective curve

$$C \subset \mathbb{P}_K^2$$

over a field K of characteristic 0, given by the homogeneous equation

$$ZX^2 = Y^3.$$

- Determine the singular points of the curve.
- Show that the assignment

$$\varphi : (S, T) \mapsto (T^3, ST^2, S^3) = (X, Y, Z)$$

gives a well-defined map

$$\varphi : \mathbb{P}^1 \longrightarrow \mathbb{P}^2.$$

- c. Show that the image points of φ lie on the curve C .
- d. Which points in $\mathbb{P}^1$ correspond to the singular points of C ?

Exercises to submit

Exercise 29.6 (3 points)

Let $m + 1$ homogeneous polynomials

$$F_0, \dots, F_m$$

in $n + 1$ variables be given, all of the same degree d . Show that there is an open set

$$U \subseteq \mathbb{P}_K^n$$

on which these polynomials define a morphism

$$\mathbb{P}_K^n \supseteq U \longrightarrow \mathbb{P}_K^m.$$

Exercise 29.7 (3 points)

Let

$$P = (a_0, \dots, a_n) \in \mathbb{P}_K^n$$

be a point in projective space. Show that projection from $\mathbb{P}_K^n$ to $\mathbb{P}_K^{n-1}$ with centre P is given by a matrix. The source does not supply that matrix. It then displays the map

$$\begin{pmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{pmatrix} \mapsto \begin{pmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Source discrepancy AGC-U29-SRC-002. The matrix block in the source is blank, and the subsequent vector is repeated unchanged, so it does not describe a projection to $\mathbb{P}_K^{n-1}$. This edition preserves both facts and does not guess the intended matrix or map.

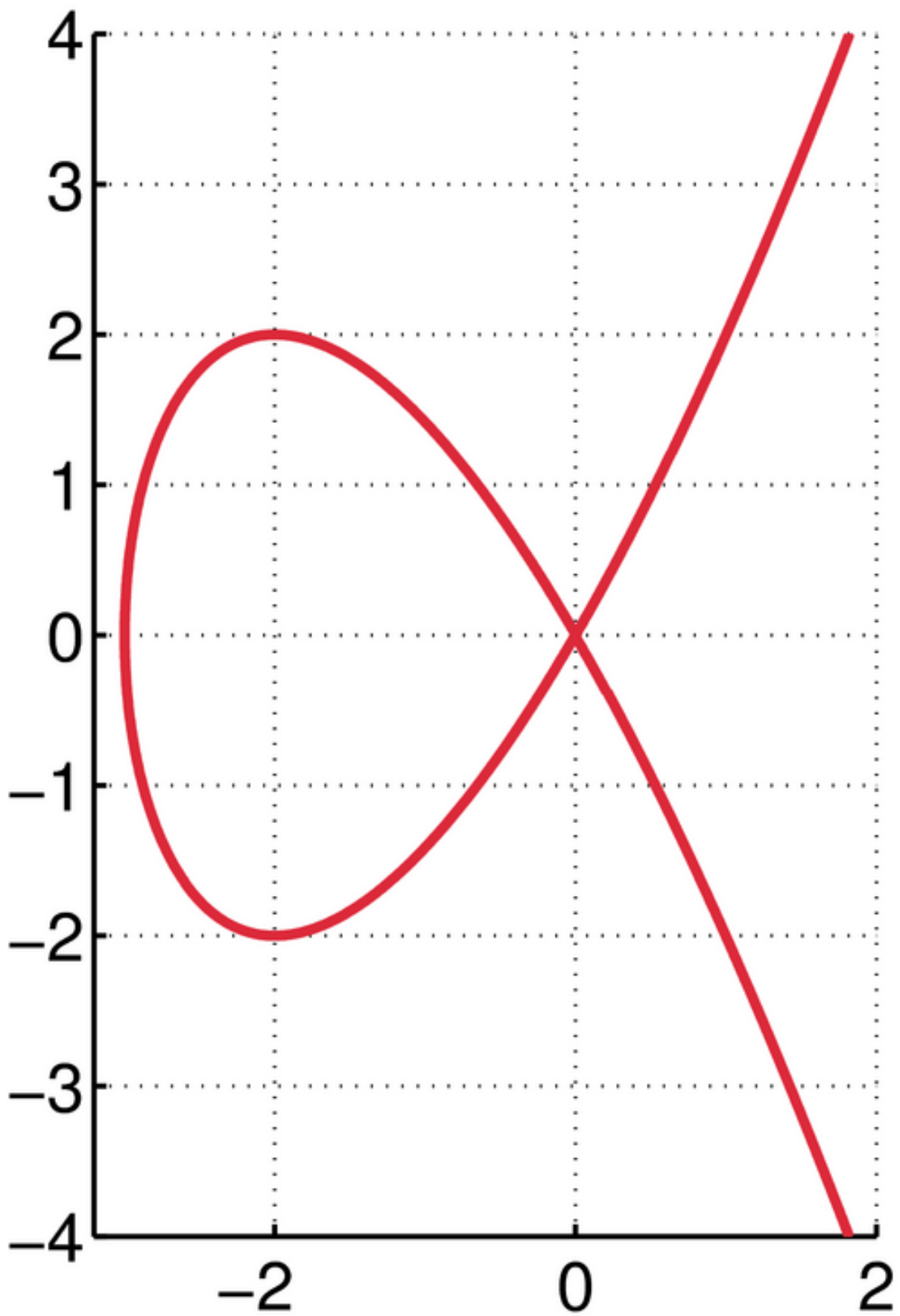


Figure 100: “Tschirnhausen cubic” according to the course’s inline caption. By Oleg Alexandrov, public domain.

Media component note. The Commons description page for this file warns that, despite its filename, the curve in the image is not a Tschirnhausen cubic because the angle of intersection at its double point differs. The image actually used by the source is retained.

Exercise 29.8 (3 points)

For the Tschirnhausen cubic given by

$$V(X^3 + 3X^2 - Y^2),$$

determine its singularities, including points at infinity. Determine the tangent lines at the singularities and at the points at infinity.

Exercise 29.9 (3 points)

For the folium of Descartes defined by

$$V(X^3 + Y^3 - 3XY),$$

determine its points at infinity in $\mathbb{P}_{\mathbb{C}}^2$, and compute the multiplicity and tangent lines at those points.

Exercise 29.10 (5 points)

Let K be an algebraically closed field of characteristic different from 2. For the projective Bernoulli lemniscate

$$V_+((X^2 + Y^2)^2 - Z^2X^2 + Z^2Y^2) \subset \mathbb{P}_K^2,$$

give a surjective morphism to a projective conic. How many points of the lemniscate map to a single point of the conic?

Source-condition note REVIEW-AK-26-30-C18 - base field. The source supplies no restriction on K . Algebraic closure is needed for the intended pointwise surjectivity, while in characteristic 2 the conic obtained from the standard squared-coordinate construction degenerates and its fibre count changes. The stated hypotheses preserve the intended smooth-conic problem.

Public Solutions to Worksheet 29

At the frozen revision boundary, the source provides public solutions only for Exercises 29.2 and 29.3. The frozen authority query reports the other eight candidate solution pages as absent. No additional solutions have been created for this edition.

Solution to Exercise 29.2

a. We add the two equations

$$2X^2 + 2Y^2 - 2 = 0 \quad \text{and} \quad X^3 - 2Y^2 + 3 = 0$$

and obtain the condition

$$X^3 + 2X^2 + 1 = 0.$$

For the possible values $x = 0, 1, 2, 3, 4$, substitution gives $1, 4, 2, 1, 2$, respectively. Thus this condition cannot be satisfied, so the intersection of the two curves in $\mathbb{A}_K^2$ is empty.

b. We seek the points in $\mathbb{P}_K^2$ satisfying simultaneously

$$X^2 + Y^2 - Z^2 = 0 \quad \text{and} \quad Z = 0.$$

This gives the condition

$$X^2 + Y^2 = 0.$$

The squares in $\mathbb{Z}/(5)$ are $0, 1, 4$. The solution $(0, 0, 0)$ is not allowed, since it does not represent a projective point, and we obtain the solutions $(\pm 1, \pm 2)$ and $(\pm 2, \pm 1)$. Since we seek projective points, the first coordinate can be normalised to 1, and the second must then be 2 or $-2 = 3$. Hence there are two points at infinity,

$$(1, 2, 0) \quad \text{and} \quad (1, 3, 0).$$

c. The two equations

$$X^3 - 2Y^2Z + 3Z^3 = 0 \quad \text{and} \quad Z = 0$$

immediately give $X^3 = 0$, and hence $X = 0$. Thus the unique point at infinity is $(0, 1, 0)$.

d. Here, as throughout this exercise, V and V_+ denote sets of K -rational zeros with their point-set Zariski topology. By definition, the projective closure is the Zariski closure. Since $V(X^2 + Y^2 - 1)$ is a finite set of K -rational points, it is already closed and equals its closure. However, by part b, $V_+(X^2 + Y^2 - Z^2)$ contains additional points. The latter set is therefore not its projective closure.

Source-convention note REVIEW-AK-26-30-C16 - finite-field point sets. The source's conclusion uses its classical K -point convention. It is not a claim that the scheme-theoretic projective closure of the affine conic is merely the finite set of its K -rational points.

Solution to Exercise 29.3

Every K -rational point $P \in \mathbb{P}_K^1$ lies on an affine line

$$P \in \mathbb{A}_K^1 = D_+(L) \subset \mathbb{P}_K^1,$$

where L is a homogeneous linear form. By translating on this affine line, we may further assume that the point in question is the origin. This can be done for every K -rational point and does not change its local ring. Therefore all these local rings are isomorphic to one another. The local ring at the origin of the affine line is the localisation

$$K[X]_{(X)}.$$

Lecture 30: Bézout's Theorem

In this lecture we prove Bézout's Theorem for the projective plane. It states that, for two projective curves in the projective plane with no common component and of degrees m and n , respectively, the sum of all their intersection multiplicities equals mn . Our presentation largely follows Fulton's treatment.

Homogeneous components of a fixed degree

For a polynomial ring P and a natural number ℓ , we shall write P_ℓ for the *homogeneous component of degree ℓ* , the space of all homogeneous polynomials of degree ℓ . We use the same notation for graded quotients of a polynomial ring, that is, quotients by homogeneous ideals. As a vector space over the base field K , this component is generated by all monomials of degree ℓ . In particular, P_ℓ is a finite-dimensional K -vector space.

Lemma 30.1: dimension of a homogeneous component of the quotient ring

Let K be a field, let

$$F, G \in K[X, Y, Z] = P$$

be homogeneous polynomials of degrees m and n , and suppose that F and G have no nonconstant common divisor. Then

$$\dim_K(P/(F, G))_\ell = mn$$

for sufficiently large ℓ .

Proof Consider the exact sequence

$$0 \longrightarrow P \xrightarrow{\begin{pmatrix} G \\ -F \end{pmatrix}} P \times P \xrightarrow{(F, G)} P \longrightarrow P/(F, G) \longrightarrow 0.$$

The first map is

$$H \mapsto (GH, -FH),$$

the next map is

$$(A, B) \mapsto AF + BG,$$

and the last map takes residue classes. All these maps are P -module homomorphisms. Injectivity at the first term is clear because P is an integral domain. The sequence is clearly exact at the last two terms; only exactness at the second term remains to be proved.

The composition of the first two maps is zero. Conversely, suppose

$$AF + BG = 0$$

in P . Since P is a unique factorisation domain and F, G are coprime, A must be a multiple of G . Write $A = QG$. The equation above becomes

$$G(QF + B) = 0,$$

so $B = -QF$. Thus

$$(A, B) = Q(G, -F)$$

comes from the map on the left.

Source correction AGC-CORR-0130 - sign in the relation. The source says that B is a multiple of F “with the same factor”, but the sign imposed by $AF + BG = 0$ gives $B = -QF$. This edition displays the minus sign explicitly, in agreement with the map $H \mapsto (GH, -FH)$ already printed in the source.

Since F and G are homogeneous of fixed degrees, the sequence restricts to homogeneous components. This gives the exact sequence

$$0 \longrightarrow P_{\ell-m-n} \xrightarrow{\begin{pmatrix} G \\ -F \end{pmatrix}} P_{\ell-m} \times P_{\ell-n} \xrightarrow{(F,G)} P_{\ell} \longrightarrow (P/(F,G))_{\ell} \longrightarrow 0,$$

where components with negative indices are defined to be 0. The restricted sequence remains exact because the homogeneous components of a homogeneous homomorphism do not mix. All the components involved are now finite-dimensional vector spaces.

If $\ell \geq m + n$, all indices are nonnegative, and

$$\dim_K(P_{\ell}) = \frac{(\ell+1)(\ell+2)}{2}.$$

By additivity of vector-space dimension in exact complexes (see Exercise 14.7), we obtain

$$\begin{aligned} \dim_K((P/(F,G))_{\ell}) &= \frac{(\ell+1)(\ell+2)}{2} - \frac{(\ell-m+1)(\ell-m+2)}{2} \\ &\quad - \frac{(\ell-n+1)(\ell-n+2)}{2} + \frac{(\ell-m-n+1)(\ell-m-n+2)}{2} \\ &= \frac{2 - (-m+1)(-m+2) - (-n+1)(-n+2) + (-m-n+1)(-m-n+2)}{2} \\ &= \frac{2mn}{2} \\ &= mn. \end{aligned}$$

□

Injectivity of multiplication by Z

Lemma 30.2: multiplication by Z in the quotient ring

Let K be an algebraically closed field and let

$$F, G \in K[X, Y, Z]$$

be homogeneous polynomials with no common projective zero on

$$V_+(Z) \subseteq \mathbb{P}_K^2.$$

Write the corresponding quotient ring as

$$R = K[X, Y, Z]/(F, G).$$

Then the map

$$\begin{aligned} R &\longrightarrow R, \\ H &\longmapsto ZH \end{aligned}$$

is injective.

Proof Take $H \in K[X, Y, Z]$ and suppose its class maps to 0. This means there are $L, M \in K[X, Y, Z]$ such that

$$ZH = LF + MG.$$

Substitute $Z = 0$ in this equation. In $K[X, Y]$, we obtain

$$0 = L(X, Y, 0)F(X, Y, 0) + M(X, Y, 0)G(X, Y, 0).$$

Since F and G have no common projective zero on $V_+(Z)$, the polynomials $F(X, Y, 0)$ and $G(X, Y, 0)$ have only $(0, 0)$ as a common zero in $\mathbb{A}_K^2$. They are therefore coprime in $K[X, Y]$. Hence there is a $Q \in K[X, Y]$ with

$$L(X, Y, 0) = QG(X, Y, 0), \quad M(X, Y, 0) = -QF(X, Y, 0).$$

Lifting back to $K[X, Y, Z]$, this means

$$L = QG(X, Y, 0) + Z\bar{L}, \quad M = -QF(X, Y, 0) + Z\bar{M}$$

for some $\bar{L}, \bar{M} \in K[X, Y, Z]$. Writing

$$F = F(X, Y, 0) + Z\bar{F}, \quad G = G(X, Y, 0) + Z\bar{G},$$

the original equation gives

$$\begin{aligned} ZH &= LF + MG \\ &= (QG(X, Y, 0) + Z\bar{L})F + (-QF(X, Y, 0) + Z\bar{M})G \\ &= Q(G - Z\bar{G})F - Q(F - Z\bar{F})G + Z\bar{L}F + Z\bar{M}G \\ &= -QZ\bar{G}F + QZ\bar{F}G + Z\bar{L}F + Z\bar{M}G \\ &= Z(-Q\bar{G}F + Q\bar{F}G + \bar{L}F + \bar{M}G). \end{aligned}$$

We can cancel Z from this equation in the polynomial ring, obtaining an expression for H as a linear combination of F and G . Thus the class of H in R is also 0. $\square$

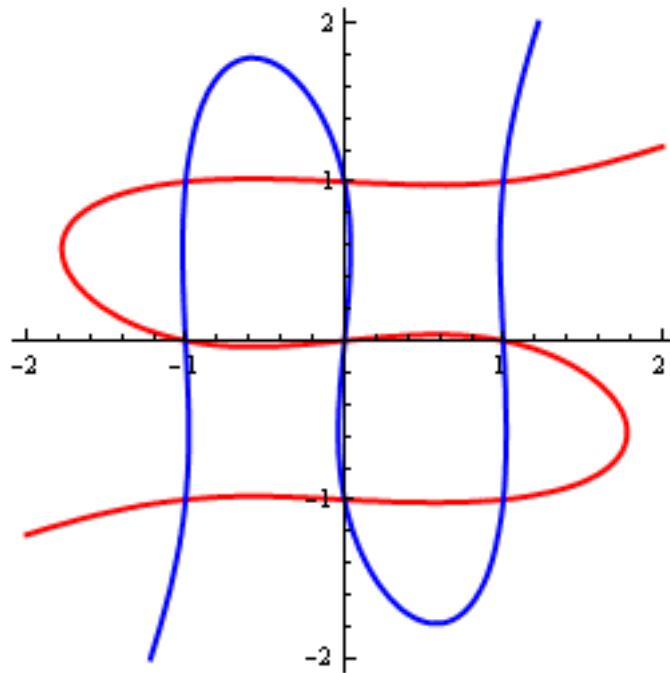


Figure 101: Two cubic curves. By Hack, created with Mathematica 6, public domain.

Bézout's Theorem

We now come to Bézout's Theorem.

Theorem 30.3: Bézout's Theorem

Let K be an algebraically closed field and let

$$F, G \in K[X, Y, Z]$$

be homogeneous polynomials of degrees m and n with no common component, with corresponding curves

$$C = V_+(F), \quad D = V_+(G) \subseteq \mathbb{P}_K^2.$$

Then

$$\sum_P \text{mult}_P(C, D) = mn.$$

Proof The intersection $C \cap D$ consists of only finitely many points. By Exercise 27.10, after a projective change of coordinates we may assume that all intersection points lie in

$$\mathbb{A}_K^2 = D_+(Z) \subseteq \mathbb{P}_K^2.$$

Let $\tilde{F}, \tilde{G} \in K[X, Y]$ be the dehomogenisations describing the affine curves $C \cap \mathbb{A}_K^2$ and $D \cap \mathbb{A}_K^2$. Then

$$\begin{aligned} \sum_{P \in \mathbb{P}_K^2} \text{mult}_P(F, G) &= \sum_{P \in \mathbb{A}_K^2} \text{mult}_P(\tilde{F}, \tilde{G}) \\ &= \sum_{P \in \mathbb{A}_K^2} \dim_K(K[X, Y]_{\mathfrak{m}_P} / (\tilde{F}, \tilde{G})) \\ &= \dim_K(K[X, Y] / (\tilde{F}, \tilde{G})). \end{aligned}$$

The last equality follows from Theorem 26.11. We shall relate the K -dimension of this inhomogeneous quotient ring to the dimension of a homogeneous component of the quotient ring

$$(K[X, Y, Z] / (F, G))_\ell.$$

By Lemma 30.1, for sufficiently large ℓ the latter component has dimension mn .

Choose a basis

$$V_1, \dots, V_{mn}$$

of $(K[X, Y, Z] / (F, G))_\ell$, with ℓ sufficiently large and fixed. We claim that the dehomogenisations

$$v_i = V_i(X, Y, 1), \quad i = 1, \dots, mn,$$

form a basis of

$$K[X, Y] / (\tilde{F}, \tilde{G}).$$

First we prove that these elements generate. Take any $q \in K[X, Y]$ and let its homogenisation

$$Q \in K[X, Y, Z]$$

have degree d . Choose an integer $e \geq 0$ such that $d + e \geq \ell$. By Lemma 30.2, for every $\lambda \geq 1$ the map

$$\begin{aligned} (K[X, Y, Z]/(F, G))_\ell &\longrightarrow (K[X, Y, Z]/(F, G))_{\ell+\lambda}, \\ H &\longmapsto Z^\lambda H \end{aligned}$$

is injective. Since both spaces have dimension mn , it is also bijective. With $\lambda = d + e - \ell$, the elements

$$Z^\lambda V_i, \quad i = 1, \dots, mn,$$

form a basis of the component of degree $\ell + \lambda = d + e$.

Source correction AGC-CORR-0131 - the boundary case $\lambda = 0$. The source chooses $d + e \geq \ell$, but states the assertion about multiplication by Z^λ only for $\lambda \geq 1$. If $d + e = \ell$, then $\lambda = 0$ and the required map is the identity; if $d + e > \ell$, the source's argument applies with $\lambda \geq 1$. Distinguishing these cases makes the basis conclusion above valid for every choice with $d + e \geq \ell$.

Consequently there are $a_1, \dots, a_{mn} \in K$ such that

$$Z^e Q = \sum_{i=1}^{mn} a_i Z^{d+e-\ell} V_i.$$

Dehomogenising this equation immediately expresses q as a linear combination of $v_1, \dots, v_{mn}$.

To prove linear independence, suppose

$$\sum_{i=1}^{mn} a_i v_i = 0$$

in the quotient ring. Then in $K[X, Y]$ there is an equation

$$\sum_{i=1}^{mn} a_i v_i = \tilde{A}\tilde{F} + \tilde{B}\tilde{G}.$$

Take homogeneous polynomials $A, B \in K[X, Y, Z]$ with dehomogenisations $\tilde{A}, \tilde{B}$, respectively. Thus we have two expressions with the same dehomogenisation: $\sum_i a_i V_i$, homogeneous of degree ℓ , and $AF + BG$, a sum of two homogeneous polynomials whose degrees may differ.

By choosing suitable nonnegative integers r, s, t , we can make

$$\sum_{i=1}^{mn} a_i Z^r V_i \quad \text{and} \quad Z^s AF + Z^t BG$$

homogeneous of the same degree. By Exercise 6.9, equality of their dehomogenisations then implies

$$\sum_{i=1}^{mn} a_i Z^r V_i = Z^s AF + Z^t BG.$$

In $K[X, Y, Z]/(F, G)$, this equation means

$$\sum_{i=1}^{mn} a_i Z^r V_i = 0.$$

Multiplication by Z^r is injective: for $r = 0$ it is the identity, whereas for $r > 0$ this follows by repeated application of Lemma 30.2. Since $V_1, \dots, V_{mn}$ form a basis, all $a_i = 0$. Thus $v_1, \dots, v_{mn}$ do indeed form a basis, so

$$\dim_K(K[X, Y]/(\tilde{F}, \tilde{G})) = mn.$$

This proves the formula in Bézout's Theorem. $\square$

Corollary 30.4: existence of an intersection point

Let K be an algebraically closed field and let

$$C, D \subseteq \mathbb{P}_K^2$$

be projective plane curves. Then

$$C \cap D \neq \emptyset.$$

Proof The assertion is clearly true if C and D have a common component. Otherwise, it follows from Theorem 30.3. $\square$

Corollary 30.5: at most mn intersection points

Let K be an algebraically closed field and let

$$F, G \in K[X, Y, Z]$$

be homogeneous polynomials of degrees m and n with no common component, with corresponding curves

$$C = V_+(F), \quad D = V_+(G) \subseteq \mathbb{P}_K^2.$$

Then C and D have at most mn intersection points.

Proof This follows directly from Theorem 30.3, since each intersection point contributes at least 1 to the sum of intersection multiplicities. $\square$

Example: the semicubical parabola and a circle**Example 30.6: five points with total multiplicity six**

In this example we work over $\mathbb{C}$. Consider the semicubical parabola

$$C = V_+(ZY^2 - X^3)$$

and the circle with centre $(1, 0, 1)$,

$$D = V_+((X - Z)^2 + Y^2 - Z^2).$$

Source correction AGC-CORR-0132 - base field of the example. The source does not specify a base field, but its calculation uses $\sqrt{2}i$, and the displayed distinct-point description requires the absence of exceptional-characteristic issues. This edition sets $K = \mathbb{C}$, so the entire calculation, the five distinct points, and the application of Bézout's Theorem lie within a valid scope. The source equations are unchanged.

By Bézout's Theorem, we expect a total intersection multiplicity of 6. Let us compute the intersection points. If $Z = 0$, the first equation gives $X = 0$, and the second then gives $Y = 0$. This does not define a projective point, so there is no intersection point on the projective line $V_+(Z)$.

We therefore consider the affine equations

$$Y^2 - X^3 = 0, \quad (X - 1)^2 + Y^2 - 1 = 0.$$

Substituting

$$Y^2 = 1 - (X - 1)^2$$

into the first equation gives

$$\begin{aligned} 1 - (X - 1)^2 - X^3 &= -X^3 - X^2 + 2X \\ &= X(-X^2 - X + 2) \\ &= -X(X - 1)(X + 2). \end{aligned}$$

Thus the intersection points are

$$(0, 0), \quad (1, 1), \quad (1, -1), \quad (-2, 2\sqrt{2}i), \quad (-2, -2\sqrt{2}i).$$

The last two points also show why the algebraically closed field hypothesis is necessary: they would be absent if we worked only over $\mathbb{R}$. Hence there are five distinct intersection points. The semicubical parabola is singular at the origin, which is also an intersection point, so the intersection multiplicity there must exceed 1. To confirm this, consider

$$\begin{aligned} &K[X, Y]_{(X, Y)} / (Y^2 - X^3, Y^2 - 1 + (X - 1)^2) \\ &= K[X, Y]_{(X, Y)} / (Y^2 - X^3, X(X - 1)(X + 2)) \\ &= K[X, Y]_{(X, Y)} / (Y^2 - X^3, X) \\ &= K[X, Y]_{(X, Y)} / (Y^2, X) \\ &= K[Y] / (Y^2). \end{aligned}$$

Here we repeat the elimination above and then use the fact that $X - 1$ and $X + 2$ are units in the local ring $K[X, Y]_{(X, Y)}$. The dimension is 2. Thus the intersection multiplicity at the origin is 2, while it is 1 at each of the other four points, as can also be checked directly. The sum is

$$2 + 1 + 1 + 1 + 1 = 6 = 3 \cdot 2,$$

exactly as asserted by Bézout's Theorem.

Worksheet 30: Bézout's Theorem

Warm-up exercises

Exercise 30.1

Give an example showing that Lemma 30.2 fails if the assumption that the base field is algebraically closed is omitted.

Exercise 30.2

Show that the two curves in Example 30.6 intersect transversely at every computed intersection point other than $(0, 0)$.

Exercise 30.3 *

Let $K = \mathbb{C}$. For the two affine curves

$$V(Y - X^3) \quad \text{and} \quad V(Y^2 - X^3),$$

determine all their intersection points and the intersection multiplicity at each point. Also examine intersection points in $\mathbb{P}_{\mathbb{C}}^2$ and verify Bézout's Theorem in this example.

Exercise 30.4 *

Let $K = \mathbb{C}$. Consider the two plane algebraic curves

$$C = V(X - Y^2) \quad \text{and} \quad D = V(Y^2 - X^5).$$

Determine all intersection points of the two curves in the affine plane and compute the intersection multiplicity at each. Also determine the points at infinity of both curves, namely the additional points on the projective closures $\overline{C}$ and $\overline{D}$, and check for intersections at infinity. Finally, verify Bézout's Theorem in this example.

Source discrepancy AGC-U30-SRC-001. The semantic source-page title names the first curve as “ $Y = X^2$ ”, whereas the formula displayed in both the exercise and the source solution is $X = Y^2$. This edition follows the formula actually displayed and does not alter the frozen source-page identity.

Exercises to submit**Exercise 30.5 (6 points)**

Let

$$C \subseteq \mathbb{P}_K^2$$

be a smooth conic, that is, a curve of degree two, over an algebraically closed field. Show that C is isomorphic to the projective line $\mathbb{P}_K^1$.

Exercise 30.6 (5 points)

Let K be an algebraically closed field and let

$$C \subset \mathbb{P}_K^2$$

be a smooth curve of degree $d \geq 2$. Show that there is a morphism

$$C \longrightarrow \mathbb{P}_K^1$$

such that each fibre consists of at most $d - 1$ points.

Exercise 30.7 (5 points)

Let

$$C = V_+(X^3 + Y^3 + Z^3) \subseteq \mathbb{P}_K^2$$

be the Fermat cubic over an algebraically closed field of characteristic different from 3. Describe explicitly a morphism

$$C \longrightarrow \mathbb{P}_K^1$$

whose fibres each contain at most two points.

Exercise 30.8 (4 points)

Let

$$C \subseteq \mathbb{P}_{\mathbb{C}}^2$$

be the complex projective closure of the unit circle. Determine an explicit bijective parametrisation

$$\mathbb{P}_{\mathbb{C}}^1 \longrightarrow C.$$

Exercise 30.9 (4 points)

Over $\mathbb{C}$, verify Bézout's Theorem for the two projective plane curves

$$C = V_+(ZY^2 - X^3)$$

and

$$D = V_+(X^2 + (Y - Z)^2 - Z^2).$$

Sketch the situation.

Exercise 30.10 (4 points)

Over $\mathbb{C}$, verify Bézout's Theorem for the two projective plane curves

$$C = V_+(ZY - X^2)$$

and

$$D = V_+(X^2 + (Y - Z)^2 - Z^2).$$

Sketch the situation.

Scope note AGC-CORR-0133. The source of Exercises 30.9 and 30.10 does not specify a base field. This edition explicitly states $\mathbb{C}$, in keeping with the geometric interpretation and the instruction to sketch; in characteristic 2, the second quadratic form degenerates to the square of a line, so the intended calculation is no longer the same.

Exercise 30.11 (4 points)

Over an algebraically closed field K , verify Bézout's Theorem for the two monomial curves given in affine form by

$$C = V(X^2 - Y^3)$$

and

$$D = V(X^5 - Y^4).$$

Source-condition note REVIEW-AK-26-30-C19 - base field. The source does not specify the base field, while Bézout's Theorem 30.3 is stated over an algebraically closed field. This edition makes that ambient hypothesis explicit without imposing an unnecessary characteristic restriction.

Exercise 30.12 (5 points)

Let R be a commutative ring and let M, N be R -modules. If

$$f : M \longrightarrow N$$

is an R -module homomorphism, then the map

$$\begin{aligned} f^* : \operatorname{Hom}(N, R) &\longrightarrow \operatorname{Hom}(M, R), \\ \varphi &\longmapsto \varphi \circ f \end{aligned}$$

is also an R -module homomorphism.

Now let

$$0 \longrightarrow M \longrightarrow N \longrightarrow P \longrightarrow 0$$

be a short exact sequence of R -modules. Show that the induced sequence

$$0 \longrightarrow \operatorname{Hom}(P, R) \longrightarrow \operatorname{Hom}(N, R) \longrightarrow \operatorname{Hom}(M, R)$$

is exact. Also give an example with $R = \mathbb{Z}$ showing that the last arrow is not surjective in general.

Public Solutions to Worksheet 30

At the frozen revision boundary, the source provides public solutions only for Exercises 30.3 and 30.4. The frozen authority query reports the other ten candidate solution pages as absent. No additional solutions have been created for this edition.

Solution to Exercise 30.3

The intersection points of the two curves in the complex affine plane are given by

$$V(Y - X^3, Y^2 - X^3).$$

Thus an intersection point (x, y) must satisfy

$$y = x^3 \quad \text{and} \quad y^2 = x^3.$$

Substituting the first equation into the second gives

$$y(y - 1) = 0.$$

Hence $y = 0$ or $y = 1$. The intersection points are therefore

$$\{(0, 0), (1, 1), (\zeta, 1), (\zeta^2, 1)\},$$

where ζ is a primitive cube root of unity. At the origin, the quotient ring is

$$\begin{aligned} \mathbb{C}[X, Y]_{(X, Y)} / (Y - X^3, Y^2 - X^3) &\cong \mathbb{C}[X]_{(X)} / (X^3, X^6 - X^3) \\ &\cong \mathbb{C}[X] / (X^3). \end{aligned}$$

Its dimension as a complex vector space is 3, so the intersection multiplicity at the origin is 3.

To determine the multiplicities at the other three points, compute the gradients in the conventional coordinate order (X, Y) . For

$$F = Y - X^3, \quad G = Y^2 - X^3,$$

we obtain

$$\nabla F = (-3x^2, 1) \quad \text{and} \quad \nabla G = (-3x^2, 2y) = (-3x^2, 2).$$

Since $x \neq 0$, these two directions are linearly independent. Thus both curves are smooth and intersect transversely at the three points; each has intersection multiplicity 1.

Source correction AGC-CORR-0134 - order of gradient components. The source writes derivative components in the implicit order (Y, X) in one part of the solution, unlike the convention (X, Y) used elsewhere. This edition writes both gradients consistently in the order (X, Y) ; the transversality test and its result are unchanged.

In projective space, we homogenise the two ideals. Thus we consider

$$V_+(YZ^2 - X^3) \quad \text{and} \quad V_+(Y^2Z - X^3).$$

Setting $Z = 0$ gives $X = 0$, so the intersection point at infinity is $(0, 1, 0)$. The affine neighbourhood $D_+(Y)$ gives the equations

$$Z^2 - X^3 \quad \text{and} \quad Z - X^3.$$

At the origin of this chart, eliminating Z again gives a local quotient of dimension 3. Hence the intersection multiplicity at this point at infinity is also 3. The sum of all intersection multiplicities is

$$3 + 3 \cdot 1 + 3 = 9,$$

in agreement with the product of the two curve degrees, $3 \cdot 3 = 9$.

Solution to Exercise 30.4

Adding the two equations immediately gives the condition

$$0 = X - X^5 = X(1 - X^4).$$

Thus the x -coordinate of an intersection point is 0 or a fourth root of unity,

$$x \in \{0, 1, -1, i, -i\}.$$

If $x = 0$, we immediately obtain $y = 0$. The local quotient ring can be written as

$$\begin{aligned} \mathbb{C}[X, Y]_{(X, Y)} / (X - Y^2, Y^2 - X^5) &\cong \mathbb{C}[Y]_{(Y)} / (Y^2 - Y^{10}) \\ &= \mathbb{C}[Y]_{(Y)} / (Y^2(1 - Y^8)) \\ &\cong \mathbb{C}[Y]_{(Y)} / (Y^2). \end{aligned}$$

Since $1 - Y^8$ is a unit in this local ring, the intersection multiplicity at $(0, 0)$ is 2.

Source correction AGC-CORR-0135 - localisation subscript. In the middle line, the source prints $\mathbb{C}[Y]_Y$, which usually means inverting powers of Y and cannot be the local ring at the origin. This edition retains the localisation specified in the preceding line, $\mathbb{C}[Y]_{(Y)}$, making the unit and local-length argument valid.

Now suppose x is a fourth root of unity. Since $y^2 = x$, the number y is an eighth root of unity. If ζ is the first primitive eighth root of unity, the other eight intersection points are

$$\begin{aligned} &(1, 1), (1, -1), (i, \zeta), (i, -\zeta), \\ &(-1, i), (-1, -i), (-i, \zeta^3), (-i, -\zeta^3). \end{aligned}$$

We show that the intersection is transverse at all eight points, so each intersection multiplicity is 1. For

$$F = X - Y^2, \quad G = Y^2 - X^5,$$

the gradients are

$$\nabla F = (1, -2Y) \quad \text{and} \quad \nabla G = (-5X^4, 2Y).$$

At each point (x, y) above, $x \neq 0$, so both curves are smooth. Since $x^4 = 1$, the second gradient has the form $(-5, 2y)$. The two directions can be linearly dependent only if $-2y = -10y$, which is impossible because $y \neq 0$ over $\mathbb{C}$. Thus all eight intersections are transverse.

Finally, consider the points at infinity. The homogenisation of the first equation is

$$\tilde{F} = XZ - Y^2,$$

so the unique point at infinity on $\overline{C} = V_+(\tilde{F})$ is $(1, 0, 0)$. The homogenisation of the second equation is

$$\tilde{G} = Y^2Z^3 - X^5,$$

so the unique point at infinity on $\overline{D} = V_+(\widetilde{G})$ is $(0, 1, 0)$. These points differ, so there is no additional intersection at infinity.

The total intersection multiplicity is therefore

$$2 + 8 \cdot 1 = 10.$$

Since the curves have degrees 2 and 5, this sum equals the product of their degrees, $2 \cdot 5 = 10$, as asserted by Bézout's Theorem.

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3. **Several straight lines** – [File:Straight_lines.svg](#); creator/attribution: Lokilech; licence: [CC BY-SA 3.0](#).
4. **A linear space** – [File:Linear_space2.png](#); creator/attribution: kmhkmh; licence: [CC BY 3.0](#).
5. **Hydrant on the island of Krk, Croatia** – [File:Hydrant_Insel_Krk_Kroatien.jpg](#); creator/attribution: Usien; licence: [CC BY-SA 3.0](#).
6. **Principal directions on a cylinder** – [File:Cylinder_principal_directions.svg](#); creator/attribution: Luca Antonelli (Luke Antony); licence: [CC BY-SA 3.0](#).
7. **Two cubic curves** – [File:Two_cubic_curves.png](#); creator/attribution: Hack; licence: Public domain.
8. **A surface and a vertical line** – [File:Non_cohen_macaulay_scheme_thumb.png](#); creator/attribution: The original uploader was Jakob.scholbach at English Wikipedia.; licence: [CC BY-SA 3.0](#).
9. **Rectangular hyperbola** – [File:Rectangular_hyperbola.svg](#); creator/attribution: Qef; licence: Public domain.

Media credits for Unit 5

The credits follow the order of the images in Lecture 5 and Worksheet 5. Each component retains the rights status of its source.

1. **A cone (the zero locus of a homogeneous polynomial)** - [File:Kuzel_obecný.svg](#); creator/attribution: Pajs at Czech Wikipedia; rights status/licence: public domain.
2. **Emmy Noether (1882-1935)** - [File:Noether.jpg](#); creator/attribution: unknown creator (photograph taken before 1910); source/publisher recorded on the Commons page; rights status/licence: public domain.
3. **Triaxial ellipsoid with semiaxes in the ratio 3:2:1** - [File:Elipsoid_trojosy321.png](#); creator/attribution: Pajs at Czech Wikipedia; rights status/licence: public domain.

Media credits for Unit 6

The credits follow the order of the images in Lecture 6 and Worksheet 6. Each component retains the rights status of its source.

1. **A parametrised curve can be pictured as the path of a moving point** - [File:Krivka parametricky.png](#); creator/attribution: Beny at Czech Wikipedia; rights status/licence: public domain.
2. **Cubic curve with a double point** - [File:Cubic with double point.svg](#); creator/attribution: Gunther at German Wikipedia; rights status/licence: public domain.
3. **The cissoid of Diocles (shown in black) admits a rational parametrisation** - [File:Dioklova kisoida.png](#); creator/attribution: Pajs at Czech Wikipedia; rights status/licence: public domain.

Media credits for Unit 7

The credits follow the nine image positions in Lecture 7. Each component retains the rights status or licence of its source. The three orbit GIFs remain animated in the HTML reader; the PDF uses their locally derived first frames.

1. **Standard cone** - [File:DoubleCone.png](#); creator/attribution: Lars H. Rohwedder (User:RokerHRO); rights status/licence: Public domain.
2. **Sections of the standard cone by affine planes** - [File:Conic sections.svg](#); creator/attribution: Anuskafm; rights status/licence: [CC BY-SA 3.0](#).
3. **First stage of the principal-axis transformation of a quadric** - [File:Hauptachsentransformation1.png](#); creator/attribution: rdb; rights status/licence: Public domain.
4. **Second stage of the principal-axis transformation of a quadric** - [File:Hauptachsentransformation2.png](#); creator/attribution: rdb; rights status/licence: Public domain.
5. **Third stage of the principal-axis transformation of a quadric** - [File:Hauptachsentransformation3.png](#); creator/attribution: rdb; rights status/licence: Public domain.
6. **Portrait of an unknown man, formerly misidentified as Johannes Kepler** - [File:Portrait Confused With Johannes Kepler 1610.jpg](#); creator/attribution: Unidentified painter; rights status/licence: Public domain.
7. **Elliptic orbit** - [File:Elliptic orbit.gif](#); creator/attribution: Brandir; rights status/licence: [CC BY-SA 3.0](#).
8. **Parabolic orbit** - [File:Parabolic orbit.gif](#); creator/attribution: Brandir; rights status/licence: [CC BY-SA 3.0](#).
9. **Hyperbolic orbit** - [File:Hyperbolic orbit.gif](#); creator/attribution: Brandir; rights status/licence: [CC BY-SA 3.0](#).

Media credits for Unit 8

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1. **File:Lemniscate Building.gif** - [File:Lemniscate Building.gif](#); creator/attribution: Zorgit; licence: [CC BY-SA 3.0](#).
2. **File:Parallele lijnen.png** - [File:Parallele lijnen.png](#); creator/attribution: Ellywa at Dutch Wikipedia; licence: [CC BY-SA 3.0](#).
3. **File:Ellipse tri.png** - [File:Ellipse tri.png](#); creator/attribution: Original uploader was David Shai at he.wikipedia; licence: [CC BY-SA 3.0](#).
4. **File:Steam engine in action.gif** - [File:Steam engine in action.gif](#); creator/attribution: Panther; licence: [CC BY-SA 3.0](#).
5. **File:Intersection of cylinders.jpg** - [File:Intersection of cylinders.jpg](#); creator/attribution: Jan Schoenke; licence: [CC BY-SA 3.0](#).
6. **File:Alg Kurven OS2008 Lsg8.10 v2.png** - [Datei:Alg Kurven OS2008 Lsg8.10 v2.png](#); creator/attribution: Christian Boberg; licence: [CC BY-SA 3.0](#).

Media credits for Unit 9

The single substantive media position in Lecture 9 retains its Commons identity and source rights status. The two PDFs referenced by the page are official build witnesses recorded in the authority manifest, not separate reader media.

1. **David Hilbert (1886)** - [File:David Hilbert 1886.jpg](#); creator: unknown; Commons uploader: Jacek Halicki; rights status/licence: Public domain.

Media credits for Unit 11

The single substantive media position in Lecture 11 retains its Commons identity and source rights status. The two PDFs referenced by the page are official build witnesses recorded in the authority manifest, not separate reader media.

1. **Disjoint ellipses** - [File:Disjoint ellipses.png](#); creator: Pmidden; source credit: own work, created with Mathematica; rights status/licence: public domain. The reader file is the 250-pixel Commons rendering at

the source's original dimensions, not a claim of byte identity with the original file; the original file's identity is preserved in `authority/RIGHTS-unit-11.csv`.

Media credits for Unit 12

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1. **Alexander Grothendieck** — [File:Alexander Grothendieck.jpg](#); photograph: Konrad Jacobs; credit: Copyright by MFO / Oberwolfach Photo Collection; licence: [CC BY-SA 2.0 de](#); reader form: original file, 268 × 326.
2. **Horizontal line** — [File:Lineline.jpg](#); creator: Astur1; rights status: public domain; reader form: original file, 290 × 138.
3. **Cartesian graph of a linear function** — [File:Lineair-cartesiaans.png](#); creator/attribution: MADe at Dutch Wikipedia; licence: [CC BY-SA 3.0](#); reader form: official 250-pixel Commons derivative, with the original file's identity and metadata retained in the rights ledger.
4. **Graph of a polynomial of degree five** — [File:Polynomialdeg5.png](#); creator/attribution: Derbeth (on the basis of copyright claims in the source); licence: [CC BY-SA 3.0](#); reader form: official 120-pixel Commons derivative, with the original file's identity and metadata retained in the rights ledger.

Media credits for Unit 13

The credits follow the order of the images in Lecture 13. Each component retains its own source licence.

1. **Graph of the hyperbola $y = 1/x$** - [File:Hyperbola one over x.svg](#); creator/attribution: No machine-readable author provided. Ktims assumed (based on copyright claims).; licence/rights status: [CC BY-SA 3.0](#).
2. **Connected topological space (red) and disconnected space (green)** - [File:Connected and disconnected spaces2.svg](#); creator/attribution: Dbc334; licence/rights status: Public domain.

Media credits for Unit 14

The credits cover the single image in Lecture 14. The component retains its own source licence.

1. **Graph of a global function on two-dimensional affine space** - [File:Monkey Saddle Surface \(Shaded\).png](#); creator/attribution: Inductiveload; licence/rights status: Public domain.

Media credits for Unit 15

The credits cover the single image in Lecture 15. The component retains its own source licence.

1. **Schematic representation of a neighbourhood filter** - [File:Concentric Circles.svg](#); creator/attribution: Andreas Pietzowski; licence/rights status: [CC BY-SA 4.0](#).

Media credits for Unit 16

The credits follow the order of the images in Lecture 16 and Public Solution 16.12. Each component retains its own source licence.

1. **A filter can be identified by what it retains** - [File:Kaffeefilter.jpg](#); creator/attribution: Elke Wetzig (Elya); licence/rights status: [CC BY-SA 3.0](#).
2. **Not every function on the cone outside a line extends to the whole affine space with that line removed** - [File:Cone intersects line.png](#); creator/attribution: Pmidden; licence/rights status: Public domain.
3. **Fibres of a map; M is the codomain, and the domain is the union of all fibres** - [File:FiberBundle 2.png](#); creator/attribution: ja:user:132ninme; licence/rights status: [CC BY-SA 3.0](#).

4. **Sketch of the union of the three coordinate axes in Solution 16.12** - [File:Draft0.svg](#); creator/attribution: Kalan; licence/rights status: [CC BY-SA 3.0](#).

Media credits for Unit 17

Lecture 17, Worksheet 17, and the four frozen public solutions contain no reader media positions. The two official upstream PDFs are retained solely as authority witnesses and retain their own component rights and attributions, as recorded in `authority/ASSET_CLOSURE-unit-17.json`. The translated text remains under CC BY-SA 4.0; no blanket licence is claimed for the mixed collection.

Media credits for Unit 18

The single substantive media position in Lecture 18 retains its Commons identity and source licence. The two official PDFs are authority witnesses, not additional reader media positions.

1. **Neil's parabola with a cuspidal singularity at the origin** - [File:Cusp.svg](#); creator/attribution: Georg-Johann; licence/rights status: [CC BY-SA 3.0](#).

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Media credits for Unit 19

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1. **Twisted cubic curve** - [File:Twisted cubic curve.png](#); creator/attribution: Claudio Rocchini; licence/rights status: [CC BY 3.0](#).

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1. **Whitney umbrella** - [File:Whitney umbrella.png](#); creator/attribution: Claudio Rocchini; licence/rights status: [CC BY 2.5](#).

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Media credits for Unit 21

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Media credits for Unit 22

The seven substantive media positions in Lecture 22 and Worksheet 22 retain the identities, attributions, and licensing routes of their source components. The two official PDFs are authority/build witnesses, not additional reader media positions.

1. **Tangent to a curve** — [File:Tangent to a curve.svg](#); creator according to the current Commons record: Jacj, with later versions by Oleg Alexandrov; rights status: public domain.
2. **Three lines meeting at one point** — [File:3 equations -5.JPG](#); creator: Cronholm144; rights status: public domain.
3. **Portrait of René Descartes** — [File:Frans Hals - Portret van René Descartes.jpg](#); work after Frans Hals, reproduction source as recorded by Commons; rights status: public domain.
4. **Folium of Descartes** — [File:Kartesisches-Blatt.svg](#); creator: Georg-Johann; licensing route used: [CC BY-SA 3.0](#).
5. **Two curve components intersecting at a triple point** — [File:Intersect3.png](#); creator: Michael Larsen; licensing route used through GFDL migration on Commons: [CC BY-SA 3.0](#).
6. **A tangent to a circle is perpendicular to the radius** — [File:Cercle tangente rayon.svg](#); original work: Christophe Dang Ngoc Chan (Cdang), SVG conversion/derivative: Hagman; licensing route used: [CC BY-SA 3.0](#).
7. **Cardioid** — [File:Cardioid.svg](#); creator: D.328; reuse route displayed by Commons: [CC BY-SA 3.0](#), with the raw GFDL/CC BY-SA 2.1 JP and migration records retained in the rights ledger.

Reconciliation note: the inline Wikiversity credit for the tangent image names AxelBoldt, whereas the frozen Commons description names Jacj and Oleg Alexandrov. Commons component metadata governs attribution in this edition; the differing Wikiversity label is retained as provenance in the freeze, not republished as an authorship fact. Raw dual-licensing and migration records for the four licensed components are retained in `authority/RIGHTS-unit-22.csv`; the edition makes no blanket-licence claim for the mixed collection.

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Media credits for Unit 23

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Rights and accessibility note: the internal boilerplate in both PDFs states CC BY-SA 3.0, whereas the frozen course statements and Commons records identify both as CC BY-SA 4.0 components. The lecture PDF has seven pages; the worksheet PDF has five pages, of which the fourth is blank. Both are untagged and lack PDF document structure. The edition retains all these differences as provenance, does not treat the PDFs as current semantic copies, and does not count them as reader media positions.

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Media credits for Unit 24

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Rights note: the file-description pages for both 2012 PDFs retain the older CC BY-SA 2.0 Germany statement alongside the current print/course version statement, CC BY-SA 4.0. The edition records both routes and does not grant a new blanket licence over the PDF files. The lecture PDF has six pages and the worksheet PDF has two pages. Both have extractable text, but are untagged, have no structure tree, and specify no document language.

The lecture PDF also contains the historical form $G = x^2 + z^2 - 1$, whereas the frozen live semantic source contains the consistent form $G = y^2 + z^2 - 1$. Both the PDF and the live semantic source write the coefficient $a_{\ell+1}$ in a proof step that mathematically requires $b_{\ell+1}$. The reader follows the semantic source for the cylinder example and openly marks the coefficient correction; the PDF is not treated as a current semantic copy or as a reader media position.

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Media credits for Unit 25

Lecture 25, Worksheet 25, and both frozen public solutions contain no substantive reader media positions. Exercise 25.13 asks the reader to draw an image using suitable software, but the source supplies no image file for that exercise. The two official upstream PDFs are retained solely as visual/build authority witnesses and retain their own component rights and attributions, as recorded in `authority/ASSET_CLOSURE-unit-25.json` and `authority/RIGHTS-unit-25.csv`.

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Media credits for Unit 26

Lecture 26 contains one substantive reader media position. Worksheet 26 and its public solution contain no additional reader media positions. The two official upstream PDFs are retained as visual/build authority witnesses, not as reader media positions.

1. **Transverse and non-transverse intersections** - [File:Intersect3.png](#), image created by Michael Larsen and uploaded to Commons by Maksim. The reader uses the official Commons thumbnail, 250 x 249 pixels, stored as `authority/assets/250px-Intersect3.png`, 5,922 bytes, SHA-256 `b29c15edf6619632fe033e0b6064c1826226abce0be6219262ca028a2a157818`. The licensing route used is [CC BY-SA 3.0](#).

The Commons metadata freezes the `File:Intersect3.png` description page at revision 475878038. The original file is 400 x 399 pixels, 5,018 bytes, and has SHA-1 `26fef135fcc9d950958068778e5805830ffa8b8e`. The inline source credits name Michael Larsen and Maksim separately; the rights ledger preserves those creator and uploader relationships without conflating them.

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Media credits for Unit 27

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2. **Projective line, illustration 1** - [File:Projektiveline1bb.jpg](#), created and uploaded by Darapti, CC BY-SA 3.0. Local file: 723,518 bytes, SHA-256 f6feb42ac3b37d3d1ed8b7826f901b88fac64c0a2fdd8245a178b18e068b61b6.
3. **Projective line, illustration 2** - [File:Projektiveline2bb.jpg](#), created and uploaded by Darapti, CC BY-SA 3.0. Local file: 777,323 bytes, SHA-256 40664e9c2f5f63fa03947df2bb120bc5427380cc7db8c14d39f4cc415c6beaea.
4. **Projective line, illustration 3** - [File:Projektiveline3bb.jpg](#), created and uploaded by Darapti, CC BY-SA 3.0. Local file: 808,222 bytes, SHA-256 ef5b0c0b0e34a7b947a946bda22d04d5ed11ecb2b88d5e0256f22dcd28e594b2.
5. **Projective plane, illustration 1** - [File:Projektiveplane1bb.jpg](#), created and uploaded by Darapti, CC BY-SA 3.0. Local file: 737,889 bytes, SHA-256 5845a36699676293242090f7499daf7f033aed559590b2e61cdab0f4c635f23c.
6. **Projective plane, illustration 2** - [File:Projektiveplane2bb.jpg](#), created and uploaded by Darapti, CC BY-SA 3.0. Local file: 789,468 bytes, SHA-256 784560087dcdcae0ef62505c865bd86db3ebf72a71fb674615ed6e55e2ca782d.
7. **Projective plane, illustration 3** - [File:Projektiveplane3bb.jpg](#), created and uploaded by Darapti, CC BY-SA 3.0. Local file: 801,410 bytes, SHA-256 7a5642c90b40cae72a66b4f2cfc1804e1a0abf4a64f9d41769b1e5694b824afb.
8. **Projective plane, illustration 4** - [File:Projektiveplane4bb.jpg](#), created and uploaded by Darapti, CC BY-SA 3.0. Local file: 839,639 bytes, SHA-256 92ad7dff6cdf2ba0fb6b361444cfd4ed9663a5ae59b1d555604533025fdafd9.
9. **Principle of perspective projection** - [File:Perspective Projection Principle.jpg](#), image by Joachim Baecker and upload by Fantagu, CC BY-SA 3.0. Local file: 53,956 bytes, SHA-256 38ced602230489102e8cd69886e3dfd609539606ac4c702d7c930b384d119a6c.
10. **Blue sphere** - [File:Blue-sphere.png](#), created by Lucas Vieira, uploaded as LucasVB, public domain. The historical course credit names user Kieff; both sides of that relationship are preserved. Local file: 123,336 bytes, SHA-256 5024f8120e040b6f5d40141cc3109c703eda56cb8cdf133d0ef0d11ee7dbb733.

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The official Lecture 27 PDF has nine pages, 171,996 bytes, SHA-256 0d4402bfae46abd09cb4719110a006287b03de31b0e620e0157a4ef9a07817f2. The official Worksheet 27 PDF has two pages, 41,952 bytes, SHA-256 e1fa608c2b54c988f16d0c0b2119f1d21440b37debbf87d89b7bbf228c6bdf9d. Neither is encrypted, and their text is extractable, but they are untagged, have no structure tree or document language, and have no bookmarks. The file-description pages retain the older CC BY-SA 2.0 Germany route and the current CC BY-SA 4.0 print/course route; this edition makes no blanket licensing claim.

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Media credits for Unit 28

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1. **Football pattern or sphere of genus zero** - [File:Soccerball.svg](#), originating from OpenClipart; the frozen Commons page does not name a creator and records the current uploader as MapGrid. This file is under [CC0 1.0](#). The older inline course credit names user Ranveig and the label PD; both records are preserved without recasting Ranveig as the creator. Local file: 1,311 bytes, SHA-256 0405cfe3c75353882ffecdbd4c8514bba49954482109c5f06f760d6a93b70e7.
2. **Torus, genus one** - [File:Torus illustration.png](#), created and uploaded by Oleg Alexandrov, public domain. Local file: 150,645 bytes, SHA-256 f5c22545e3dbdf4e056c4439d63bdd41f029589893e178502d460576c44d78b7.
3. **Double torus, genus two** - [File:Double torus illustration.png](#), created and uploaded by Oleg Alexandrov, public domain. Local file: 266,030 bytes, SHA-256 1a7664a61899dd83245760b2842c6f1b8c2f18887bb507fdb5405acd1d8a038.
4. **Sphere with three handles, genus three** - [File:Sphere with three handles.png](#), work by Oleg Alexandrov created using MATLAB, public domain. Local file: 398,740 bytes, SHA-256 49398059697841332186226bccf87e46032289c1a8a02063fc2600070c51a311.

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83,502 bytes, SHA-256 f3dda9da65db9e431f25ea77eb83f51aed2eff1c191dc1206e0759561ee613c7, at authority/assets/Tschirnhausen_cubic-500.png.

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